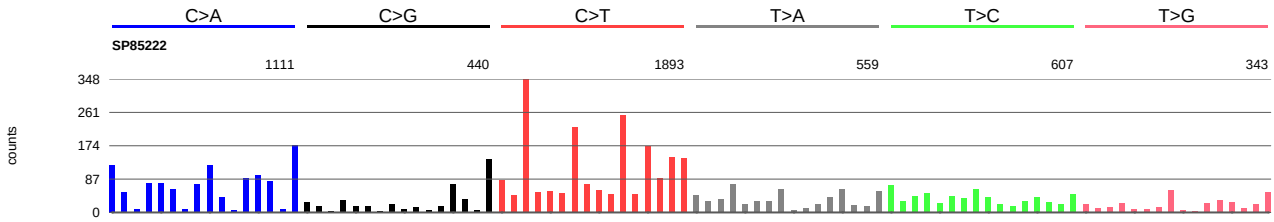
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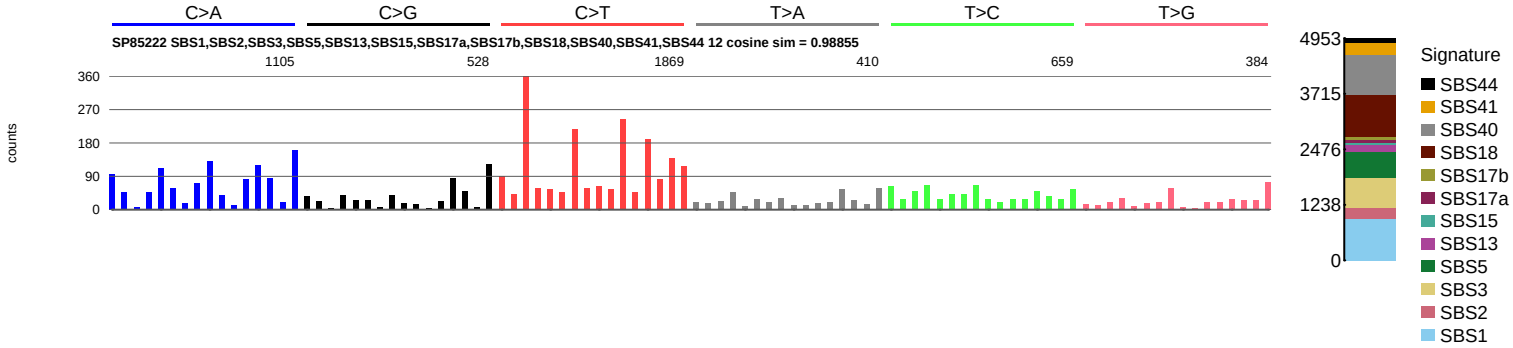
SP85222
1111
440
1893
559
607
343
counts
348
261
174
87
0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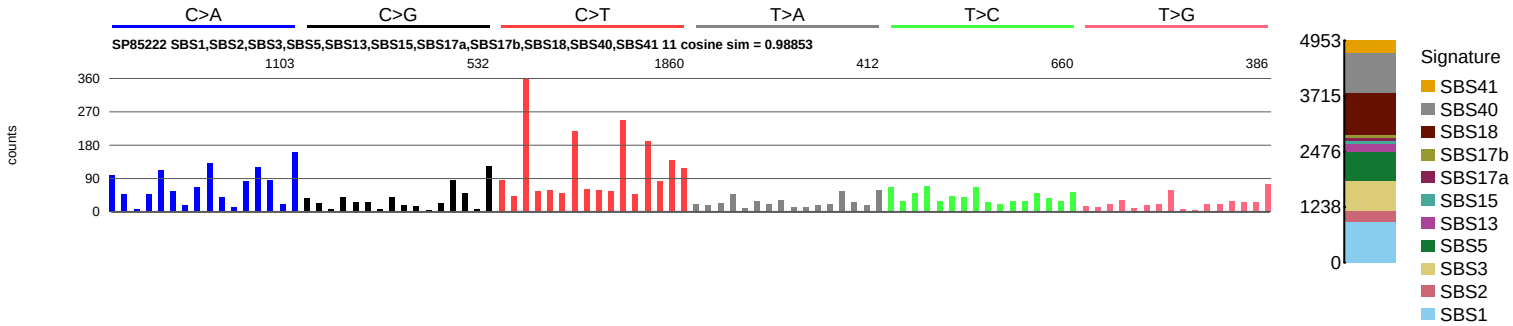
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 12 cosine sim = 0.98855
1105
528
1869
410
659
384
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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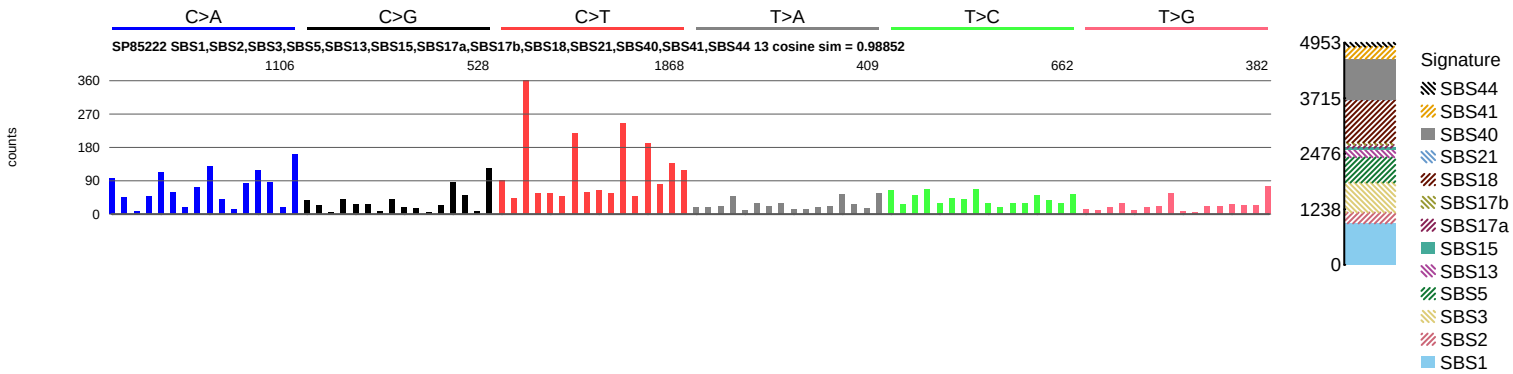
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 11 cosine sim = 0.98853
1103
532
1860
412
660
386
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 13 cosine sim = 0.98852
1106
528
1868
409
662
382
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
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SBS17b
SBS17a
SBS15
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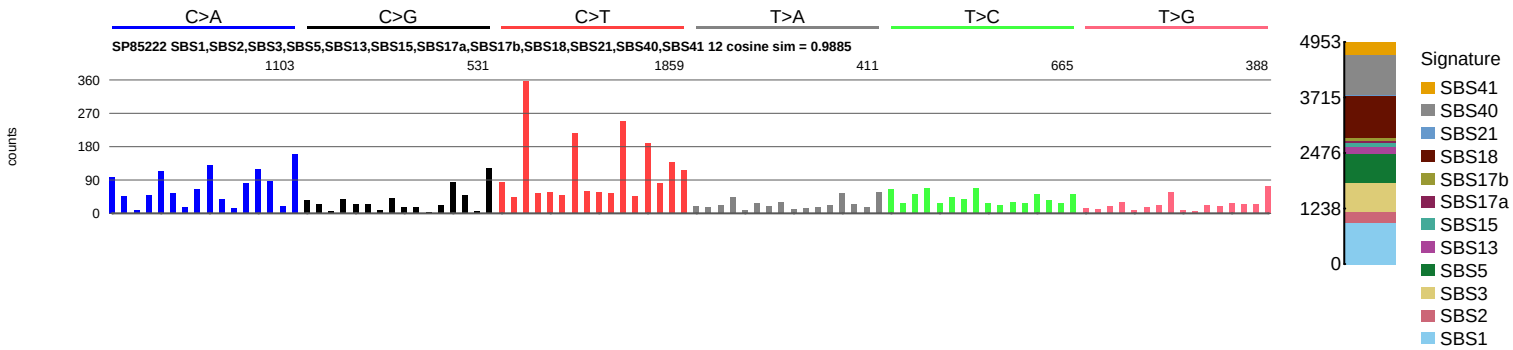
SBS3
SBS2
SBS1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1103
531
1859
411
665
388
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
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SBS18
SBS17b
SBS17a
SBS15
SBS13
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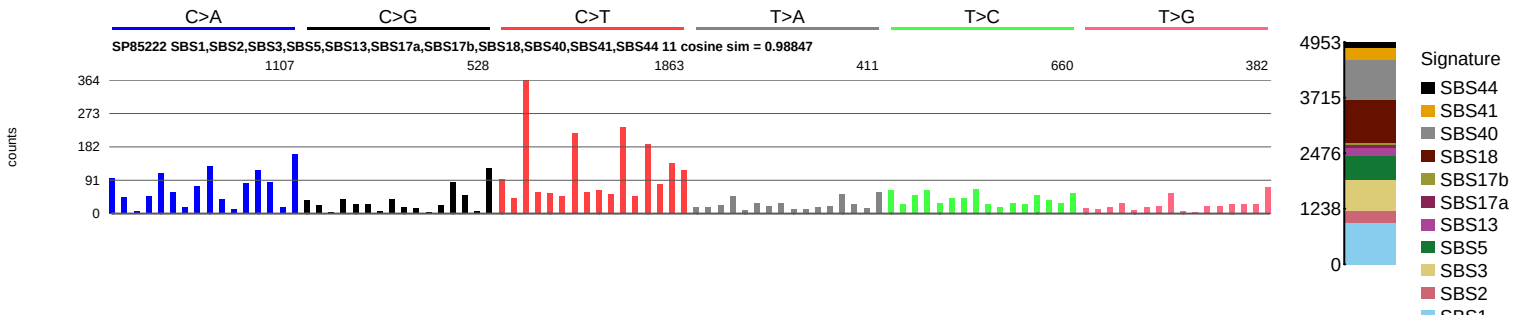
SBS3
SBS2
SBS1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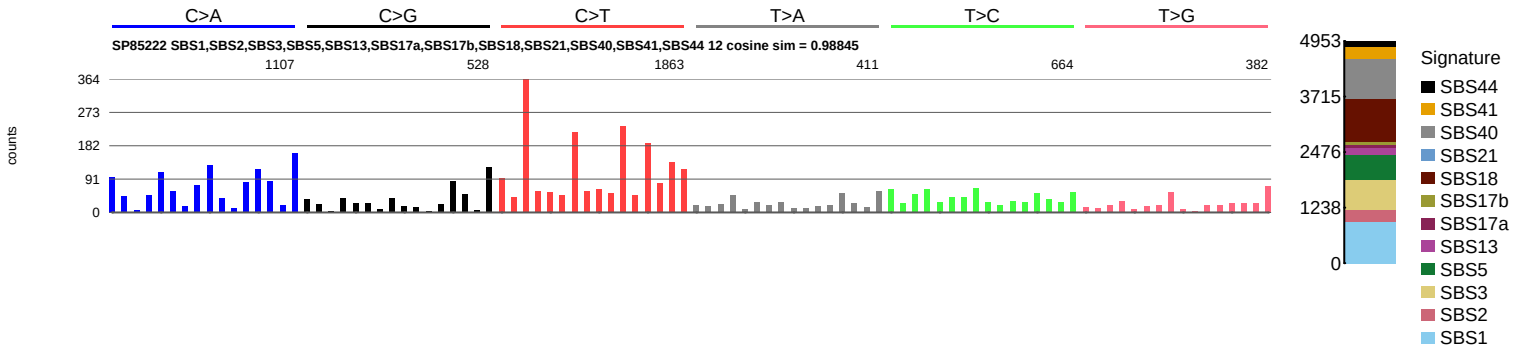
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98847
1107
528
1863
411
660
382
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2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C>G
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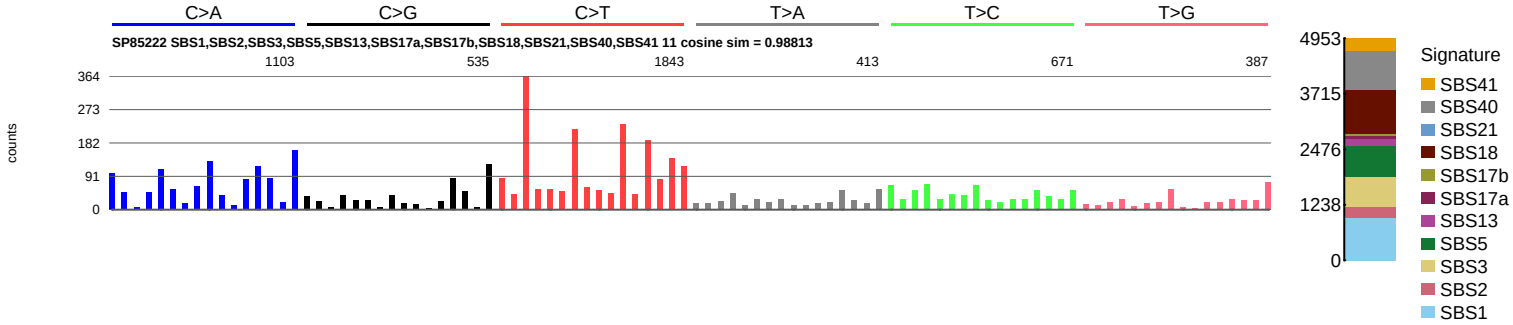
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98845
1107
528
1863
411
664
382
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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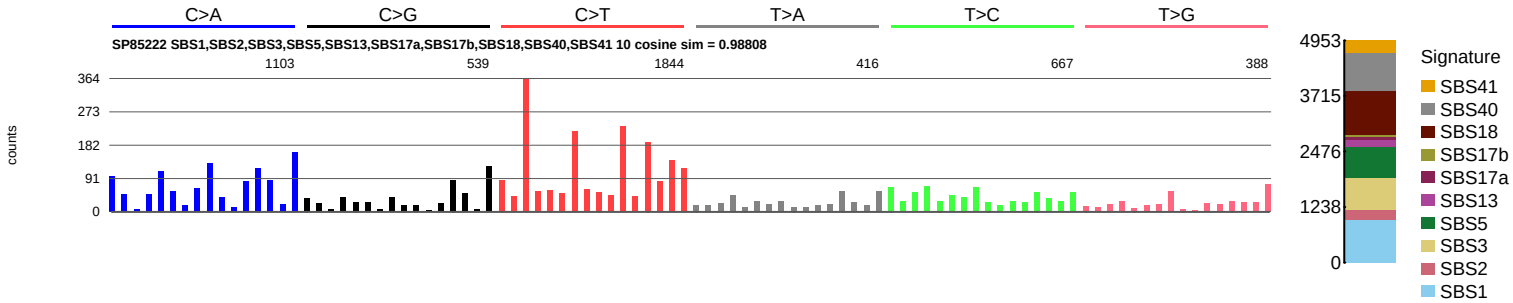
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98813
1103
535
1843
413
671
387
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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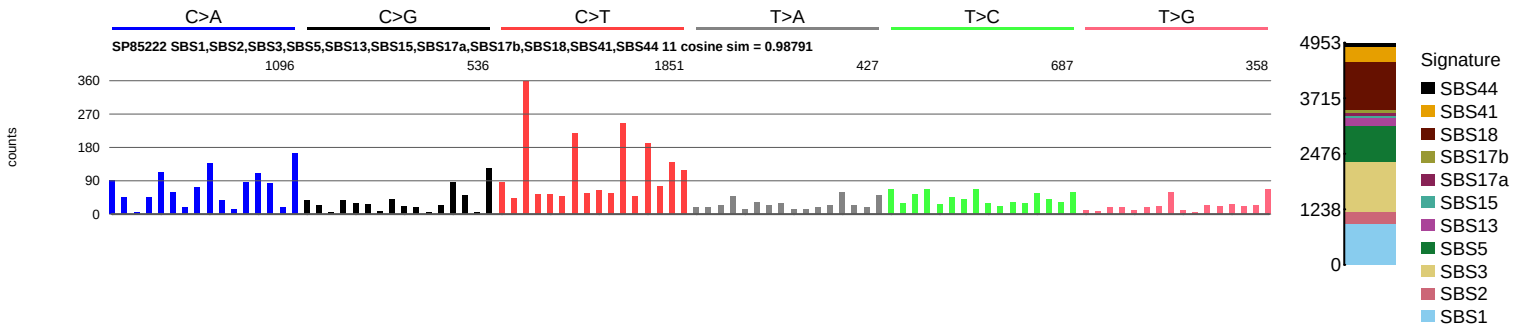
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98808
1103
539
1844
416
667
388
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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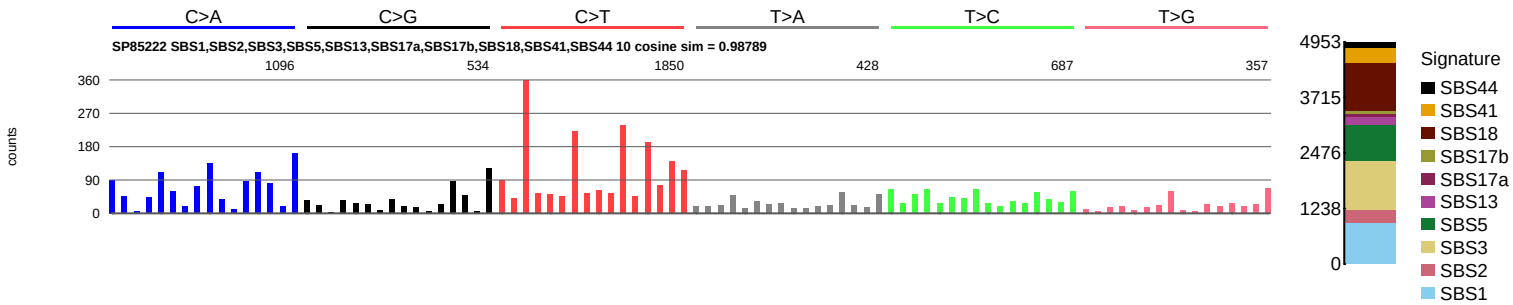
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 11 cosine sim = 0.98791
1096
536
1851
427
687
358
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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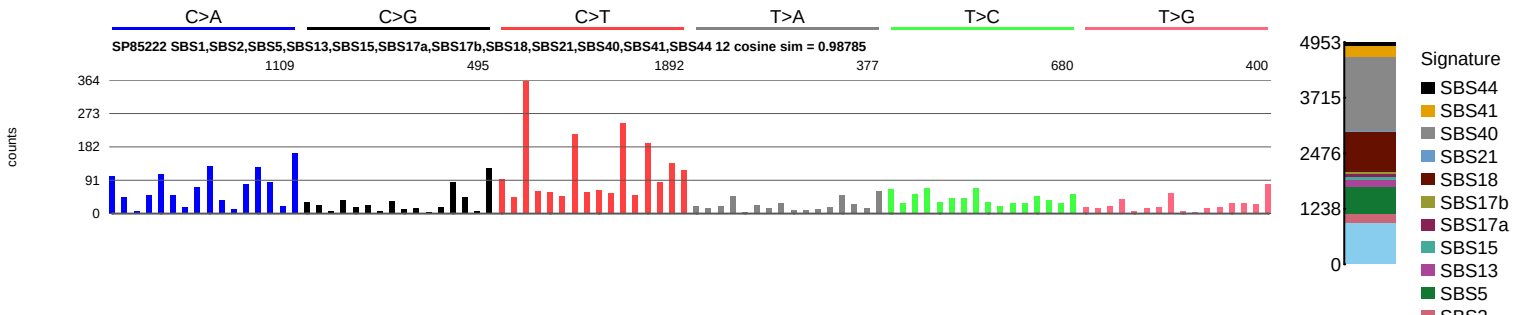
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98789
1096
534
1850
428
687
357
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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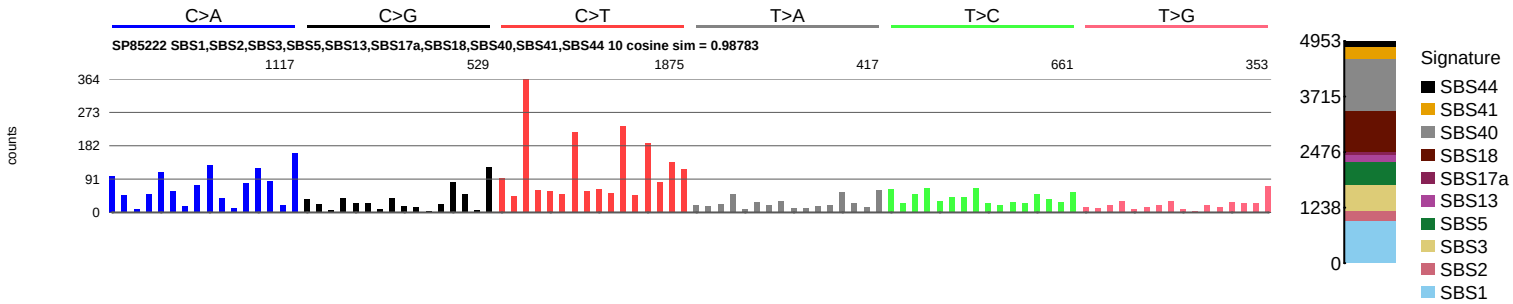
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98785
1109
495
1892
377
680
400
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5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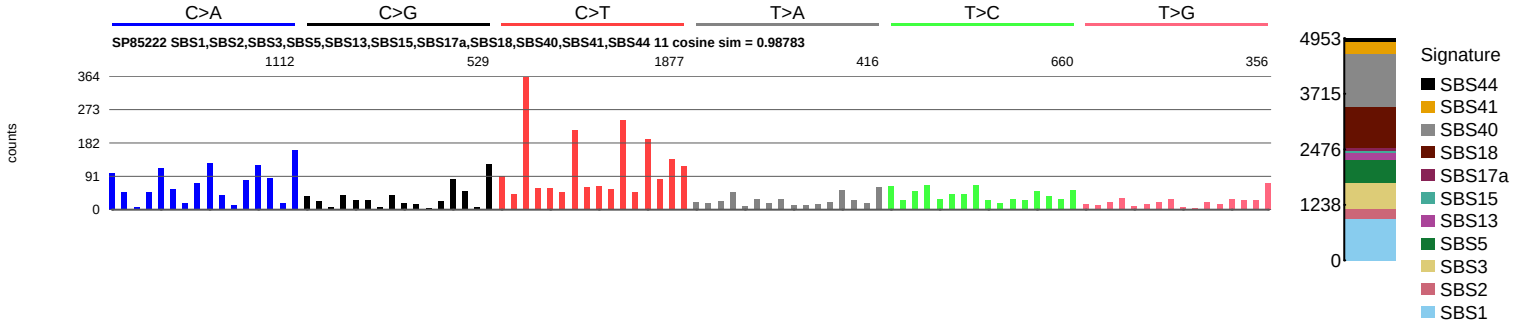
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98783
1117
529
1875
417
661
353
364
273
182
91
0
counts
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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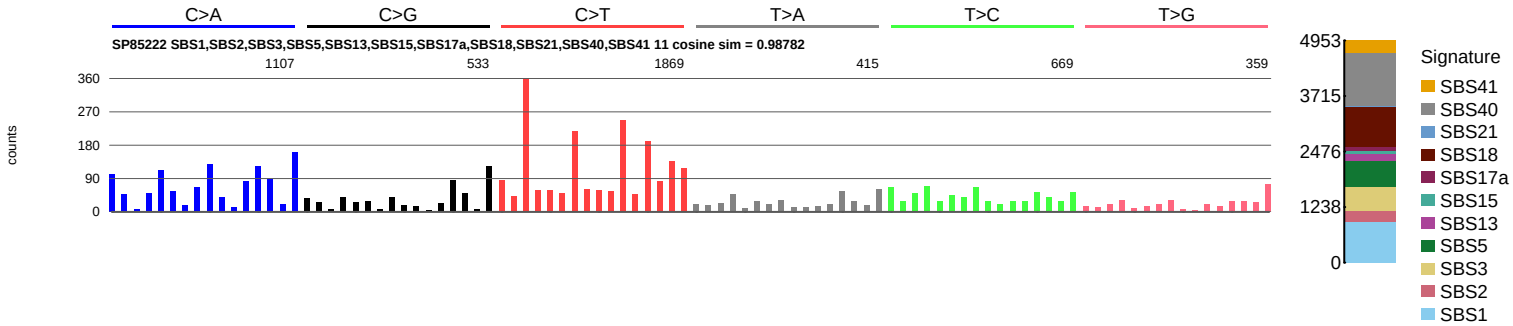
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98783
1112
529
1877
416
660
356
364
273
182
91
0
counts
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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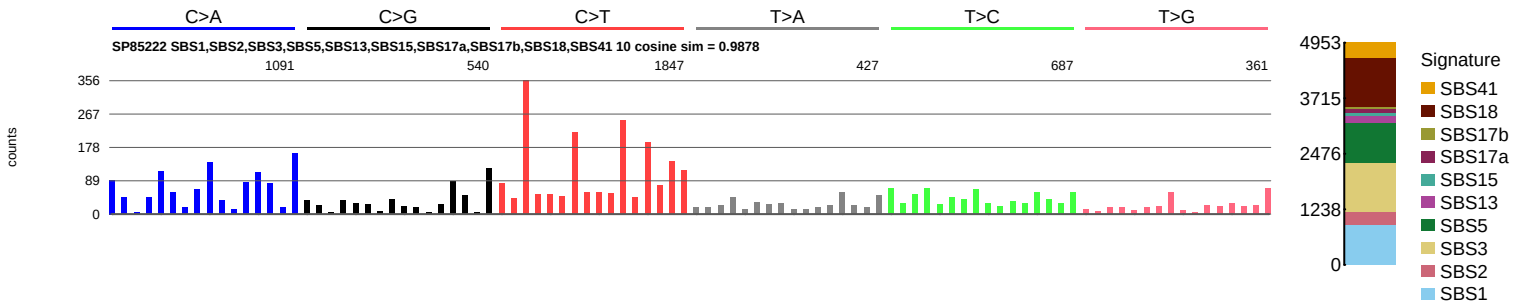
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98782
1107
533
1869
415
669
359
360
270
180
90
0
counts
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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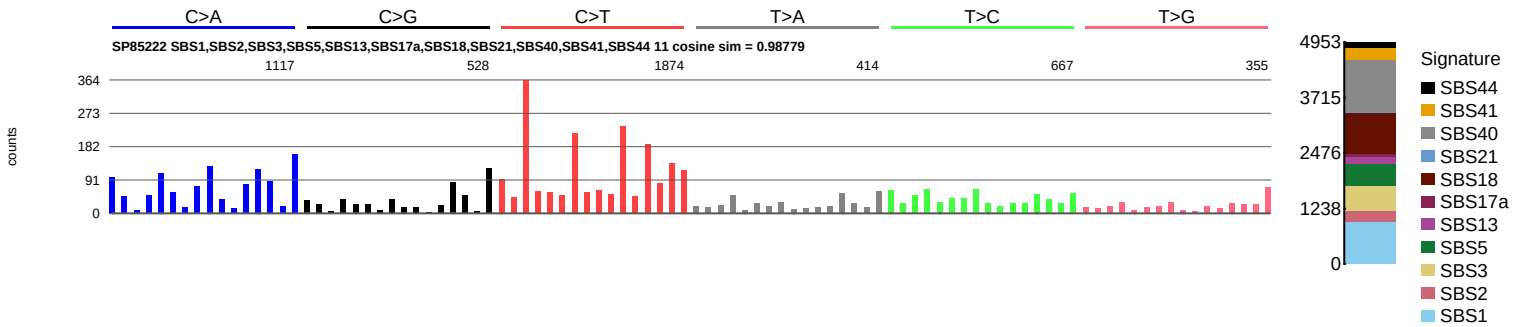
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.9878
1091
540
1847
427
687
361
356
267
178
89
0
counts
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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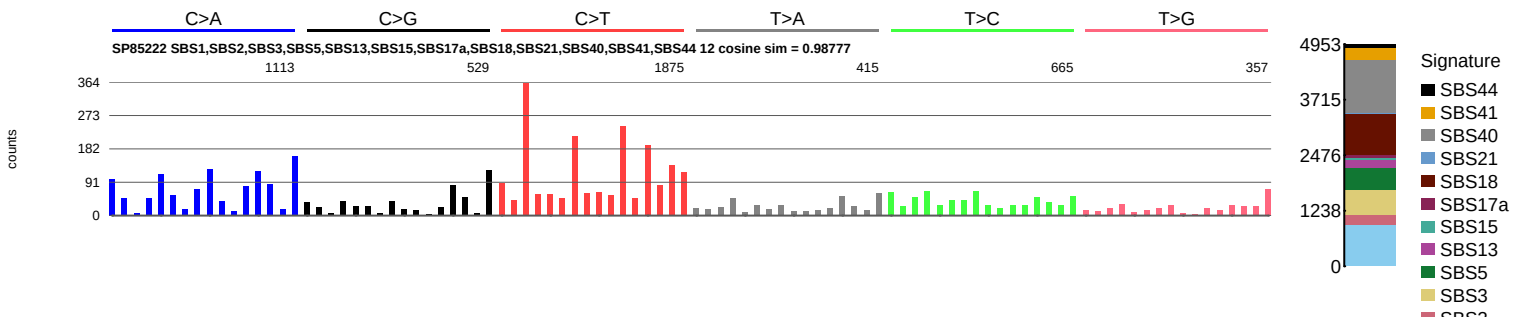
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98779
1117
528
1874
414
667
355
364
273
182
91
0
counts
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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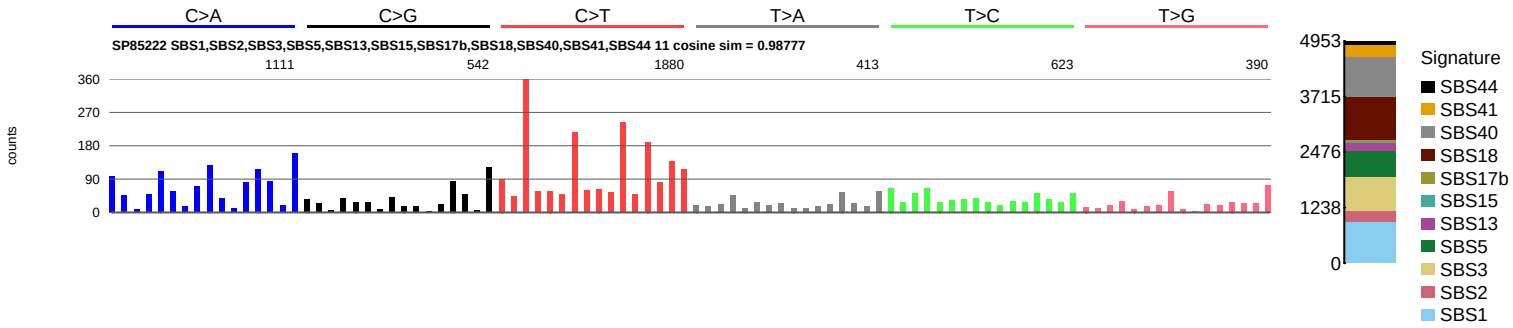
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98777
1113
529
1875
415
665
357
364
273
182
91
0
counts
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3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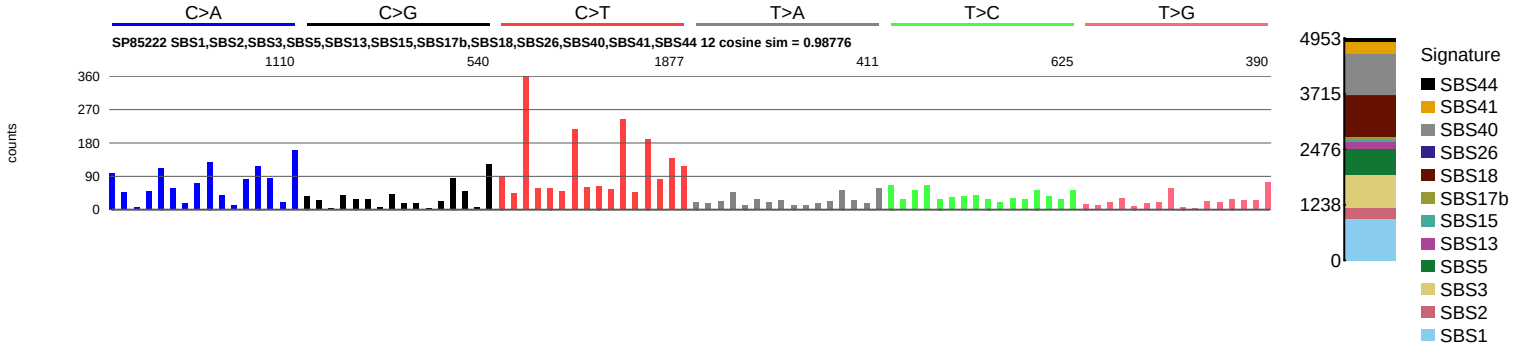
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98777
1111
542
1880
413
623
390
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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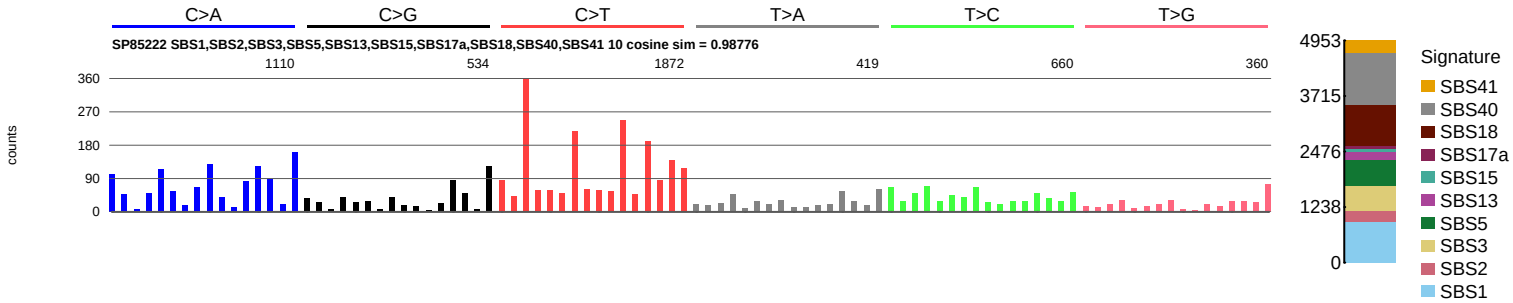
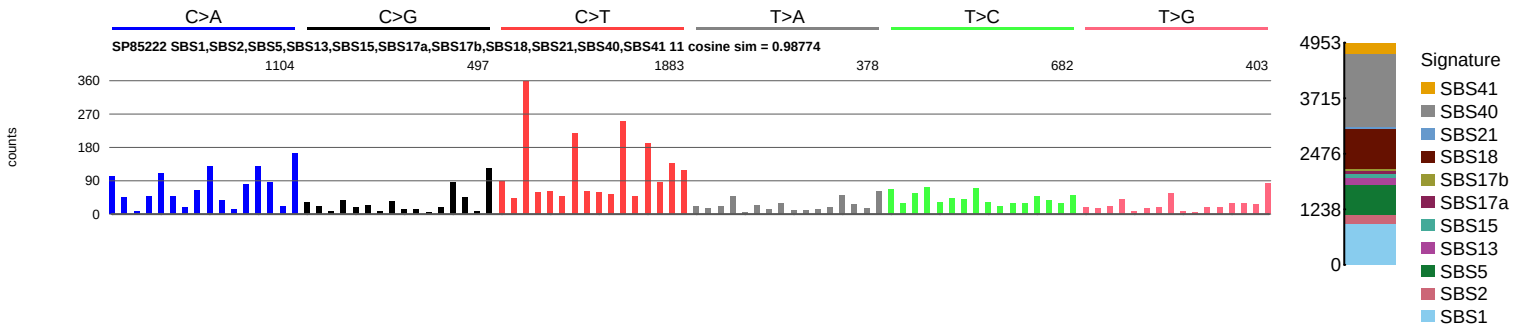
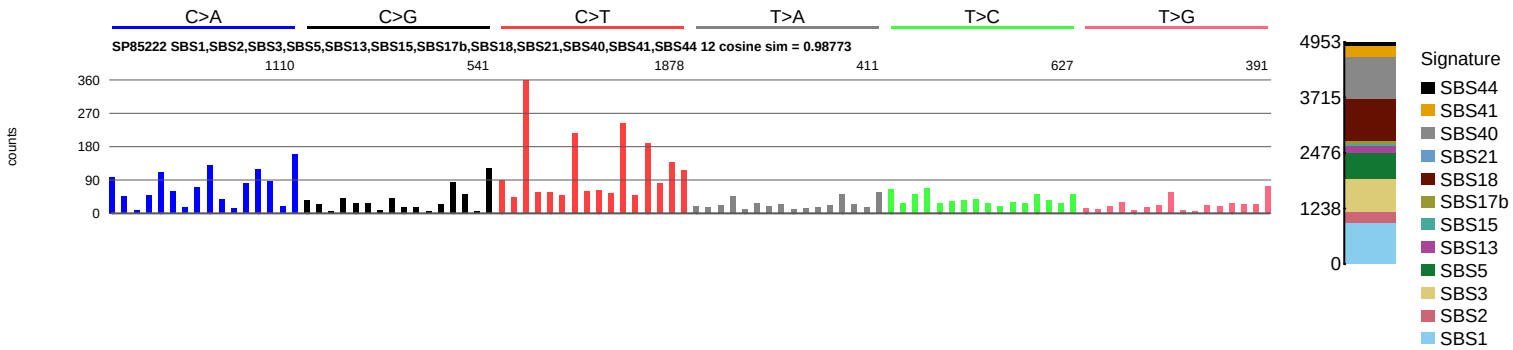
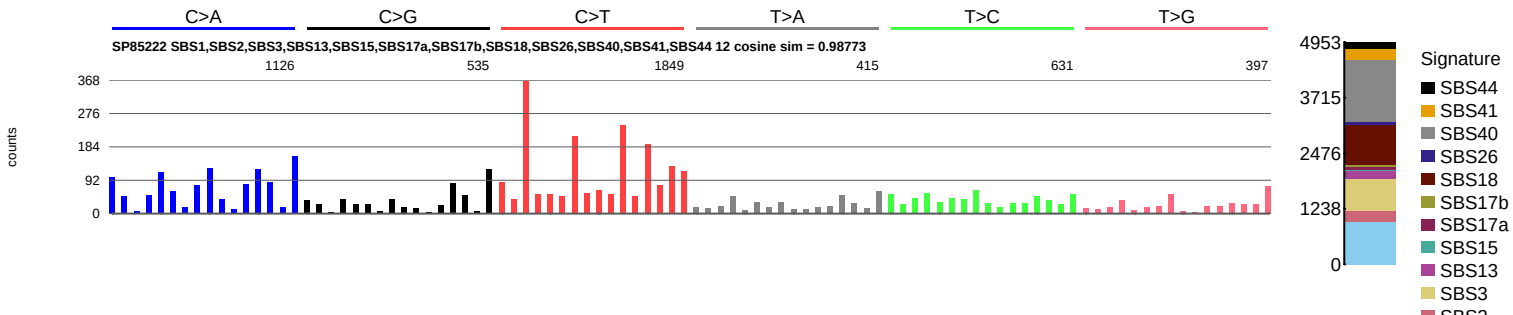
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98776
1110
540
1877
411
625
390
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41 10 cosine sim = 0.98776
1110
534
1872
419
660
360
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98774
1104
497
1883
378
682
403
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98773
1110
541
1878
411
627
391
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98773
1126
535
1849
415
631
397
counts
368
276
184
92
0
4953
3715
2476
1238
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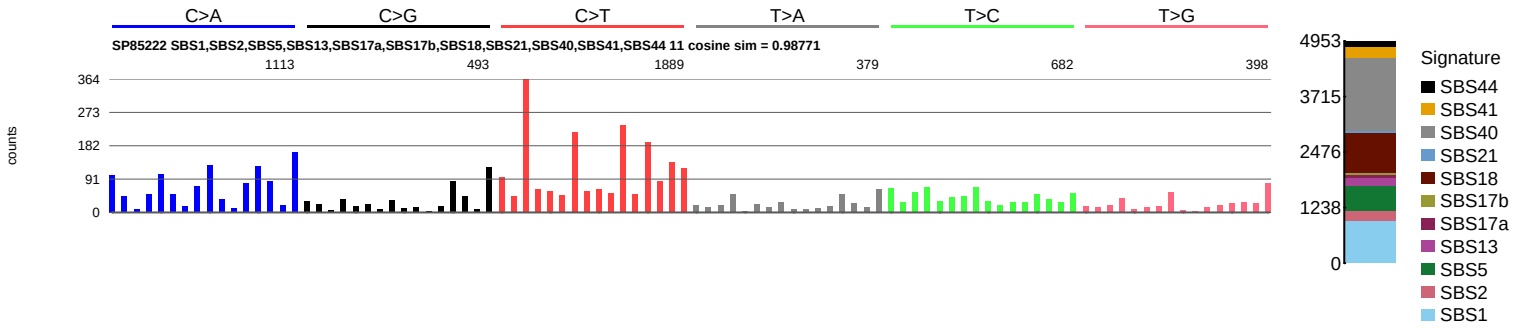
SBS3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SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98771
1113
493
1889
379
682
398
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
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SBS21
SBS18
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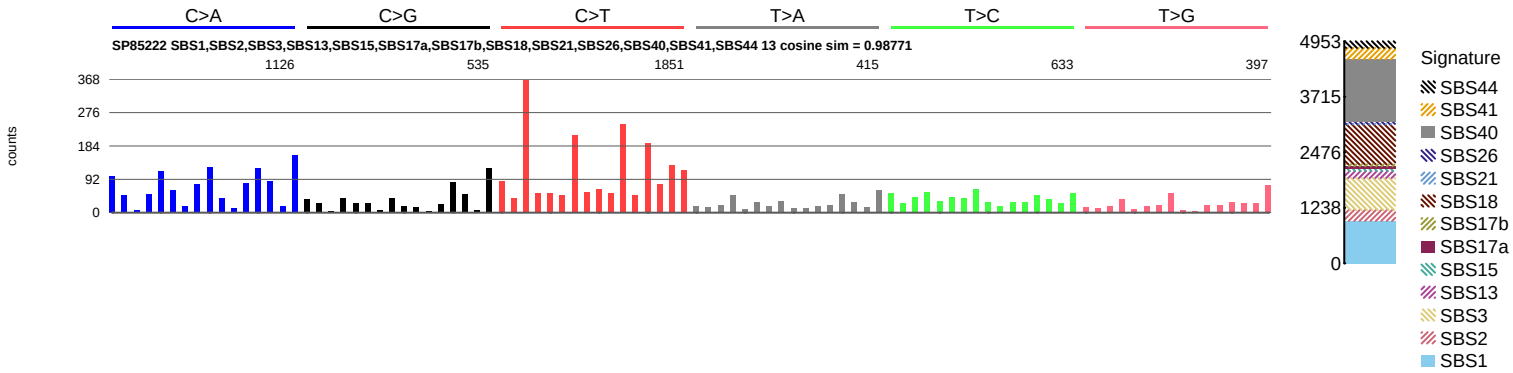
SBS17a
SBS13
SBS5
SBS2
SBS1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SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 13 cosine sim = 0.98771
1126
535
1851
415
633
397
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
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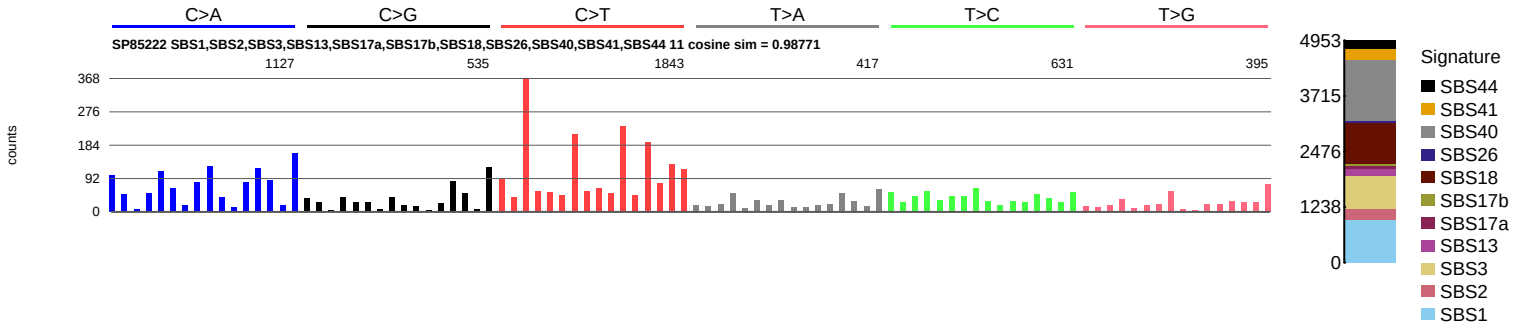
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98771
1127
535
1843
417
631
395
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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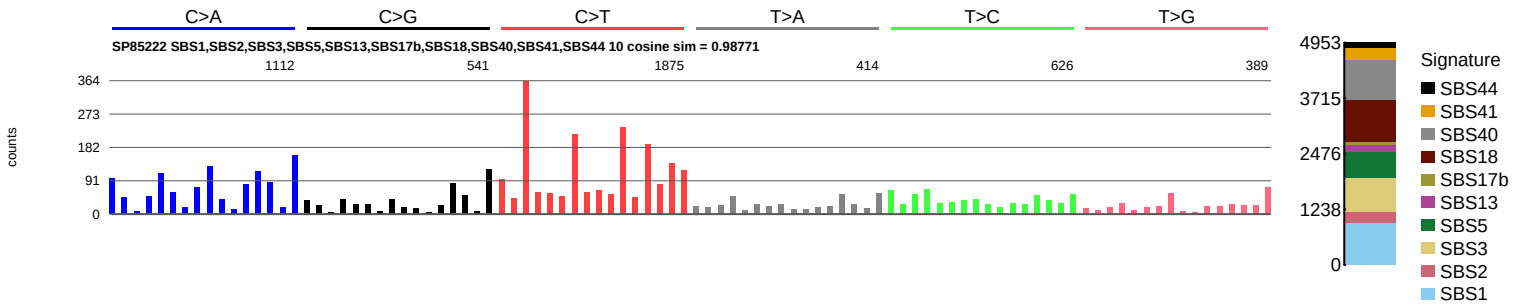
SBS2
SBS1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T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98771
1112
541
1875
414
626
389
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
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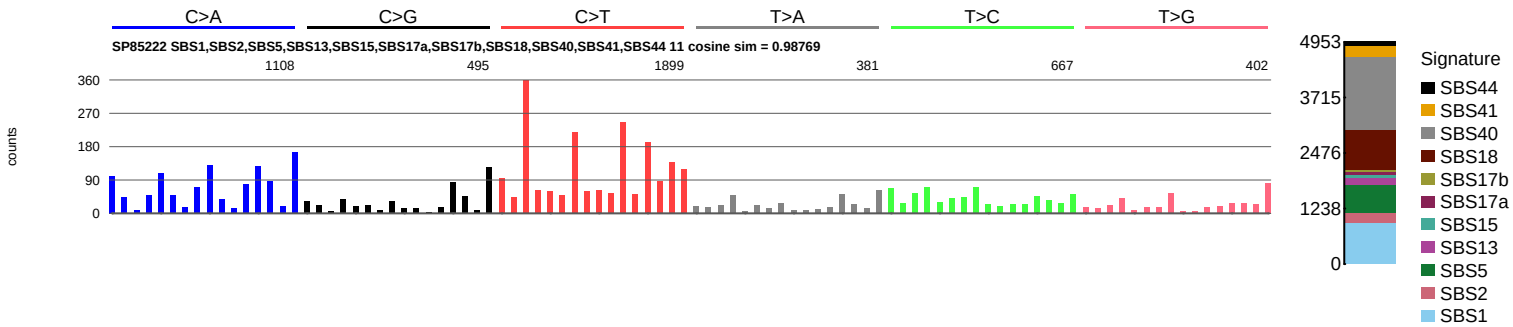
SBS2
SBS1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SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98769
1108
495
1899
381
667
402
counts
360
270
180
90
0
4953
3715
2476
1238
0
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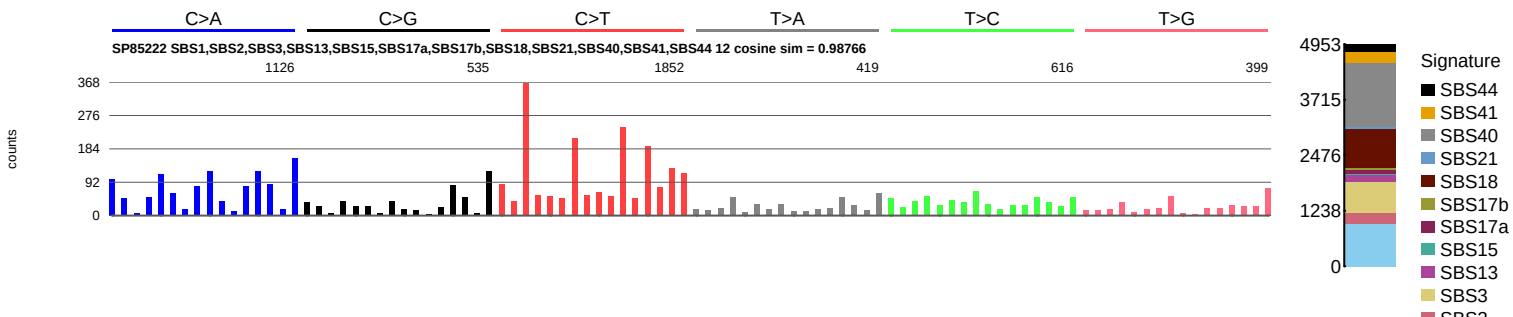
SBS2
SBS1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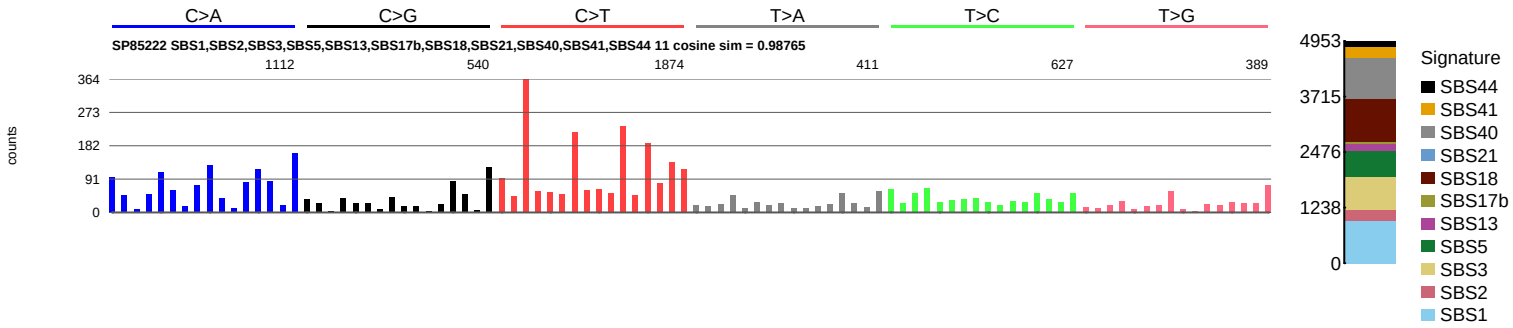
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98766
1126
535
1852
419
616
399
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98765
1112
540
1874
411
627
389
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
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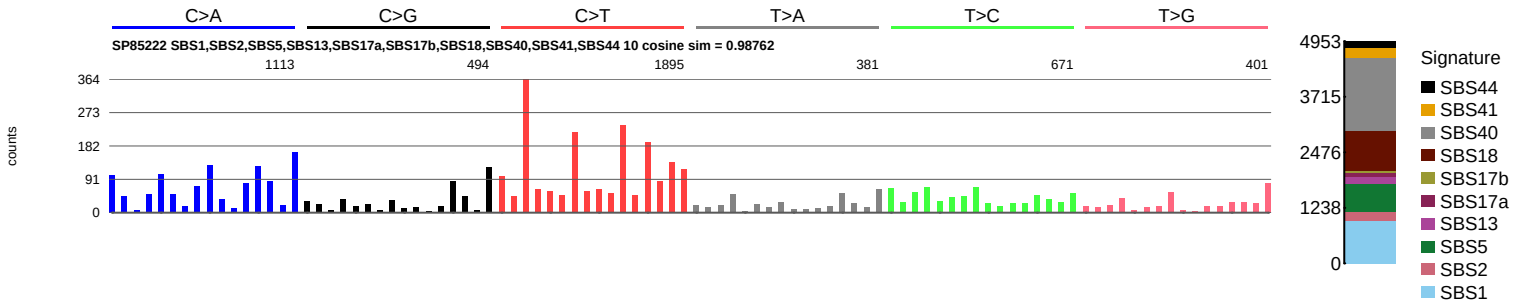
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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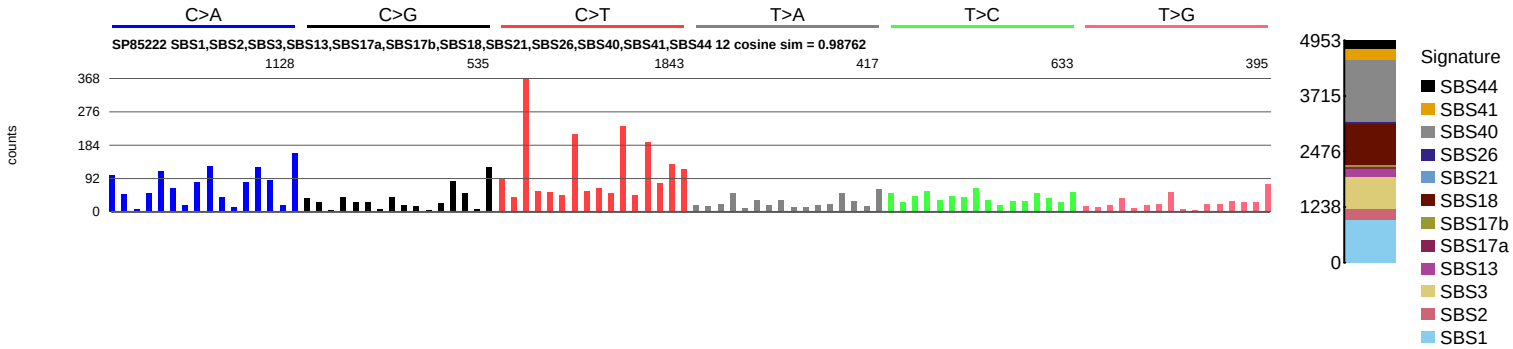
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98762
1113
494
1895
381
671
401
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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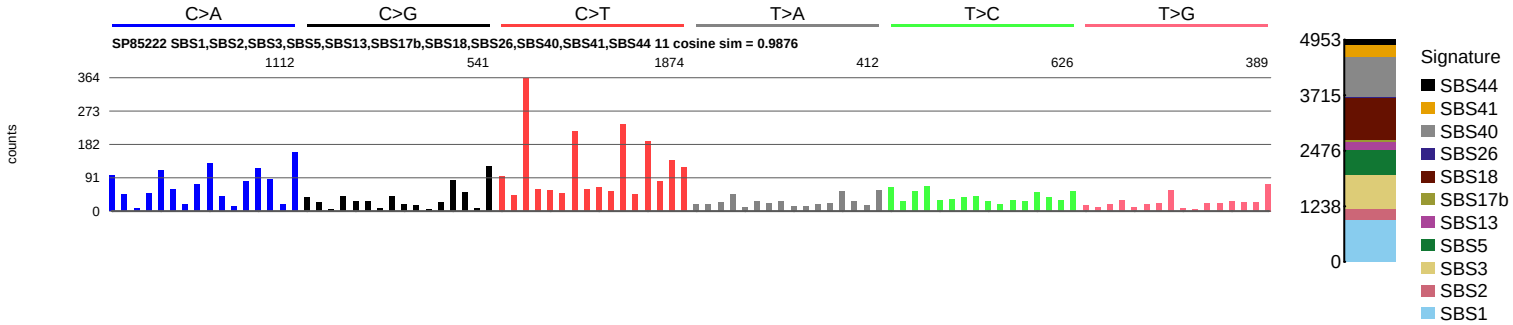
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98762
1128
535
1843
417
633
395
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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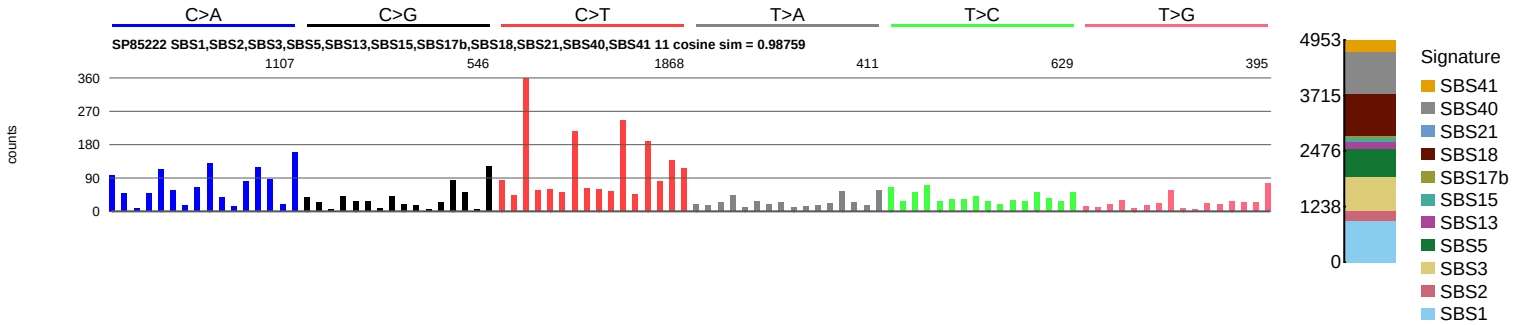
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.9876
1112
541
1874
412
626
389
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98759
1107
546
1868
411
629
395
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
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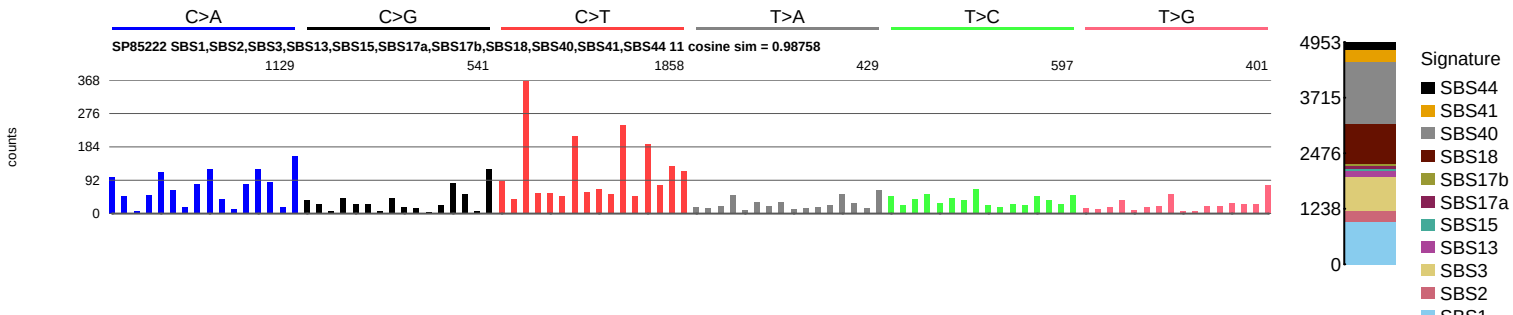
SBS13
SBS5
SBS3
SBS2
SBS1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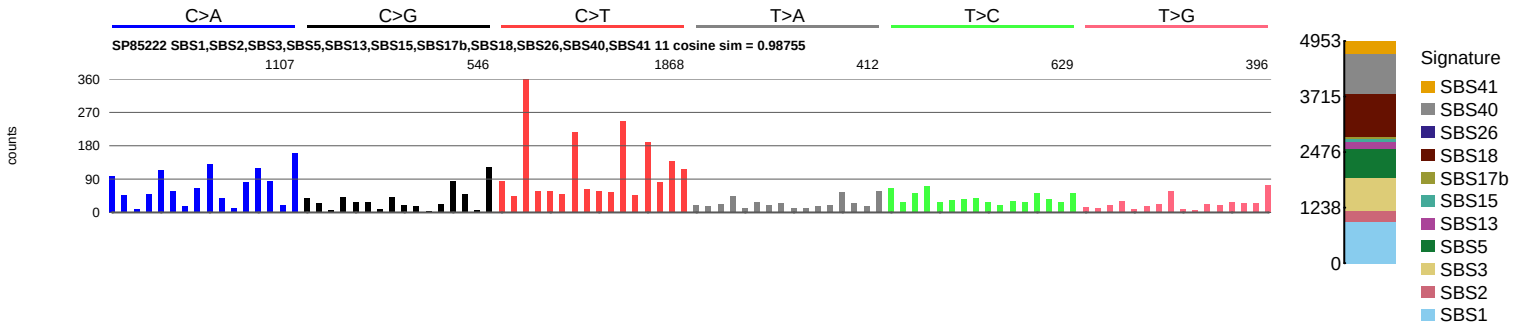
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98758
1129
541
1858
429
597
401
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2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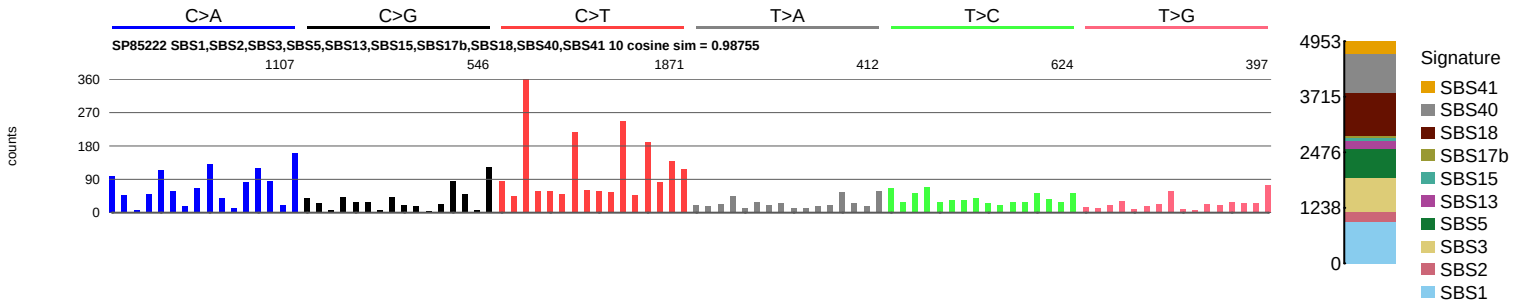
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.98755
1107
546
1868
412
629
396
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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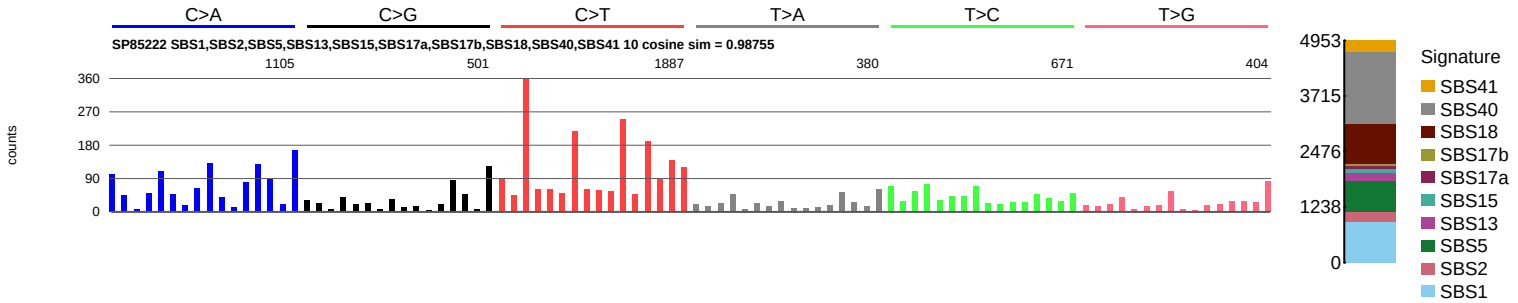
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98755
1107
546
1871
412
624
397
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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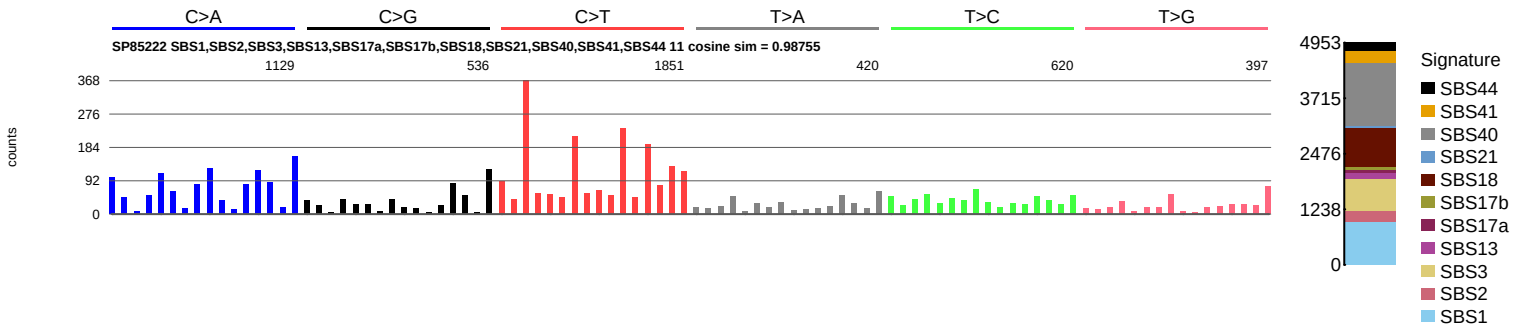
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98755
1105
501
1887
380
671
404
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98755
1129
536
1851
420
620
397
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
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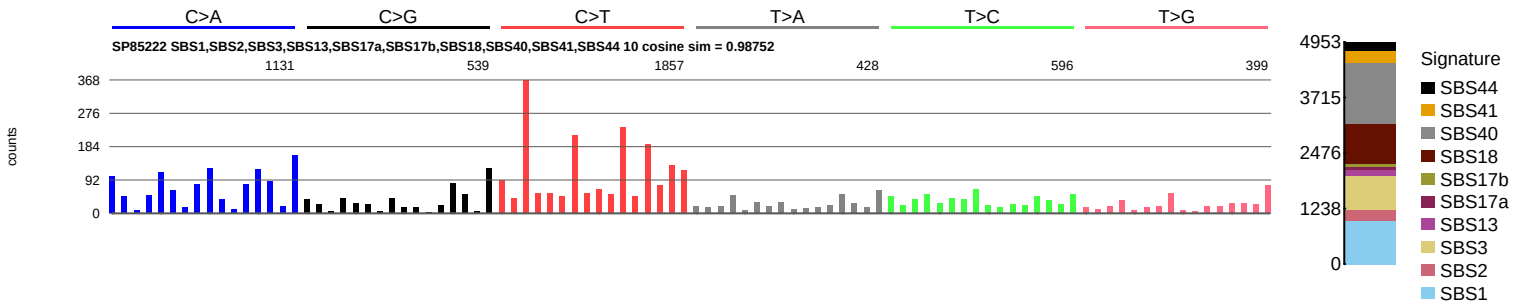
SBS3
SBS2
SBS1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T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98752
1131
539
1857
428
596
399
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
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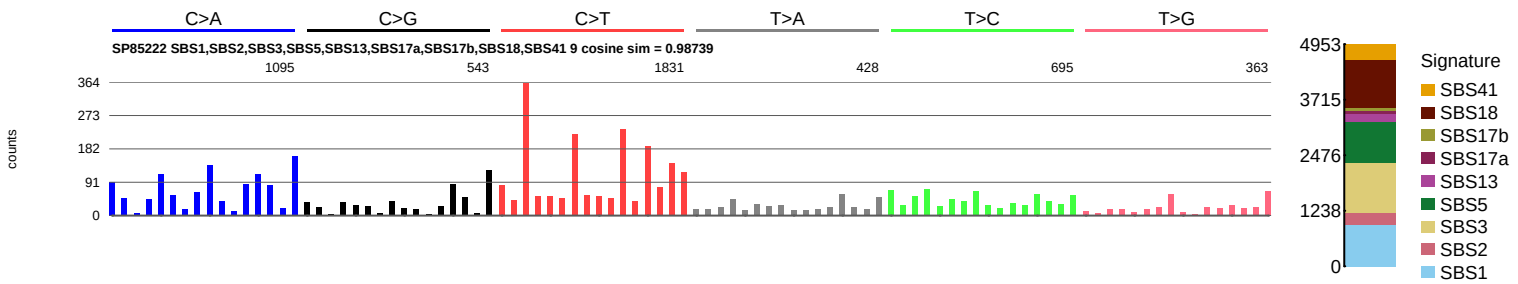
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98739
1095
543
1831
428
695
363
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
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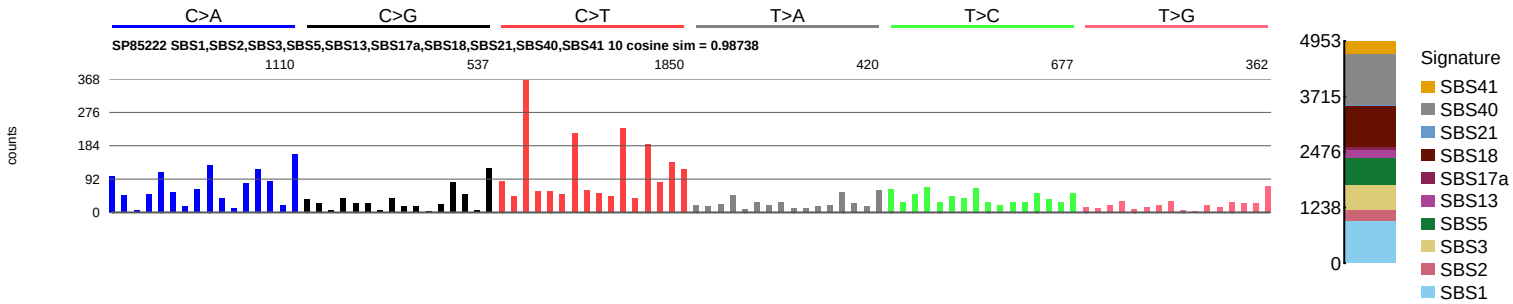
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98738
1110
537
1850
420
677
362
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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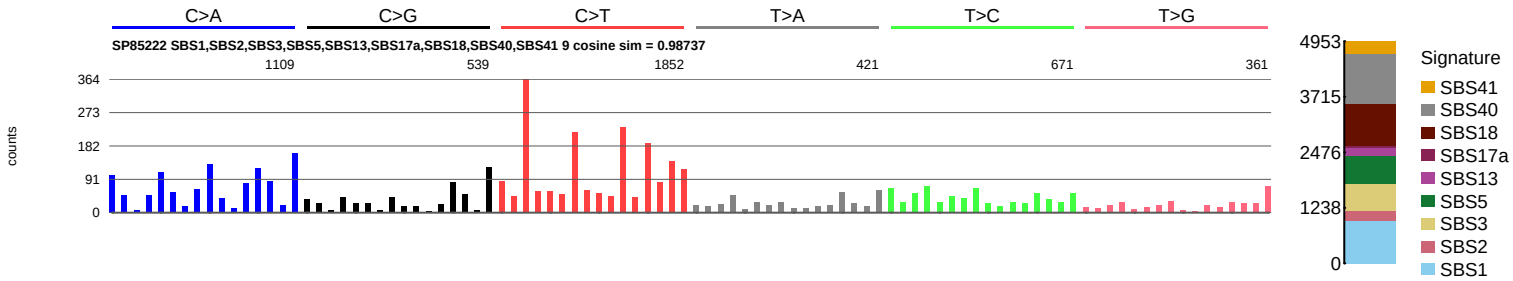
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41 9 cosine sim = 0.98737
1109
539
1852
421
671
361
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
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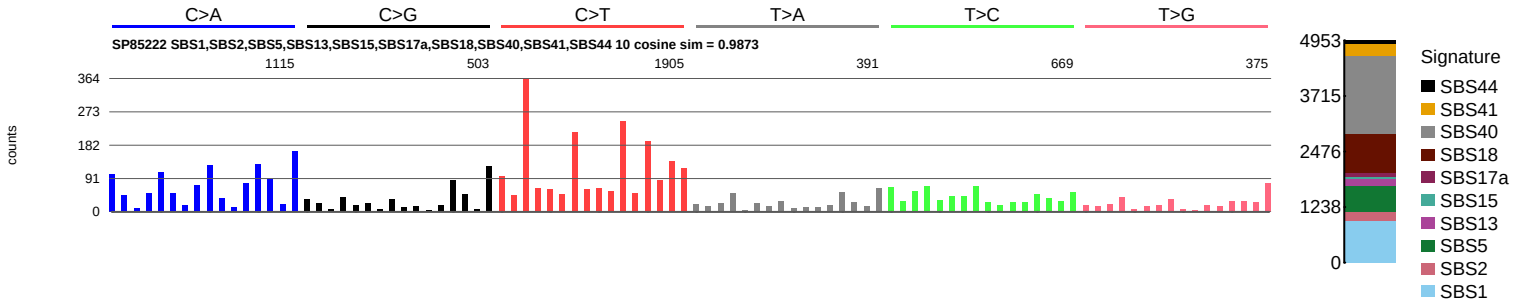
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.9873
1115
503
1905
391
669
375
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
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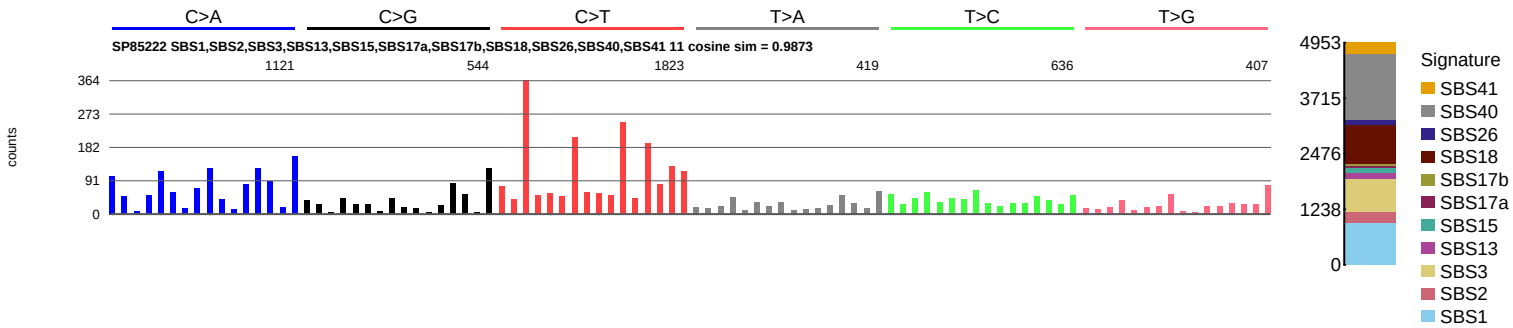
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.9873
1121
544
1823
419
636
407
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
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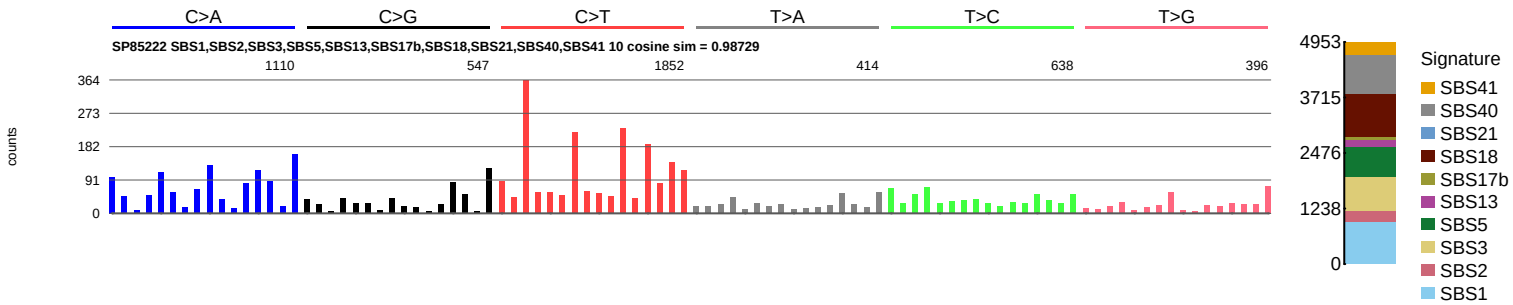
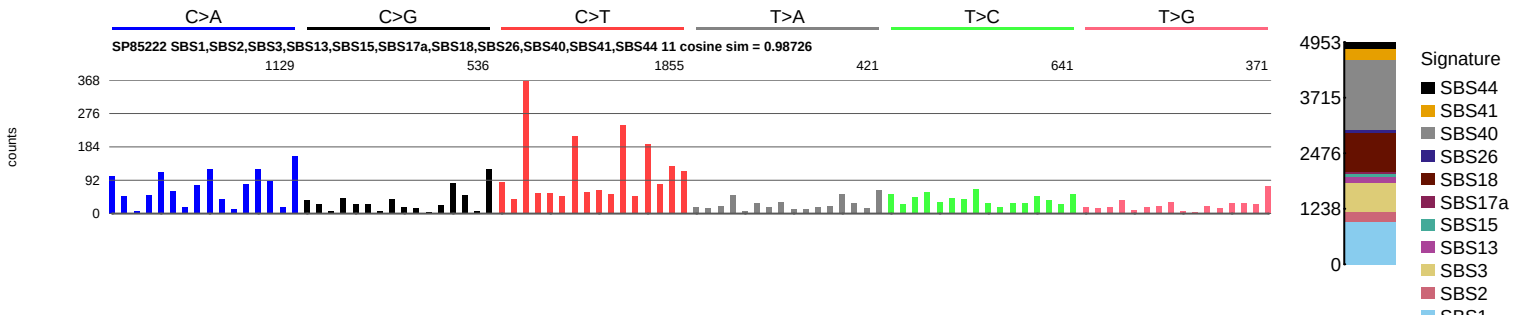
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98729
1110
547
1852
414
638
396
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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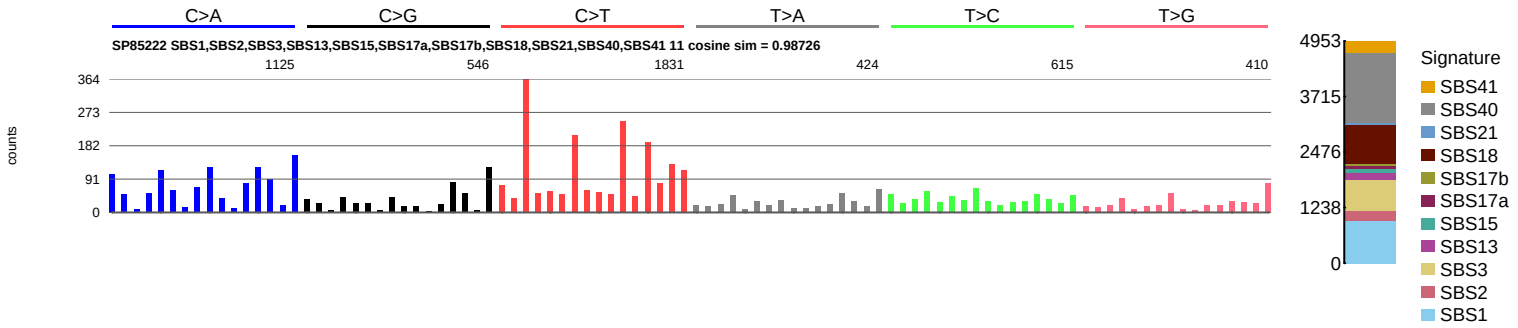
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98726
1129
536
1855
421
641
371
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2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T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98726
1125
546
1831
424
615
410
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
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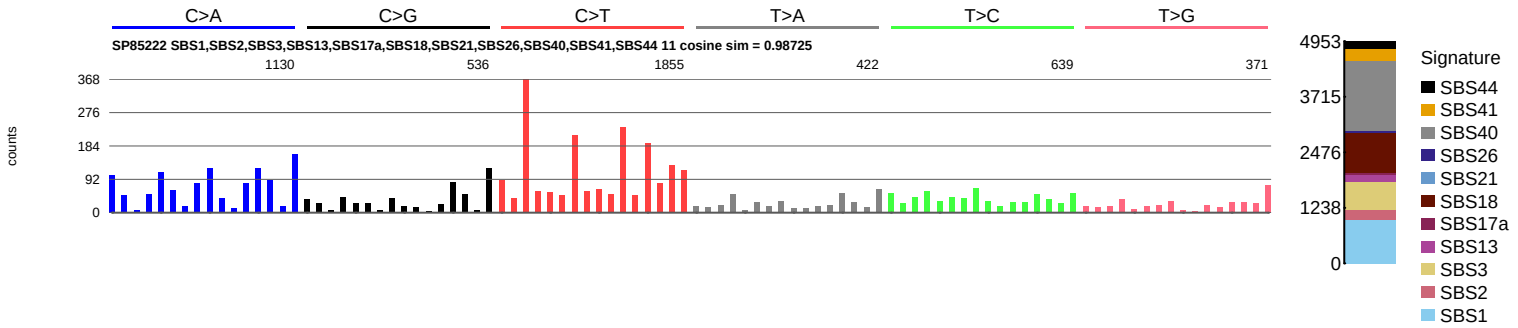
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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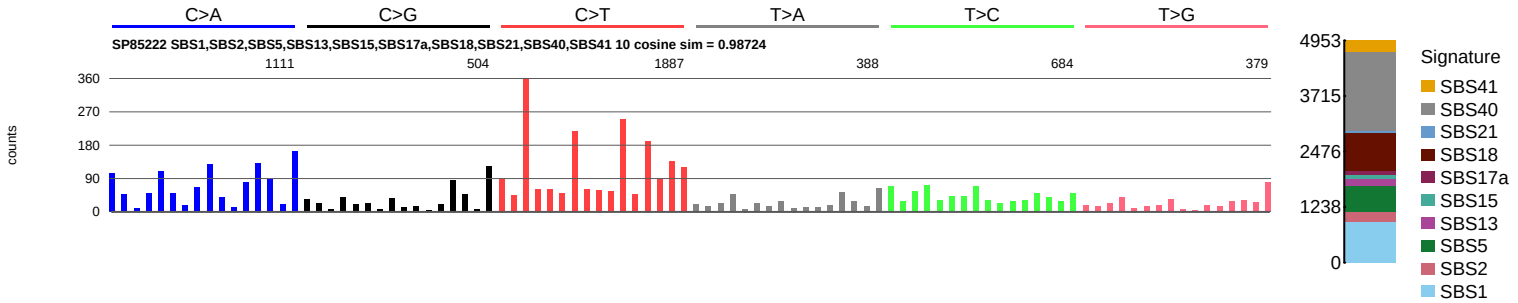
1130
536
1855
422
639
371
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1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1111
504
1887
388
684
379
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
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SBS21
SBS18
SBS17a
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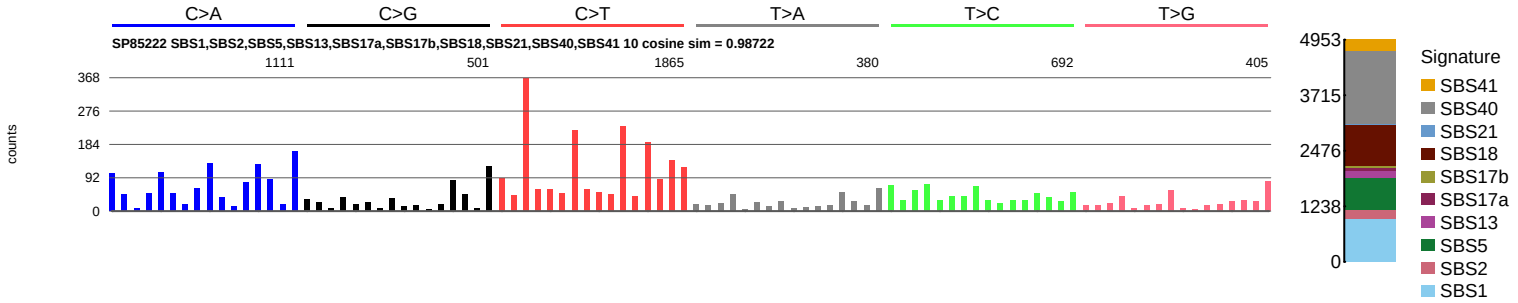
SBS13
SBS5
SBS2
SBS1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T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98722
1111
501
1865
380
692
405
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
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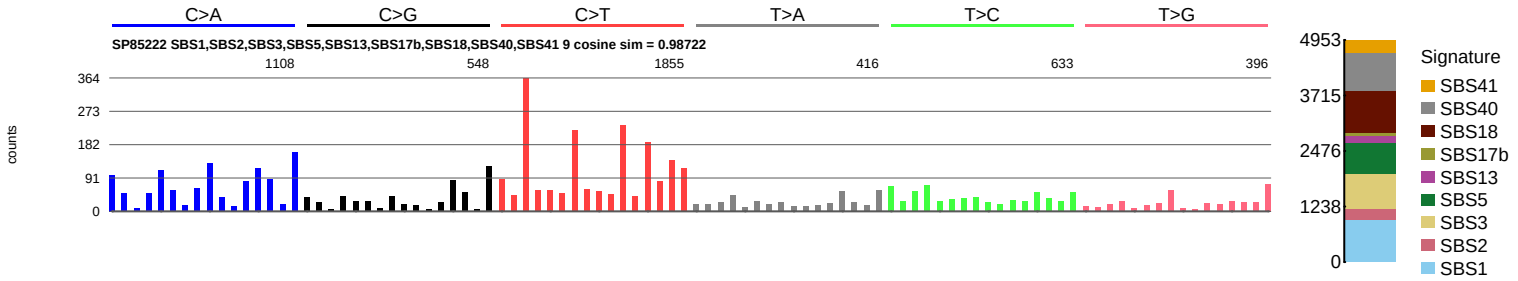
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98722
1108
548
1855
416
633
396
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
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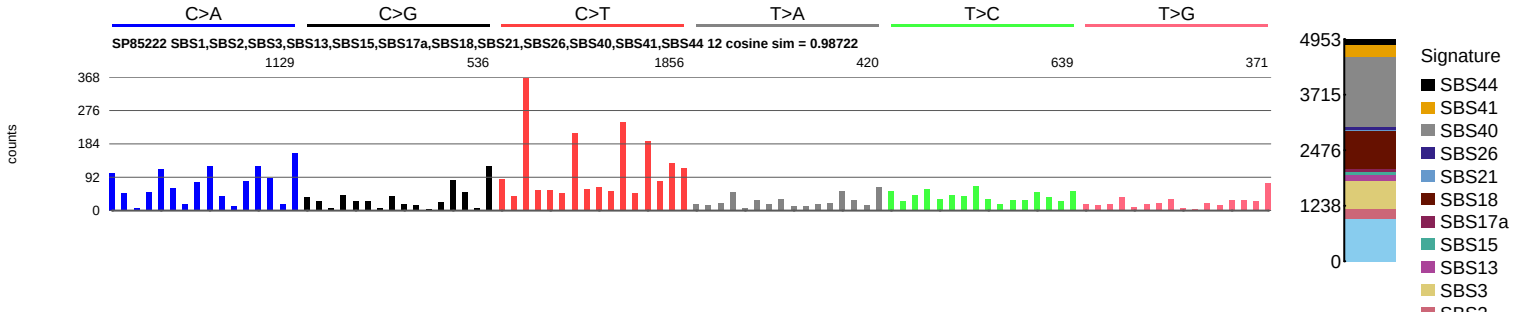
SBS2
SBS1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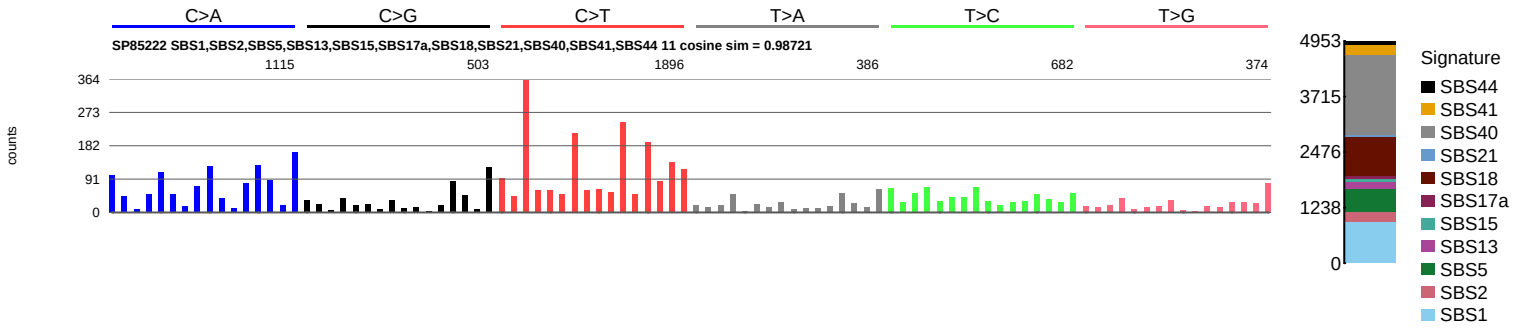
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98722
1129
536
1856
420
639
371
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3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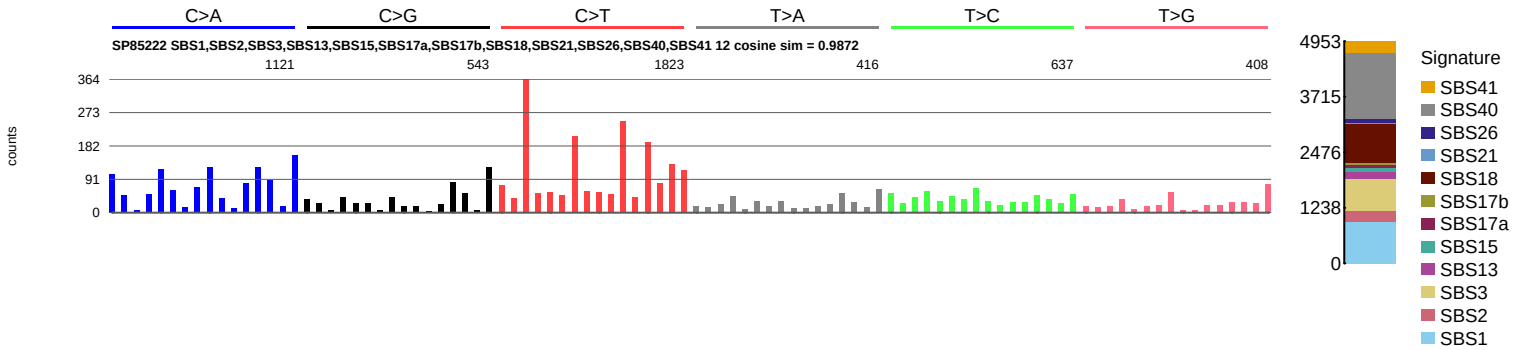
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98721
1115
503
1896
386
682
374
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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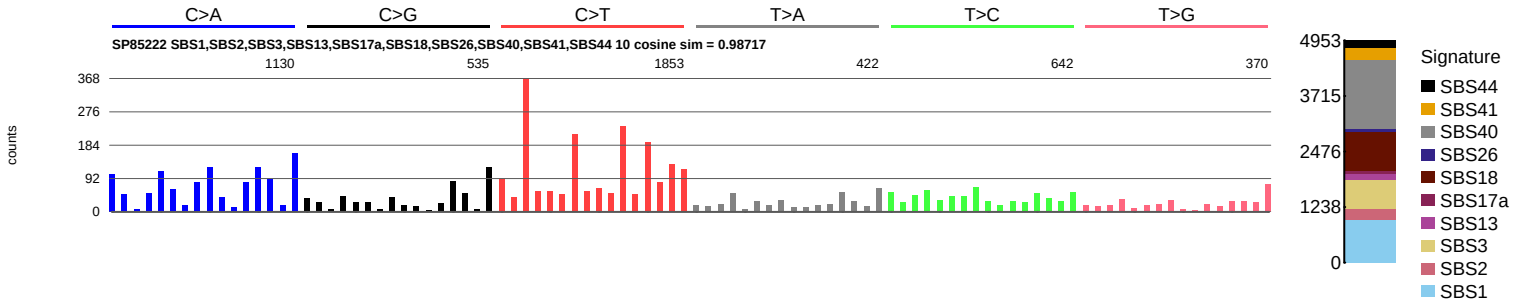
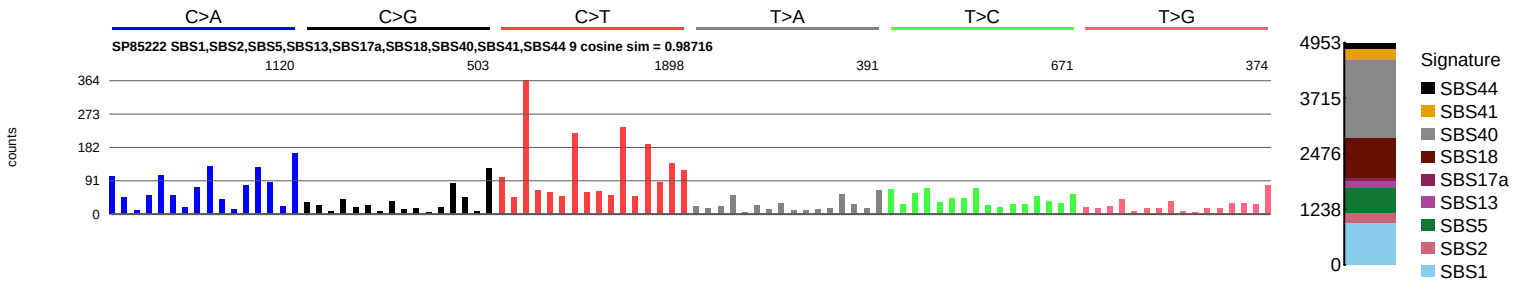
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 12 cosine sim = 0.9872
1121
543
1823
416
637
408
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98717
1130
535
1853
422
642
370
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98716
1120
503
1898
391
671
374
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
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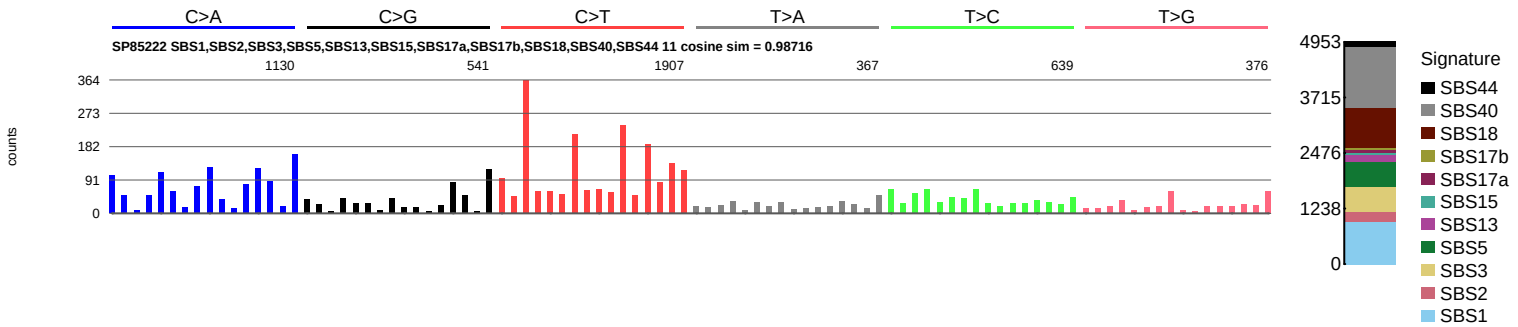
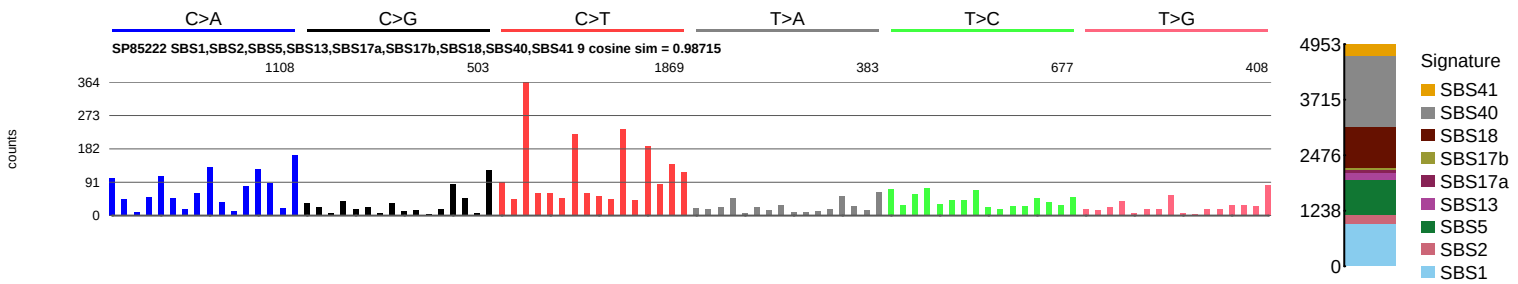
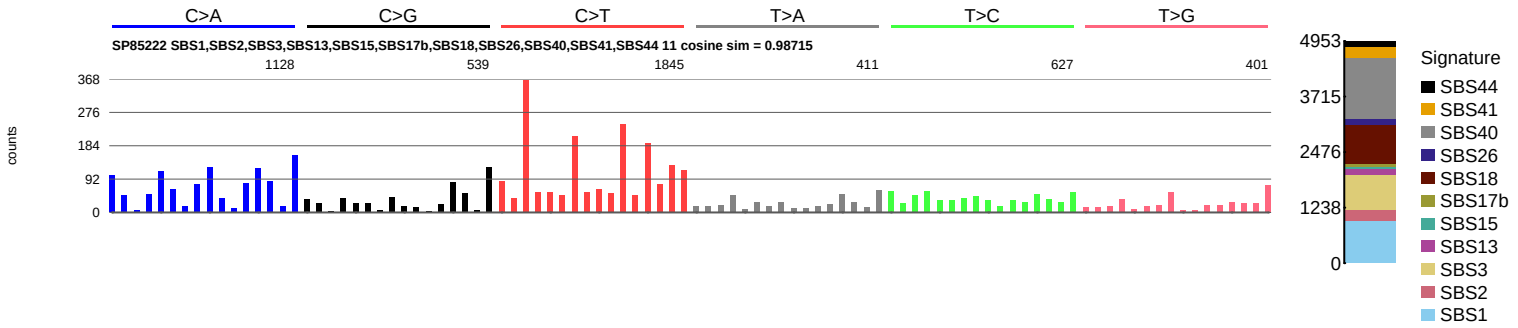
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 11 cosine sim = 0.98716
1130
541
1907
367
639
376
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98715
1108
503
1869
383
677
408
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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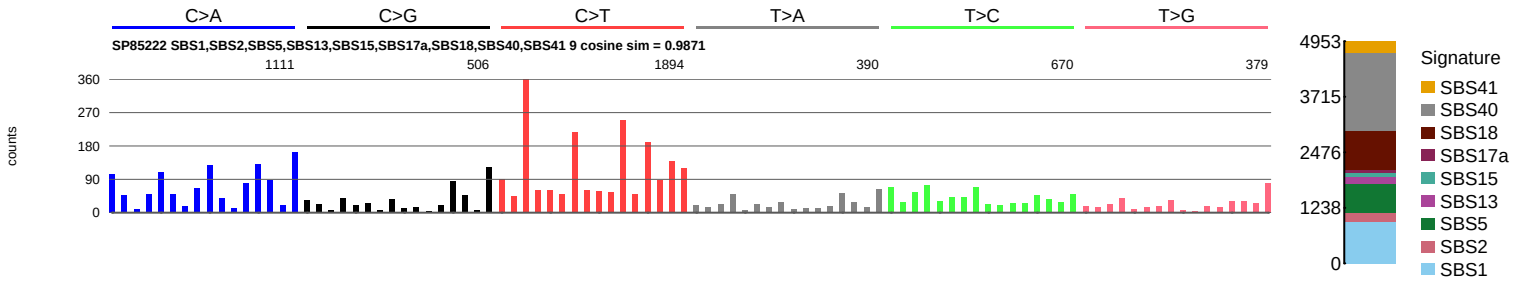
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98715
1128
539
1845
411
627
401
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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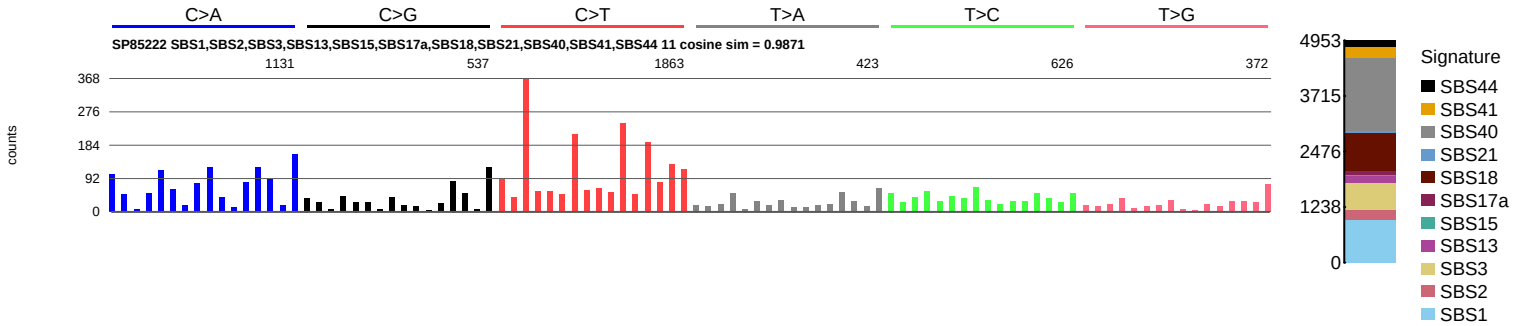
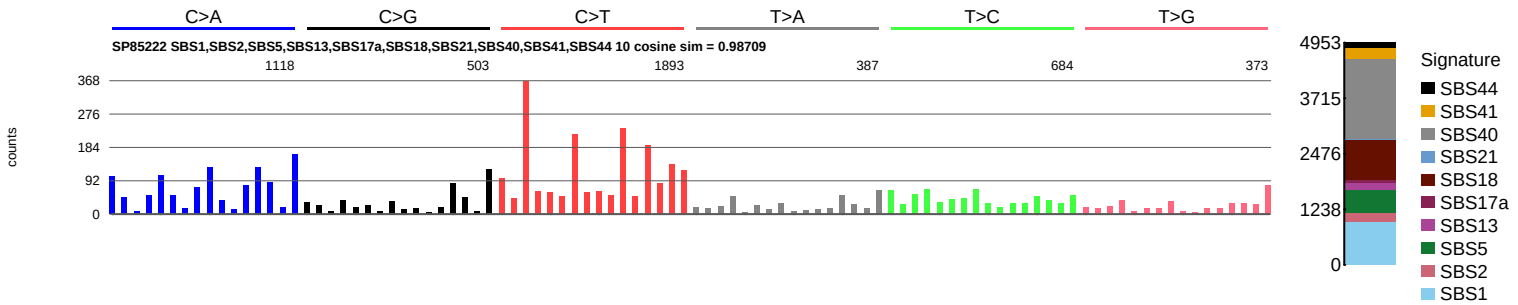
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41 9 cosine sim = 0.9871
1111
506
1894
390
670
379
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.9871
1131
537
1863
423
626
372
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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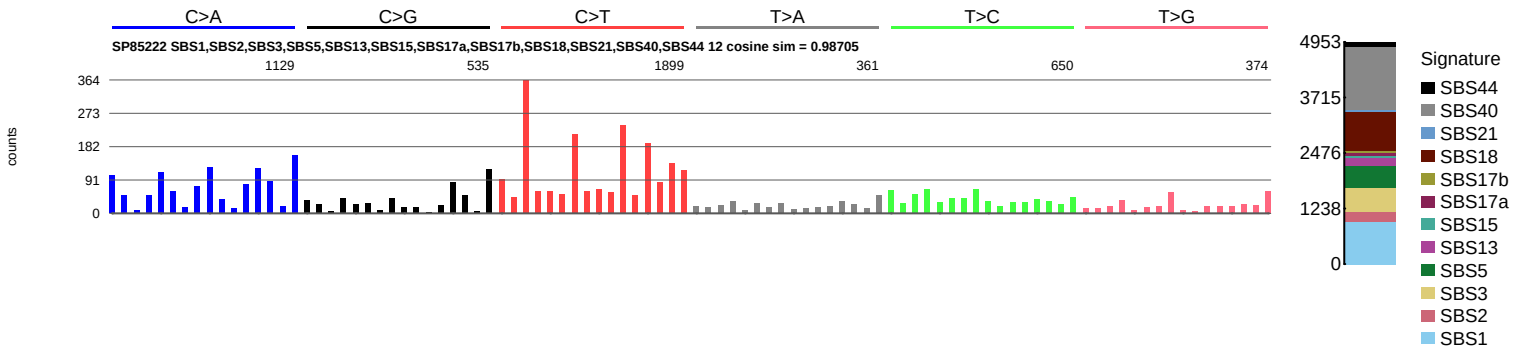
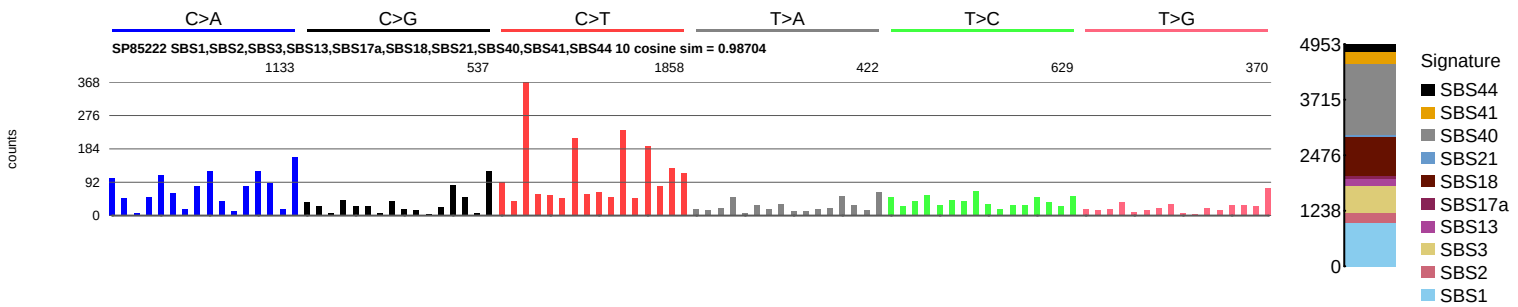
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98709
1118
503
1893
387
684
373
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 12 cosine sim = 0.98705
1129
535
1899
361
650
374
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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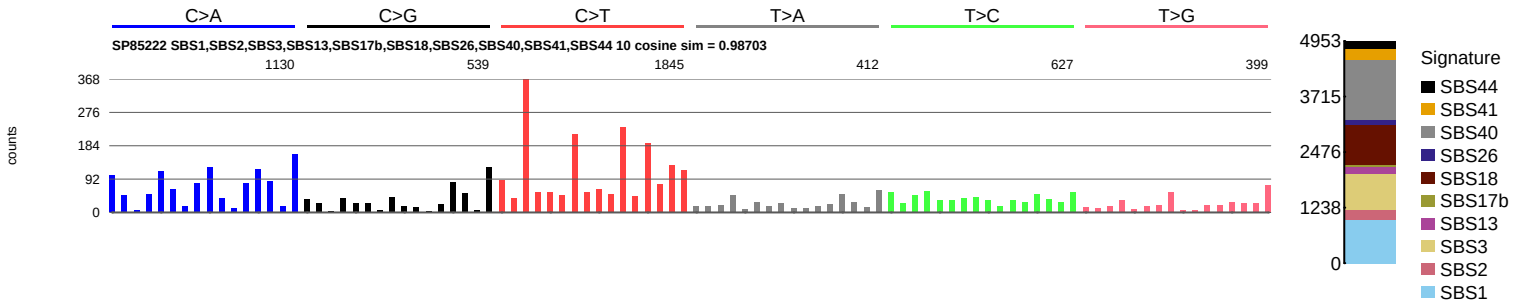
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98704
1133
537
1858
422
629
370
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98703
1130
539
1845
412
627
399
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
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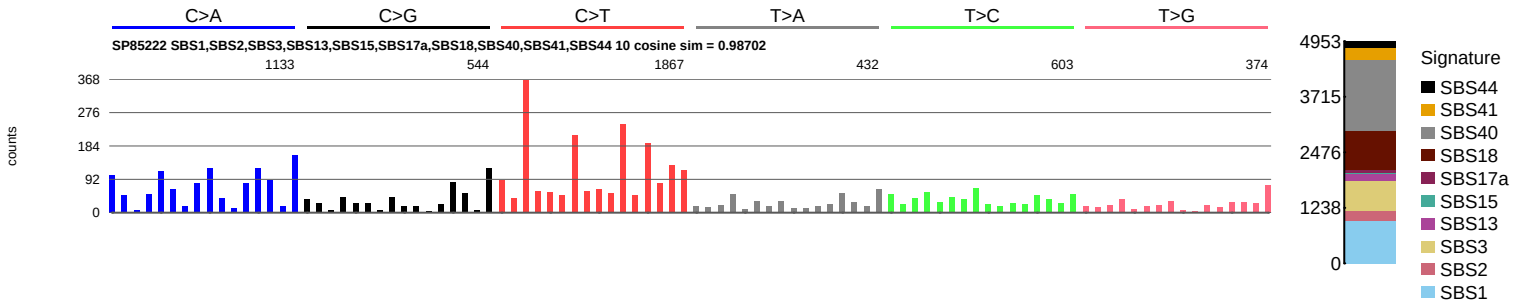
SBS3
SBS2
SBS1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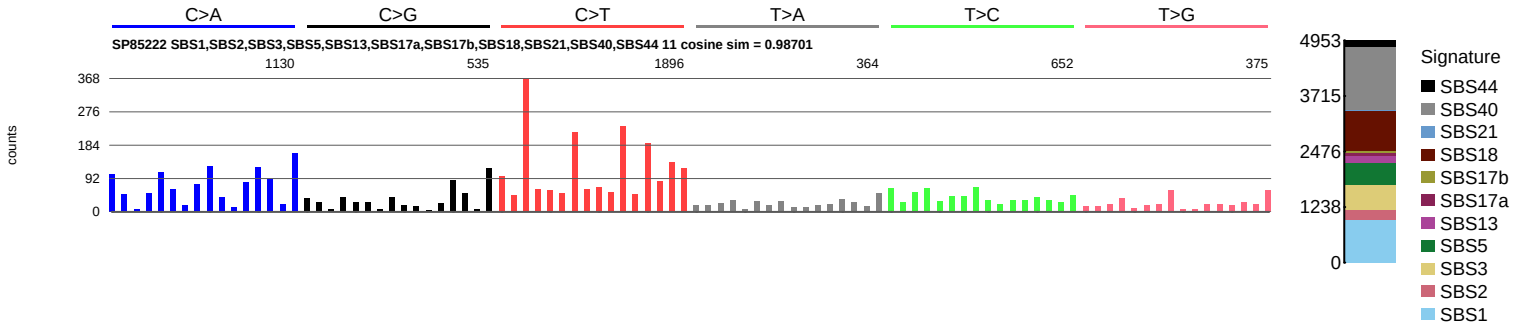
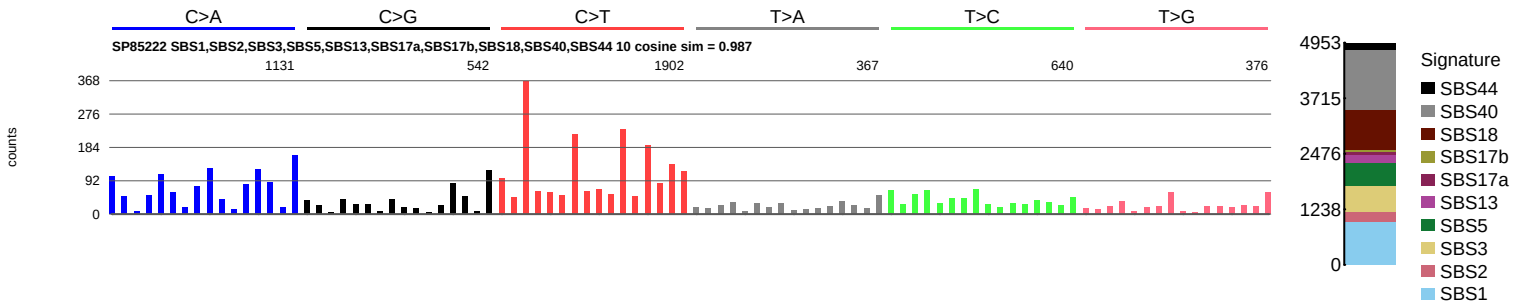
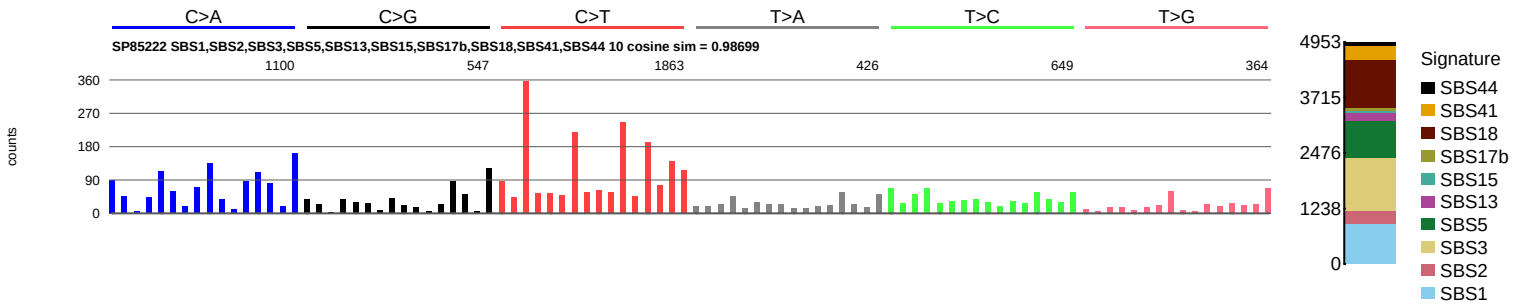
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98702
1133
544
1867
432
603
374
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98701
1130
535
1896
364
652
375
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.987
1131
542
1902
367
640
376
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98699
1100
547
1863
426
649
364
counts
360
270
180
90
0
4953
3715
2476
1238
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Signature
SBS44
SBS41
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SBS17b
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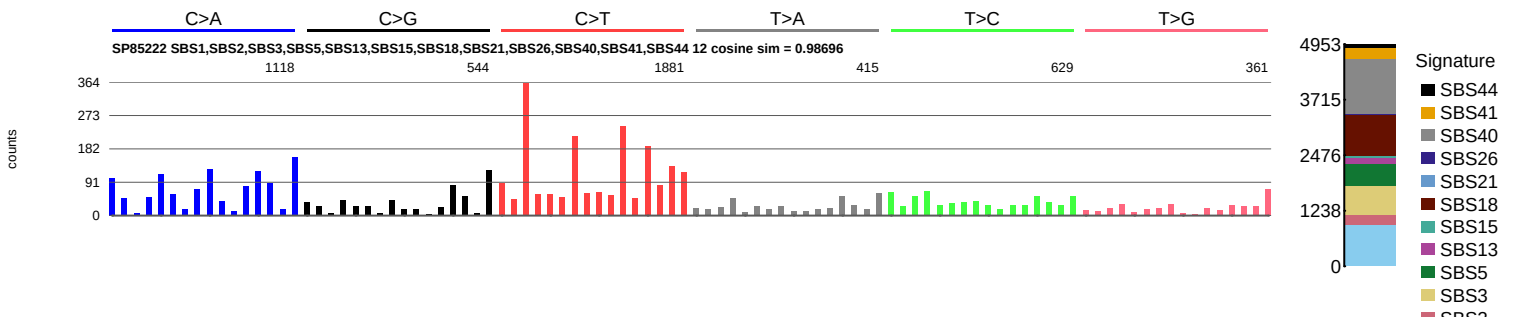
SBS1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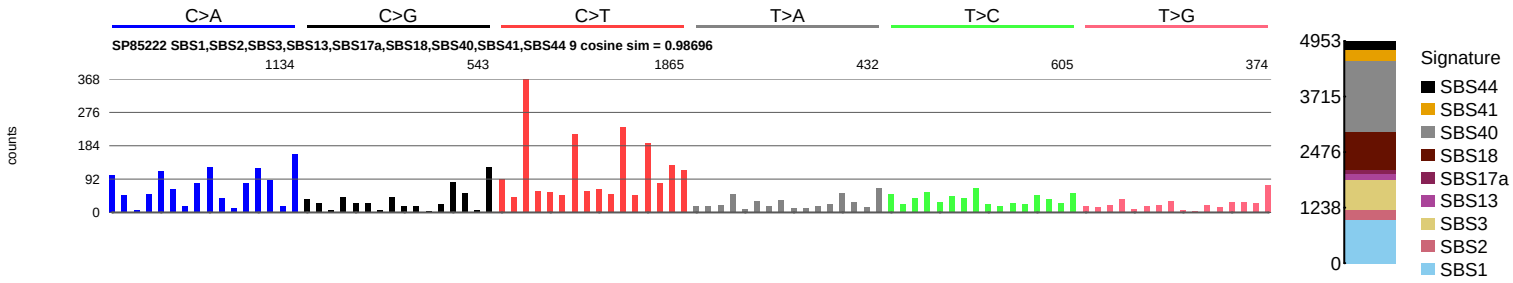
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98696
1118
544
1881
415
629
361
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS5
SBS3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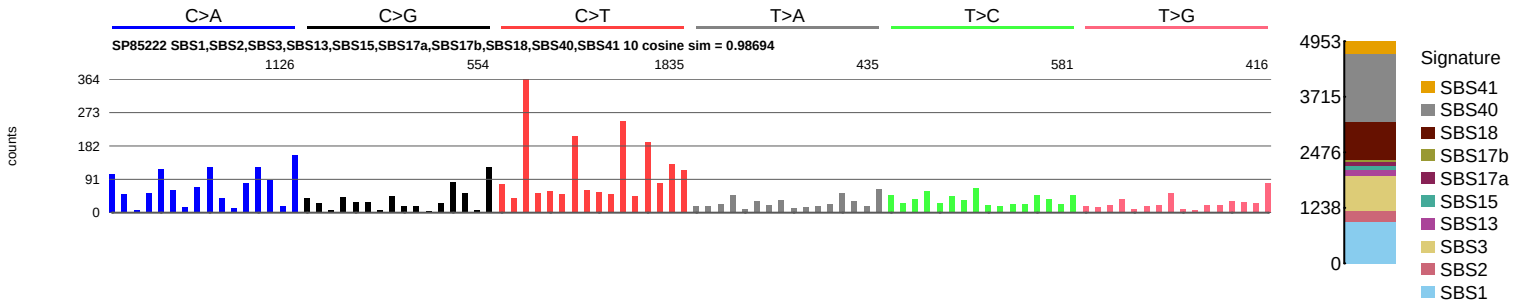
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98696
1134
543
1865
432
605
374
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1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T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98694
1126
554
1835
435
581
416
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
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SBS18
SBS17b
SBS17a
SBS15
SBS13
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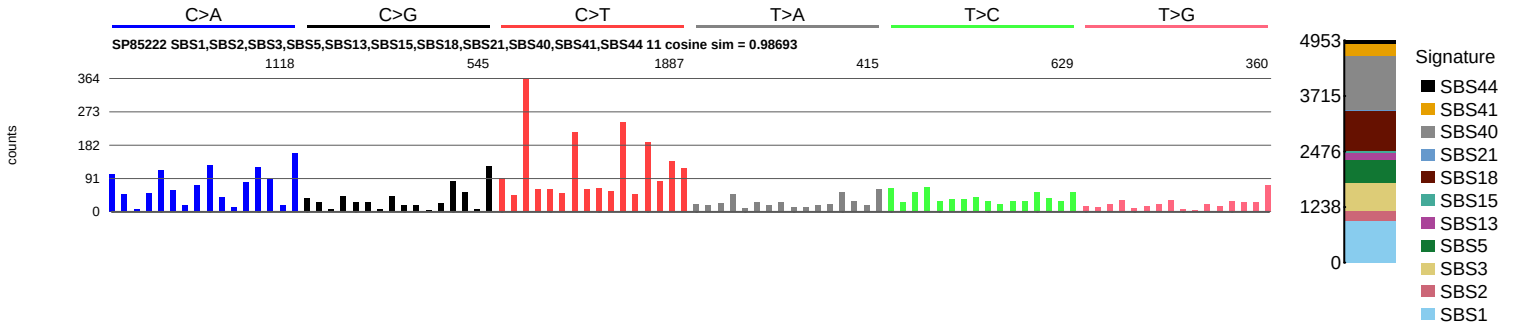
SBS2
SBS1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T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98693
1118
545
1887
415
629
360
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
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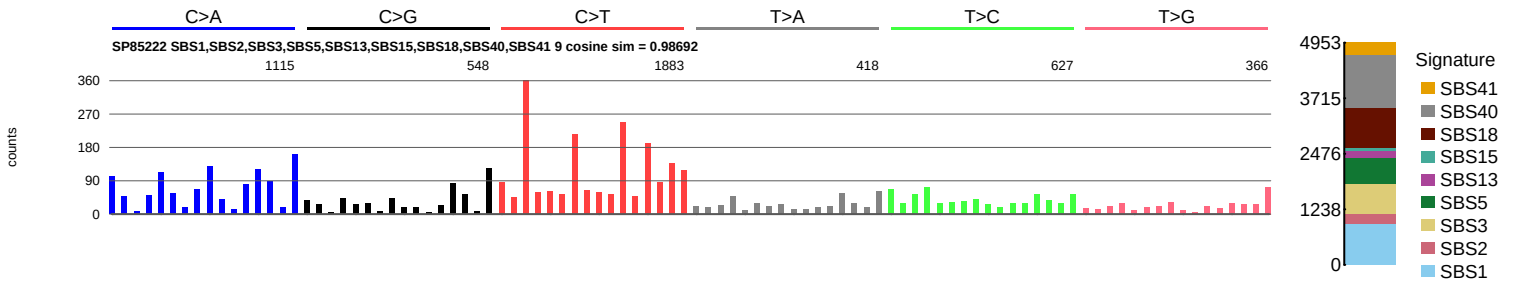
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41 9 cosine sim = 0.98692
1115
548
1883
418
627
366
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
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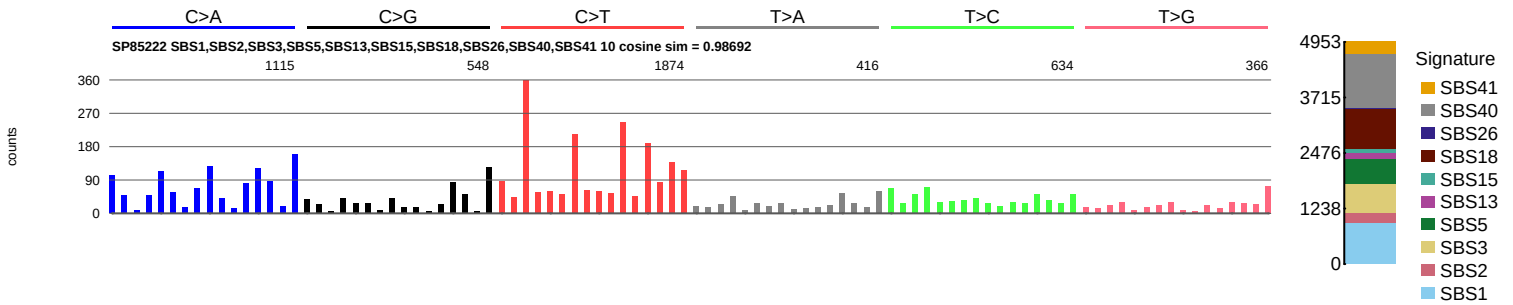
SBS2
SBS1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T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98692
1115
548
1874
416
634
366
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
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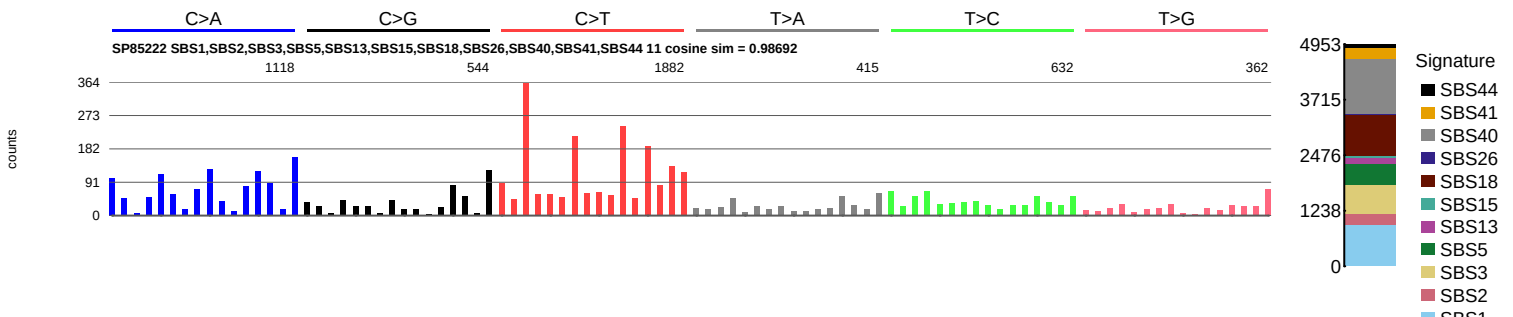
SBS2
SBS1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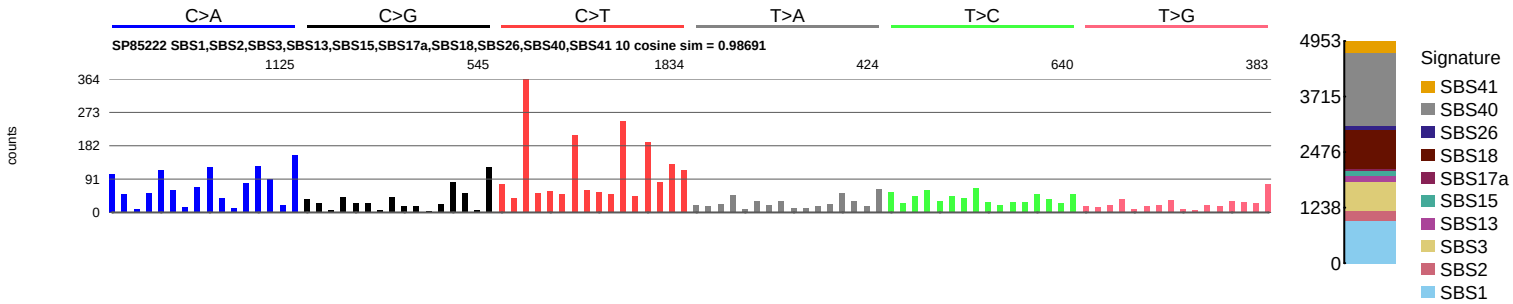
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98692
1118
544
1882
415
632
362
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98691
1125
545
1834
424
640
383
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
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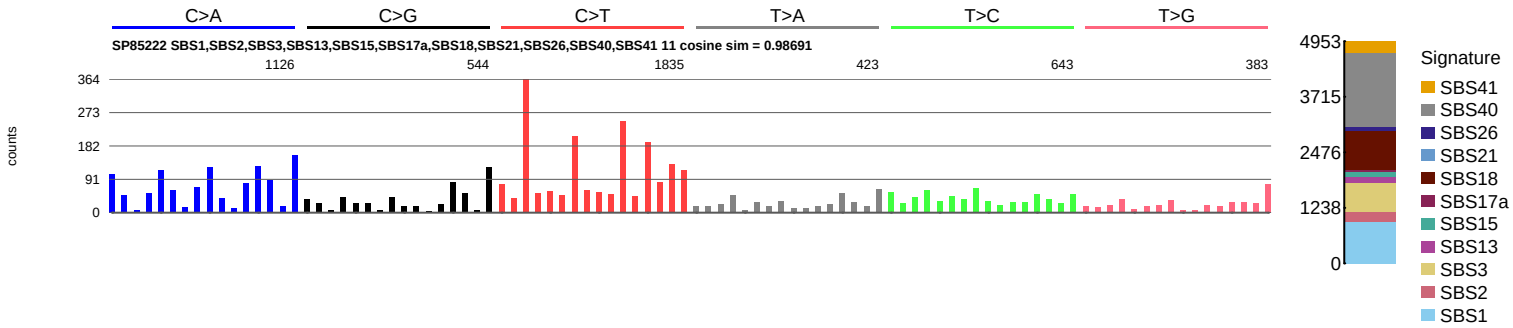
SBS2
SBS1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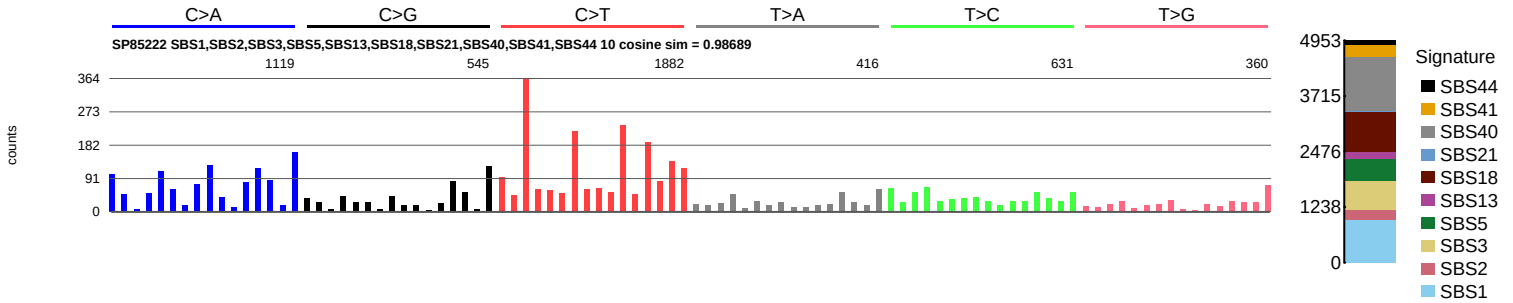
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98691
1126
544
1835
423
643
383
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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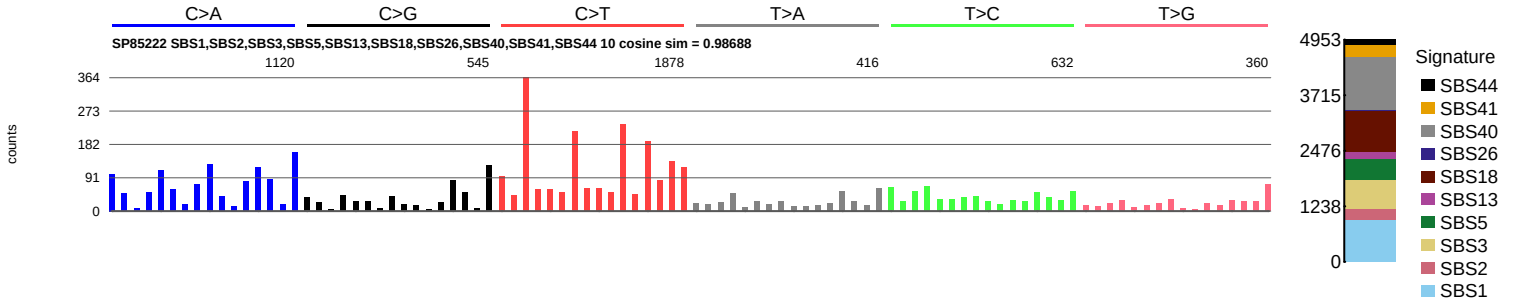
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98689
1119
545
1882
416
631
360
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98688
1120
545
1878
416
632
360
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
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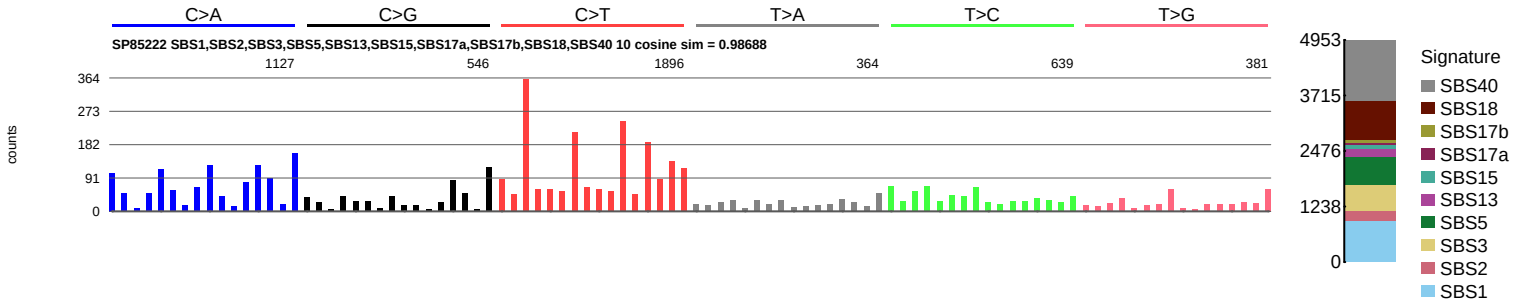
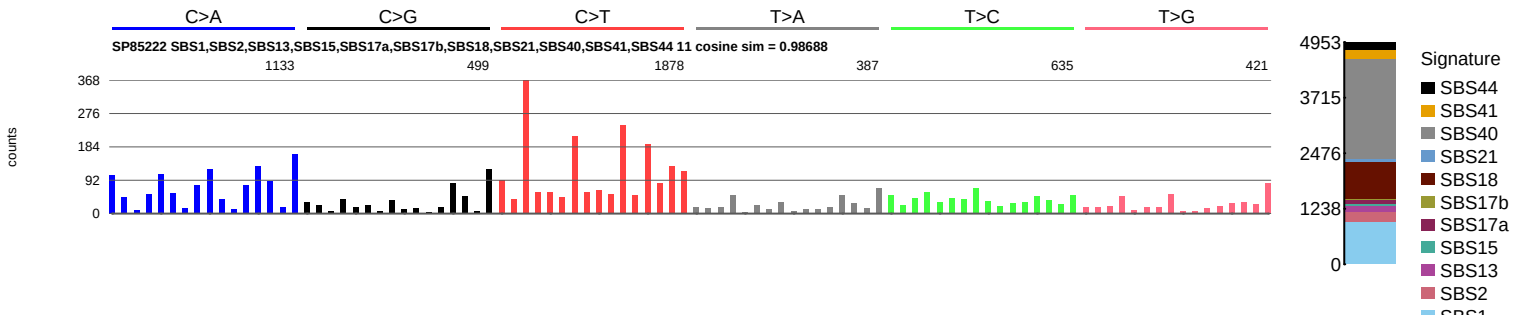
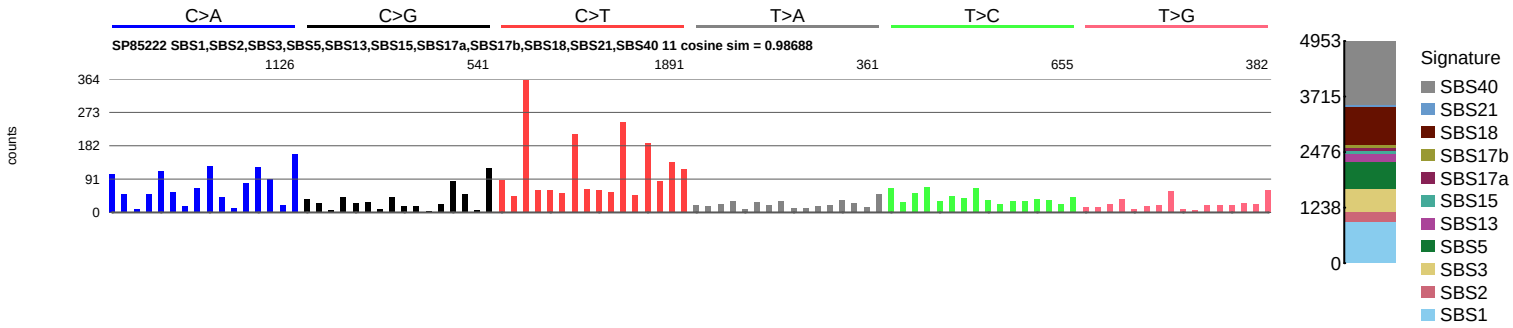
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 10 cosine sim = 0.98688
1127
546
1896
364
639
381
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98688
1133
499
1878
387
635
421
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2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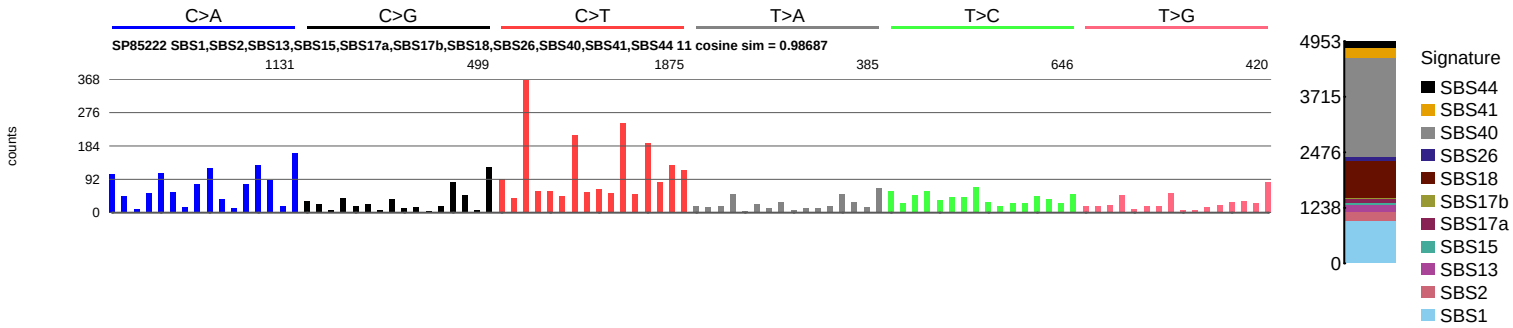
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 11 cosine sim = 0.98688
1126
541
1891
361
655
382
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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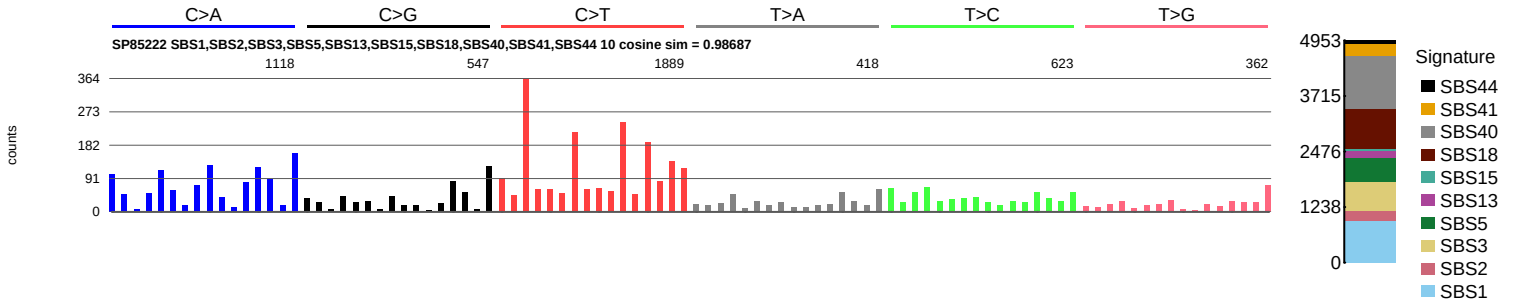
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98687
1131
499
1875
385
646
420
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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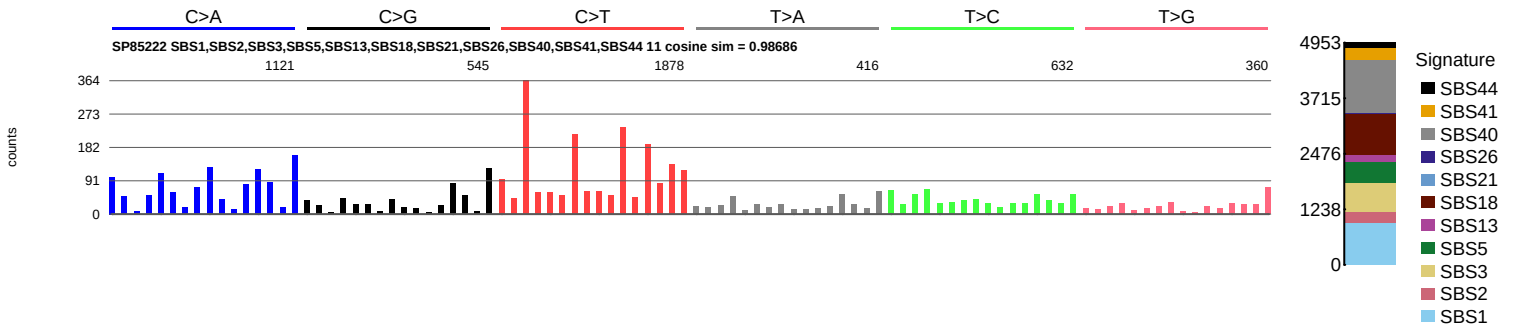
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98687
1118
547
1889
418
623
362
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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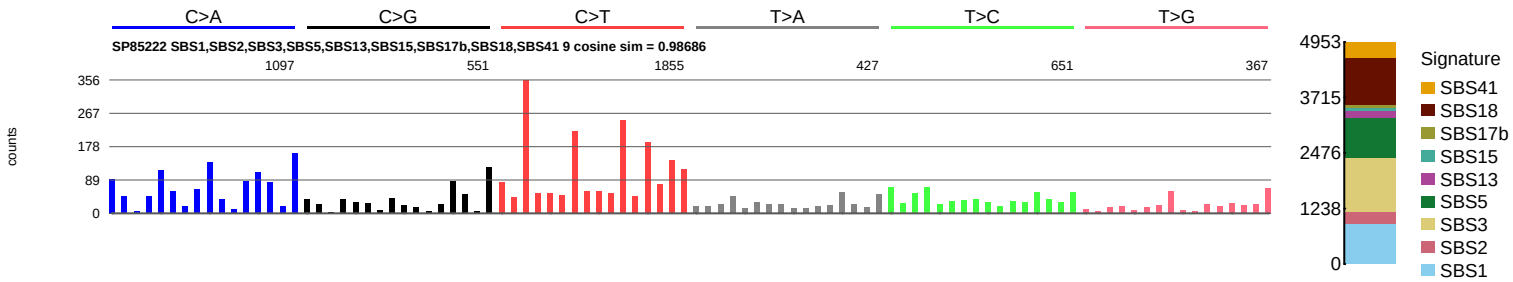
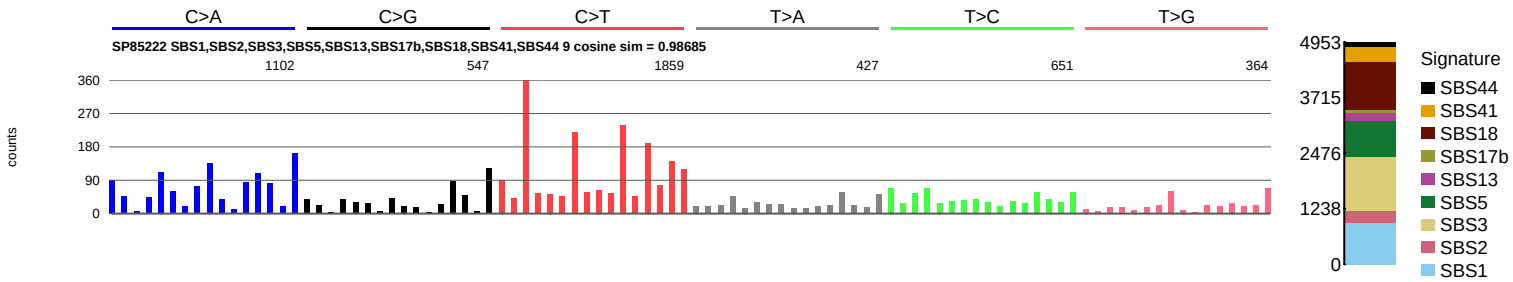
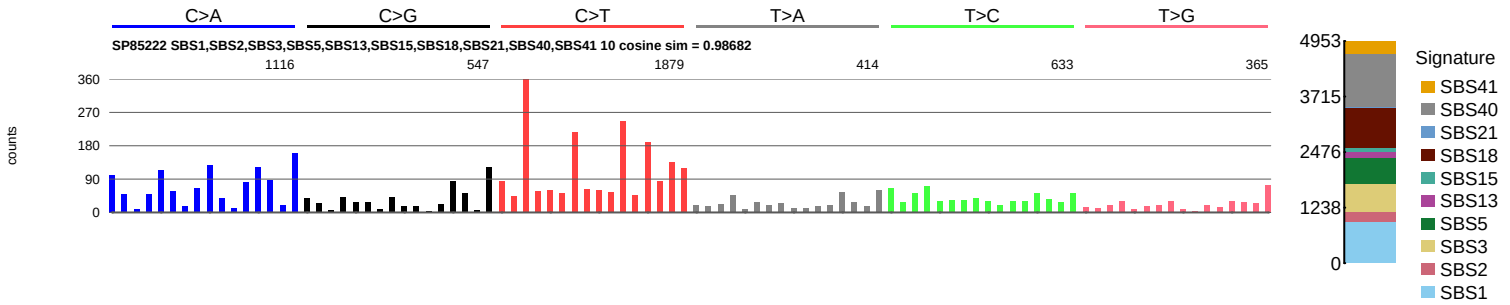
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98686
1121
545
1878
416
632
360
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 9 cosine sim = 0.98686
1097
551
1855
427
651
367
counts
356
267
178
89
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98685
1102
547
1859
427
651
364
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98682
1116
547
1879
414
633
365
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
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SBS40
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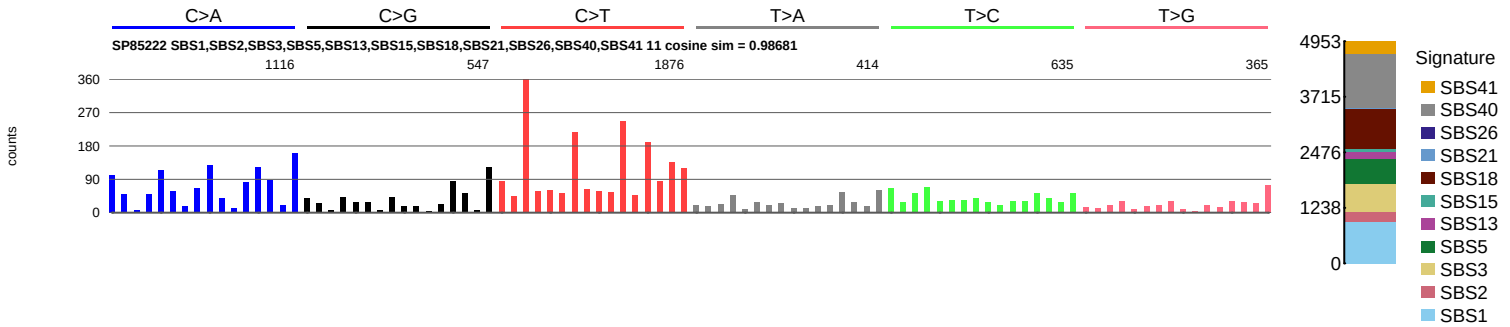
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98681
1116
547
1876
414
635
365
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
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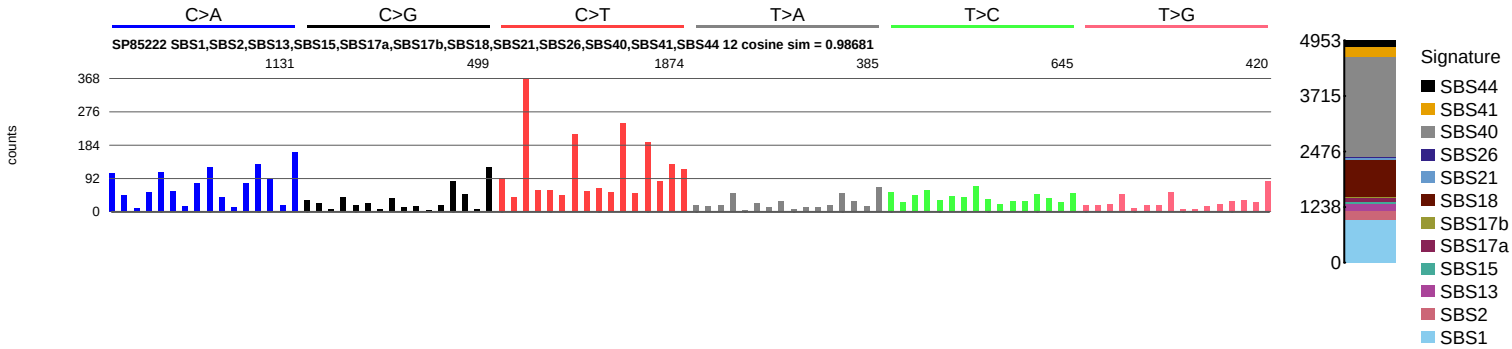
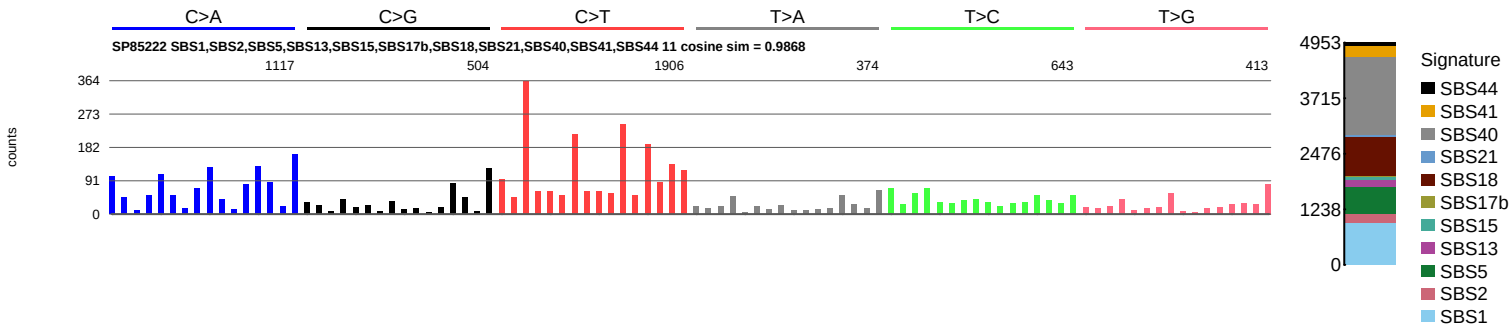
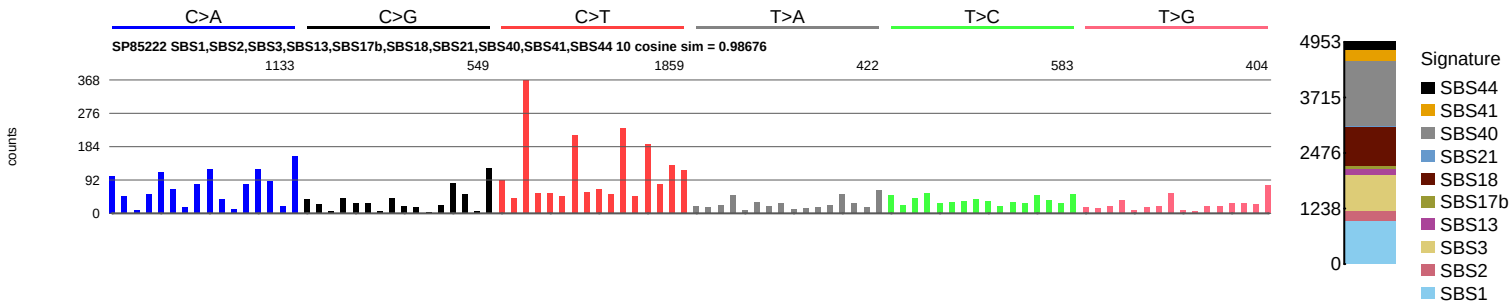
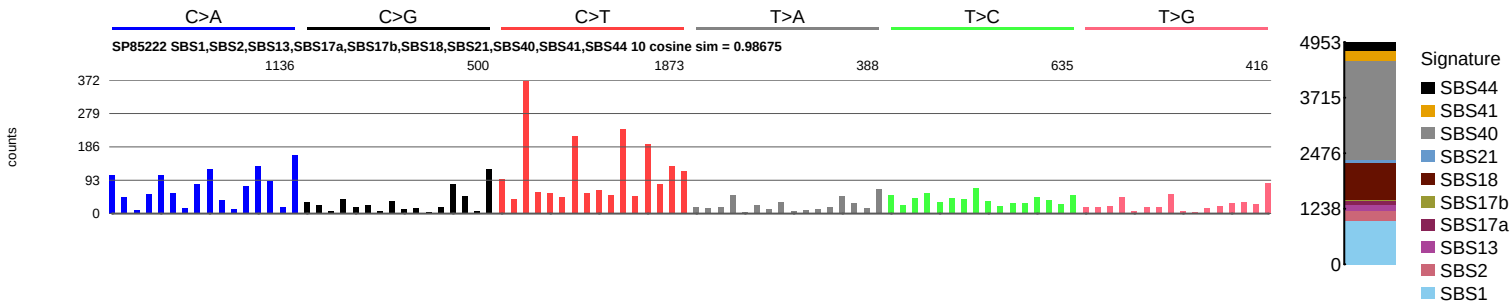
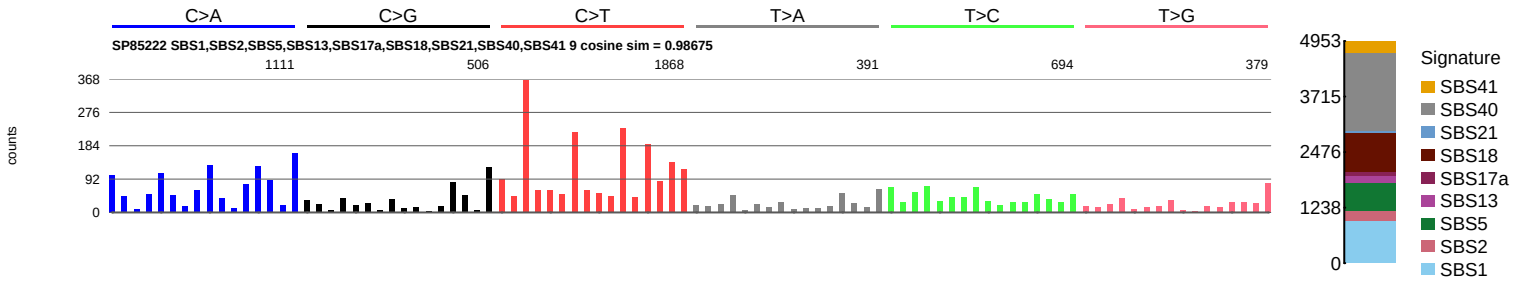
SBS3
SBS2
SBS1

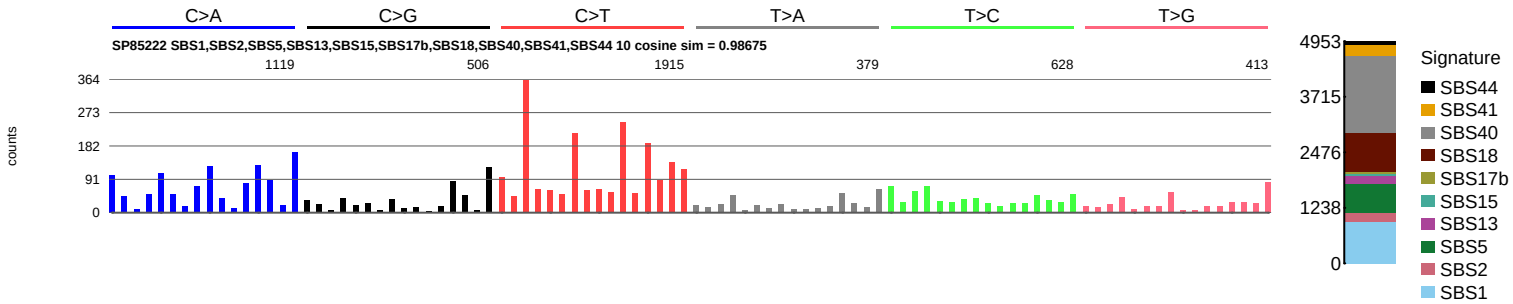

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98681
1131
499
1874
385
645
420
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.9868
1117
504
1906
374
643
413
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98676
1133
549
1859
422
583
404
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98675
1136
500
1873
388
635
416
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98675
1111
506
1868
391
694
379
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98675
1119
506
1915
379
628
413
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
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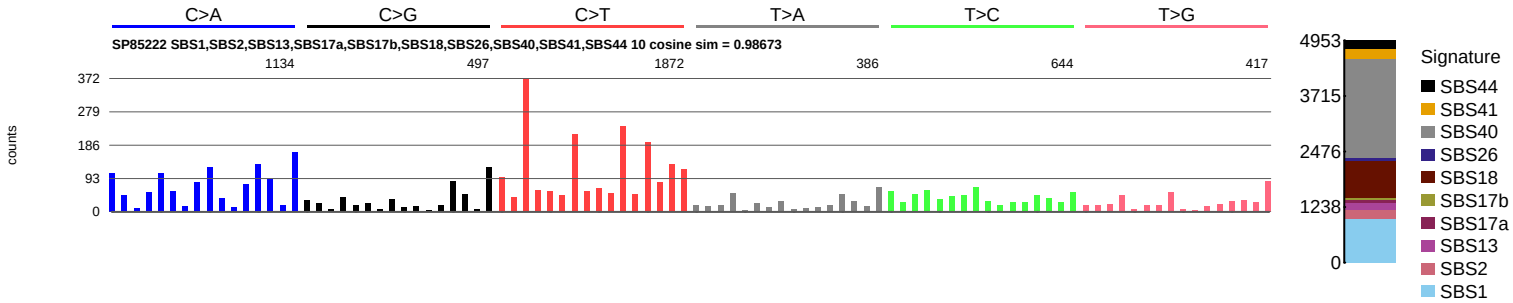
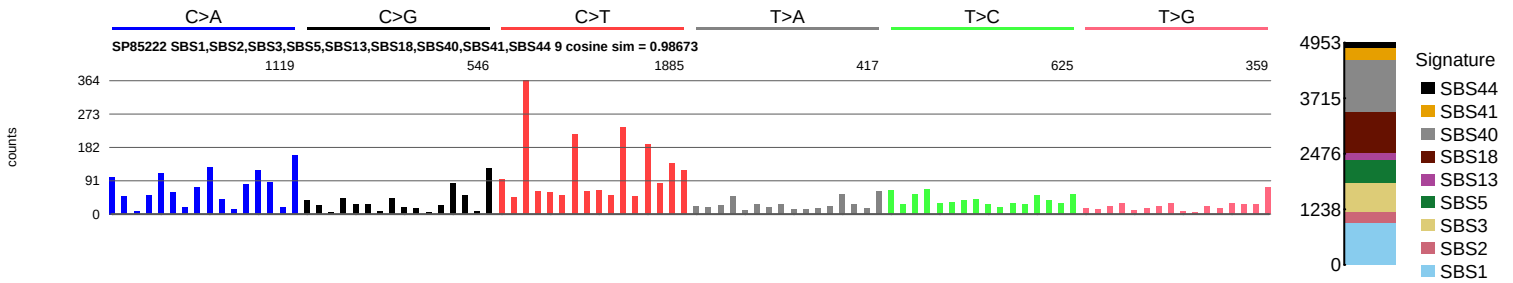
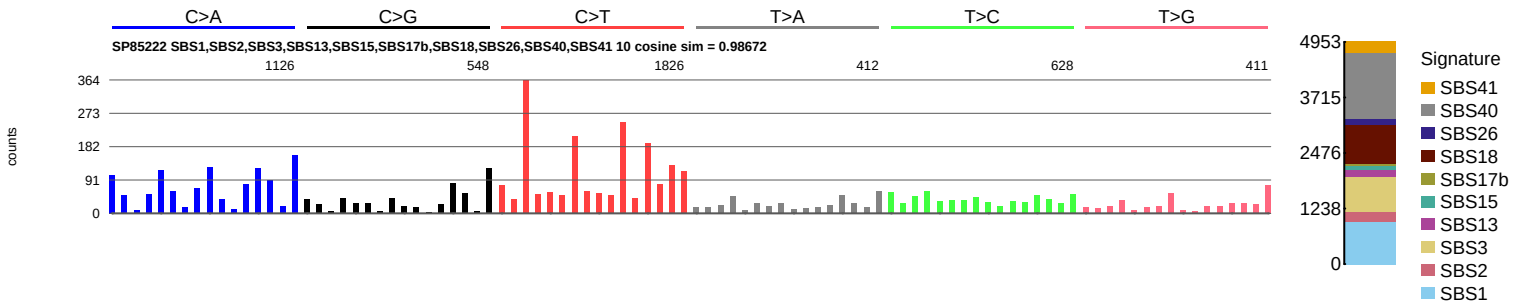
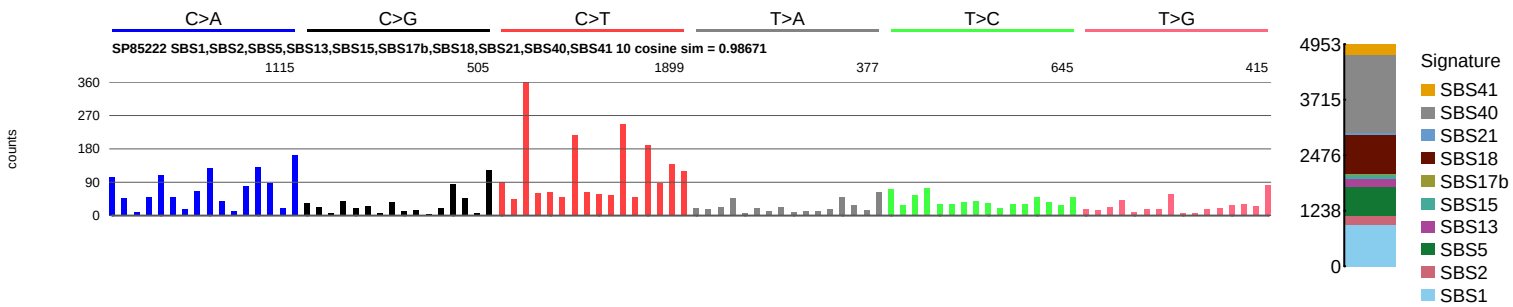
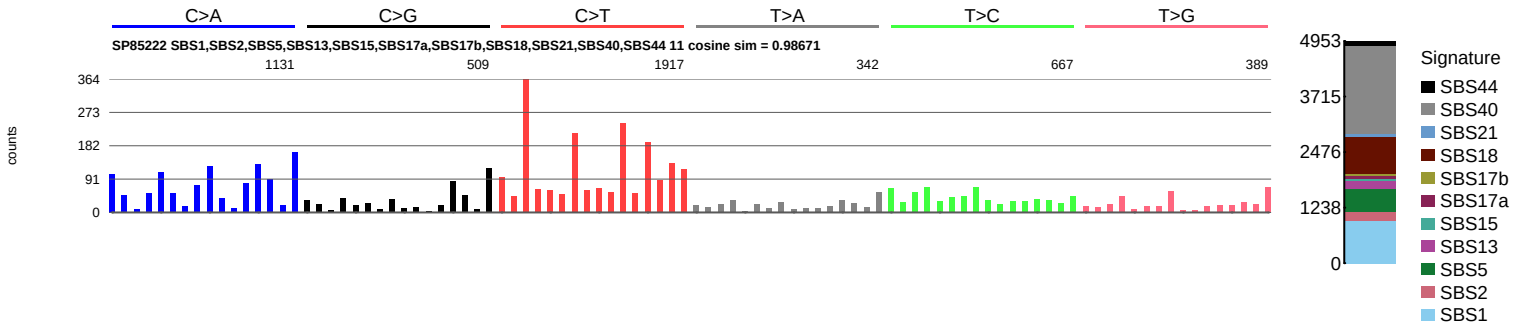
SBS13
SBS5
SBS2
SBS1

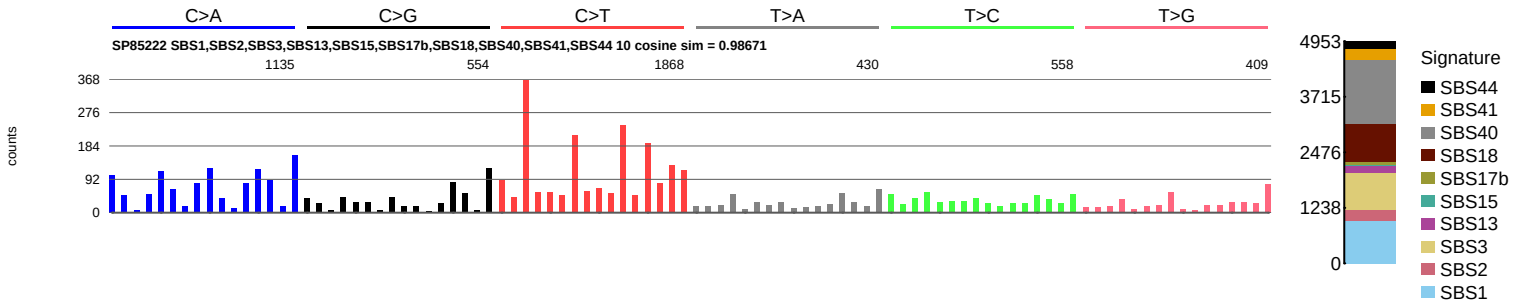

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98673
1134
497
1872
386
644
417
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

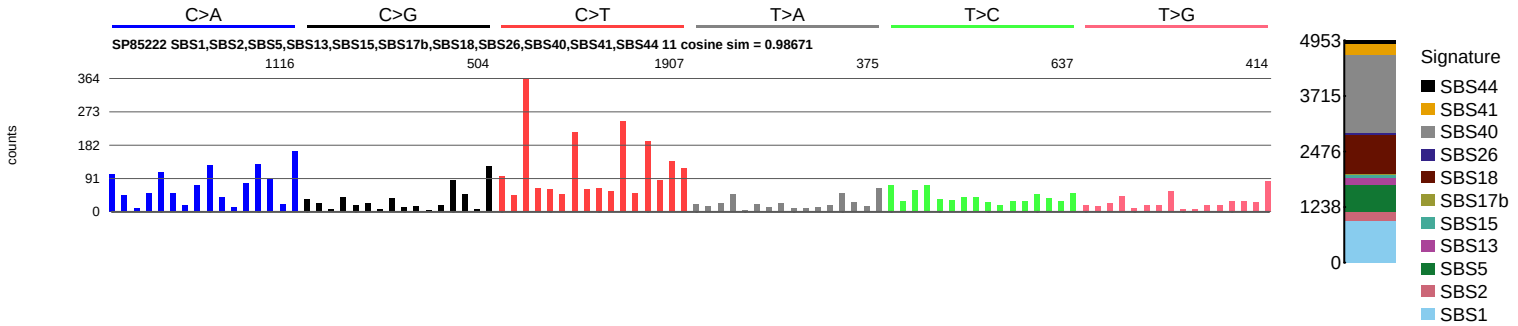

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98673
1119
546
1885
417
625
359
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1

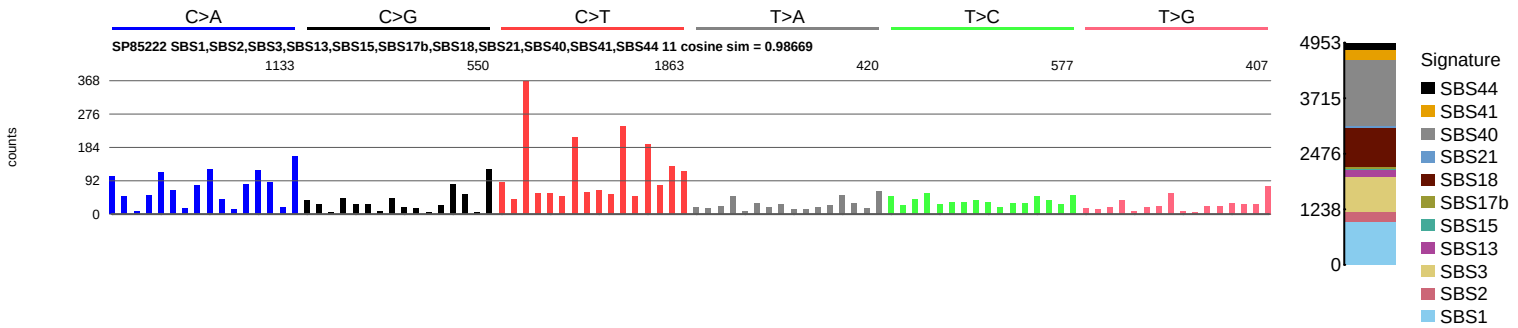

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98672
1126
548
1826
412
628
411
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

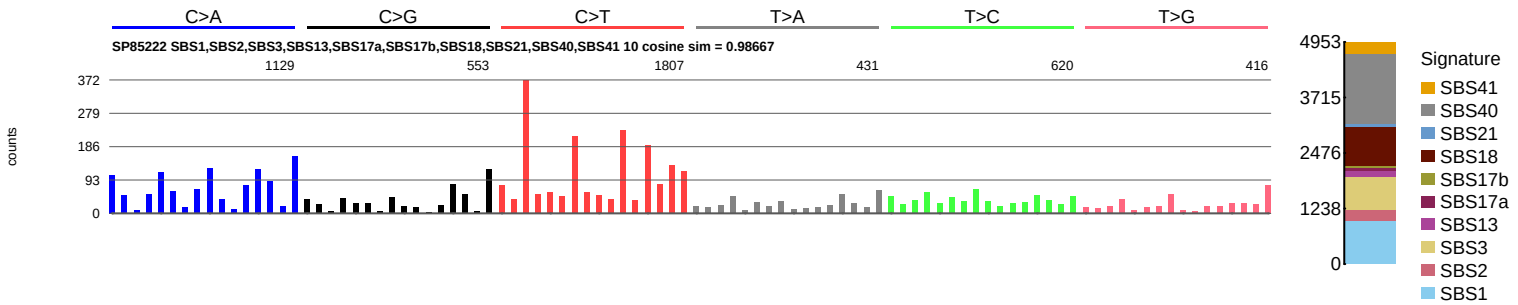

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98671
1115
505
1899
377
645
415
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

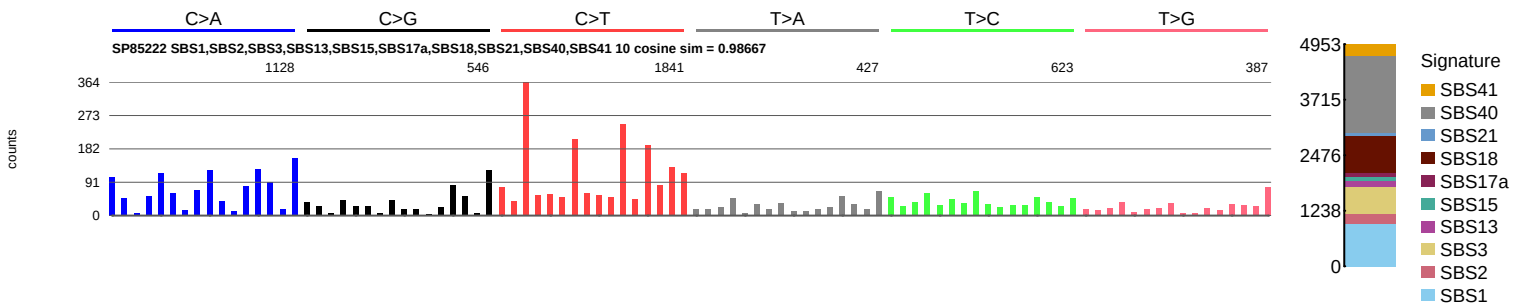

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98671
1131
509
1917
342
667
389
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

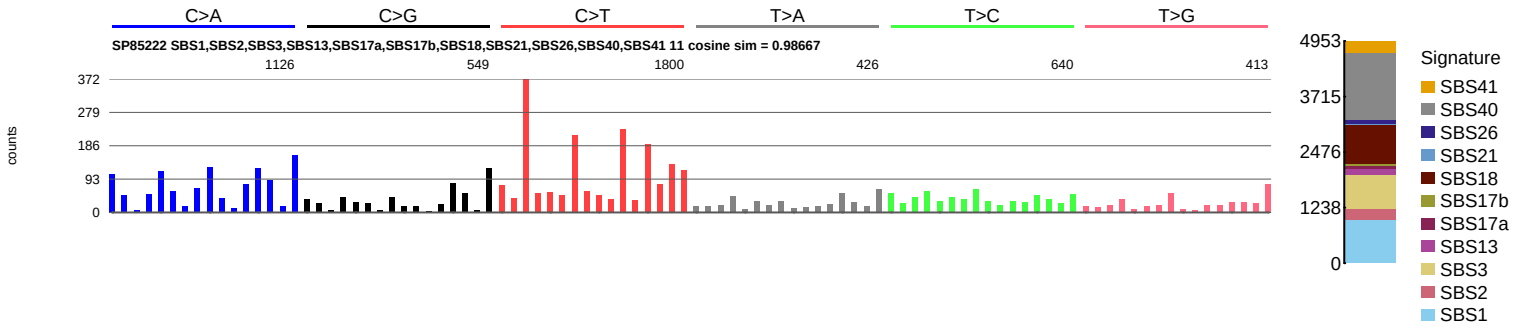

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98671
1135
554
1868
430
558
409
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

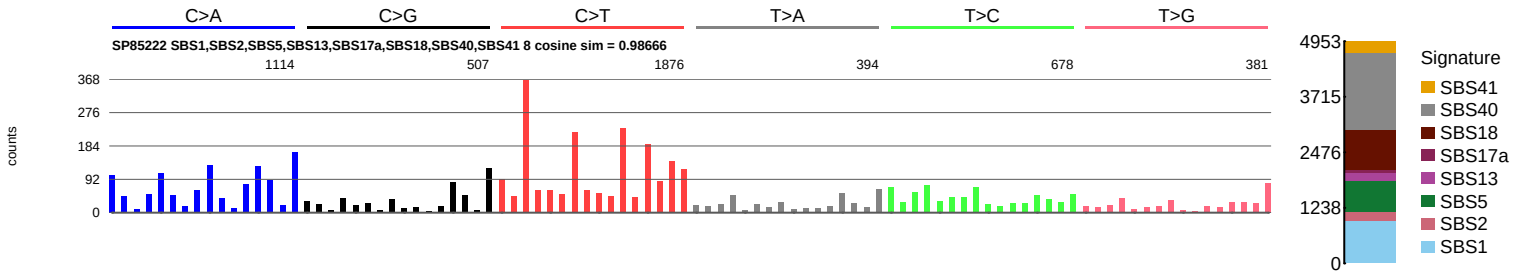

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98671
1116
504
1907
375
637
414
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

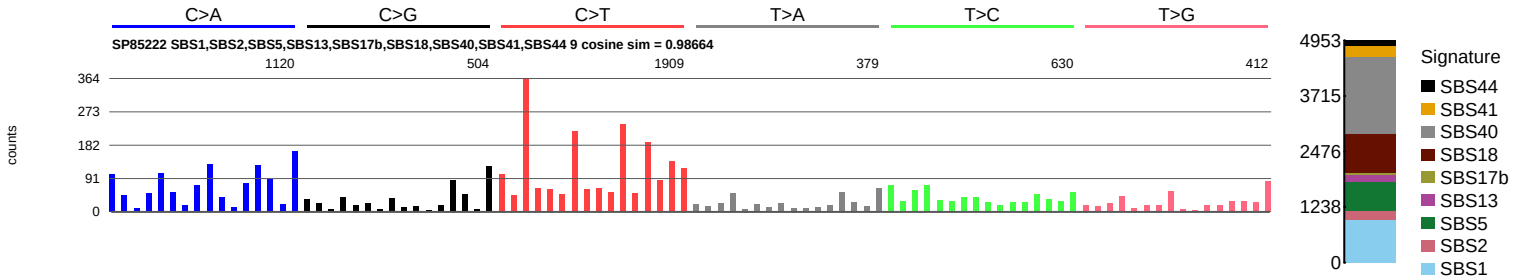

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98669
1133
550
1863
420
577
407
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

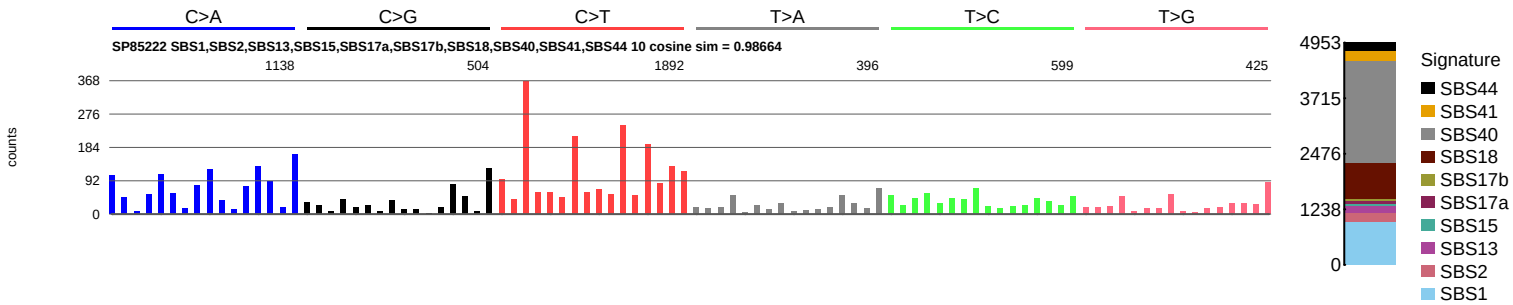

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98667
1129
553
1807
431
620
416
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

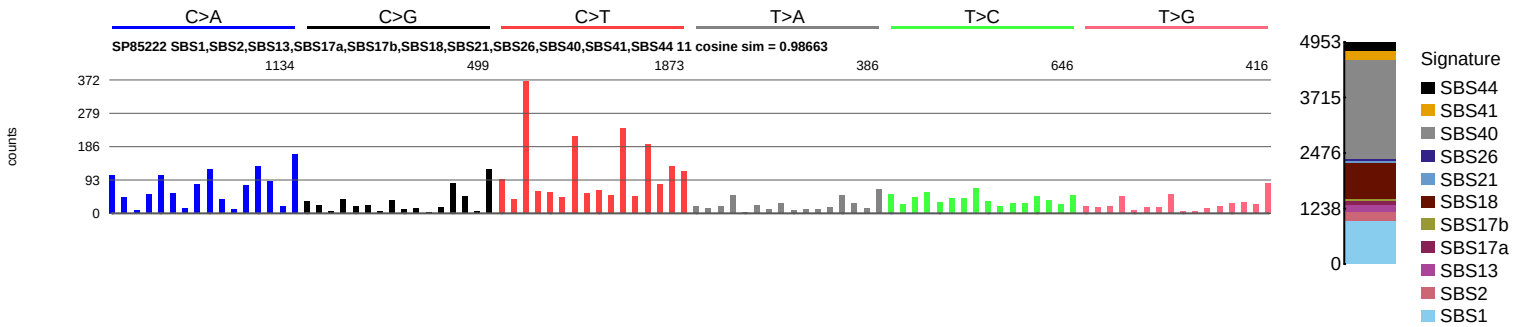

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98667
1128
546
1841
427
623
387
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

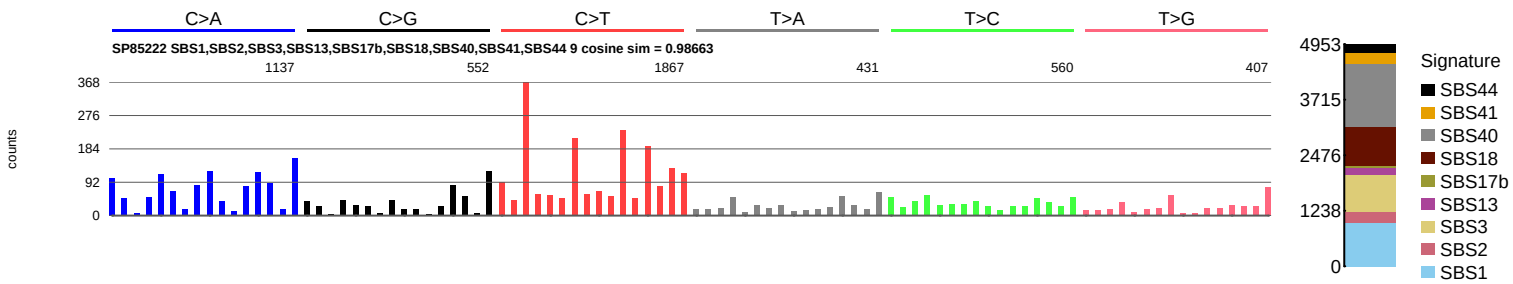

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98667
1126
549
1800
426
640
413
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

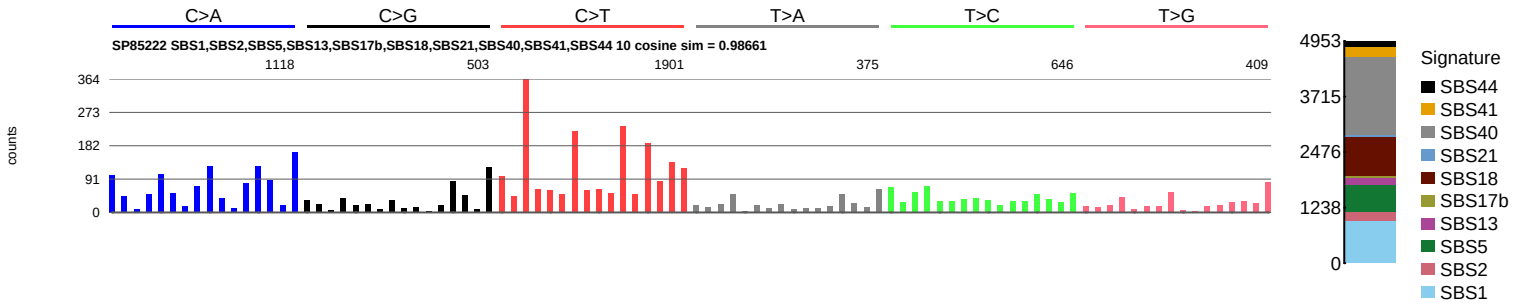

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98666
1114
507
1876
394
678
381
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1

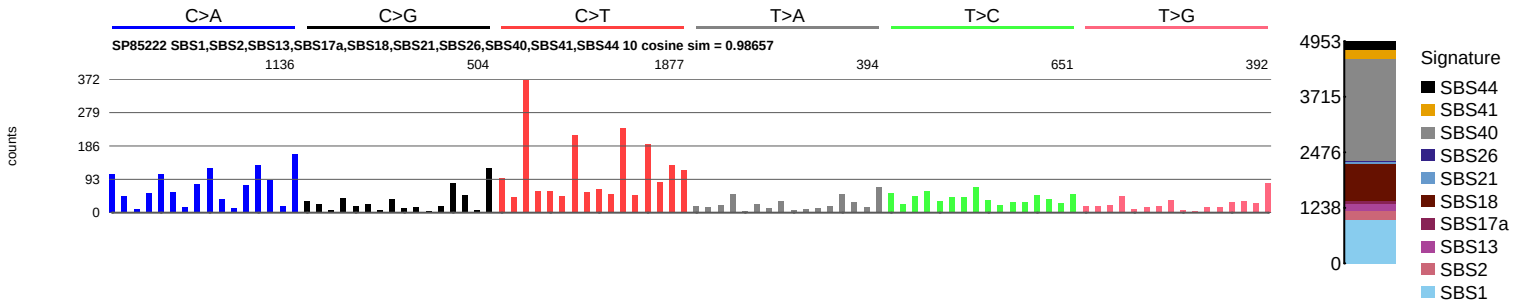

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98664
1120
504
1909
379
630
412
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

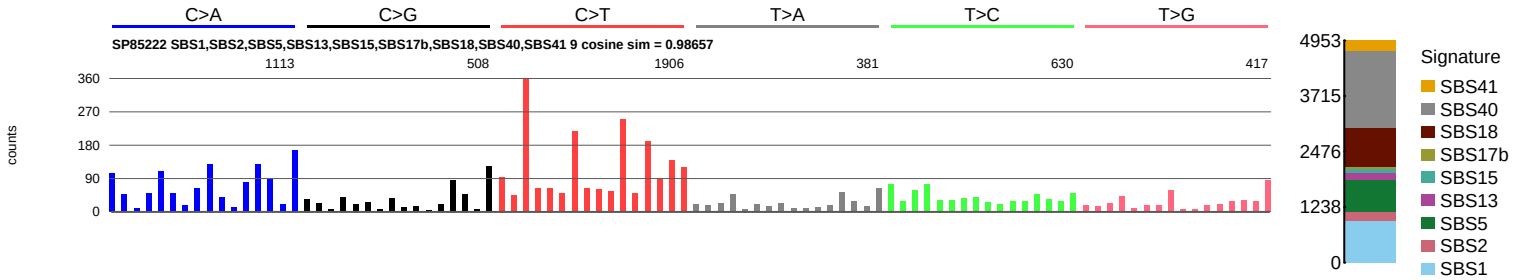

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98664
1138
504
1892
396
599
425
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

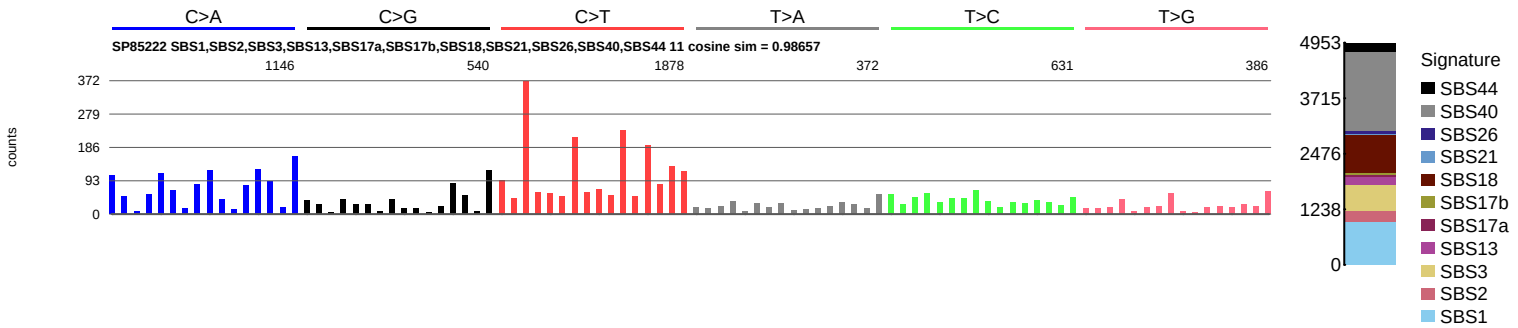

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98663
1134
499
1873
386
646
416
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

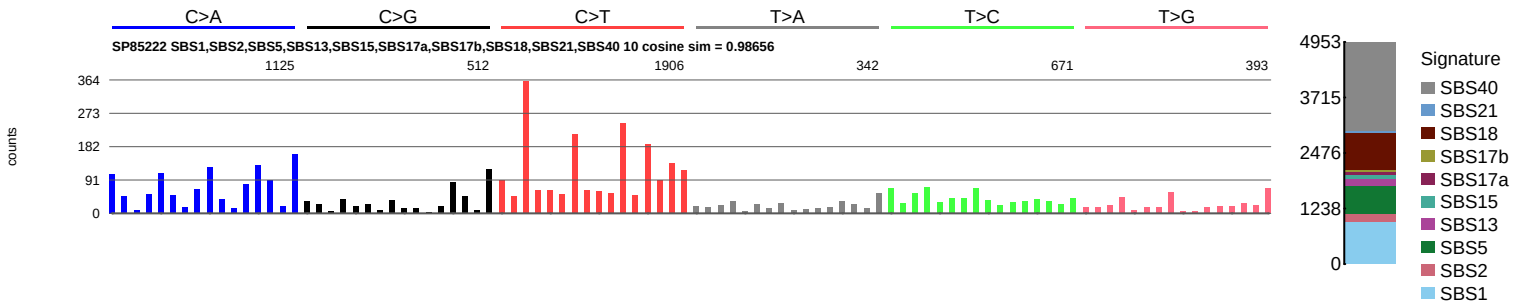

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98663
1137
552
1867
431
560
407
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

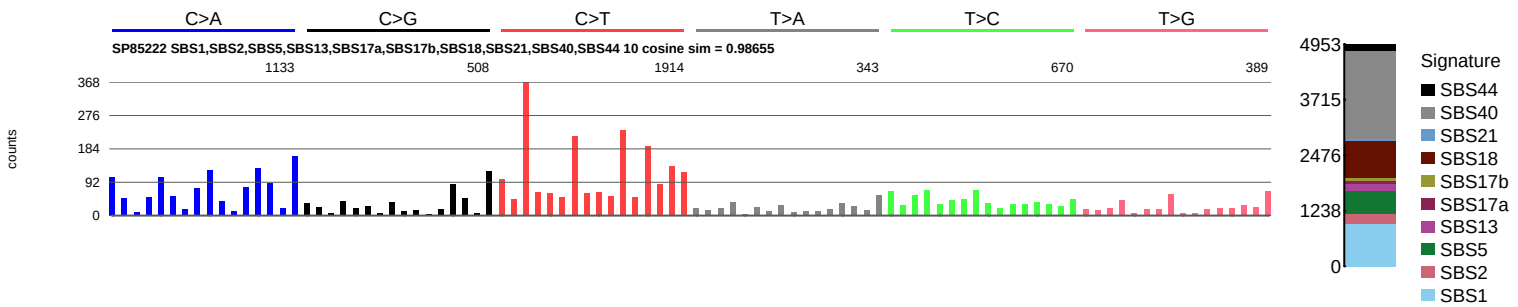

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98661
1118
503
1901
375
646
409
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

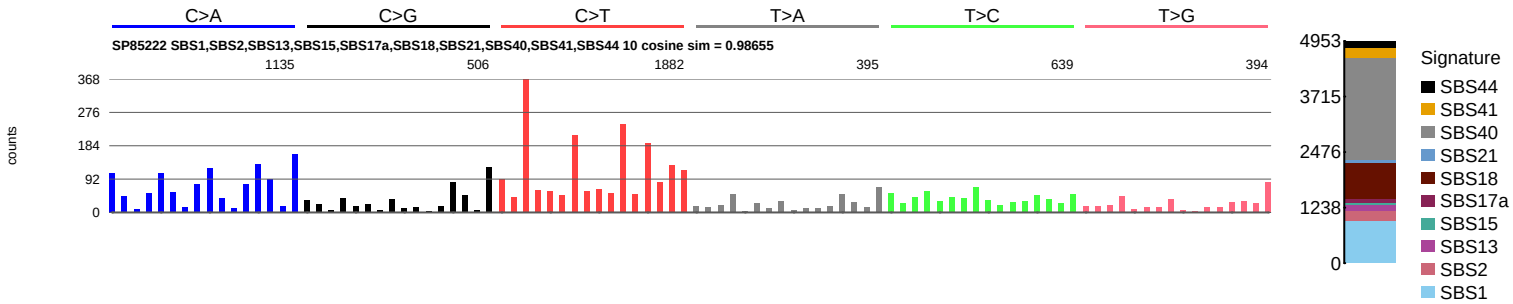

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98657
1136
504
1877
394
651
392
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98657
1113
508
1906
381
630
417
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98657
1146
540
1878
372
631
386
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98656
1125
512
1906
342
671
393
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98655
1133
508
1914
343
670
389
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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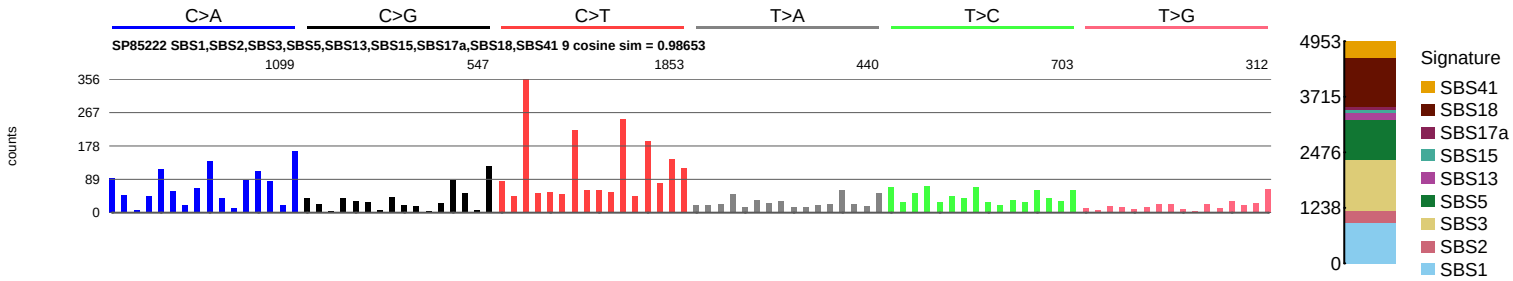
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98655
1135
506
1882
395
639
394
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1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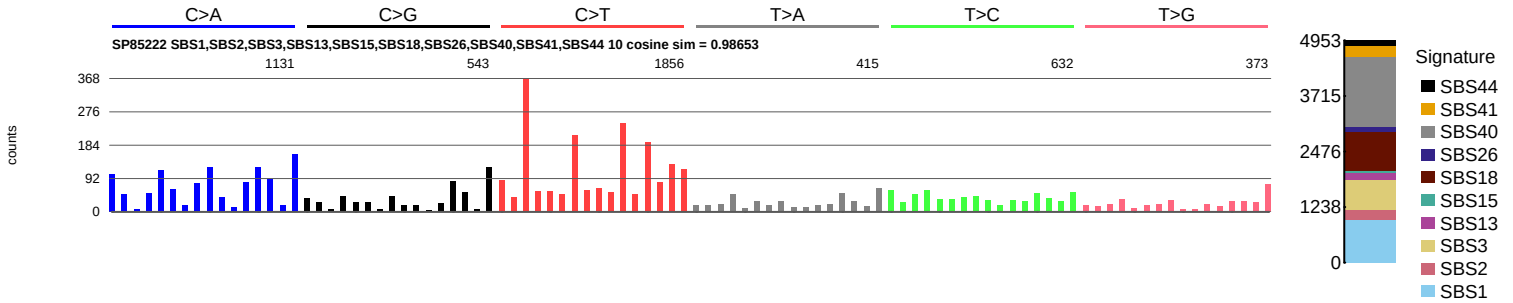
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41 9 cosine sim = 0.98653
1099
547
1853
440
703
312
counts
356
267
178
89
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98653
1131
543
1856
415
632
373
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
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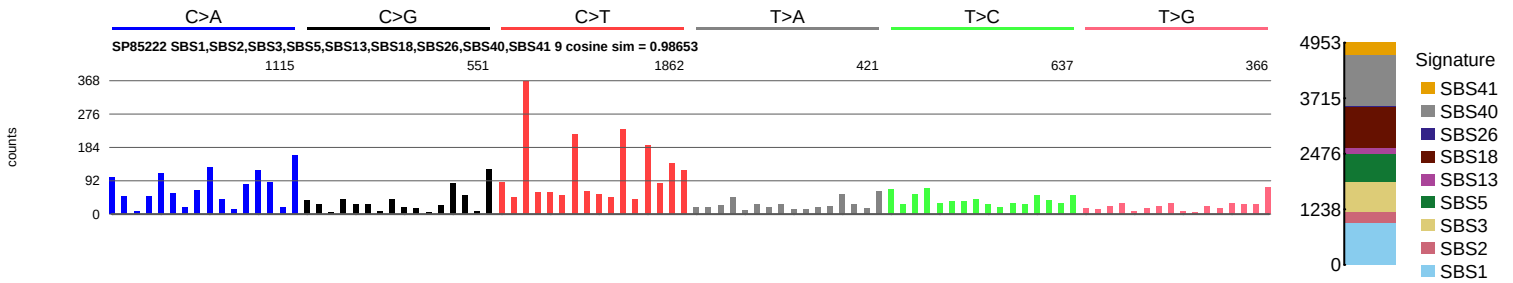
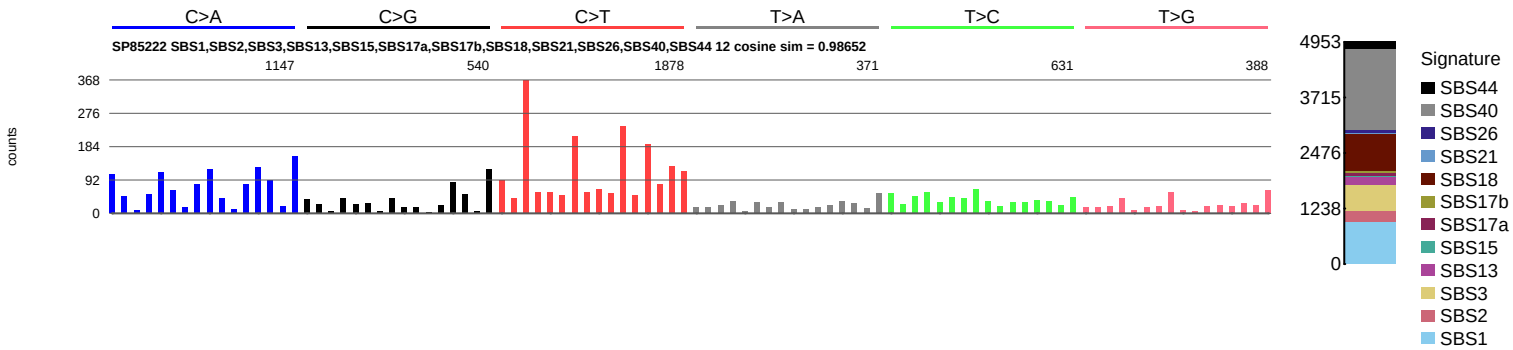
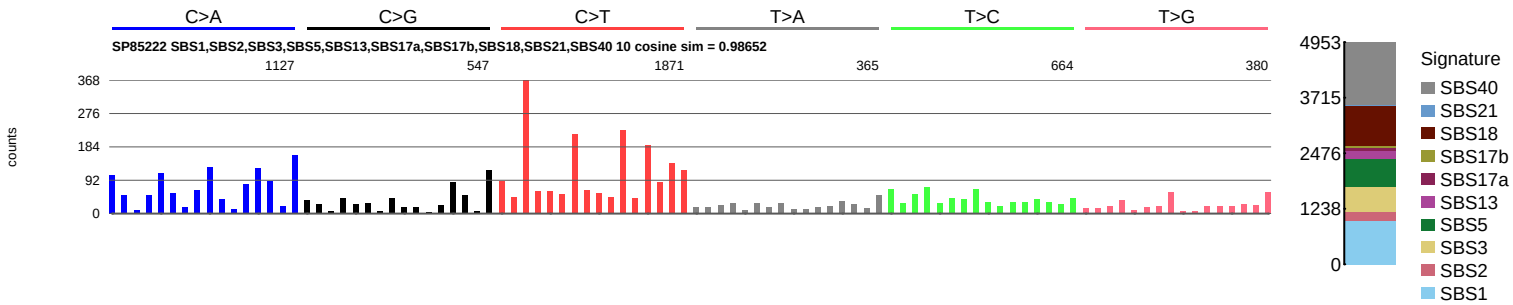
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98653
1115
551
1862
421
637
366
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 12 cosine sim = 0.98652
1147
540
1878
371
631
388
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98652
1127
547
1871
365
664
380
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
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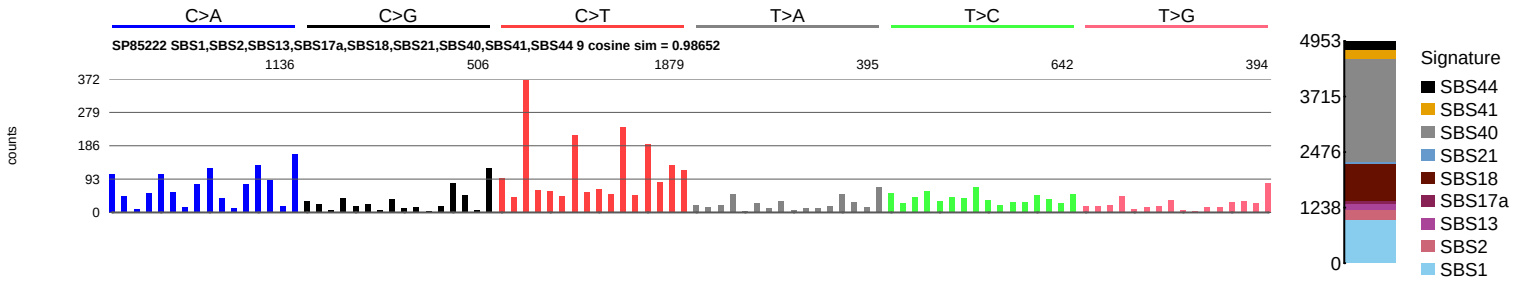
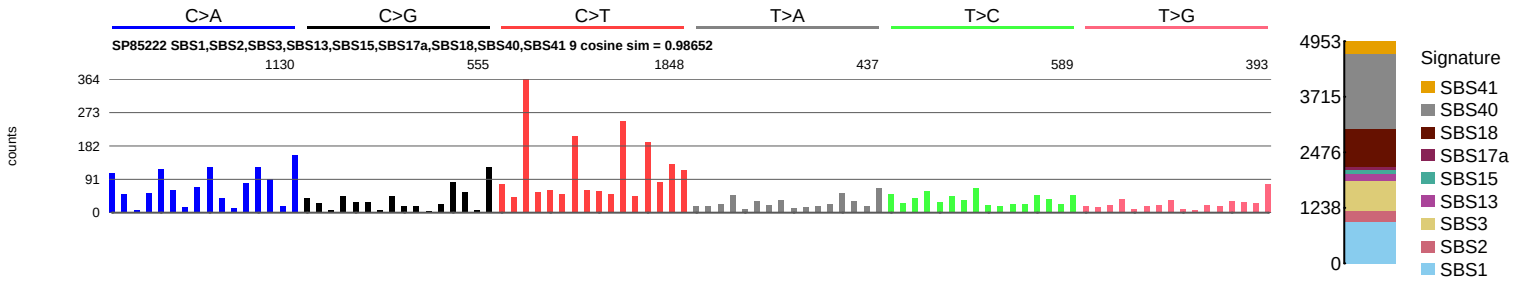
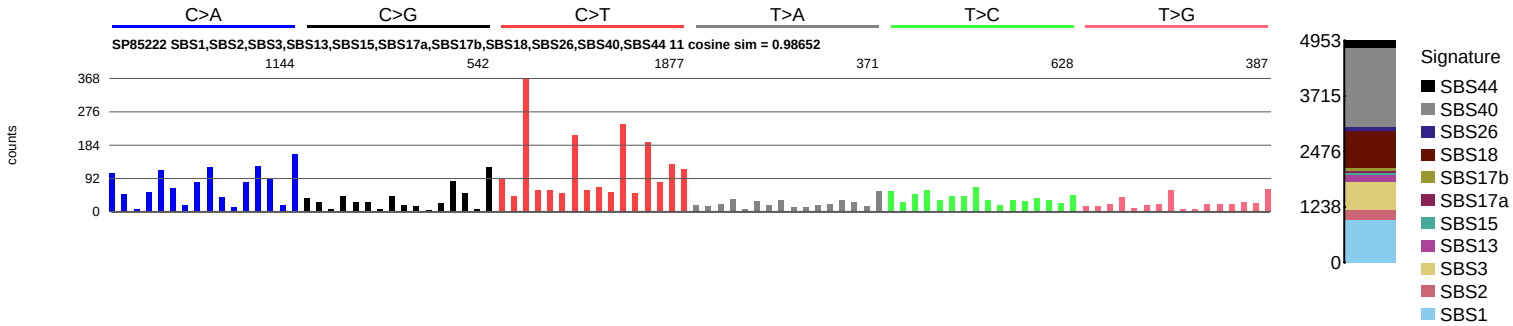
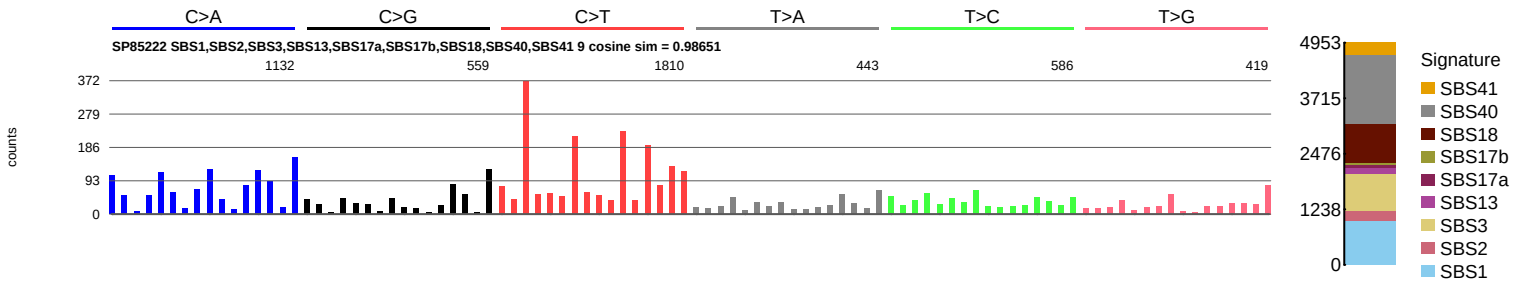
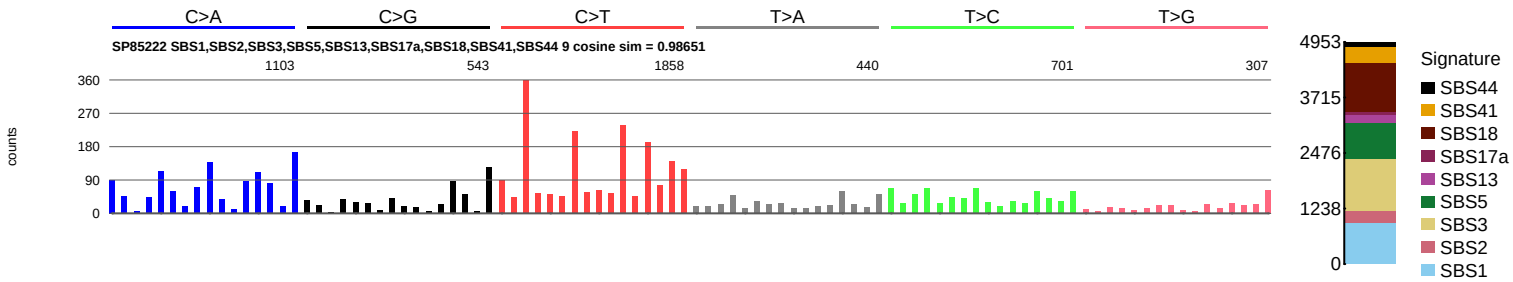
SBS3
SBS2
SBS1

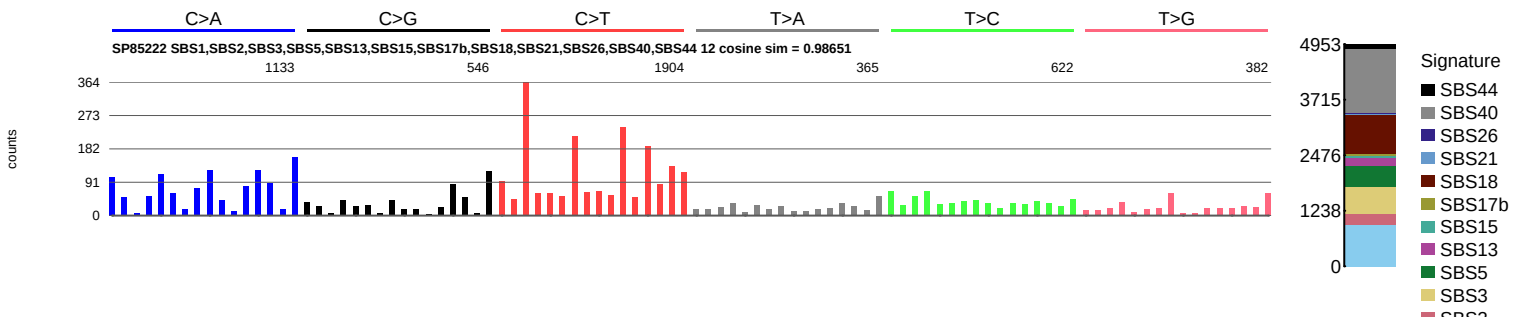

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98652
1136
506
1879
395
642
394
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1

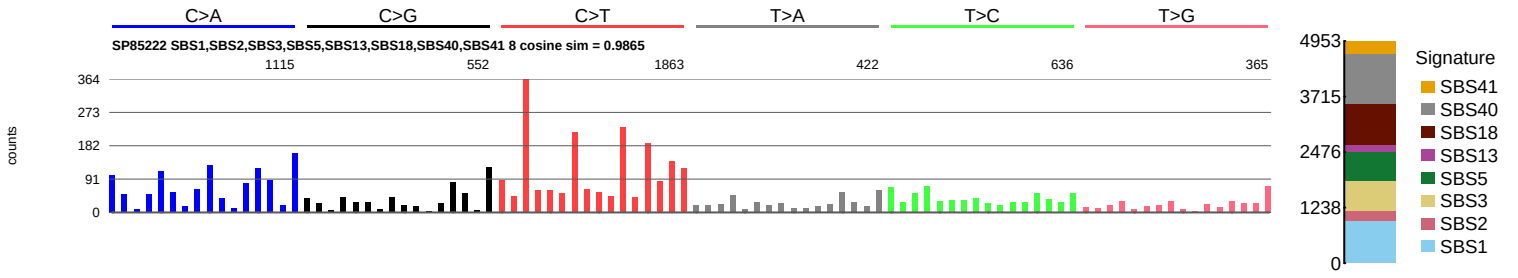

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41 9 cosine sim = 0.98652
1130
555
1848
437
589
393
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

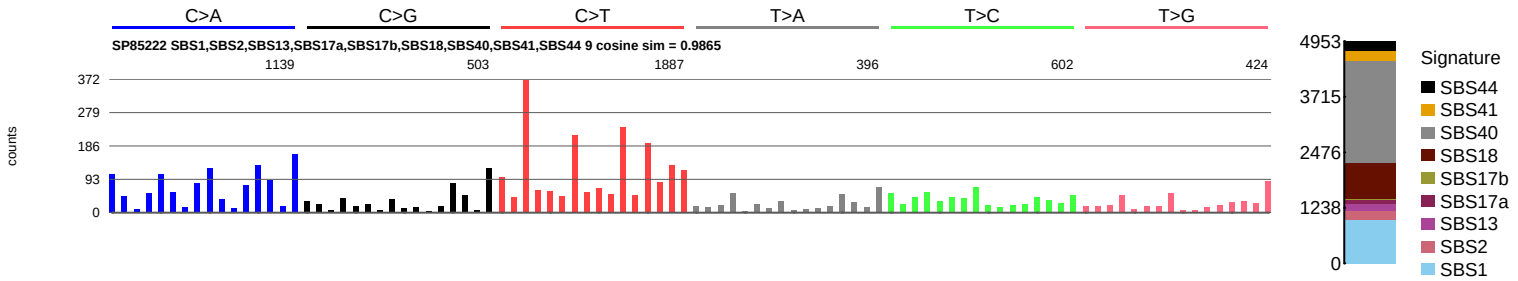

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98652
1144
542
1877
371
628
387
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

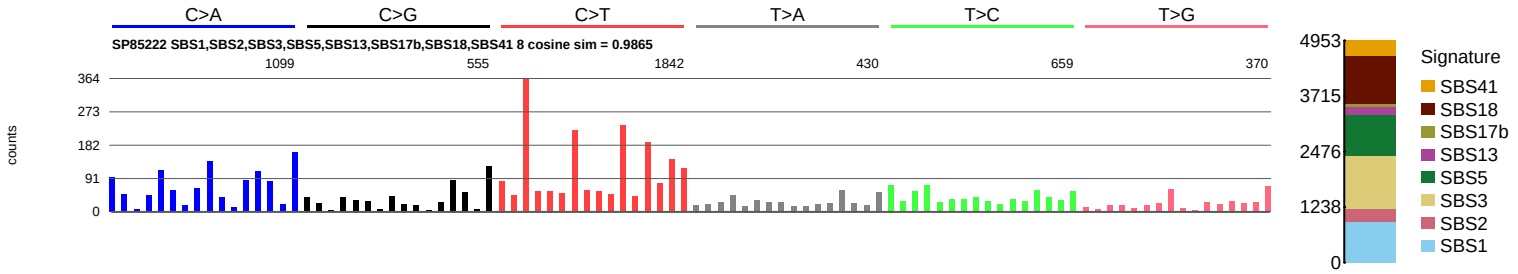

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98651
1132
559
1810
443
586
419
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

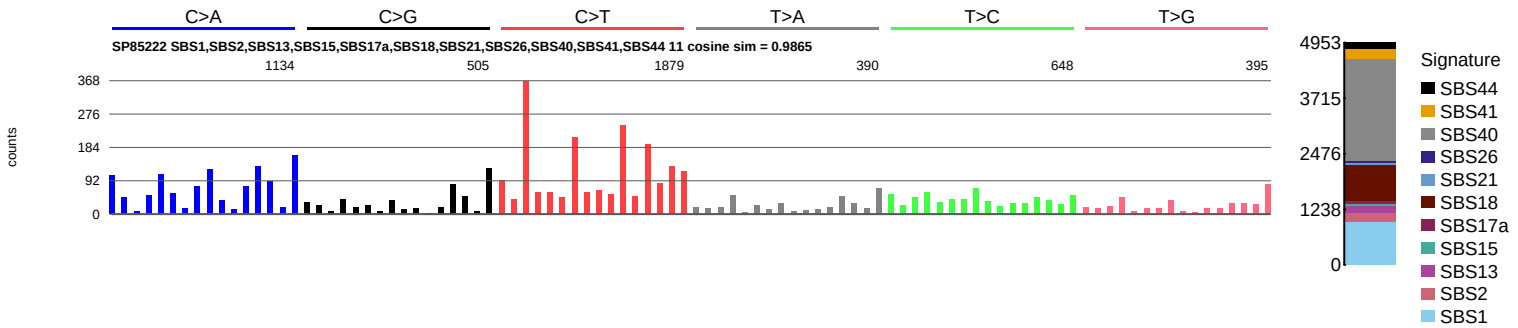

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS41,SBS44 9 cosine sim = 0.98651
1103
543
1858
440
701
307
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

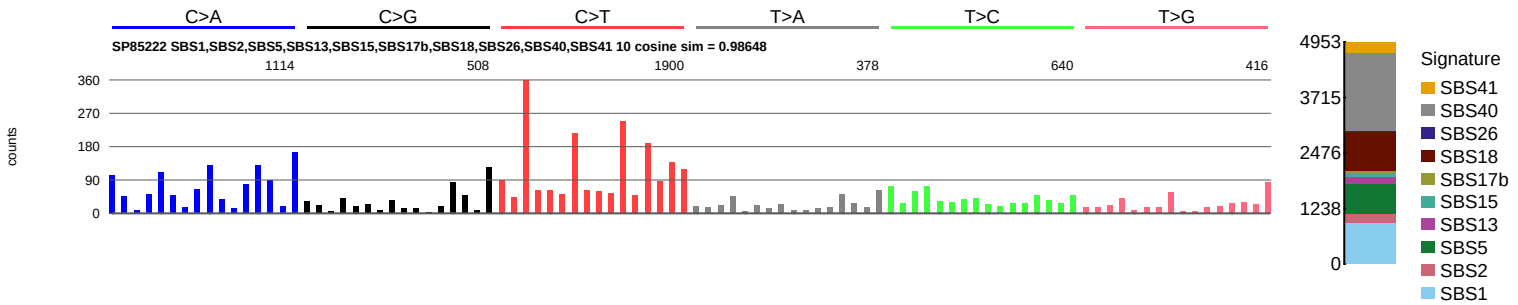

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 12 cosine sim = 0.98651
1133
546
1904
365
622
382
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3

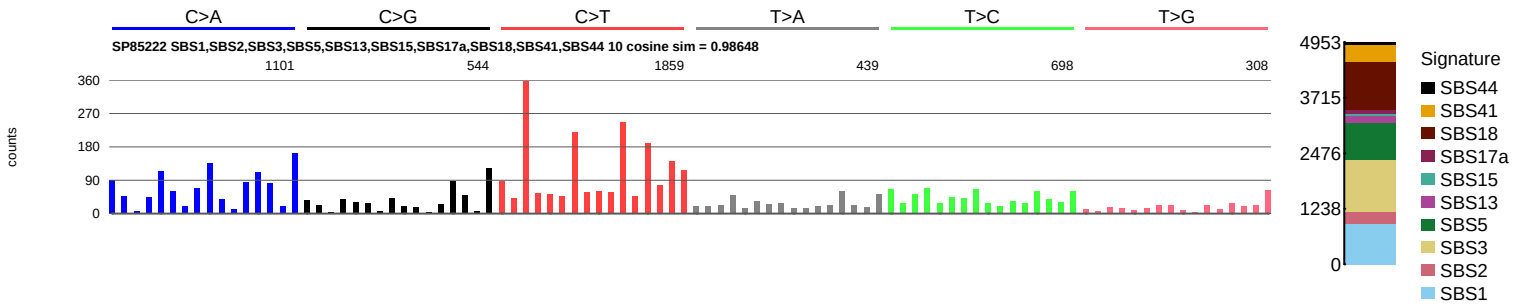

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40,SBS41 8 cosine sim = 0.9865
1115
552
1863
422
636
365
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.9865
1139
503
1887
396
602
424
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41 8 cosine sim = 0.9865
1099
555
1842
430
659
370
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.9865
1134
505
1879
390
648
395
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98648
1114
508
1900
378
640
416
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41,SBS44 10 cosine sim = 0.98648
1101
544
1859
439
698
308
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS44
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SBS18
SBS17a
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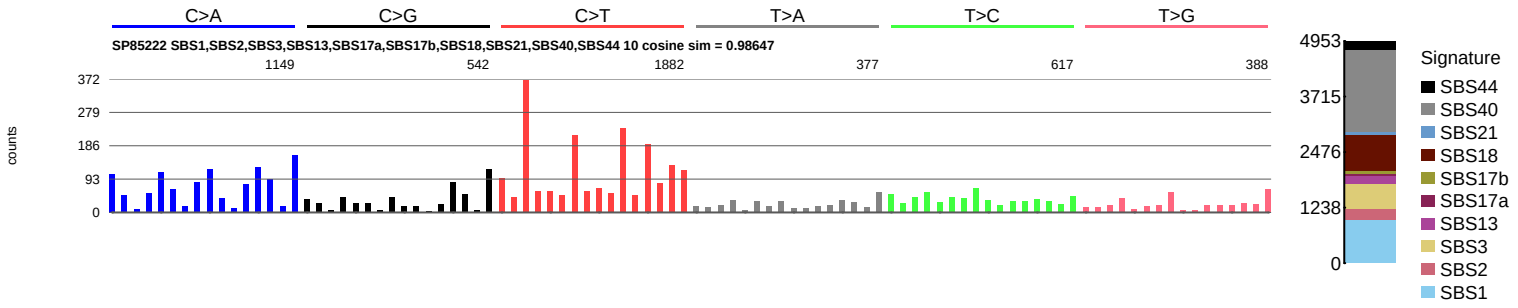
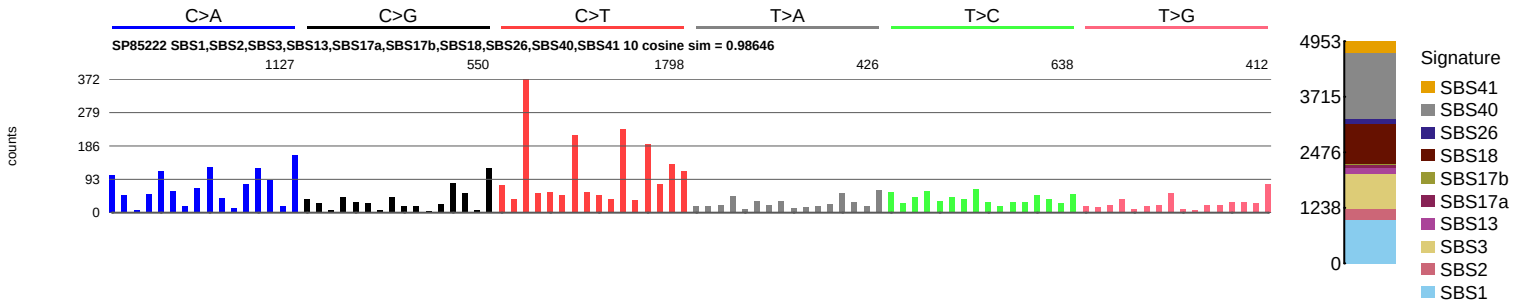
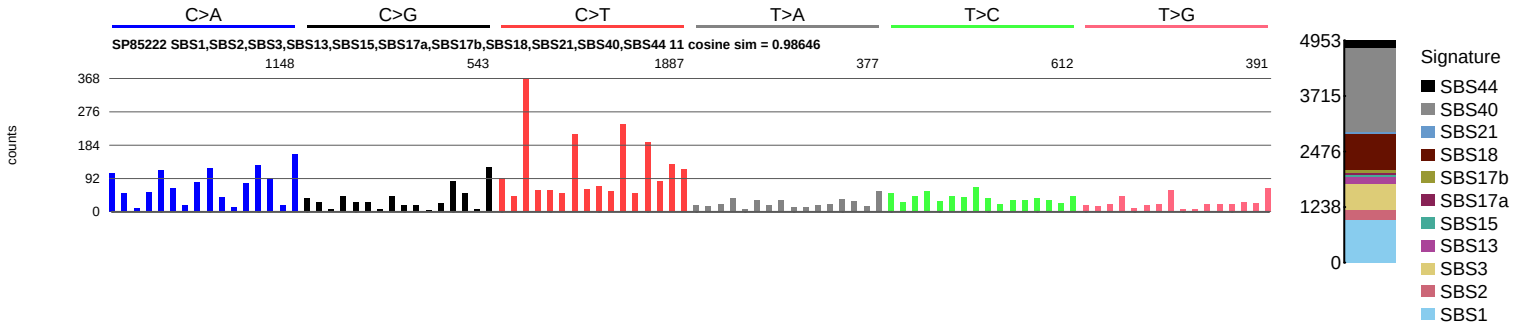
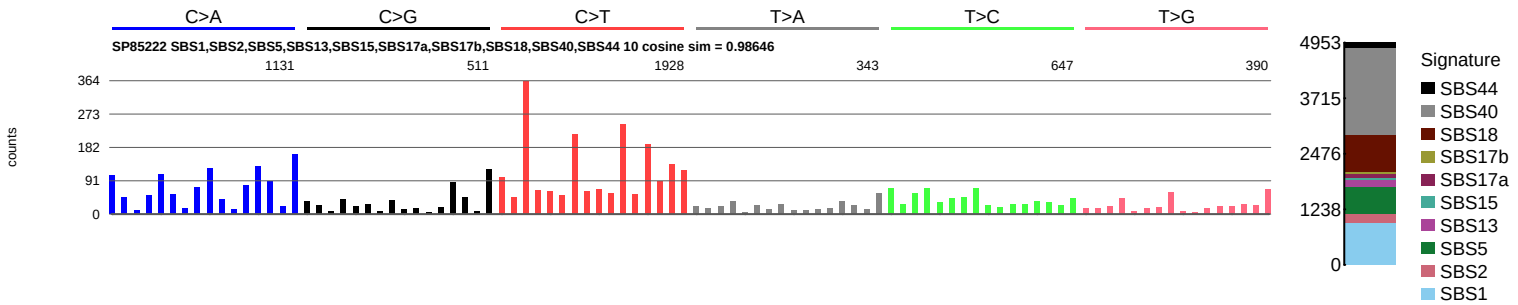
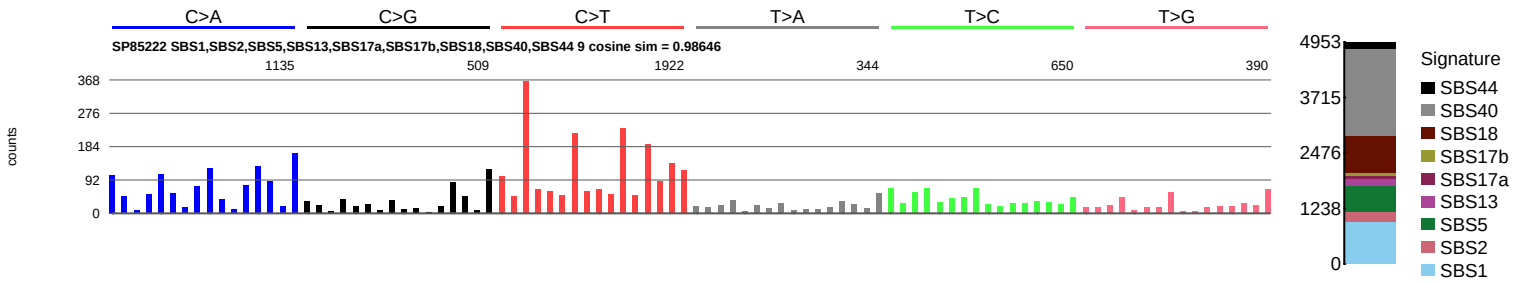
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98647
1149
542
1882
377
617
388
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98646
1127
550
1798
426
638
412
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98646
1148
543
1887
377
612
391
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98646
1131
511
1928
343
647
390
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98646
1135
509
1922
344
650
390
counts
368
276
184
92
0
4953
3715
2476
1238
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SBS44
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SBS18
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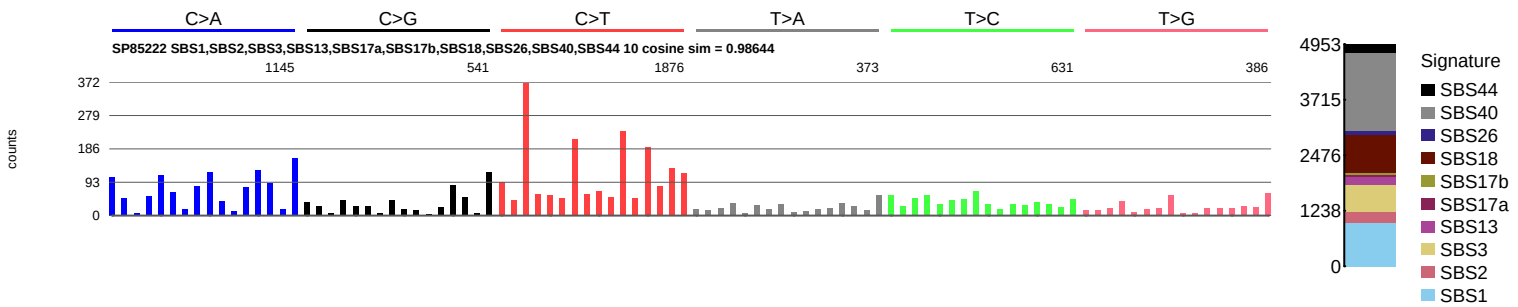
SBS2
SBS1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T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98644
1145
541
1876
373
631
386
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
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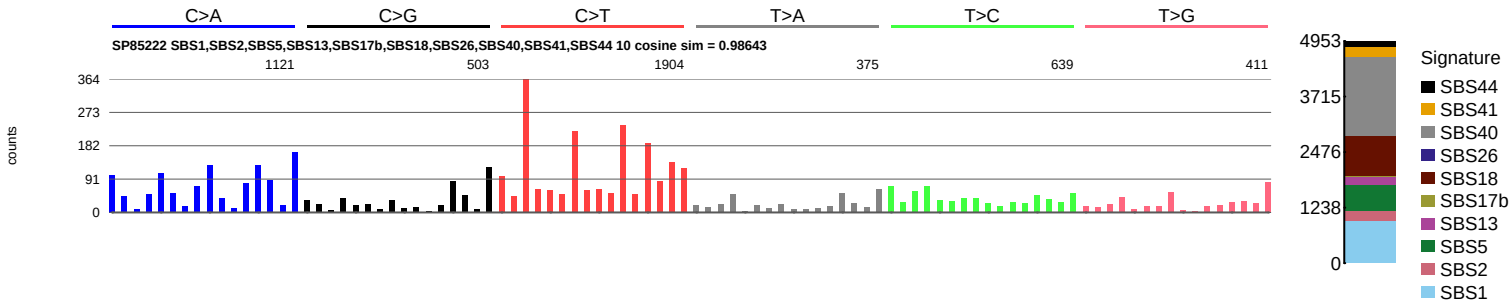
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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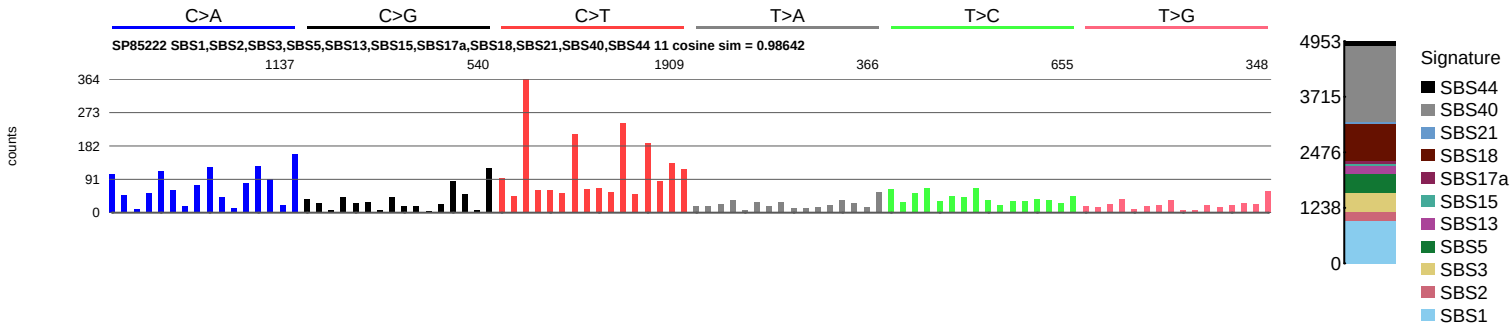
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98643
1121
503
1904
375
639
411
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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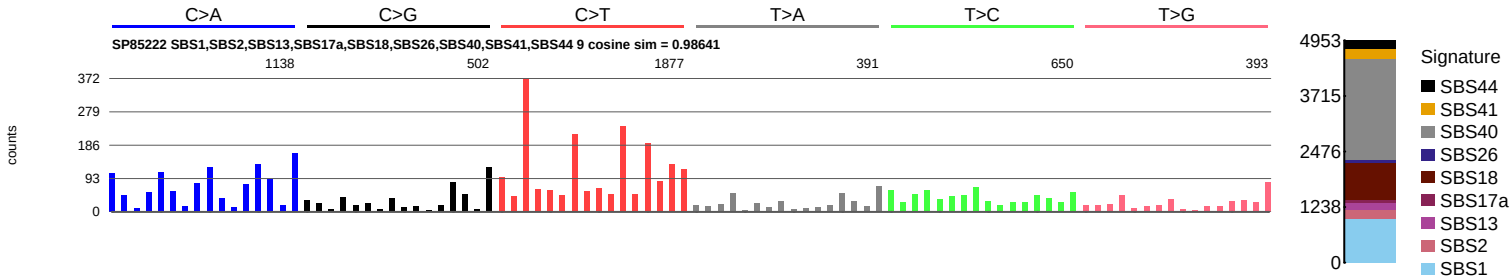
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98642
1137
540
1909
366
655
348
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>T
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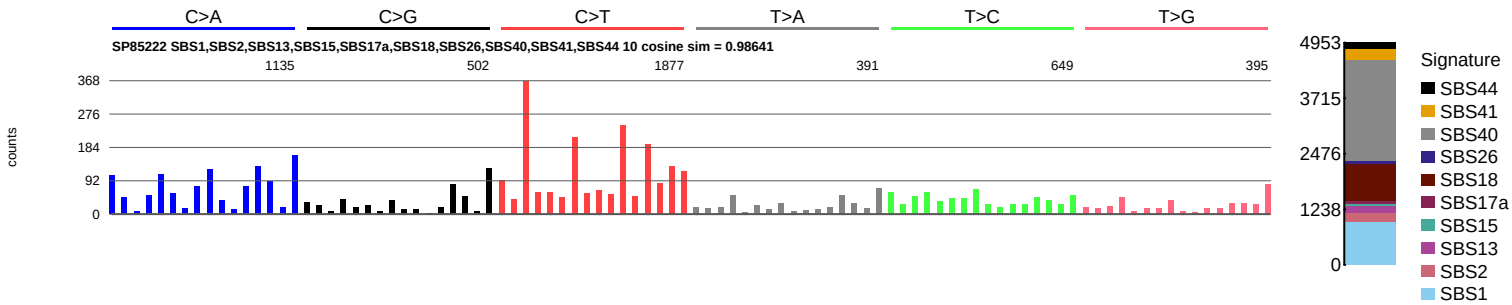
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98641
1138
502
1877
391
650
393
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98641
1135
502
1877
391
649
395
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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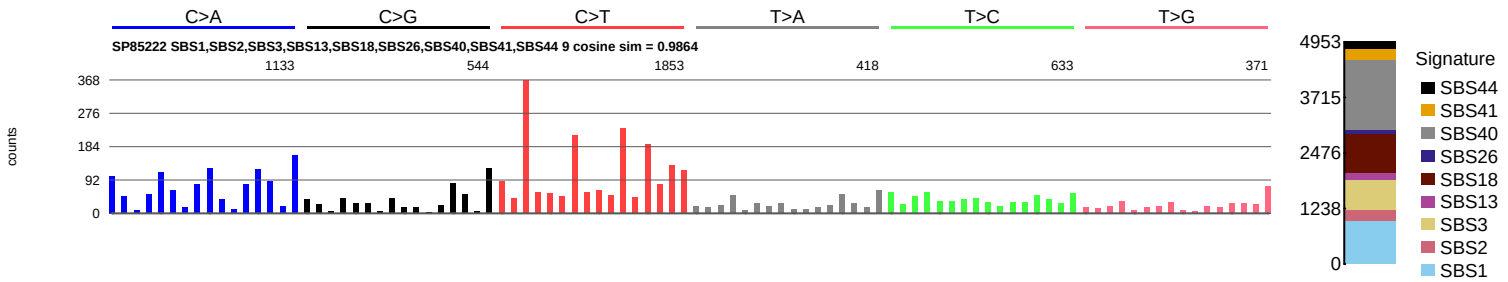
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1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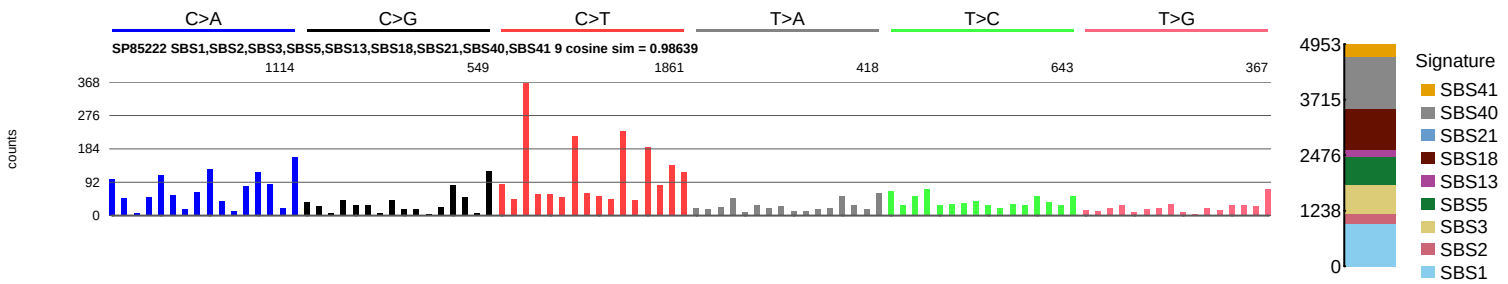
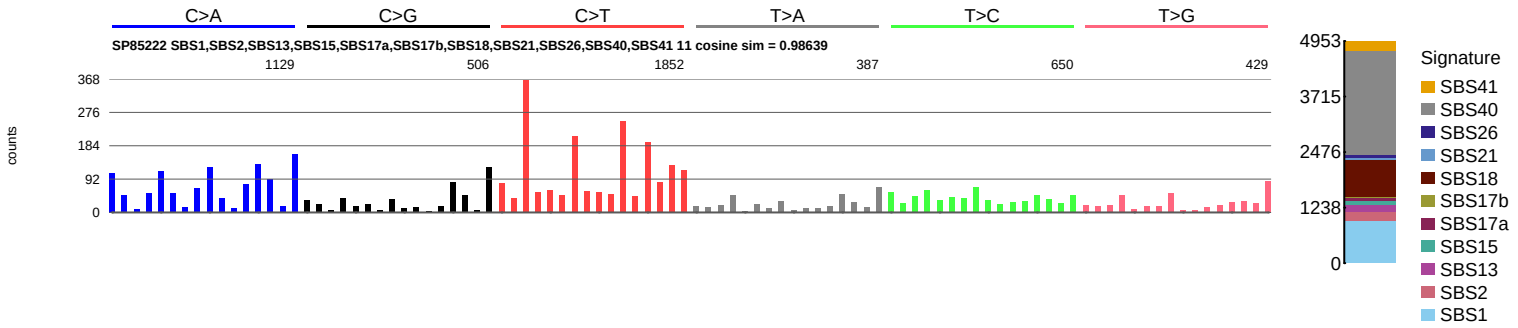
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9864
1133
544
1853
418
633
371
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98639
1114
549
1861
418
643
367
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98639
1129
506
1852
387
650
429
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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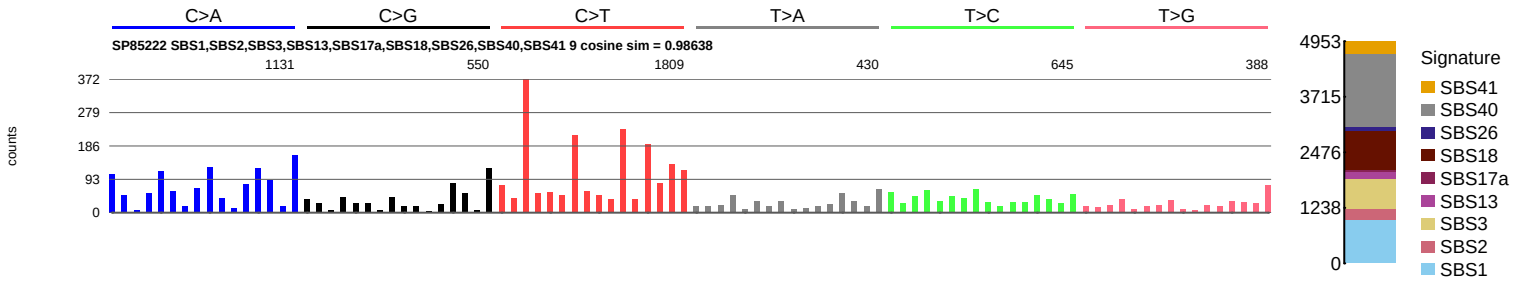
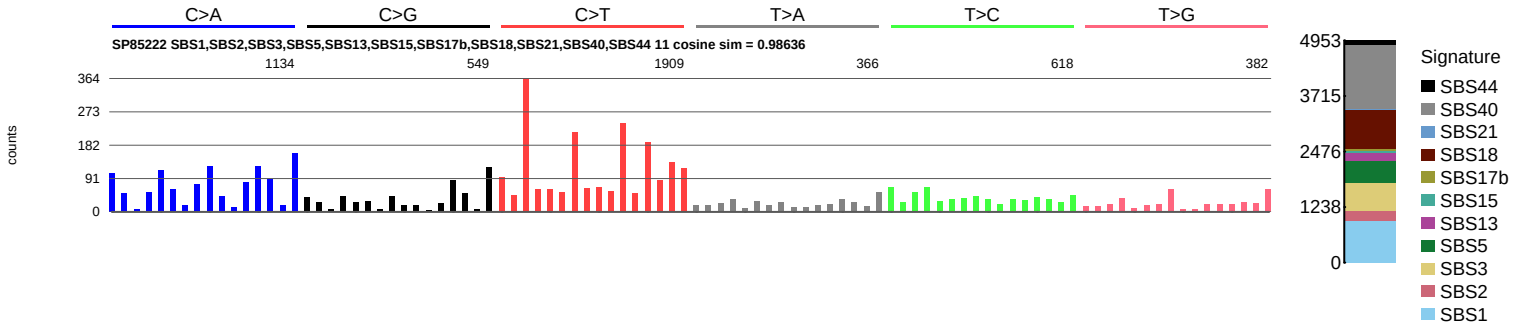
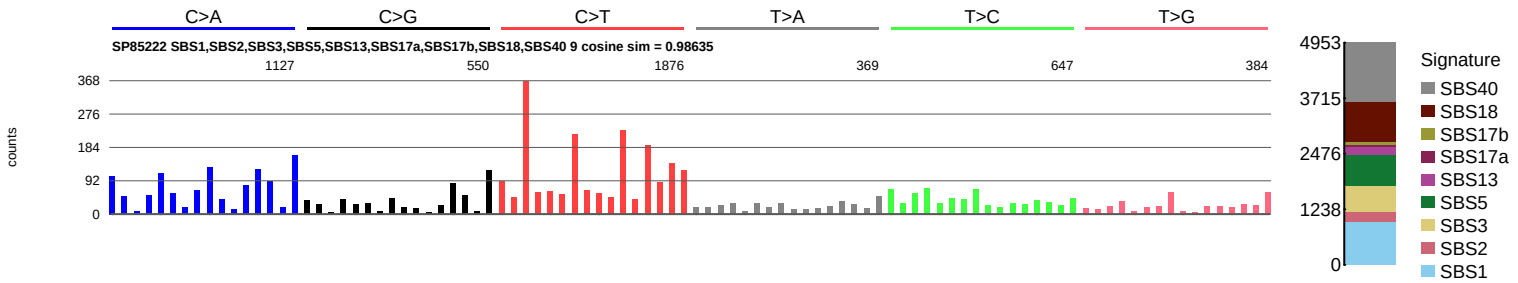
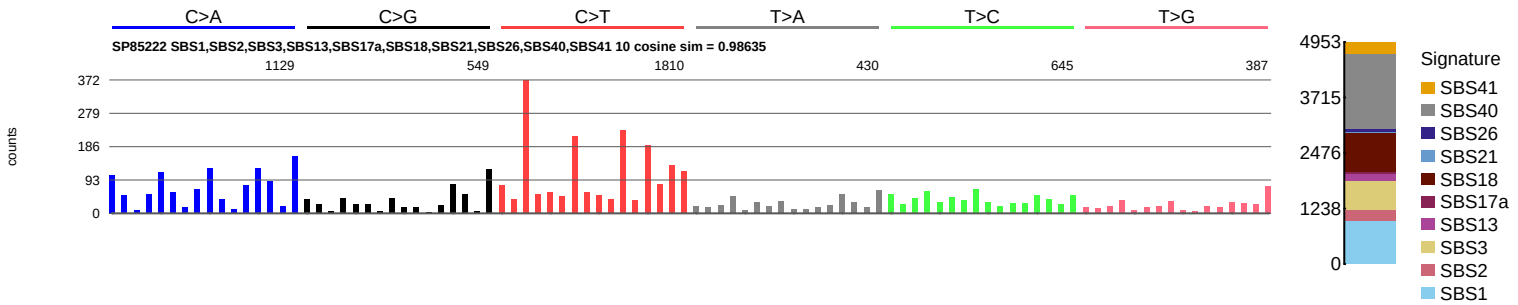
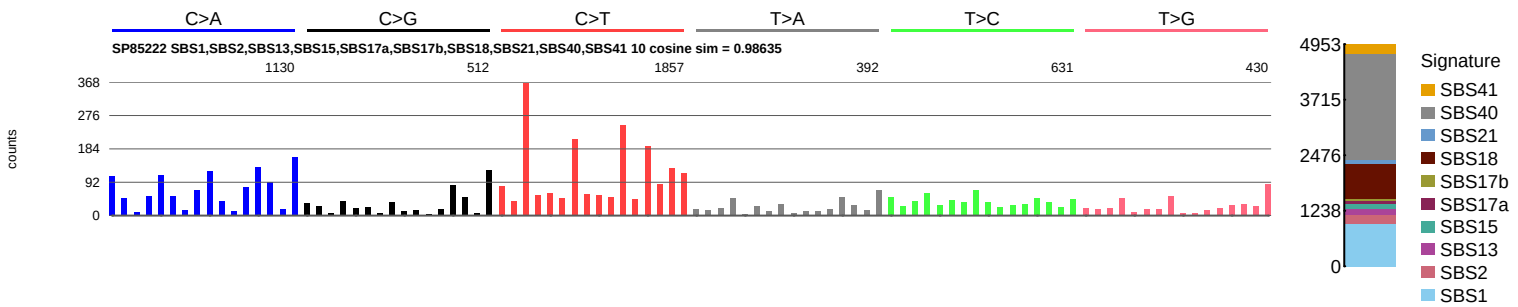
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98638
1131
550
1809
430
645
388
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98636
1134
549
1909
366
618
382
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98635
1127
550
1876
369
647
384
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98635
1129
549
1810
430
645
387
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1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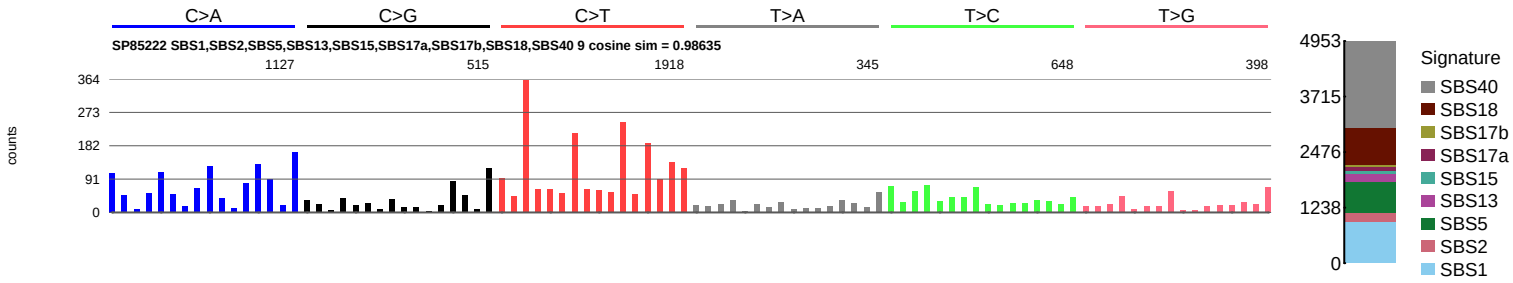
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98635
1130
512
1857
392
631
430
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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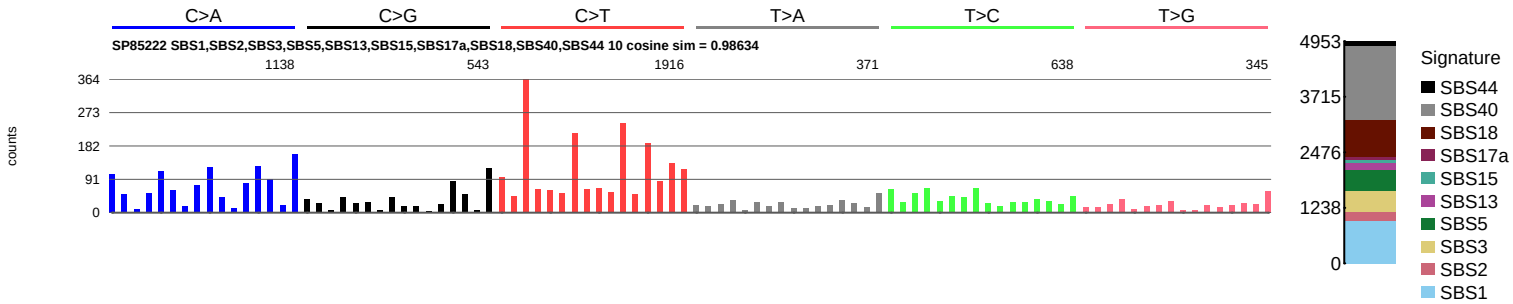
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98635
1127
515
1918
345
648
398
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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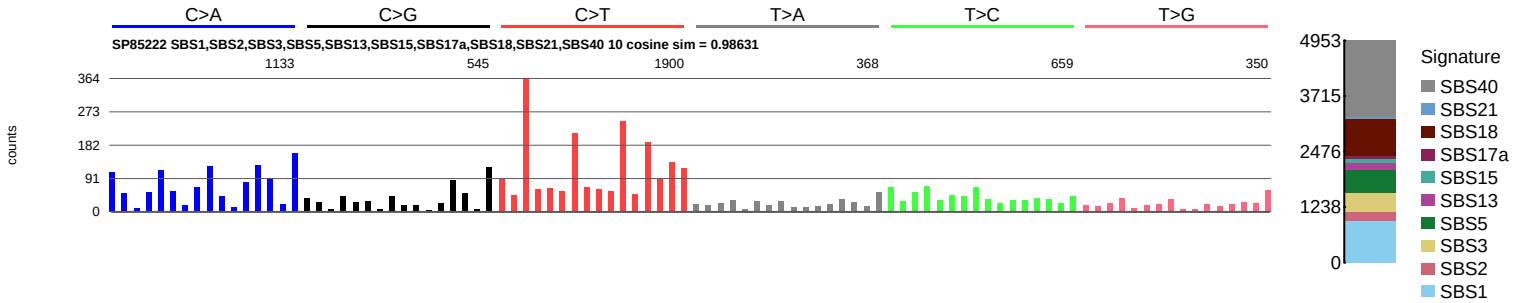
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS44 10 cosine sim = 0.98634
1138
543
1916
371
638
345
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40 10 cosine sim = 0.98631
1133
545
1900
368
659
350
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
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SBS18
SBS17a
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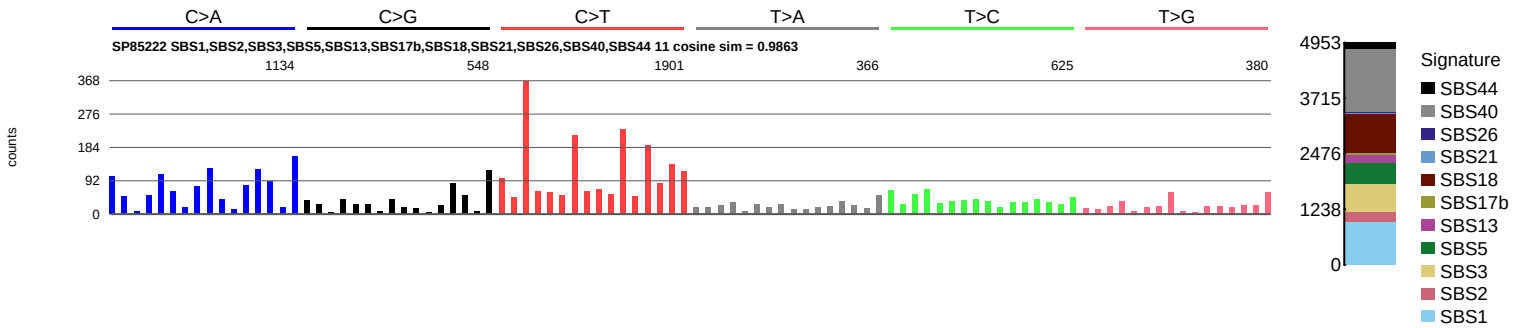
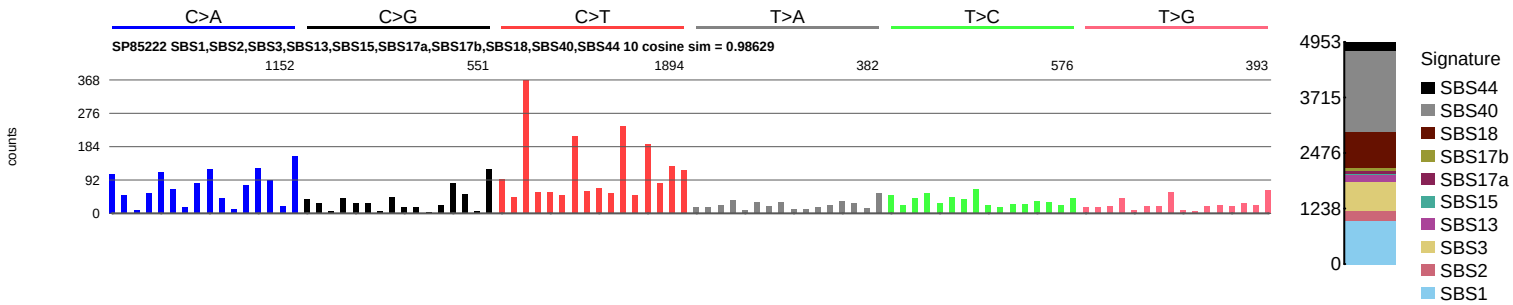
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.9863
1134
548
1901
366
625
380
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98629
1152
551
1894
382
576
393
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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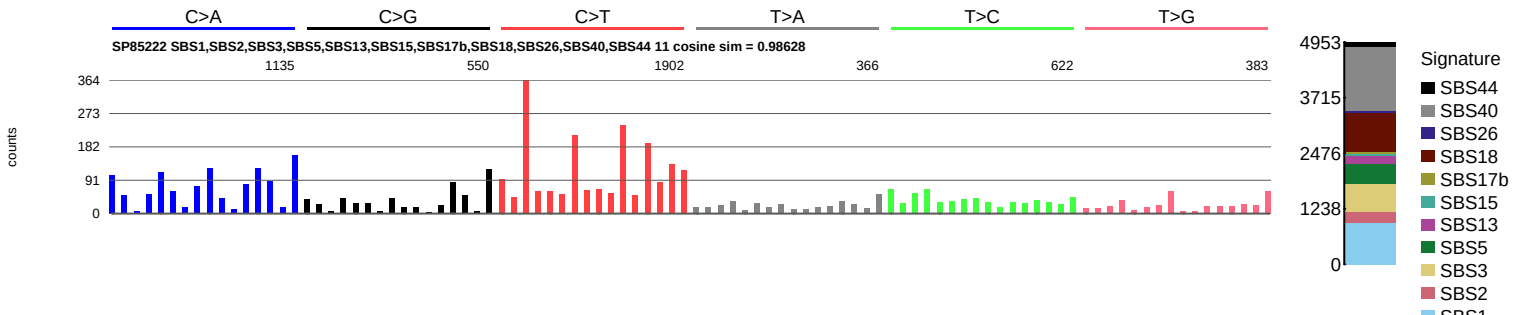
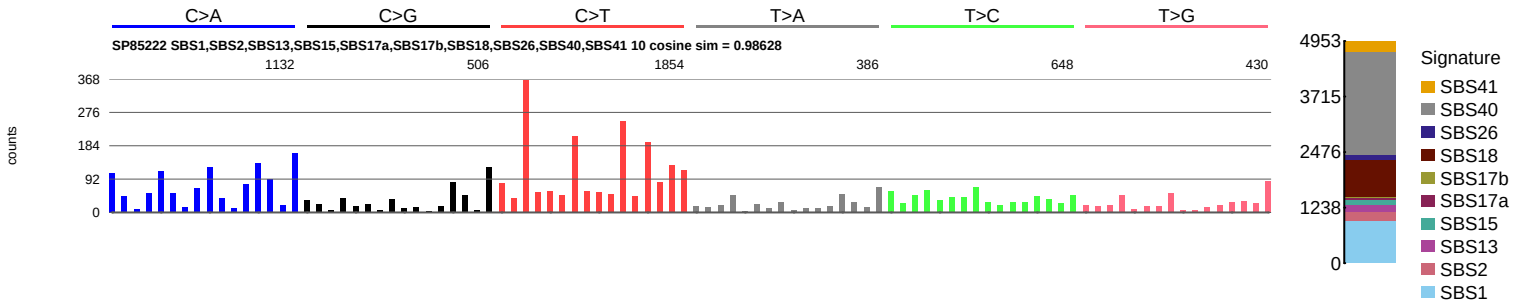
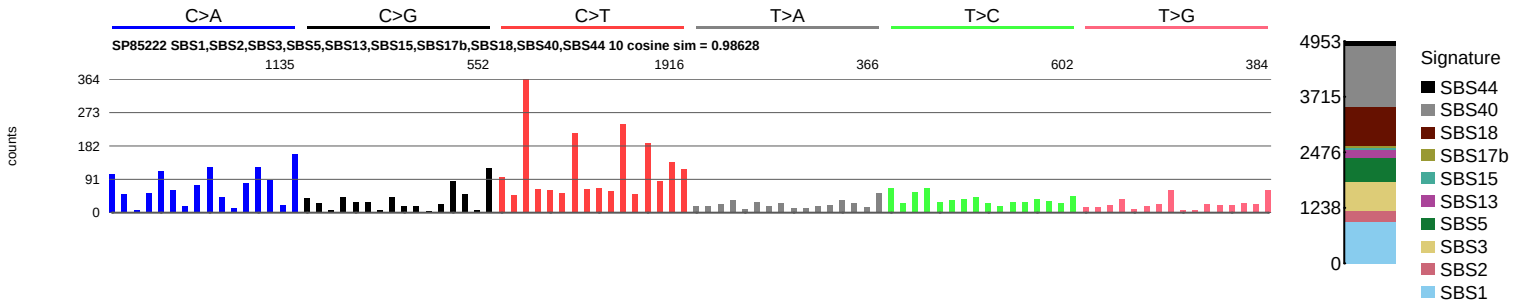
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98628
1135
550
1902
366
622
383
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98628
1132
506
1854
386
648
430
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98628
1135
552
1916
366
602
384
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
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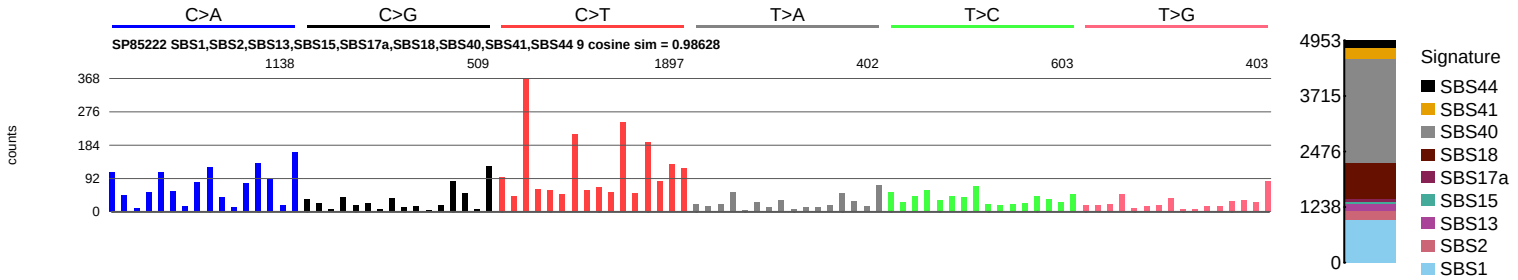
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98628
1138
509
1897
402
603
403
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
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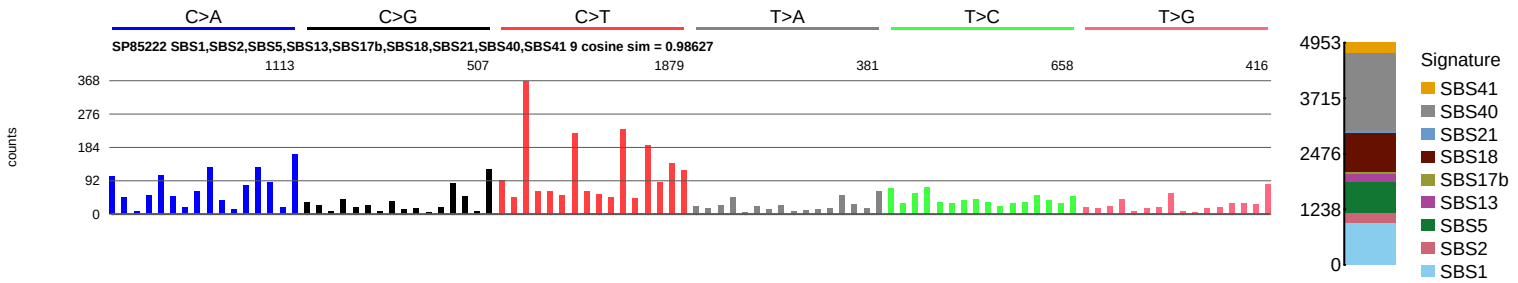
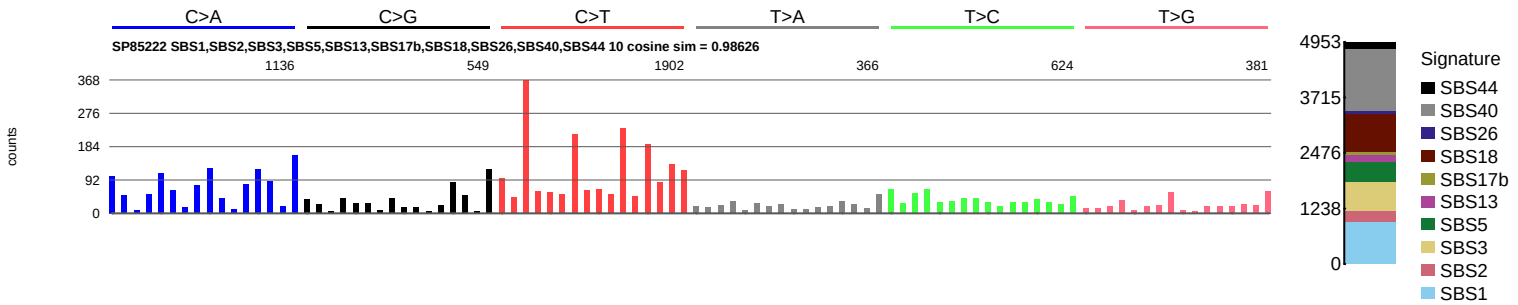
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98627
1113
507
1879
381
658
416
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98626
1136
549
1902
366
624
381
counts
368
276
184
92
0
4953
3715
2476
1238
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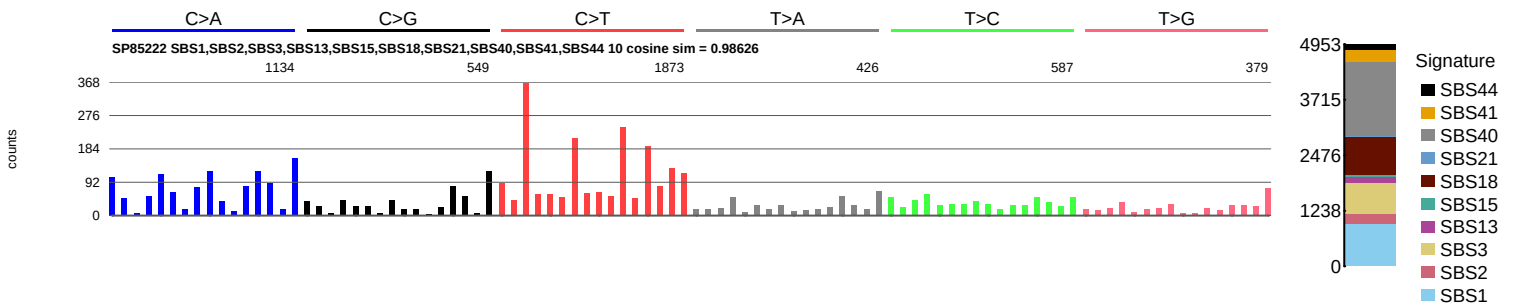
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1134
549
1873
426
587
379
counts
368
276
184
92
0
4953
3715
2476
1238
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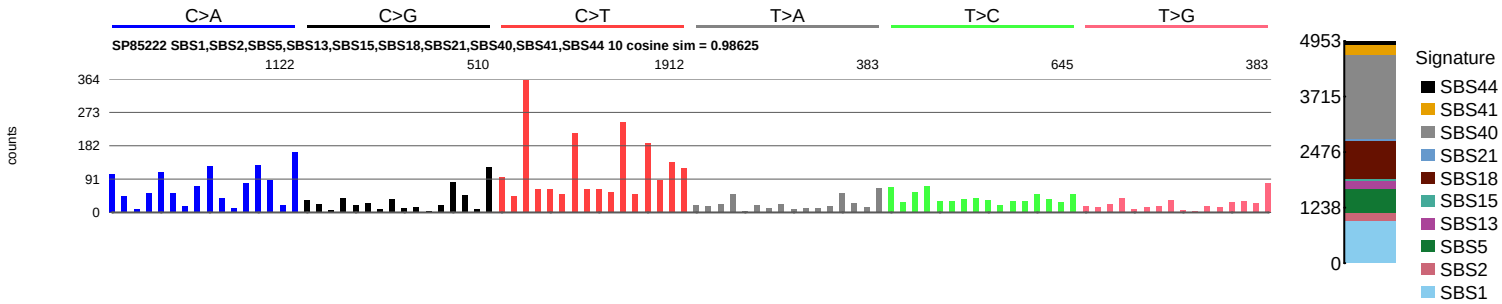
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C>T
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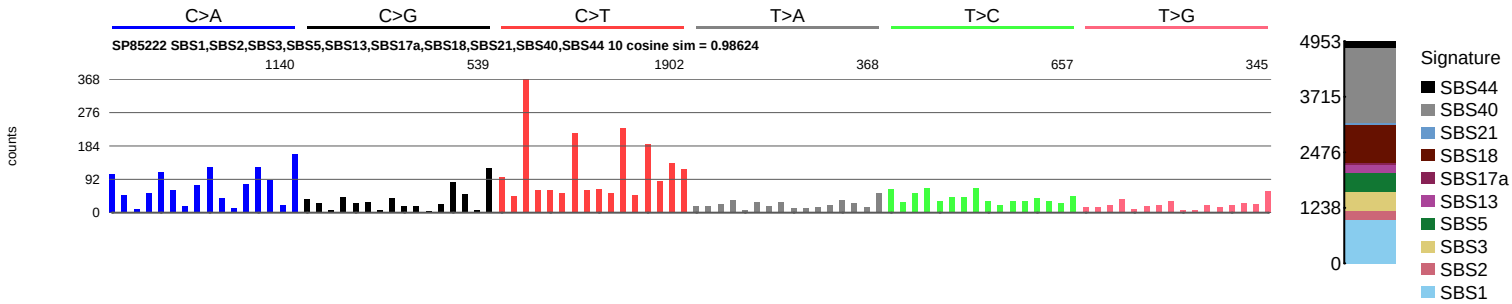
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98625
1122
510
1912
383
645
383
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1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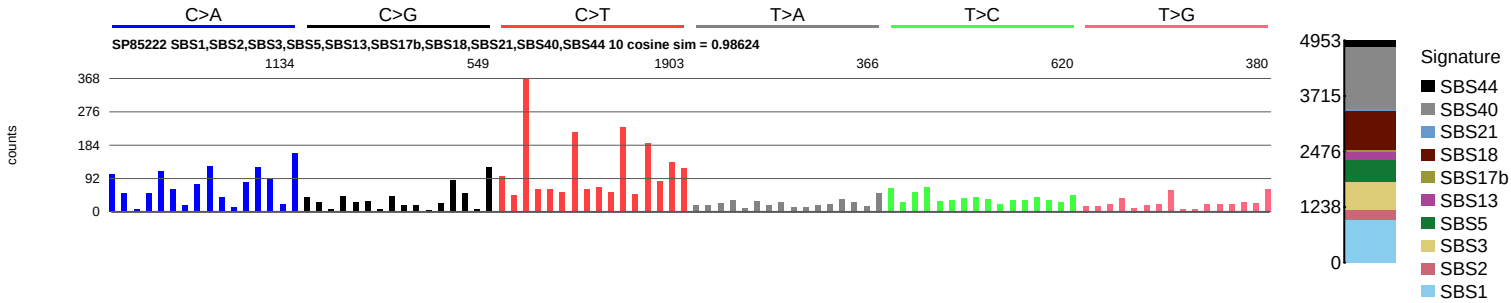
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98624
1140
539
1902
368
657
345
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98624
1134
549
1903
366
620
380
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
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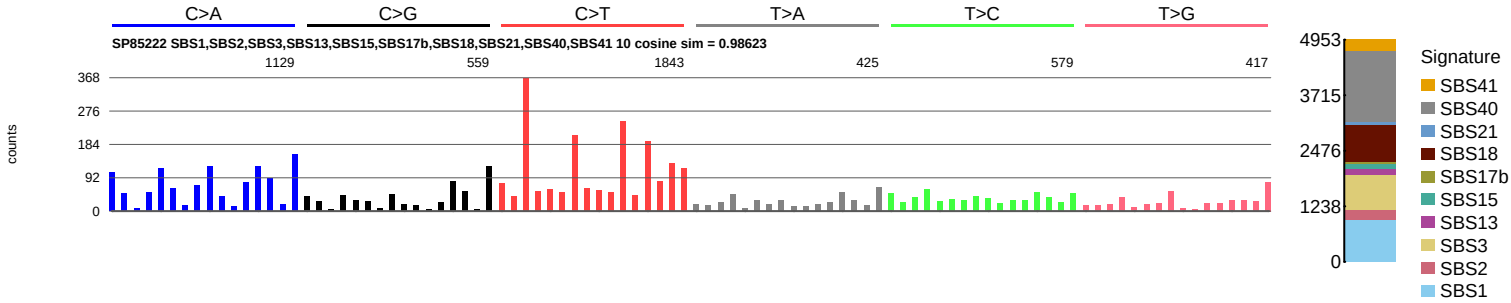
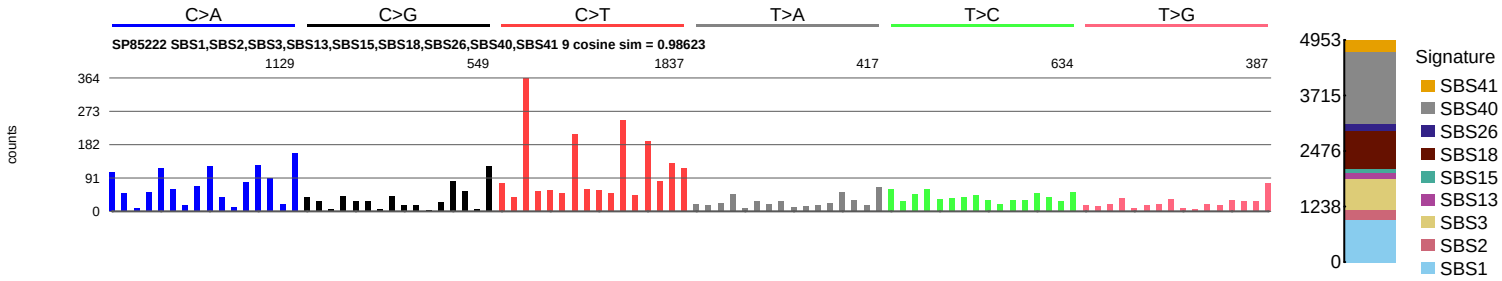
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98623
1129
559
1843
425
579
417
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1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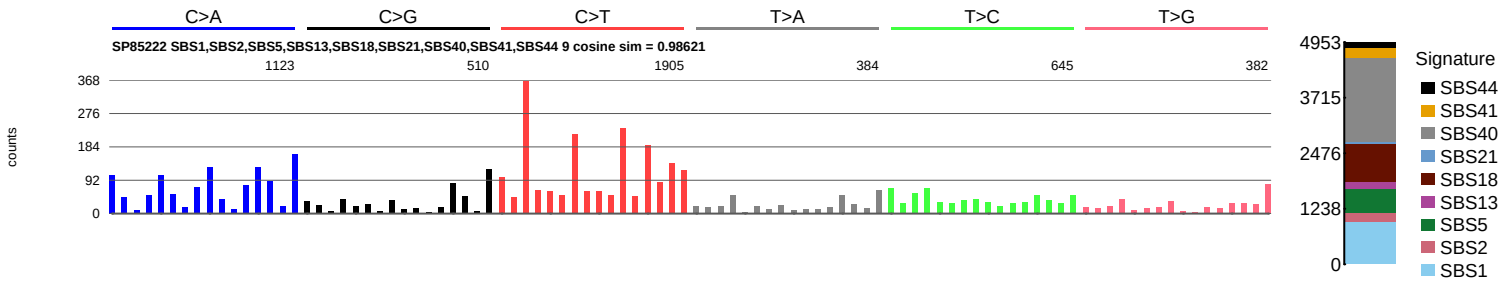
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98623
1129
549
1837
417
634
387
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98621
1123
510
1905
384
645
382
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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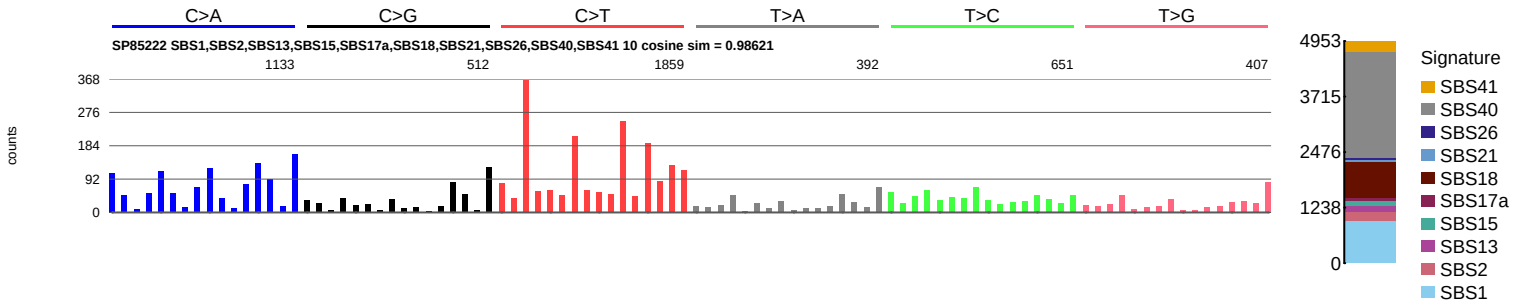
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98621
1133
512
1859
392
651
407
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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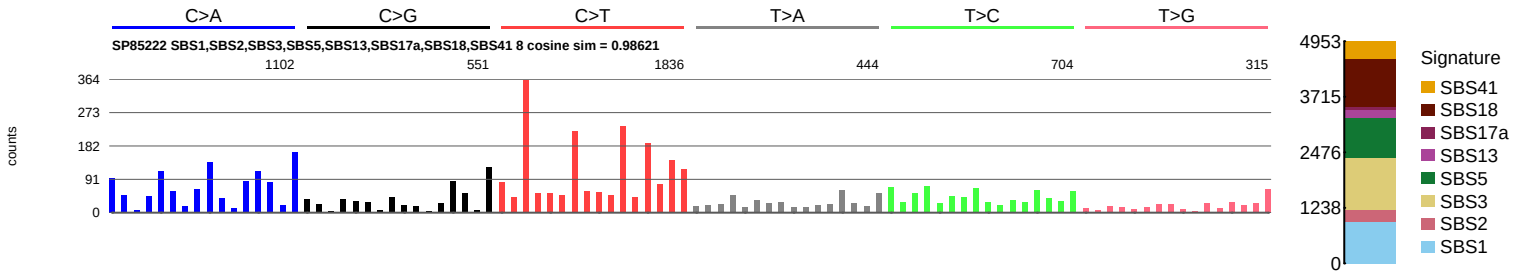
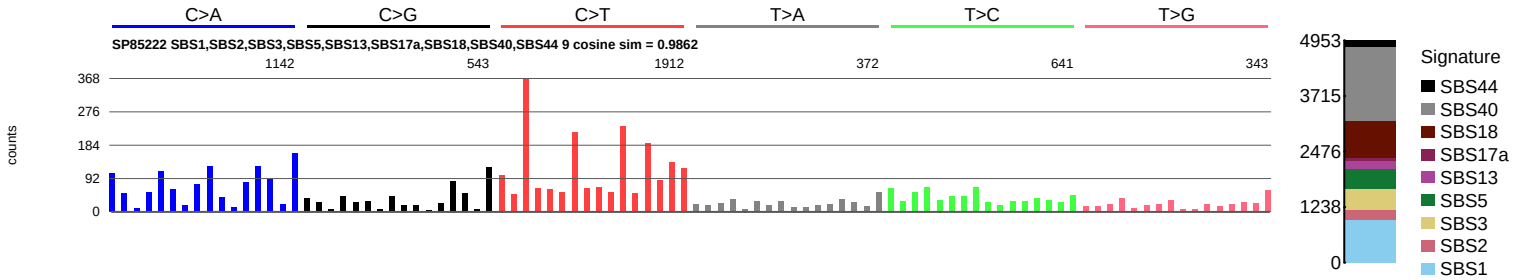
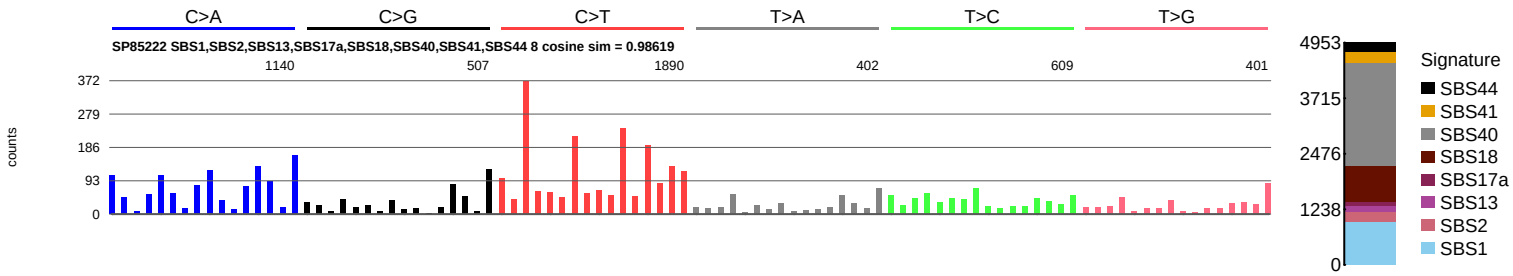
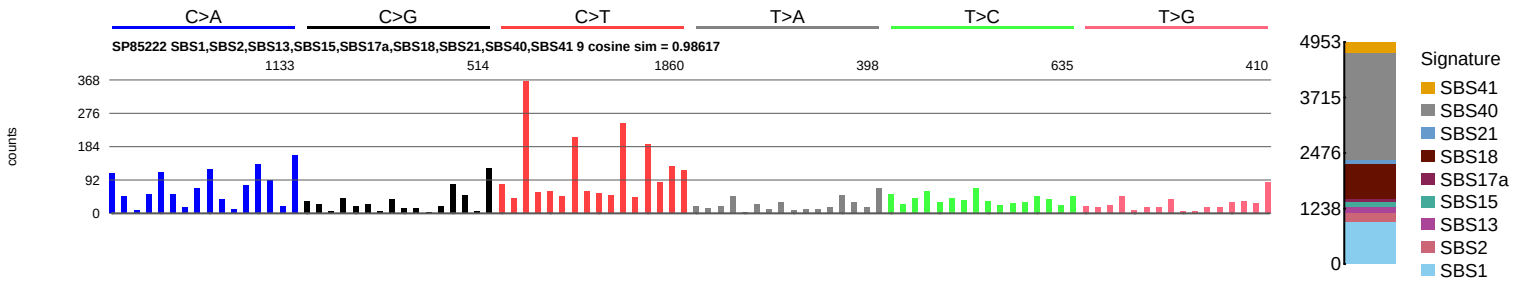
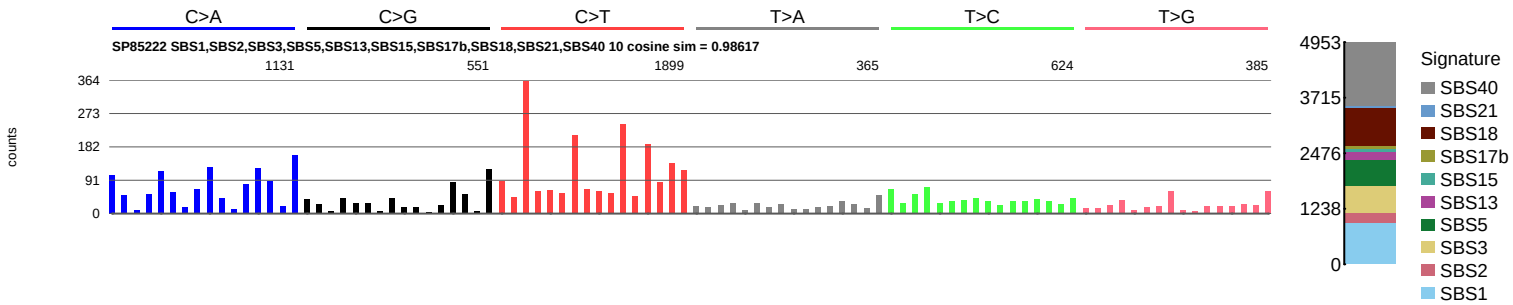
SBS2
SBS1

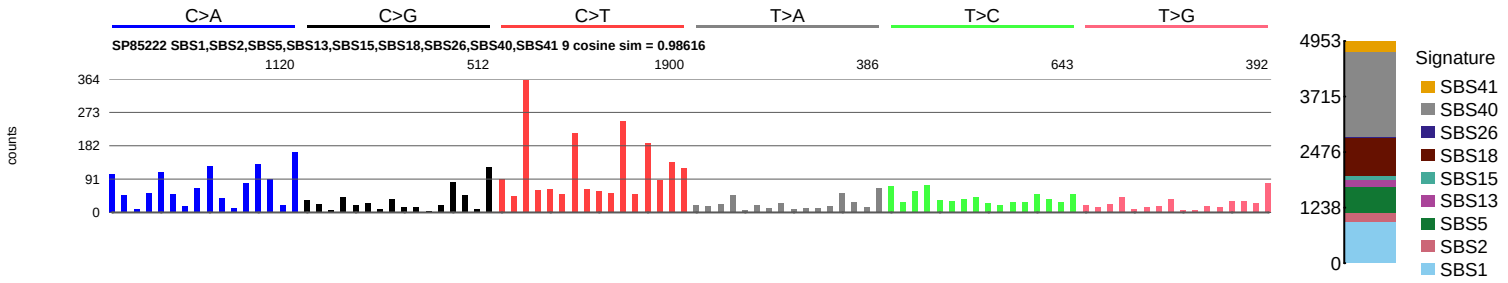

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS41 8 cosine sim = 0.98621
1102
551
1836
444
704
315
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

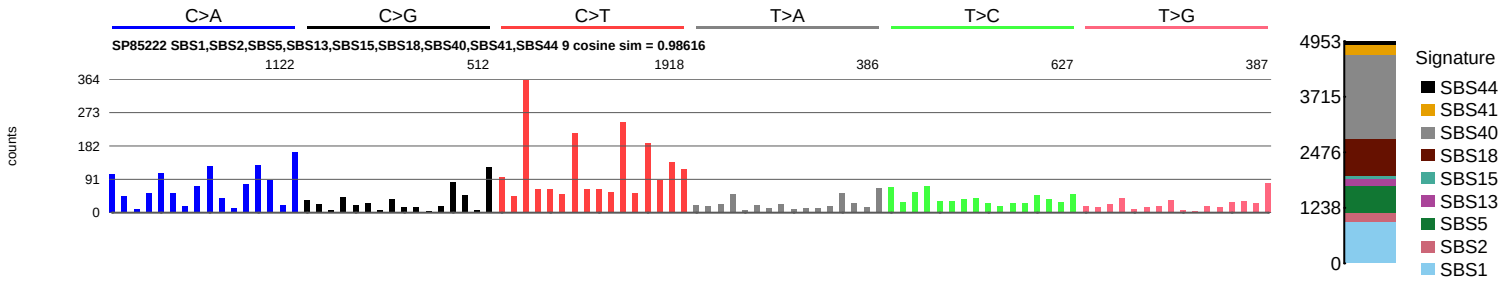

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS44 9 cosine sim = 0.9862
1142
543
1912
372
641
343
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

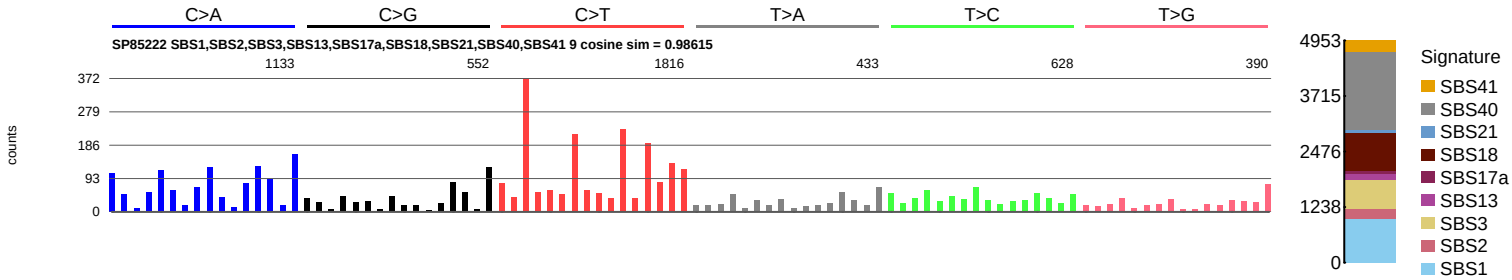

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98619
1140
507
1890
402
609
401
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98617
1133
514
1860
398
635
410
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98617
1131
551
1899
365
624
385
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98616
1120
512
1900
386
643
392
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98616
1122
512
1918
386
627
387
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1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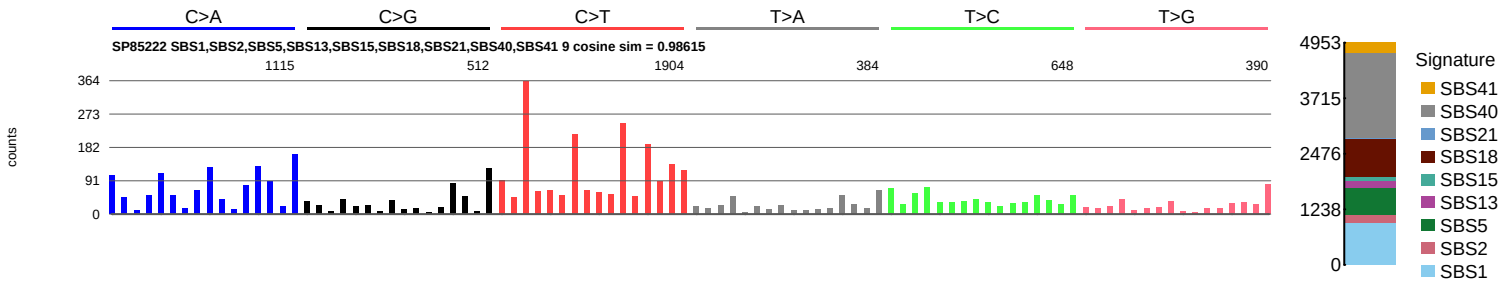
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98615
1133
552
1816
433
628
390
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98615
1115
512
1904
384
648
390
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
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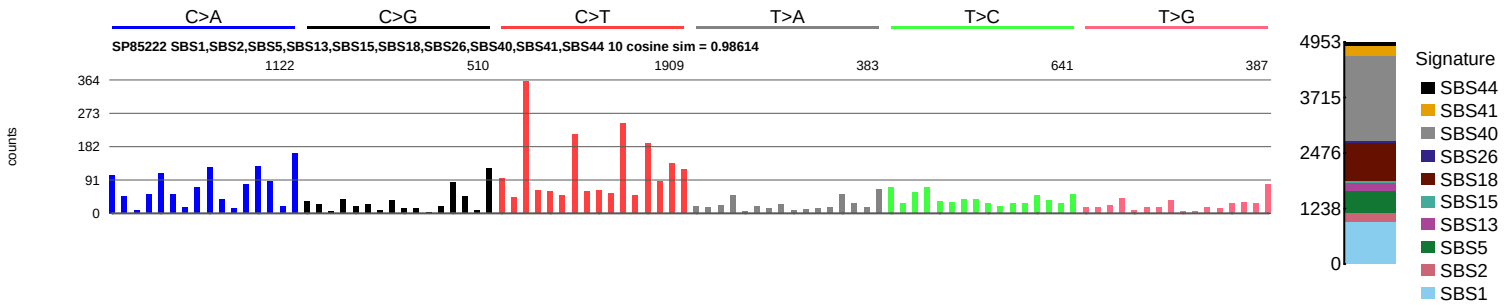
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98614
1122
510
1909
383
641
387
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
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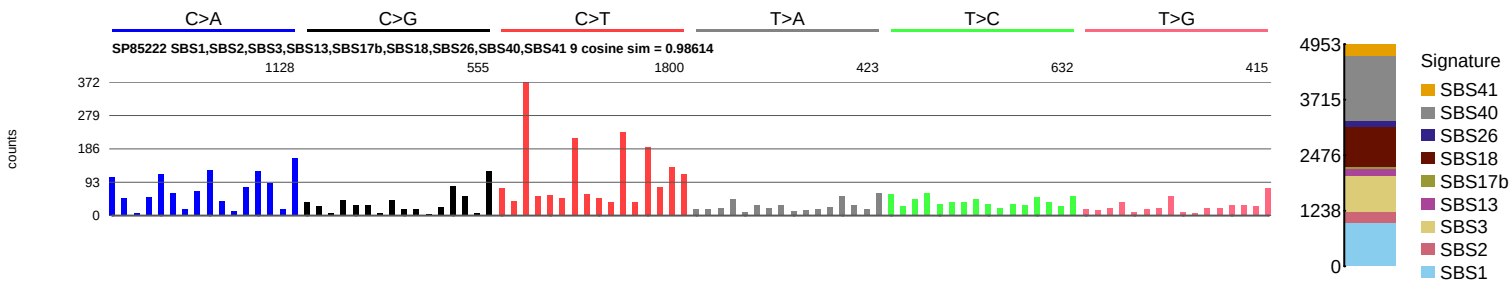
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98614
1128
555
1800
423
632
415
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
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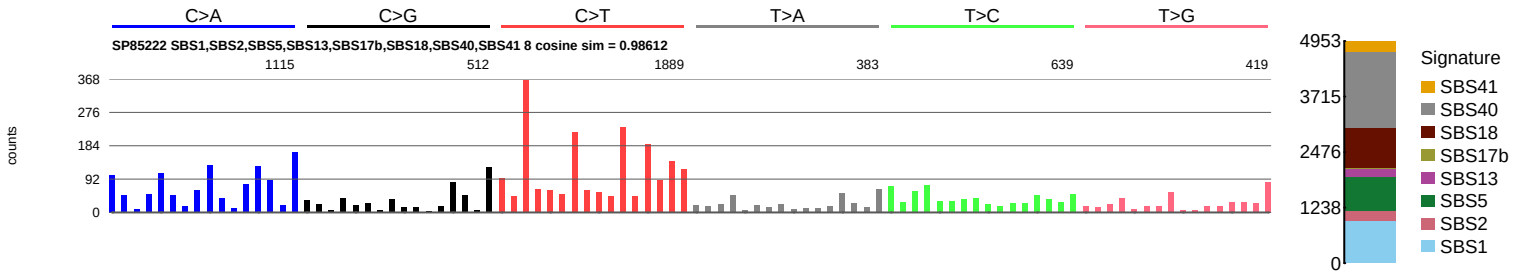
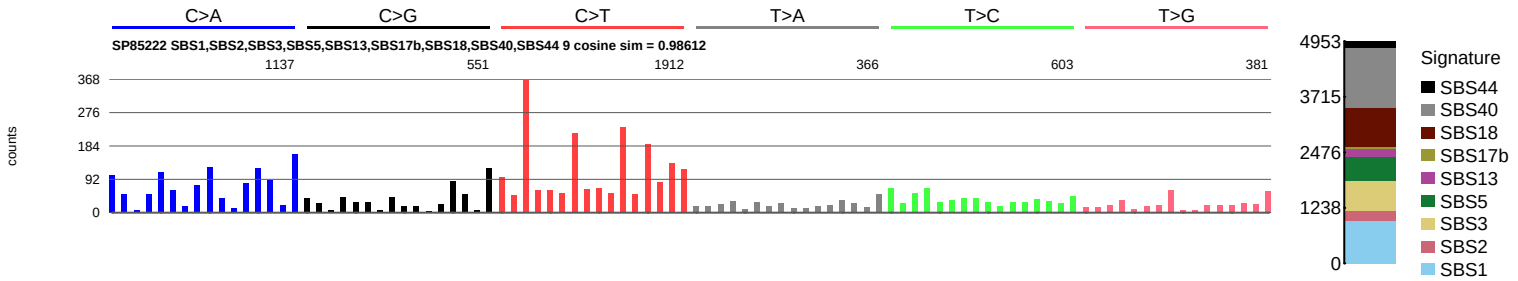
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98612
1115
512
1889
383
639
419
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98612
1137
551
1912
366
603
381
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
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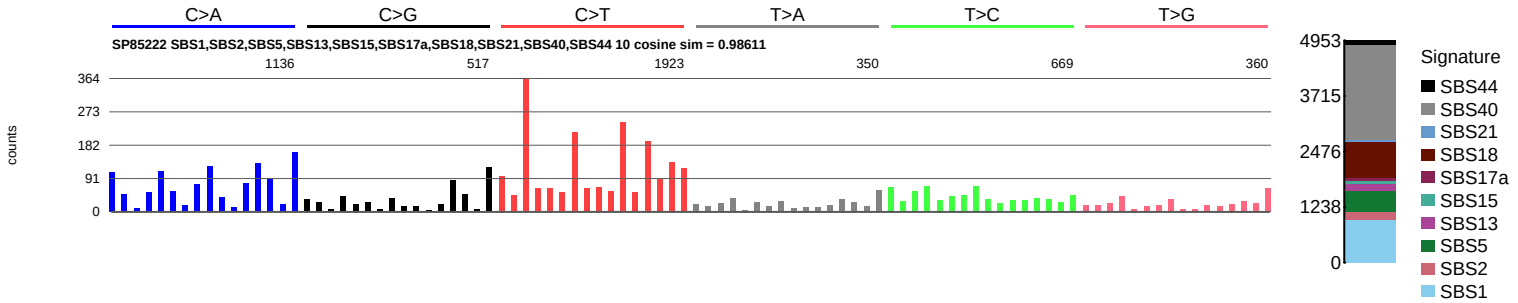
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98611
1136
517
1923
350
669
360
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
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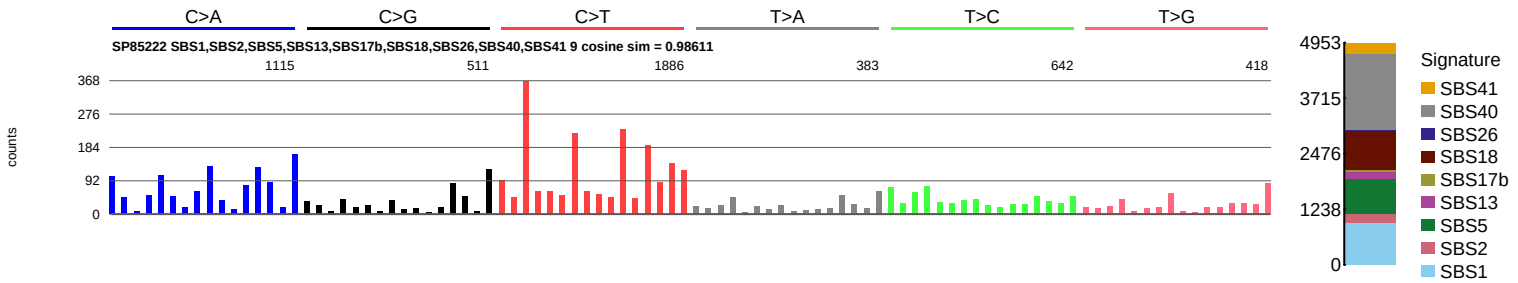
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98611
1115
511
1886
383
642
418
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
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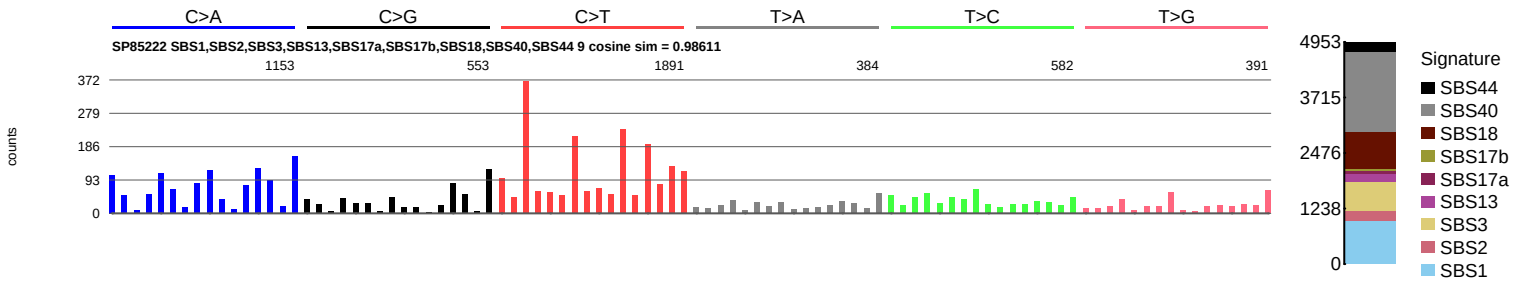
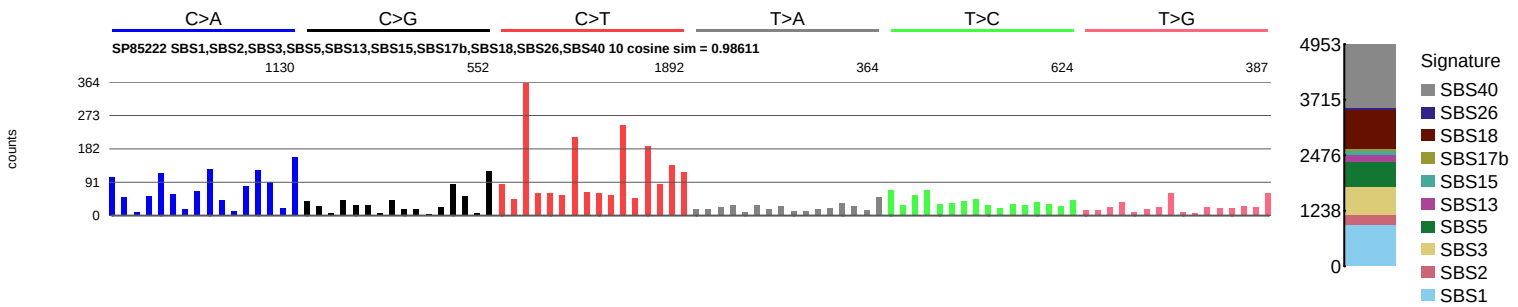
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98611
1153
553
1891
384
582
391
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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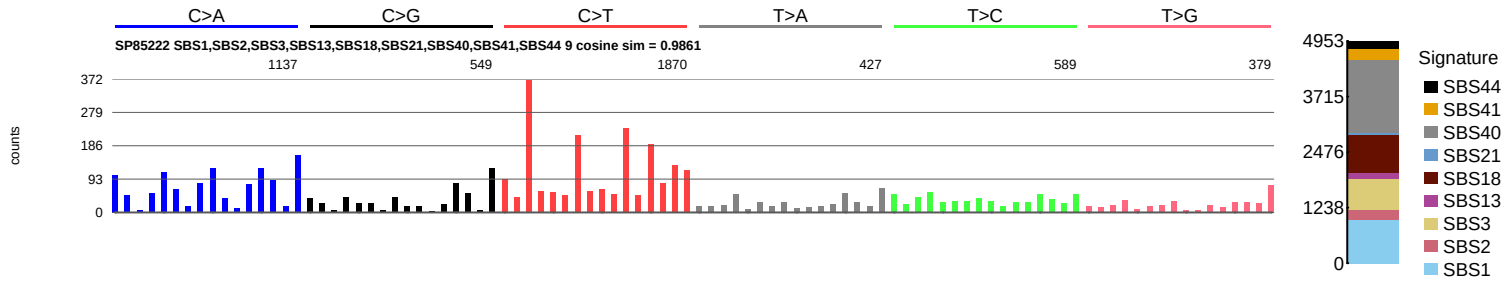
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98611
1130
552
1892
364
624
387
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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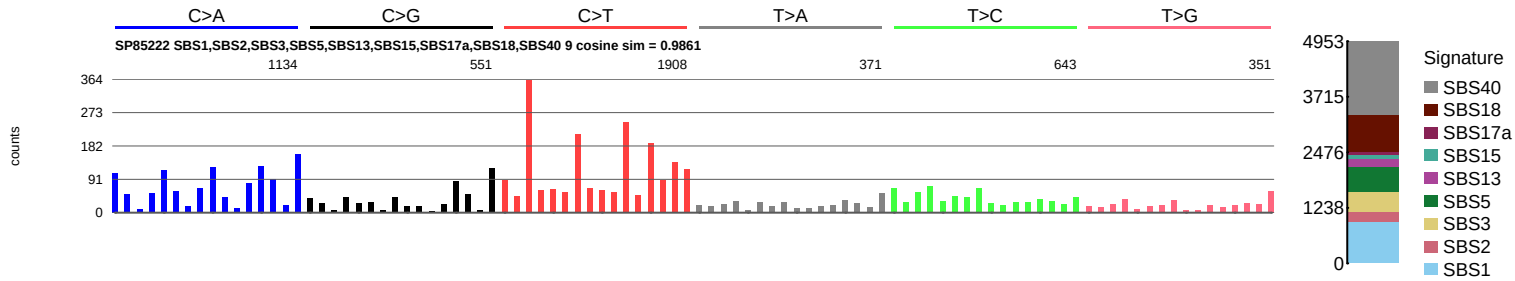
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9861
1137
549
1870
427
589
379
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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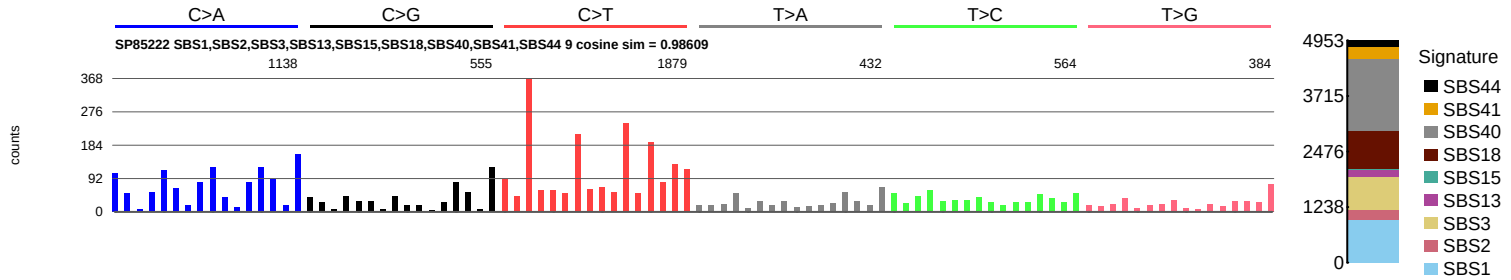
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40 9 cosine sim = 0.9861
1134
551
1908
371
643
351
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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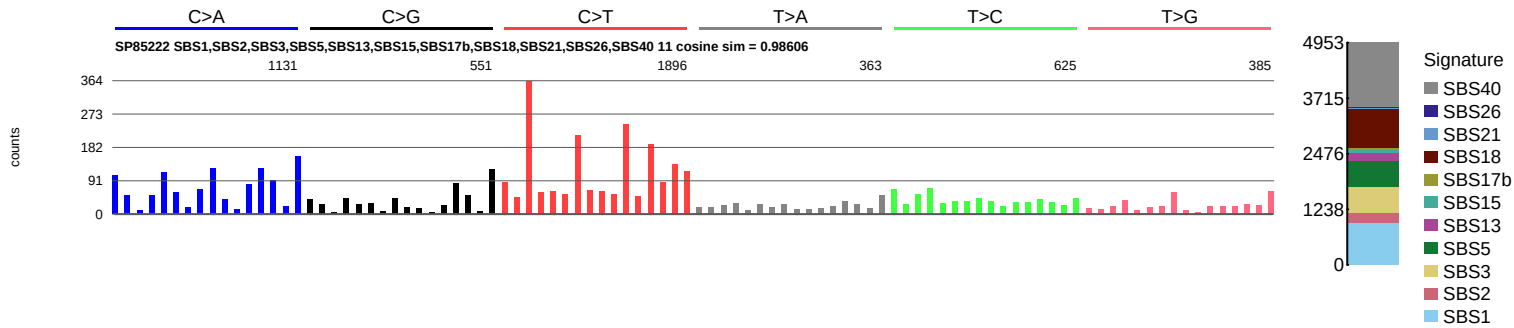
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98609
1138
555
1879
432
564
384
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1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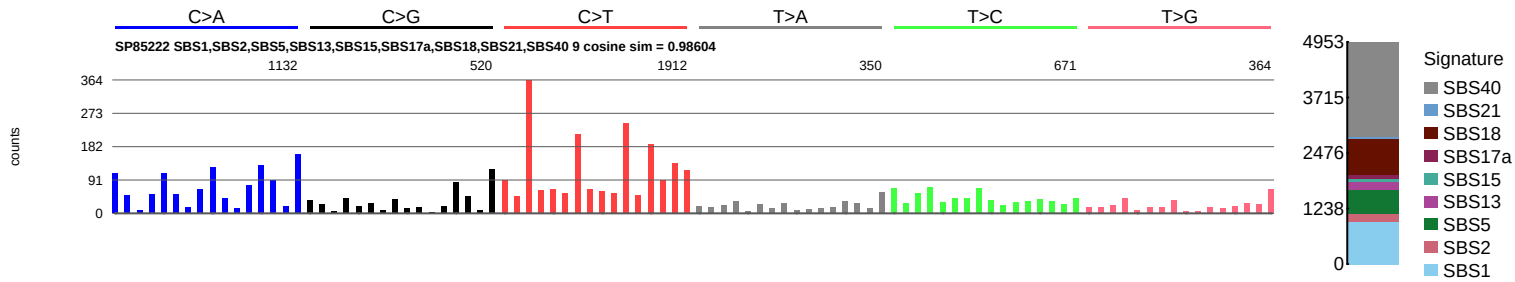
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 11 cosine sim = 0.98606
1131
551
1896
363
625
385
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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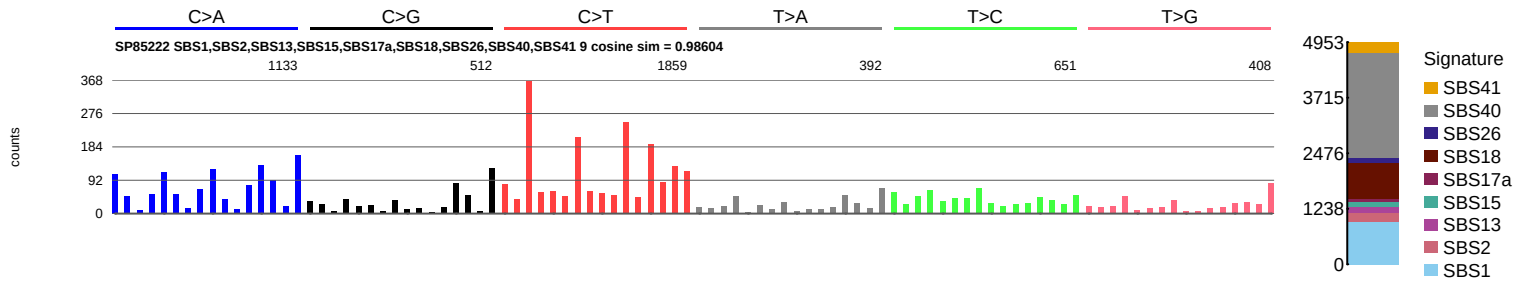
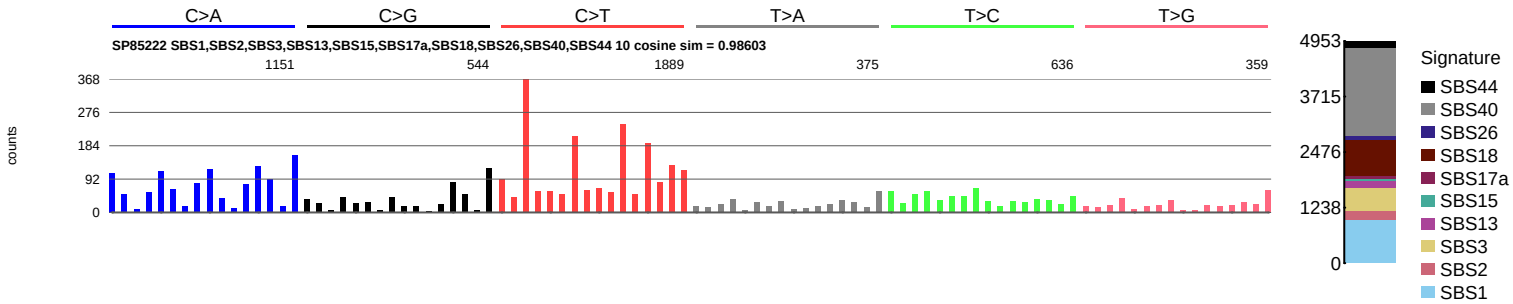
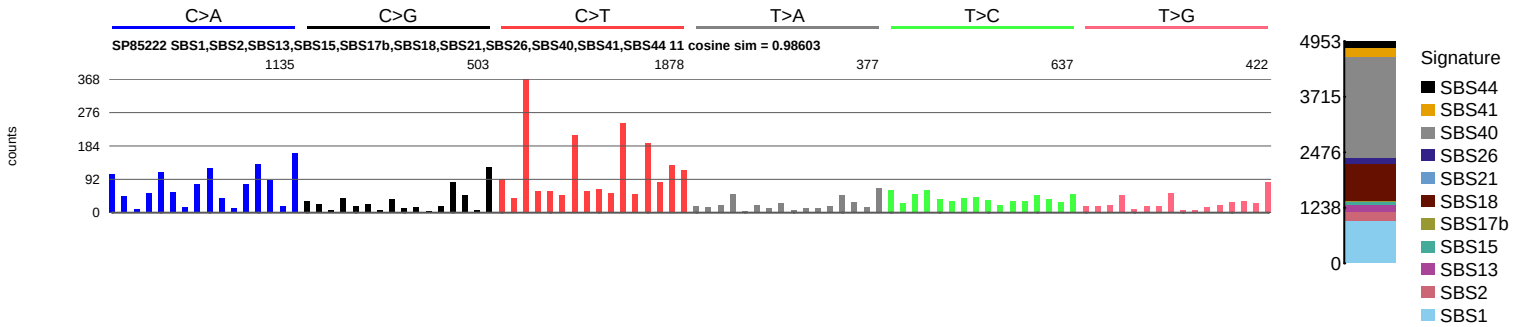
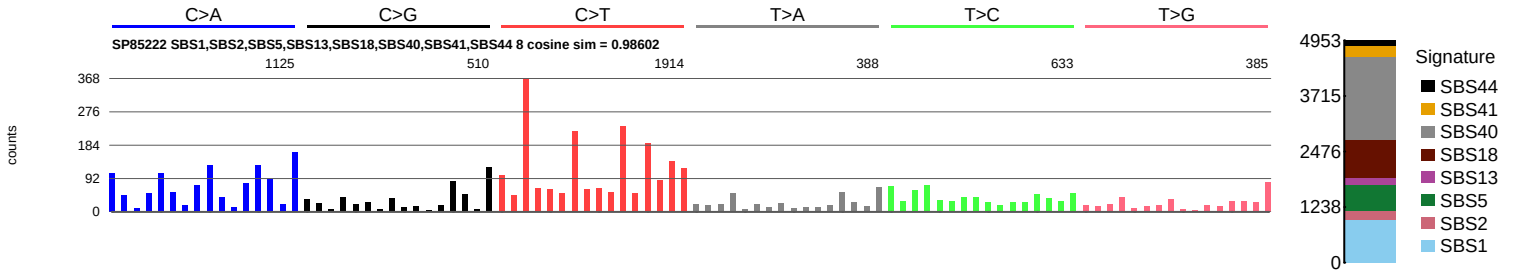
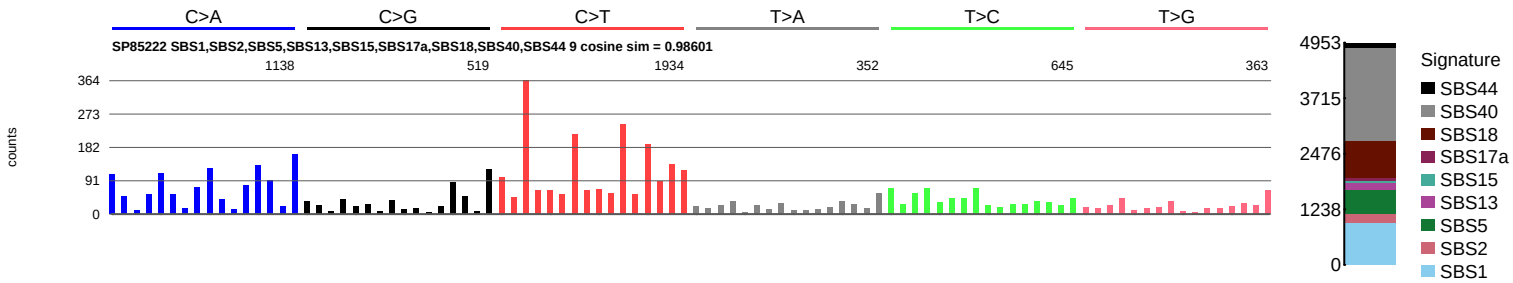
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40 9 cosine sim = 0.98604
1132
520
1912
350
671
364
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

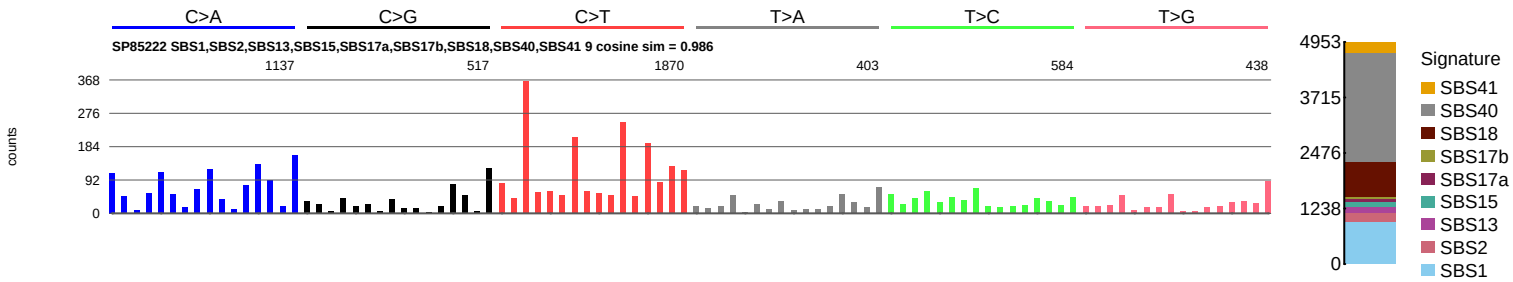

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98604
1133
512
1859
392
651
408
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98603
1151
544
1889
375
636
359
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98603
1135
503
1878
377
637
422
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98602
1125
510
1914
388
633
385
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS44 9 cosine sim = 0.98601
1138
519
1934
352
645
363
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.986
1137
517
1870
403
584
438
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
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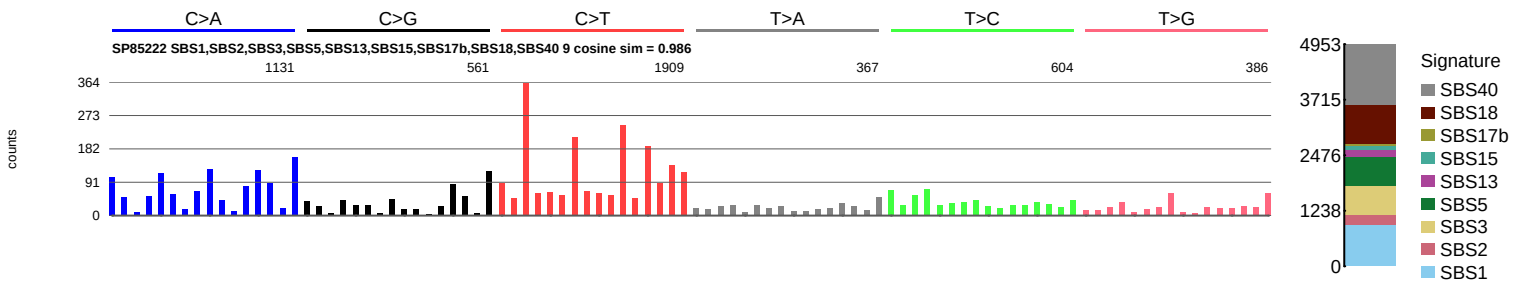
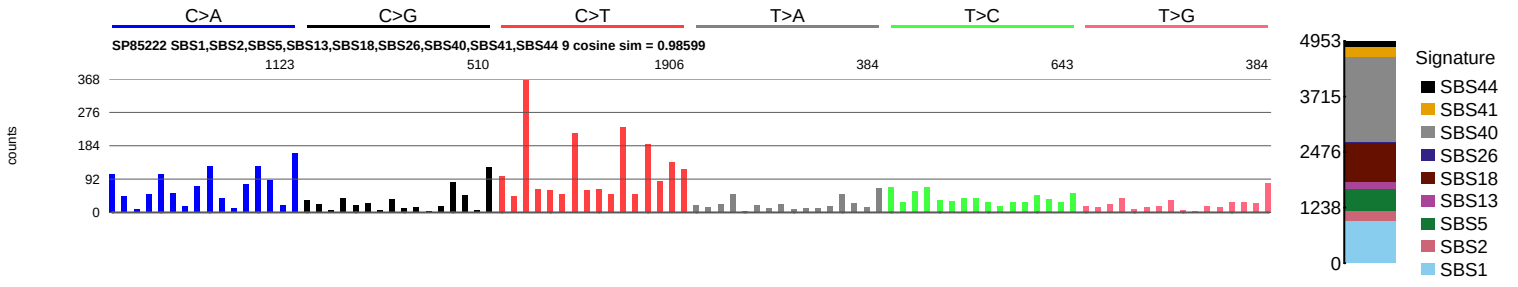
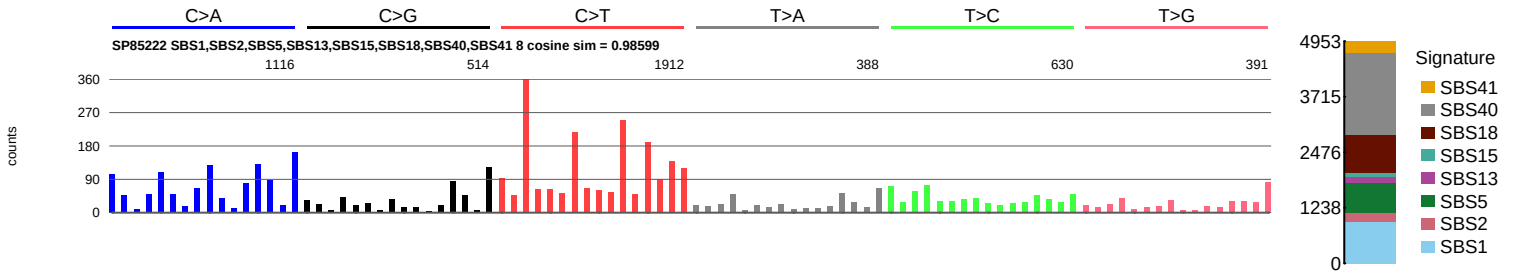
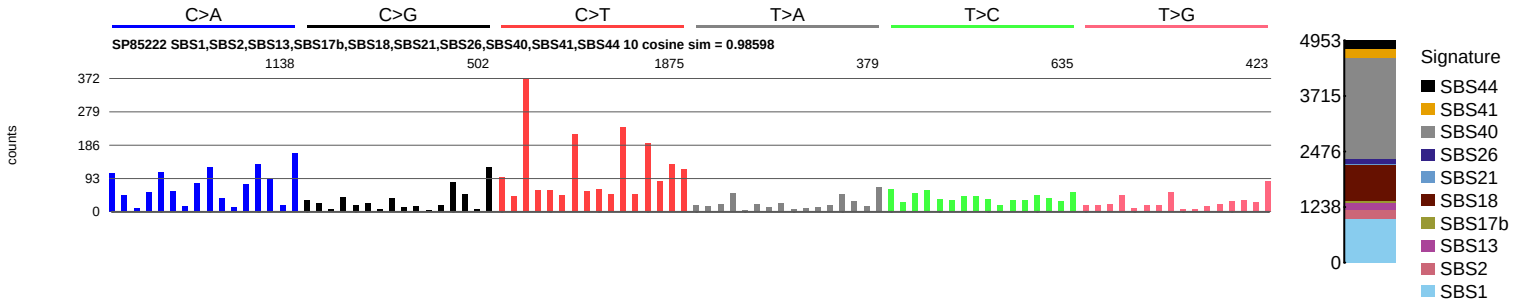
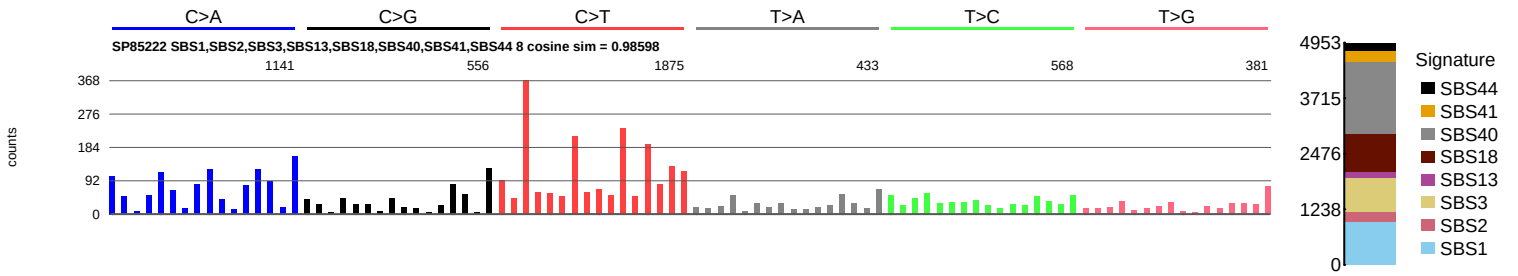
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 9 cosine sim = 0.986
1131
561
1909
367
604
386
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98599
1123
510
1906
384
643
384
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41 8 cosine sim = 0.98599
1116
514
1912
388
630
391
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98598
1138
502
1875
379
635
423
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1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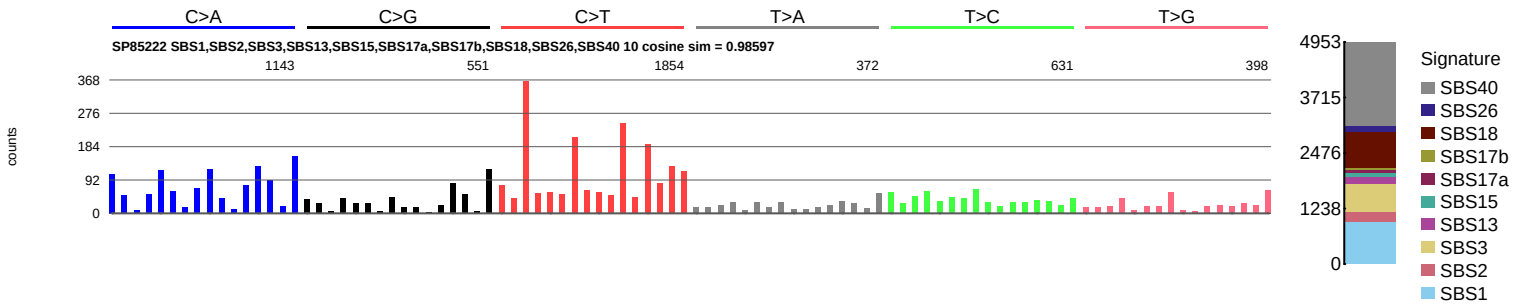
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98598
1141
556
1875
433
568
381
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98597
1143
551
1854
372
631
398
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
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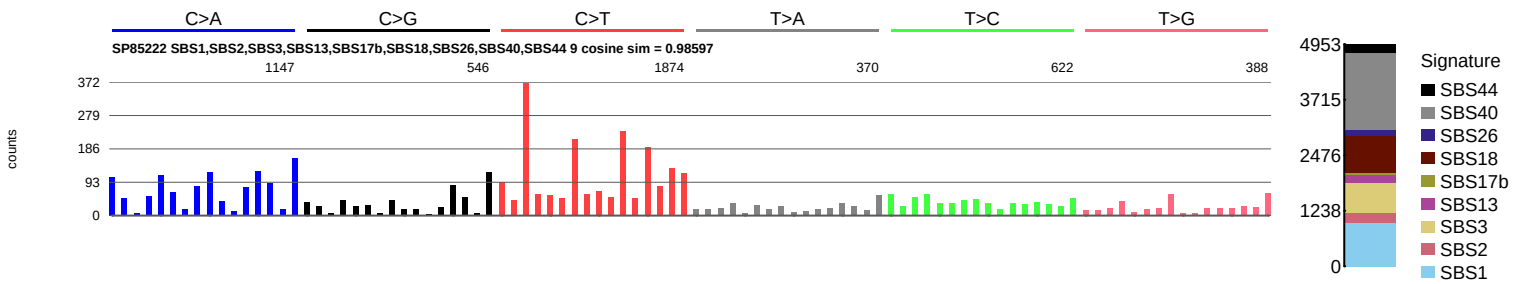
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98597
1147
546
1874
370
622
388
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
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SBS18
SBS17b
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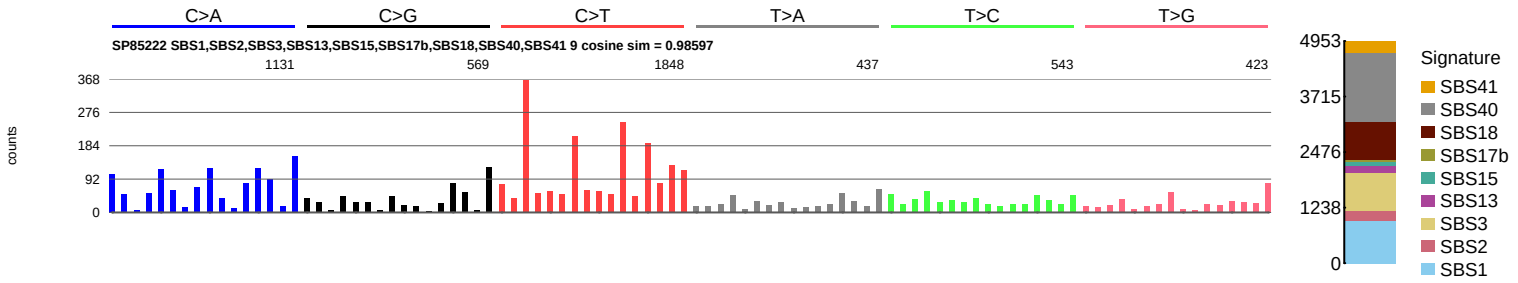
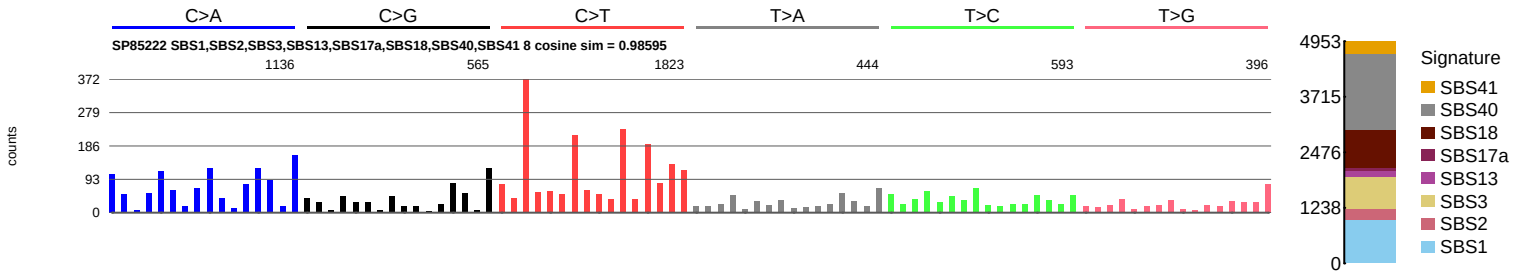
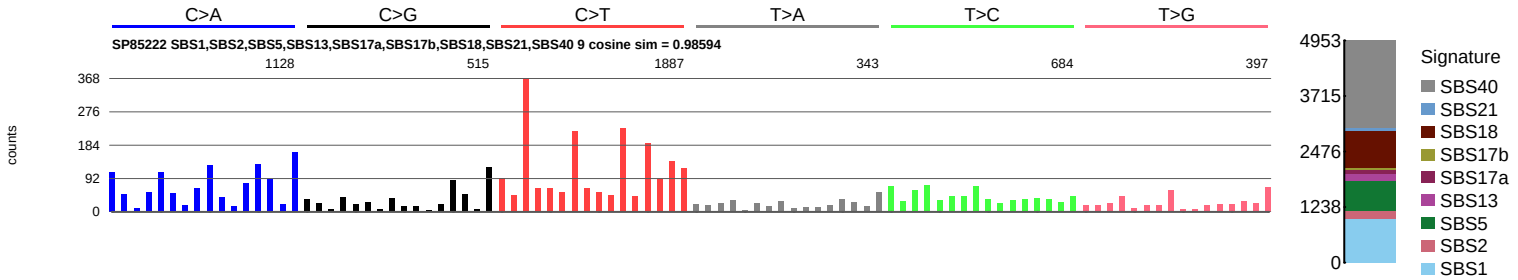
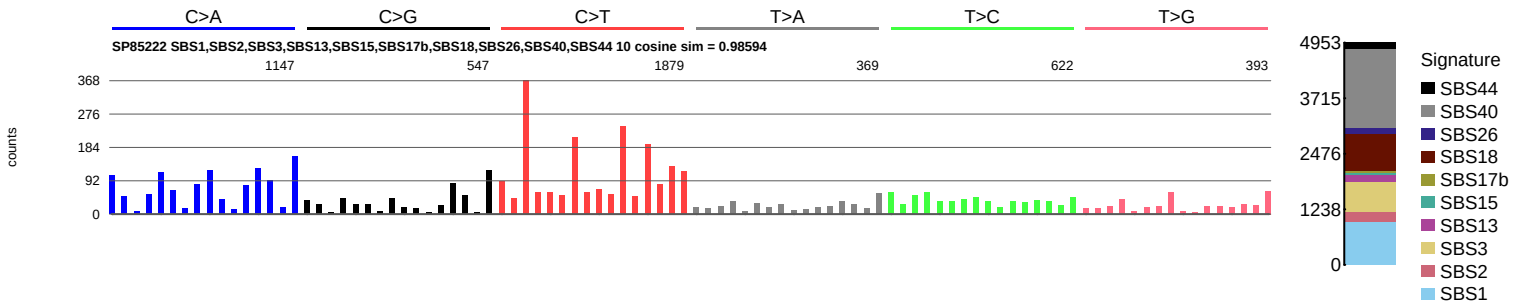
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98597
1131
569
1848
437
543
423
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98595
1136
565
1823
444
593
396
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98594
1128
515
1887
343
684
397
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98594
1147
547
1879
369
622
393
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
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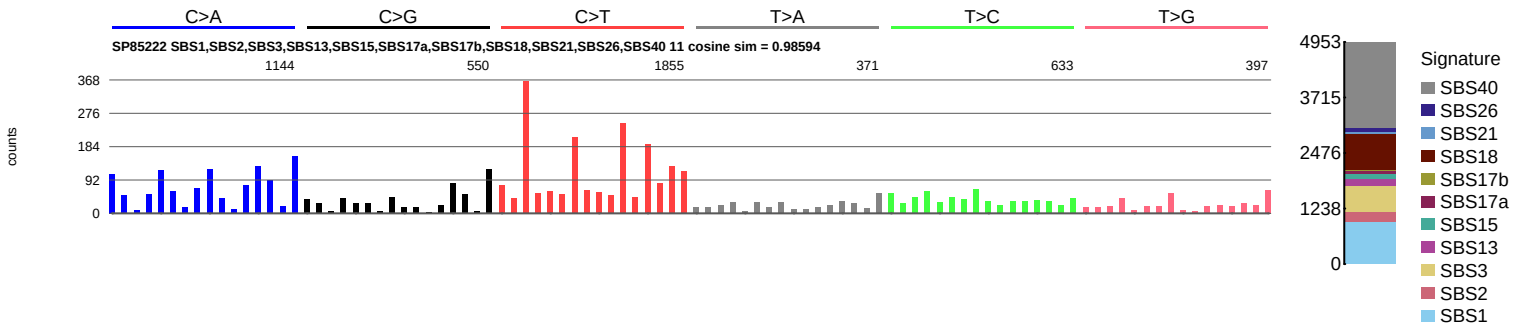
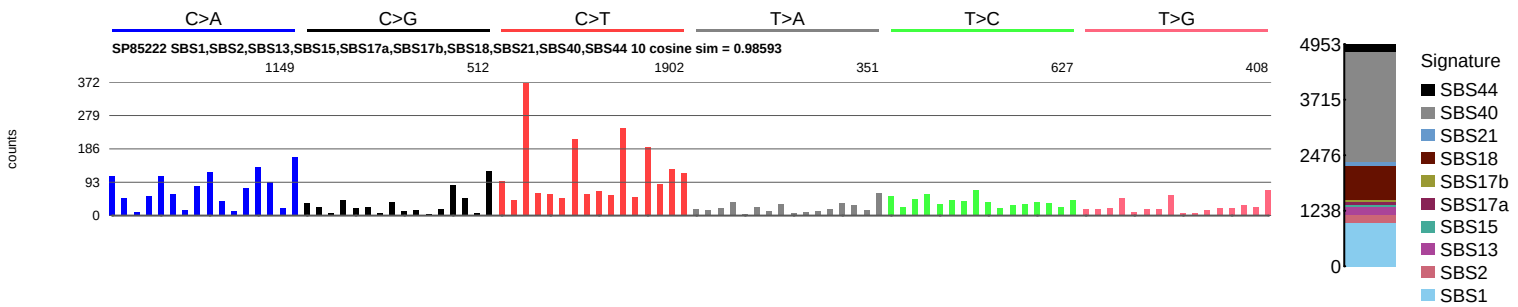
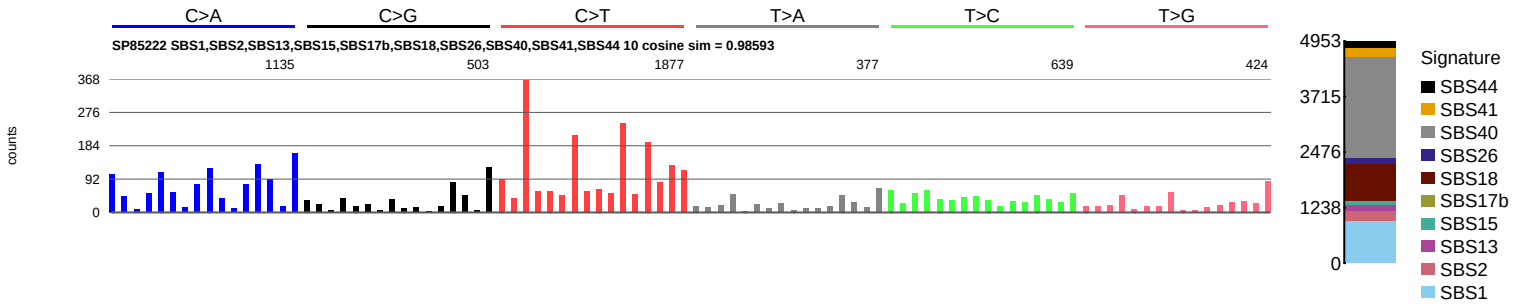
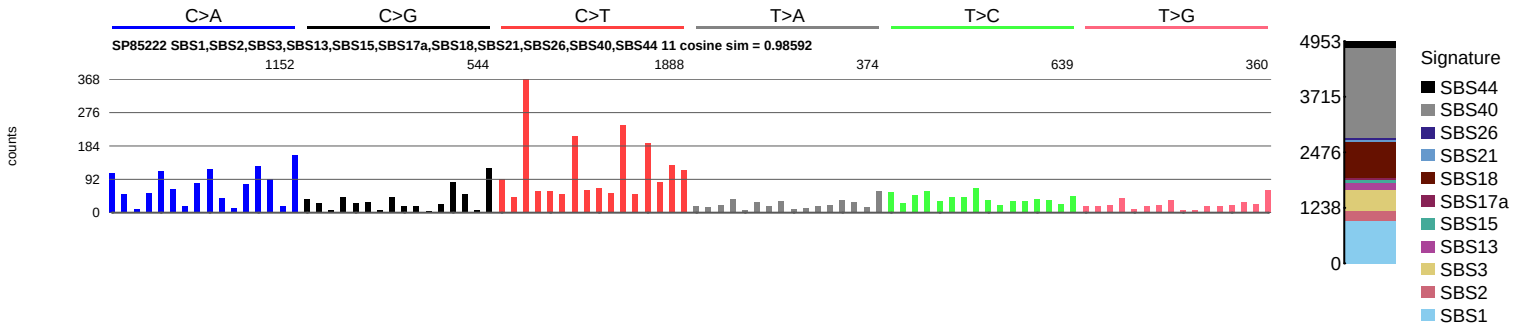
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 11 cosine sim = 0.98594
1144
550
1855
371
633
397
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98593
1149
512
1902
351
627
408
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98593
1135
503
1877
377
639
424
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1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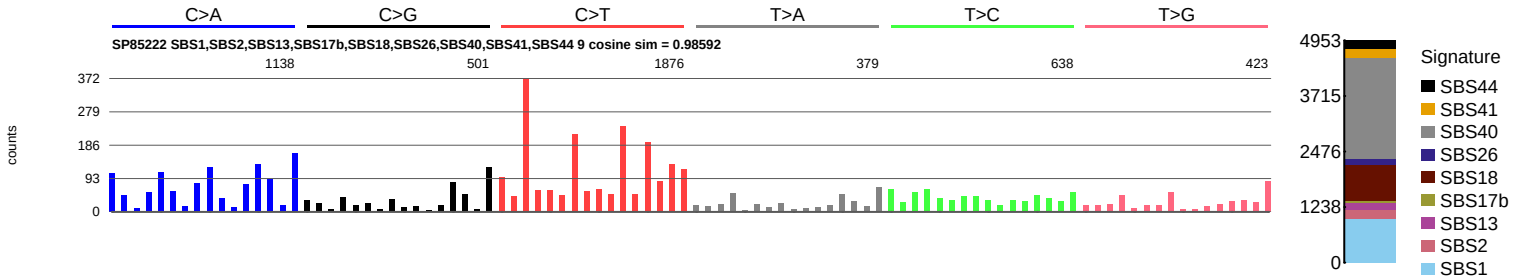
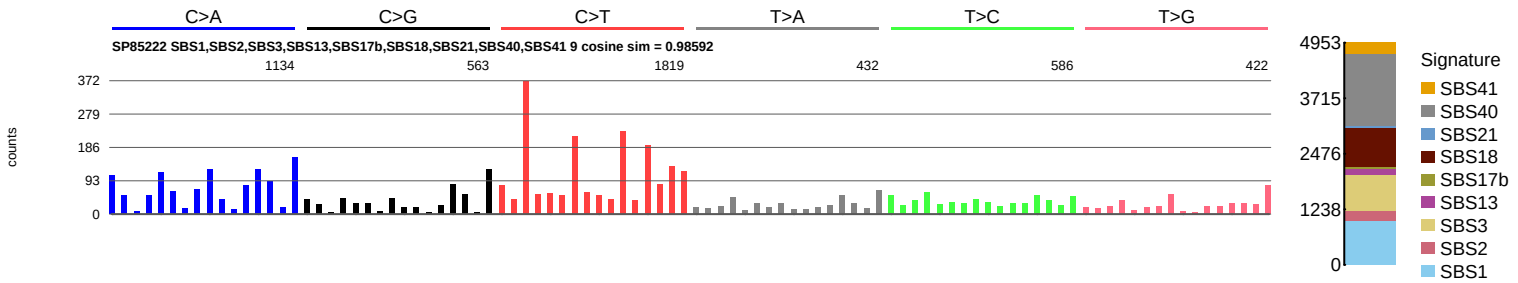
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98592
1152
544
1888
374
639
360
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98592
1138
501
1876
379
638
423
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98592
1134
563
1819
432
586
422
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
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SBS18
SBS17b
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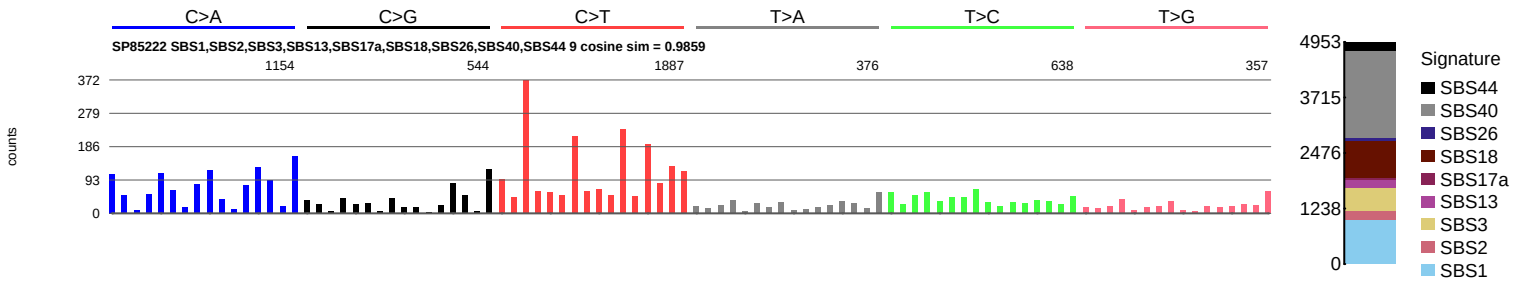
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.9859
1154
544
1887
376
638
357
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
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SBS18
SBS17a
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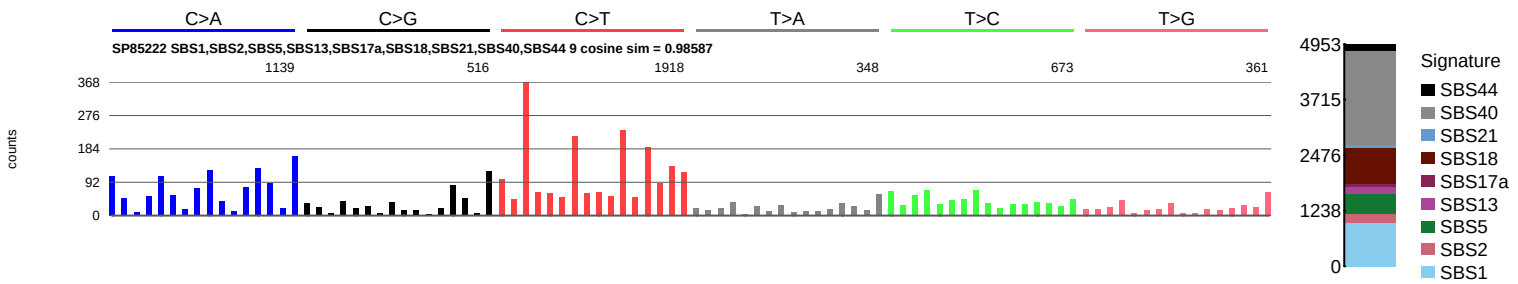
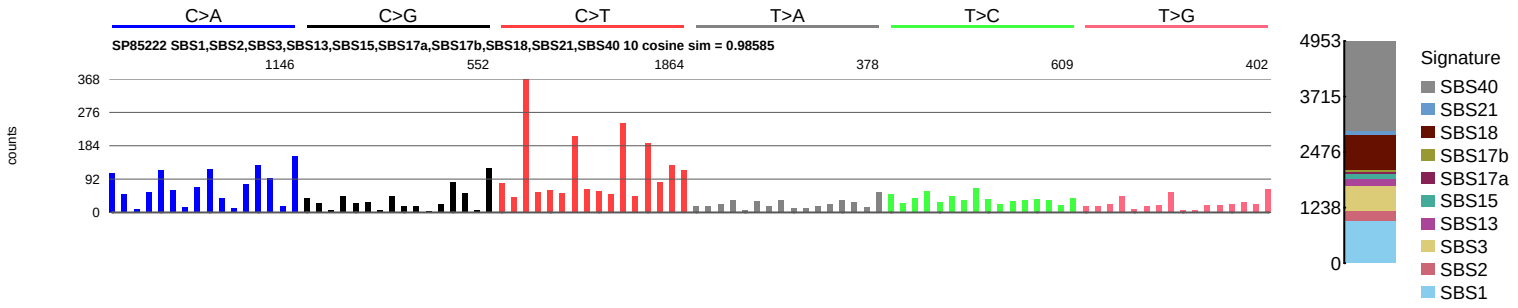
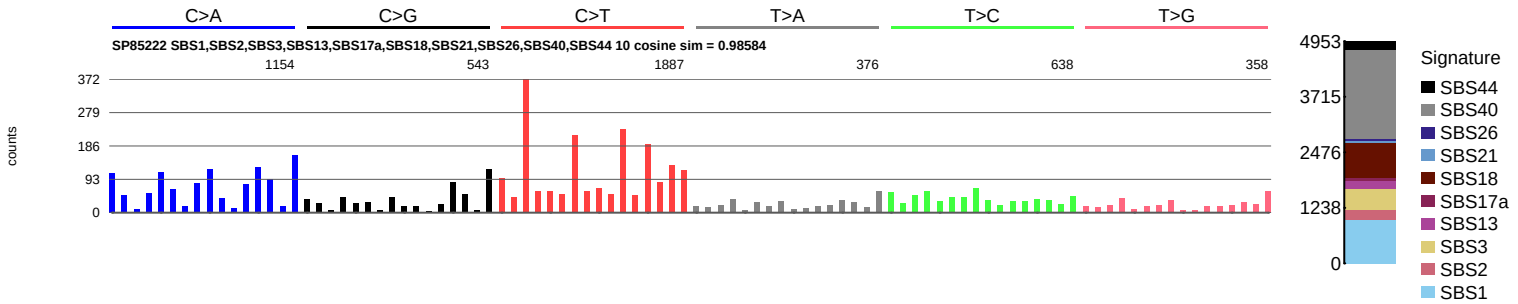
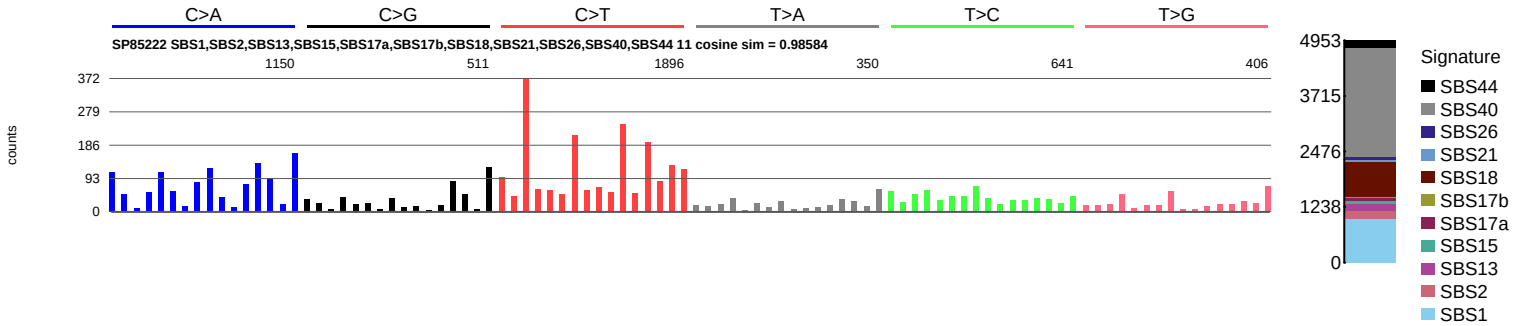
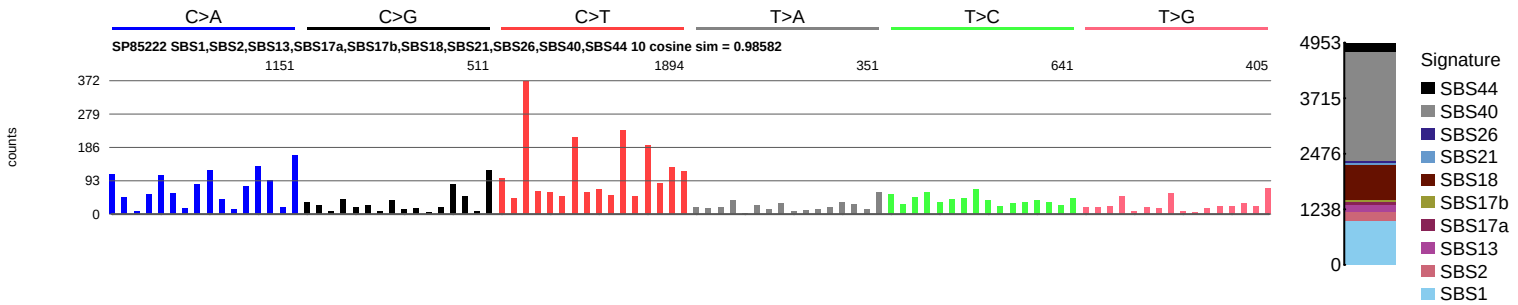
SBS3
SBS2
SBS1

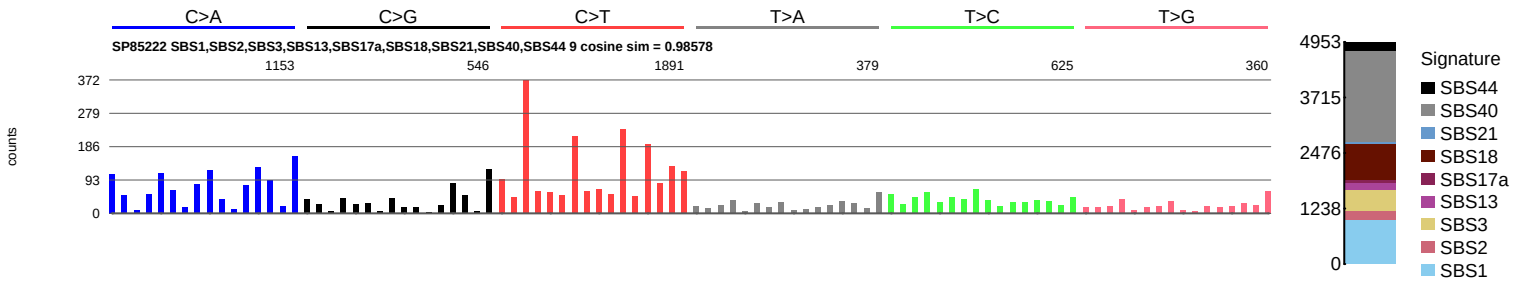

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98587
1139
516
1918
348
673
361
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1

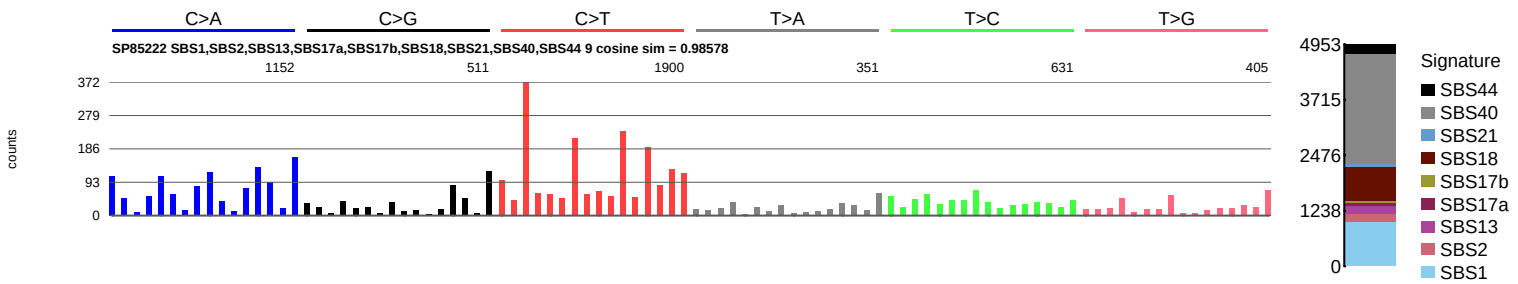

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98585
1146
552
1864
378
609
402
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

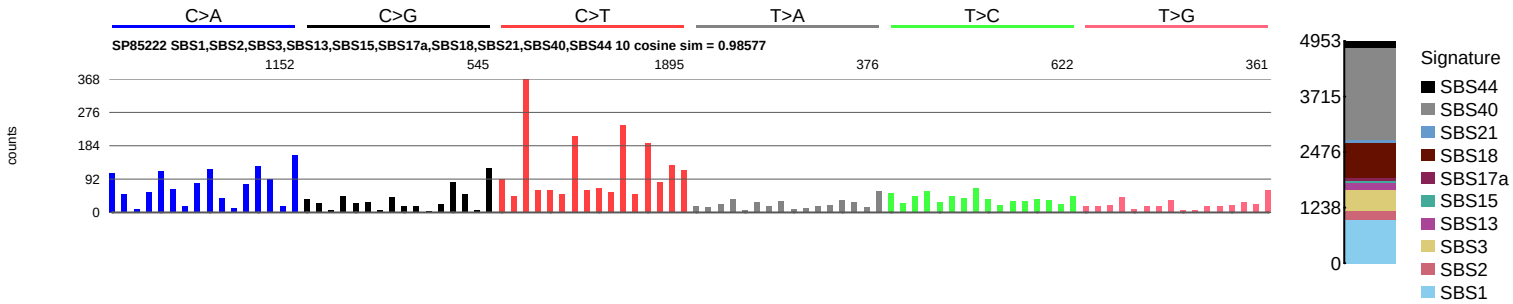

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98584
1154
543
1887
376
638
358
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

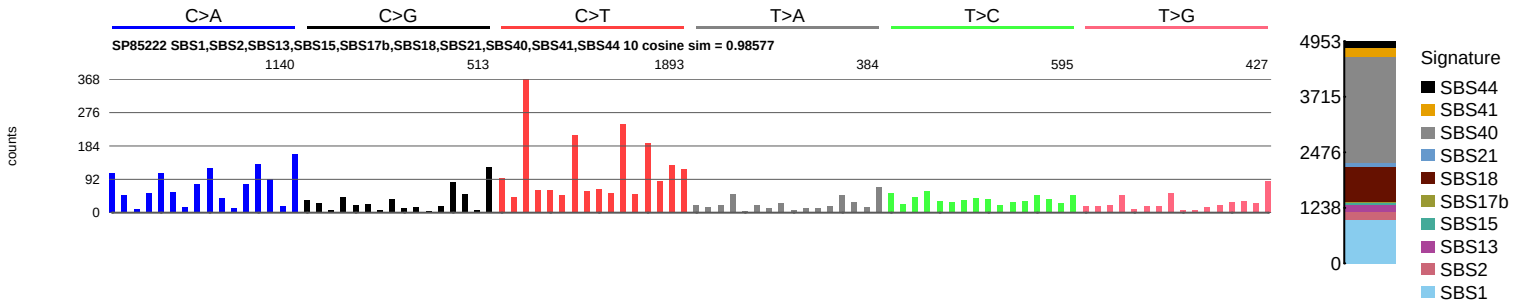

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98584
1150
511
1896
350
641
406
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

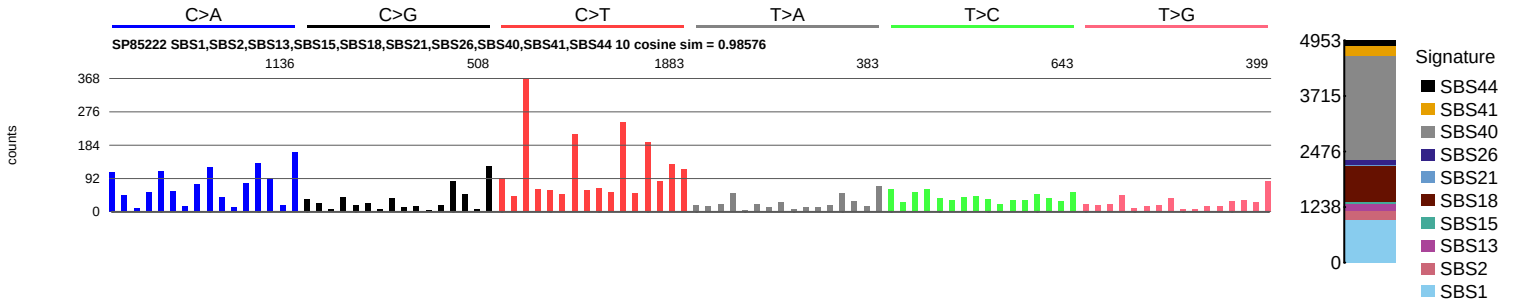

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98582
1151
511
1894
351
641
405
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98578
1153
546
1891
379
625
360
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98578
1152
511
1900
351
631
405
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98577
1152
545
1895
376
622
361
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98577
1140
513
1893
384
595
427
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98576
1136
508
1883
383
643
399
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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SBS26
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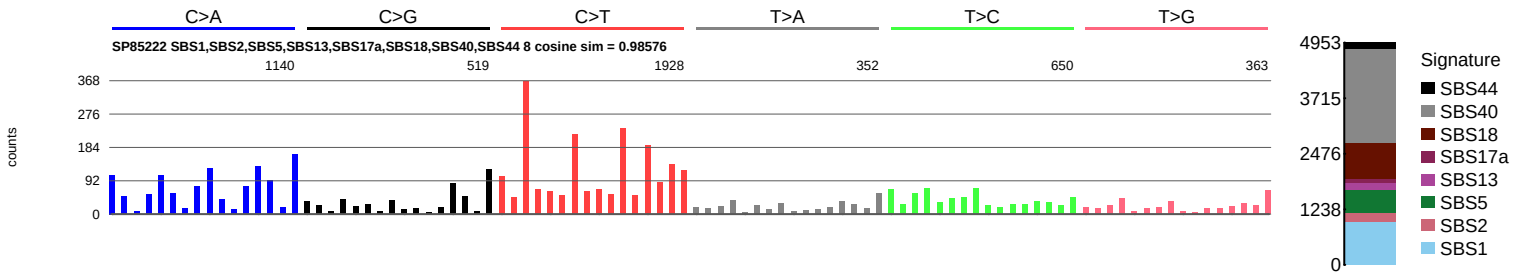
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.98576
1140
519
1928
352
650
363
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
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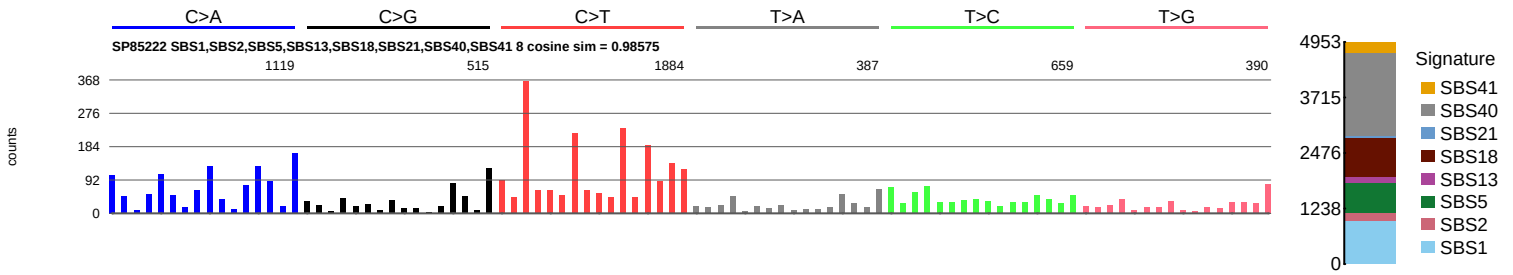
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98575
1119
515
1884
387
659
390
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
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SBS18
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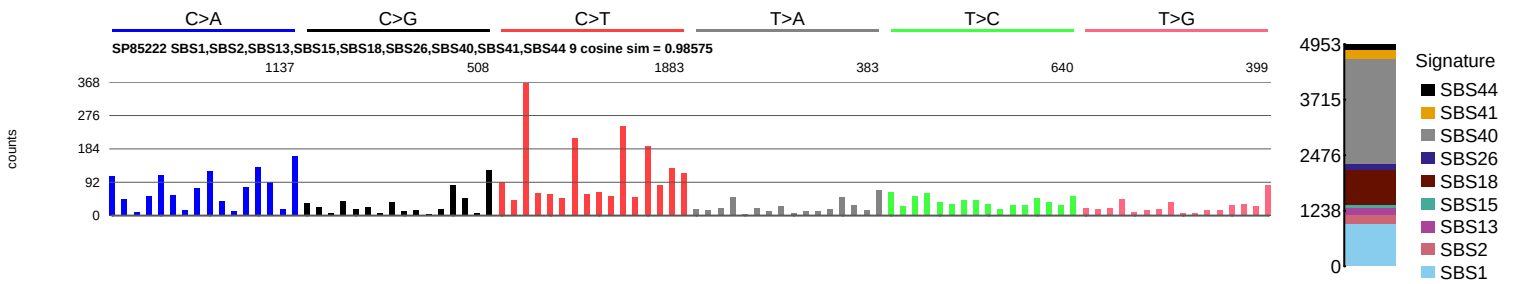
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98575
1137
508
1883
383
640
399
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
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SBS40
SBS26
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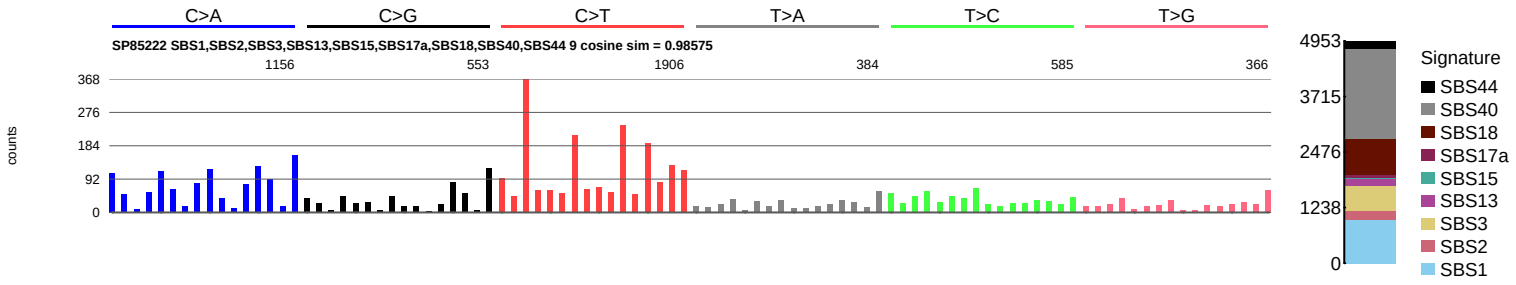
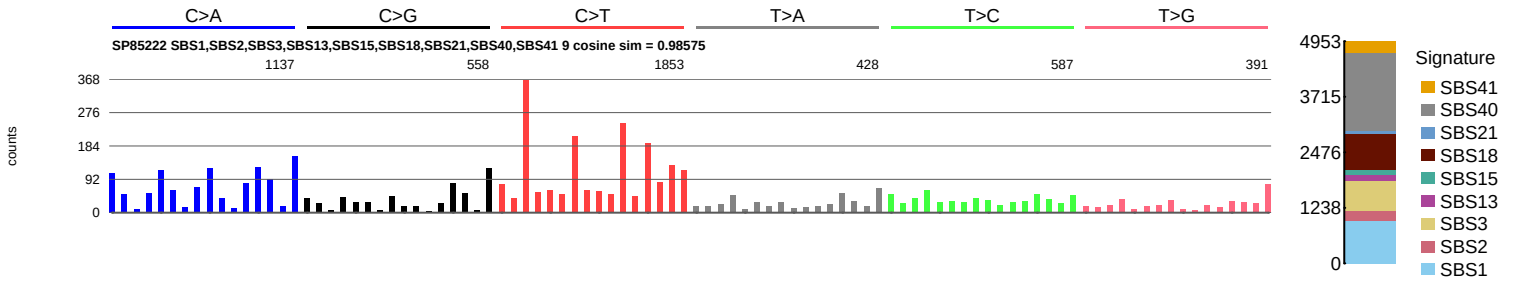
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS44 9 cosine sim = 0.98575
1156
553
1906
384
585
366
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98575
1137
558
1853
428
587
391
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
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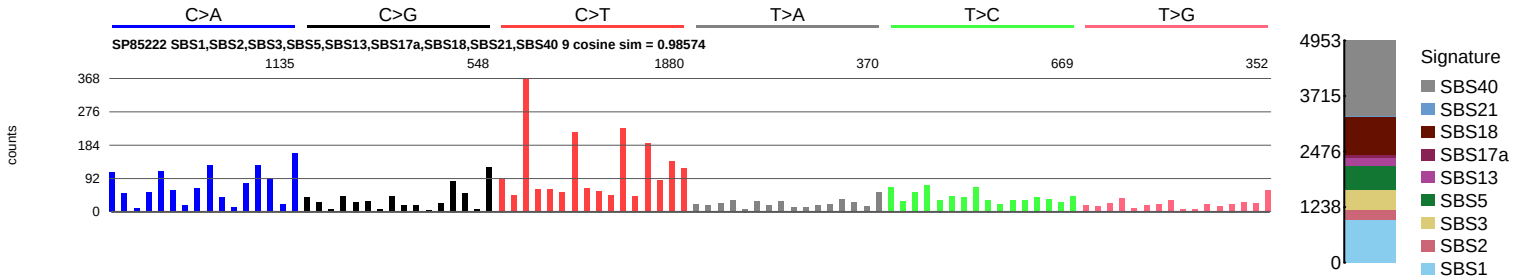
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40 9 cosine sim = 0.98574
1135
548
1880
370
669
352
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS13
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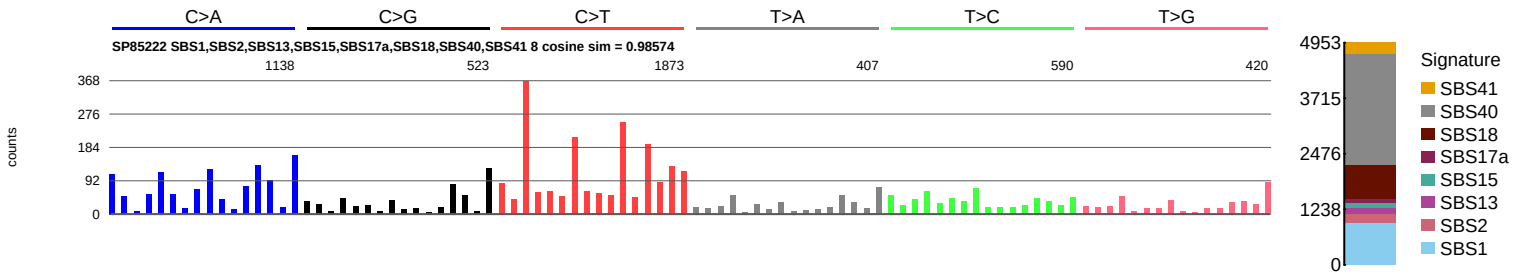
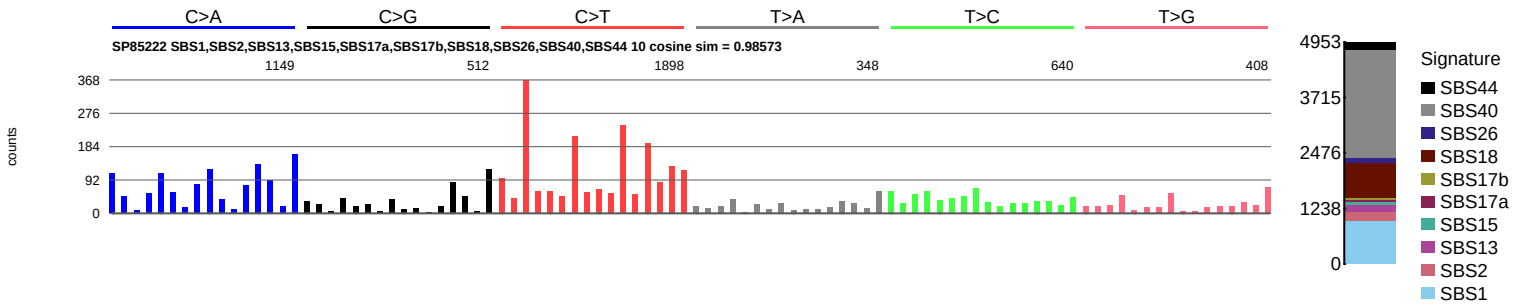
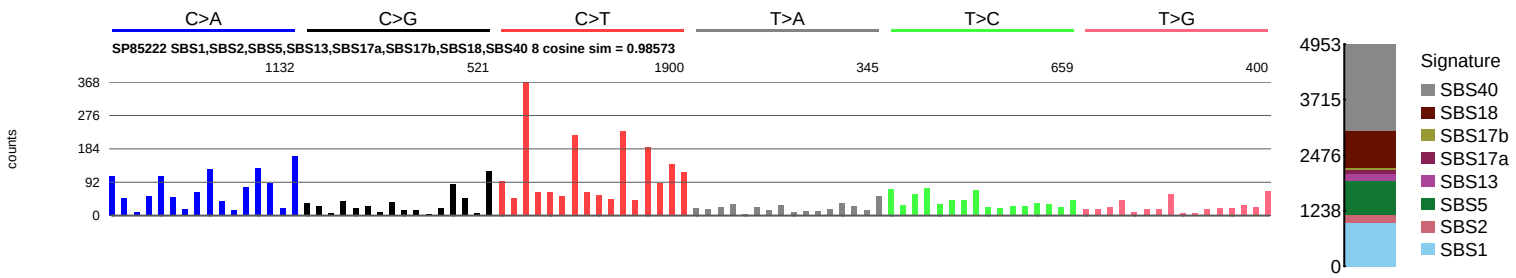
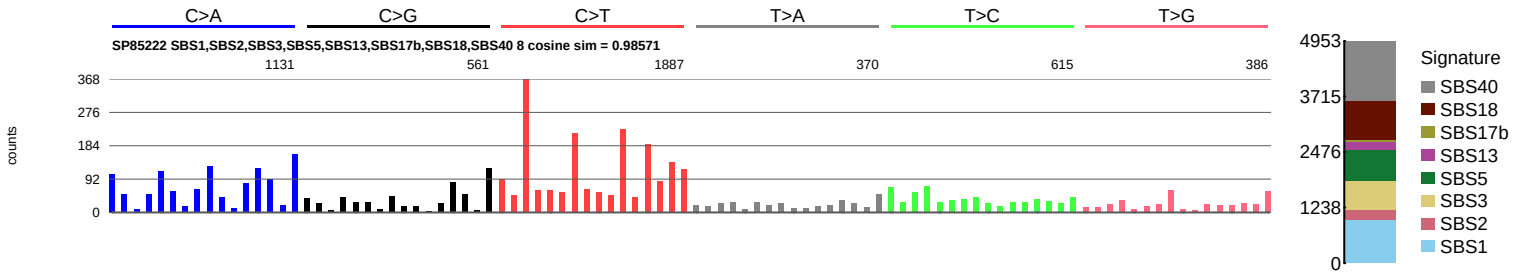
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98574
1138
523
1873
407
590
420
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98573
1149
512
1898
348
640
408
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98573
1132
521
1900
345
659
400
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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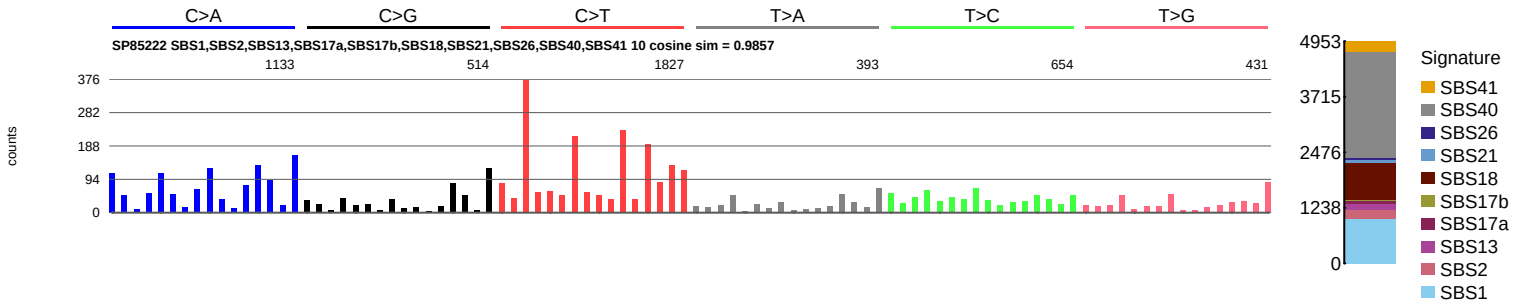
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40 8 cosine sim = 0.98571
1131
561
1887
370
615
386
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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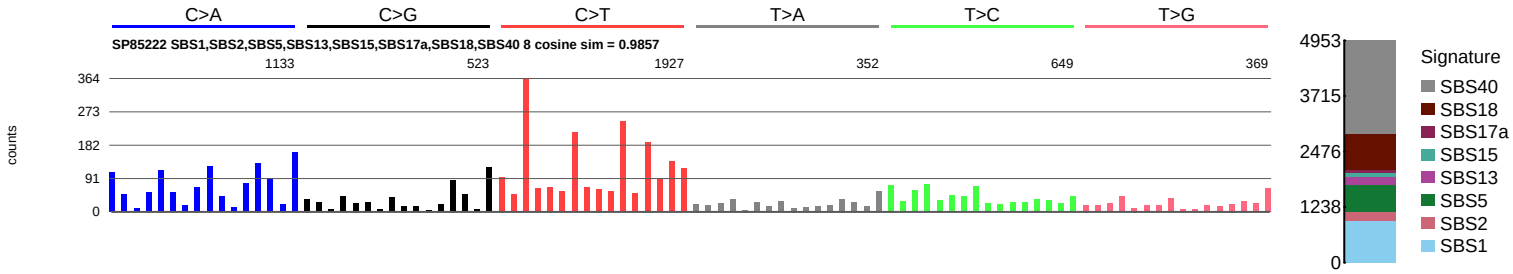
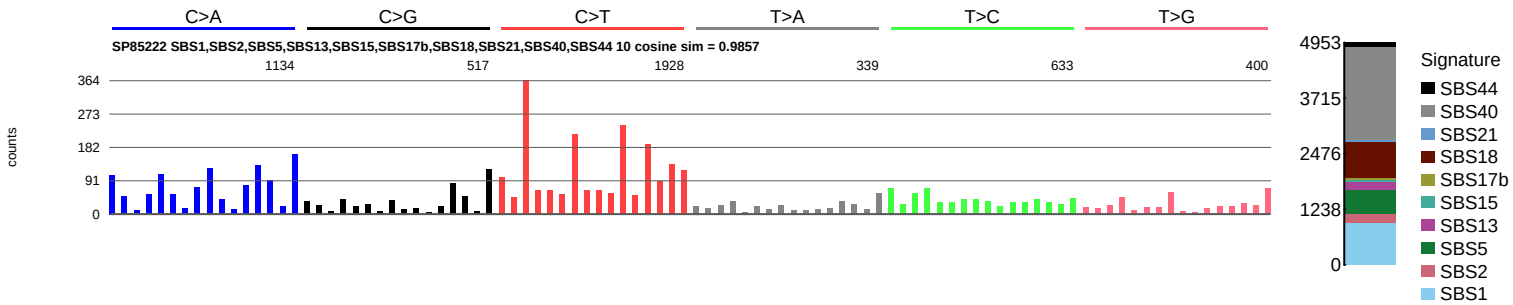
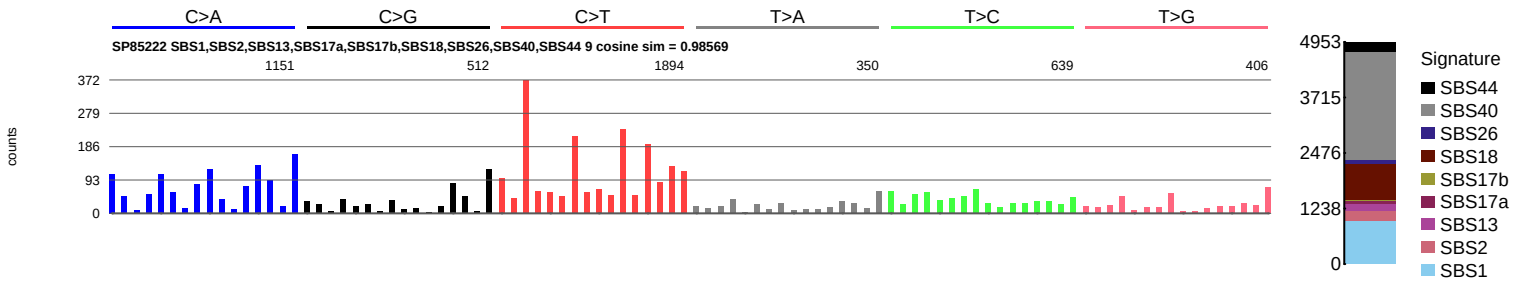
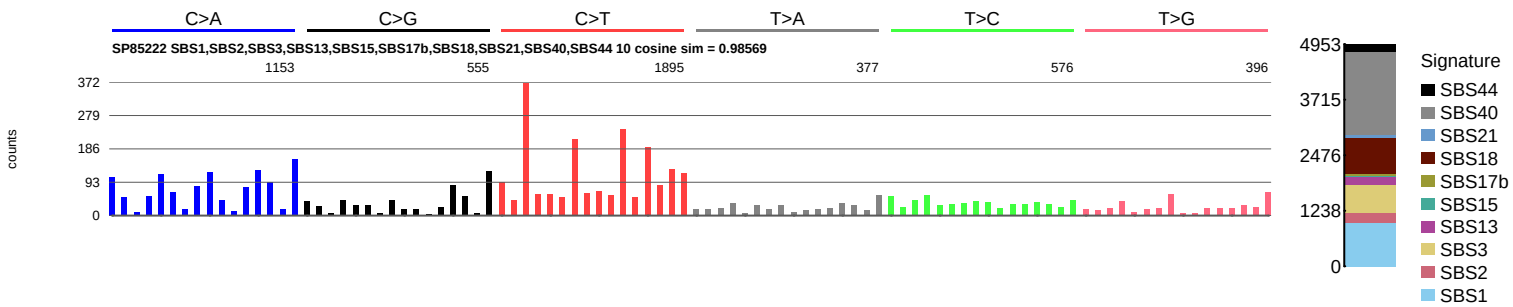
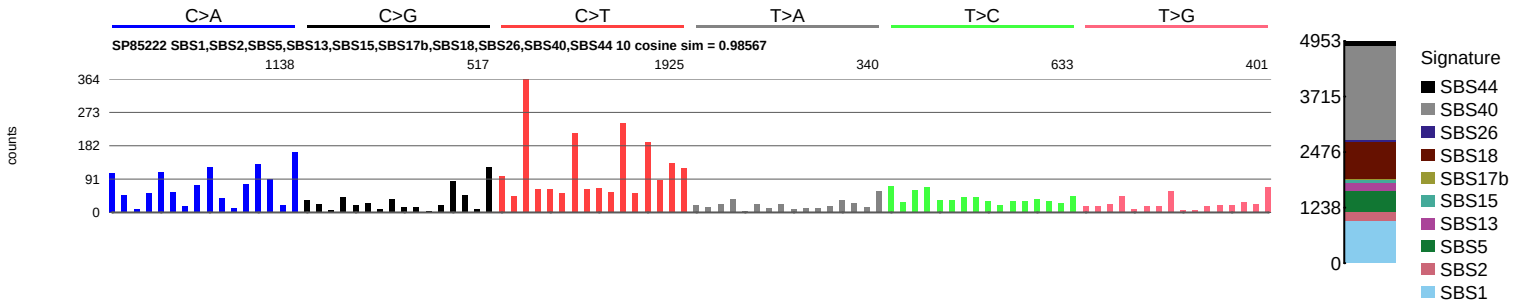
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.9857
1133
514
1827
393
654
431
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

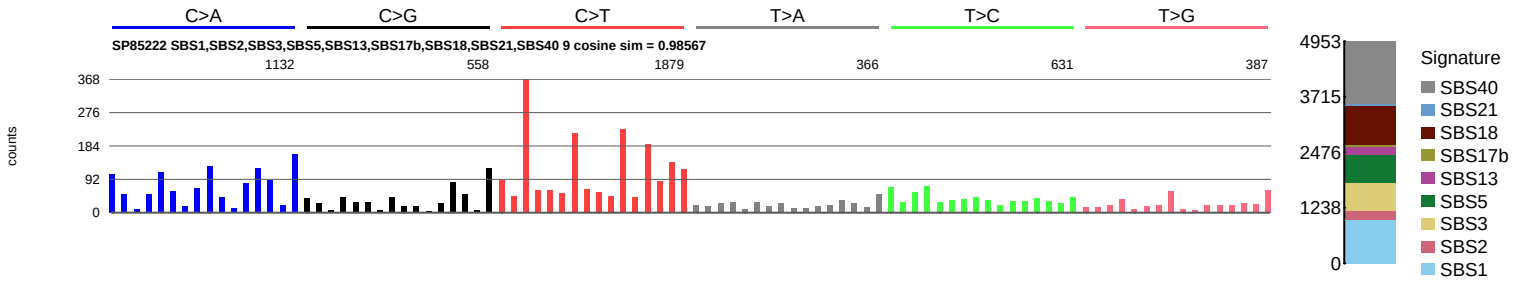

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40 8 cosine sim = 0.9857
1133
523
1927
352
649
369
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

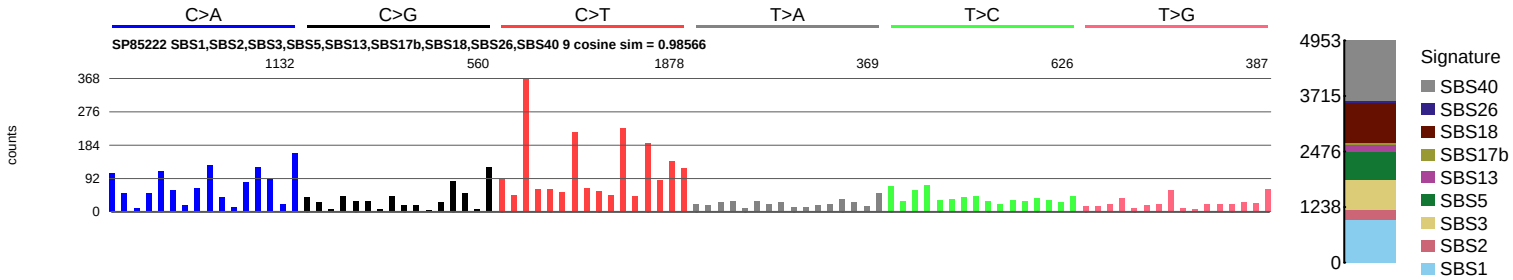

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.9857
1134
517
1928
339
633
400
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

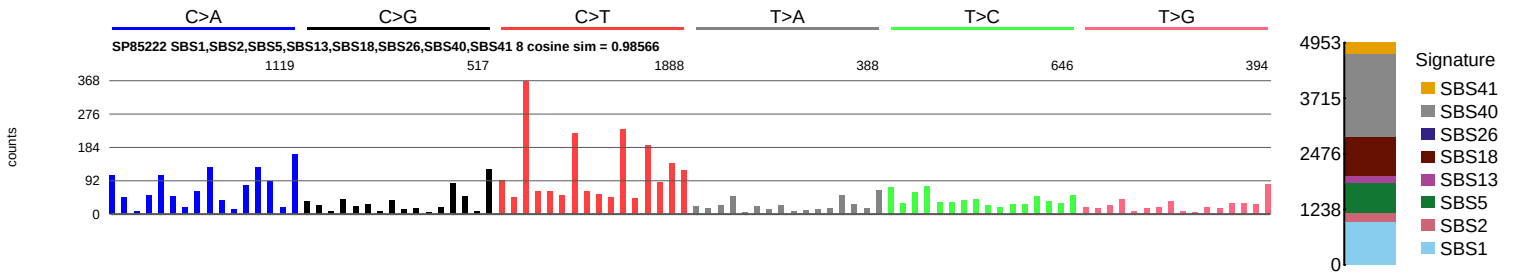

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98569
1151
512
1894
350
639
406
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

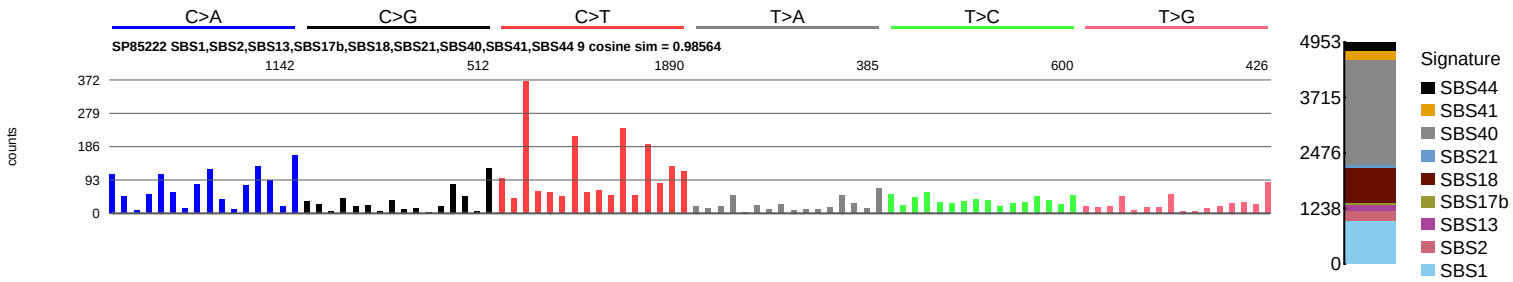

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98569
1153
555
1895
377
576
396
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

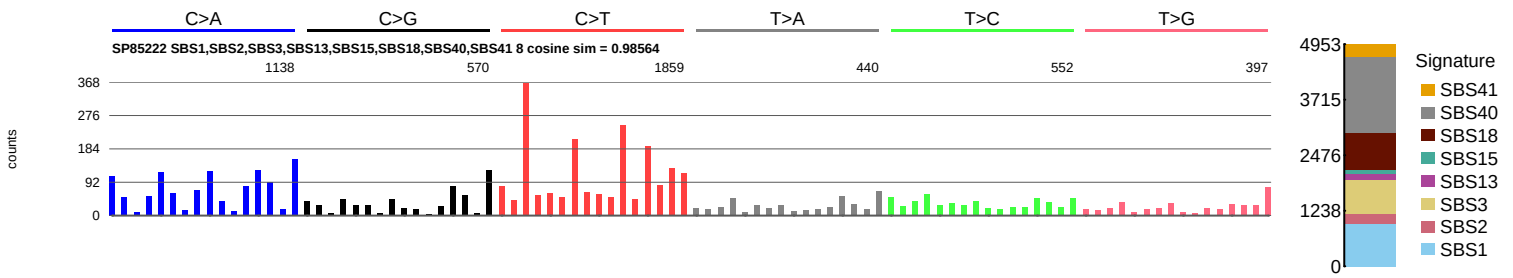

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98567
1138
517
1925
340
633
401
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

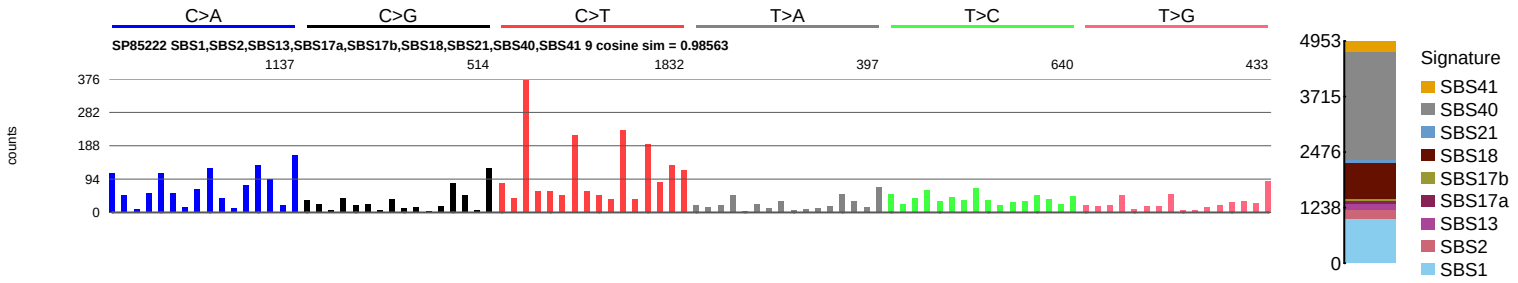

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98567
1132
558
1879
366
631
387
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

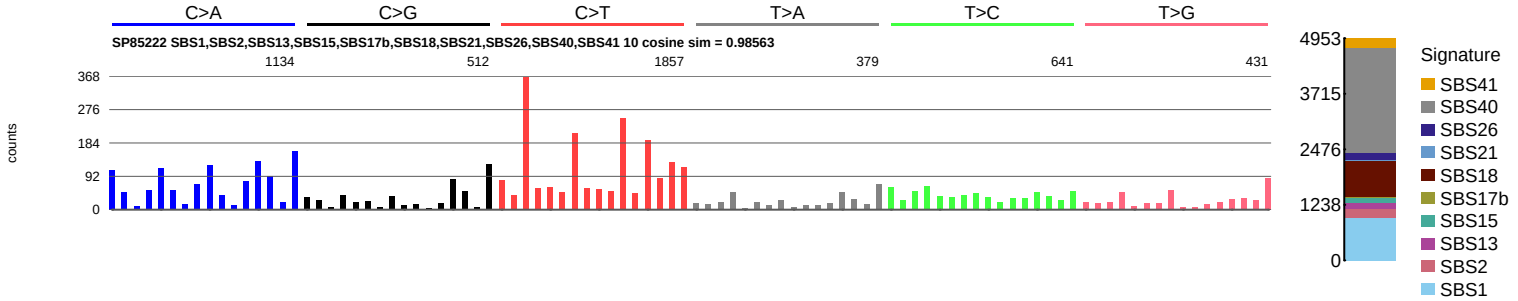

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98566
1132
560
1878
369
626
387
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

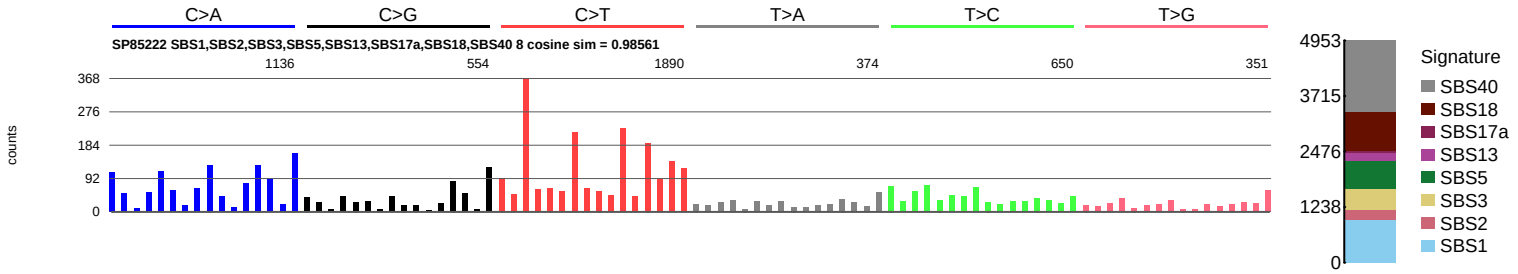

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98566
1119
517
1888
388
646
394
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98564
1142
512
1890
385
600
426
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS40,SBS41 8 cosine sim = 0.98564
1138
570
1859
440
552
397
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98563
1137
514
1832
397
640
433
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98563
1134
512
1857
379
641
431
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40 8 cosine sim = 0.98561
1136
554
1890
374
650
351
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
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SBS17a
SBS13
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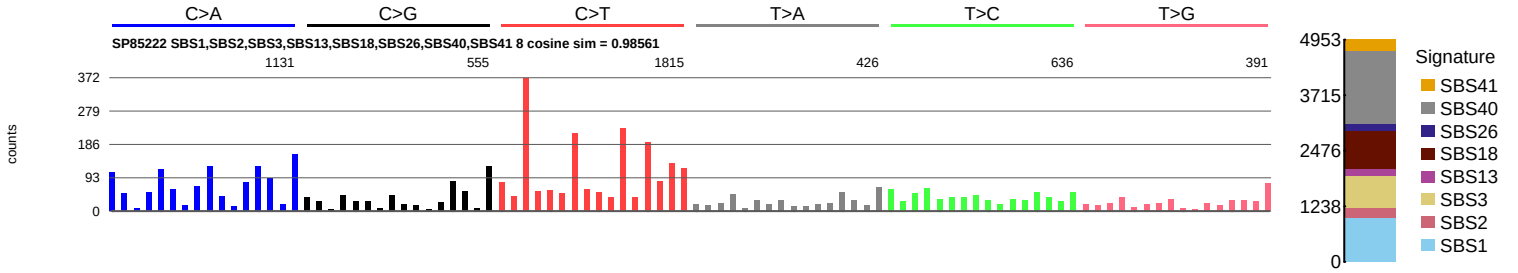
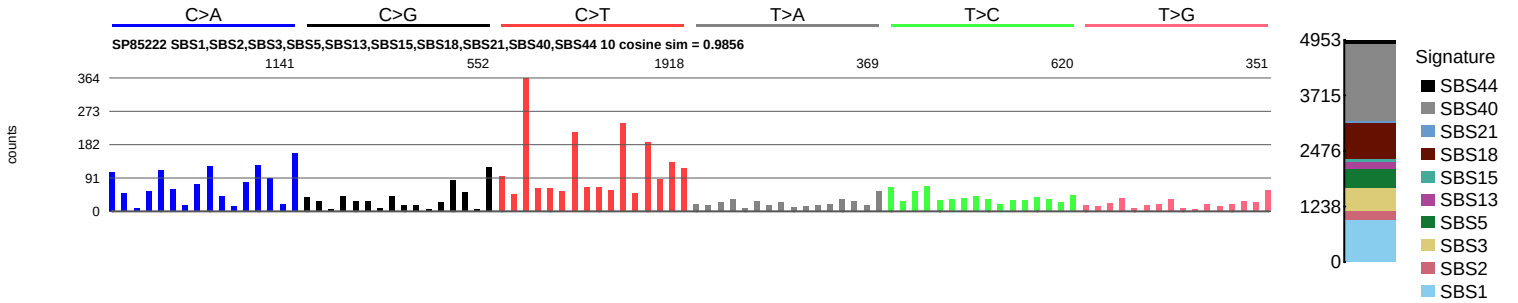
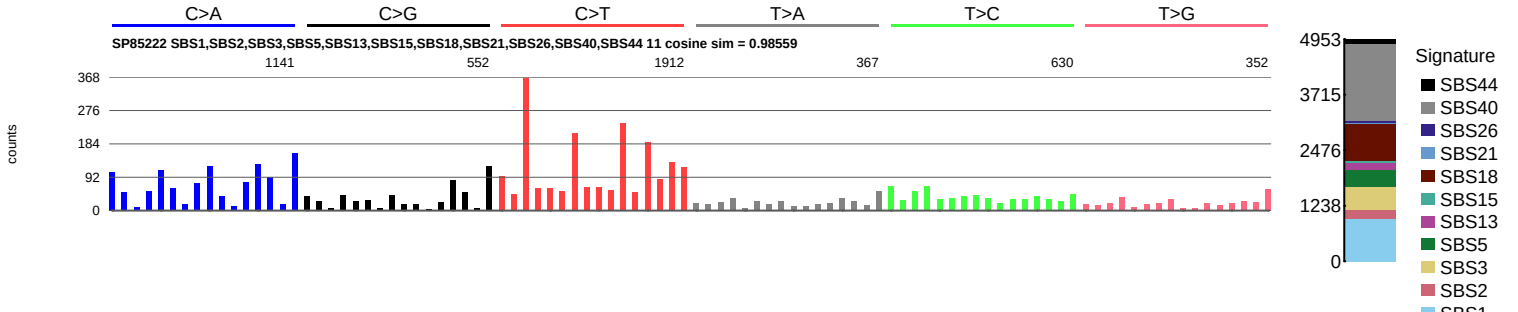
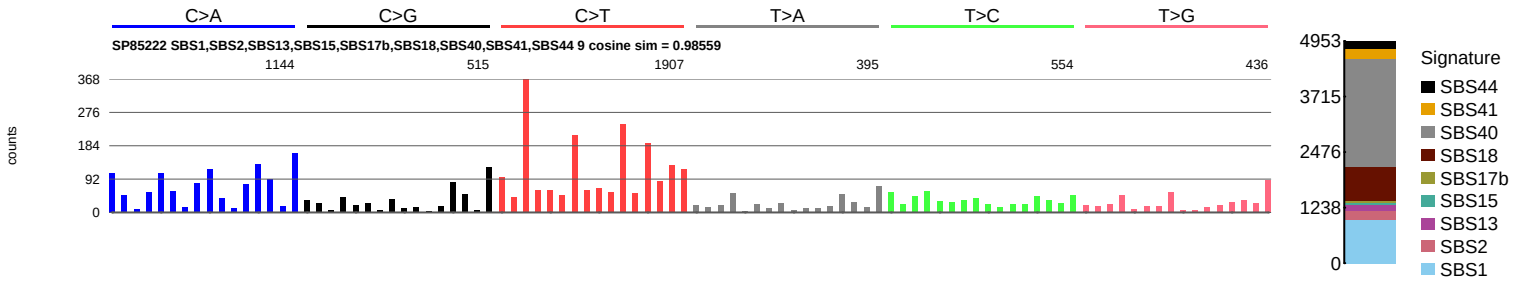
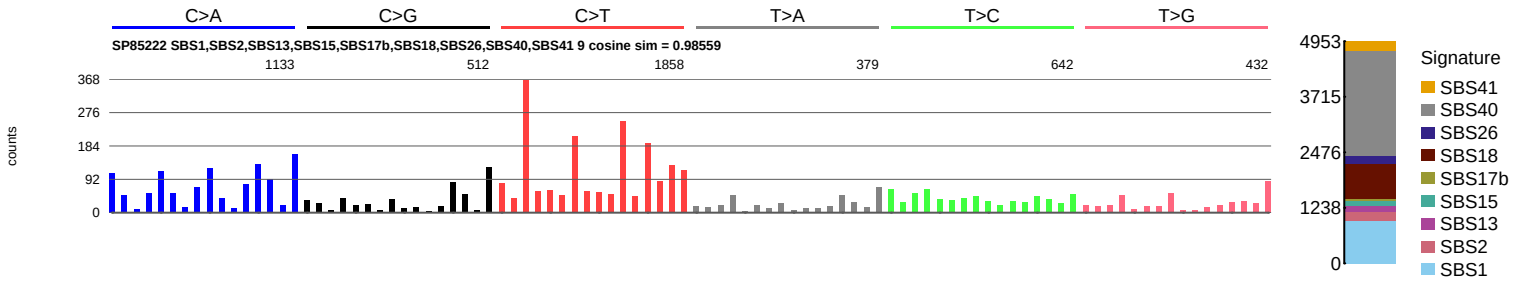
SBS3
SBS2
SBS1

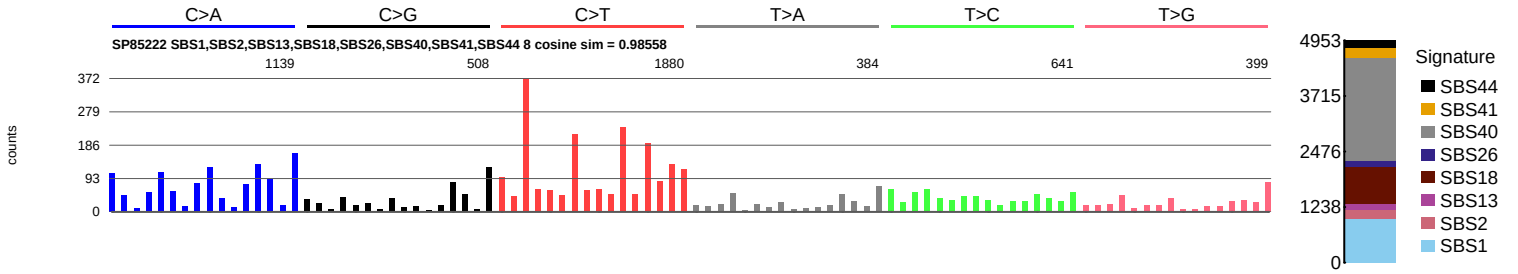

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98561
1131
555
1815
426
636
391
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS13
SBS3
SBS2
SBS1

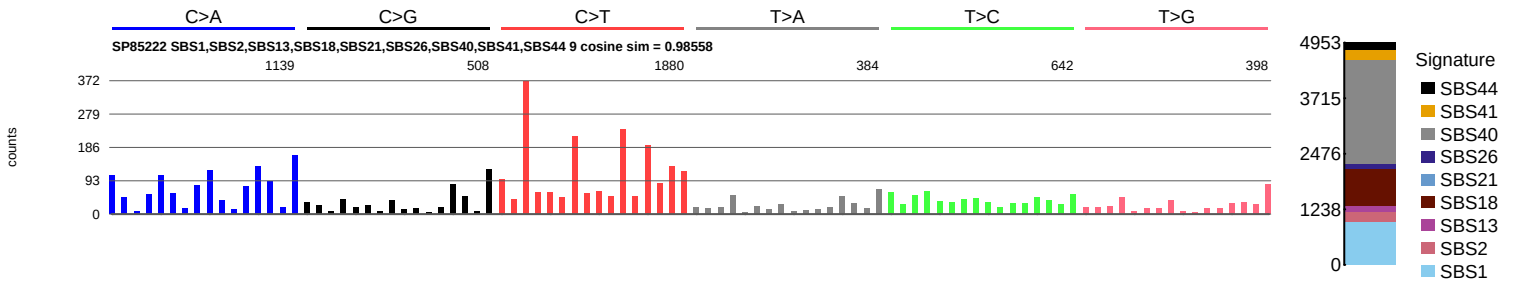

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.9856
1141
552
1918
369
620
351
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

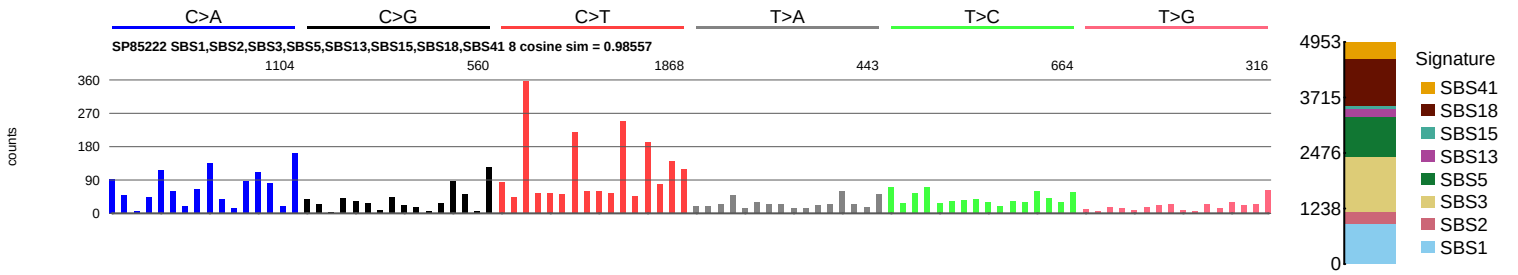

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98559
1141
552
1912
367
630
352
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2

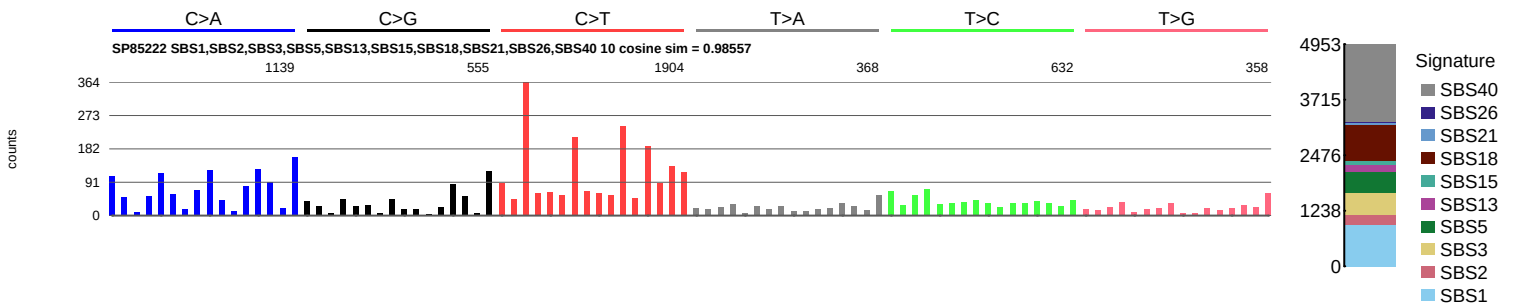

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98559
1144
515
1907
395
554
436
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98559
1133
512
1858
379
642
432
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98558
1139
508
1880
384
641
399
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98558
1139
508
1880
384
642
398
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS41 8 cosine sim = 0.98557
1104
560
1868
443
664
316
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98557
1139
555
1904
368
632
358
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS15
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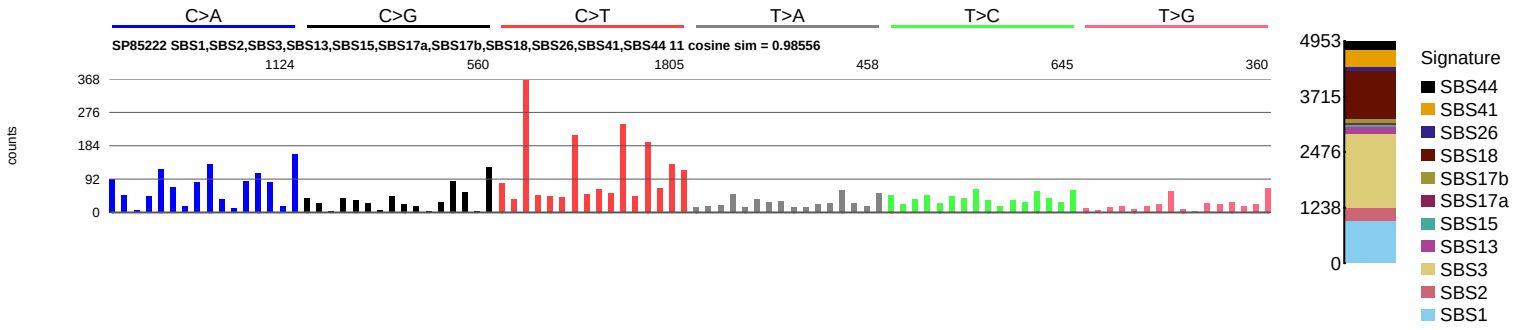
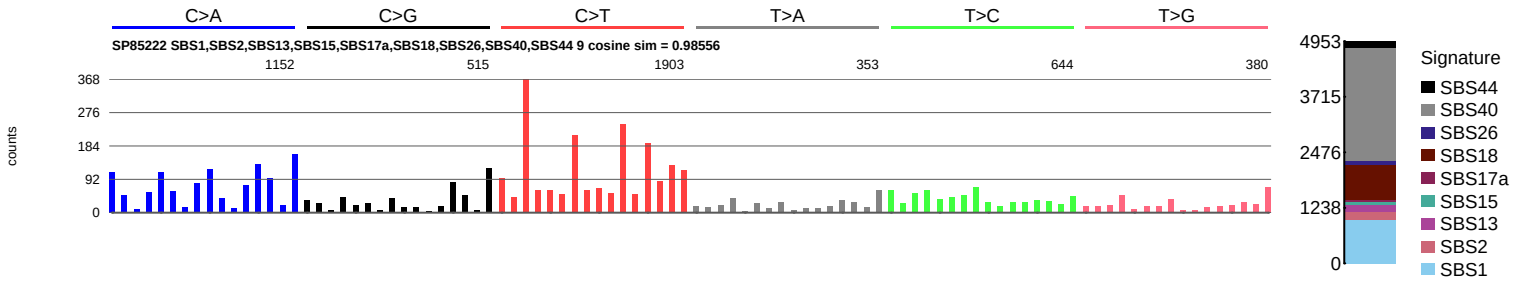
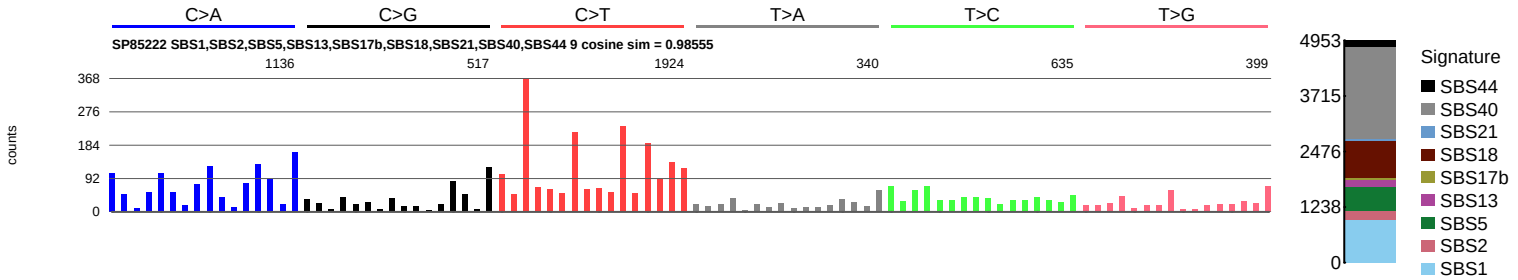
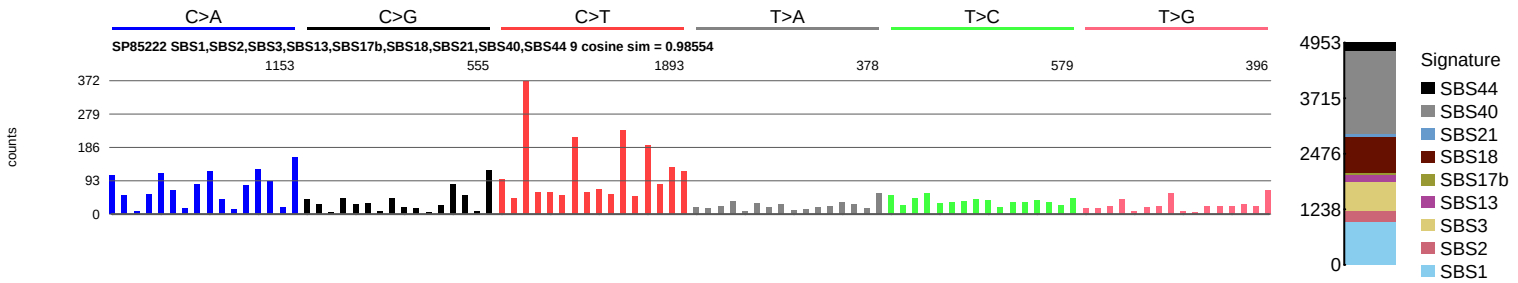
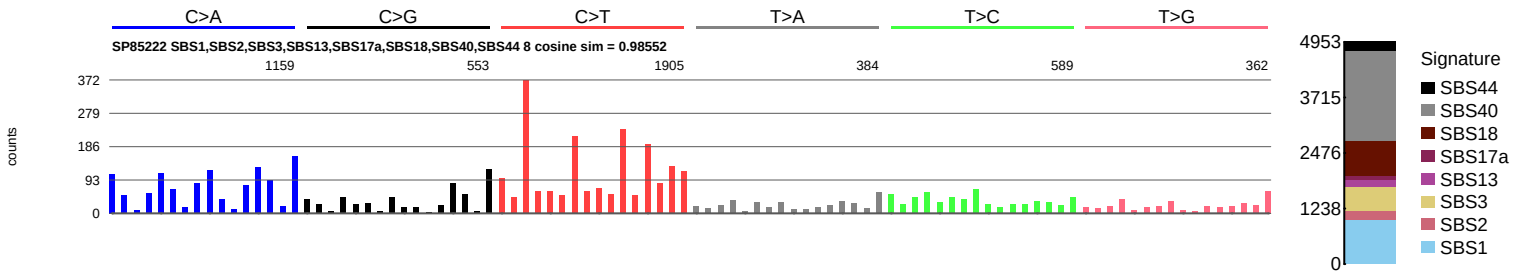
SBS5
SBS3
SBS2
SBS1

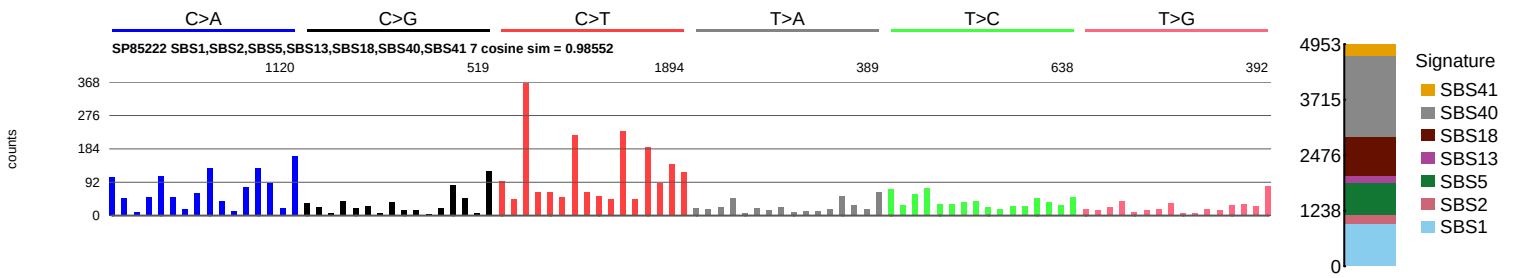

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 11 cosine sim = 0.98556
1124
560
1805
458
645
360
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

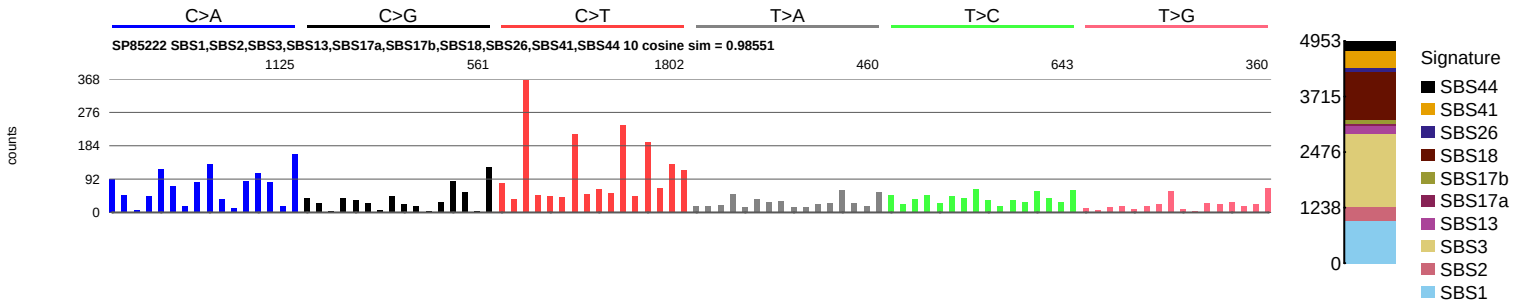

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98556
1152
515
1903
353
644
380
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

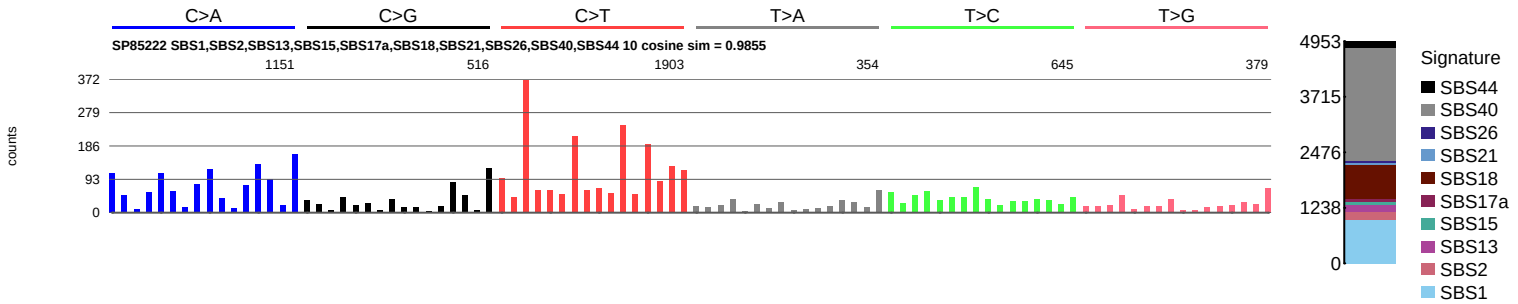

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98555
1136
517
1924
340
635
399
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98554
1153
555
1893
378
579
396
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.98552
1159
553
1905
384
589
362
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40,SBS41 7 cosine sim = 0.98552
1120
519
1894
389
638
392
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98551
1125
561
1802
460
643
360
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.9855
1151
516
1903
354
645
379
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
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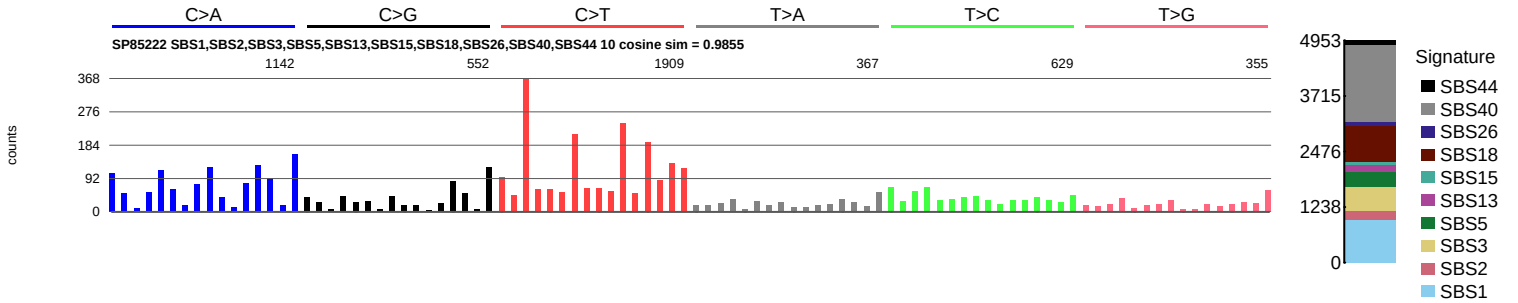
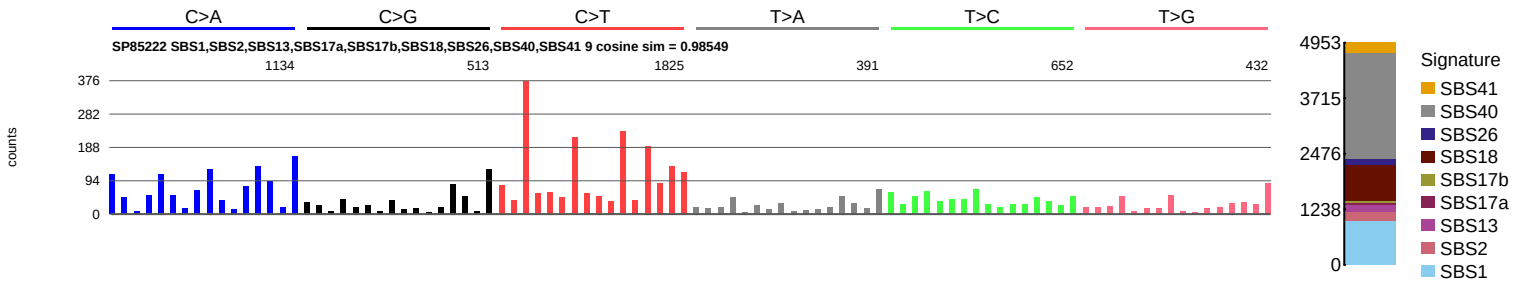
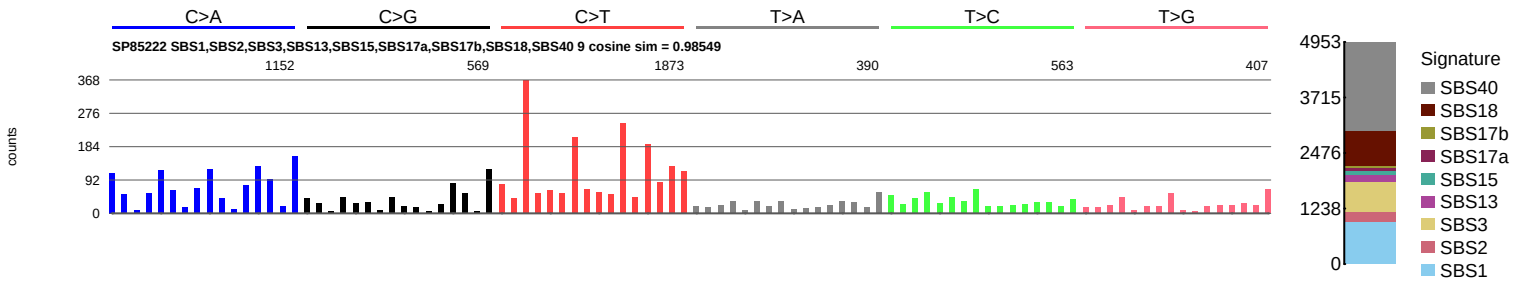
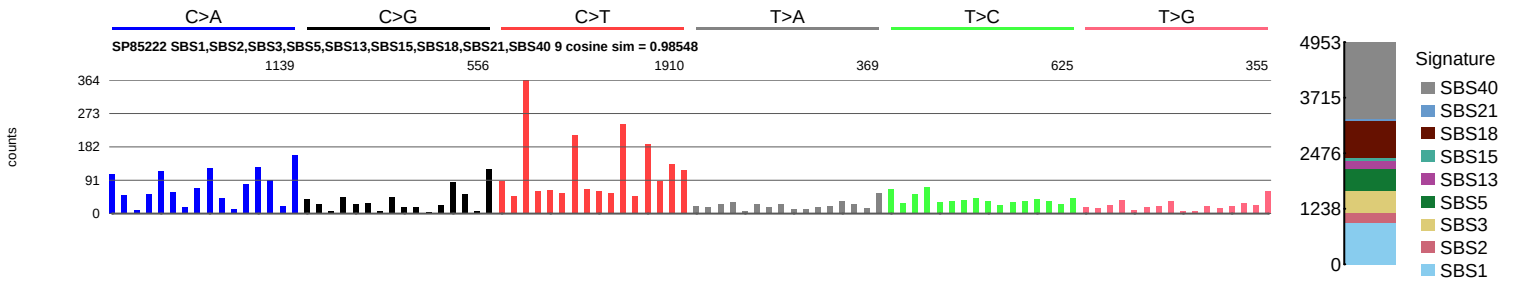
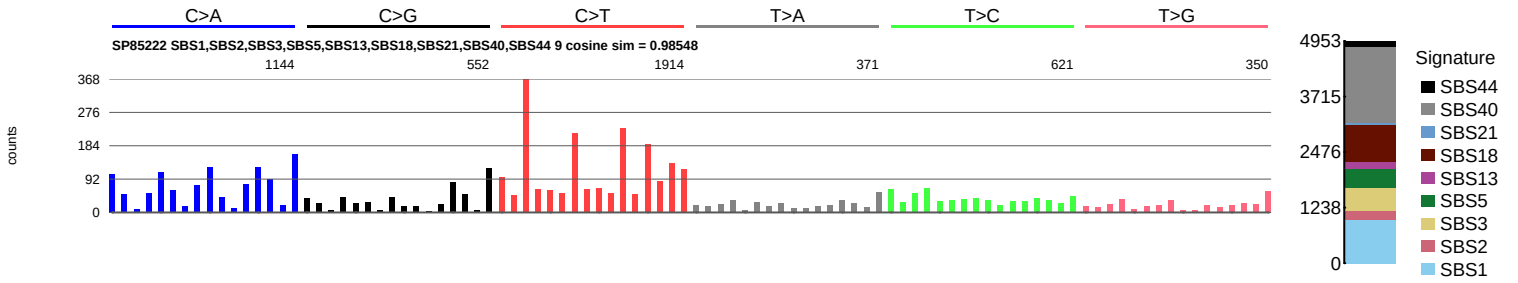
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.9855
1142
552
1909
367
629
355
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98549
1134
513
1825
391
652
432
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98549
1152
569
1873
390
563
407
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40 9 cosine sim = 0.98548
1139
556
1910
369
625
355
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98548
1144
552
1914
371
621
350
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
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SBS21
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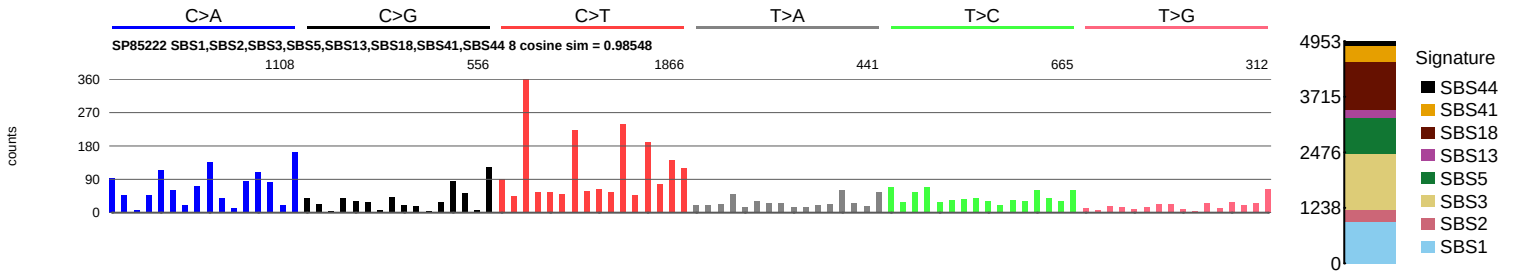
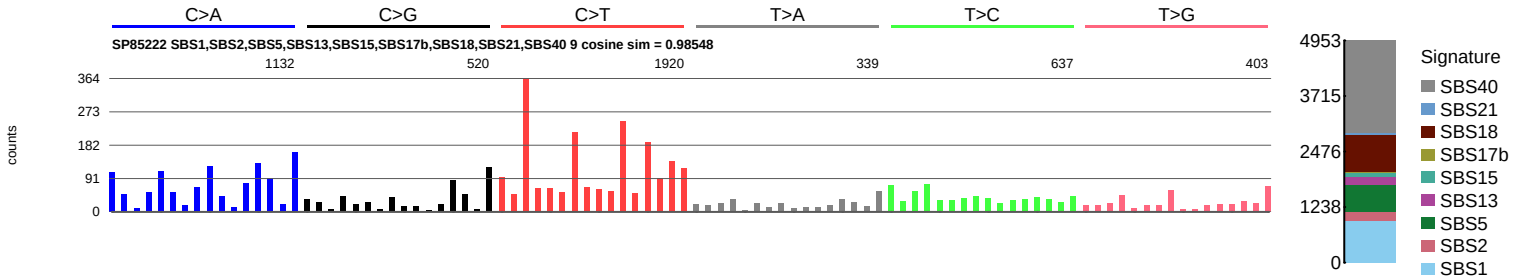
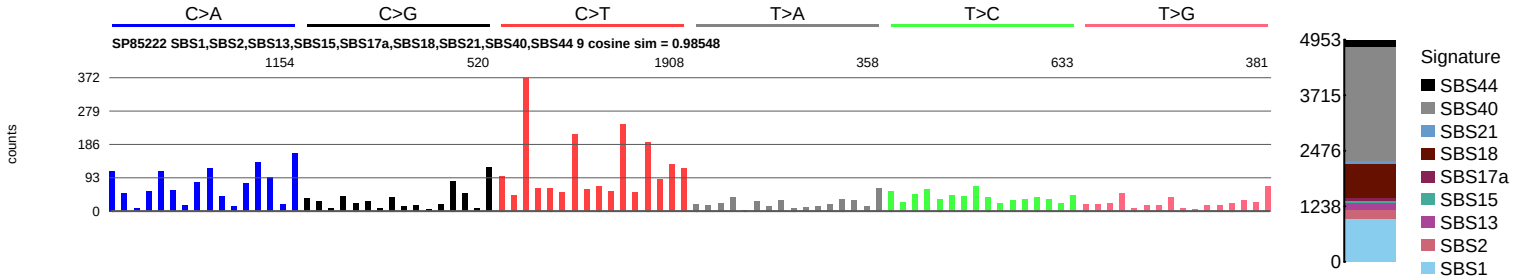
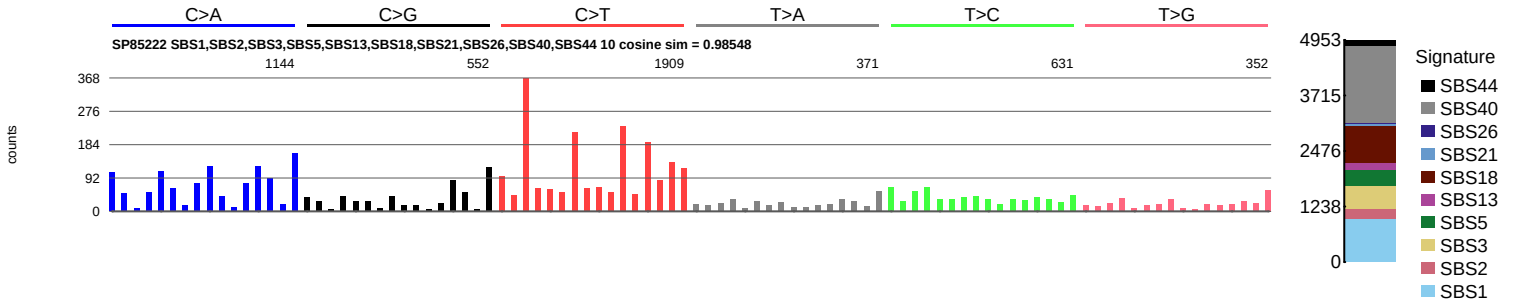
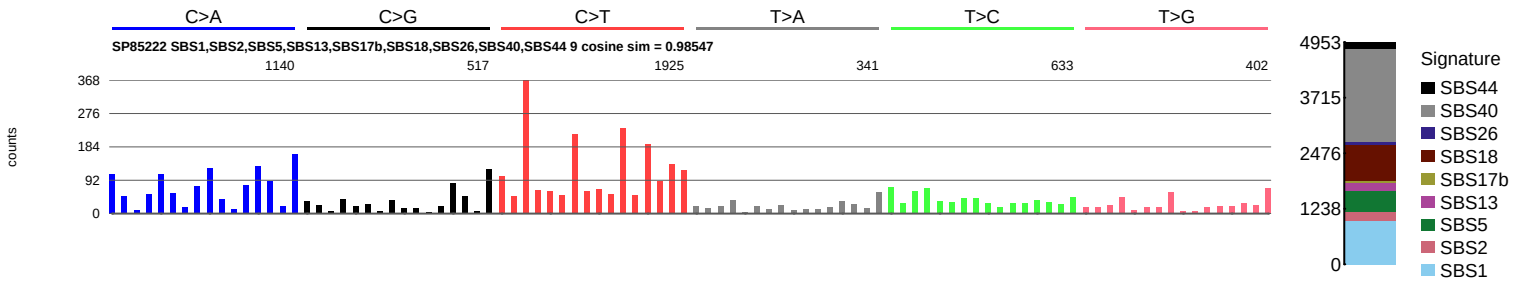
SBS3
SBS2
SBS1

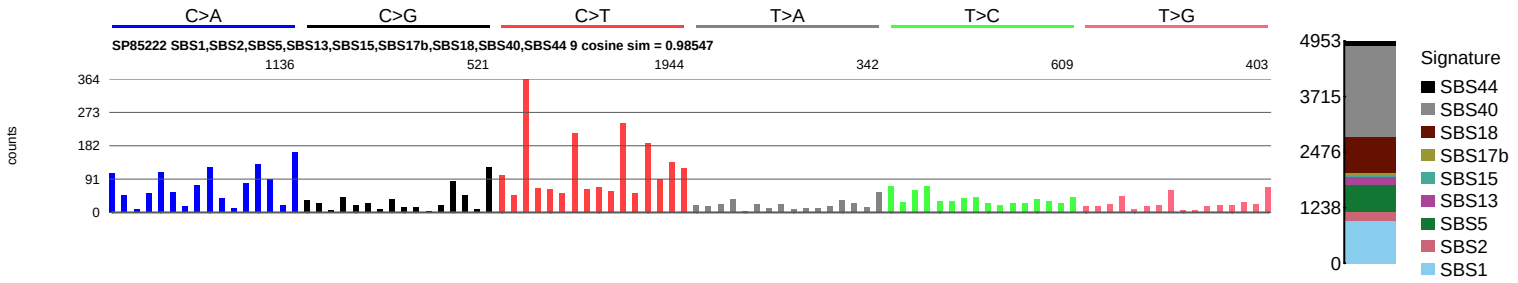

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS41,SBS44 8 cosine sim = 0.98548
1108
556
1866
441
665
312
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98548
1132
520
1920
339
637
403
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98548
1154
520
1908
358
633
381
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98548
1144
552
1909
371
631
352
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98547
1140
517
1925
341
633
402
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98547
1136
521
1944
342
609
403
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
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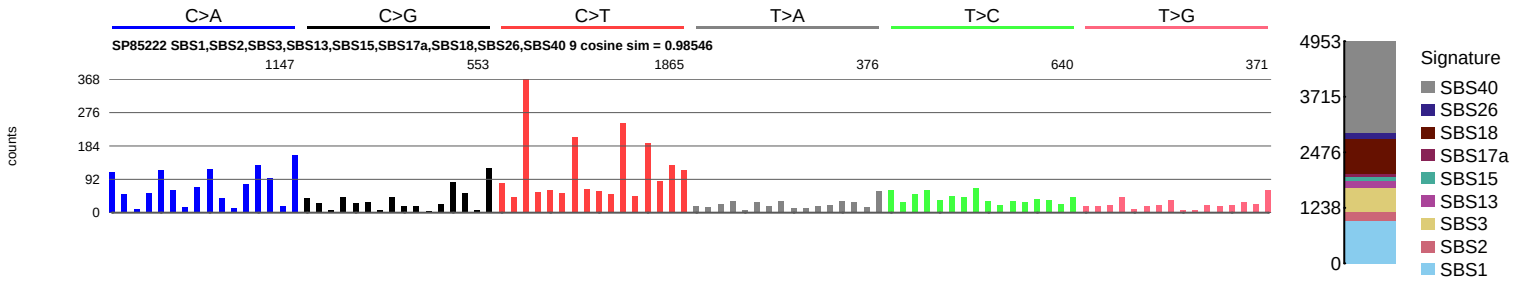
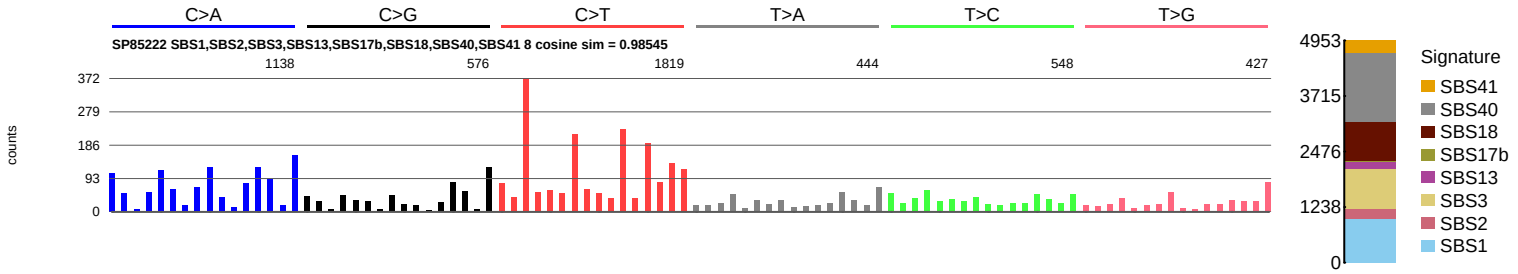
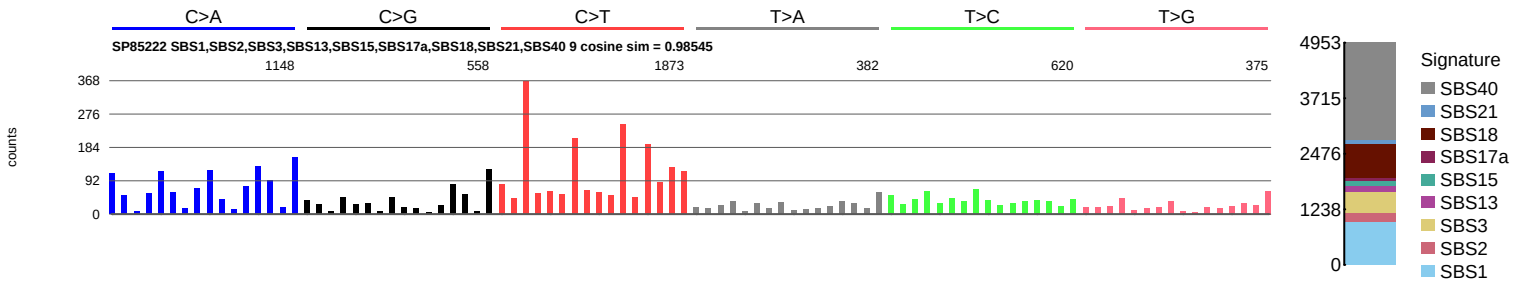
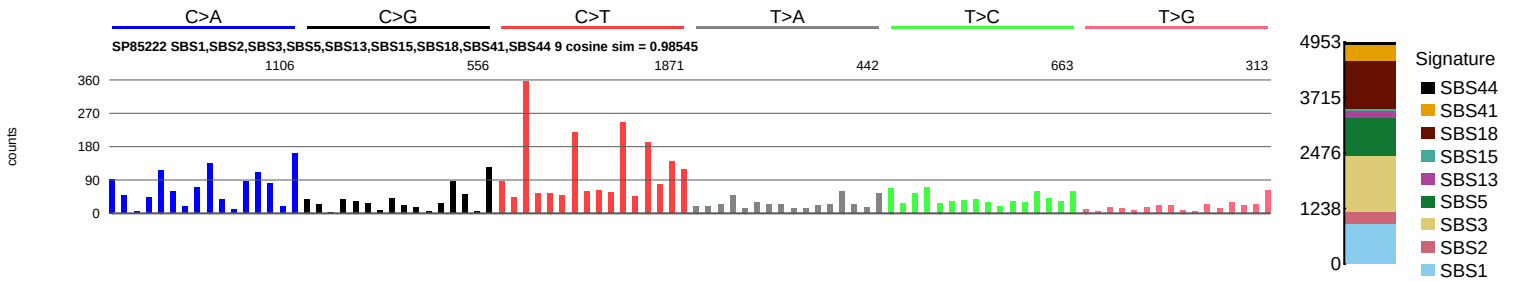
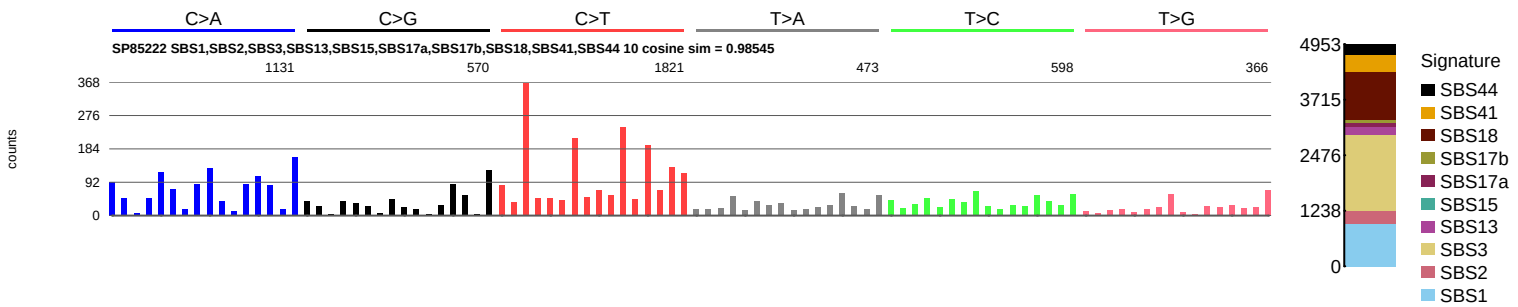
SBS2
SBS1

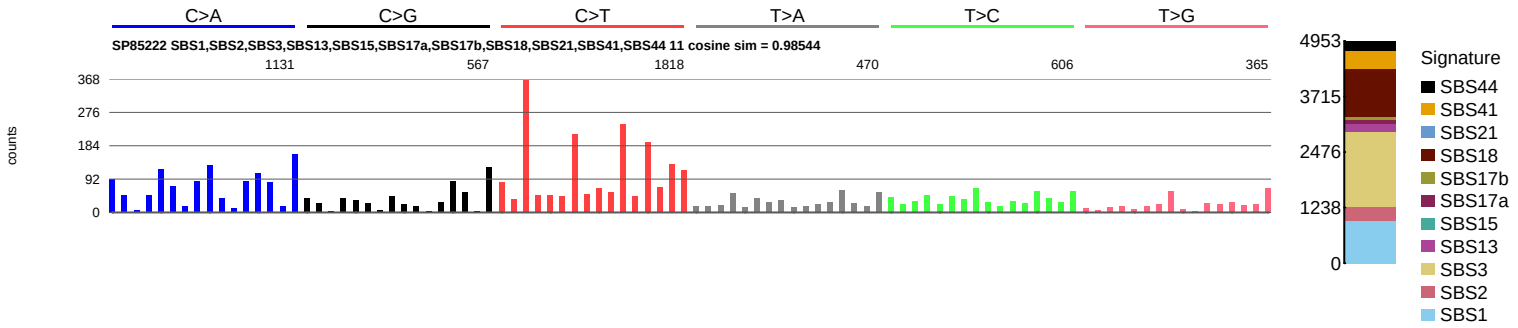

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40 9 cosine sim = 0.98546
1147
553
1865
376
640
371
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

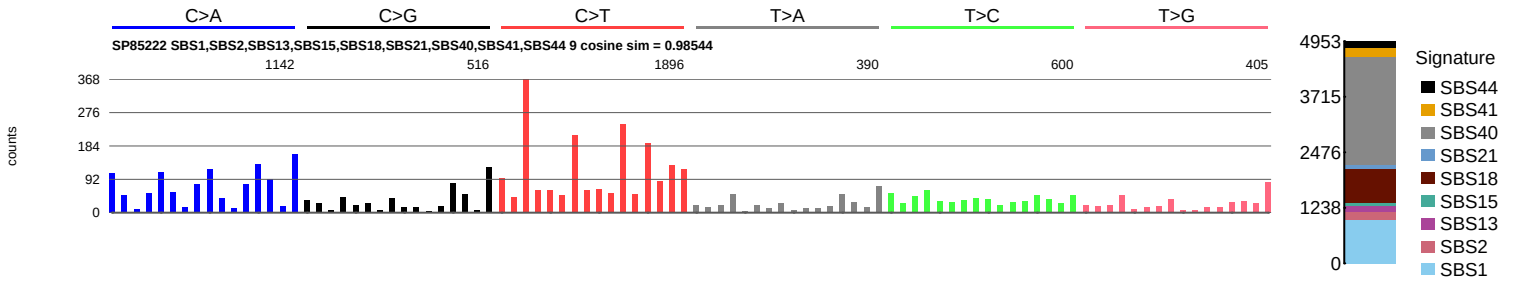

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98545
1138
576
1819
444
548
427
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40 9 cosine sim = 0.98545
1148
558
1873
382
620
375
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS41,SBS44 9 cosine sim = 0.98545
1106
556
1871
442
663
313
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98545
1131
570
1821
473
598
366
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98544
1131
567
1818
470
606
365
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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C>G
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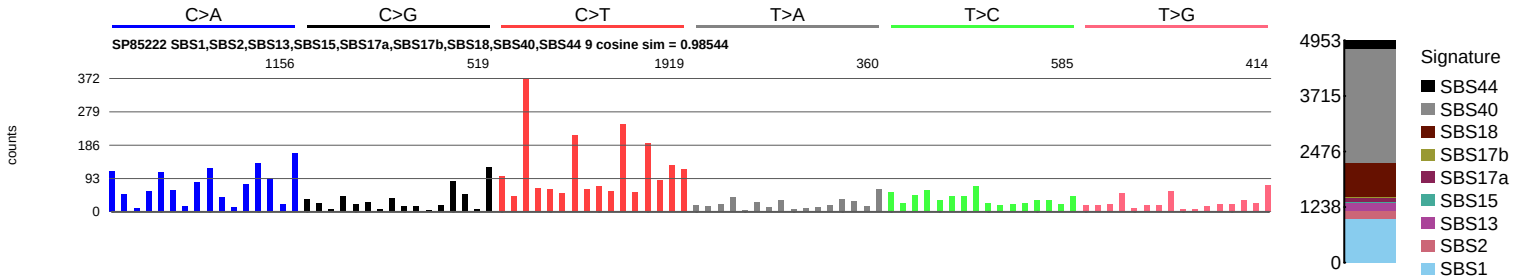
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98544
1142
516
1896
390
600
405
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1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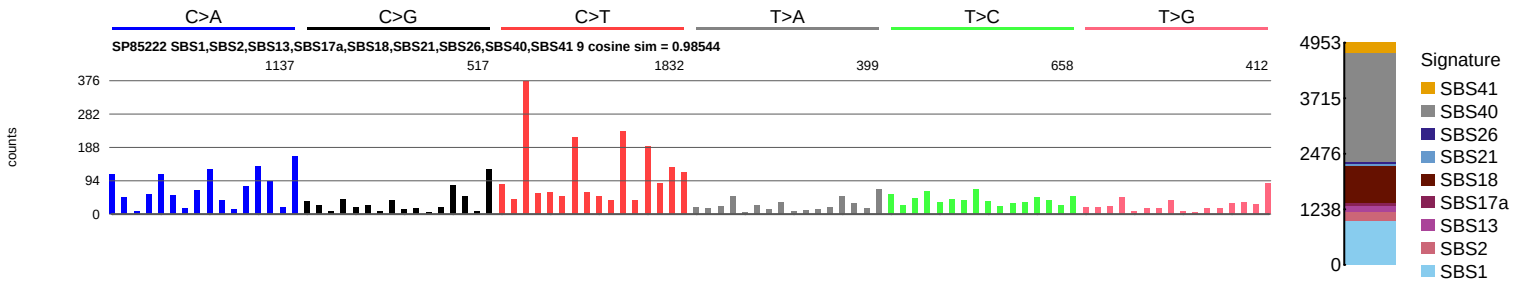
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98544
1156
519
1919
360
585
414
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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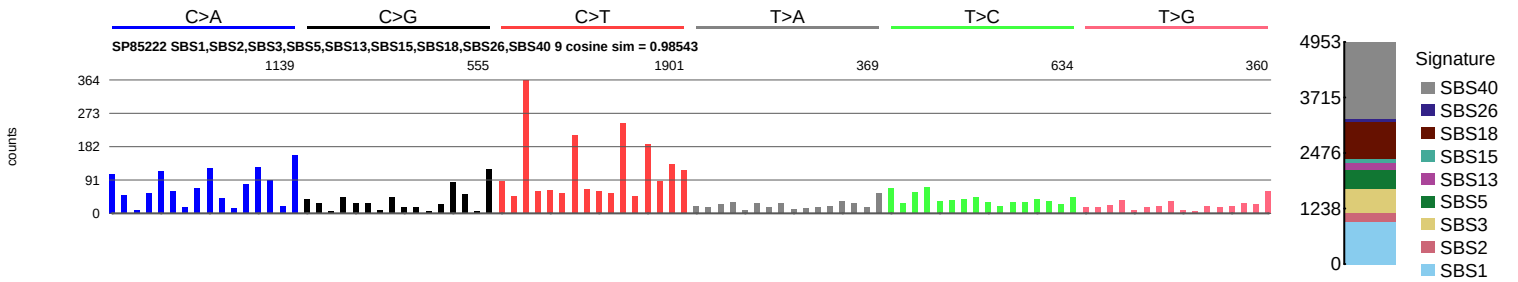
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98544
1137
517
1832
399
658
412
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1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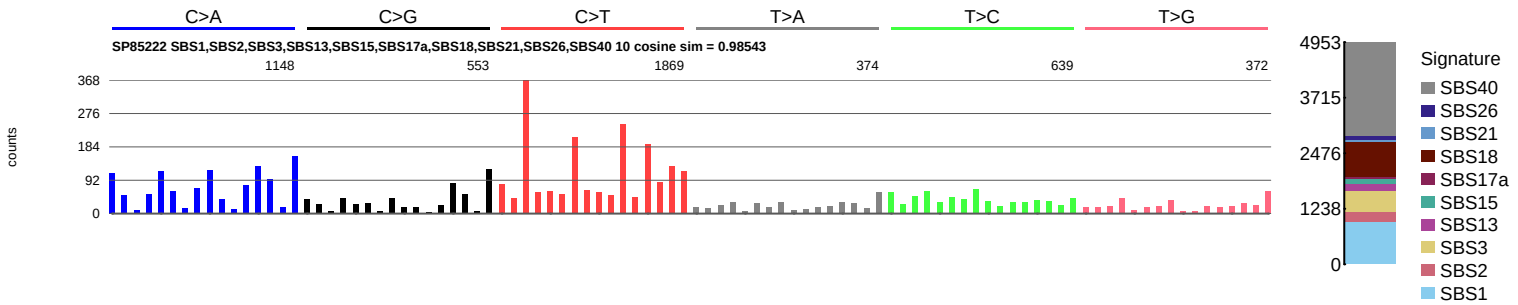
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40 9 cosine sim = 0.98543
1139
555
1901
369
634
360
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98543
1148
553
1869
374
639
372
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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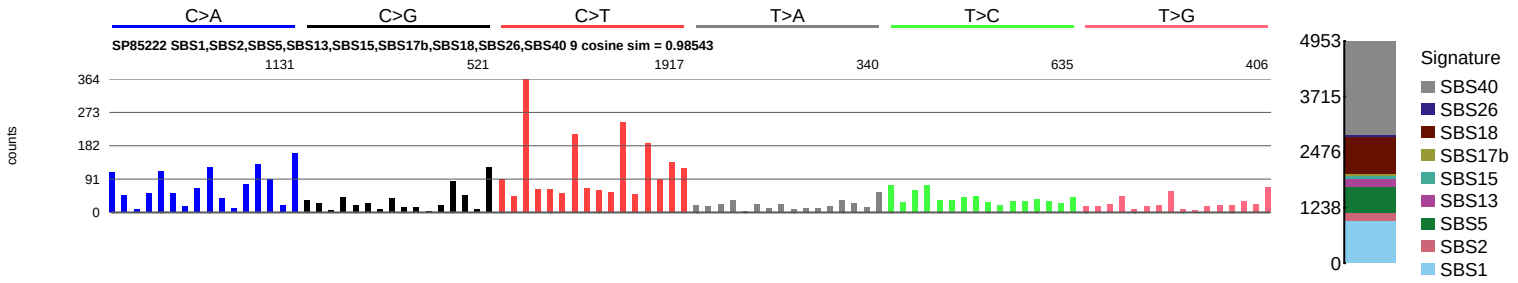
SBS3
SBS2
SBS1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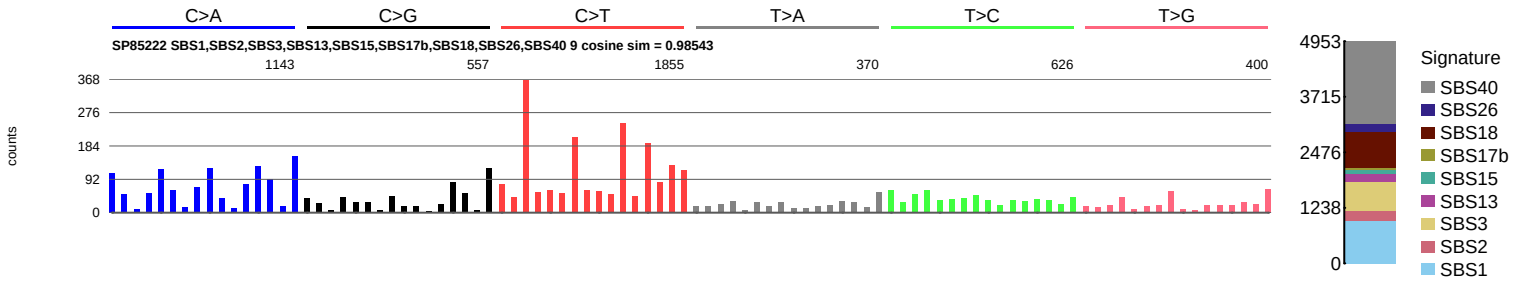
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98543
1131
521
1917
340
635
406
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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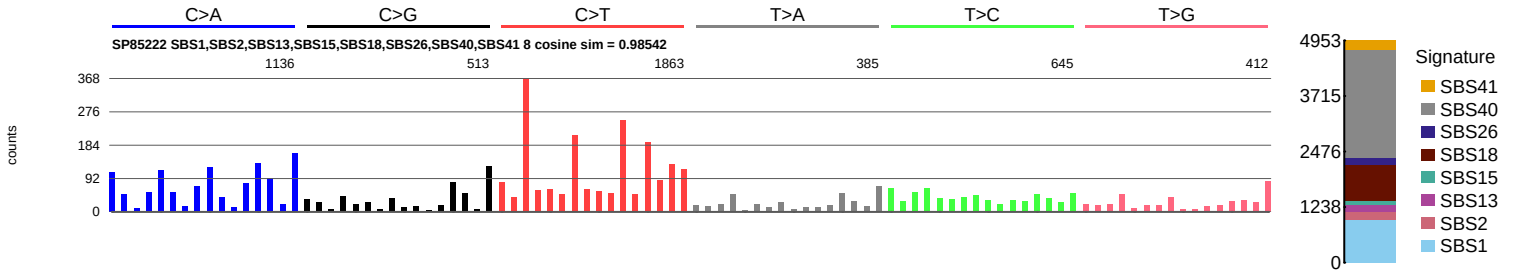
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98543
1143
557
1855
370
626
400
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1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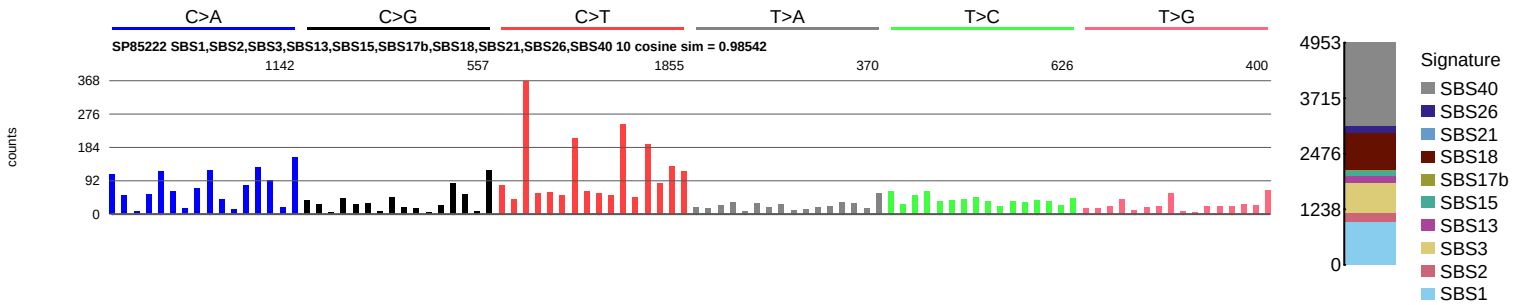
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98542
1136
513
1863
385
645
412
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS13
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98542
1142
557
1855
370
626
400
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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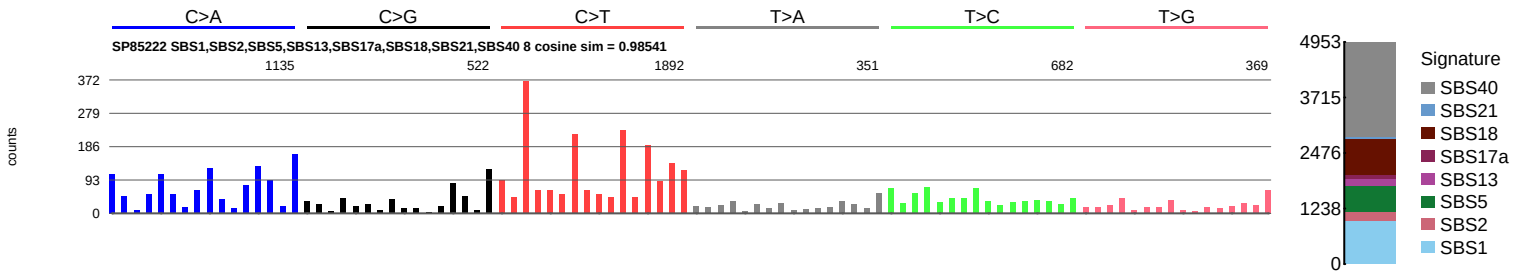
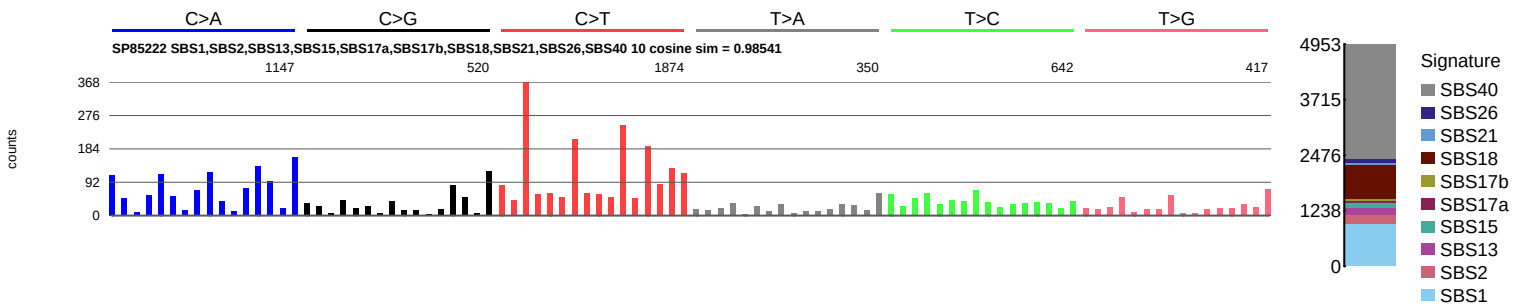
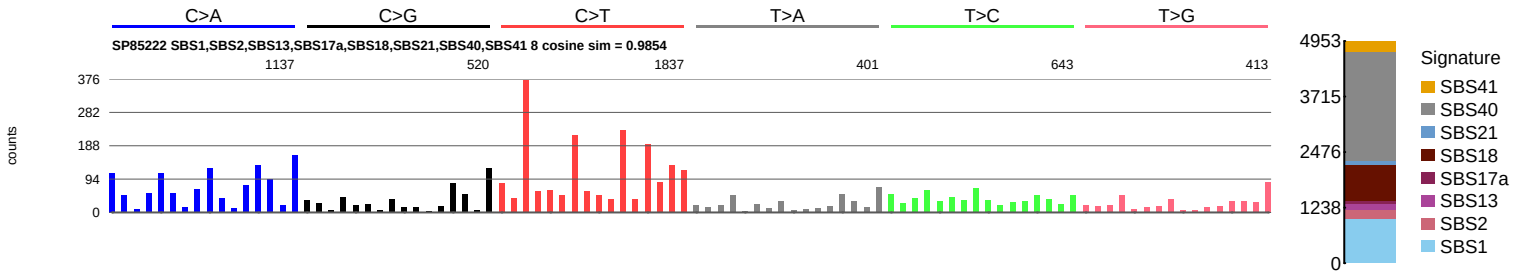
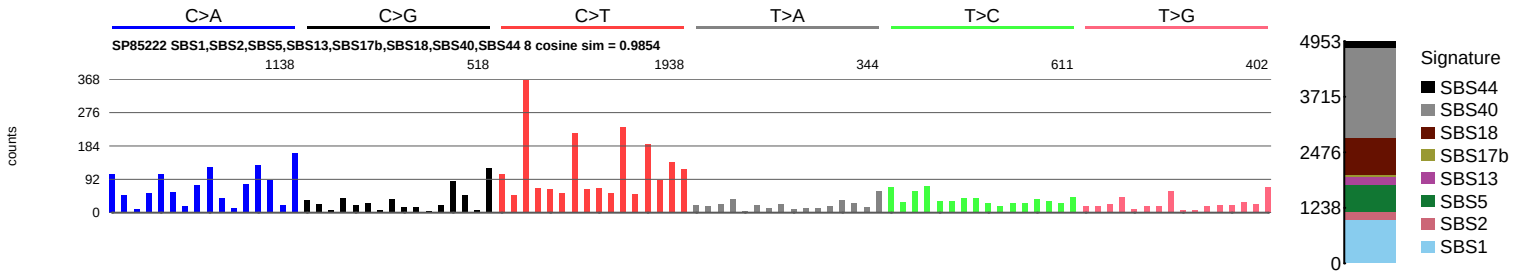
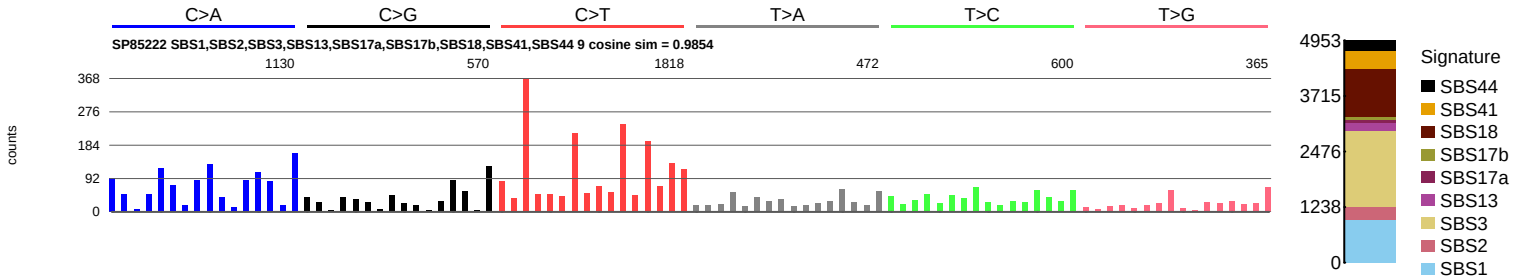
SBS2
SBS1

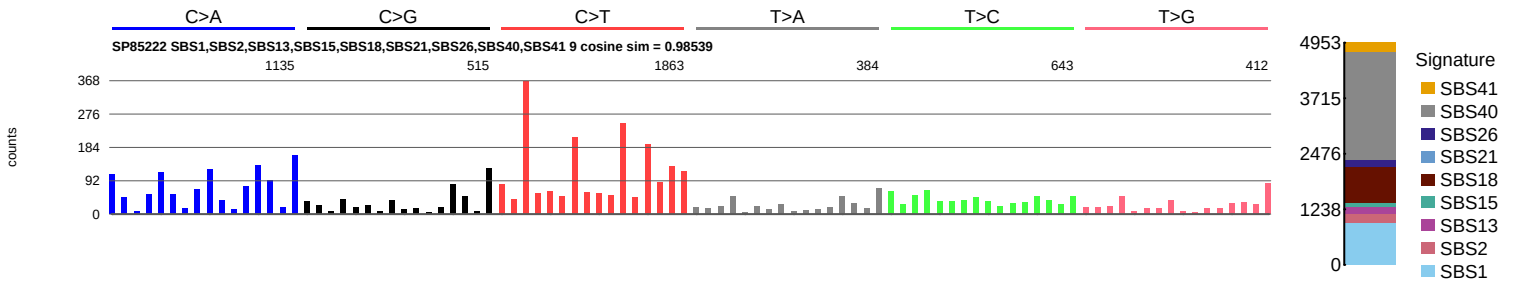

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40 8 cosine sim = 0.98541
1135
522
1892
351
682
369
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98541
1147
520
1874
350
642
417
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.9854
1137
520
1837
401
643
413
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.9854
1138
518
1938
344
611
402
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.9854
1130
570
1818
472
600
365
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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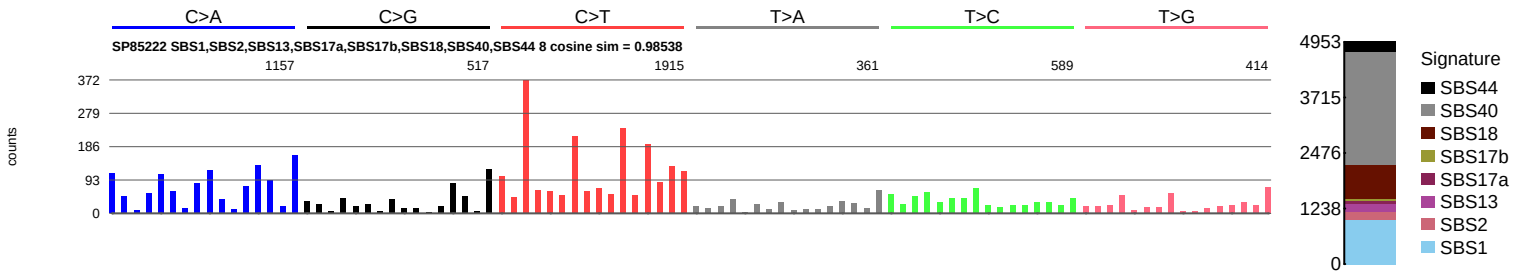
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98539
1135
515
1863
384
643
412
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1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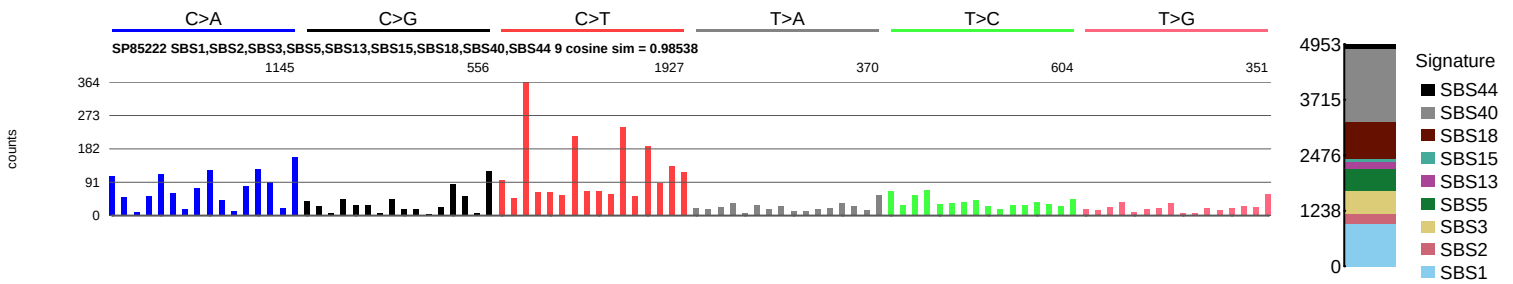
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98538
1157
517
1915
361
589
414
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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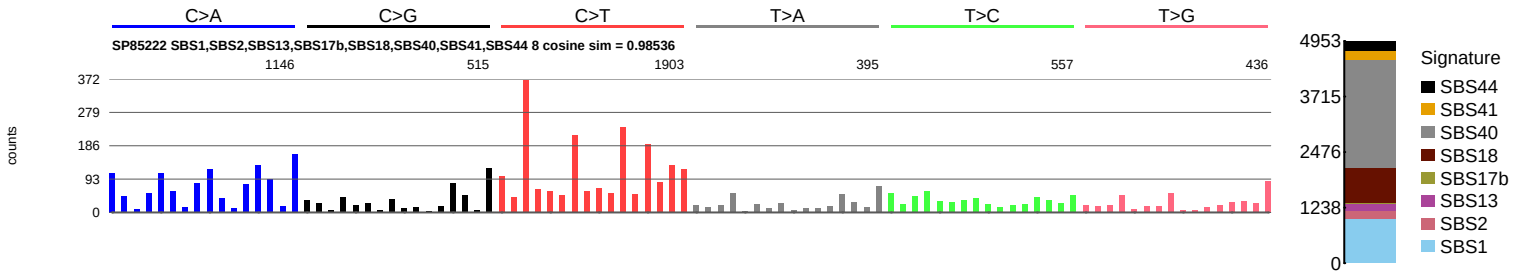
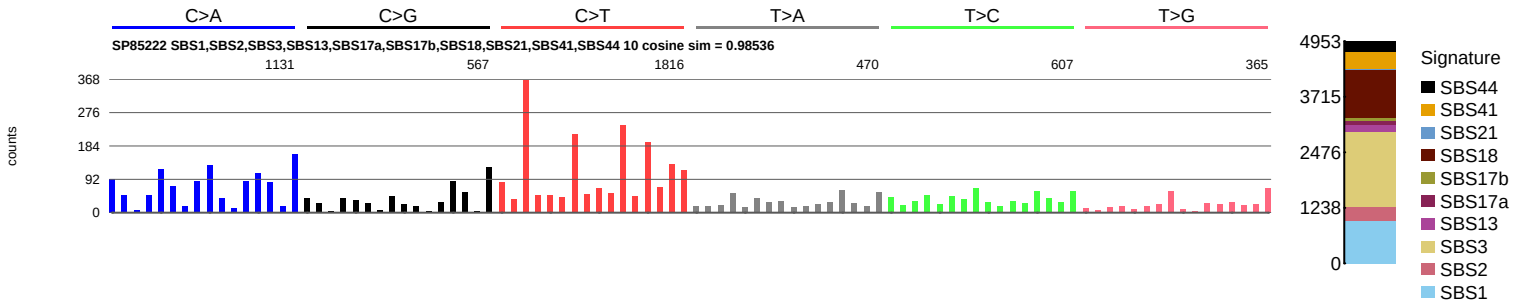
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS40,SBS44 9 cosine sim = 0.98538
1145
556
1927
370
604
351
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98536
1146
515
1903
395
557
436
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98536
1131
567
1816
470
607
365
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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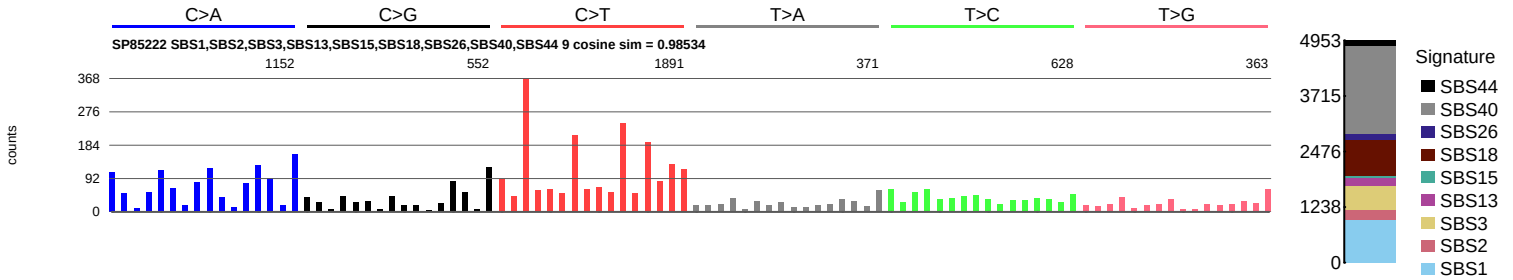
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98534
1152
552
1891
371
628
363
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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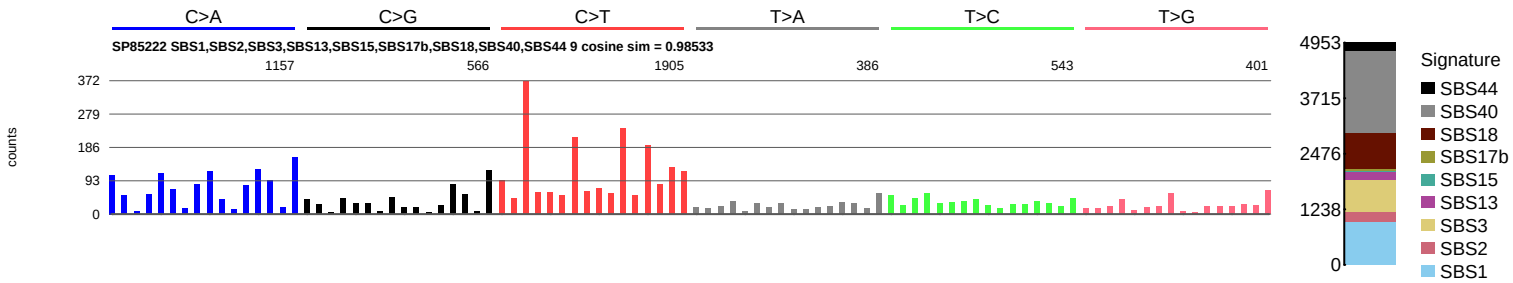
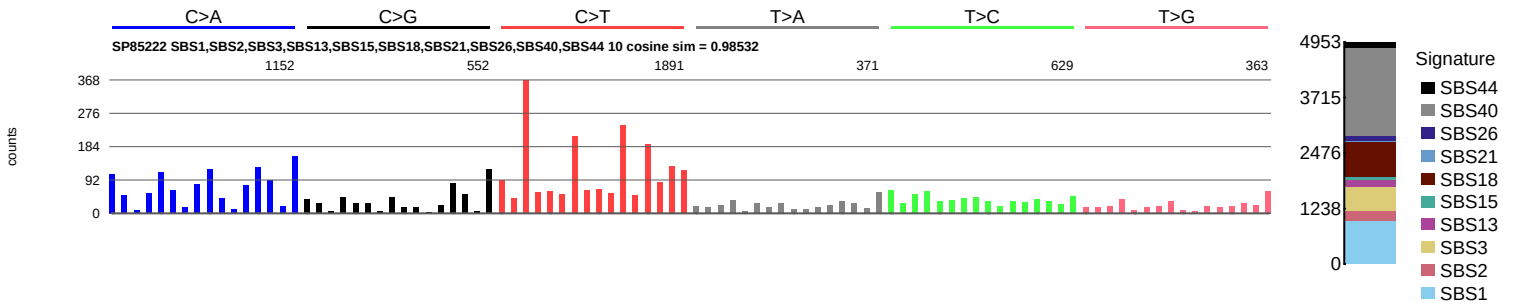
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98533
1157
566
1905
386
543
401
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98532
1152
552
1891
371
629
363
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
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SBS26
SBS21
SBS18
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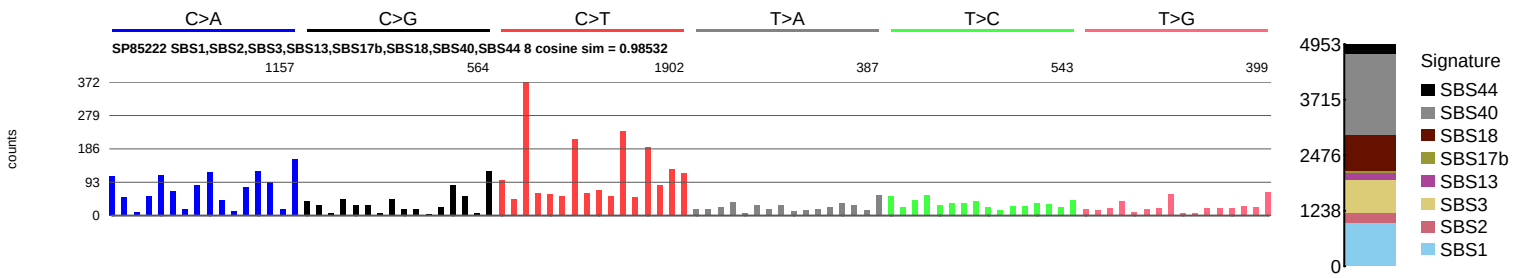
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98532
1157
564
1902
387
543
399
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
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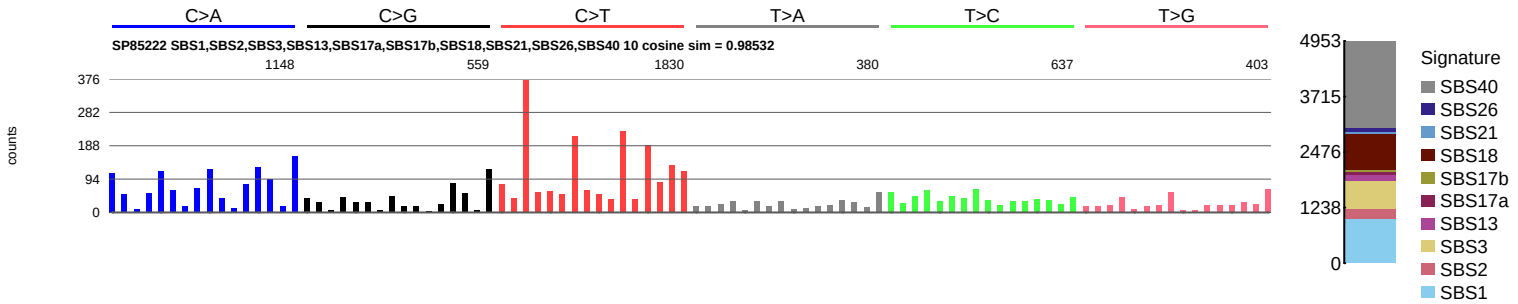
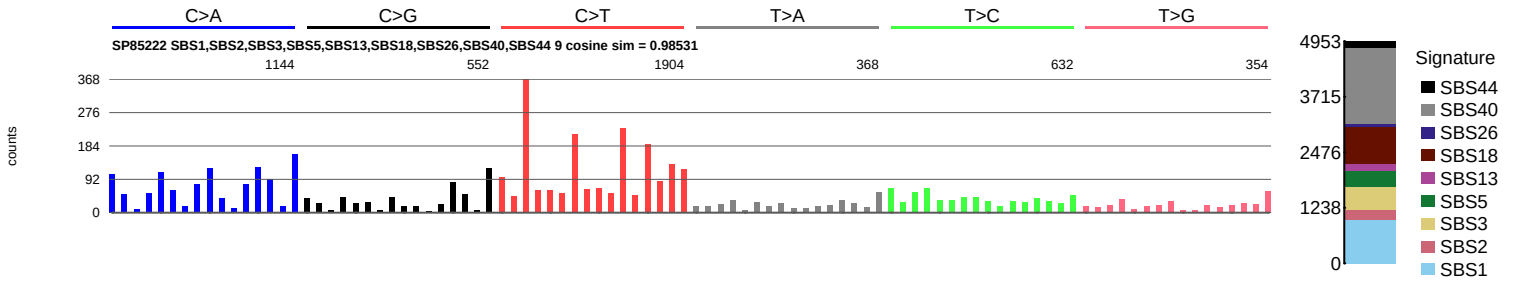
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98532
1148
559
1830
380
637
403
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98531
1144
552
1904
368
632
354
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS13
SBS5
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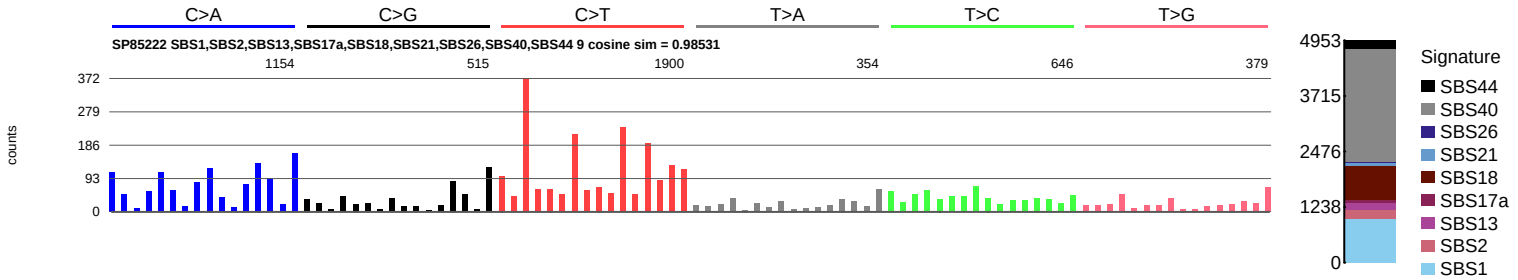
SBS2
SBS1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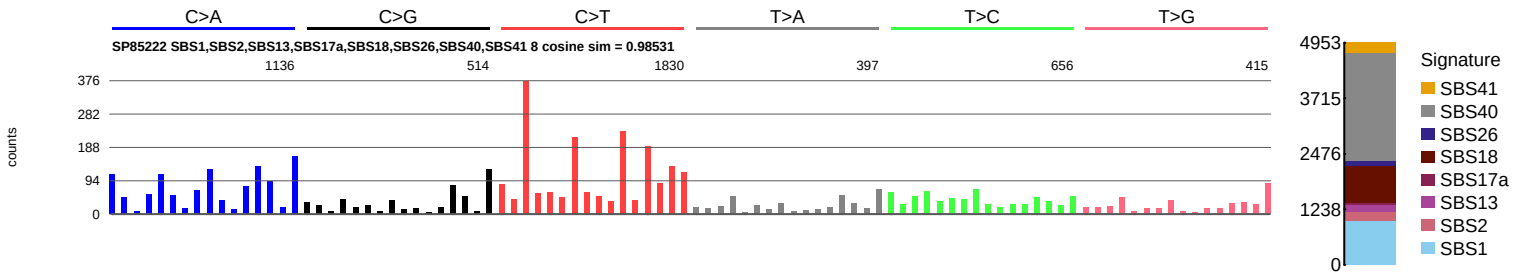
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98531
1154
515
1900
354
646
379
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1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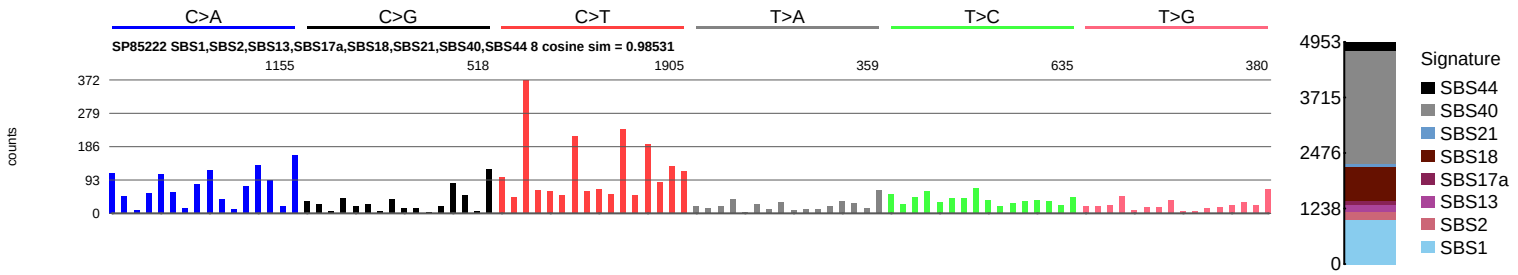
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98531
1136
514
1830
397
656
415
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS2
SBS1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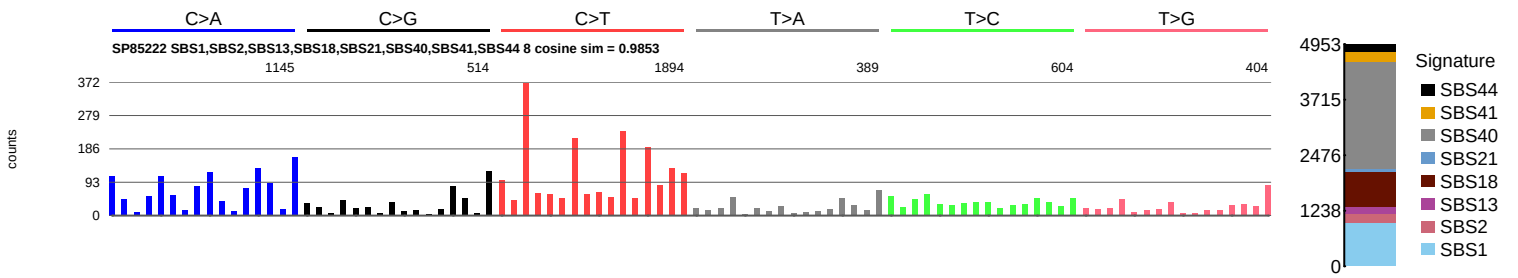
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98531
1155
518
1905
359
635
380
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1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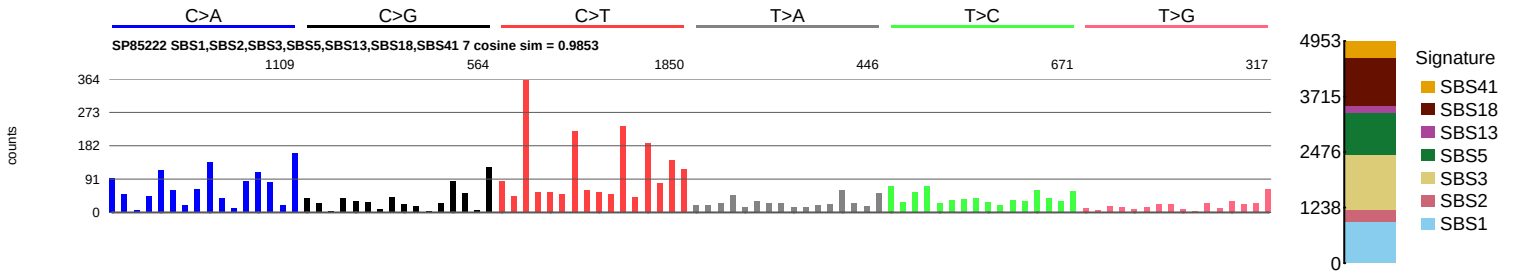
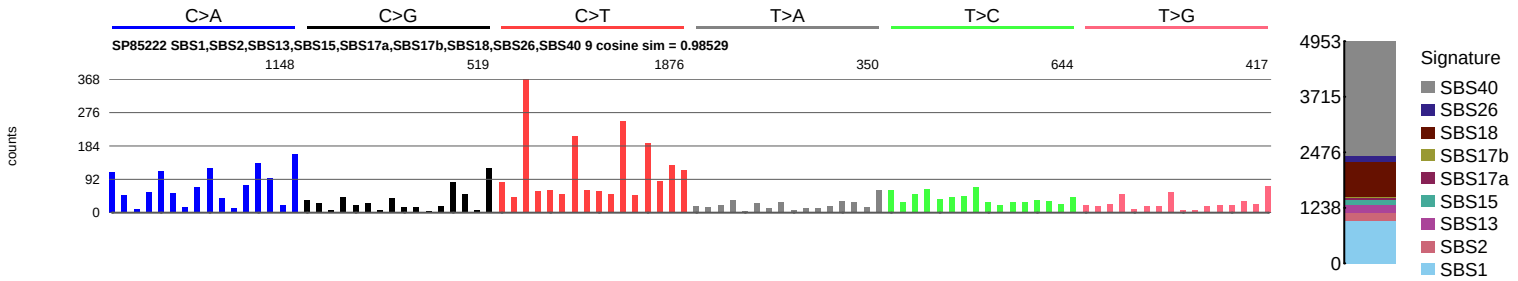
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.9853
1145
514
1894
389
604
404
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS41 7 cosine sim = 0.9853
1109
564
1850
446
671
317
364
273
182
91
0
counts
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1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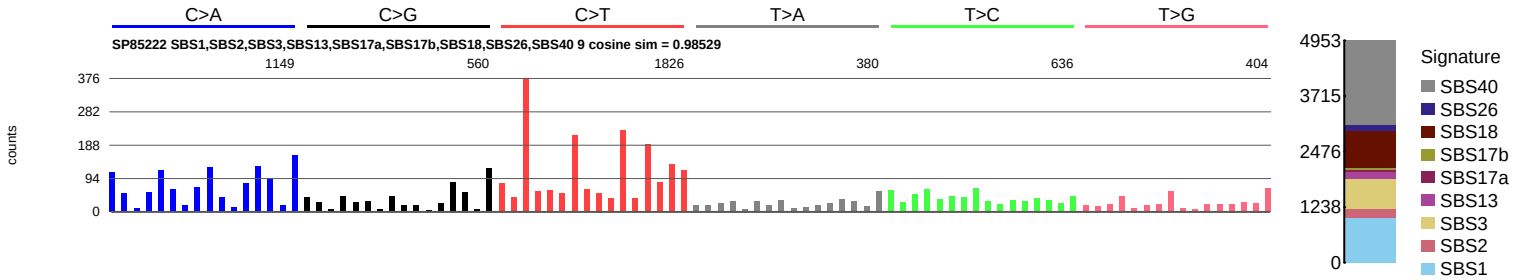
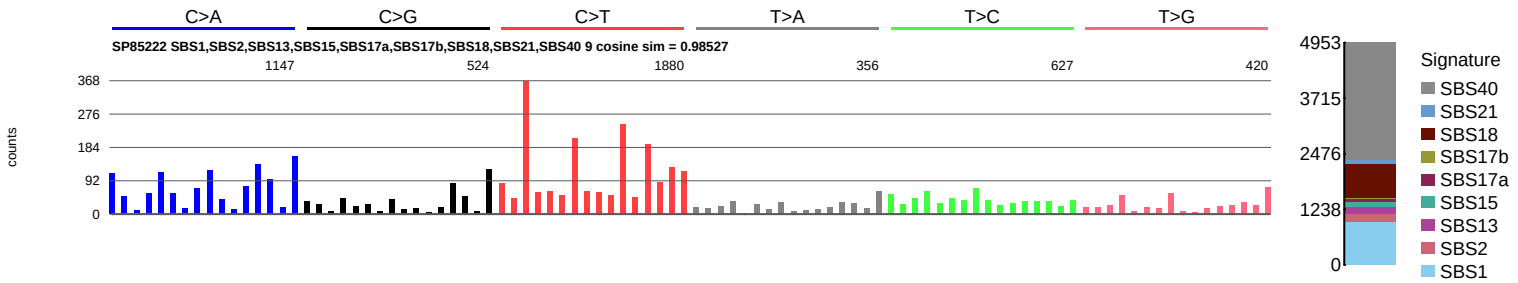
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98529
1148
519
1876
350
644
417
368
276
184
92
0
counts
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98529
1149
560
1826
380
636
404
376
282
188
94
0
counts
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98527
1147
524
1880
356
627
420
368
276
184
92
0
counts
4953
3715
2476
1238
0
Signature
SBS40
SBS21
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SBS17b
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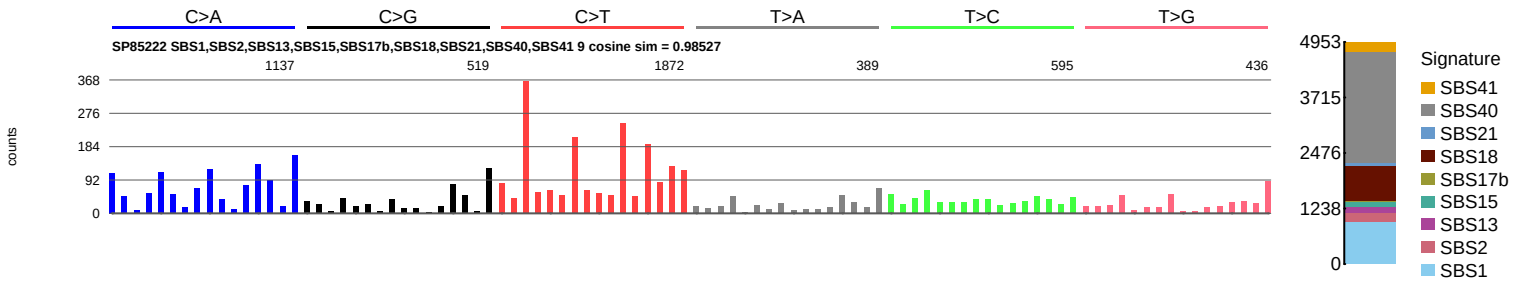
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98527
1137
519
1872
389
595
436
368
276
184
92
0
counts
4953
3715
2476
1238
0
Signature
SBS41
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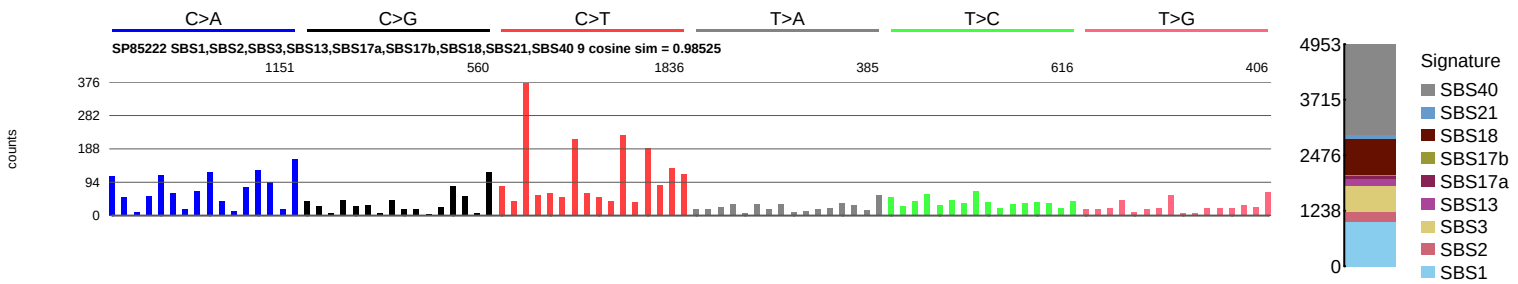
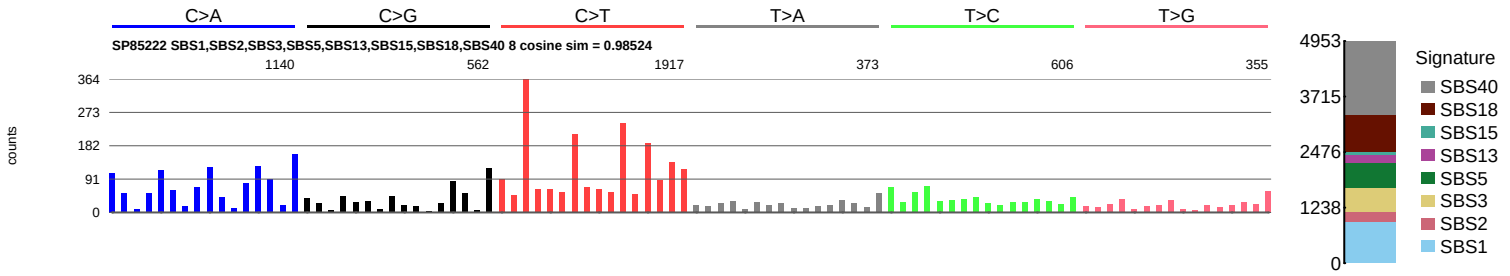
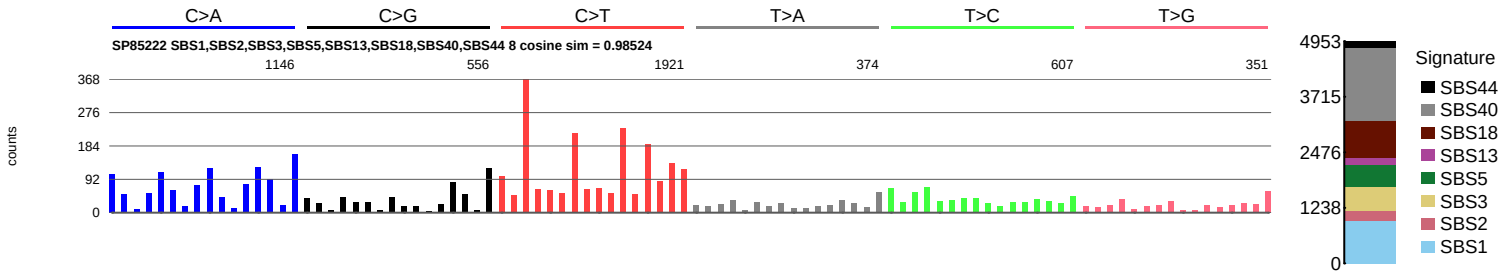
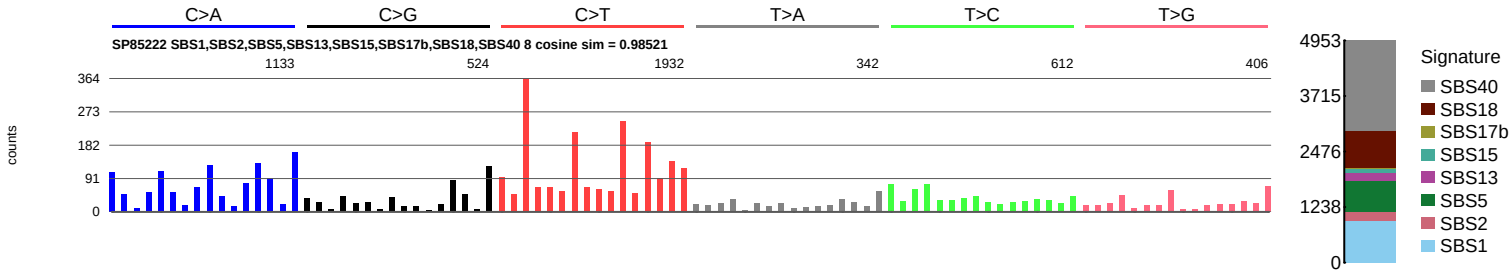
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98525
1151
560
1836
385
616
406
376
282
188
94
0
counts
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS40 8 cosine sim = 0.98524
1140
562
1917
373
606
355
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40,SBS44 8 cosine sim = 0.98524
1146
556
1921
374
607
351
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.98521
1133
524
1932
342
612
406
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
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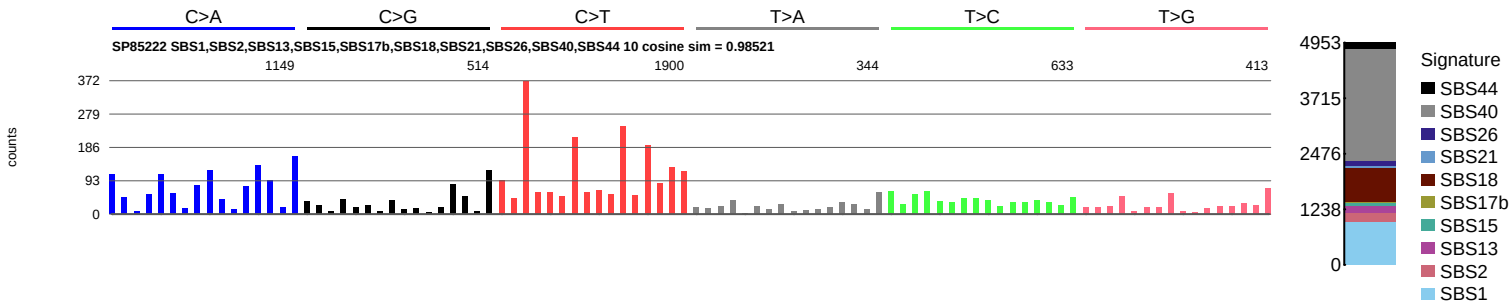
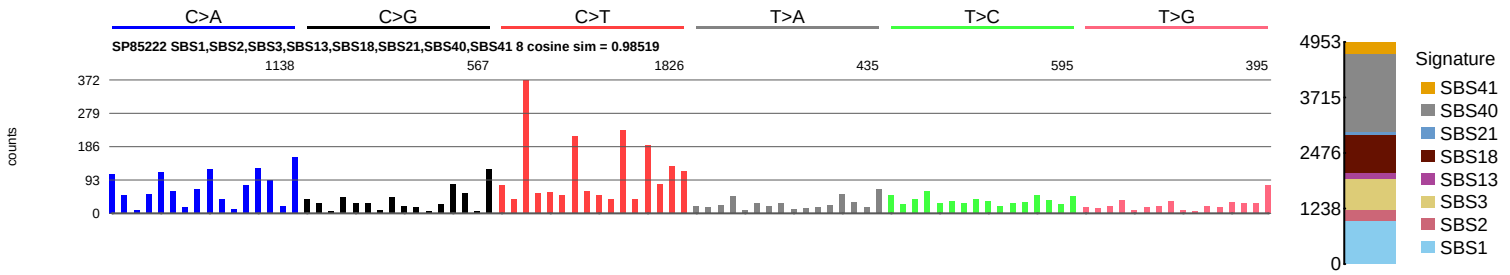
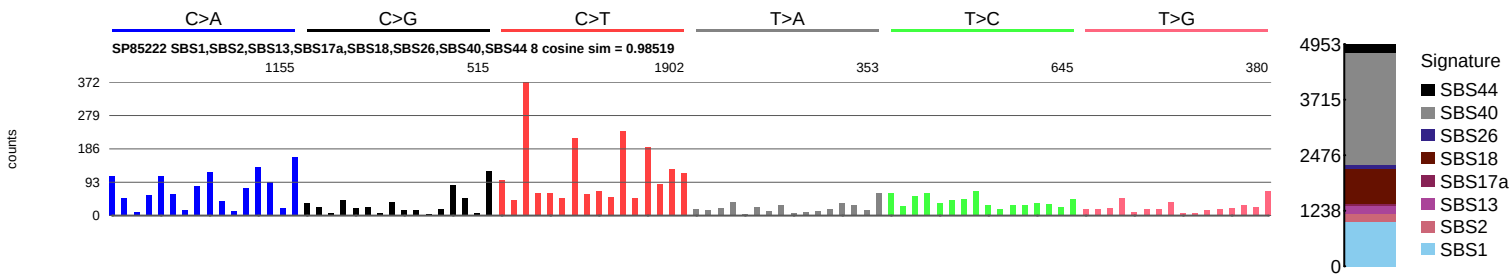
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98521
1149
514
1900
344
633
413
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98519
1138
567
1826
435
595
395
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98519
1155
515
1902
353
645
380
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
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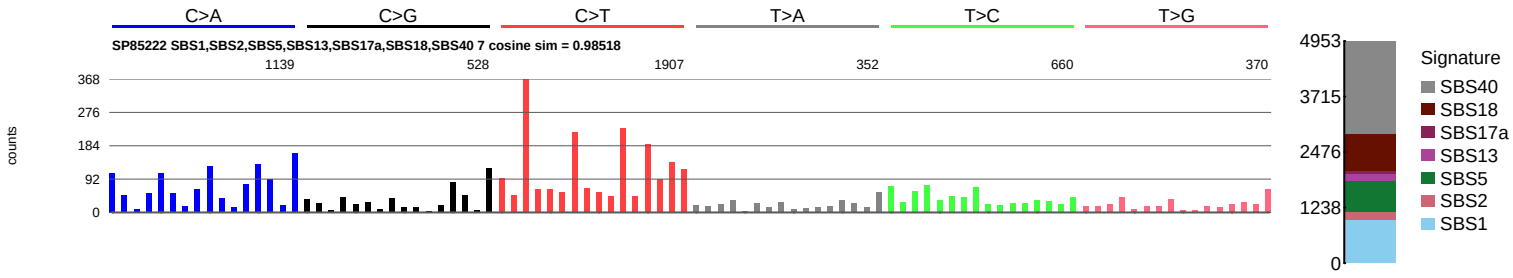
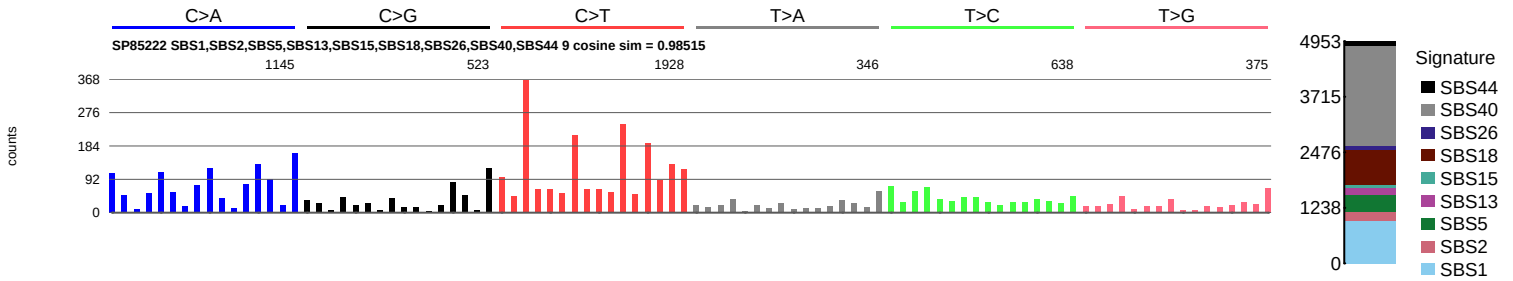
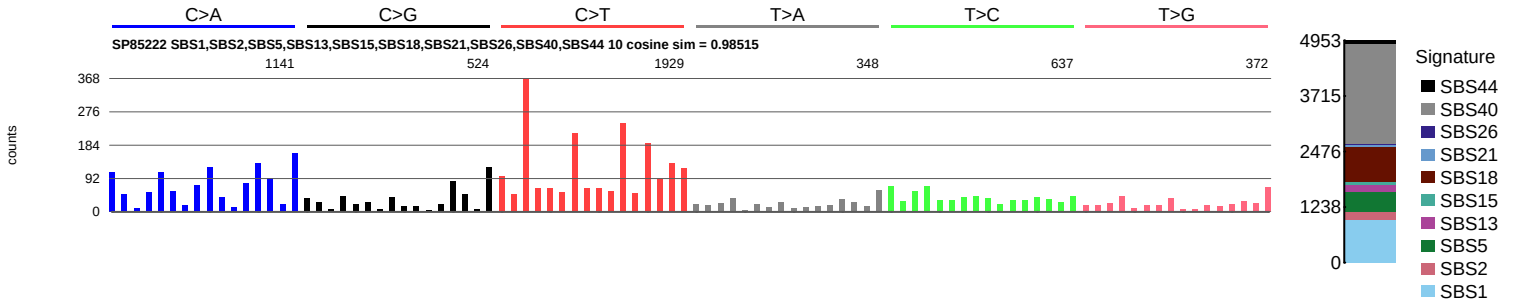
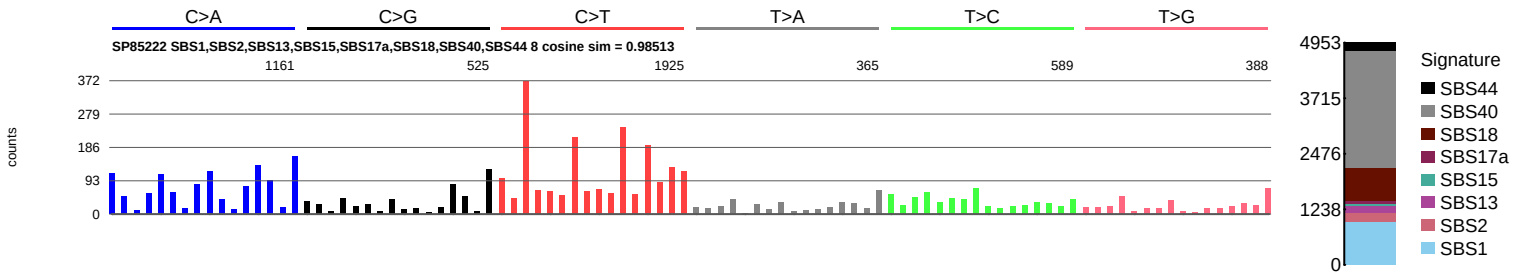
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40 7 cosine sim = 0.98518
1139
528
1907
352
660
370
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98515
1145
523
1928
346
638
375
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98515
1141
524
1929
348
637
372
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.98513
1161
525
1925
365
589
388
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
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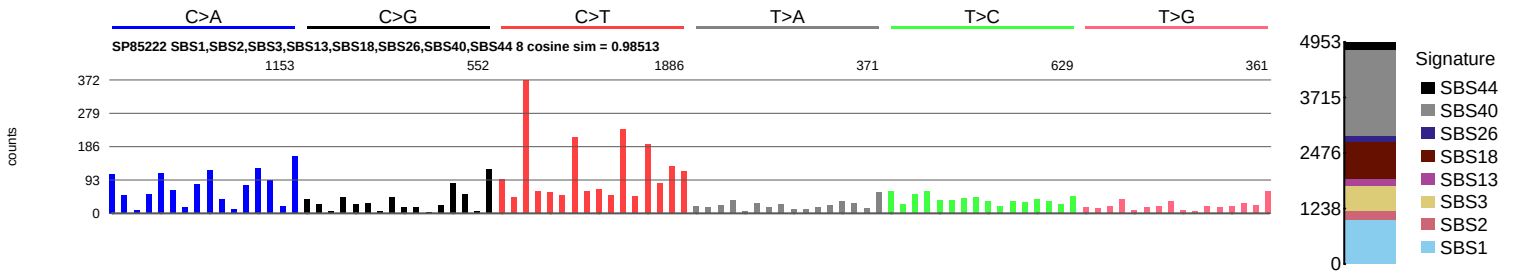
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98513
1153
552
1886
371
629
361
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
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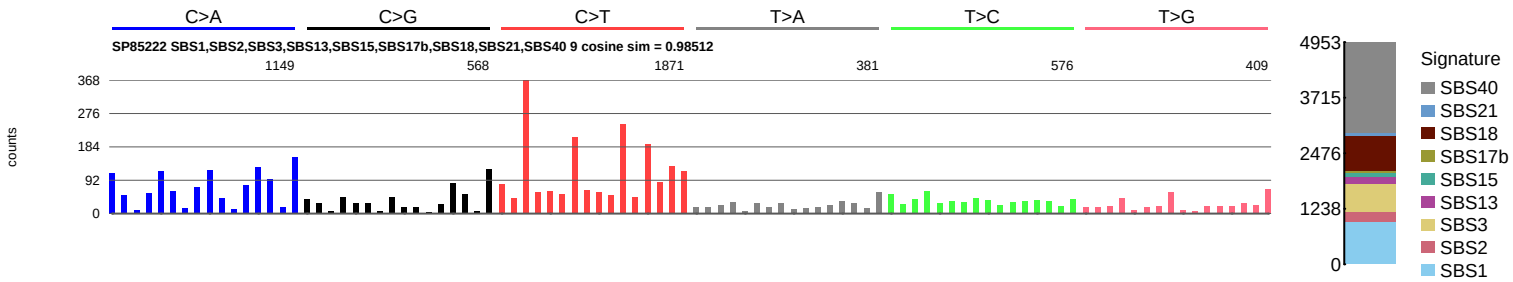
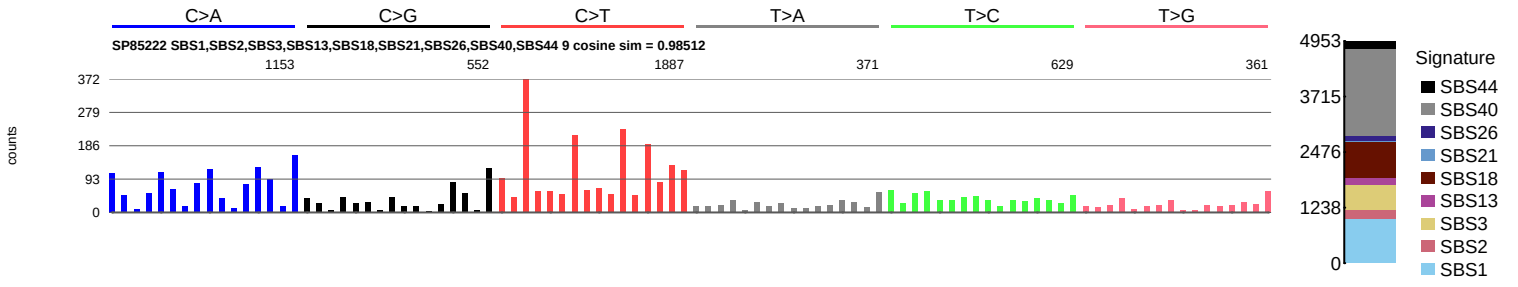
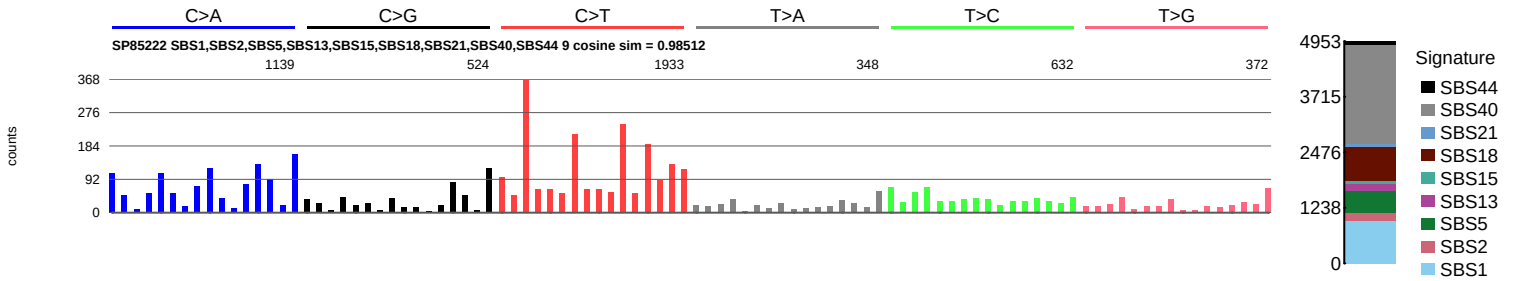
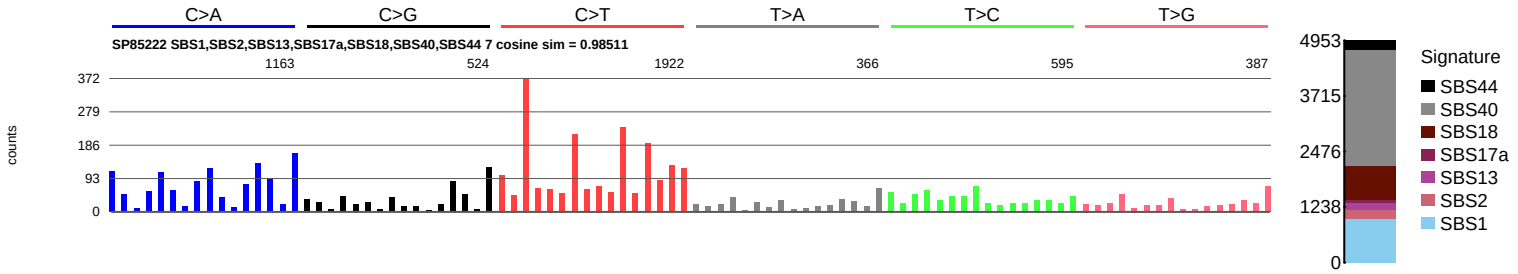
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98512
1149
568
1871
381
576
409
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98512
1153
552
1887
371
629
361
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98512
1139
524
1933
348
632
372
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS40,SBS44 7 cosine sim = 0.98511
1163
524
1922
366
595
387
counts
372
279
186
93
0
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SBS44
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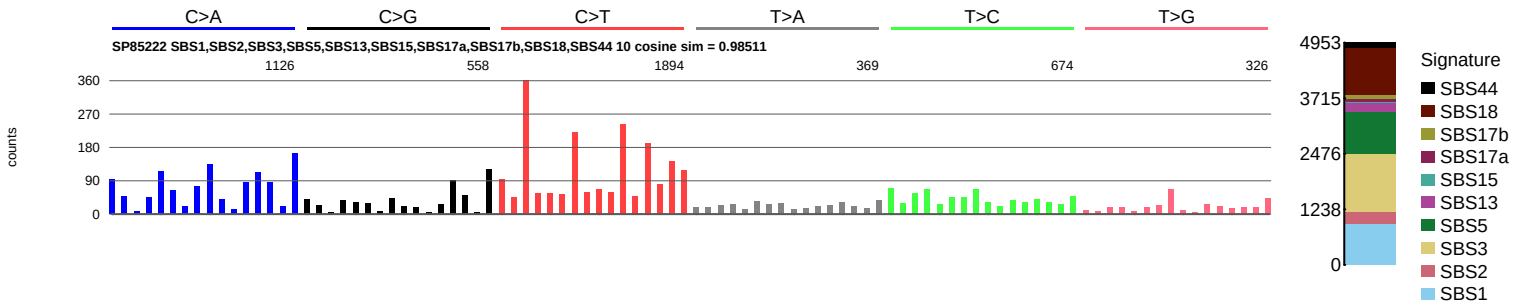
SBS2
SBS1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T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 10 cosine sim = 0.98511
1126
558
1894
369
674
326
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
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SBS17b
SBS17a
SBS15
SBS13
SBS5
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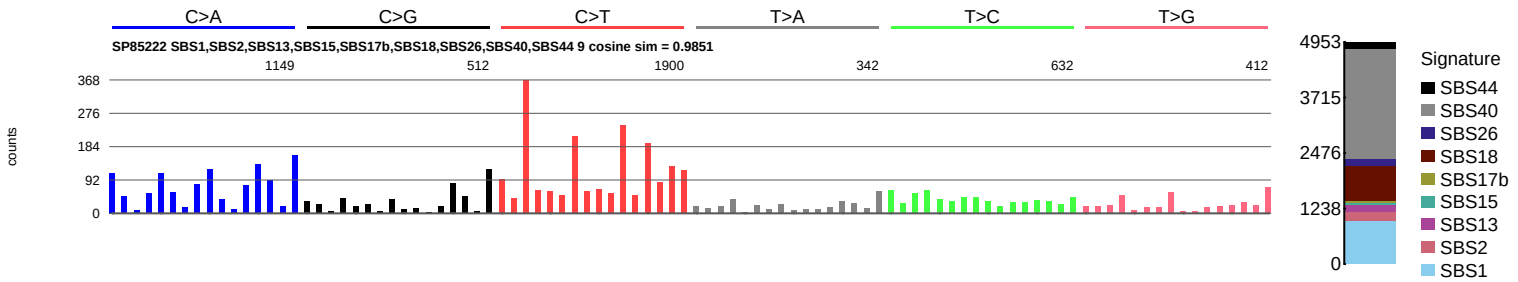
SBS2
SBS1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T>A
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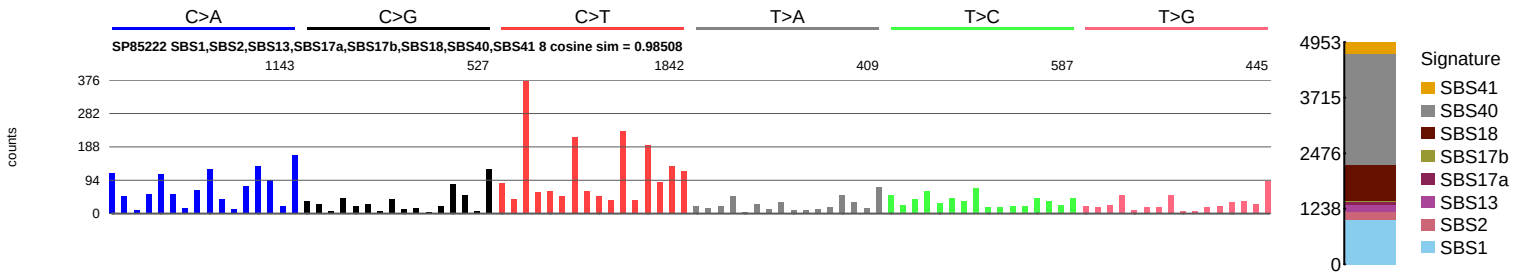
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.9851
1149
512
1900
342
632
412
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1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C>T
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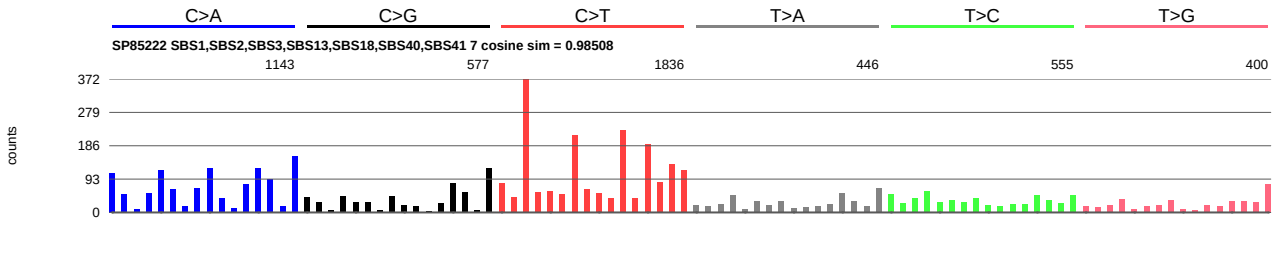
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98508
1143
527
1842
409
587
445
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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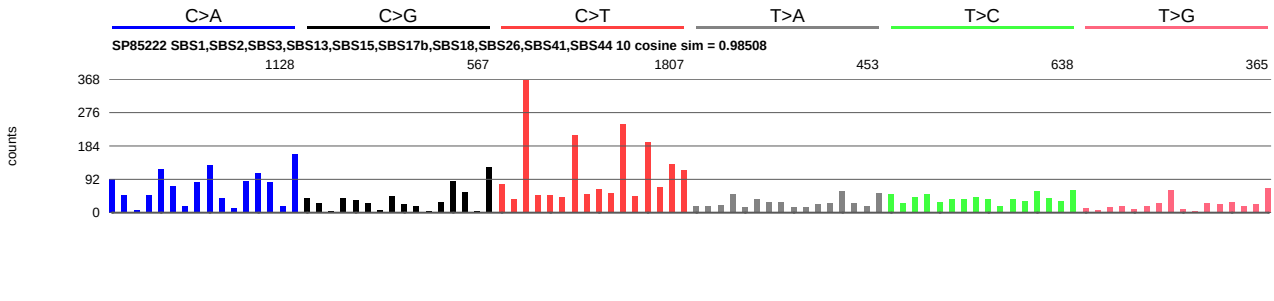
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS40,SBS41 7 cosine sim = 0.98508
1143
577
1836
446
555
400
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS13
SBS3
SBS2
SBS1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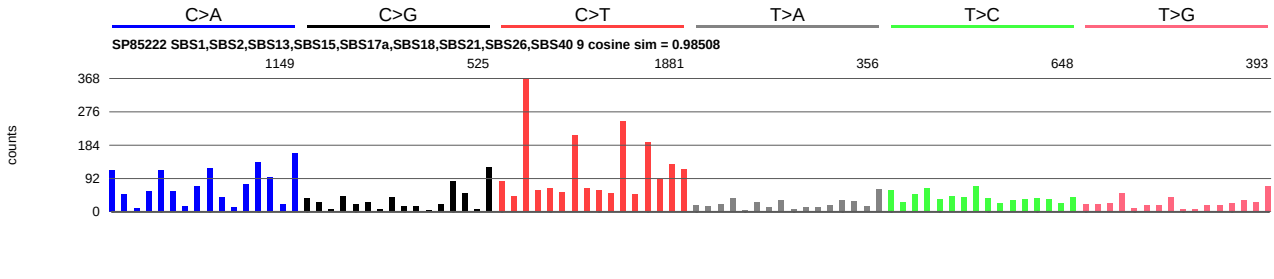
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98508
1128
567
1807
453
638
365
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1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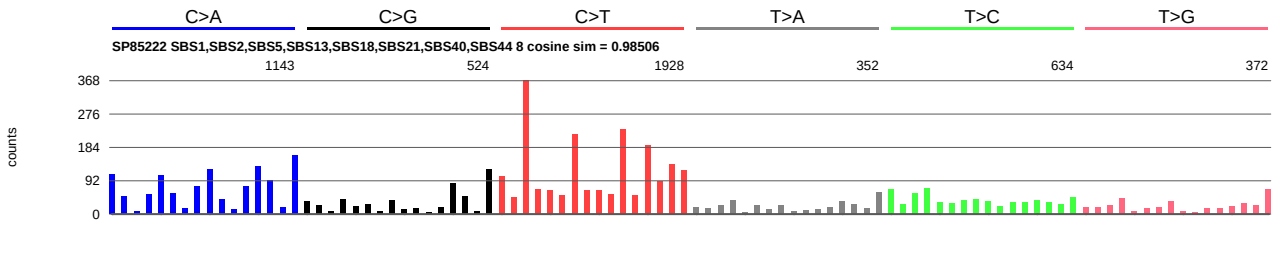
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98508
1149
525
1881
356
648
393
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1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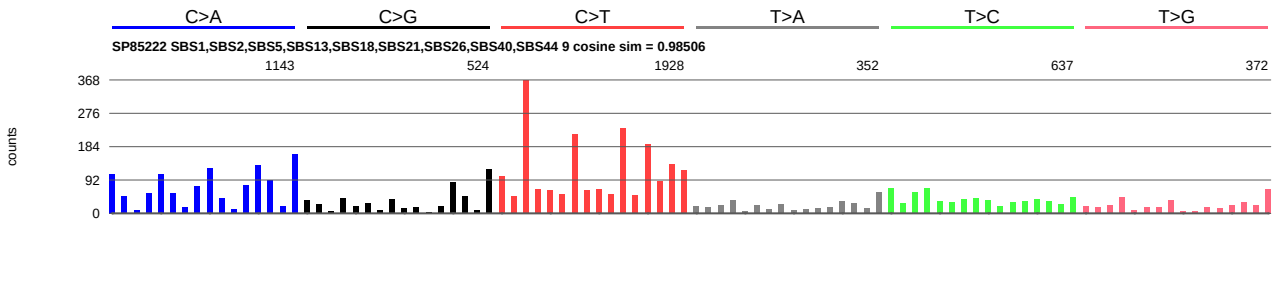
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98506
1143
524
1928
352
634
372
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1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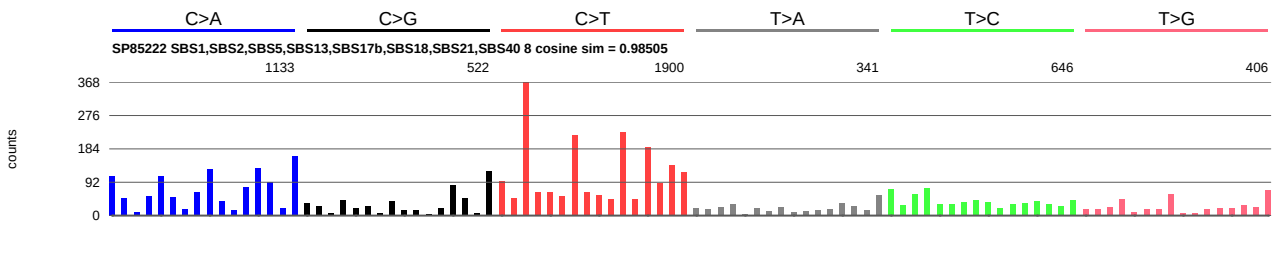
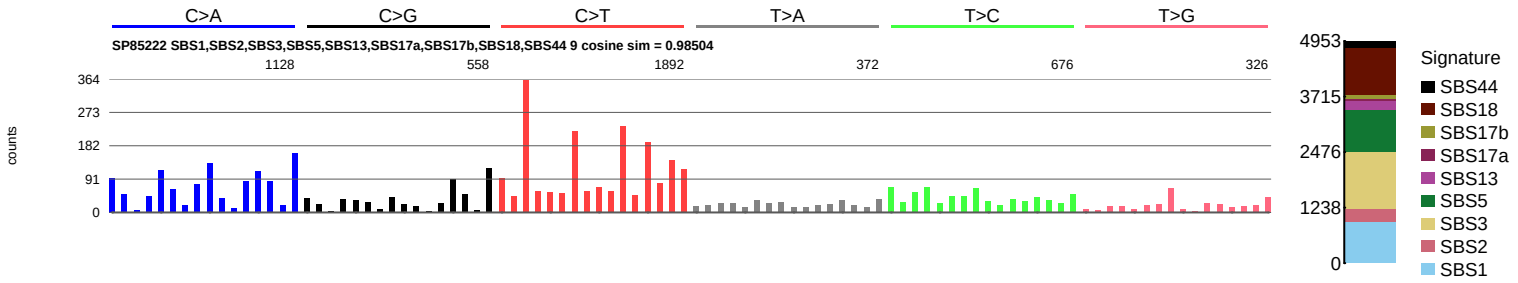
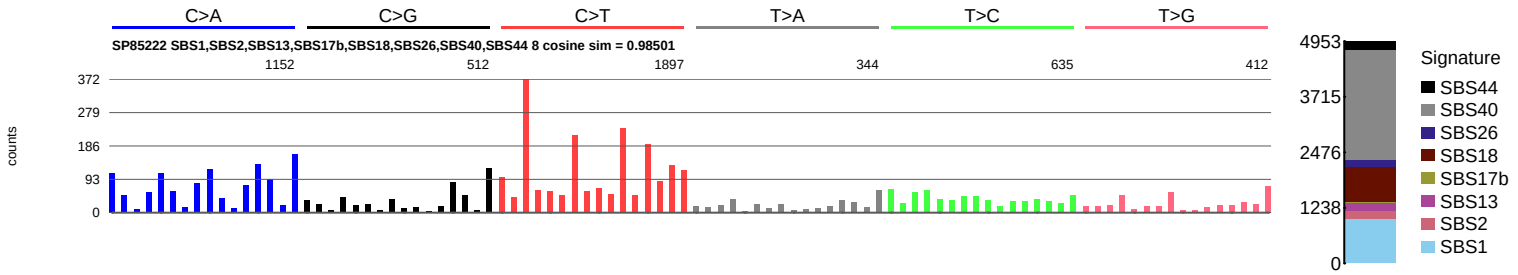
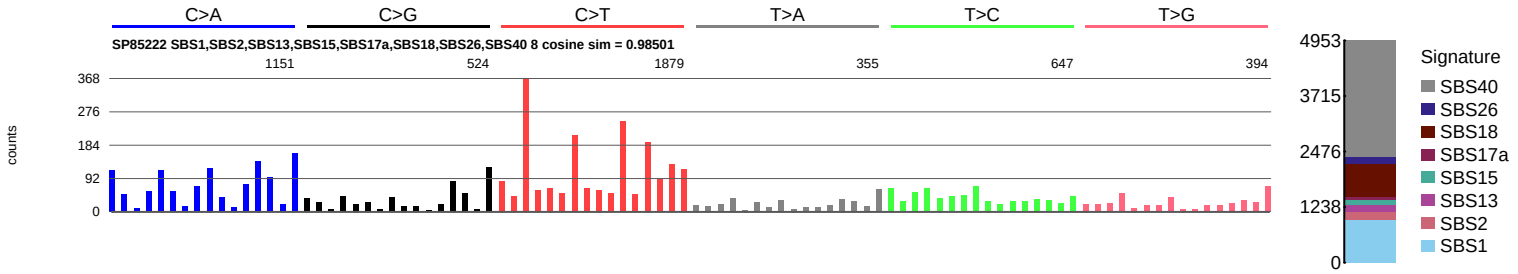
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98506
1143
524
1928
352
637
372
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98505
1133
522
1900
341
646
406
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98504
1128
558
1892
372
676
326
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98501
1152
512
1897
344
635
412
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40 8 cosine sim = 0.98501
1151
524
1879
355
647
394
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17a
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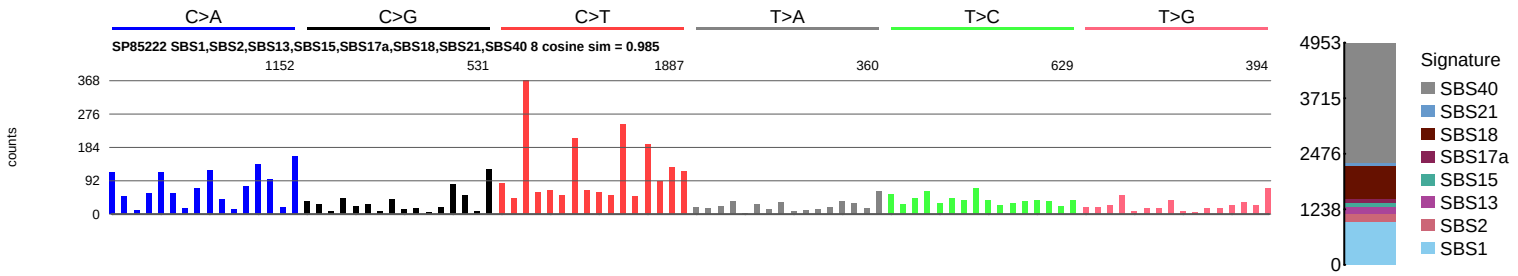
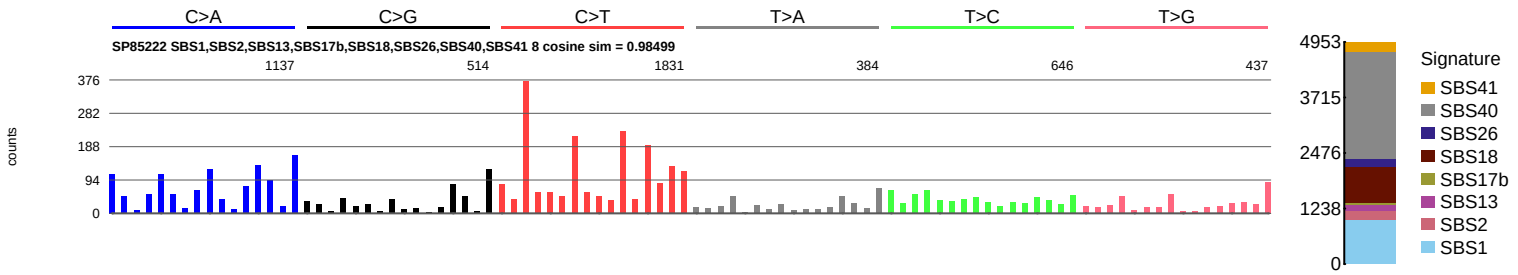
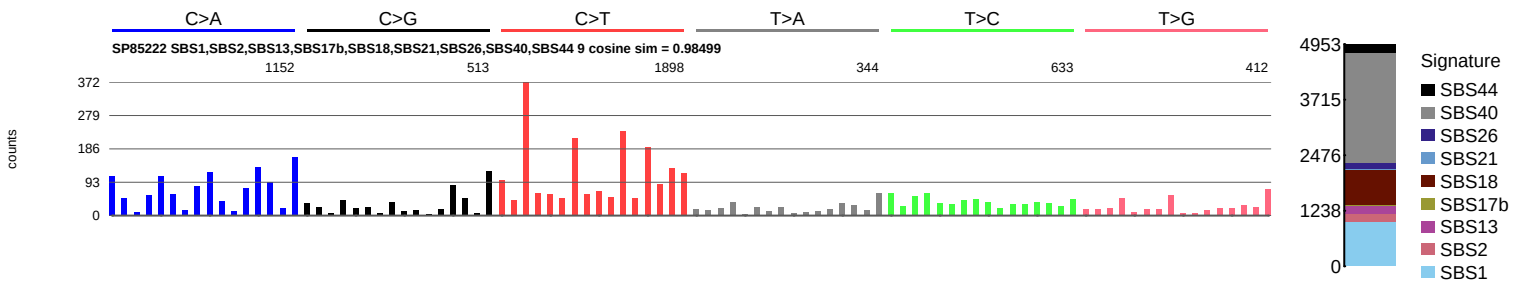
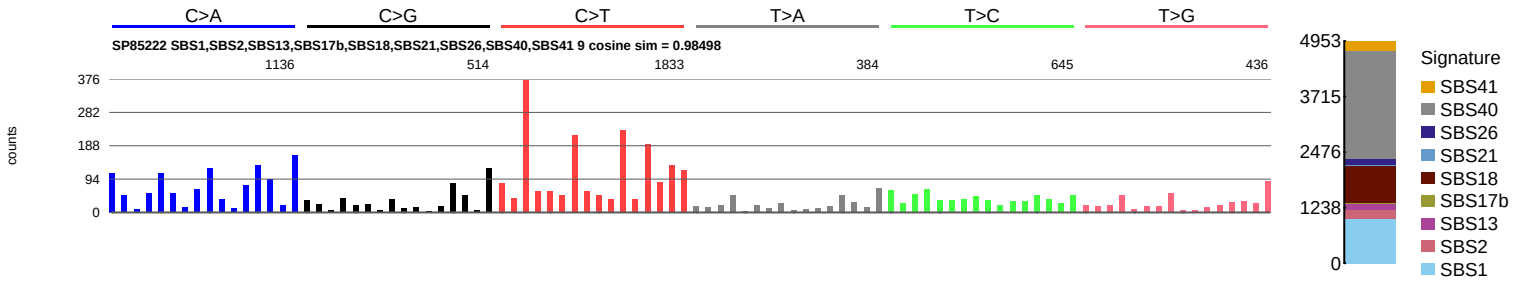
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40 8 cosine sim = 0.985
1152
531
1887
360
629
394
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98499
1137
514
1831
384
646
437
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98499
1152
513
1898
344
633
412
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1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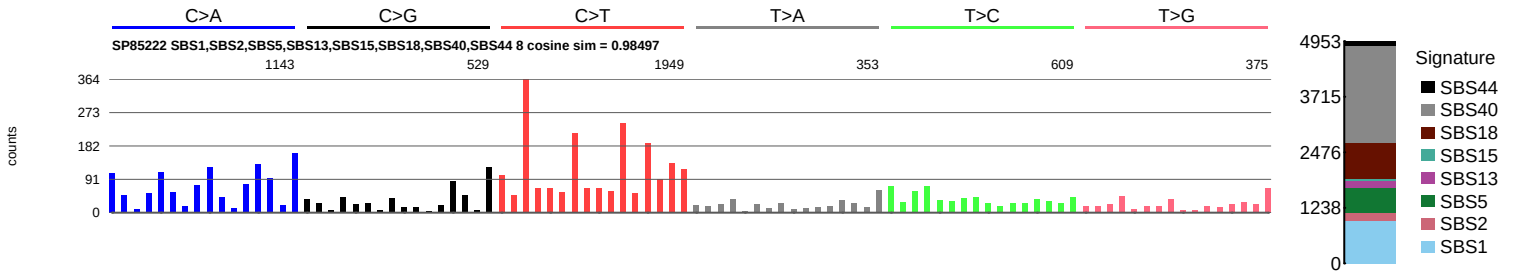
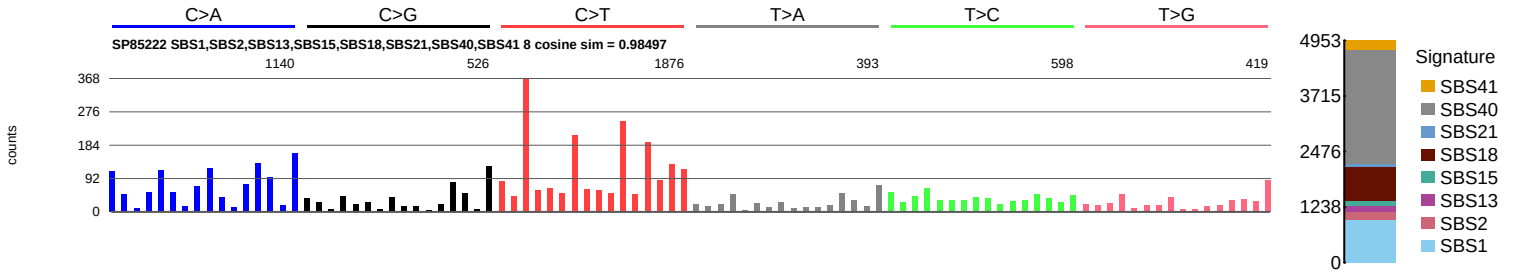
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98498
1136
514
1833
384
645
436
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40,SBS44 8 cosine sim = 0.98497
1143
529
1949
353
609
375
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98497
1140
526
1876
393
598
419
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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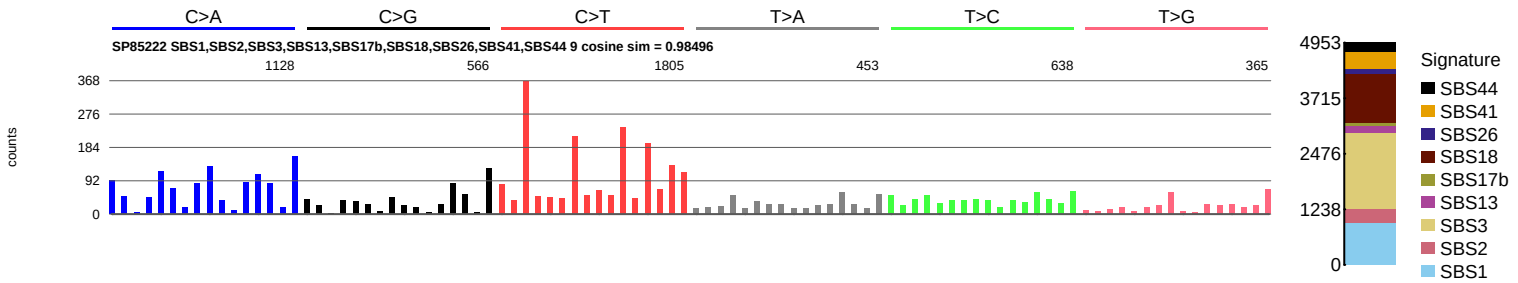
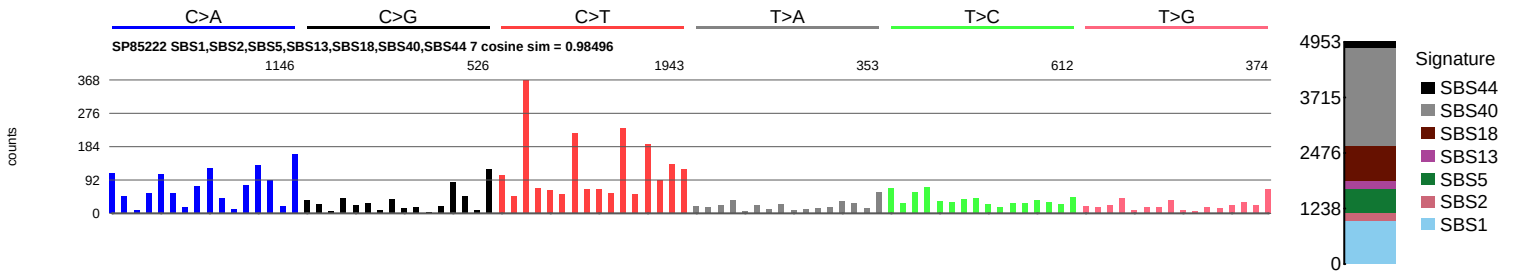
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98496
1128
566
1805
453
638
365
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40,SBS44 7 cosine sim = 0.98496
1146
526
1943
353
612
374
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
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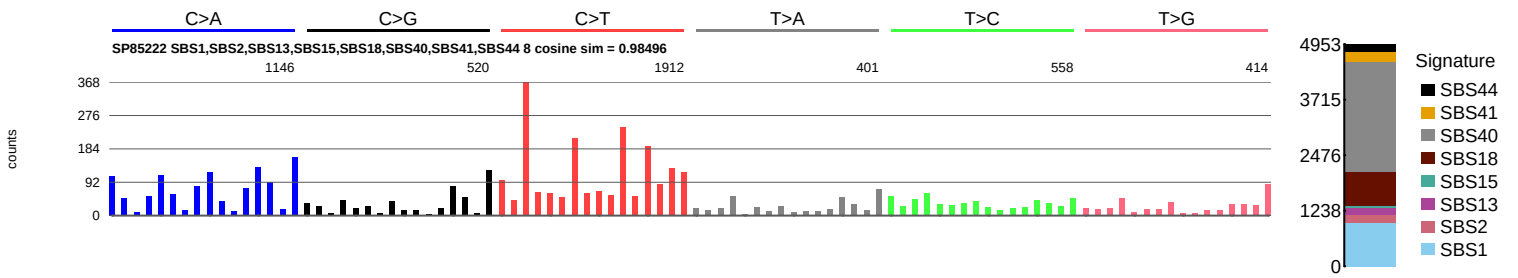
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98496
1146
520
1912
401
558
414
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
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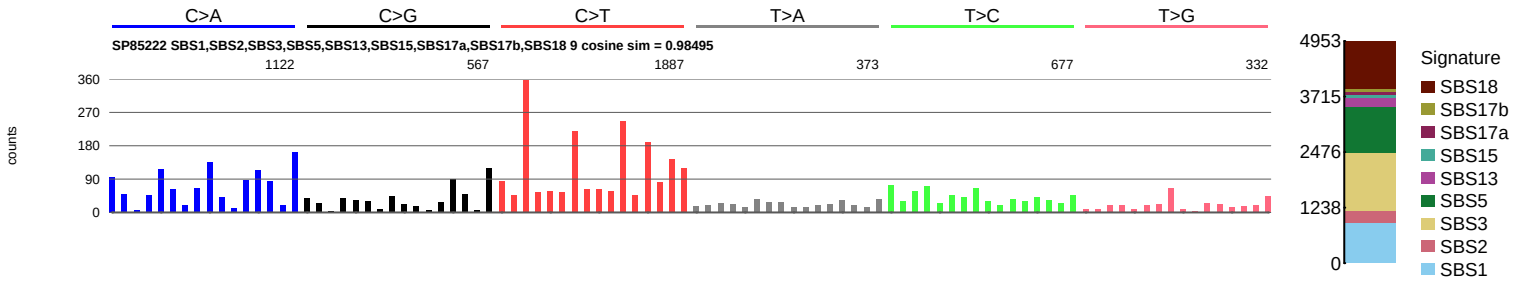
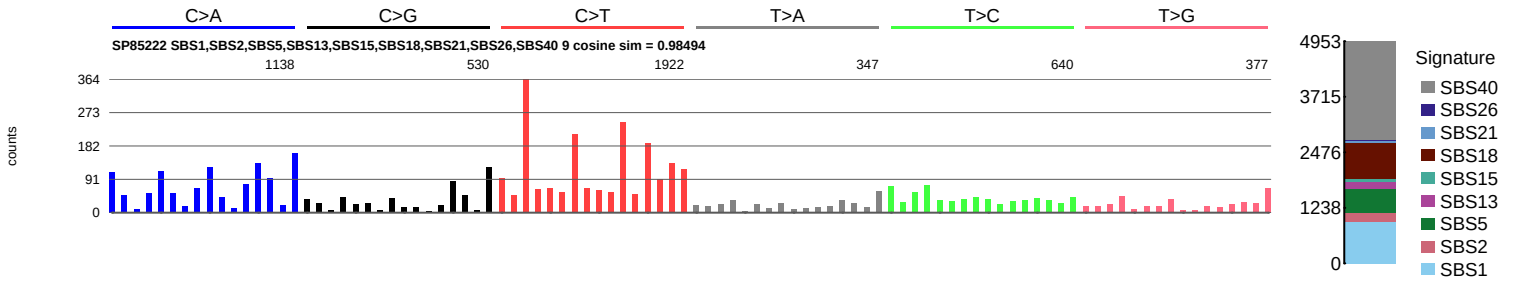
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.98495
1122
567
1887
373
677
332
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98494
1138
530
1922
347
640
377
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
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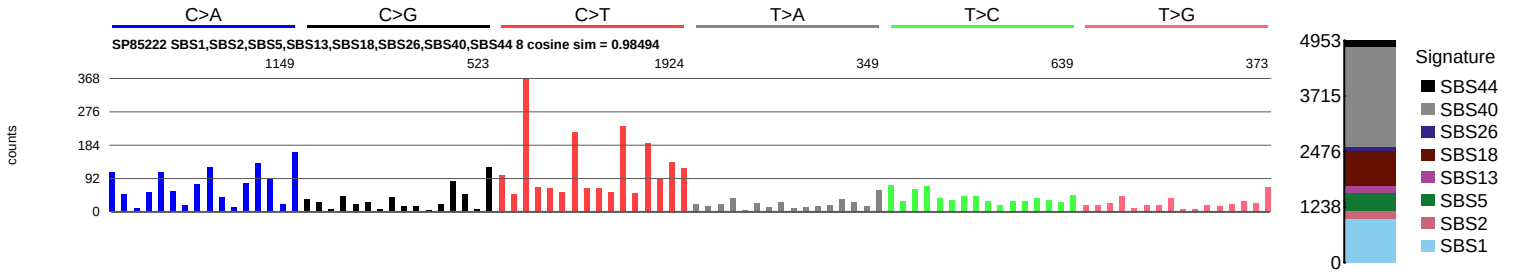
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98494
1149
523
1924
349
639
373
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
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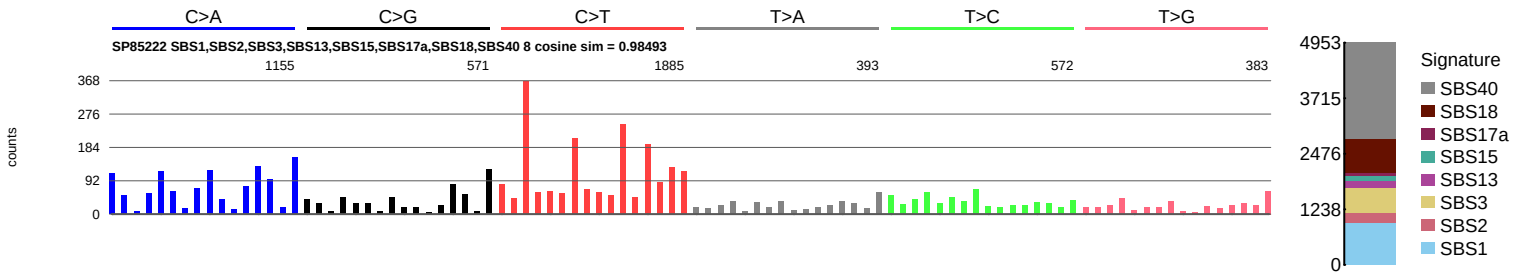
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS40 8 cosine sim = 0.98493
1155
571
1885
393
572
383
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17a
SBS15
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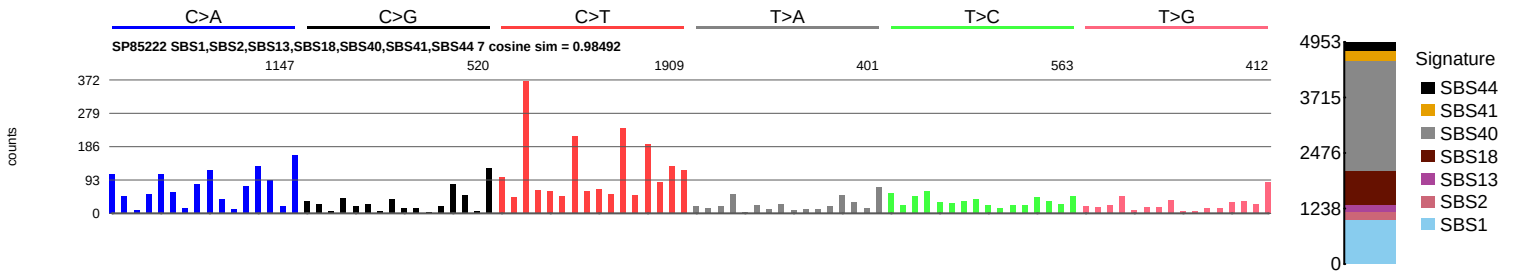
SBS2
SBS1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C>G
C>T
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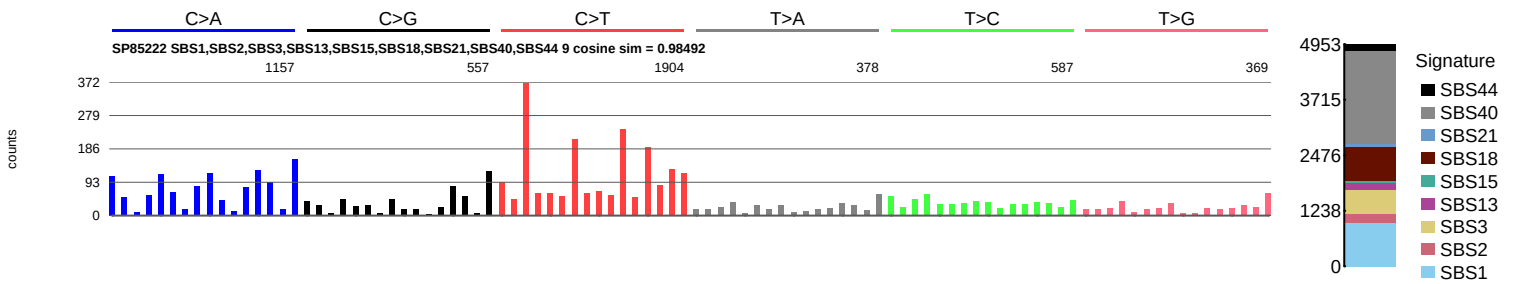
T>G
SP85222 SBS1,SBS2,SBS13,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98492
1147
520
1909
401
563
412
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS2
SBS1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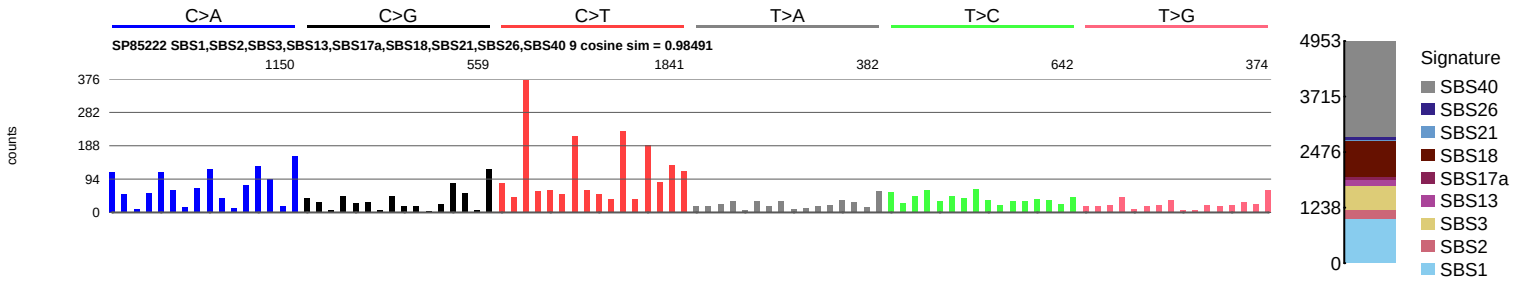
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98492
1157
557
1904
378
587
369
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1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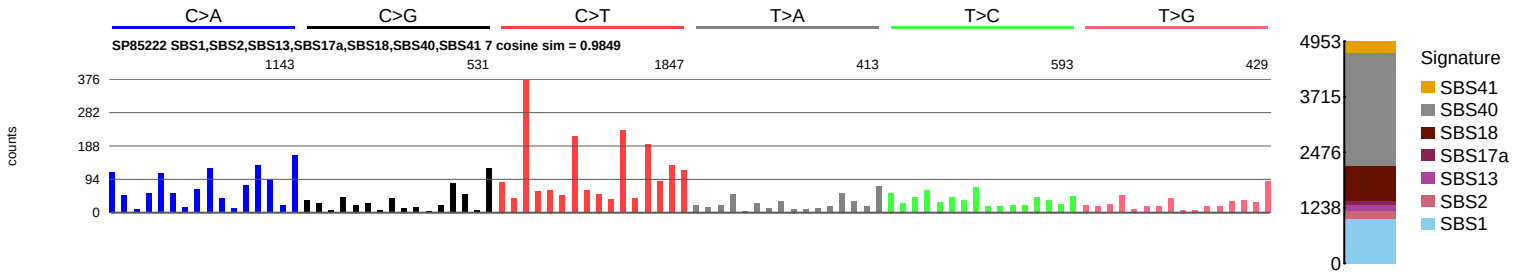
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98491
1150
559
1841
382
642
374
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1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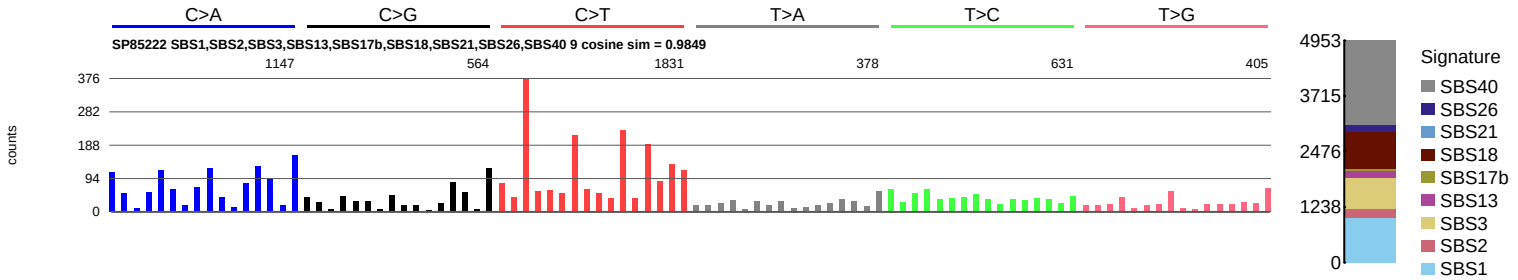
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS40,SBS41 7 cosine sim = 0.9849
1143
531
1847
413
593
429
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS2
SBS1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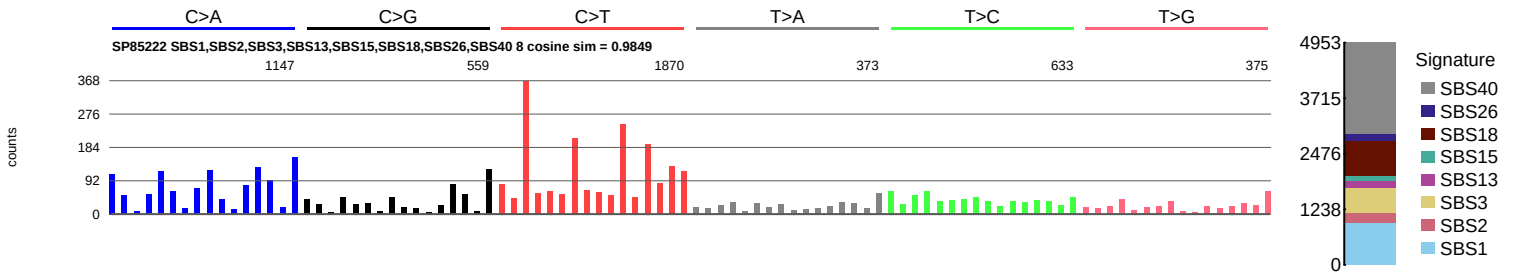
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.9849
1147
564
1831
378
631
405
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS26,SBS40 8 cosine sim = 0.9849
1147
559
1870
373
633
375
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
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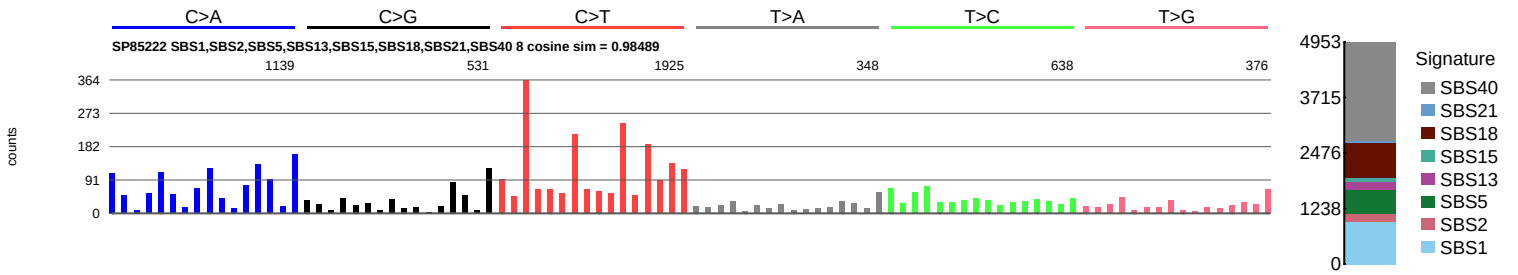
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40 8 cosine sim = 0.98489
1139
531
1925
348
638
376
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
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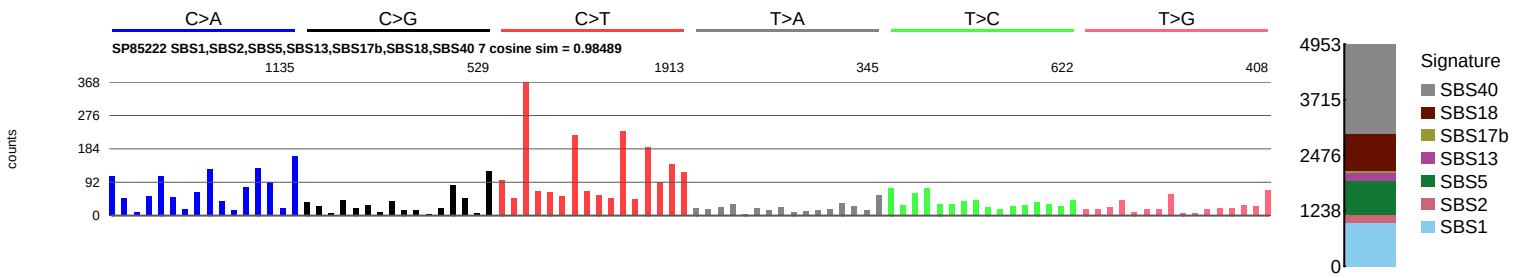
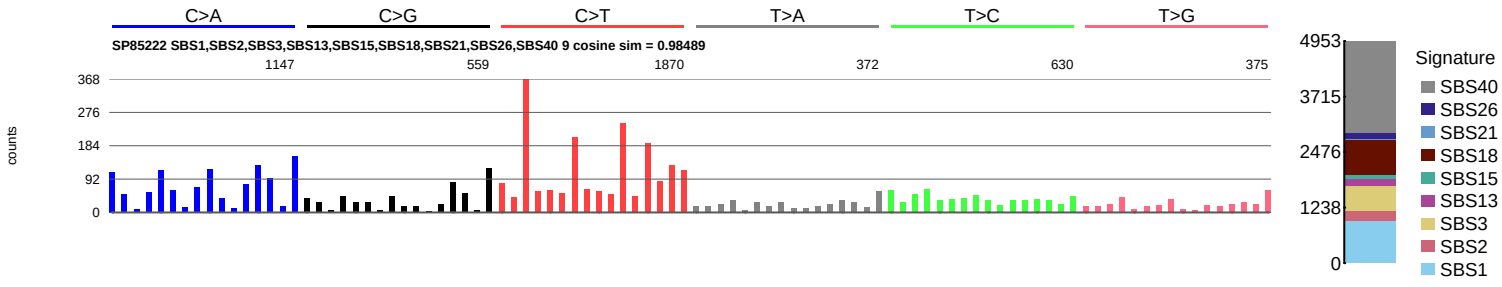
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40 7 cosine sim = 0.98489
1135
529
1913
345
622
408
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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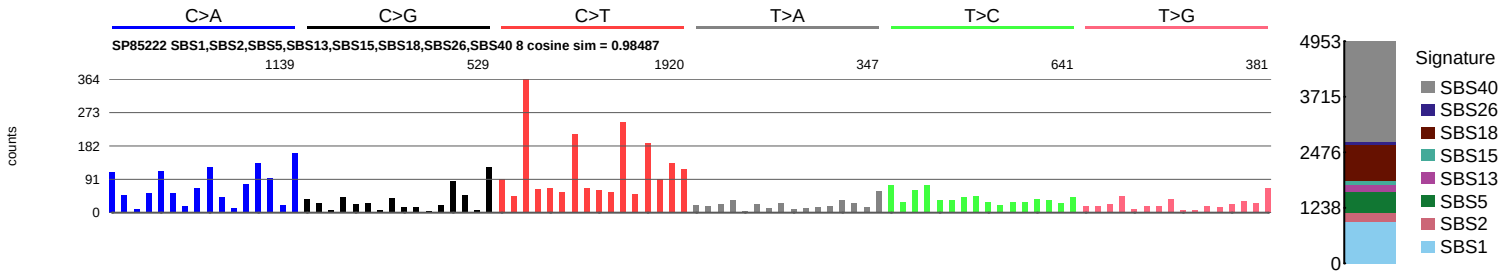
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98489
1147
559
1870
372
630
375
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1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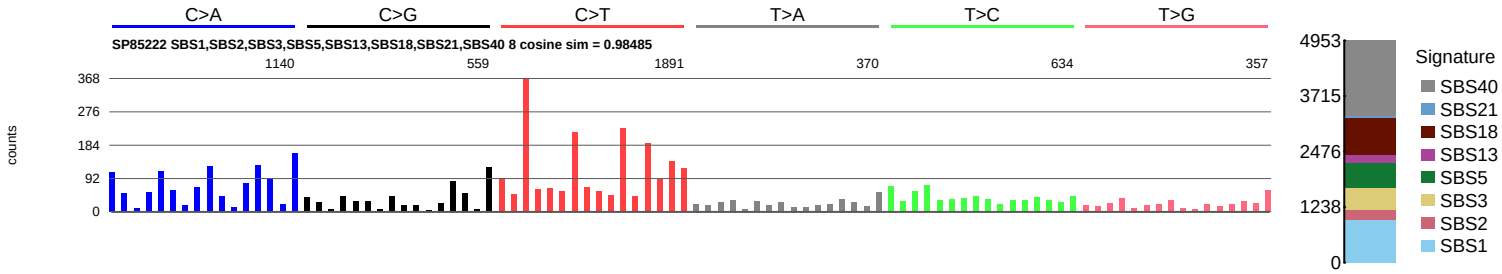
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40 8 cosine sim = 0.98487
1139
529
1920
347
641
381
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS40 8 cosine sim = 0.98485
1140
559
1891
370
634
357
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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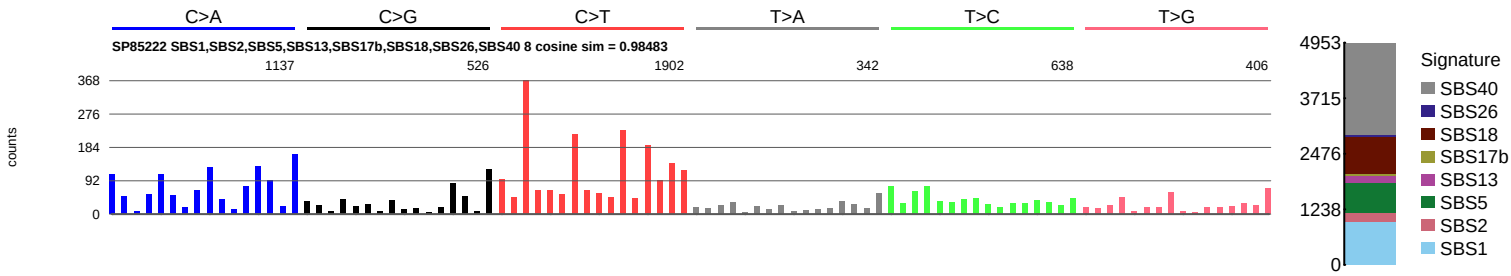
SBS2
SBS1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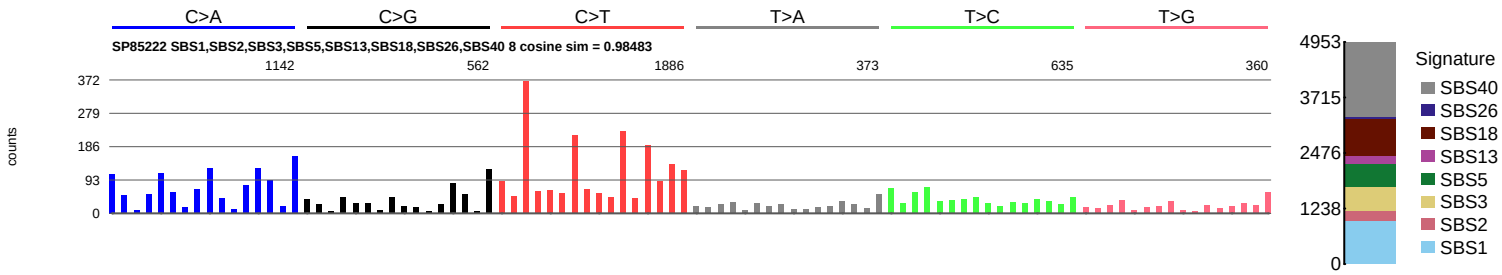
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98483
1137
526
1902
342
638
406
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS26,SBS40 8 cosine sim = 0.98483
1142
562
1886
373
635
360
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS40
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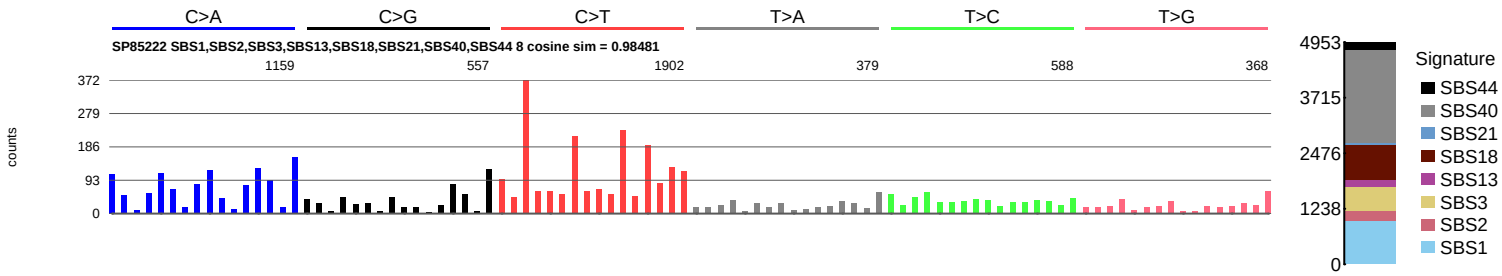
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98481
1159
557
1902
379
588
368
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
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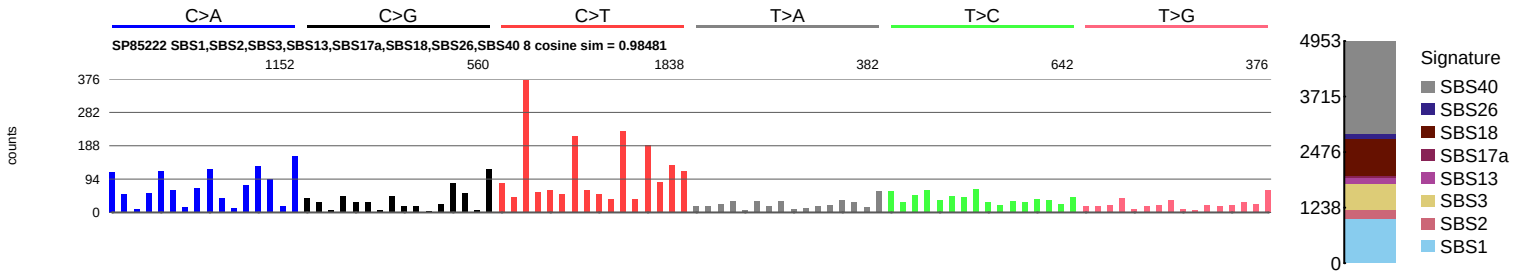
SBS3
SBS2
SBS1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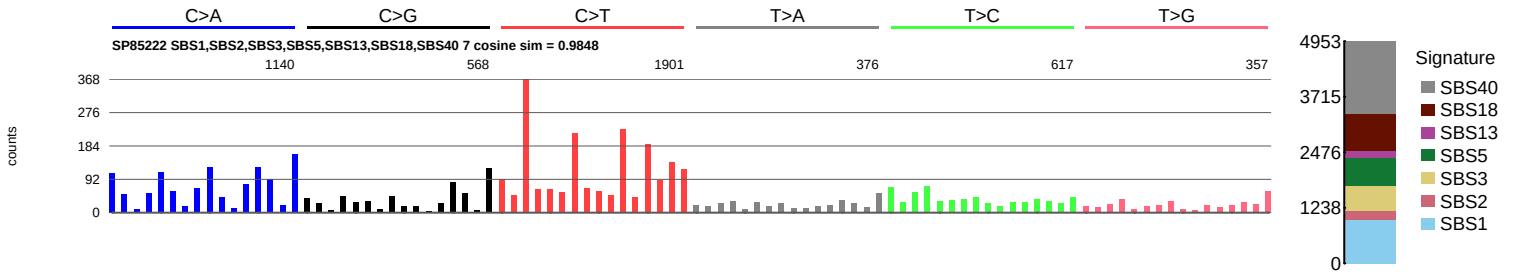
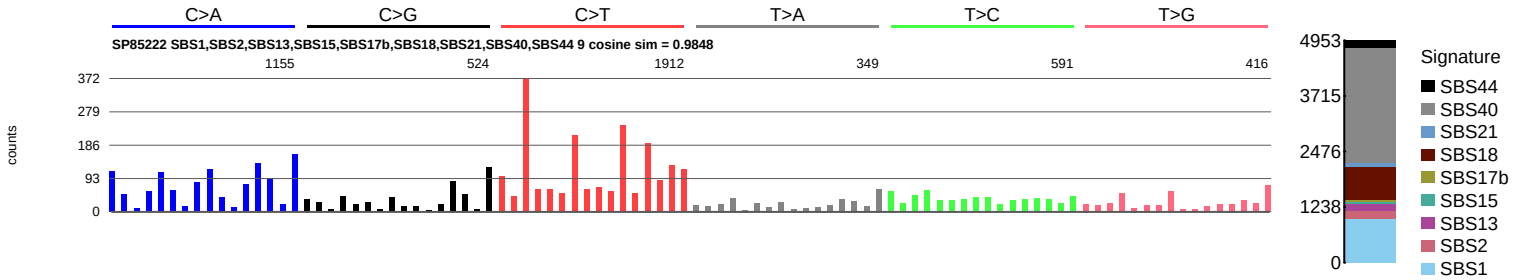
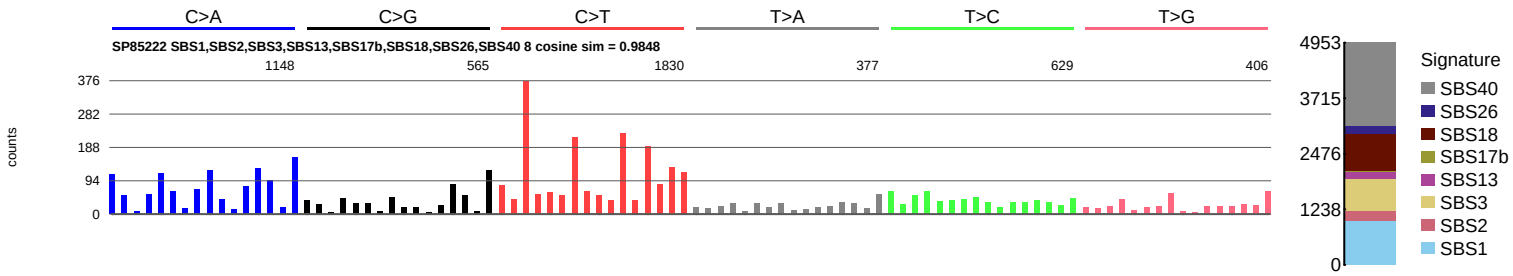
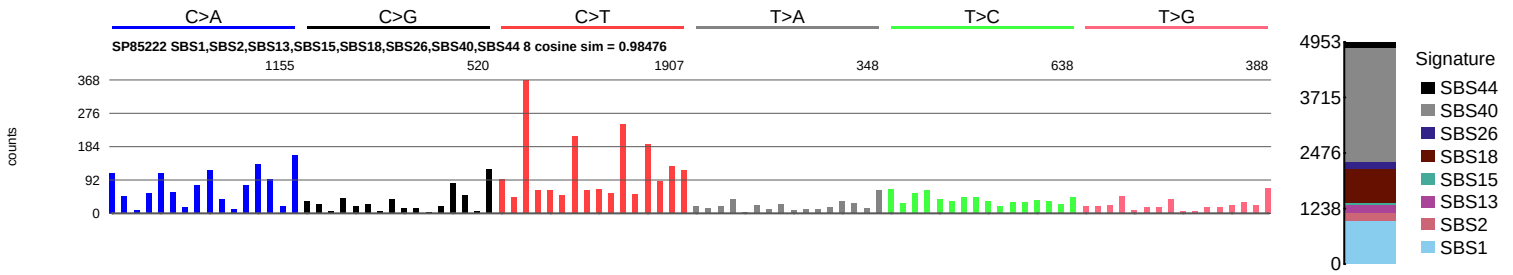
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS26,SBS40 8 cosine sim = 0.98481
1152
560
1838
382
642
376
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40 7 cosine sim = 0.9848
1140
568
1901
376
617
357
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.9848
1155
524
1912
349
591
416
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.9848
1148
565
1830
377
629
406
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98476
1155
520
1907
348
638
388
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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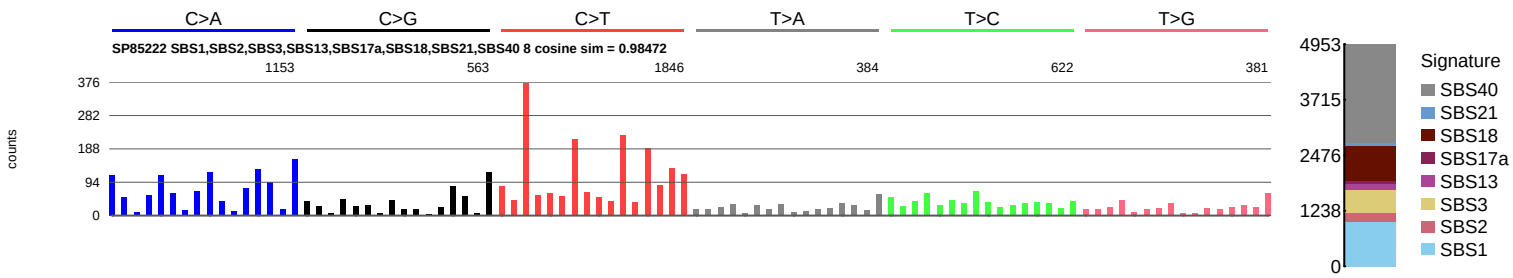
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS40 8 cosine sim = 0.98472
1153
563
1846
384
622
381
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
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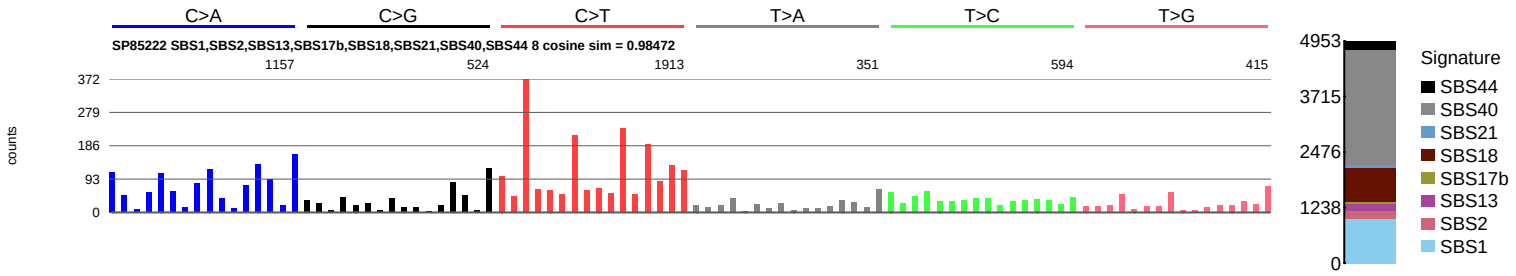
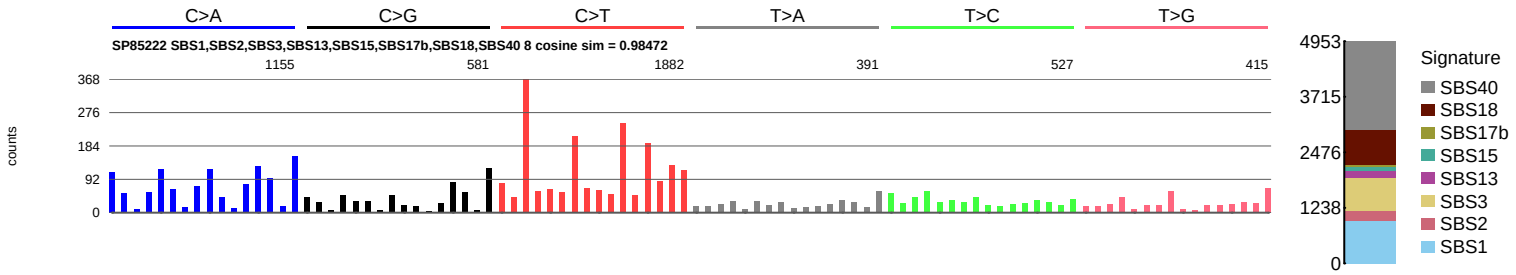
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98472
1157
524
1913
351
594
415
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.98472
1155
581
1882
391
527
415
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
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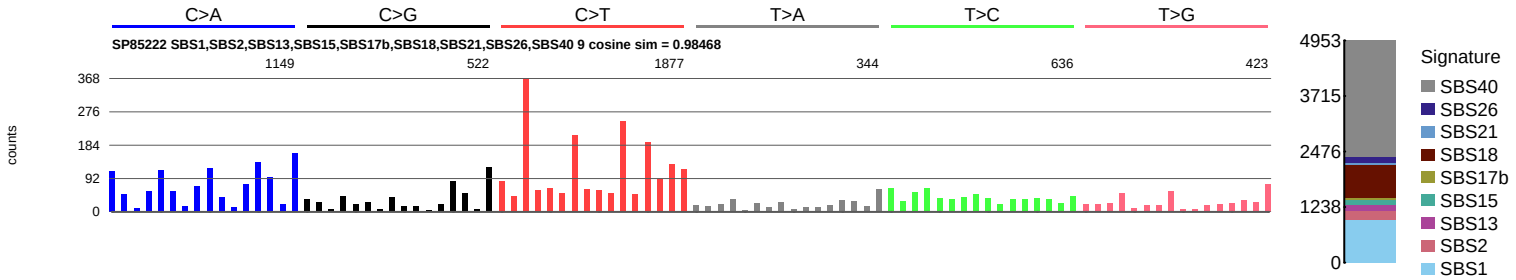
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98468
1149
522
1877
344
636
423
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
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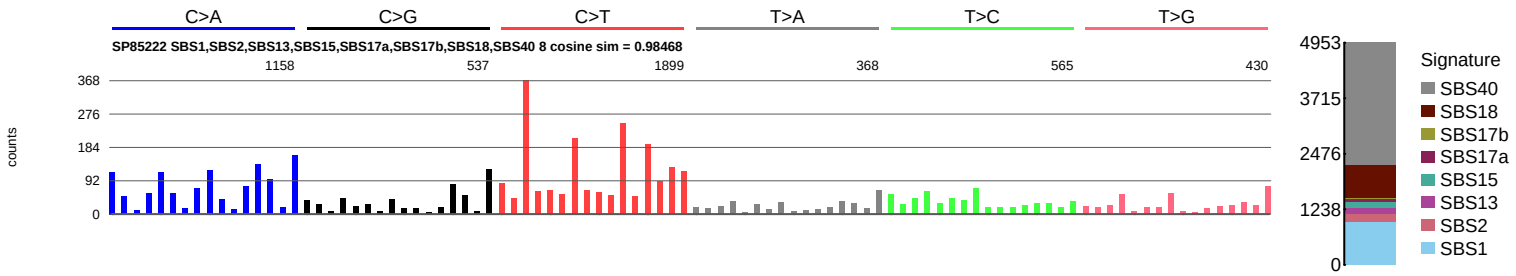
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98468
1158
537
1899
368
565
430
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS17a
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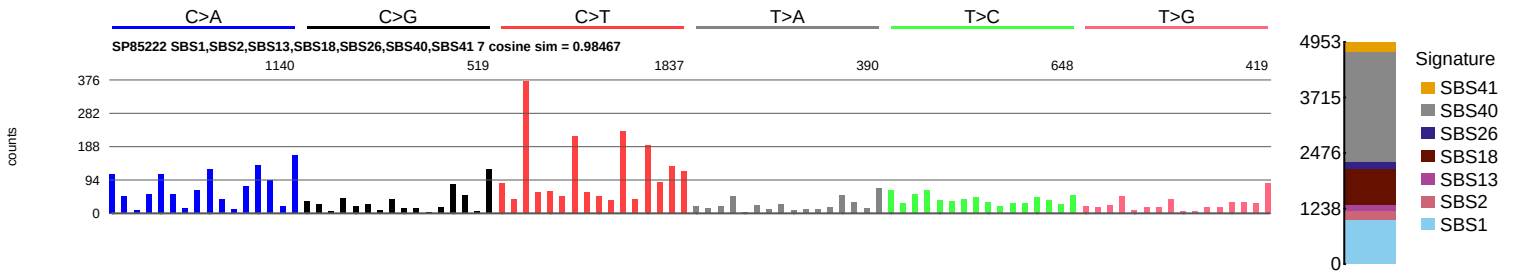
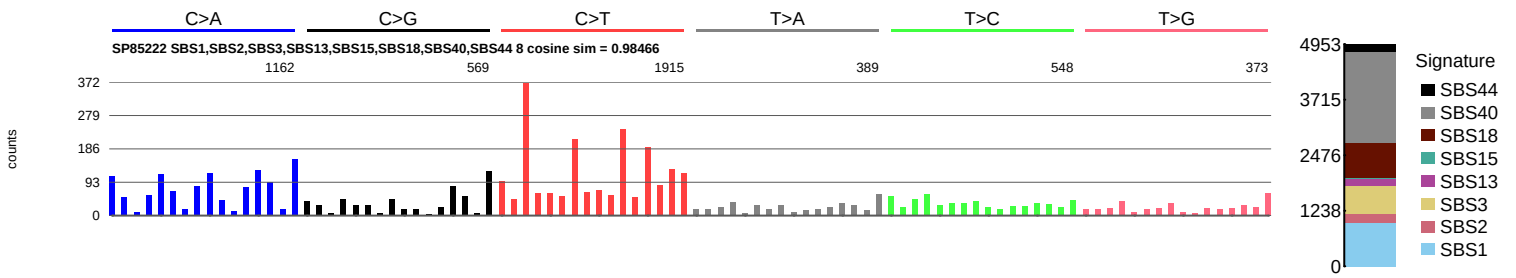
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98467
1140
519
1837
390
648
419
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS13
SBS2
SBS1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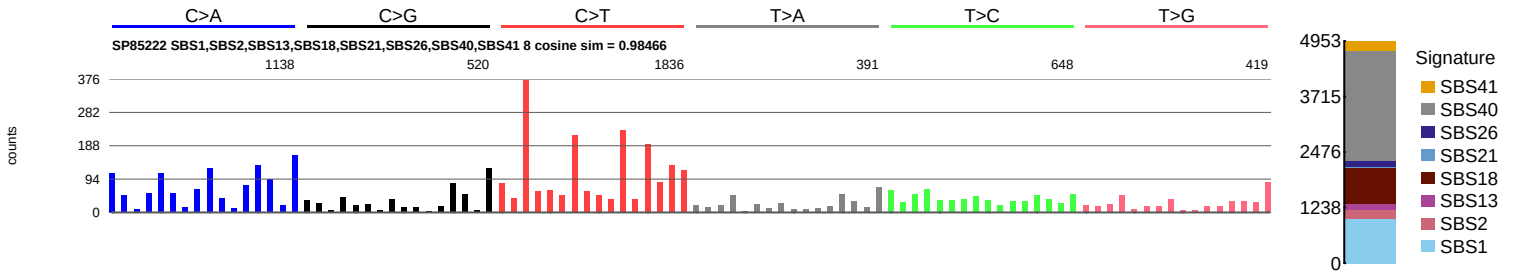
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS40,SBS44 8 cosine sim = 0.98466
1162
569
1915
389
548
373
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1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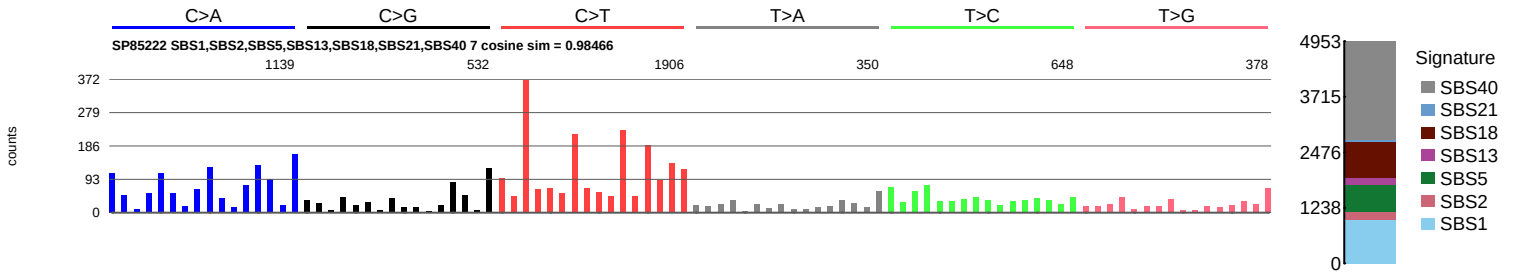
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98466
1138
520
1836
391
648
419
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS13
SBS2
SBS1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C>G
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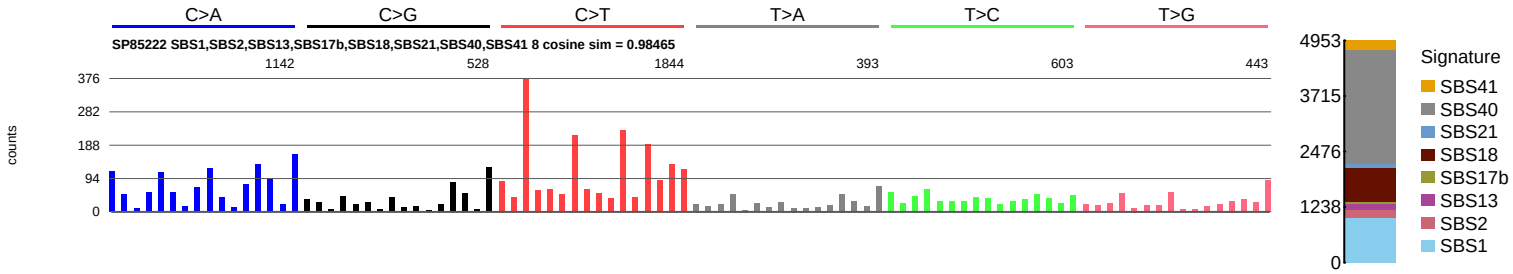
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40 7 cosine sim = 0.98466
1139
532
1906
350
648
378
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1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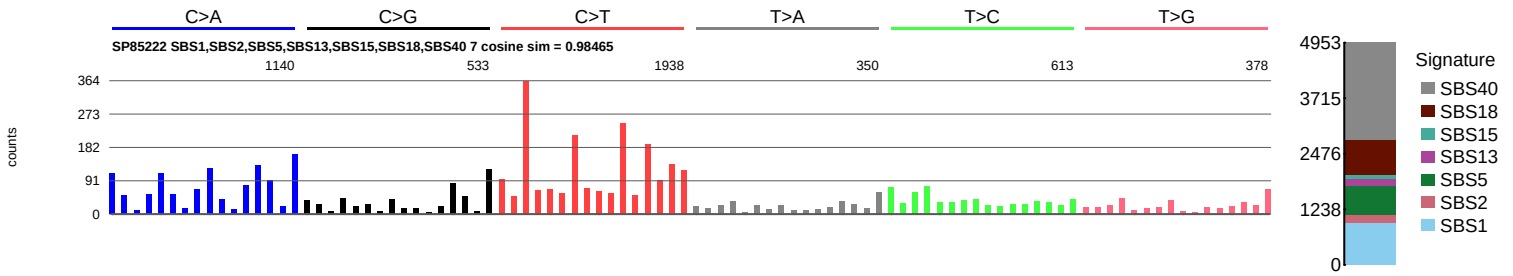
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98465
1142
528
1844
393
603
443
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1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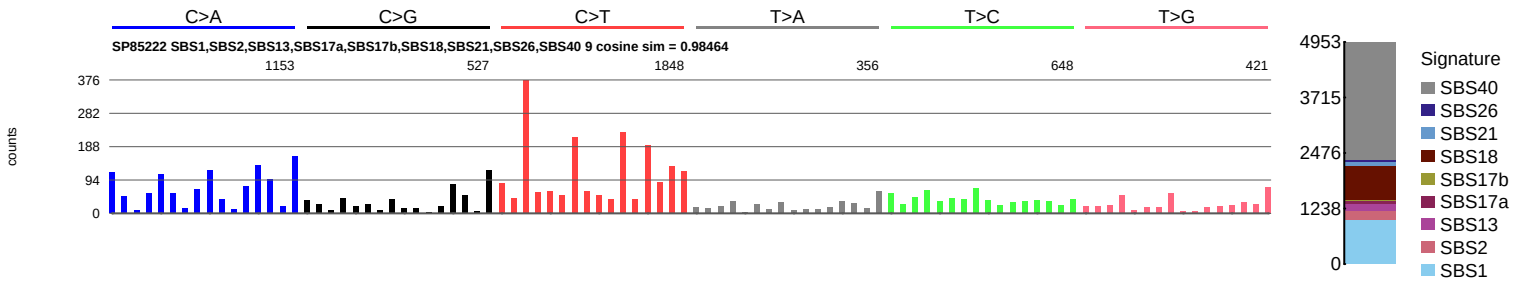
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40 7 cosine sim = 0.98465
1140
533
1938
350
613
378
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1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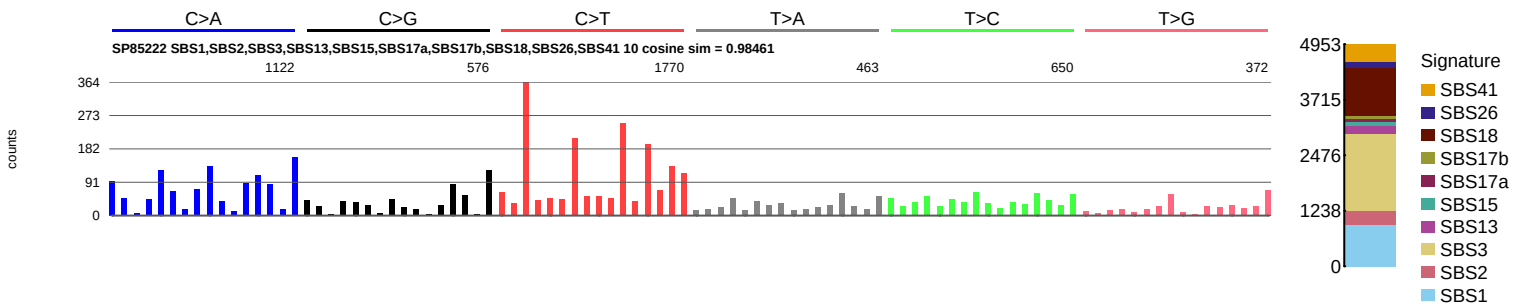
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98464
1153
527
1848
356
648
421
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
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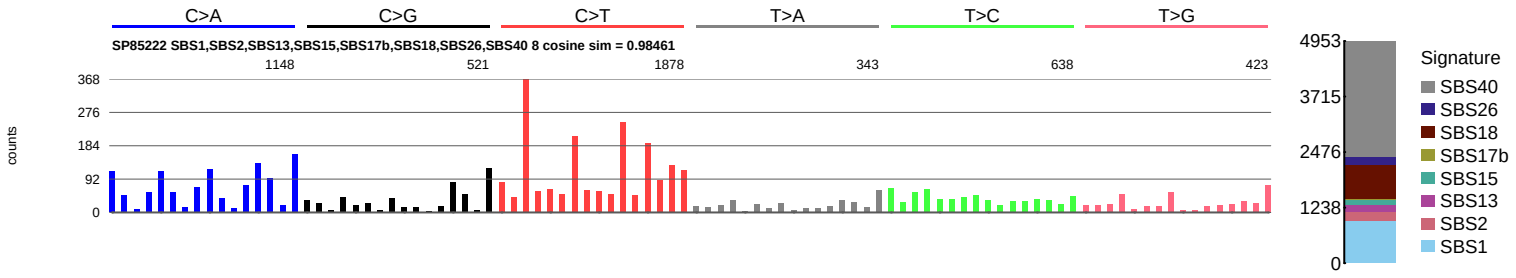
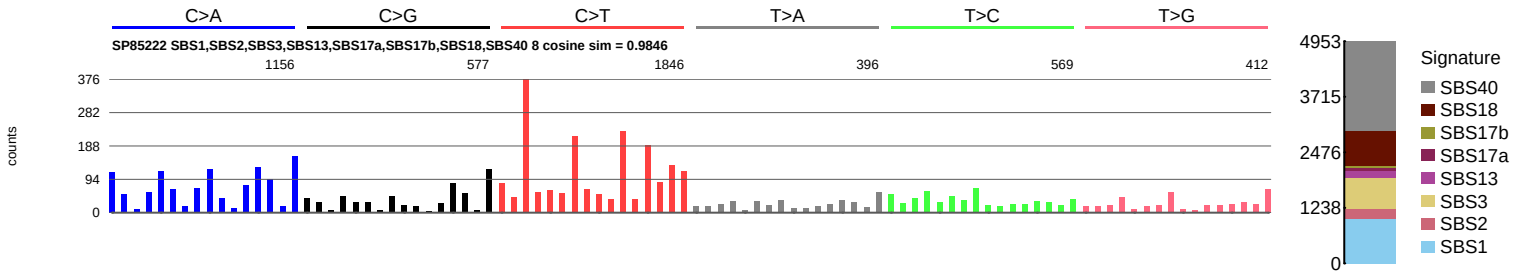
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98461
1122
576
1770
463
650
372
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98461
1148
521
1878
343
638
423
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1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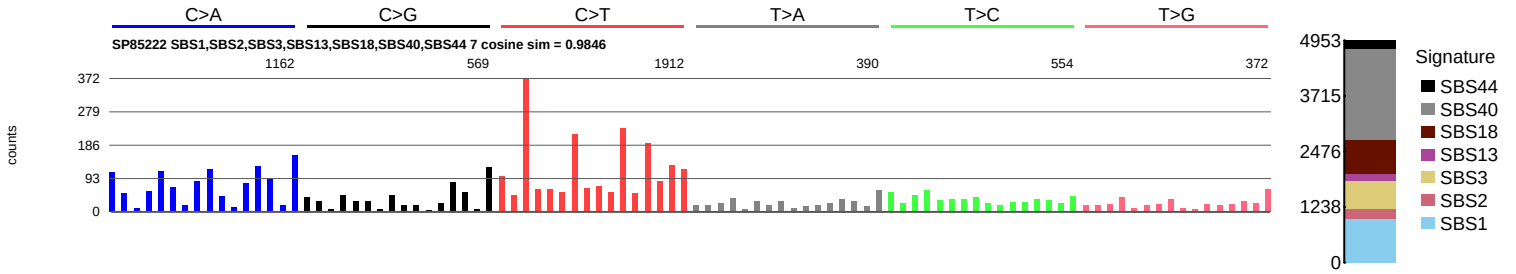
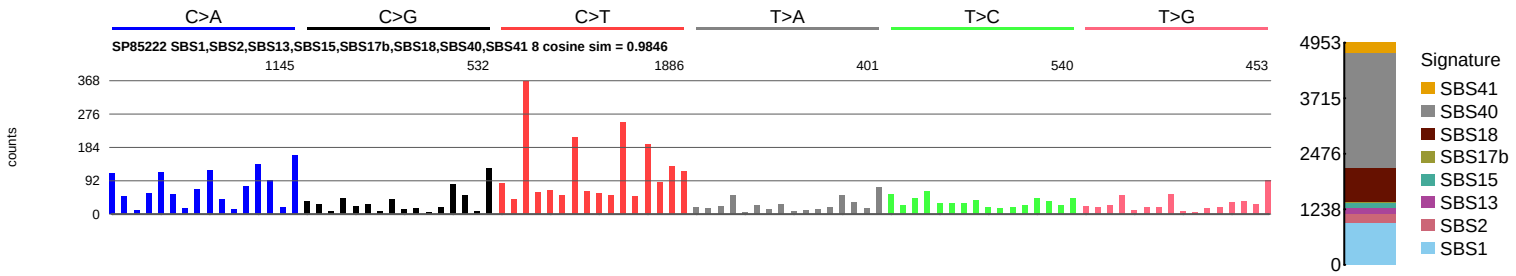
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.9846
1156
577
1846
396
569
412
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS40,SBS44 7 cosine sim = 0.9846
1162
569
1912
390
554
372
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.9846
1145
532
1886
401
540
453
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
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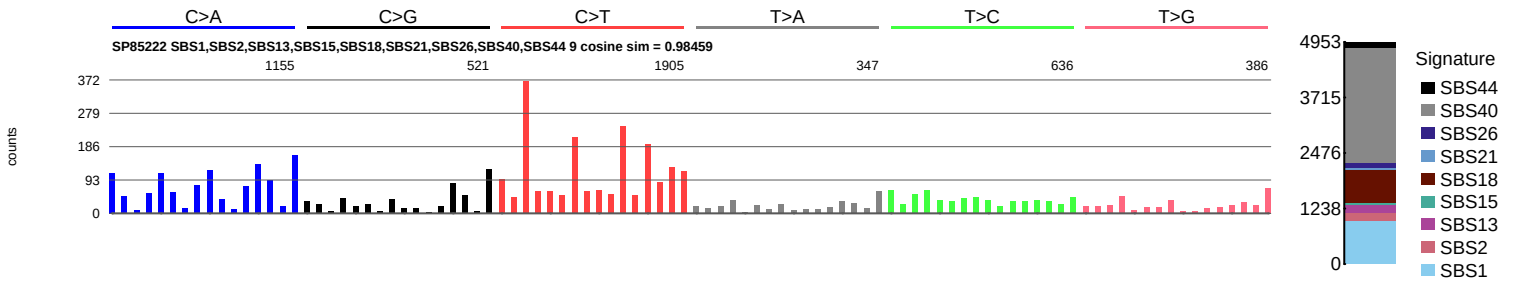
SBS2
SBS1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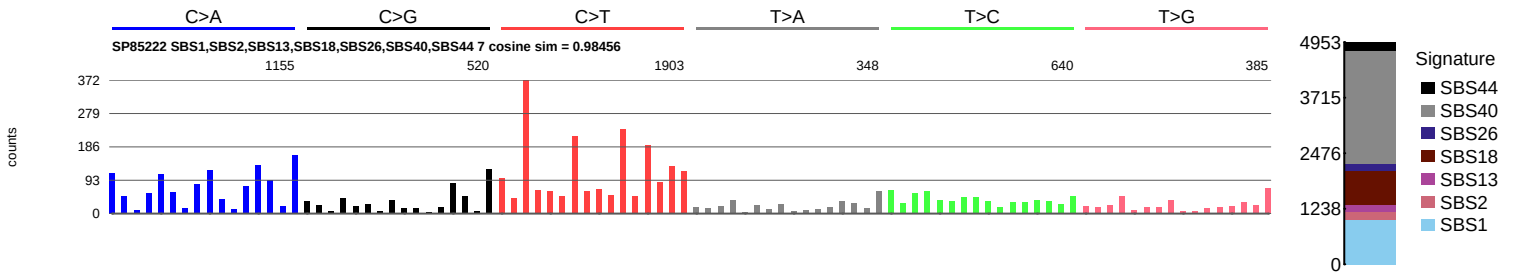
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98459
1155
521
1905
347
636
386
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1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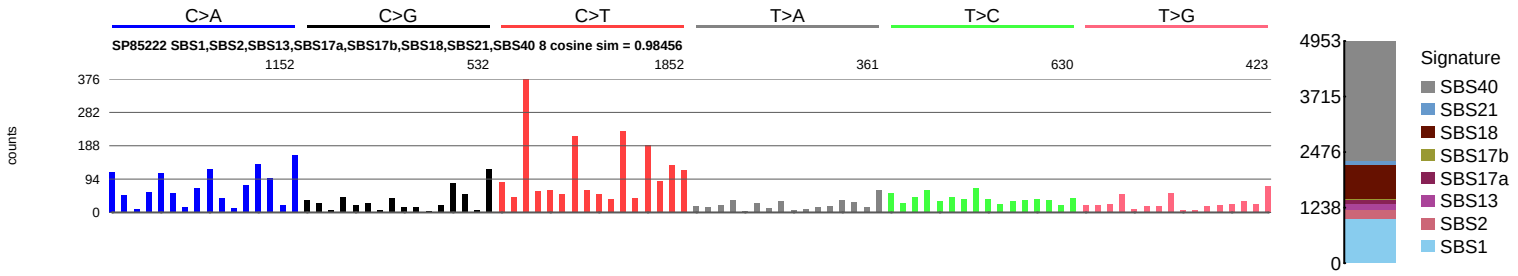
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98456
1155
520
1903
348
640
385
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS13
SBS2
SBS1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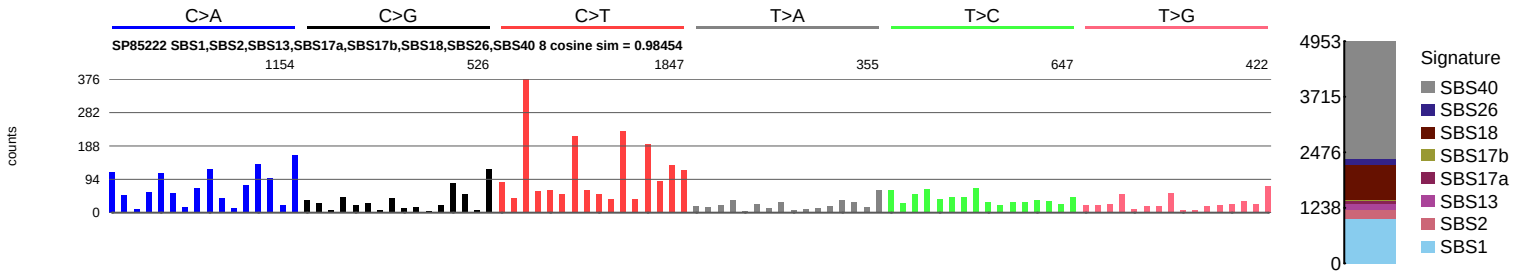
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98456
1152
532
1852
361
630
423
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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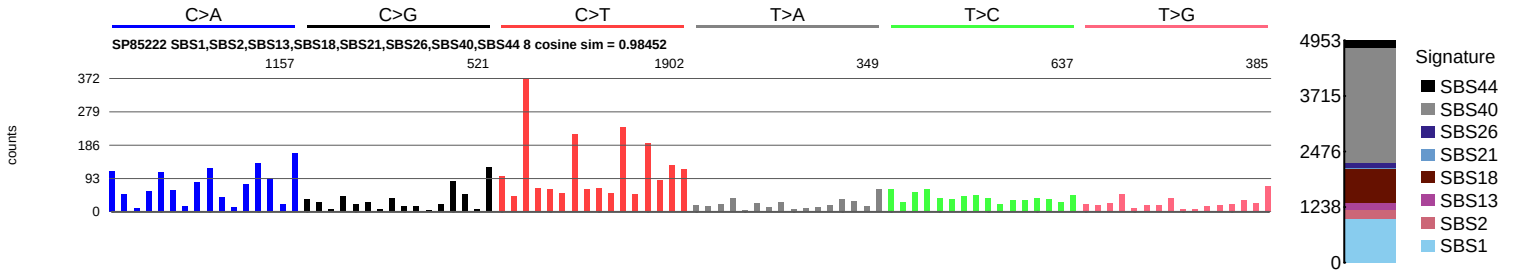
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98454
1154
526
1847
355
647
422
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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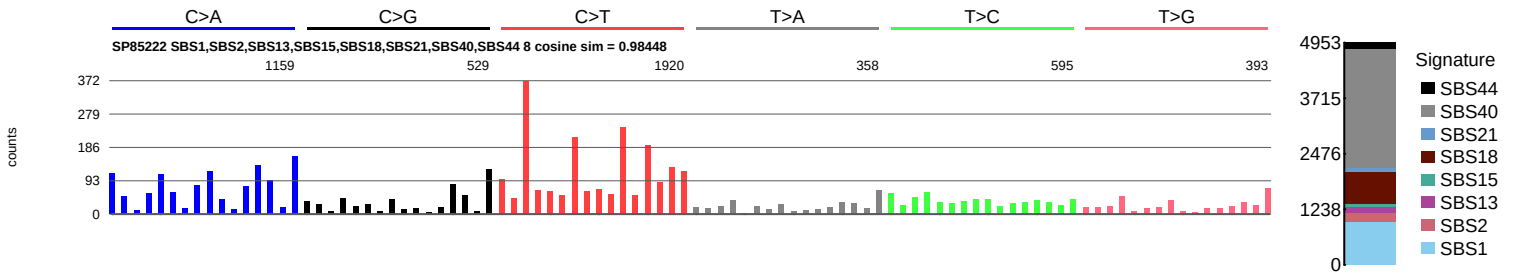
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98452
1157
521
1902
349
637
385
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98448
1159
529
1920
358
595
393
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
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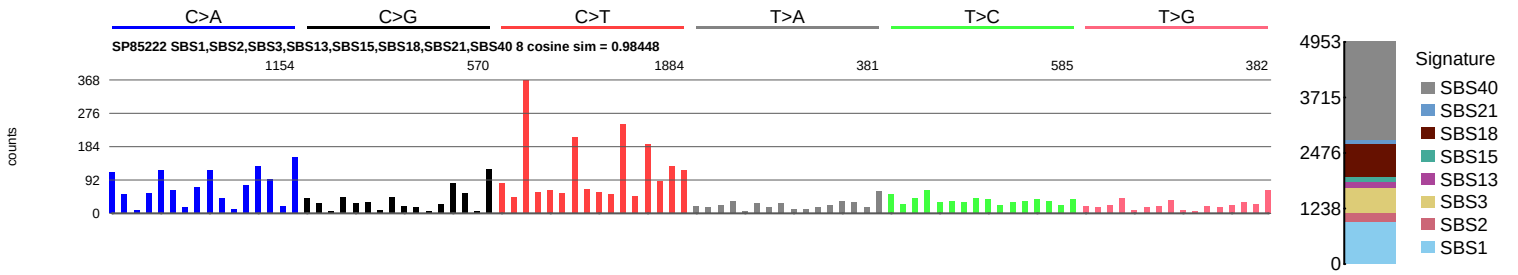
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS40 8 cosine sim = 0.98448
1154
570
1884
381
585
382
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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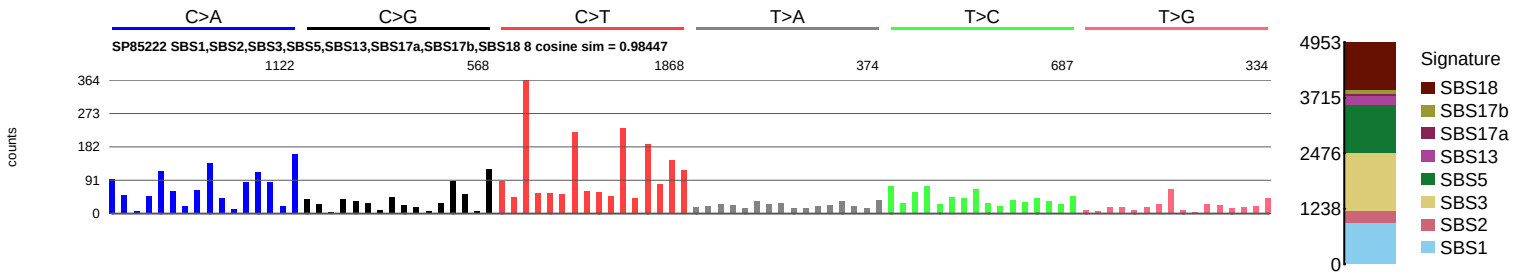
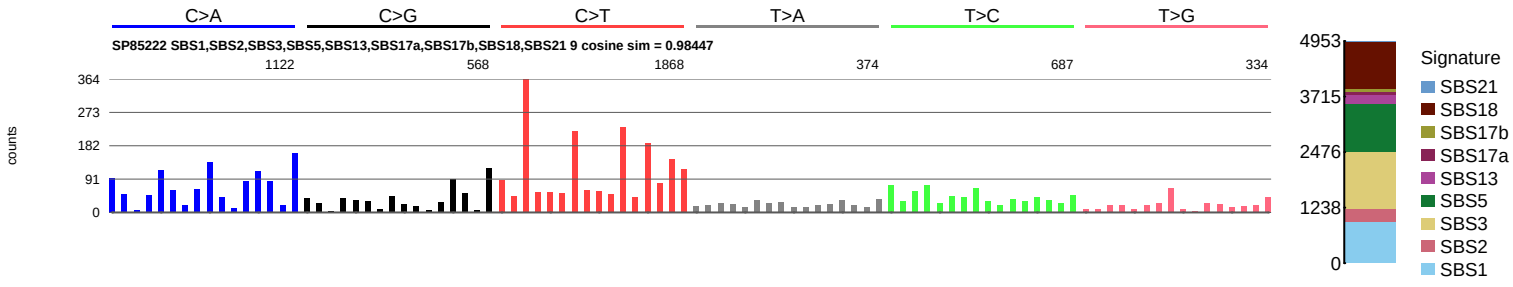
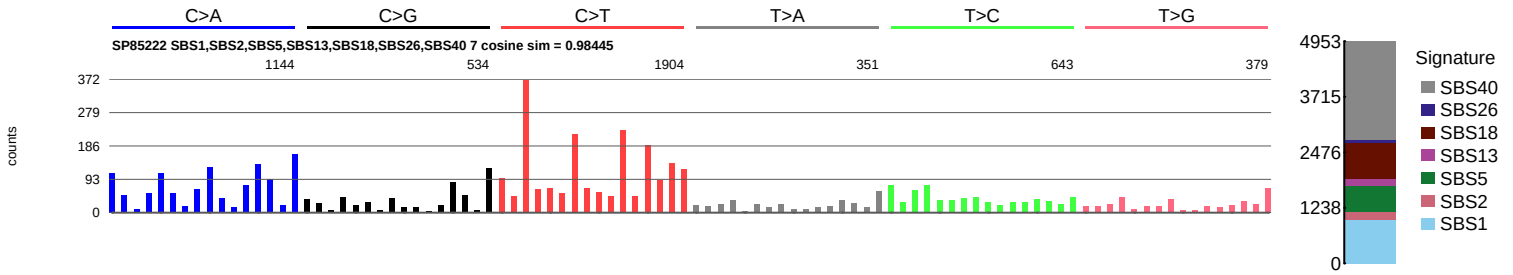
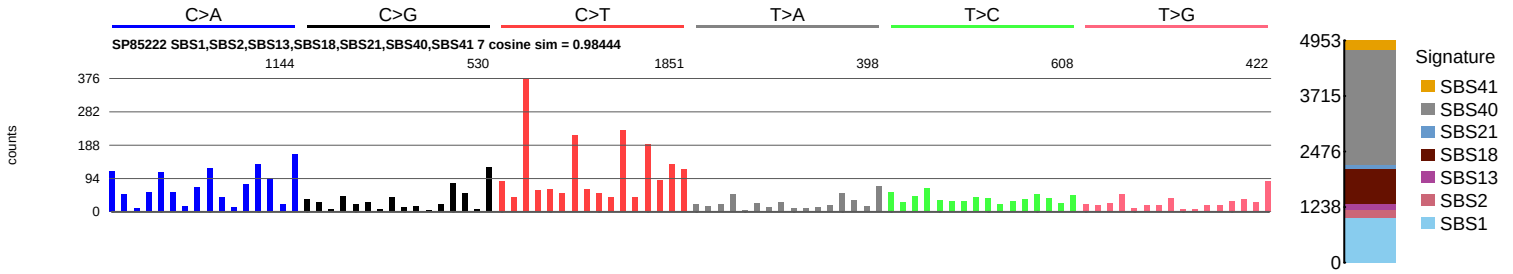
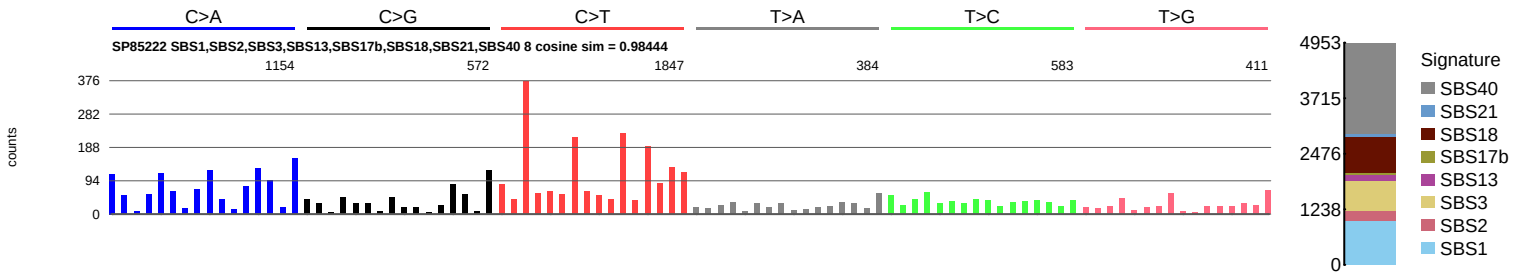
SBS2
SBS1

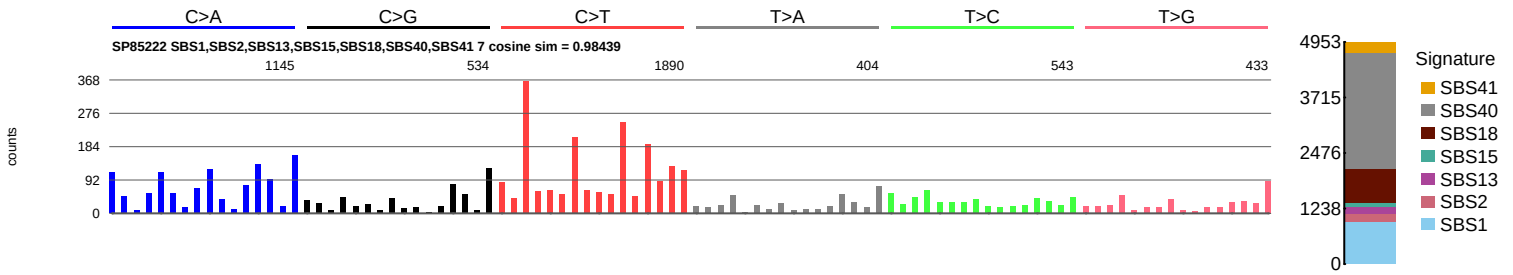

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98447
1122
568
1868
374
687
334
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98447
1122
568
1868
374
687
334
364
273
182
91
0
counts
4953
3715
2476
1238
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS26,SBS40 7 cosine sim = 0.98445
1144
534
1904
351
643
379
372
279
186
93
0
counts
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98444
1144
530
1851
398
608
422
376
282
188
94
0
counts
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98444
1154
572
1847
384
583
411
376
282
188
94
0
counts
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS40,SBS41 7 cosine sim = 0.98439
1145
534
1890
404
543
433
368
276
184
92
0
counts
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
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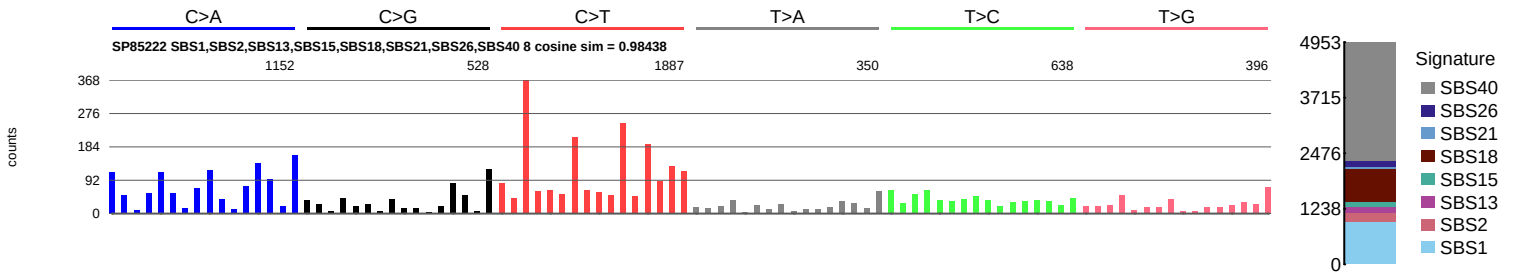
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98438
1152
528
1887
350
638
396
368
276
184
92
0
counts
4953
3715
2476
1238
0
Signature
SBS40
SBS26
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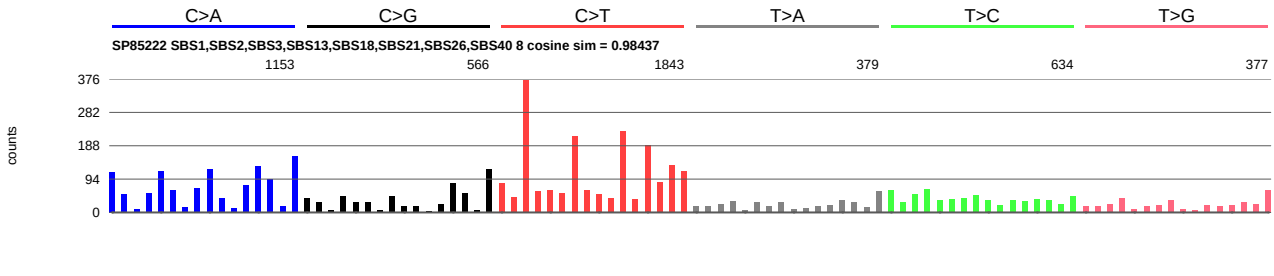
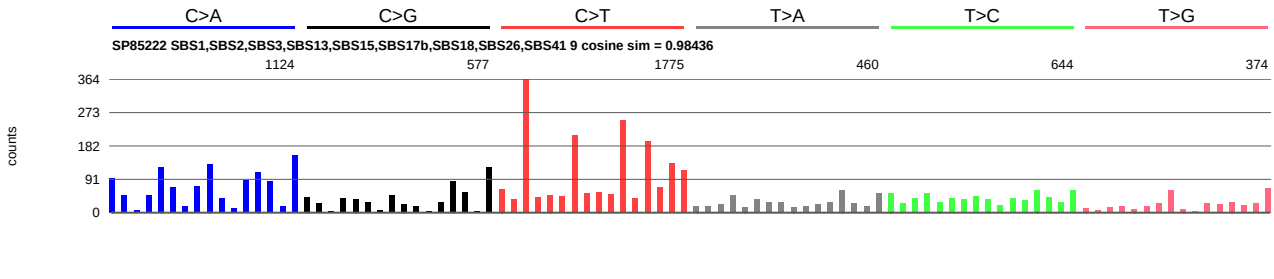
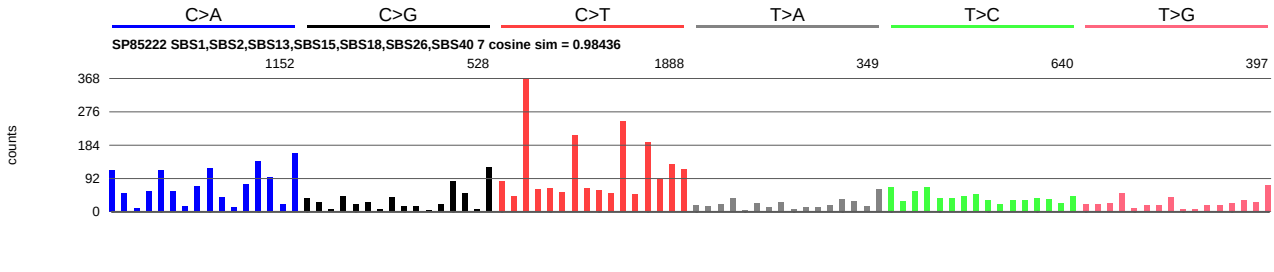
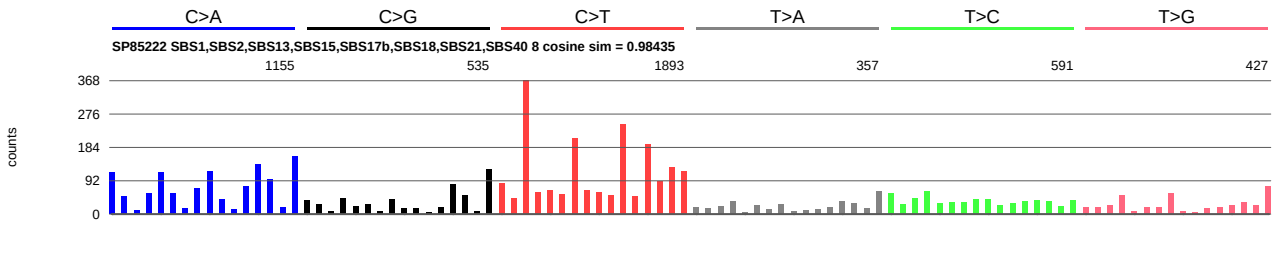
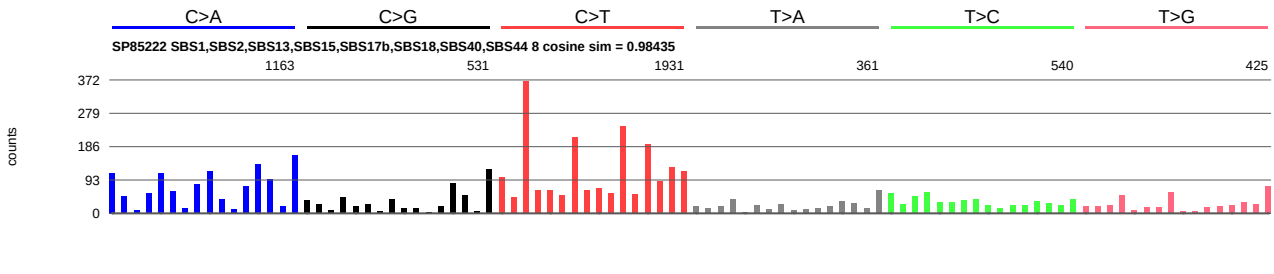
SBS2
SBS1

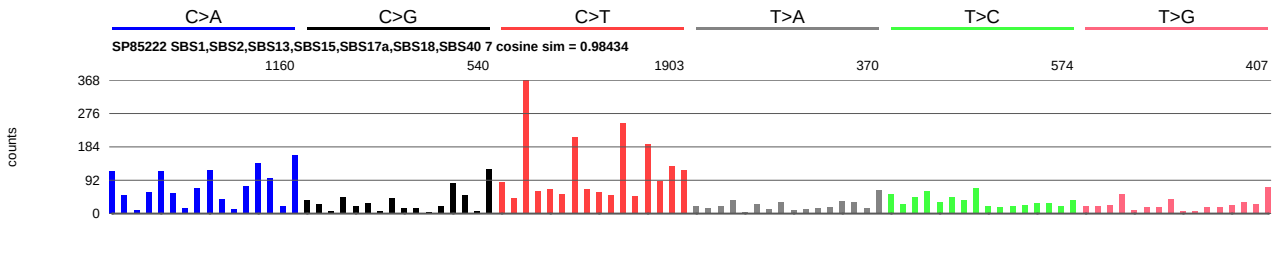

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98437
1153
566
1843
379
634
377
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98436
1124
577
1775
460
644
374
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS40 7 cosine sim = 0.98436
1152
528
1888
349
640
397
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98435
1155
535
1893
357
591
427
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98435
1163
531
1931
361
540
425
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS40 7 cosine sim = 0.98434
1160
540
1903
370
574
407
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
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SBS17a
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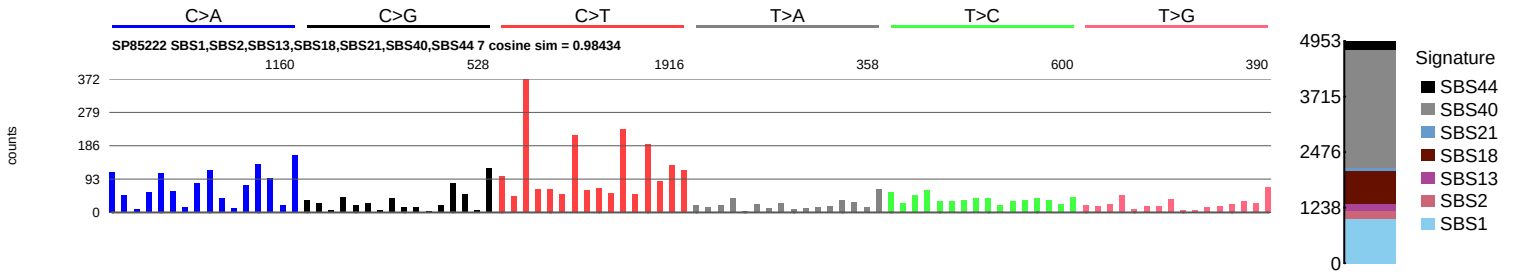
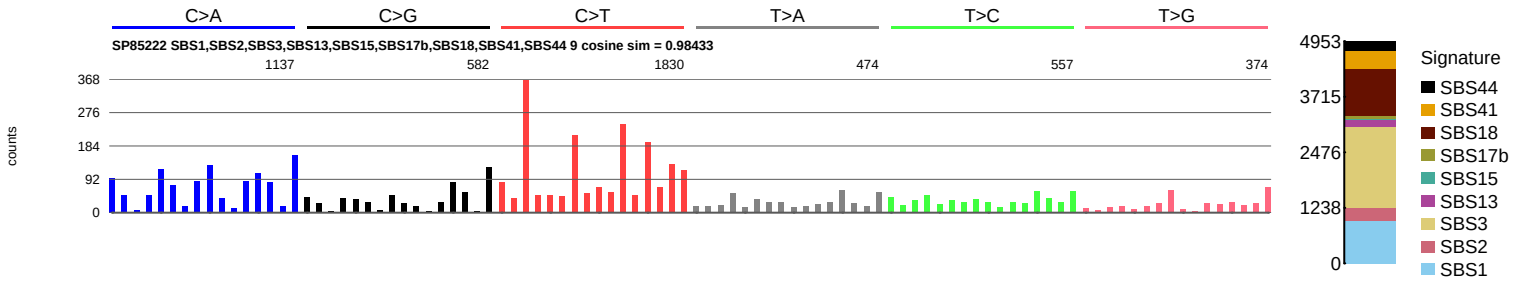
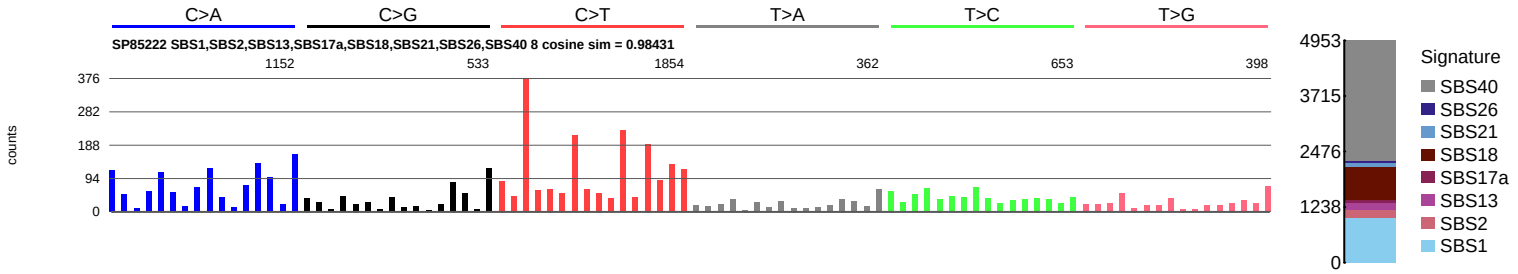
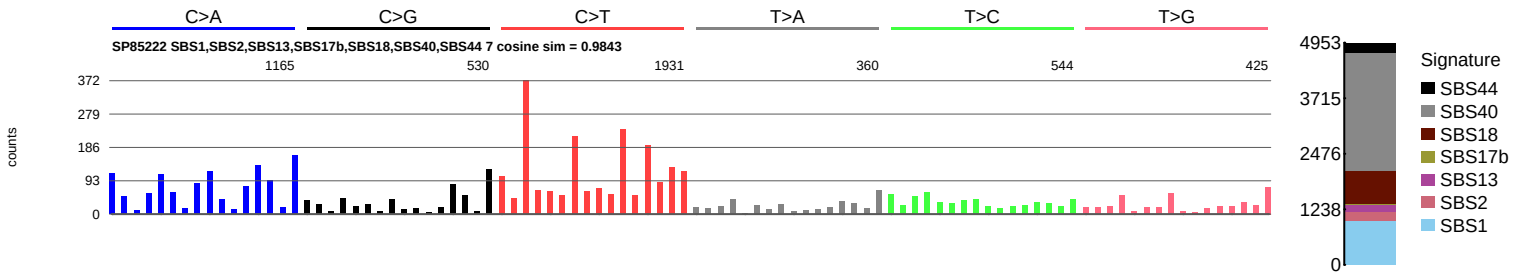
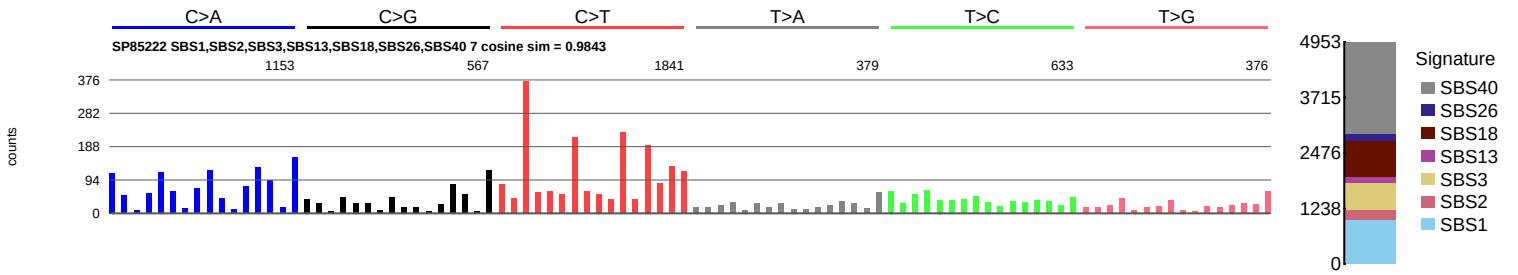
SBS13
SBS2
SBS1

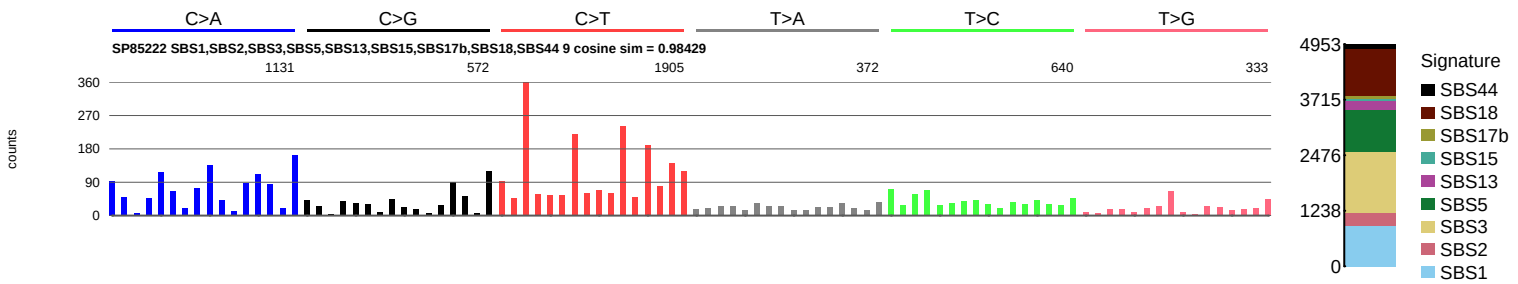

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98434
1160
528
1916
358
600
390
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS13
SBS2
SBS1

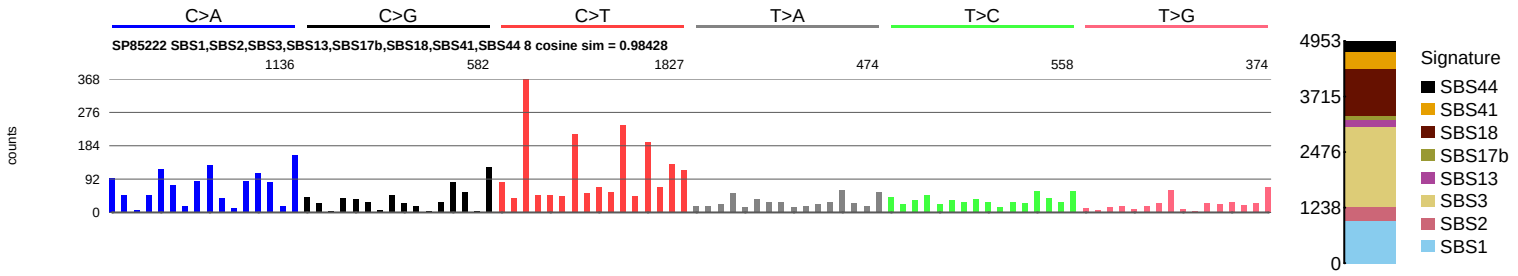

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98433
1137
582
1830
474
557
374
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

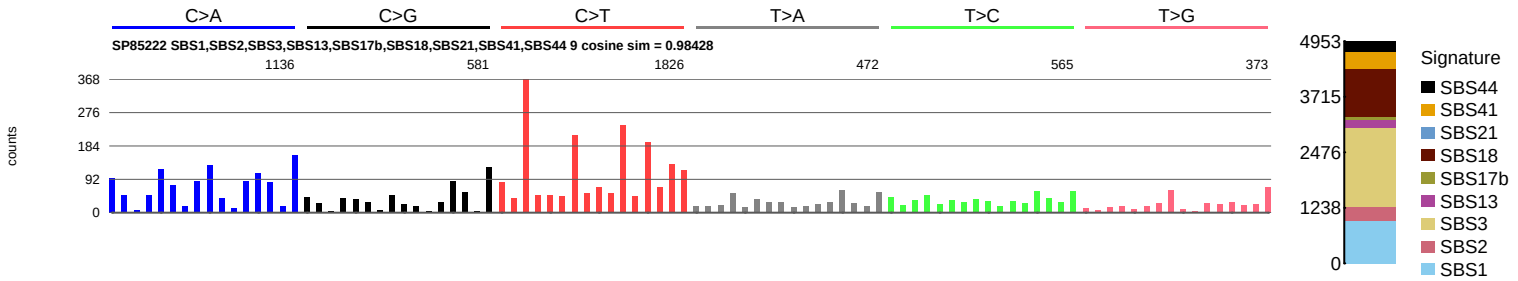

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98431
1152
533
1854
362
653
398
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1

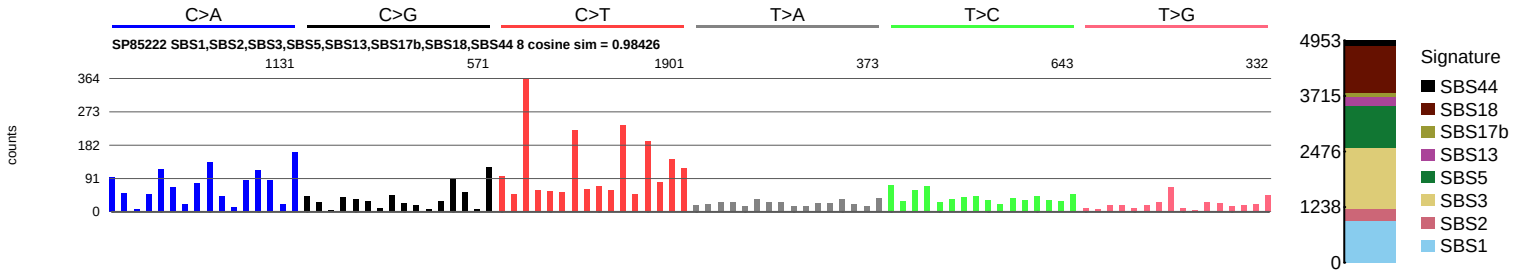

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.9843
1165
530
1931
360
544
425
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS2
SBS1

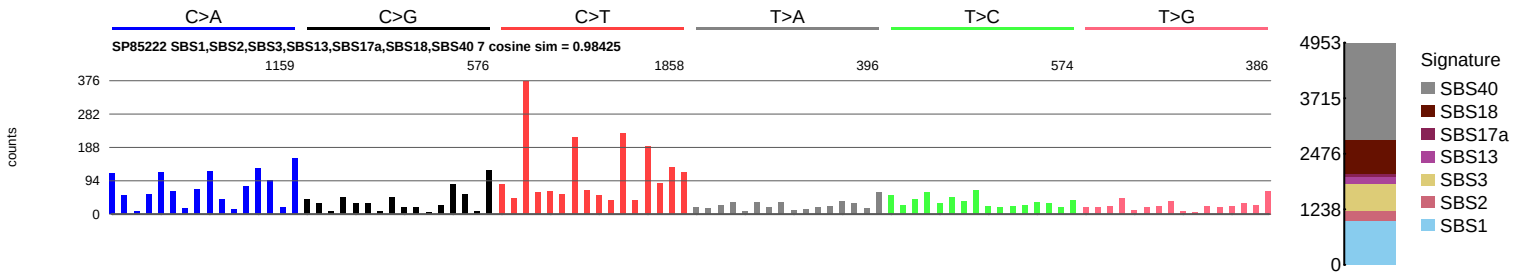

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS26,SBS40 7 cosine sim = 0.9843
1153
567
1841
379
633
376
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS13
SBS3
SBS2
SBS1

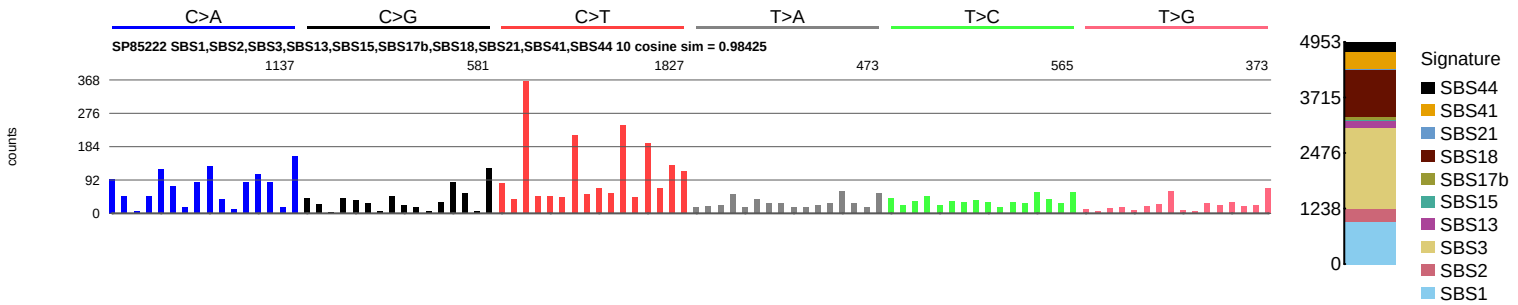

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS44 9 cosine sim = 0.98429
1131
572
1905
372
640
333
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS44
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

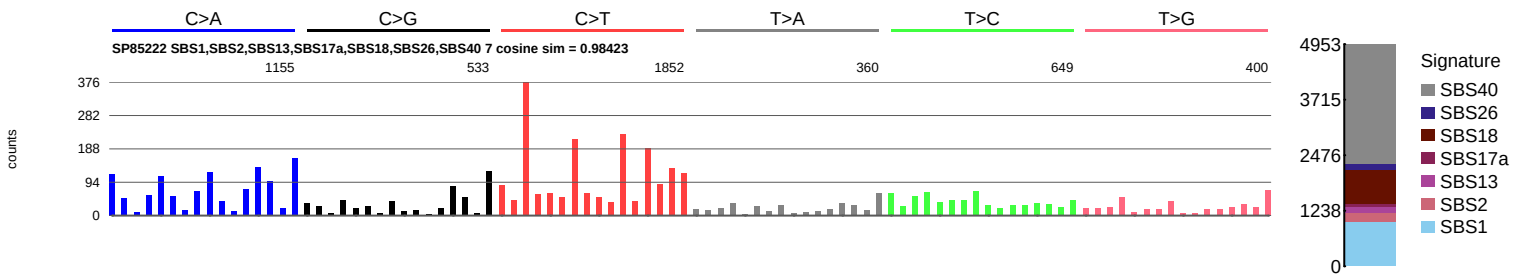

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98428
1136
582
1827
474
558
374
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

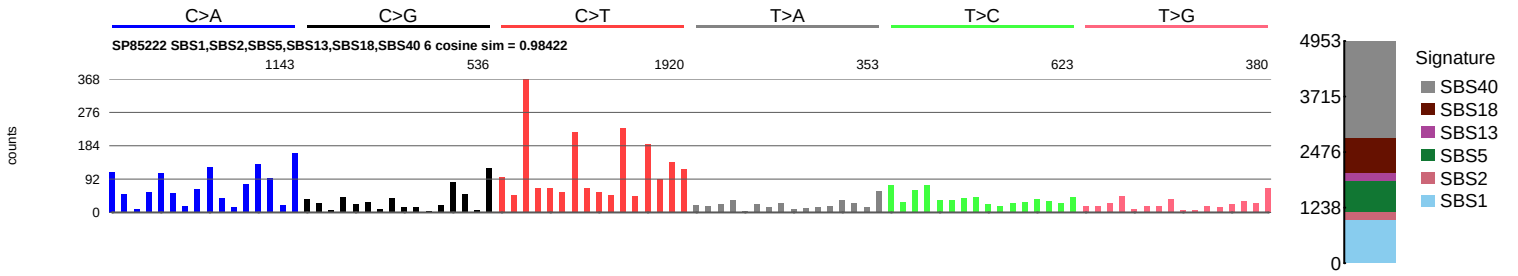

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98428
1136
581
1826
472
565
373
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS44 8 cosine sim = 0.98426
1131
571
1901
373
643
332
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS40 7 cosine sim = 0.98425
1159
576
1858
396
574
386
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98425
1137
581
1827
473
565
373
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS26,SBS40 7 cosine sim = 0.98423
1155
533
1852
360
649
400
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40 6 cosine sim = 0.98422
1143
536
1920
353
623
380
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
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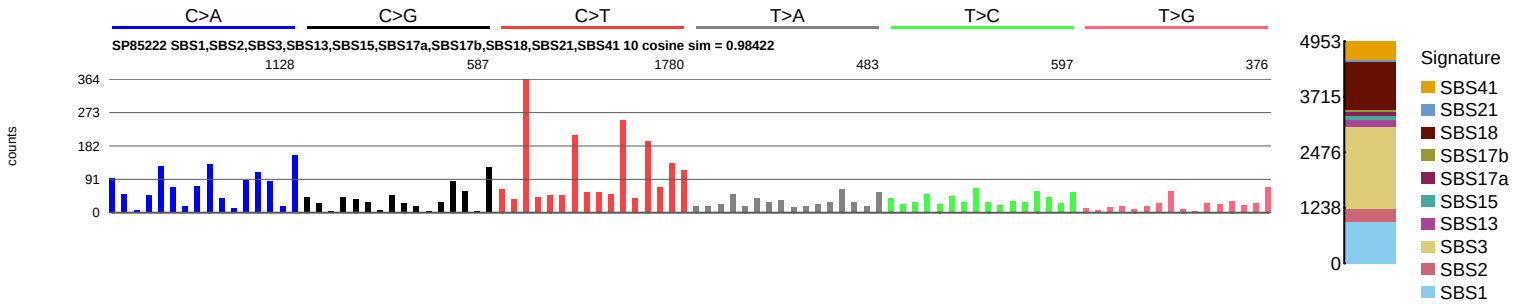
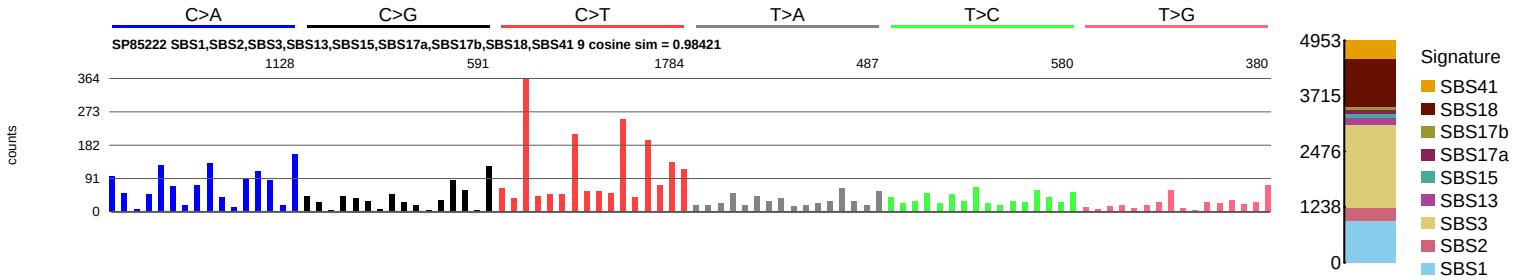
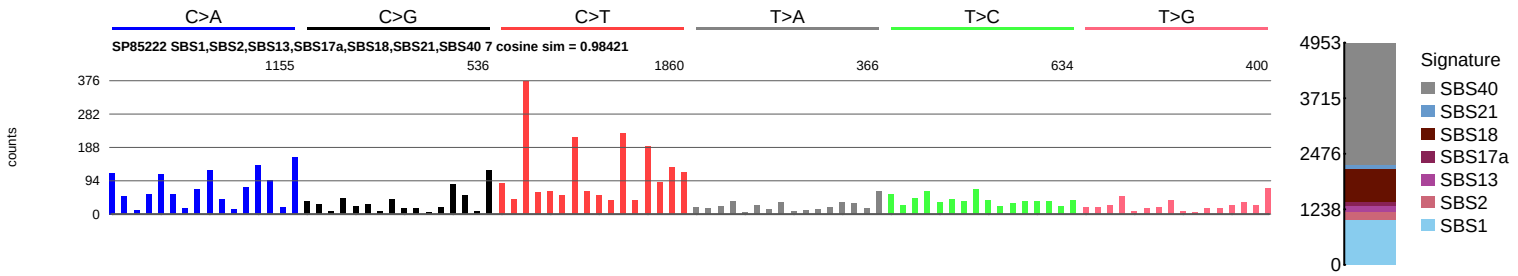
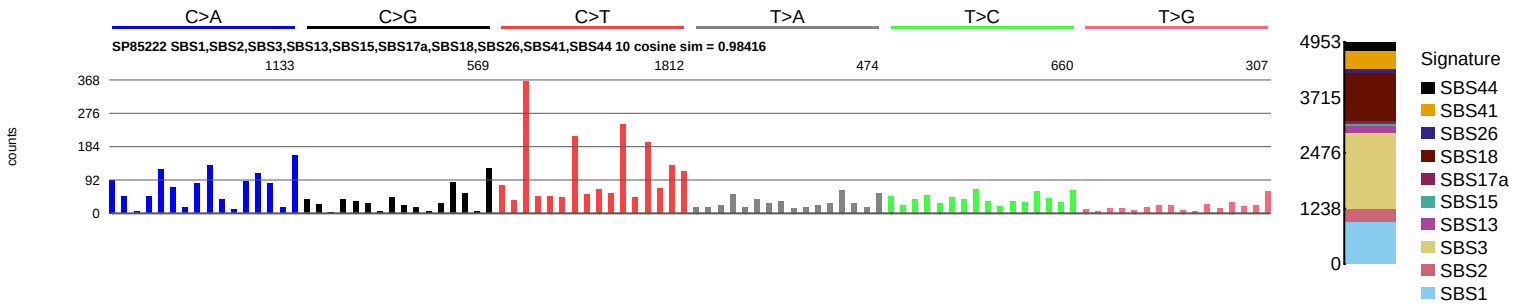
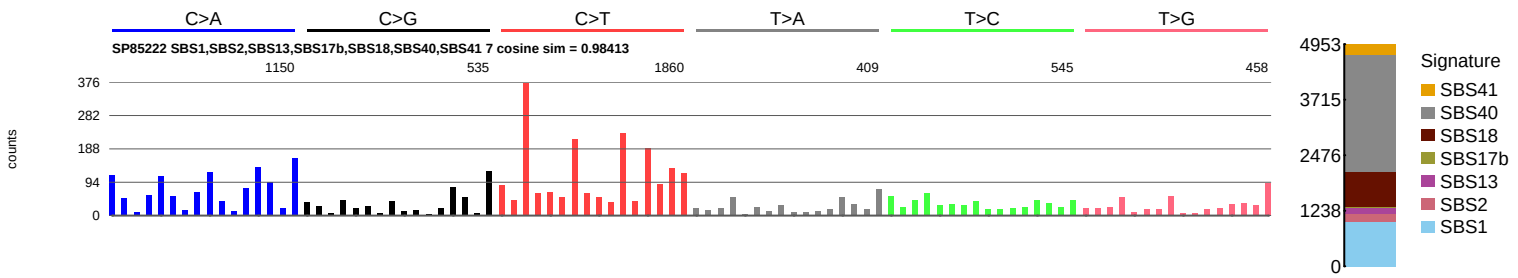
SBS5
SBS2
SBS1

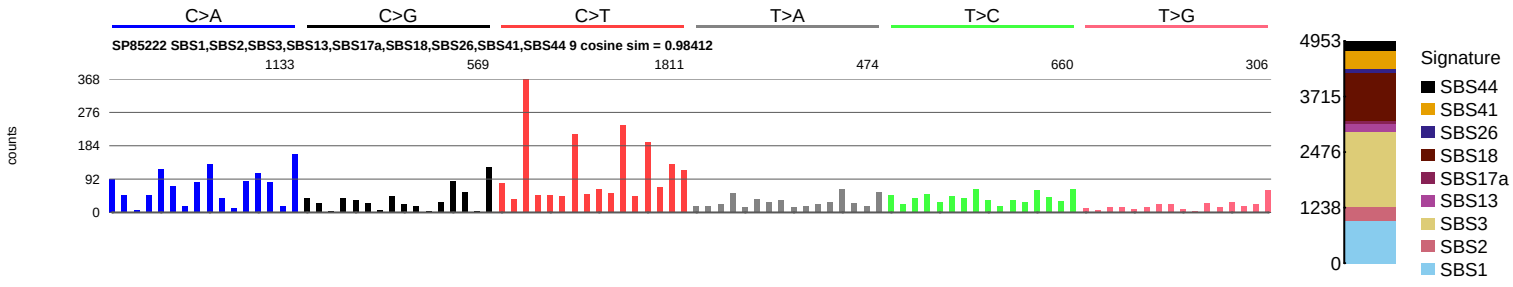

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98422
1128
587
1780
483
597
376
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

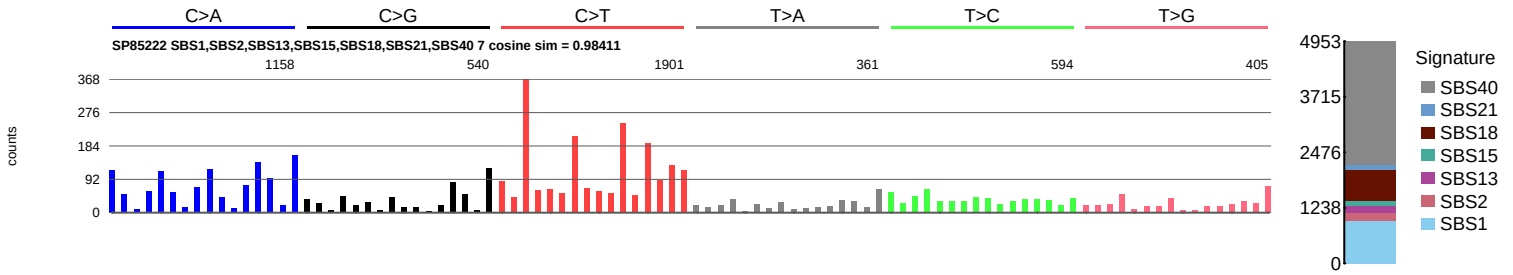

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98421
1128
591
1784
487
580
380
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS40 7 cosine sim = 0.98421
1155
536
1860
366
634
400
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98416
1133
569
1812
474
660
307
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98413
1150
535
1860
409
545
458
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98412
1133
569
1811
474
660
306
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS40 7 cosine sim = 0.98411
1158
540
1901
361
594
405
counts
368
276
184
92
0
4953
3715
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SBS40
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SBS18
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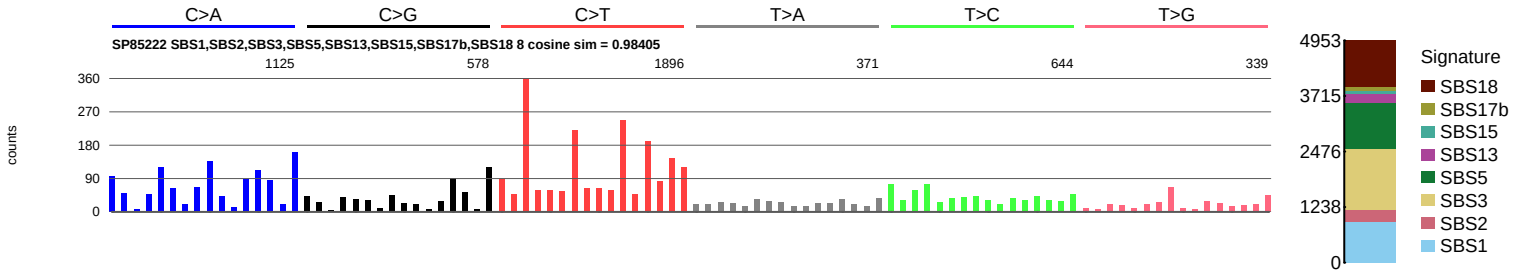
SBS2
SBS1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C>G
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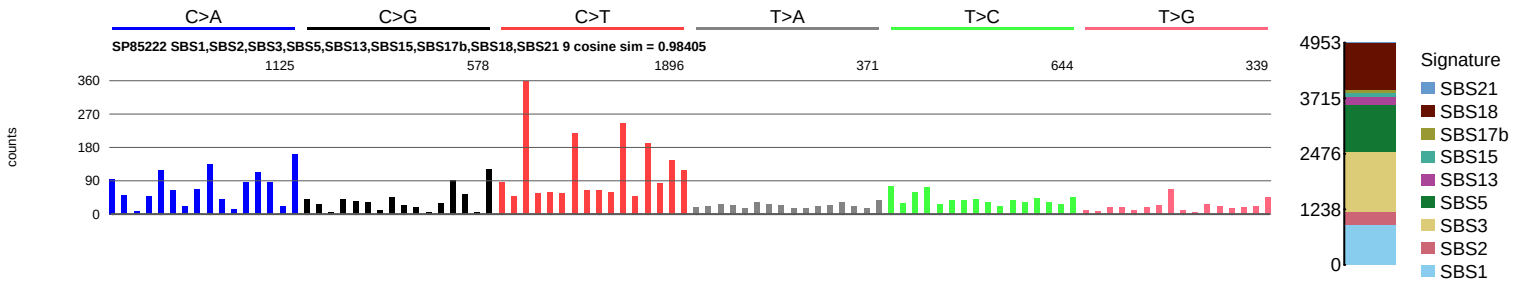
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18 8 cosine sim = 0.98405
1125
578
1896
371
644
339
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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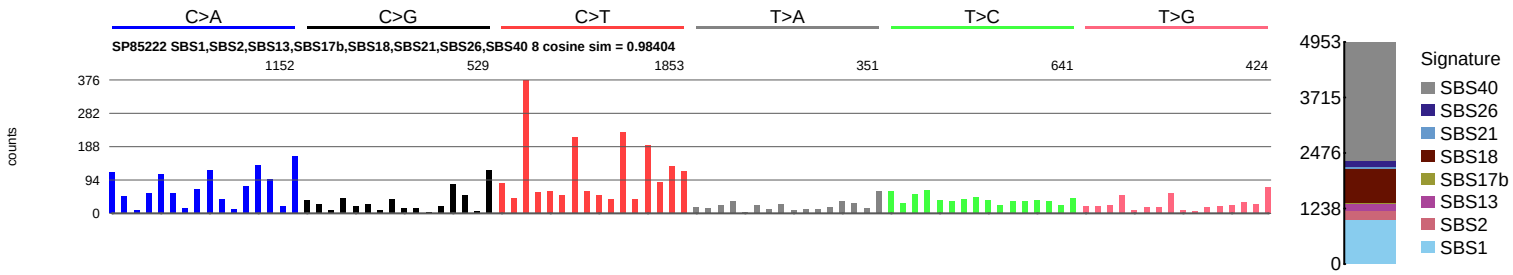
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 9 cosine sim = 0.98405
1125
578
1896
371
644
339
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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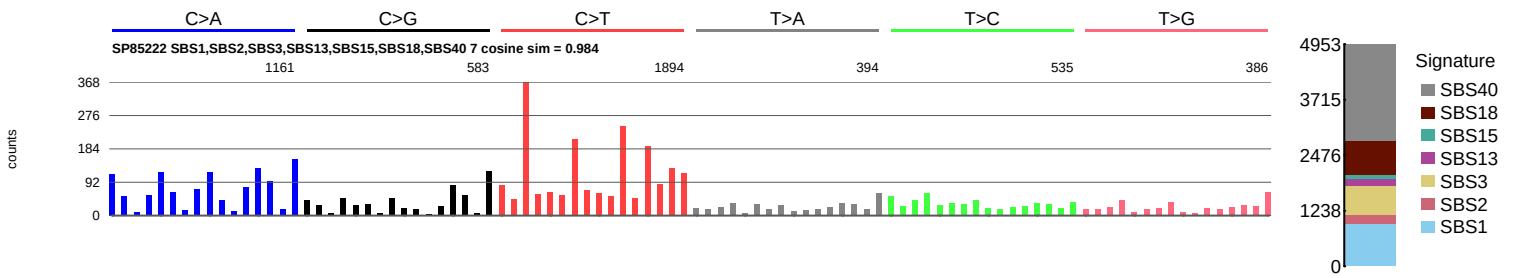
T>G
SP85222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98404
1152
529
1853
351
641
424
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1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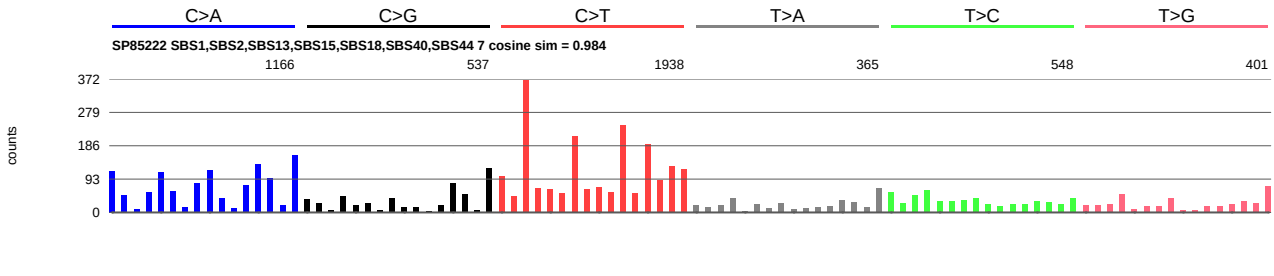
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS40 7 cosine sim = 0.984
1161
583
1894
394
535
386
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS40,SBS44 7 cosine sim = 0.984
1166
537
1938
365
548
401
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
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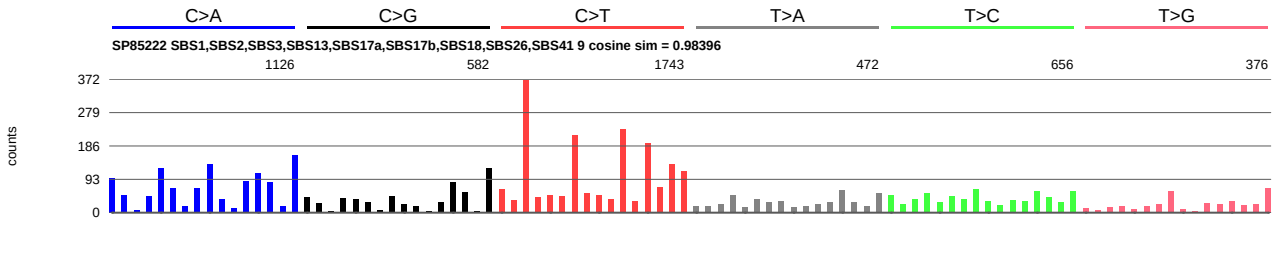
SBS2
SBS1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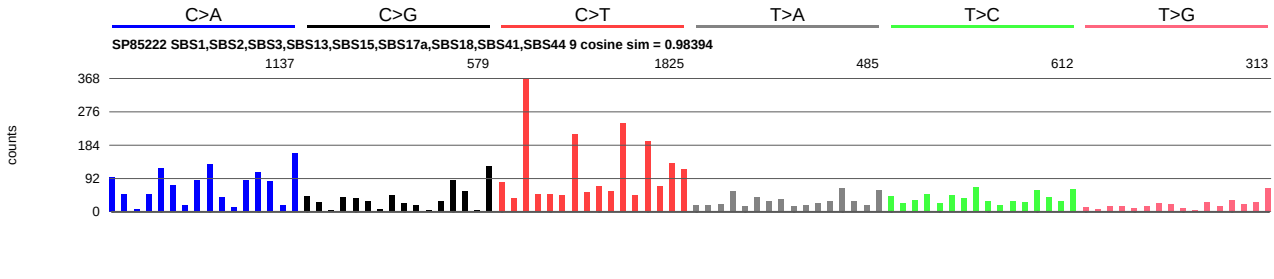
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98396
1126
582
1743
472
656
376
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS41,SBS44 9 cosine sim = 0.98394
1137
579
1825
485
612
313
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS15
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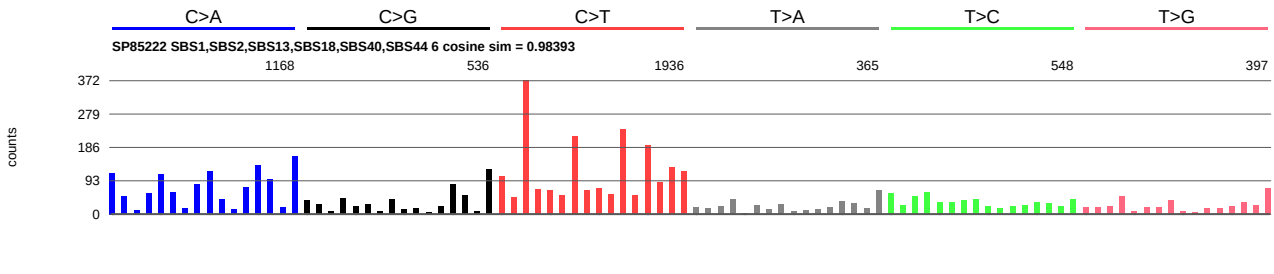
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS18,SBS40,SBS44 6 cosine sim = 0.98393
1168
536
1936
365
548
397
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
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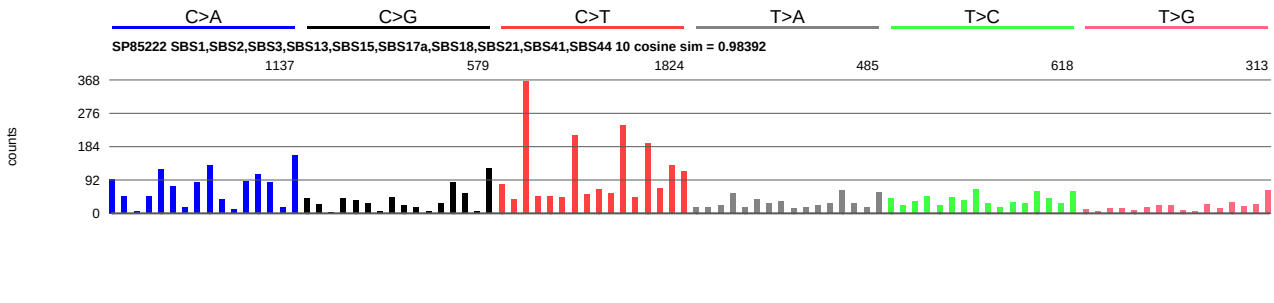
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98392
1137
579
1824
485
618
313
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
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SBS18
SBS17a
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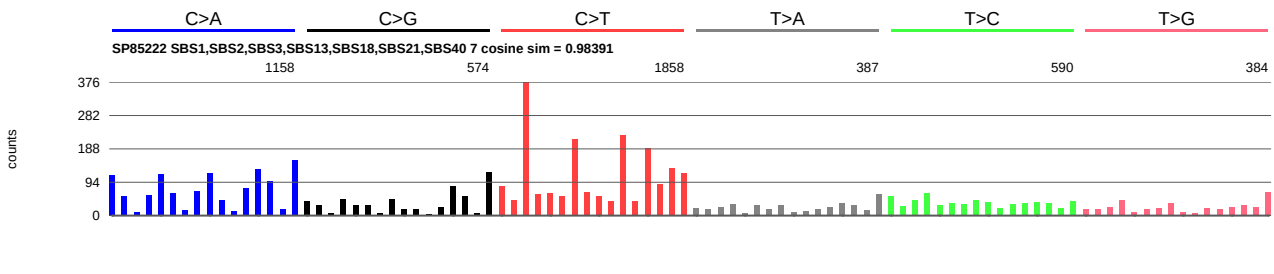
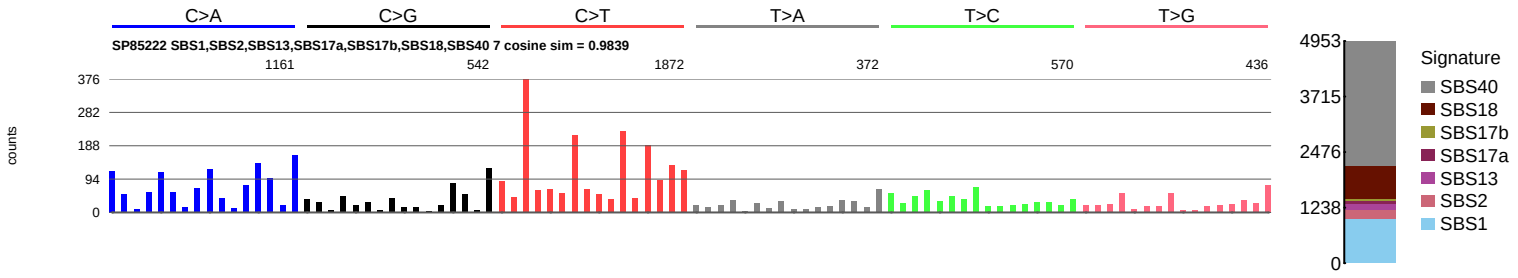
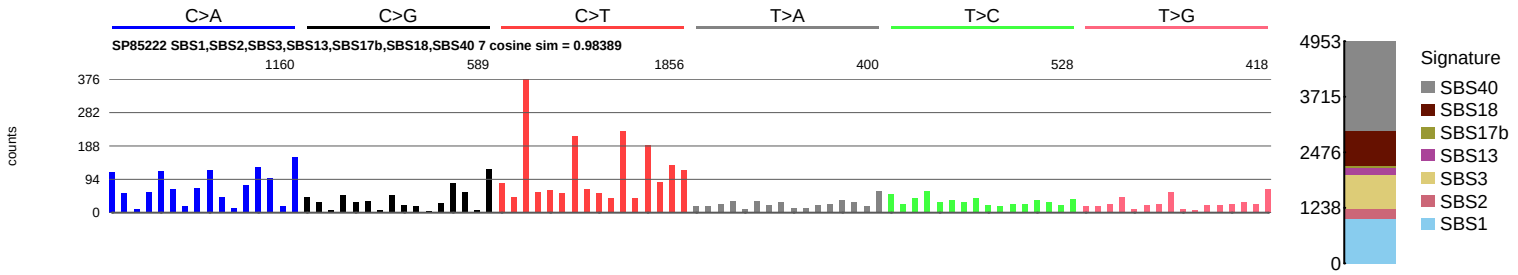
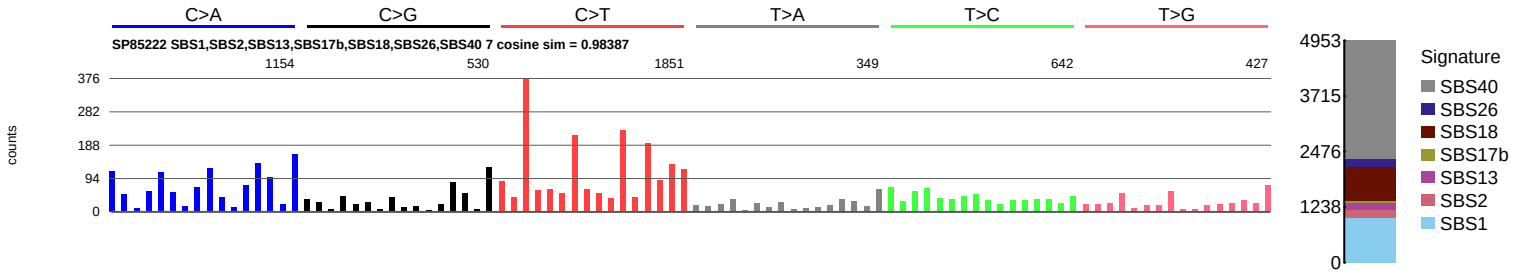
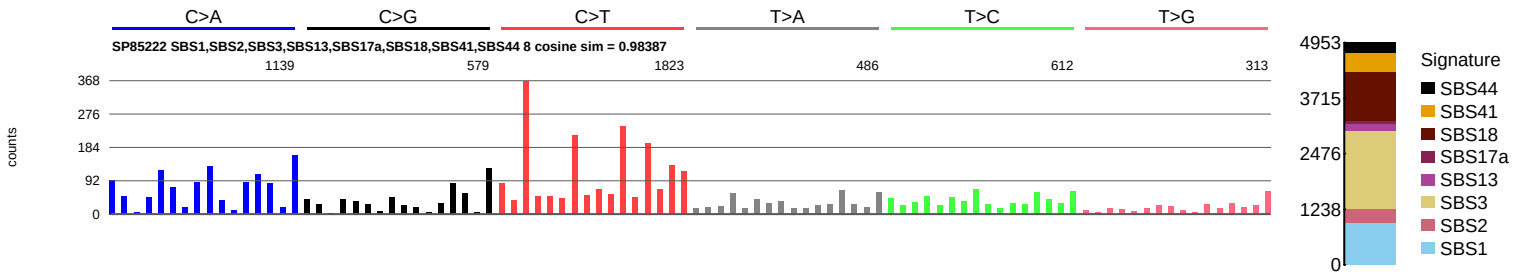
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS40 7 cosine sim = 0.98391
1158
574
1858
387
590
384
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.9839
1161
542
1872
372
570
436
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS40 7 cosine sim = 0.98389
1160
589
1856
400
528
418
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98387
1154
530
1851
349
642
427
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS41,SBS44 8 cosine sim = 0.98387
1139
579
1823
486
612
313
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
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SBS17a
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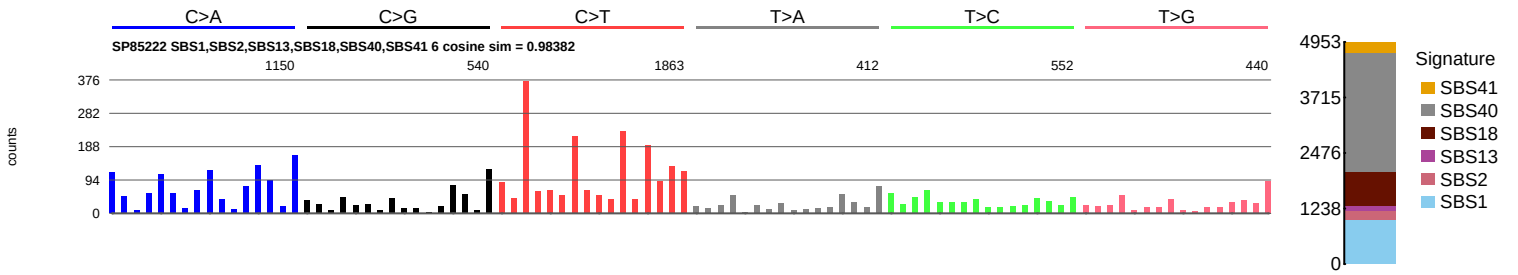
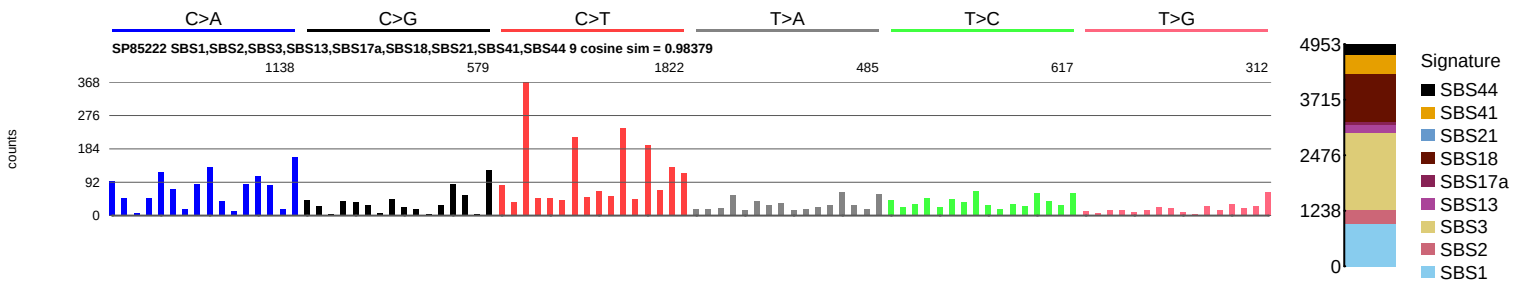
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS18,SBS40,SBS41 6 cosine sim = 0.98382
1150
540
1863
412
552
440
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS13
SBS2
SBS1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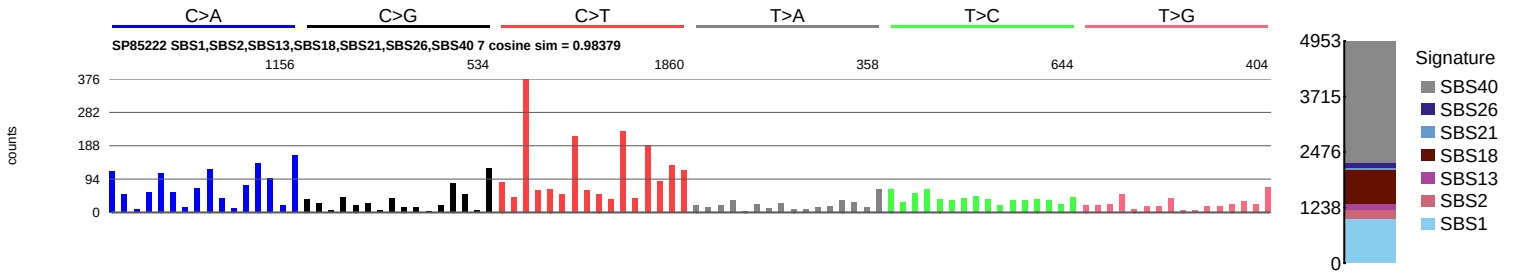
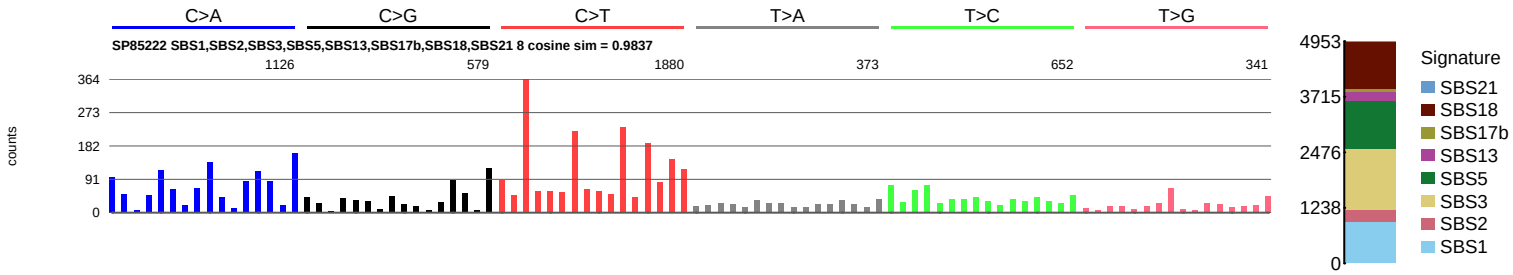
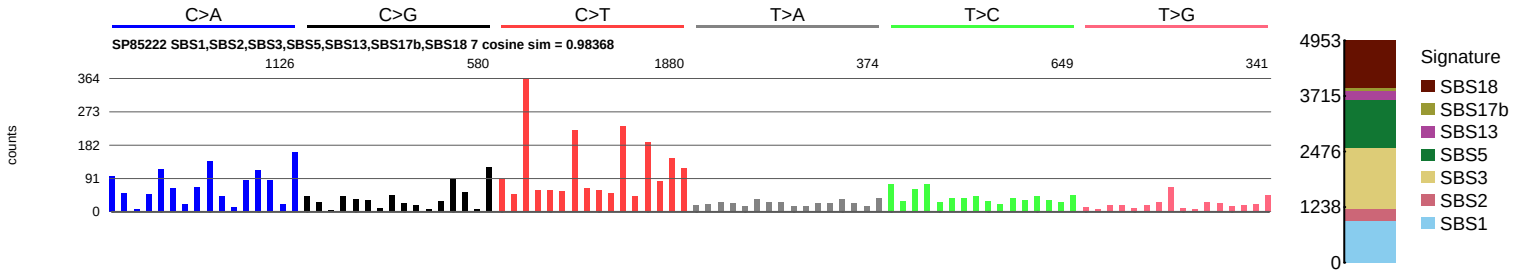
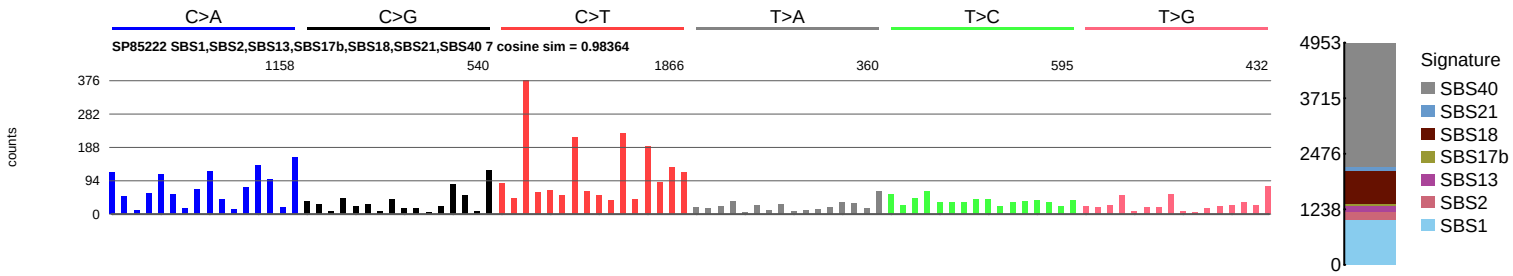
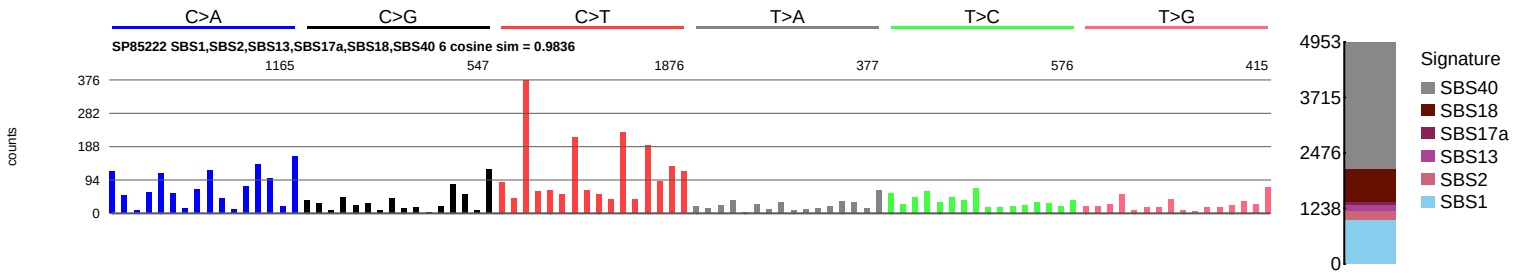
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98379
1138
579
1822
485
617
312
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

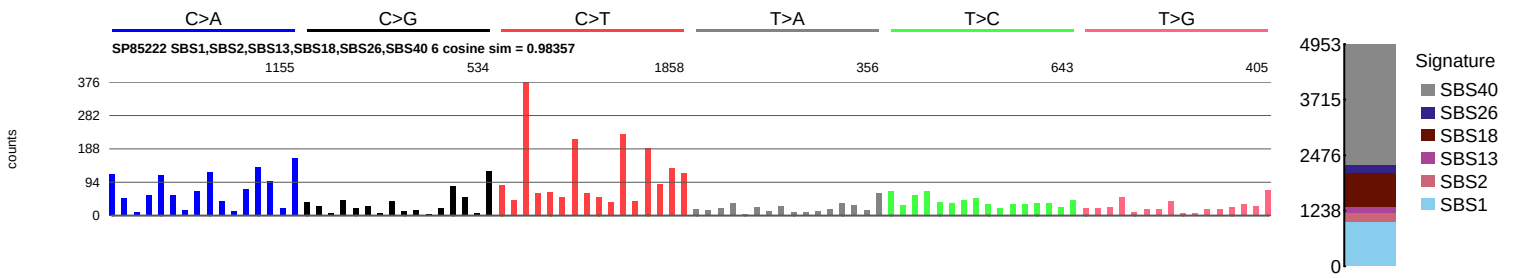

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98379
1156
534
1860
358
644
404
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS13
SBS2
SBS1

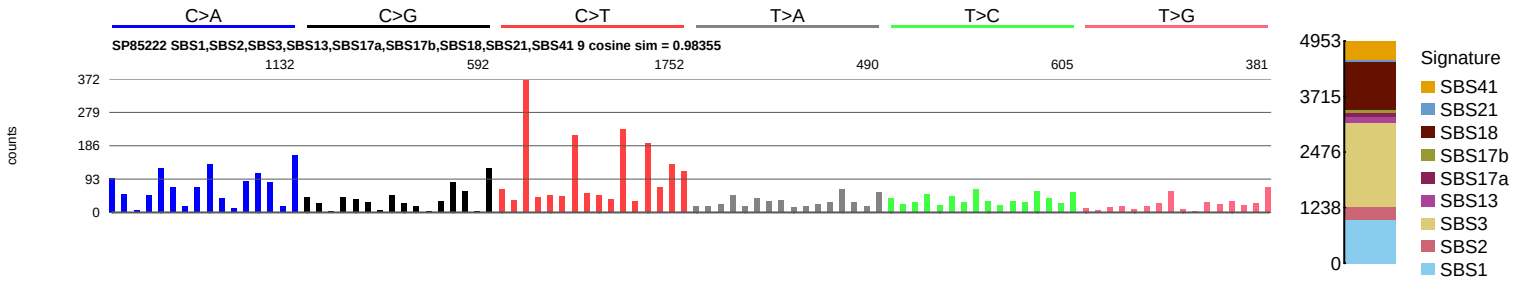

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21 8 cosine sim = 0.9837
1126
579
1880
373
652
341
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18 7 cosine sim = 0.98368
1126
580
1880
374
649
341
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98364
1158
540
1866
360
595
432
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS40 6 cosine sim = 0.9836
1165
547
1876
377
576
415
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS18,SBS26,SBS40 6 cosine sim = 0.98357
1155
534
1858
356
643
405
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98355
1132
592
1752
490
605
381
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
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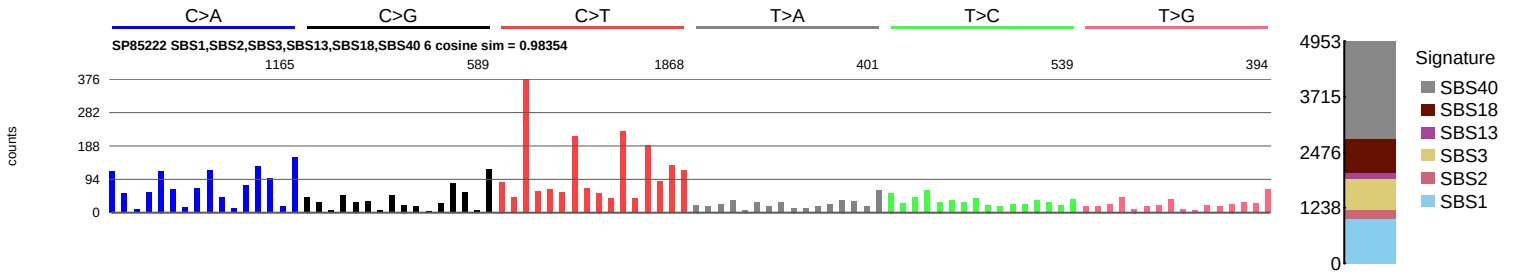
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS40 6 cosine sim = 0.98354
1165
589
1868
401
539
394
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
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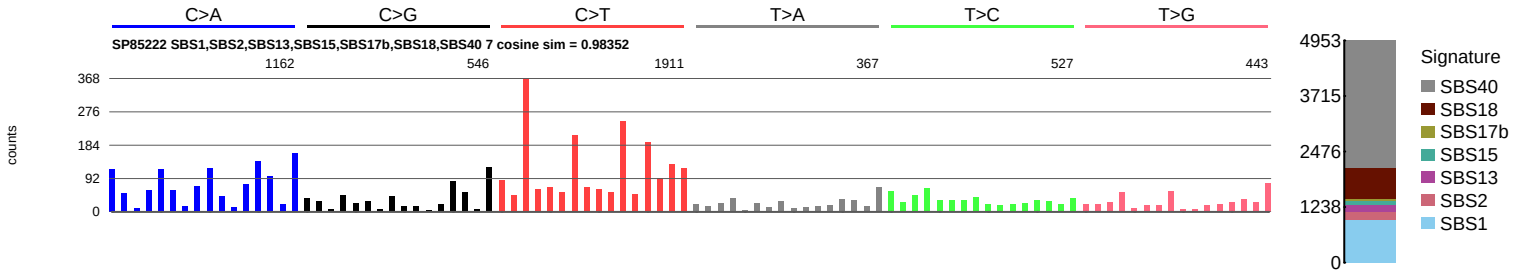
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS40 7 cosine sim = 0.98352
1162
546
1911
367
527
443
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
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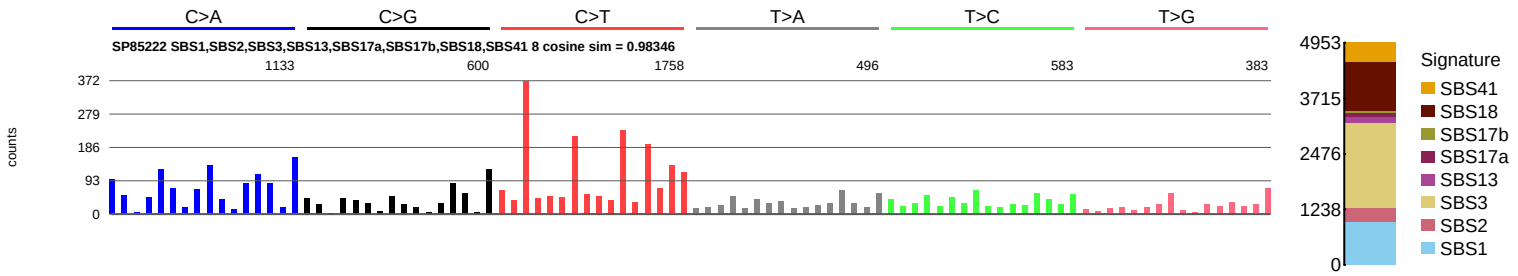
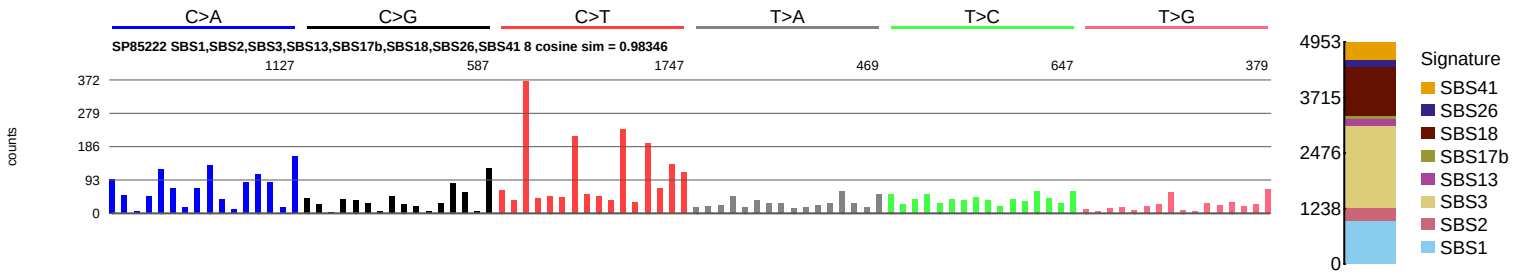
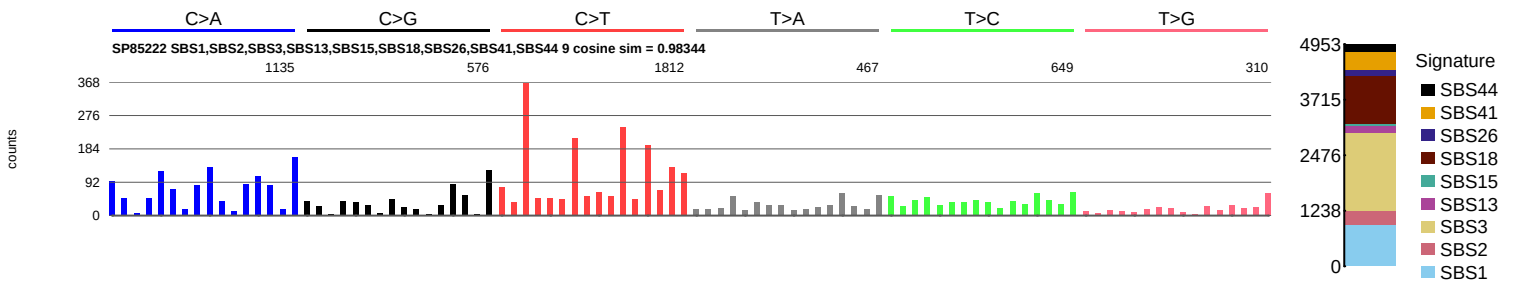
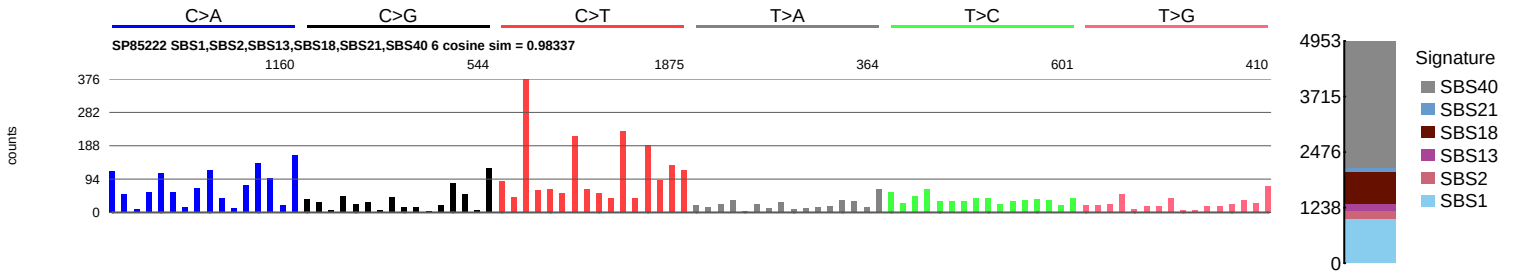
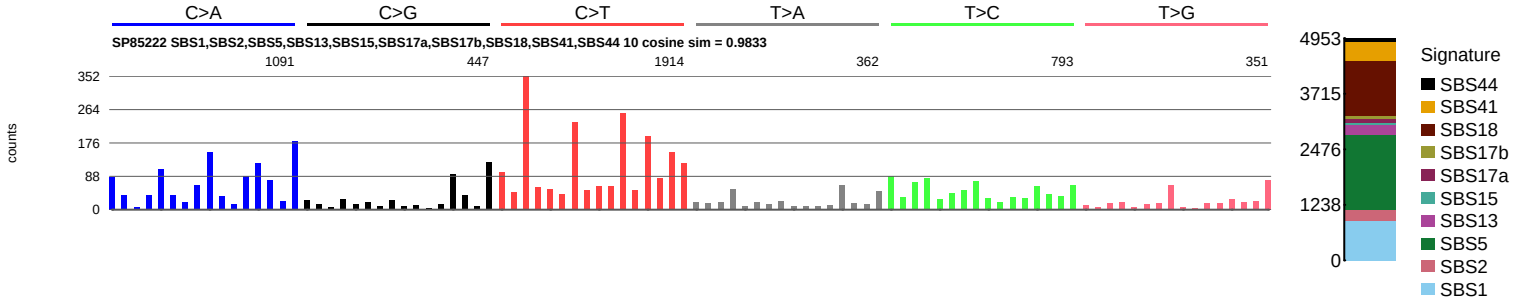
SBS2
SBS1

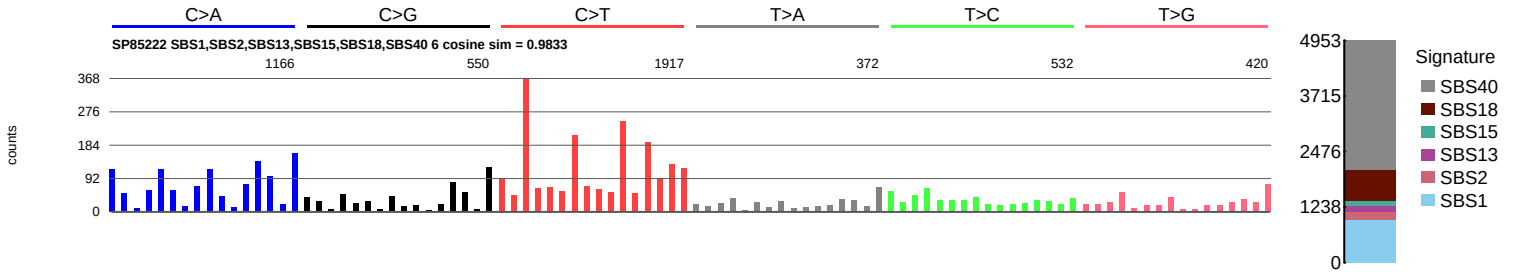

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98346
1133
600
1758
496
583
383
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

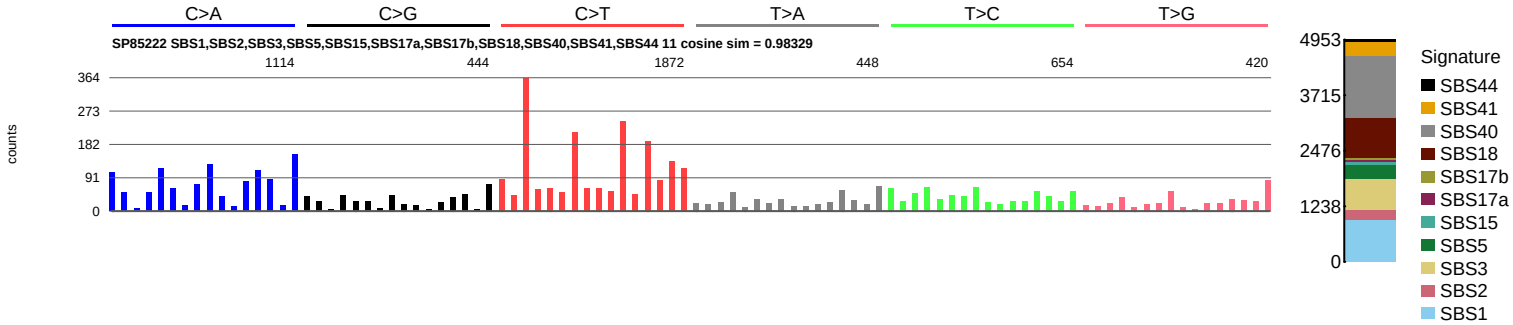

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98346
1127
587
1747
469
647
379
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

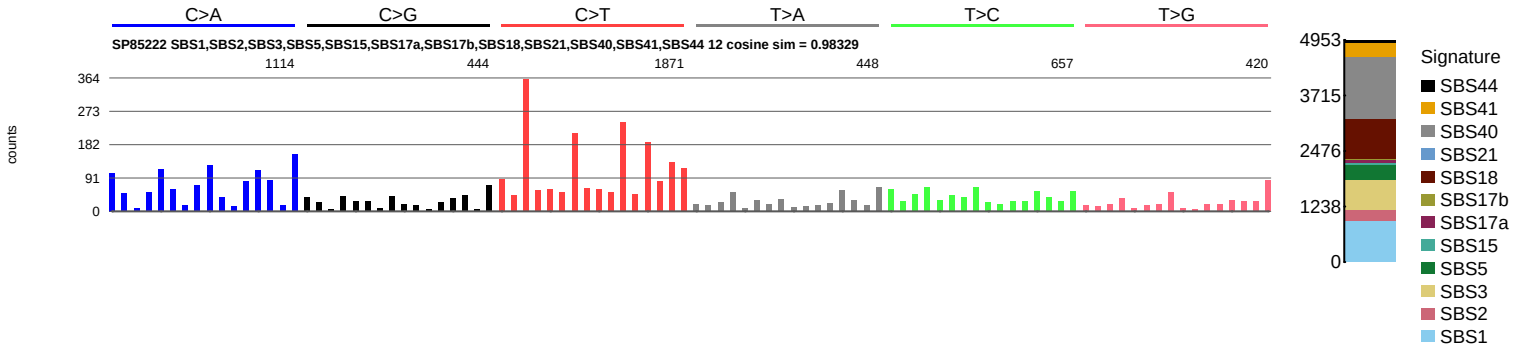

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98344
1135
576
1812
467
649
310
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

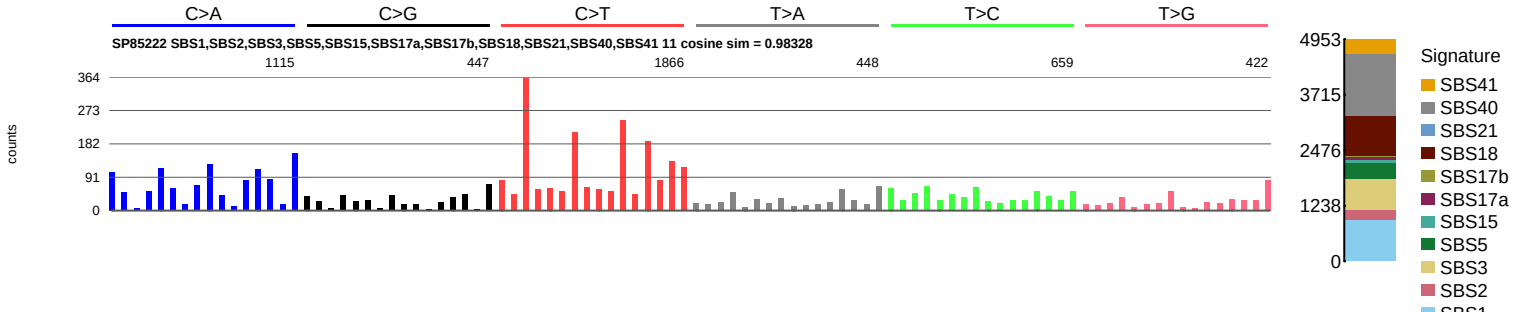

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS18,SBS21,SBS40 6 cosine sim = 0.98337
1160
544
1875
364
601
410
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS13
SBS2
SBS1

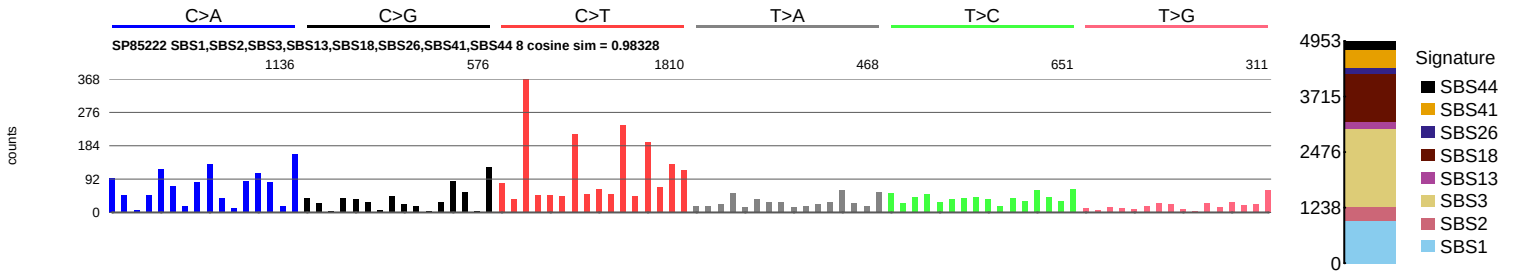

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.9833
1091
447
1914
362
793
351
counts
352
264
176
88
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS40 6 cosine sim = 0.9833
1166
550
1917
372
532
420
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98329
1114
444
1872
448
654
420
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98329
1114
444
1871
448
657
420
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98328
1115
447
1866
448
659
422
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98328
1136
576
1810
468
651
311
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
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SBS26
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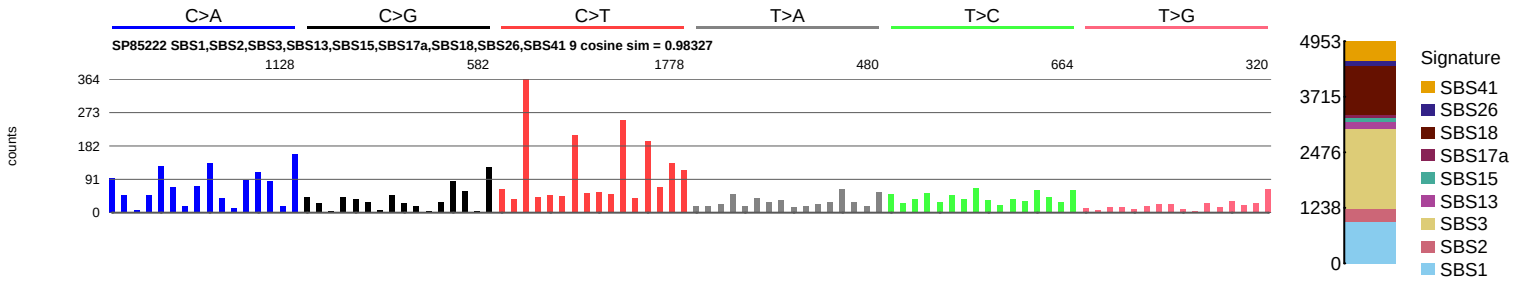
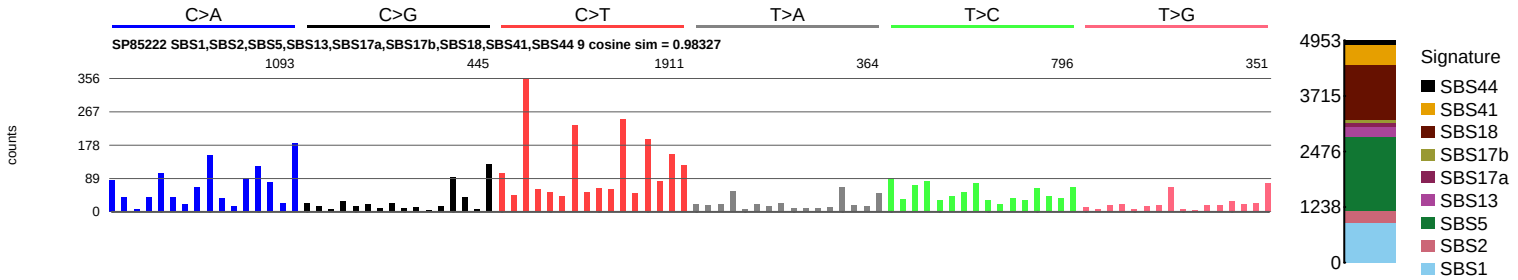
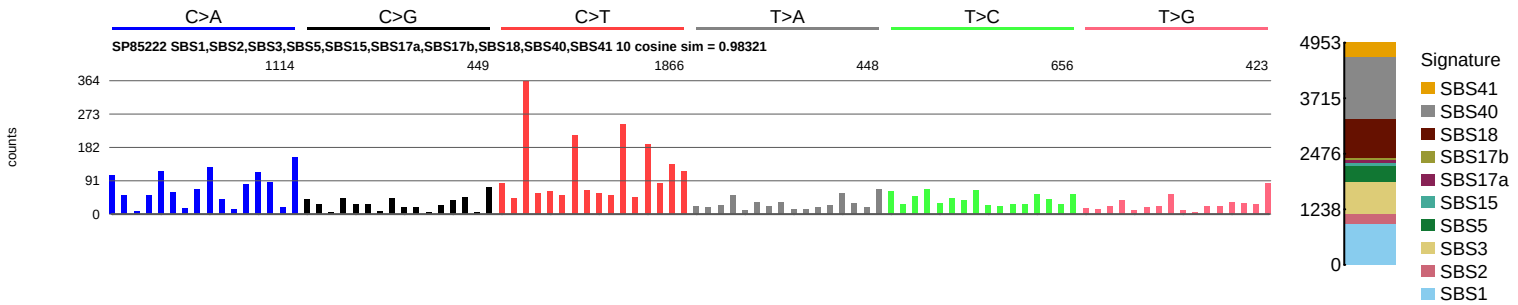
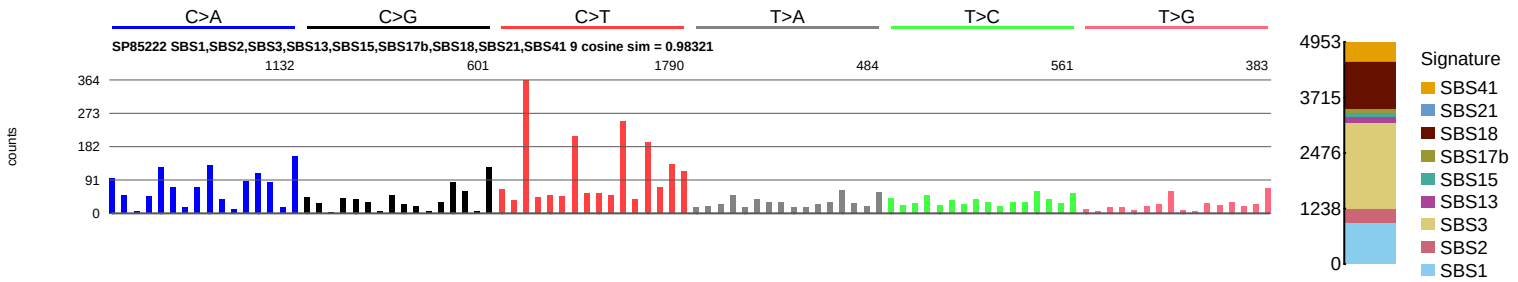
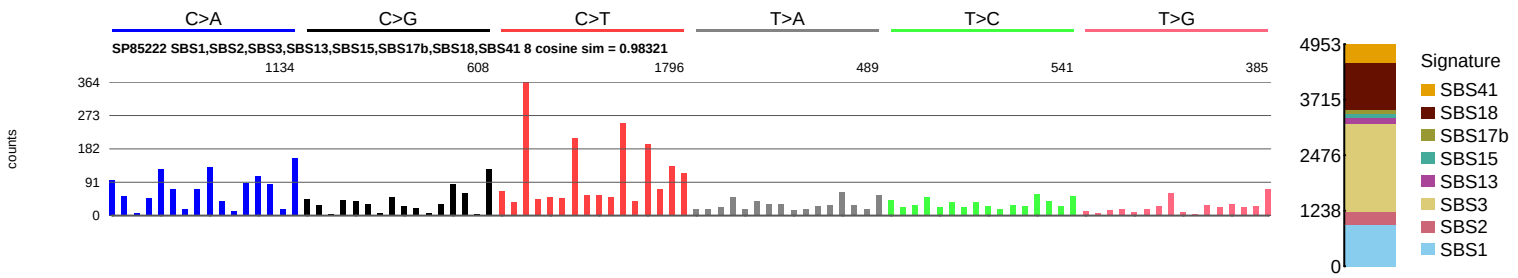
SBS2
SBS1

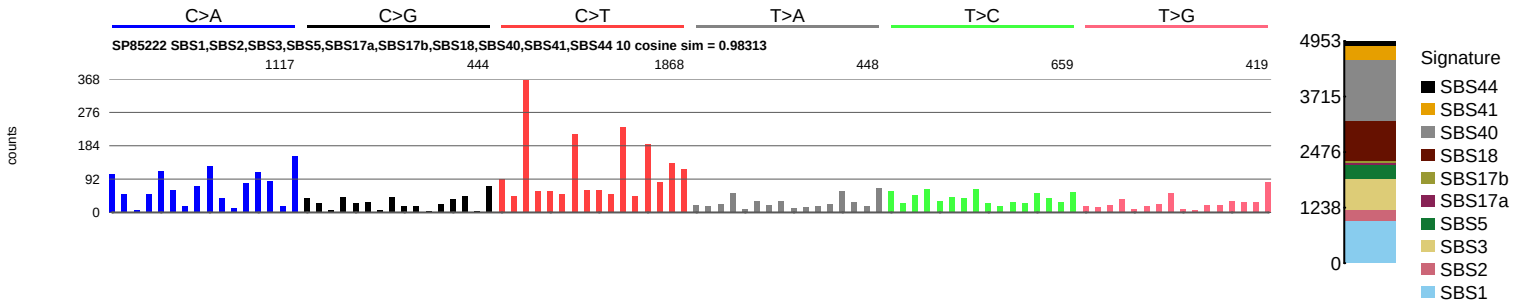

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS41 9 cosine sim = 0.98327
1128
582
1778
480
664
320
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

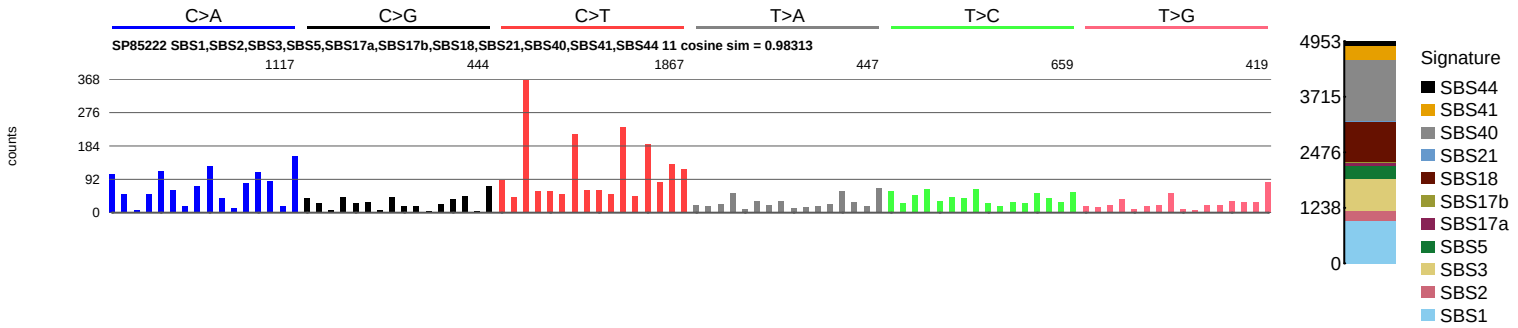

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98327
1093
445
1911
364
796
351
counts
356
267
178
89
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

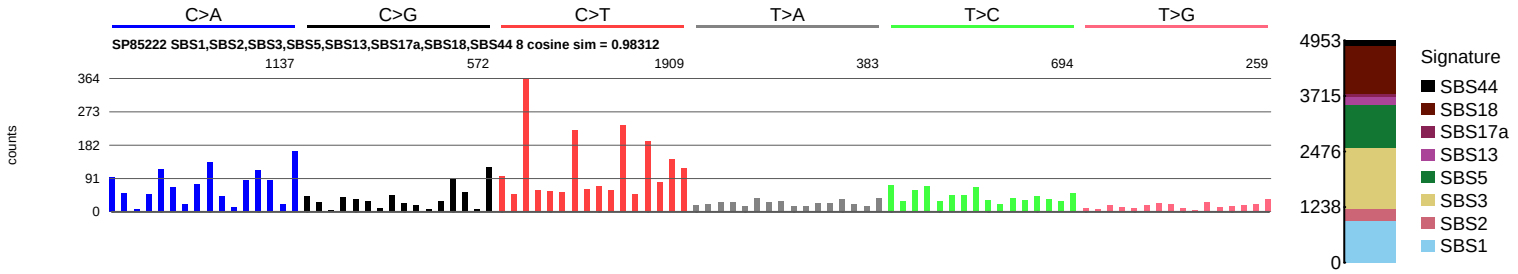

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98321
1114
449
1866
448
656
423
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

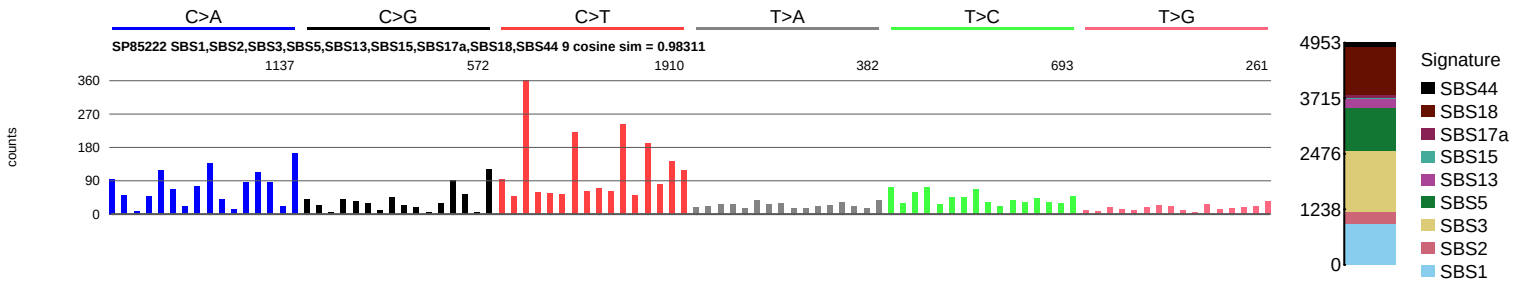

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98321
1132
601
1790
484
561
383
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

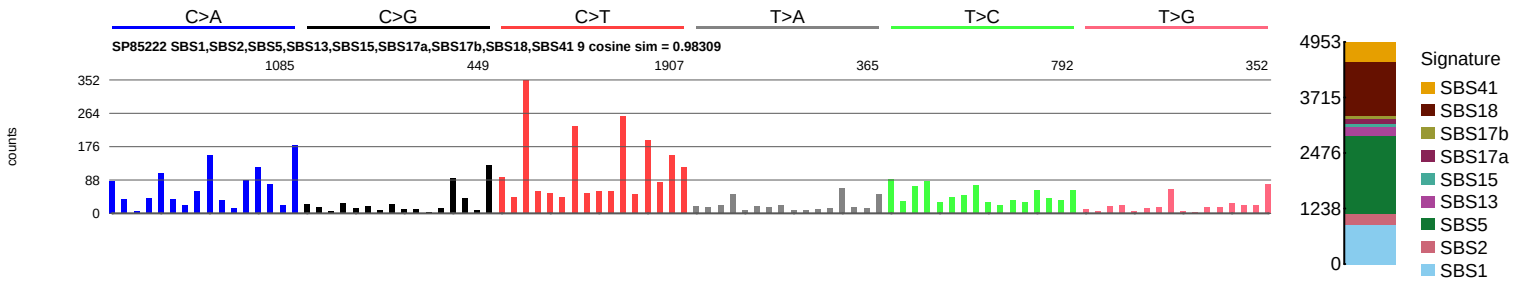

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.98321
1134
608
1796
489
541
385
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

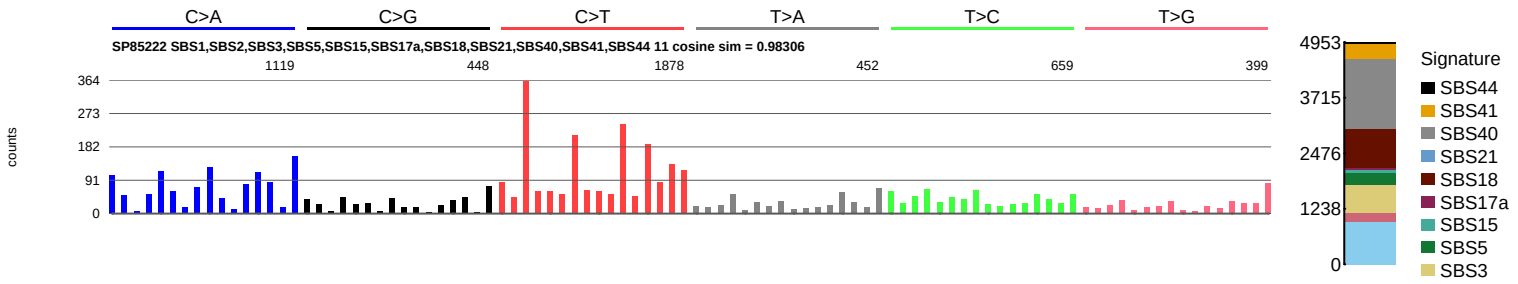

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98313
1117
444
1868
448
659
419
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

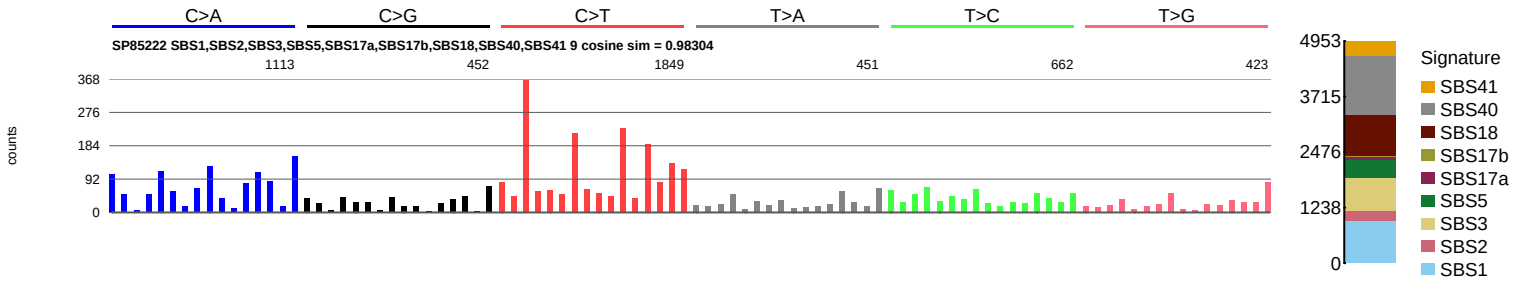

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98313
1117
444
1867
447
659
419
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS44 8 cosine sim = 0.98312
1137
572
1909
383
694
259
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS44 9 cosine sim = 0.98311
1137
572
1910
382
693
261
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS44
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98309
1085
449
1907
365
792
352
counts
352
264
176
88
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98306
1119
448
1878
452
659
399
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98304
1113
452
1849
451
662
423
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
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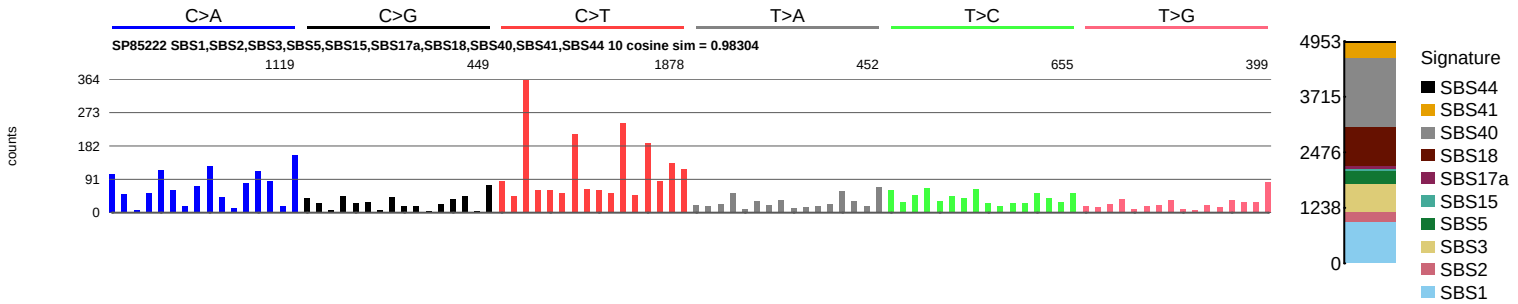
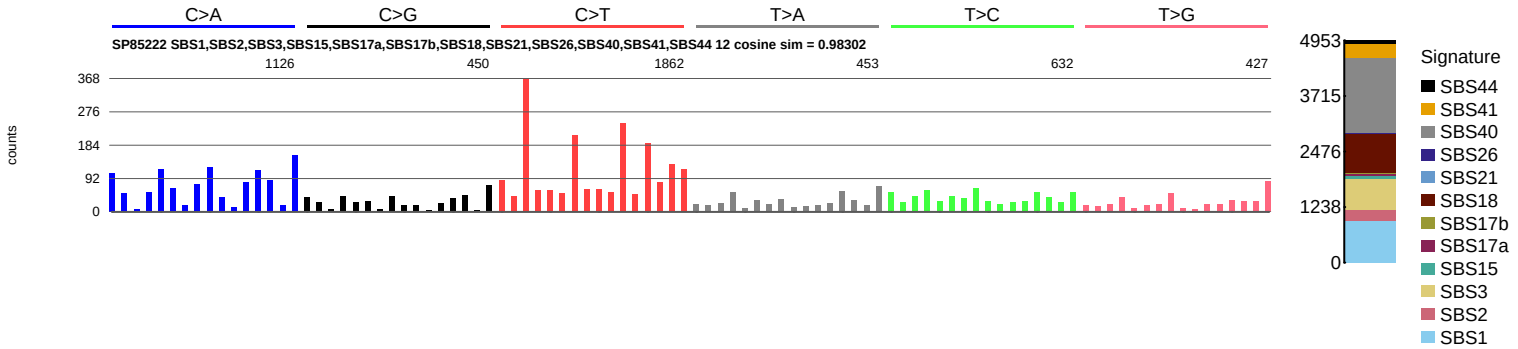
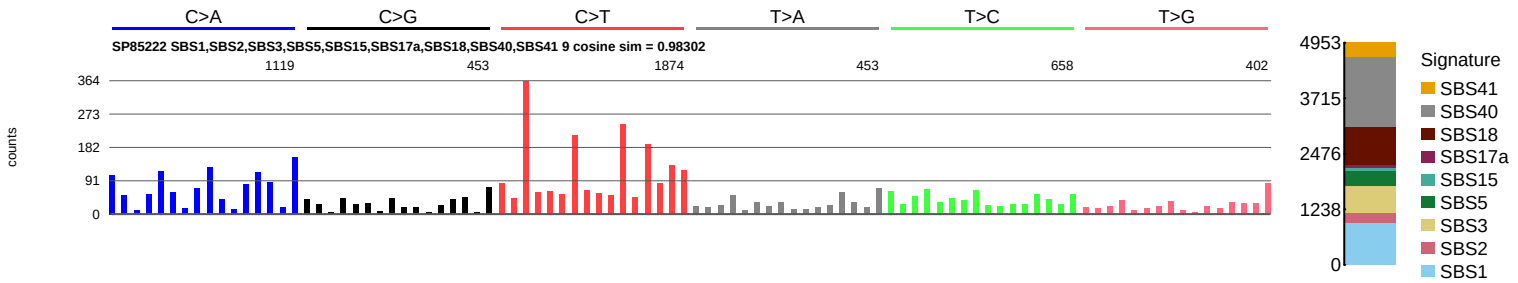
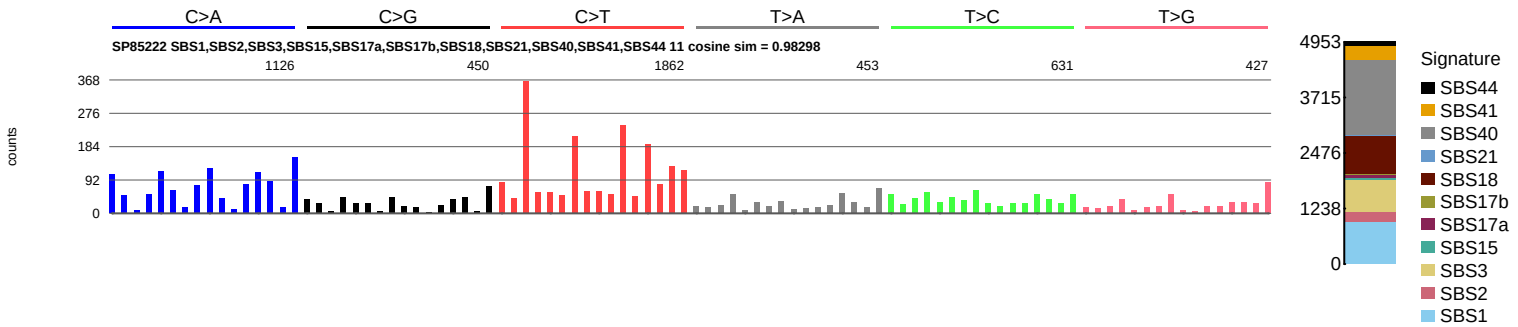
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98304
1119
449
1878
452
655
399
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98302
1126
450
1862
453
632
427
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS40,SBS41 9 cosine sim = 0.98302
1119
453
1874
453
658
402
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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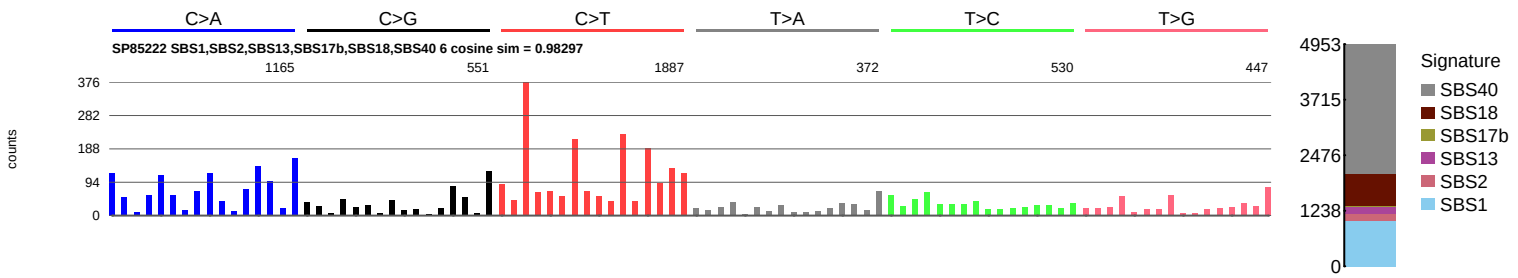
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98298
1126
450
1862
453
631
427
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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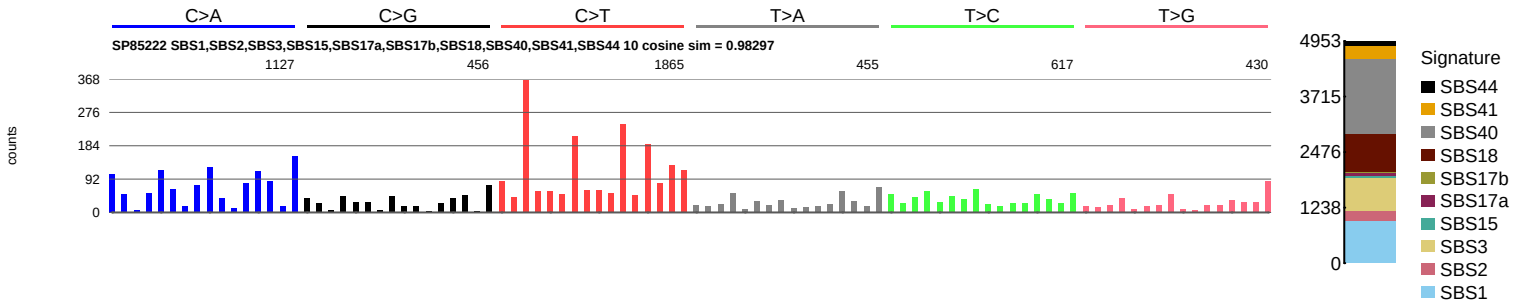
SP85222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS40 6 cosine sim = 0.98297
1165
551
1887
372
530
447
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS2
SBS1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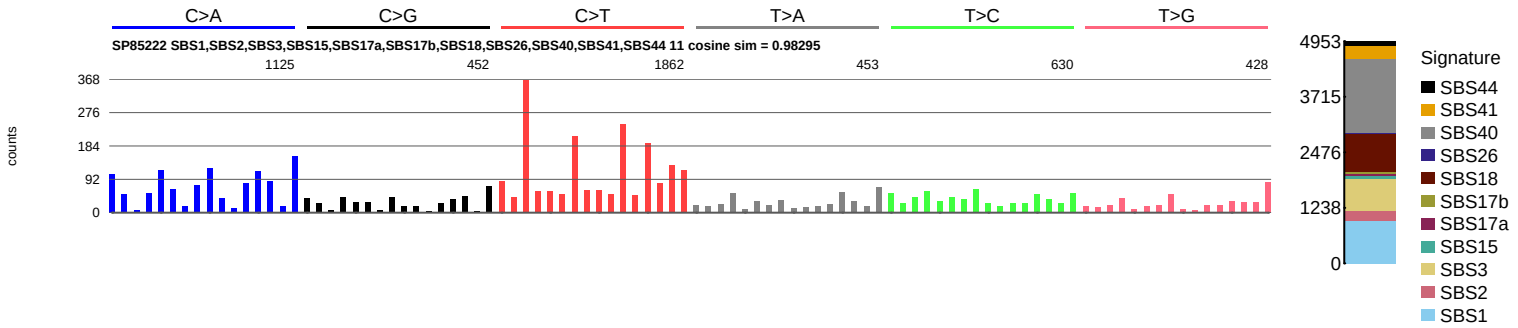
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98297
1127
456
1865
455
617
430
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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T>C
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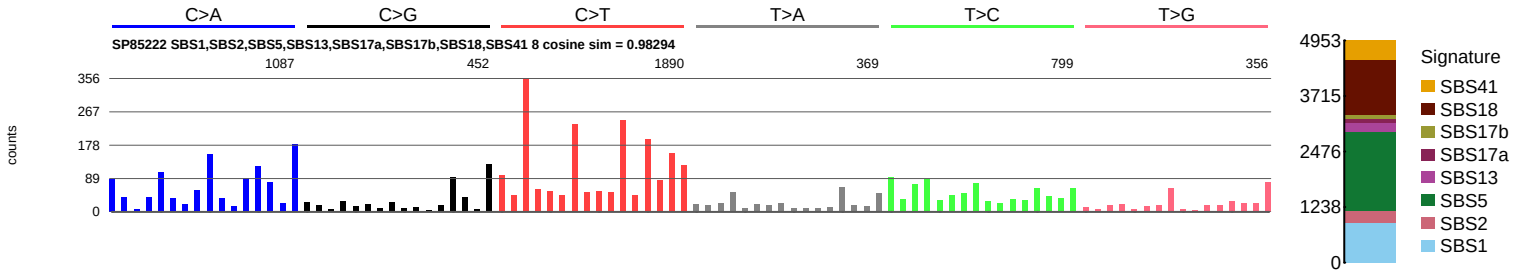
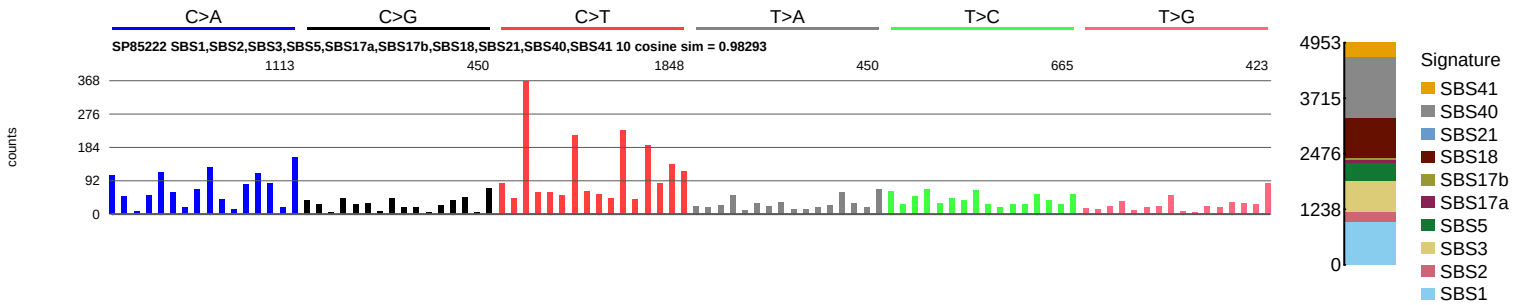
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98295
1125
452
1862
453
630
428
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98294
1087
452
1890
369
799
356
counts
356
267
178
89
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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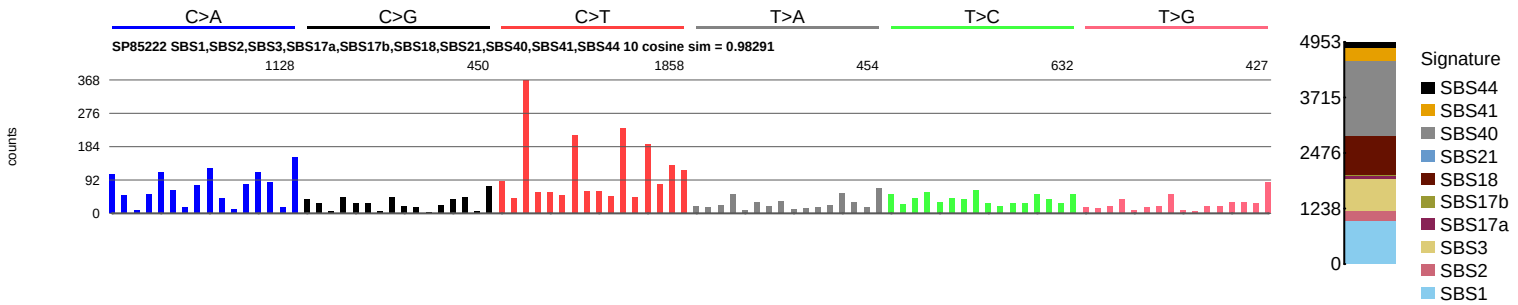
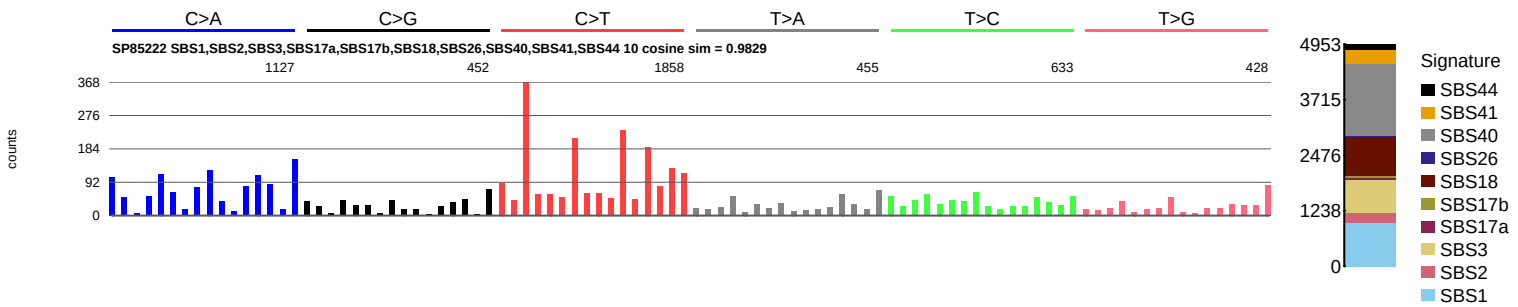
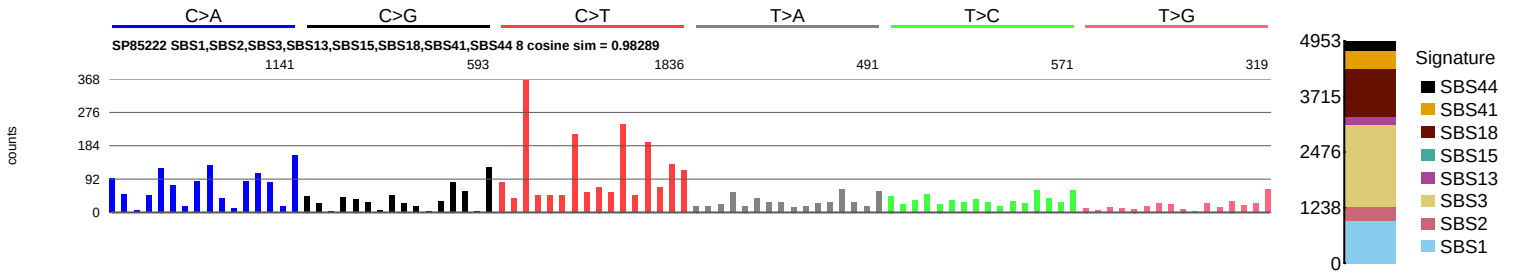
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98293
1113
450
1848
450
665
423
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98291
1128
450
1858
454
632
427
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9829
1127
452
1858
455
633
428
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS41,SBS44 8 cosine sim = 0.98289
1141
593
1836
491
571
319
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS15
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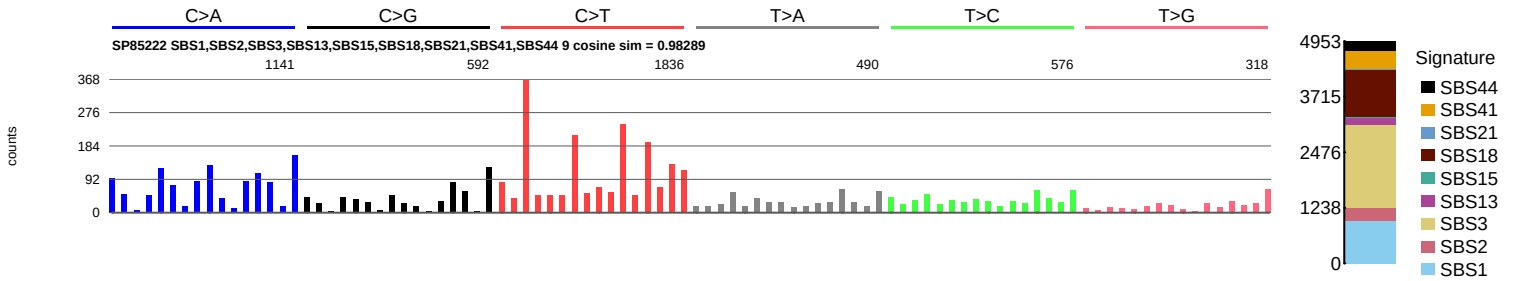
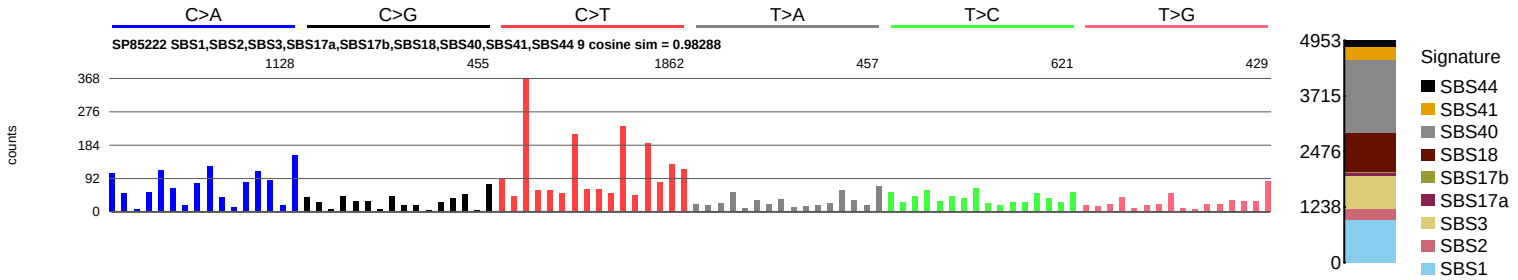
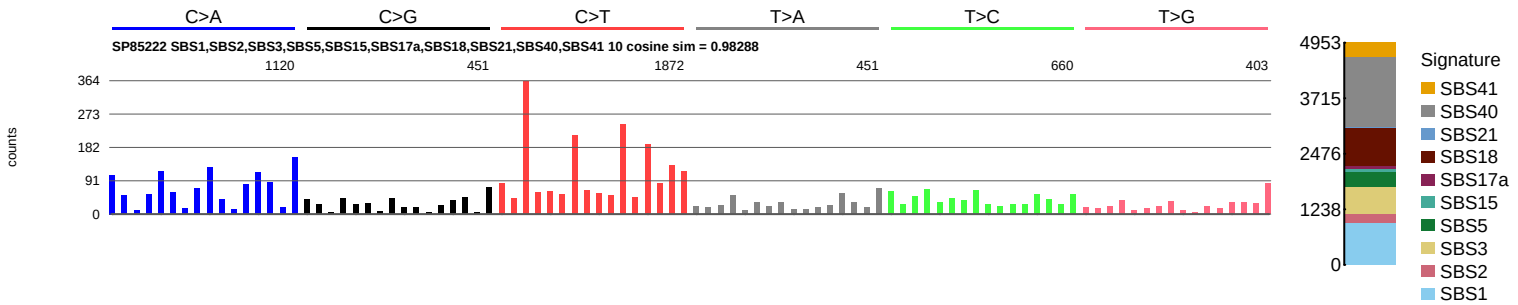
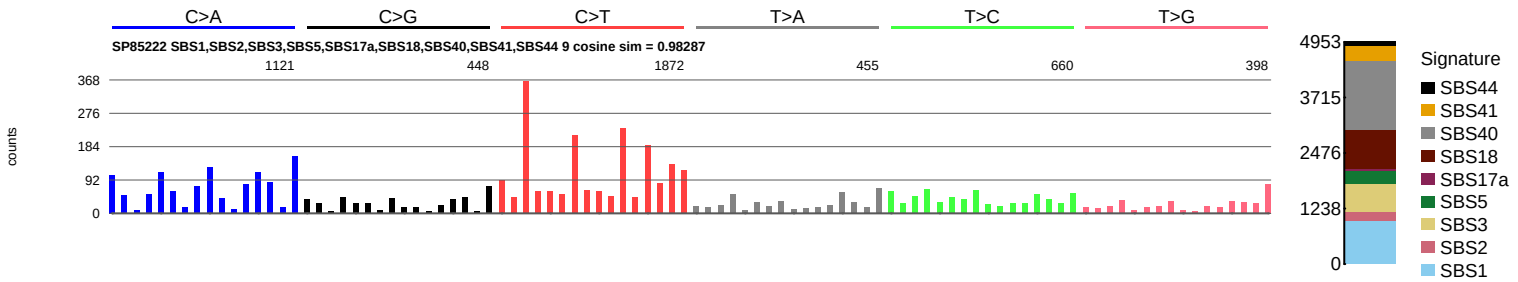
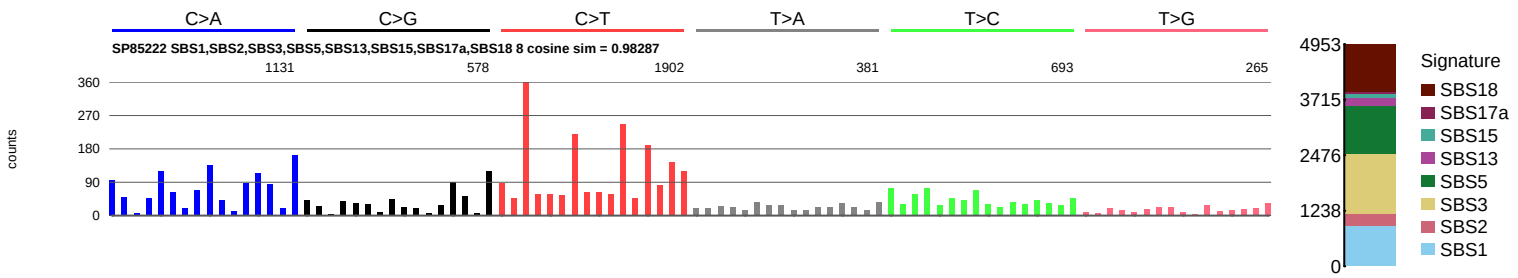
SBS3
SBS2
SBS1

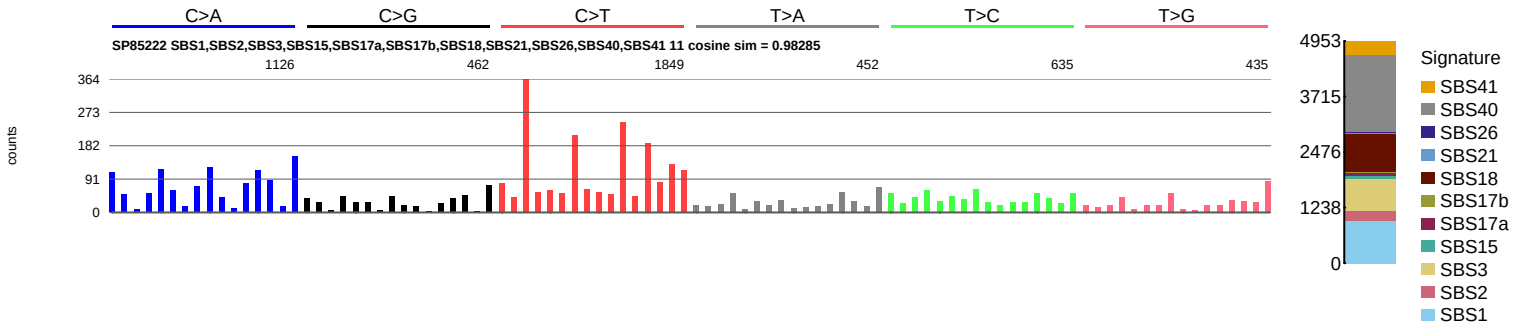

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98289
1141
592
1836
490
576
318
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

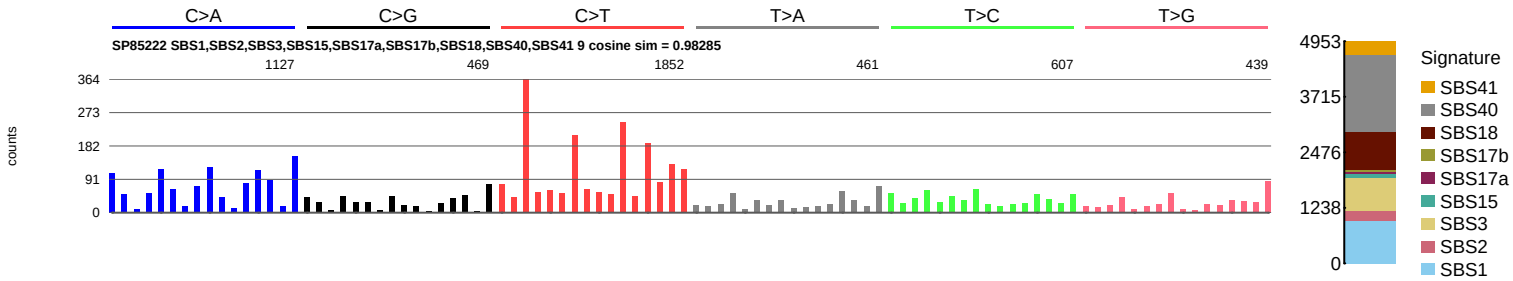

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98288
1128
455
1862
457
621
429
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

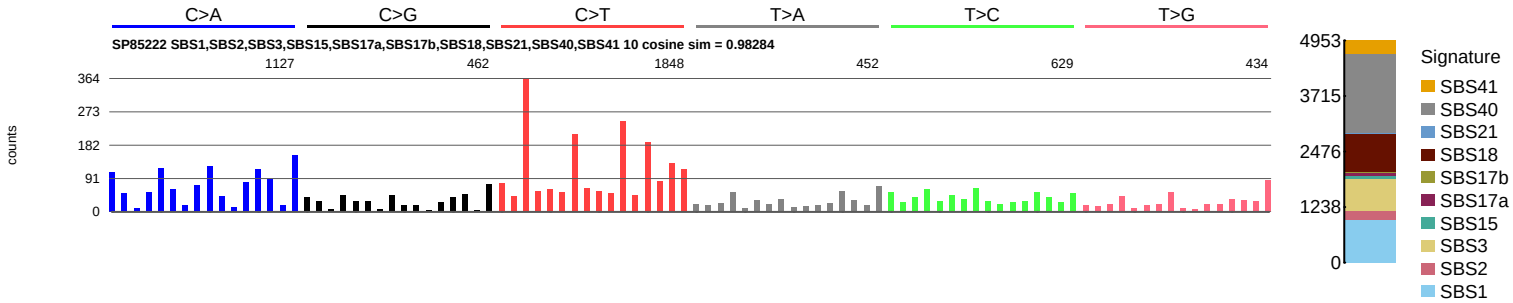

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98288
1120
451
1872
451
660
403
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98287
1121
448
1872
455
660
398
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18 8 cosine sim = 0.98287
1131
578
1902
381
693
265
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98285
1126
462
1849
452
635
435
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98285
1127
469
1852
461
607
439
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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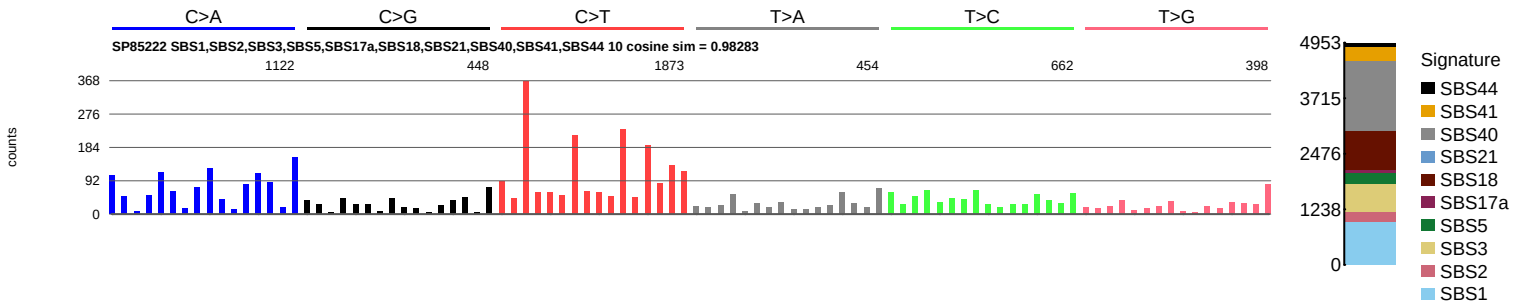
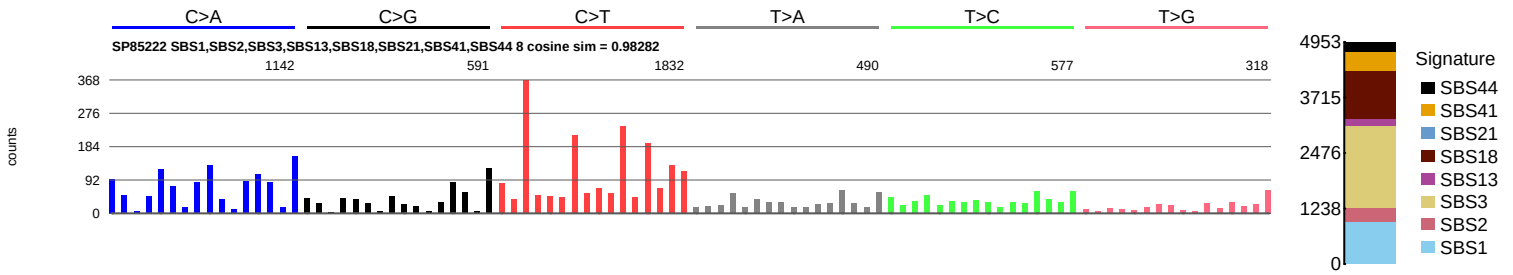
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98284
1127
462
1848
452
629
434
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98283
1122
448
1873
454
662
398
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98282
1142
591
1832
490
577
318
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
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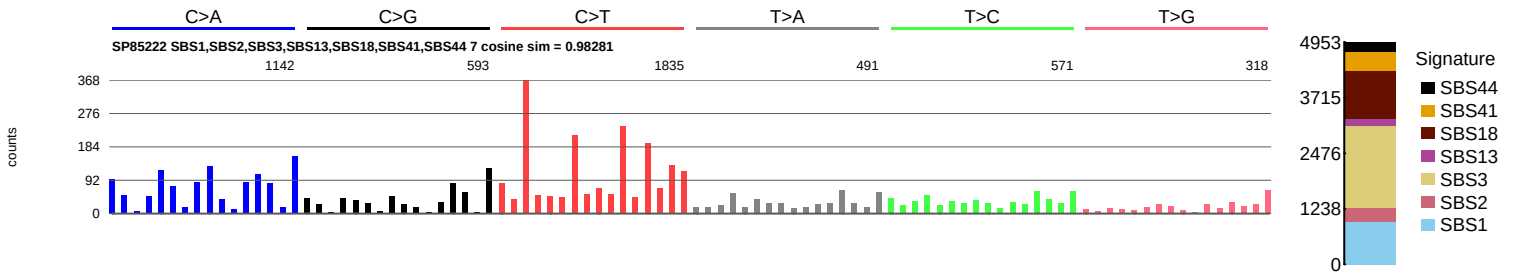
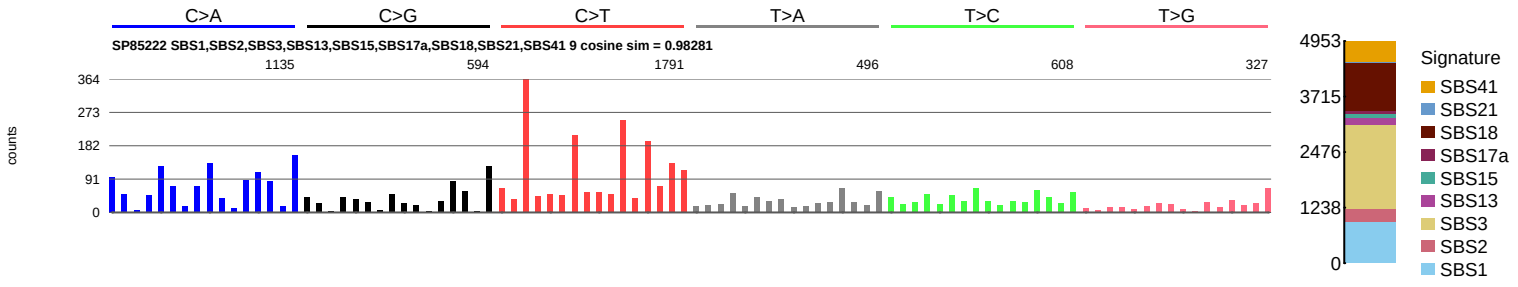
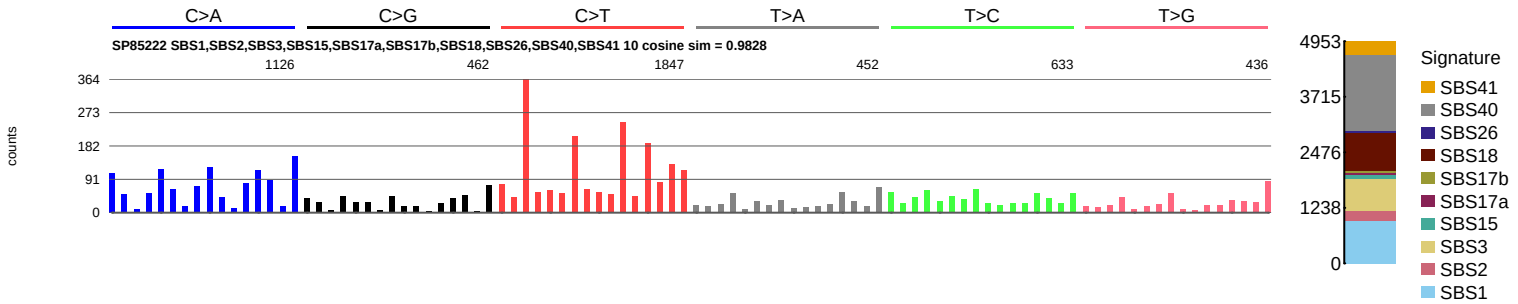
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS41,SBS44 7 cosine sim = 0.98281
1142
593
1835
491
571
318
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS41 9 cosine sim = 0.98281
1135
594
1791
496
608
327
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.9828
1126
462
1847
452
633
436
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
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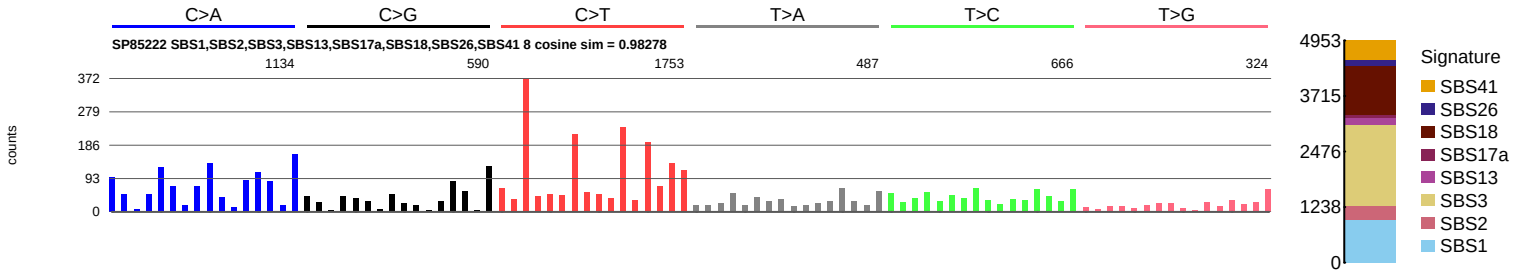
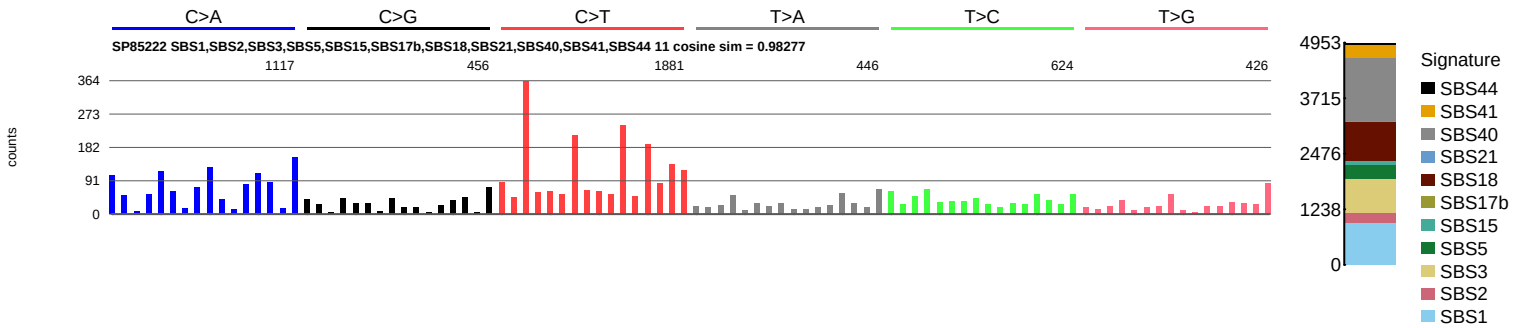
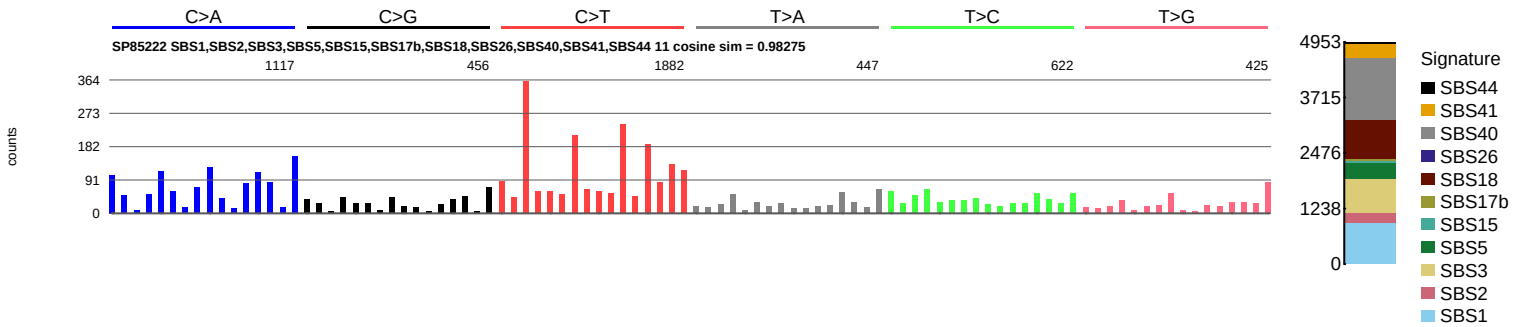
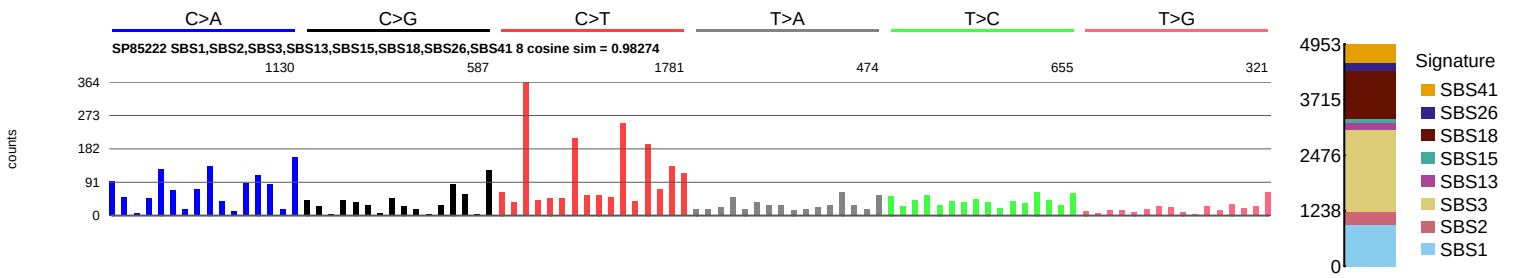
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS26,SBS41 8 cosine sim = 0.98278
1134
590
1753
487
666
324
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS26
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98277
1117
456
1881
446
624
426
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98275
1117
456
1882
447
622
425
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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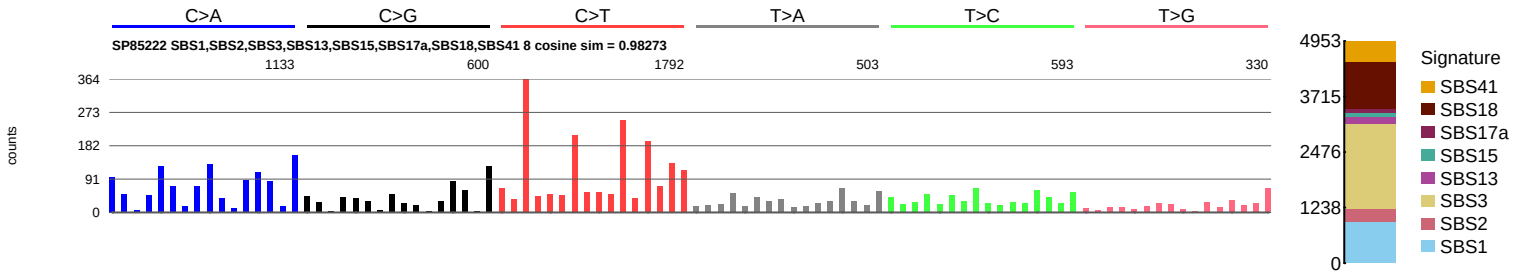
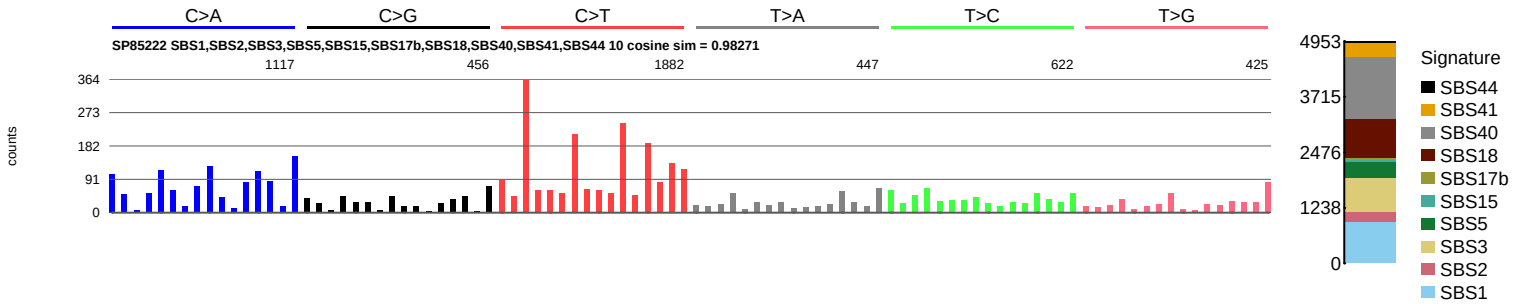
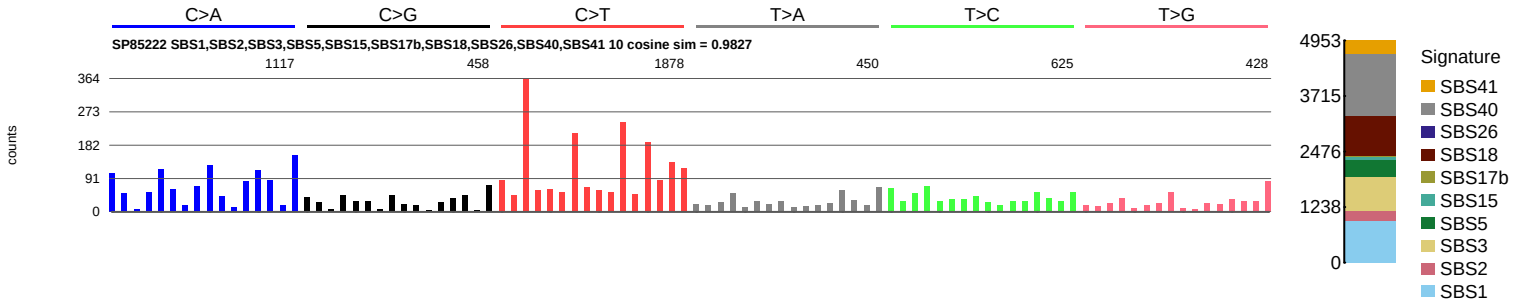
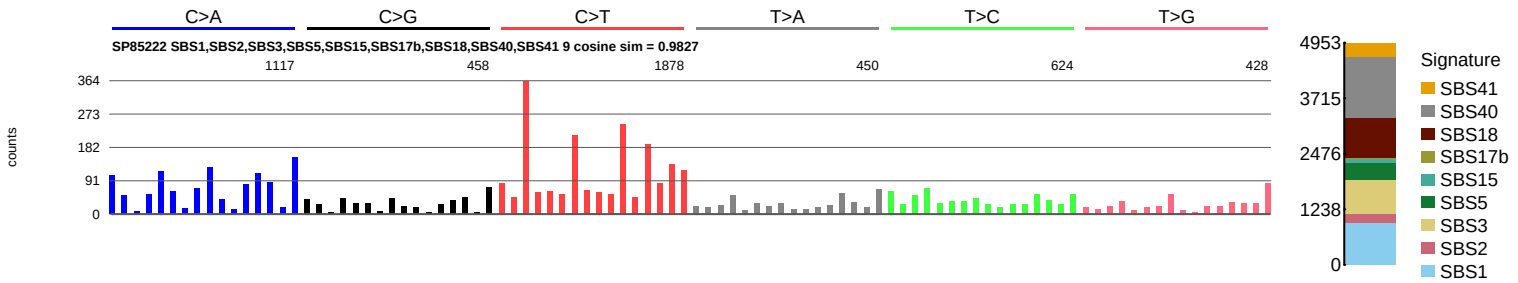
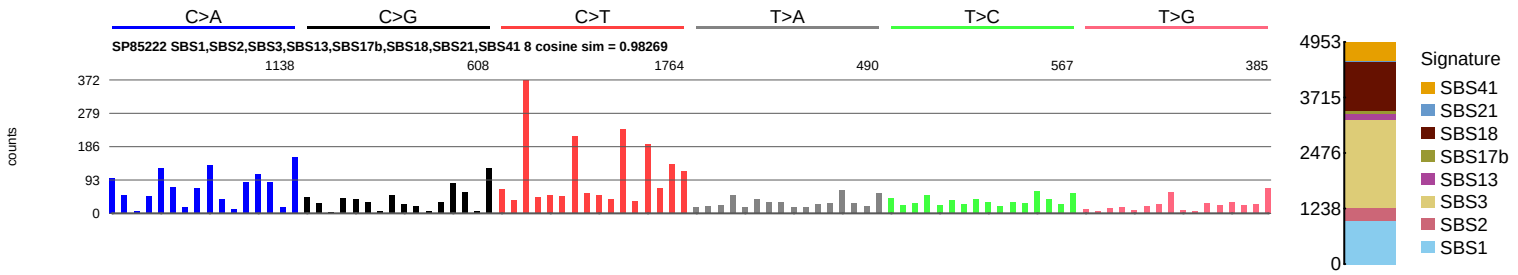
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS26,SBS41 8 cosine sim = 0.98274
1130
587
1781
474
655
321
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS26
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

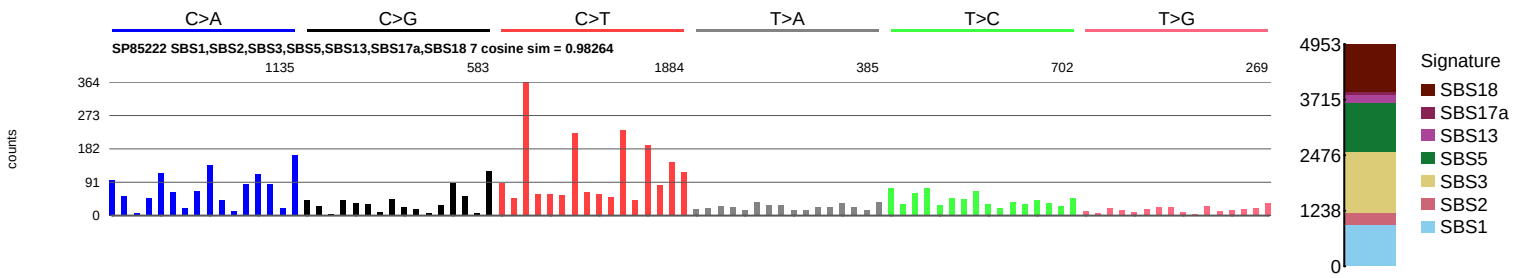

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS41 8 cosine sim = 0.98273
1133
600
1792
503
593
330
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

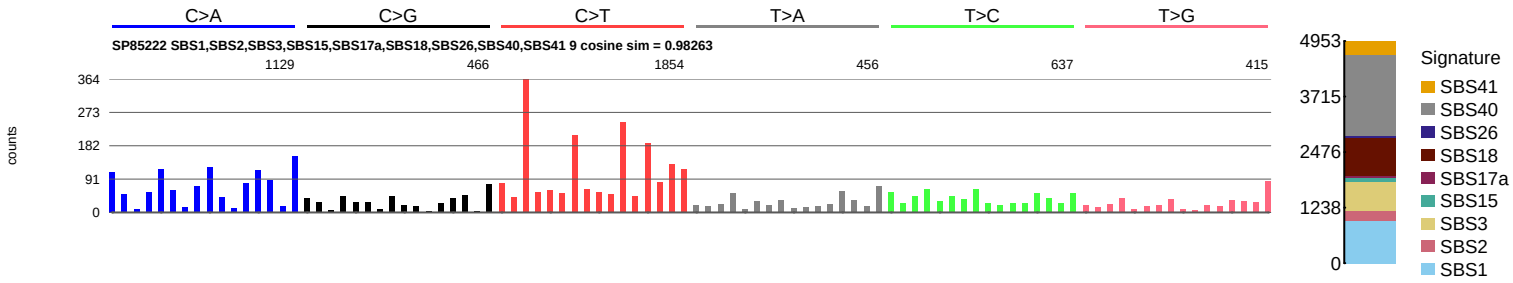

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98271
1117
456
1882
447
622
425
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

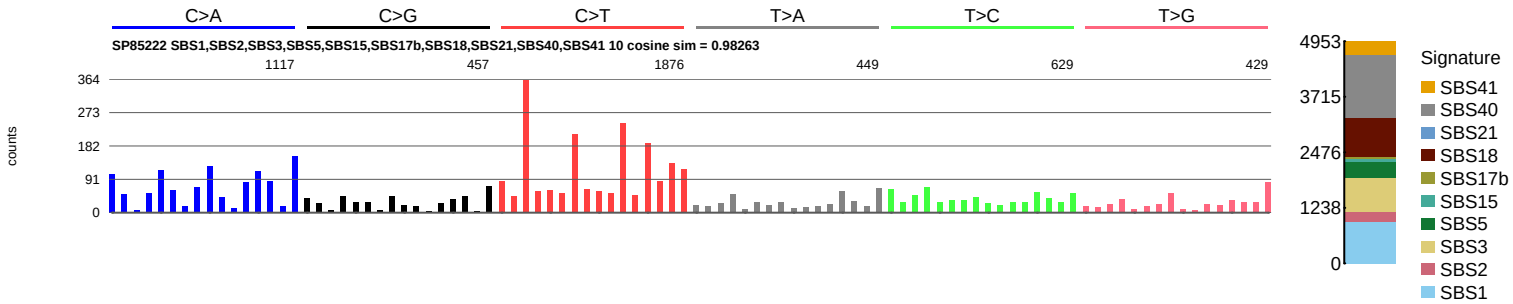

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.9827
1117
458
1878
450
625
428
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

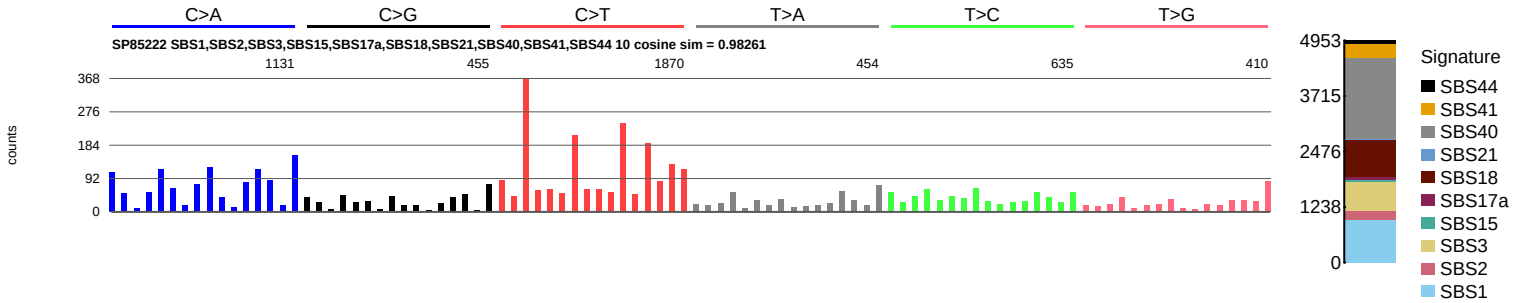

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.9827
1117
458
1878
450
624
428
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

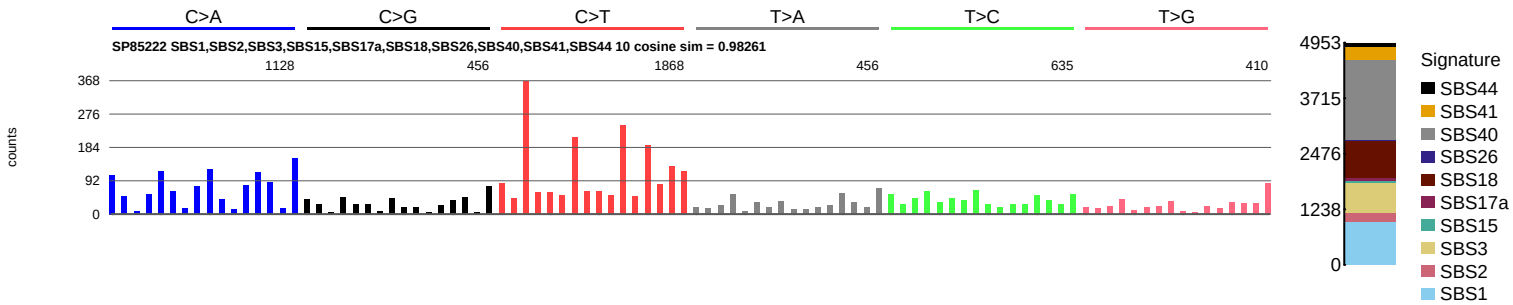

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98269
1138
608
1764
490
567
385
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18 7 cosine sim = 0.98264
1135
583
1884
385
702
269
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98263
1129
466
1854
456
637
415
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98263
1117
457
1876
449
629
429
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98261
1131
455
1870
454
635
410
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98261
1128
456
1868
456
635
410
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
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SBS40
SBS26
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SBS17a
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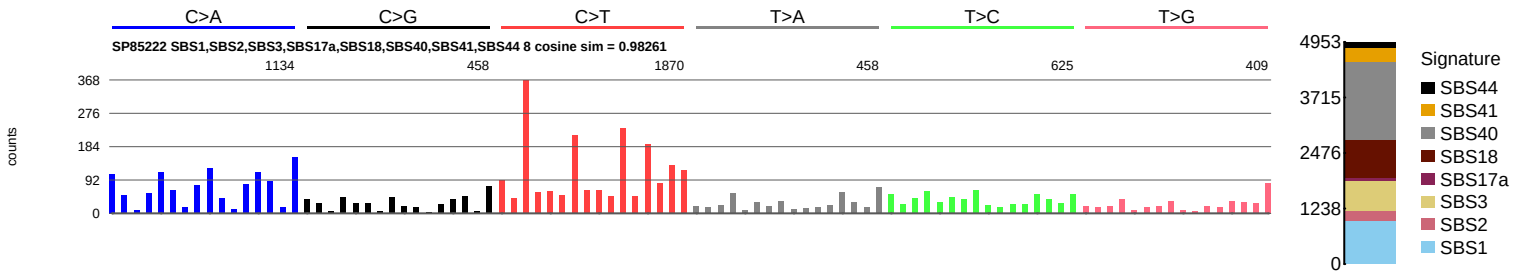
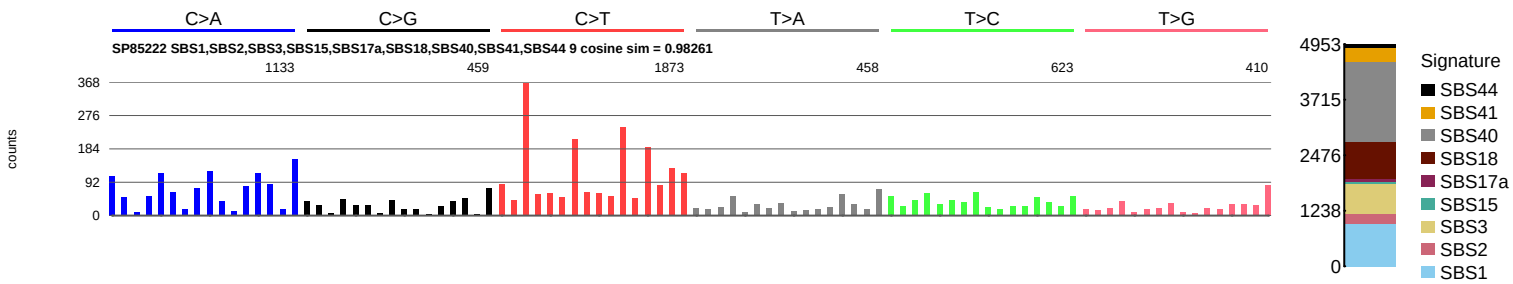
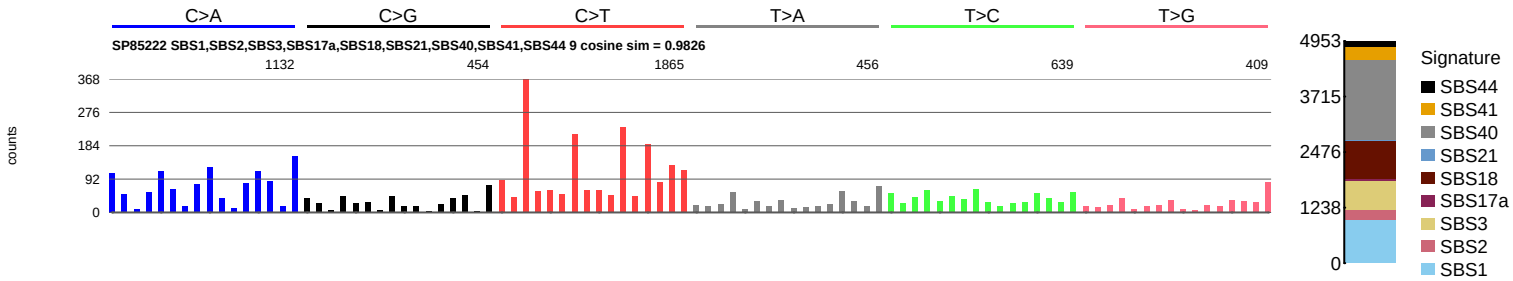
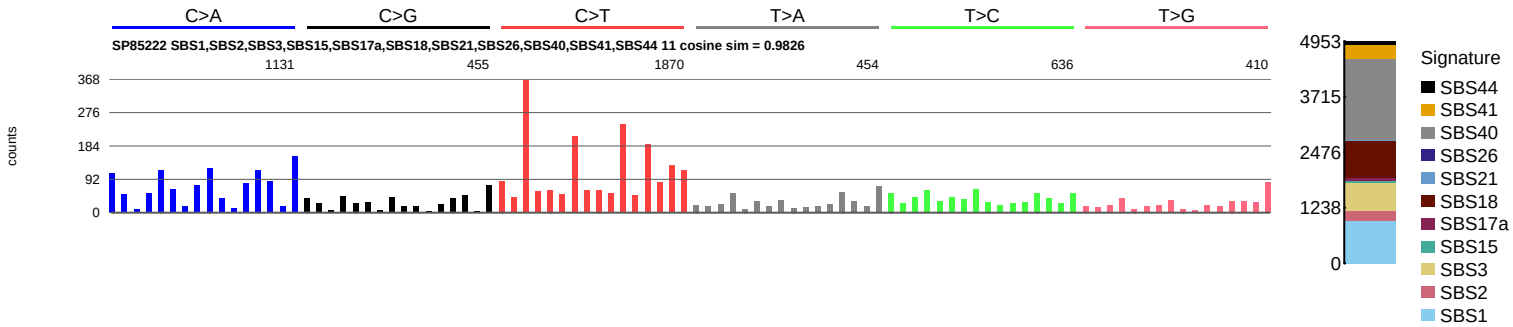
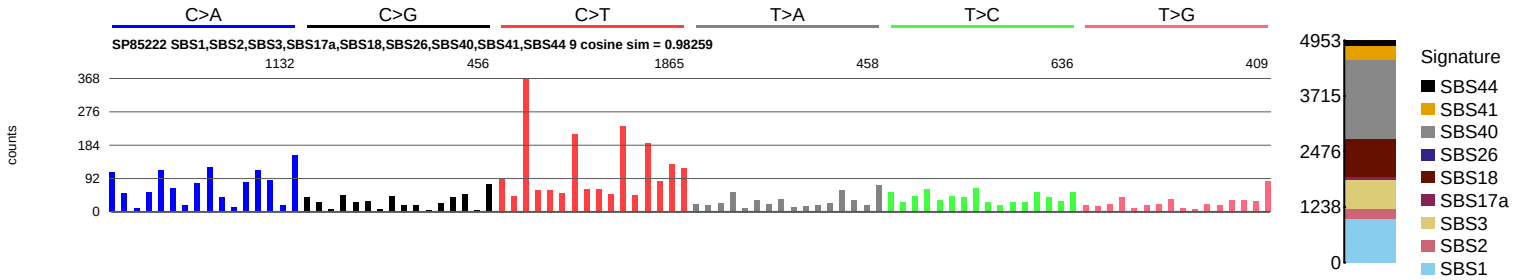
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98261
1134
458
1870
458
625
409
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98261
1133
459
1873
458
623
410
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9826
1132
454
1865
456
639
409
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.9826
1131
455
1870
454
636
410
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98259
1132
456
1865
458
636
409
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
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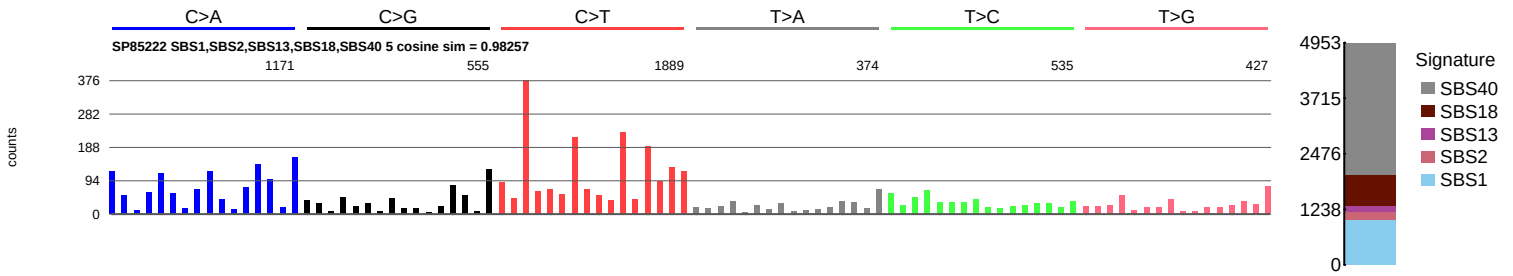
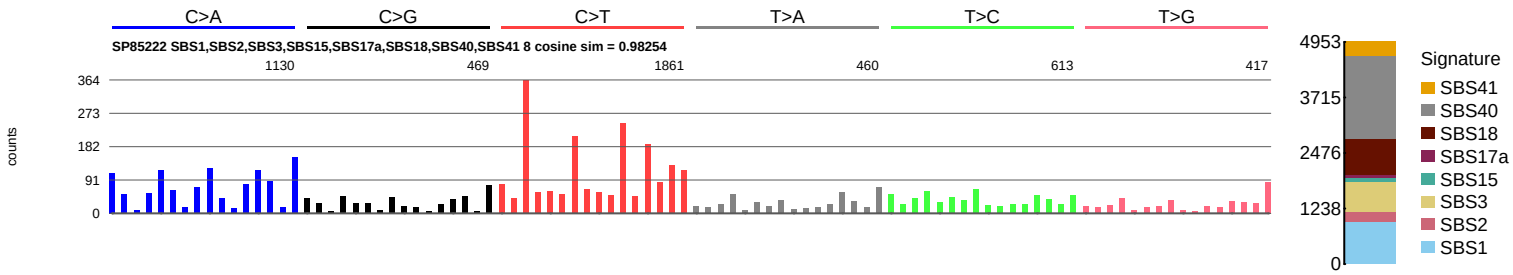
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS13,SBS18,SBS40 5 cosine sim = 0.98257
1171
555
1889
374
535
427
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98254
1130
469
1861
460
613
417
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
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SBS18
SBS17a
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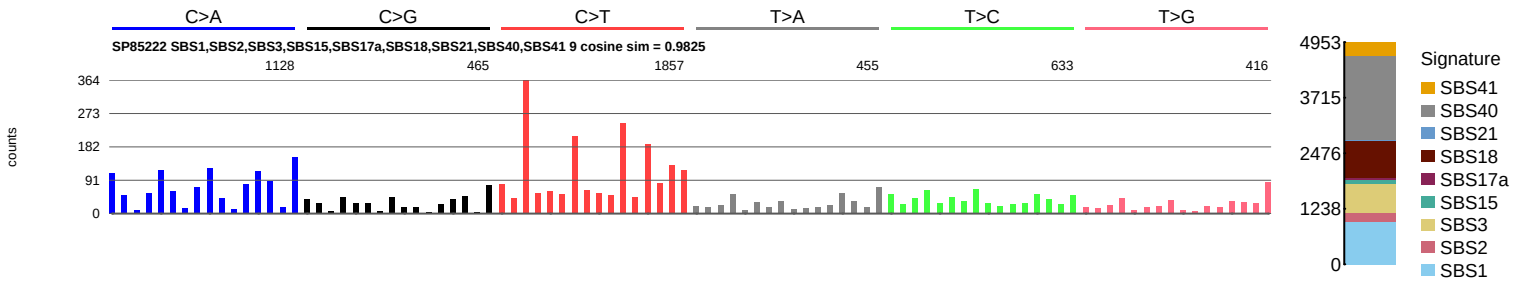
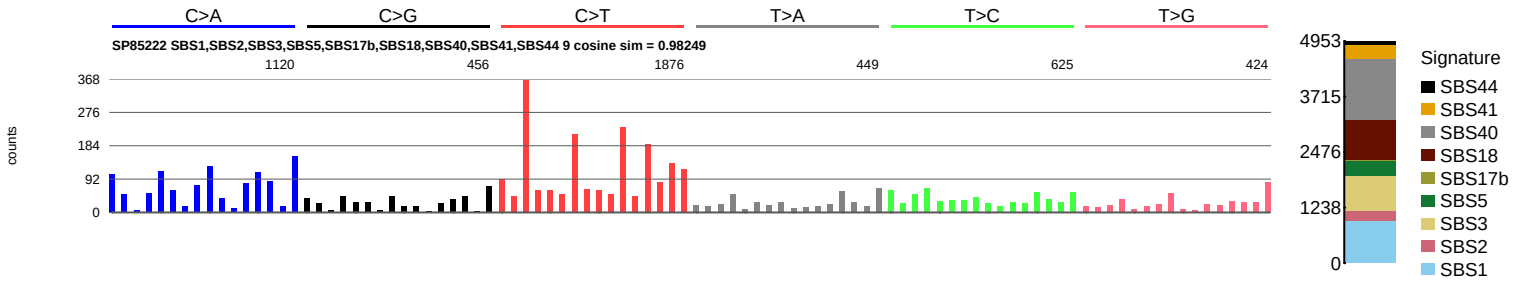
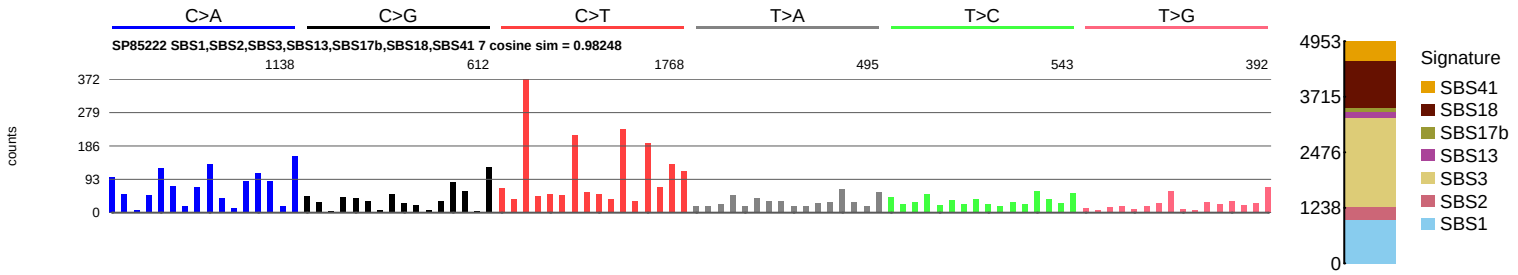
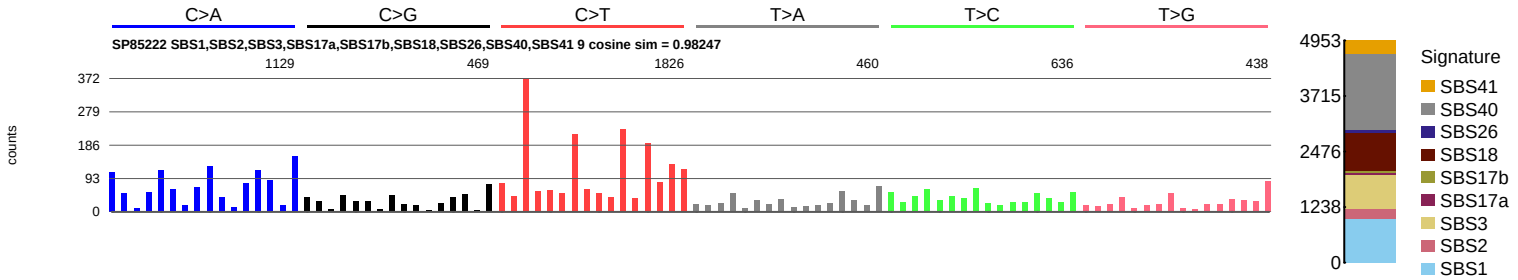
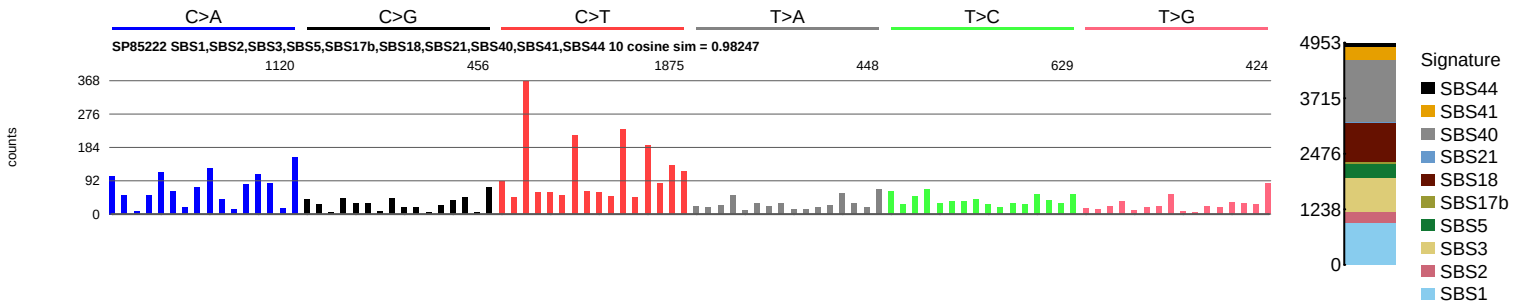
SBS3
SBS2
SBS1

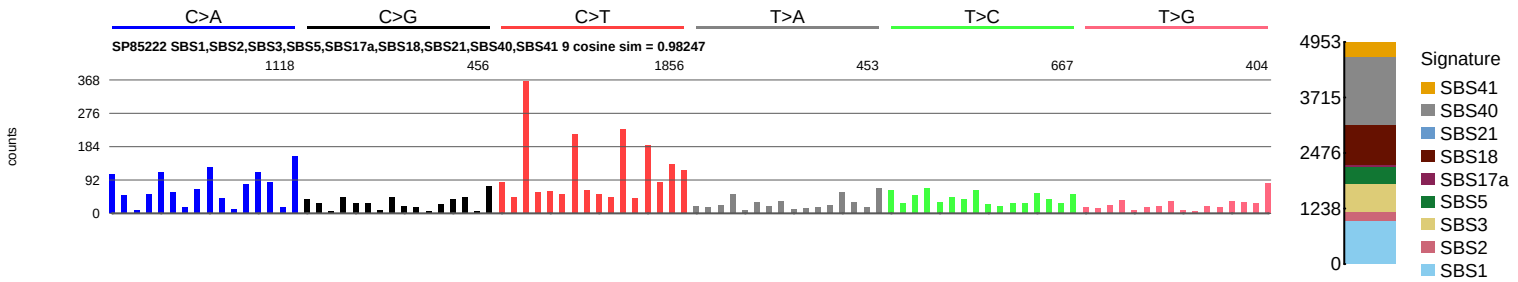

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.9825
1128
465
1857
455
633
416
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1

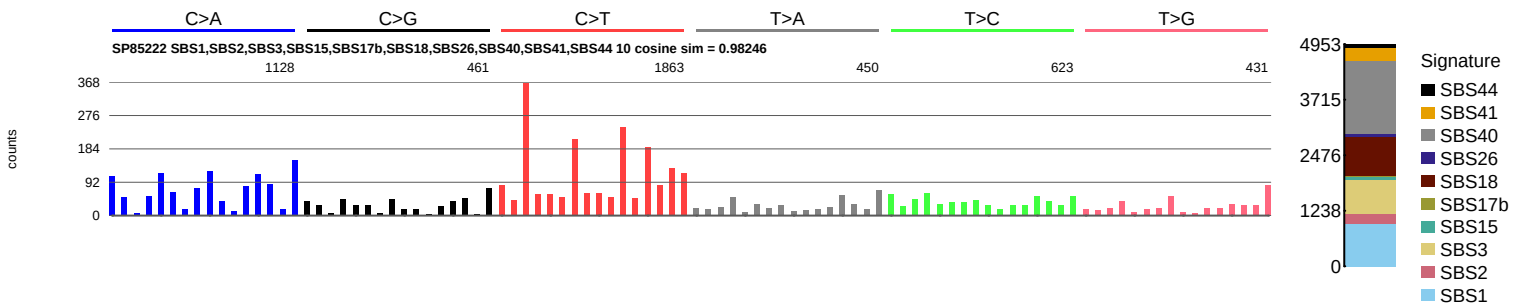

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98249
1120
456
1876
449
625
424
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS41 7 cosine sim = 0.98248
1138
612
1768
495
543
392
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98247
1129
469
1826
460
636
438
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98247
1120
456
1875
448
629
424
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98247
1118
456
1856
453
667
404
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1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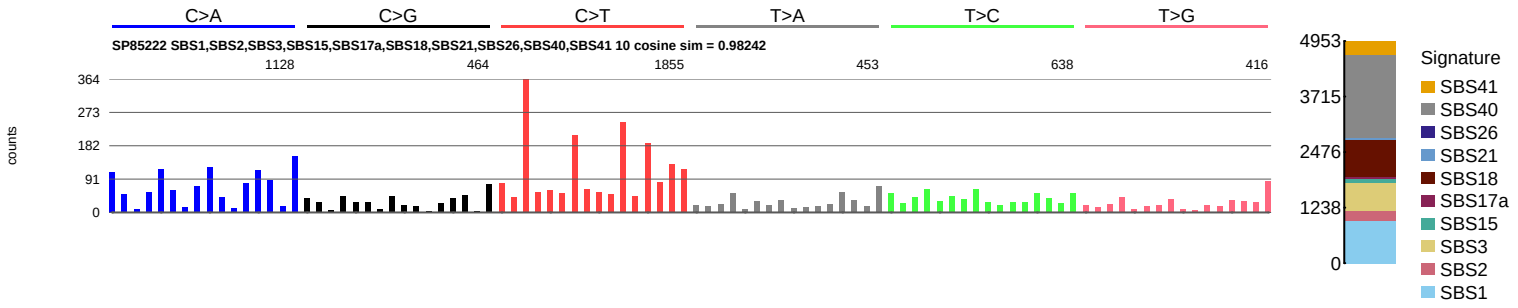
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98246
1128
461
1863
450
623
431
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1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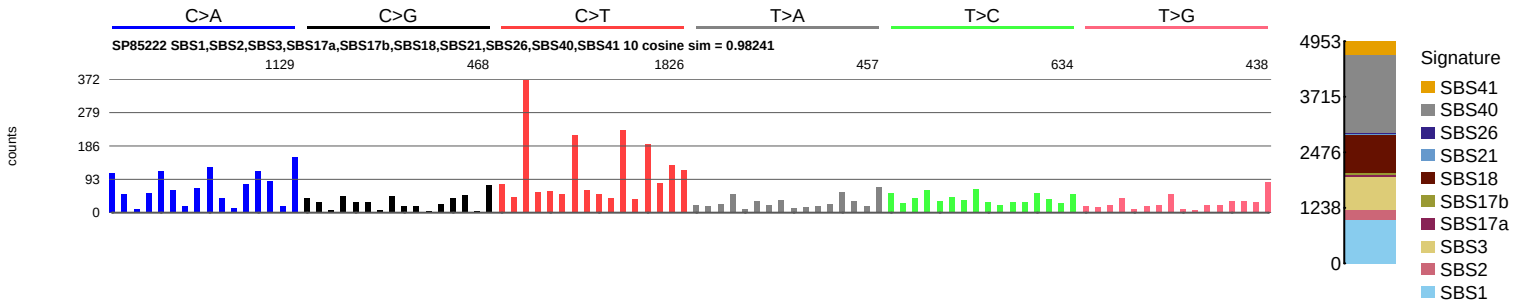
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98242
1128
464
1855
453
638
416
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1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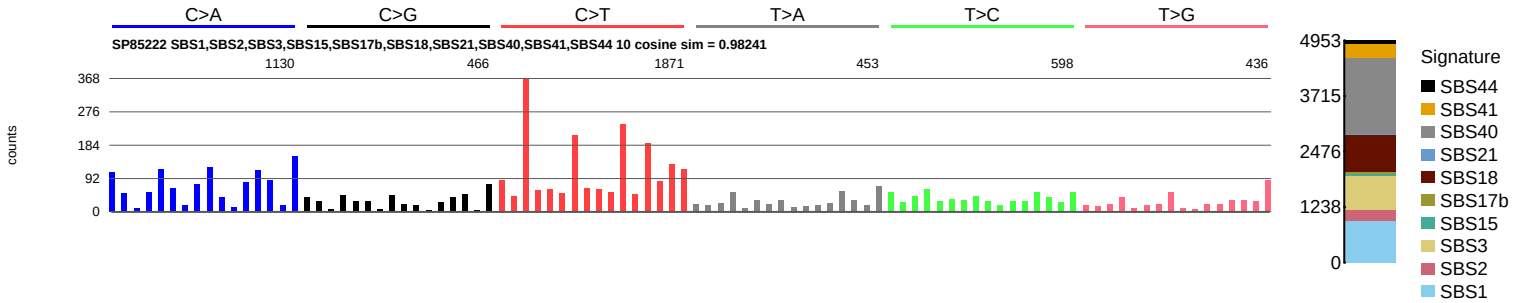
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98241
1129
468
1826
457
634
438
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
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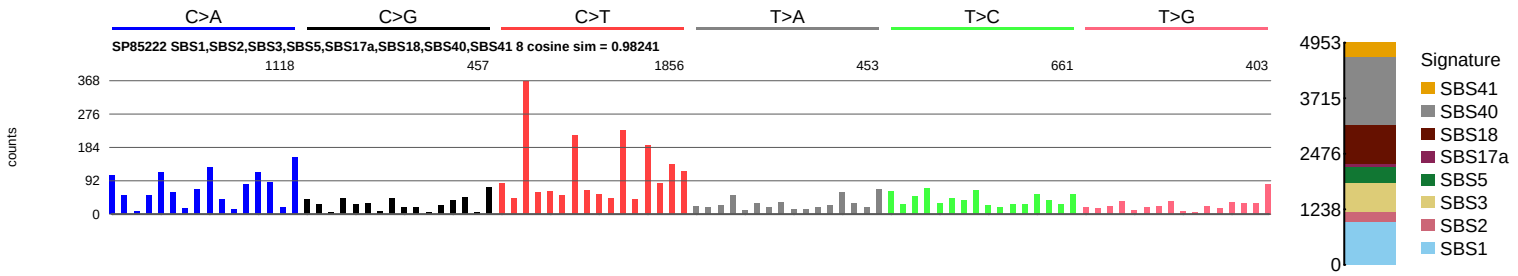
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98241
1130
466
1871
453
598
436
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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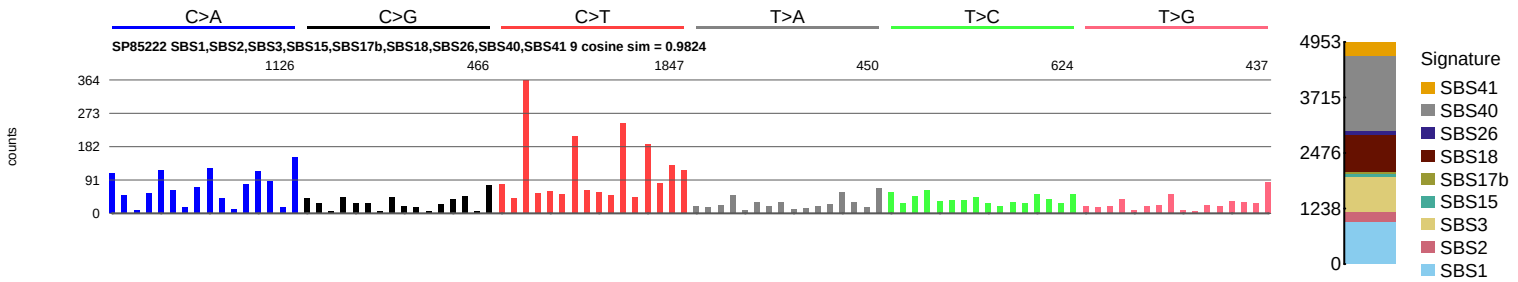
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98241
1118
457
1856
453
661
403
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.9824
1126
466
1847
450
624
437
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
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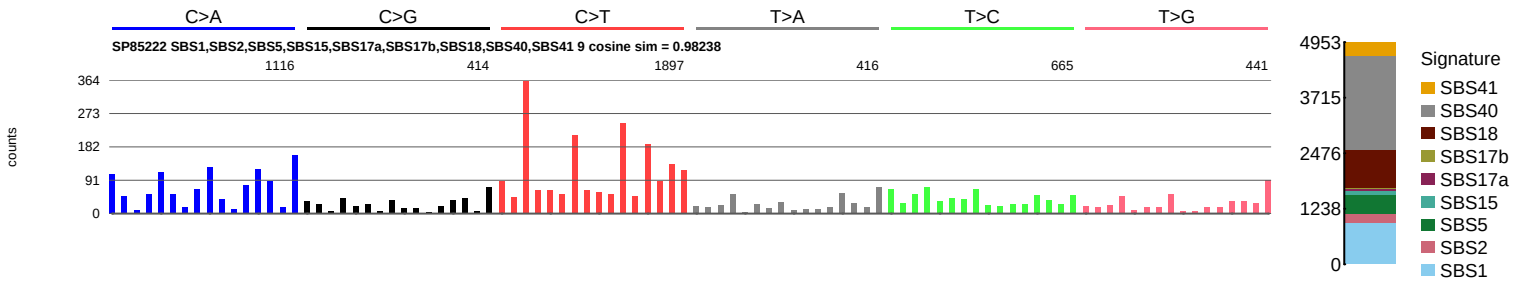
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98238
1116
414
1897
416
665
441
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
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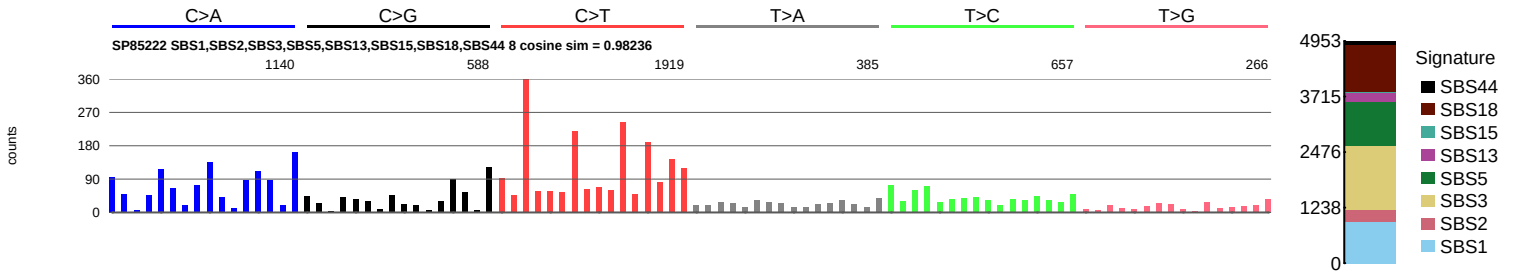
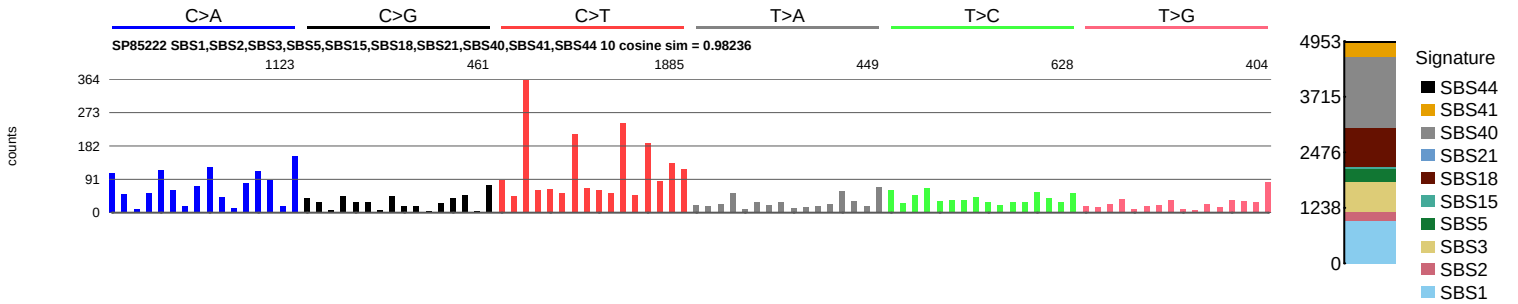
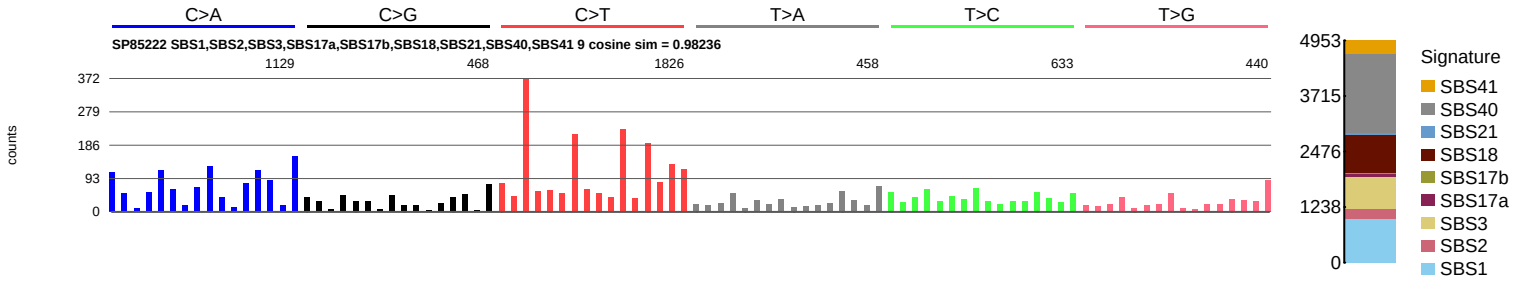
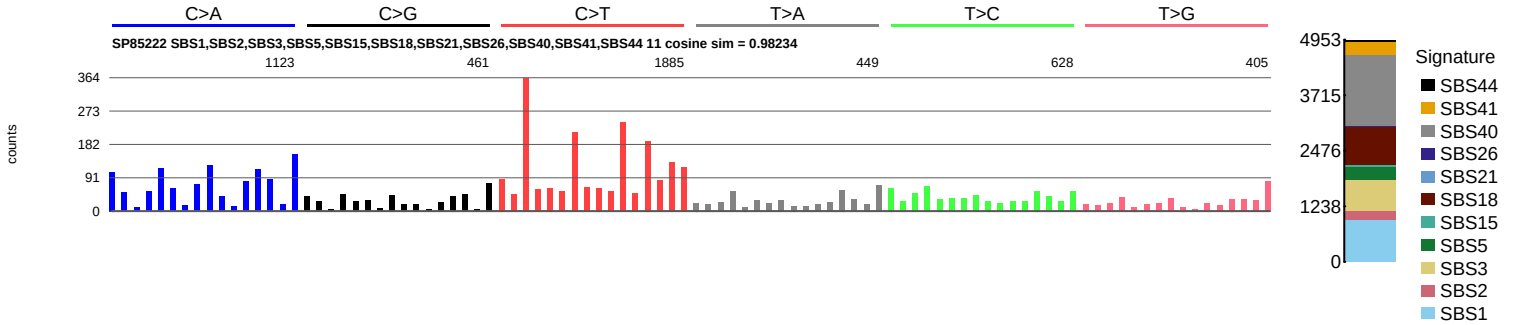
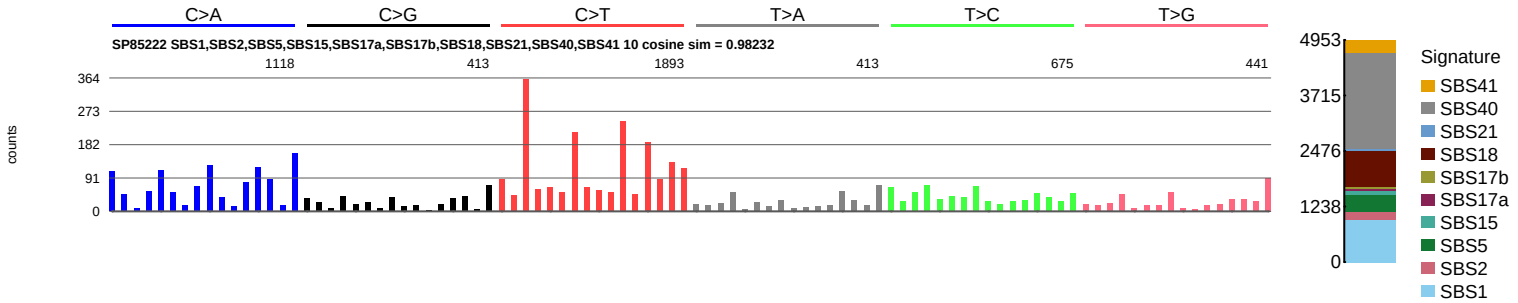
SBS5
SBS2
SBS1

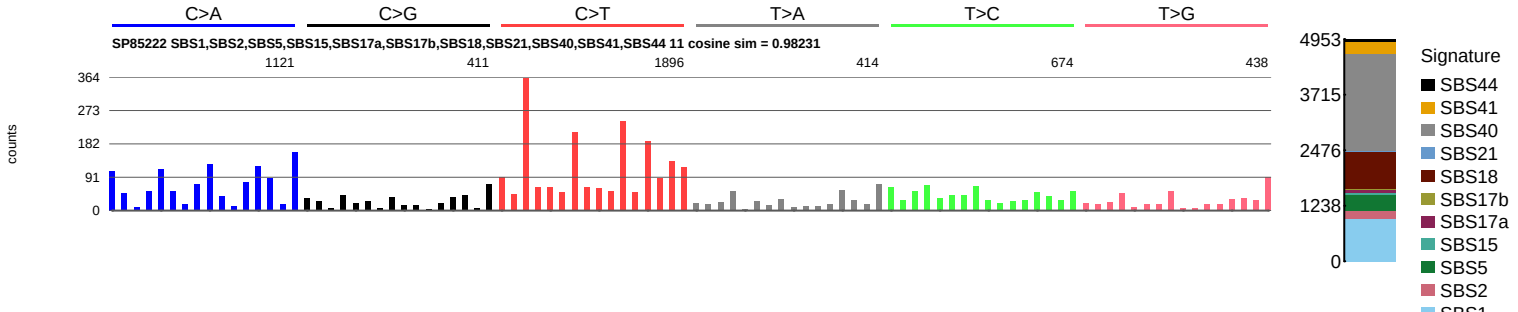

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS44 8 cosine sim = 0.98236
1140
588
1919
385
657
266
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS44
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

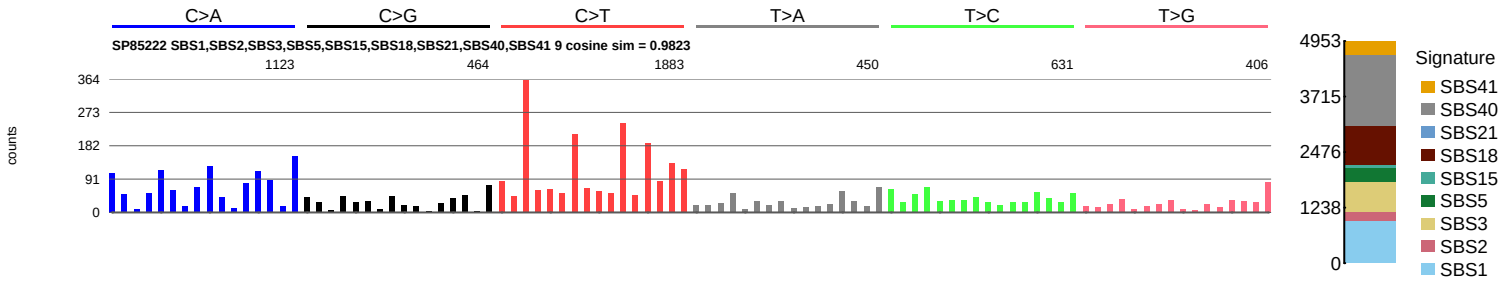

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98236
1123
461
1885
449
628
404
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS5
SBS3
SBS2
SBS1

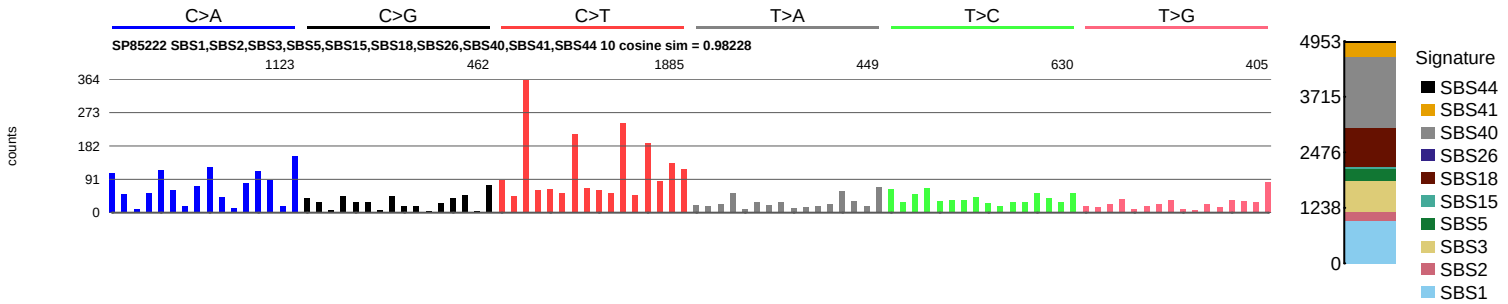

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98236
1129
468
1826
458
633
440
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98234
1123
461
1885
449
628
405
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98232
1118
413
1893
413
675
441
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98231
1121
411
1896
414
674
438
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.9823
1123
464
1883
450
631
406
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS5
SBS3
SBS2
SBS1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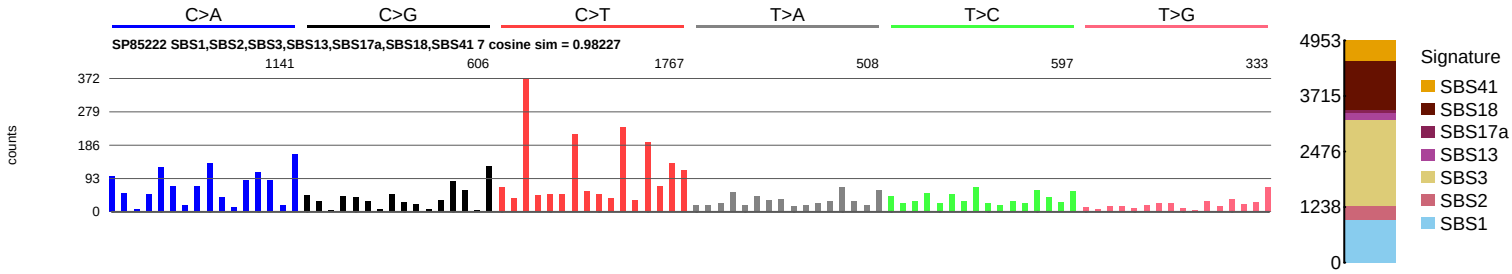
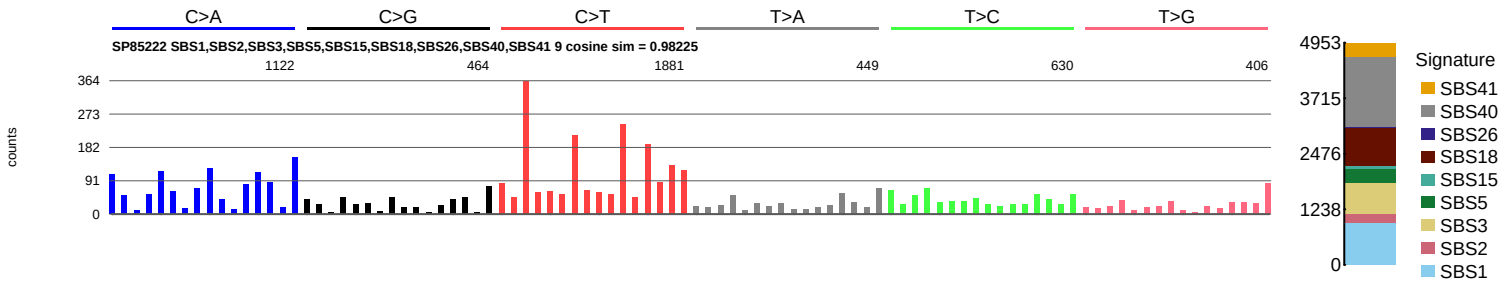
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98228
1123
462
1885
449
630
405
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS41 7 cosine sim = 0.98227
1141
606
1767
508
597
333
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1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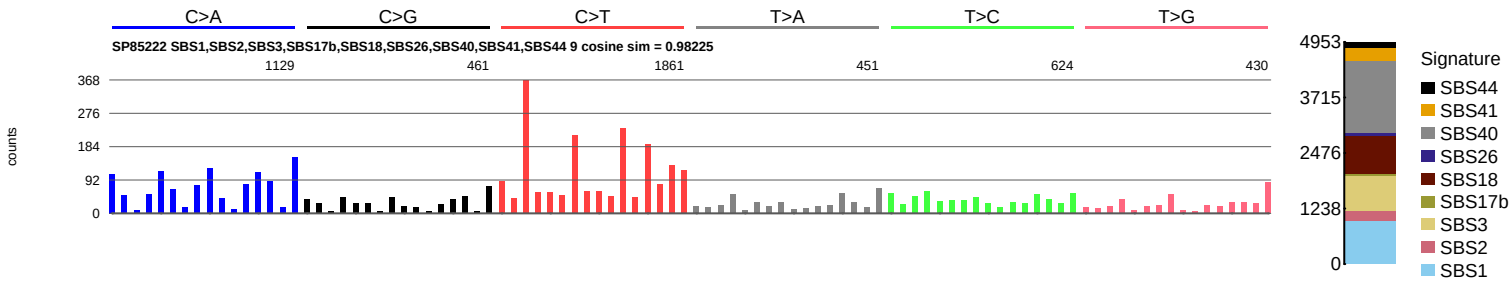
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98225
1122
464
1881
449
630
406
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS5
SBS3
SBS2
SBS1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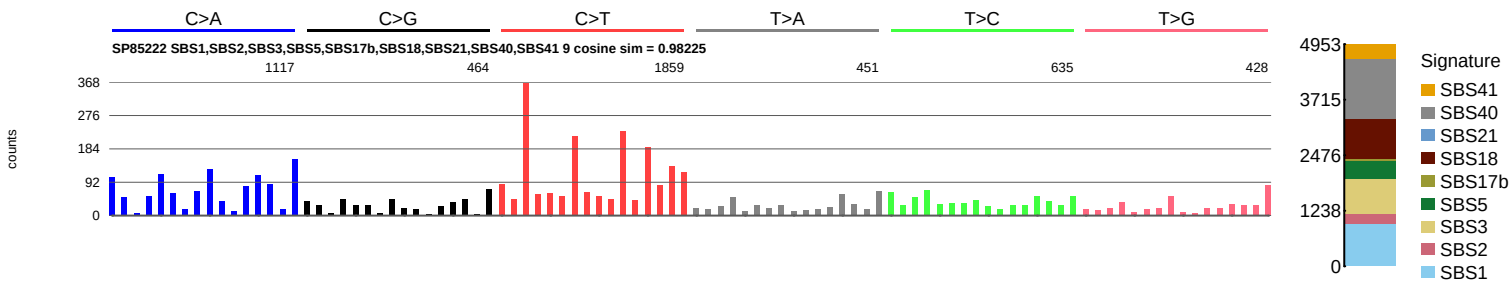
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98225
1129
461
1861
451
624
430
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1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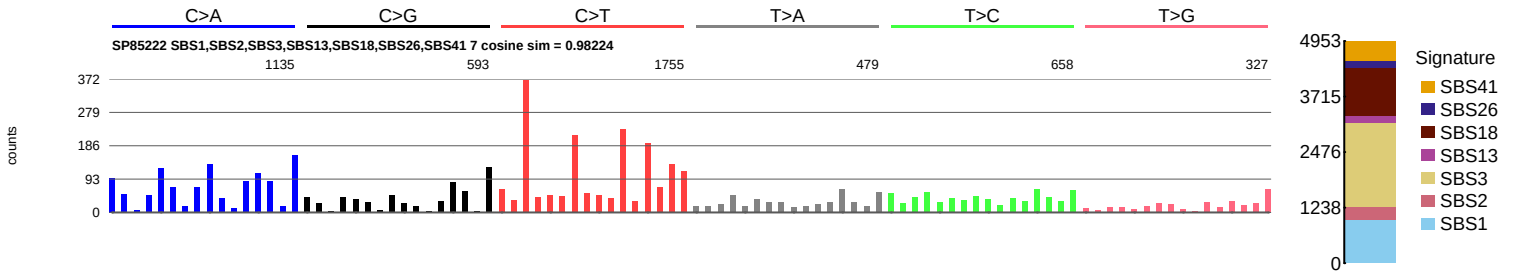
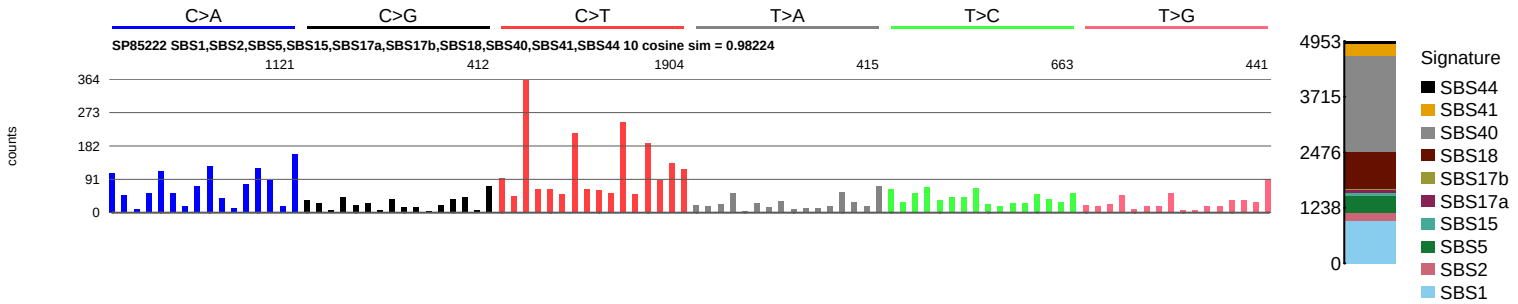
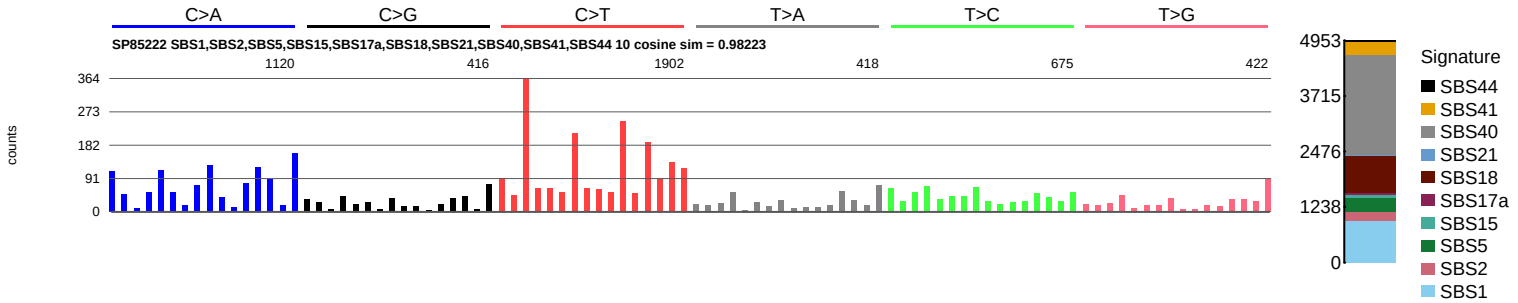
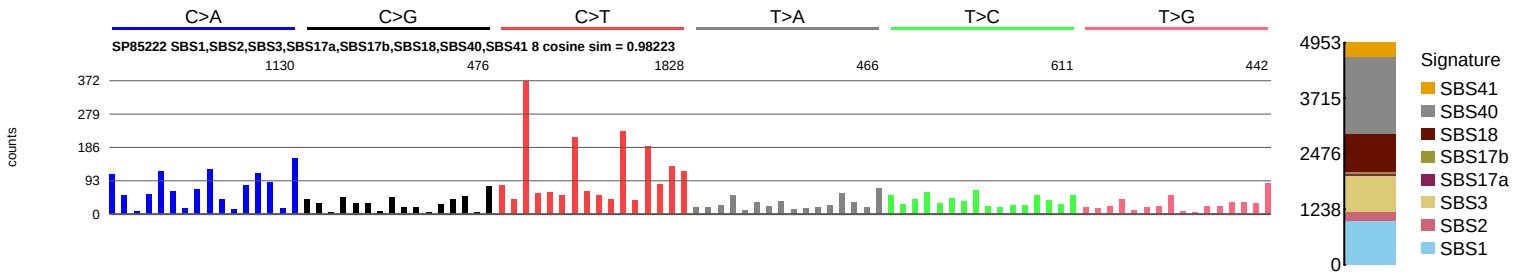
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98225
1117
464
1859
451
635
428
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS26,SBS41 7 cosine sim = 0.98224
1135
593
1755
479
658
327
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS26
SBS18
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98224
1121
412
1904
415
663
441
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98223
1120
416
1902
418
675
422
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98223
1130
476
1828
466
611
442
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
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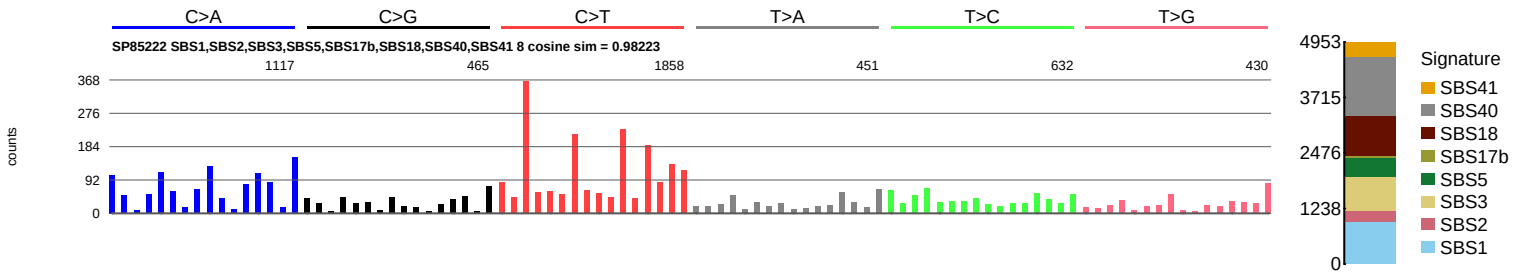
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98223
1117
465
1858
451
632
430
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
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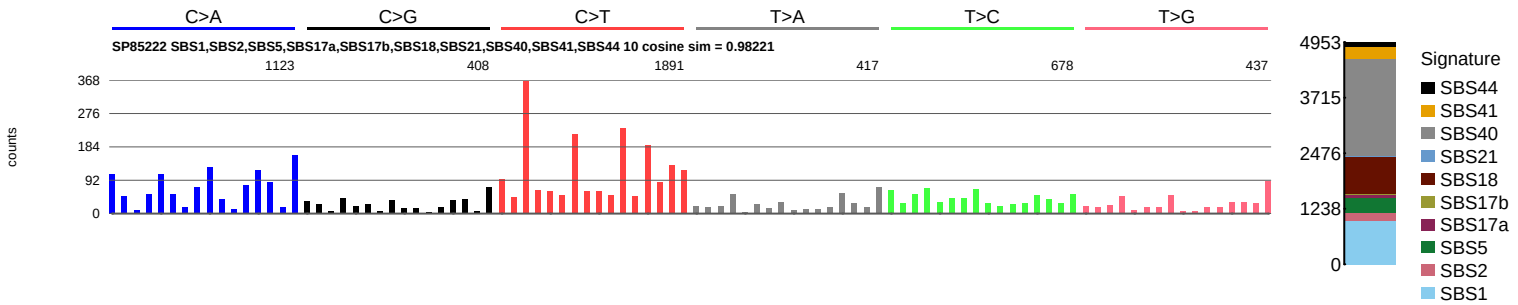
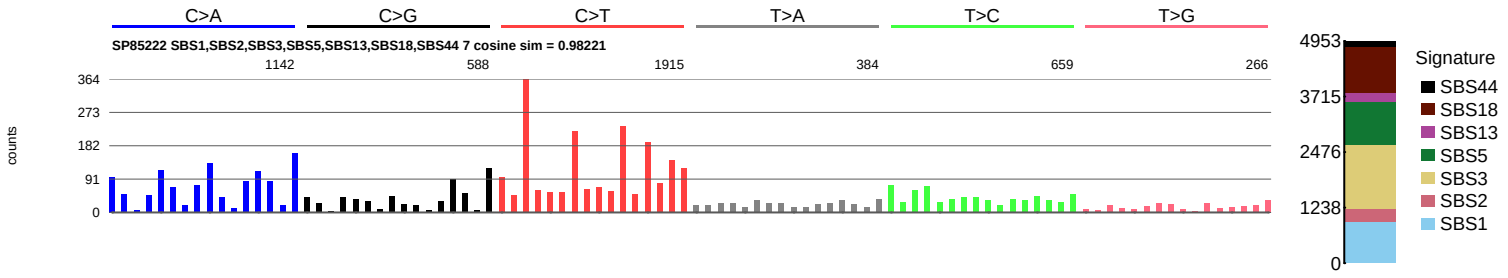
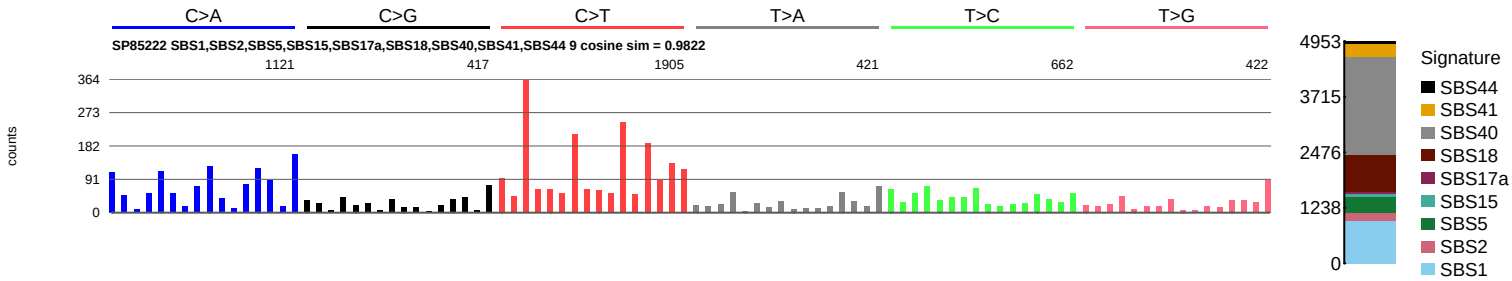
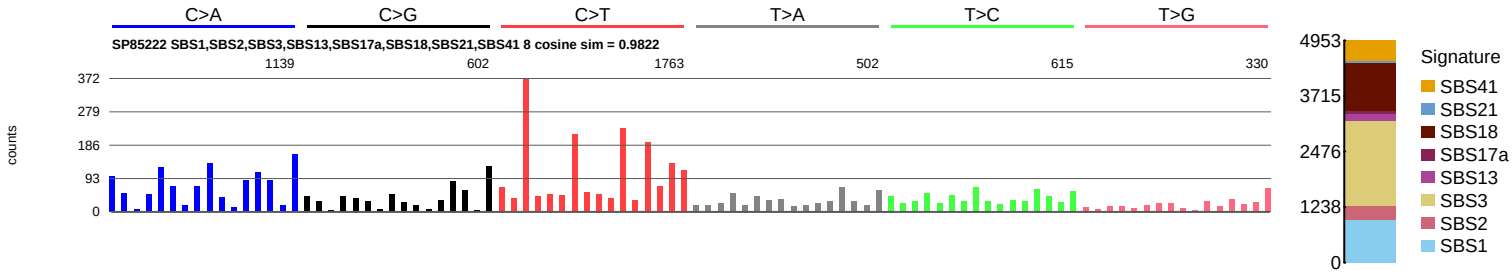
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98221
1123
408
1891
417
678
437
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS44 7 cosine sim = 0.98221
1142
588
1915
384
659
266
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.9822
1121
417
1905
421
662
422
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS41 8 cosine sim = 0.9822
1139
602
1763
502
615
330
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS21
SBS18
SBS17a
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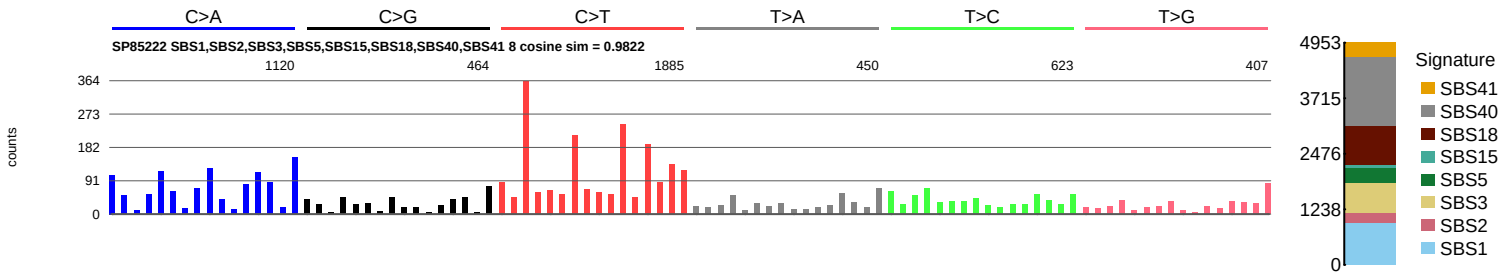
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS40,SBS41 8 cosine sim = 0.9822
1120
464
1885
450
623
407
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
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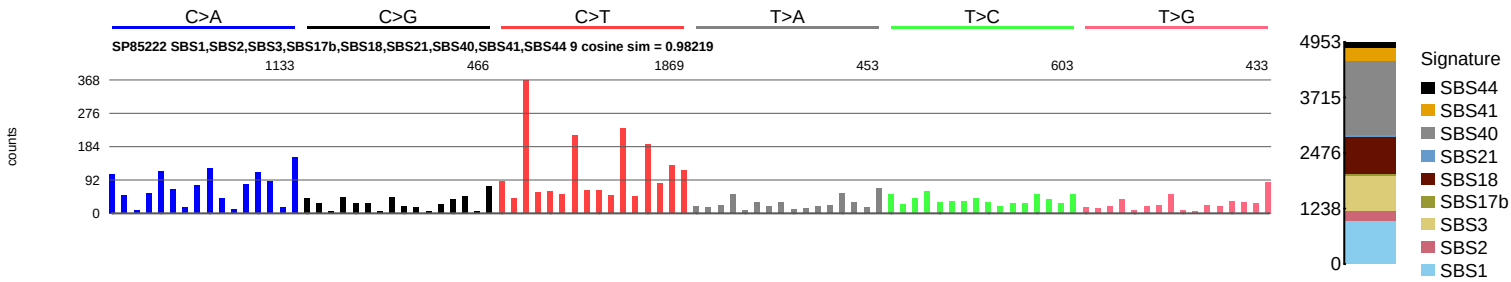
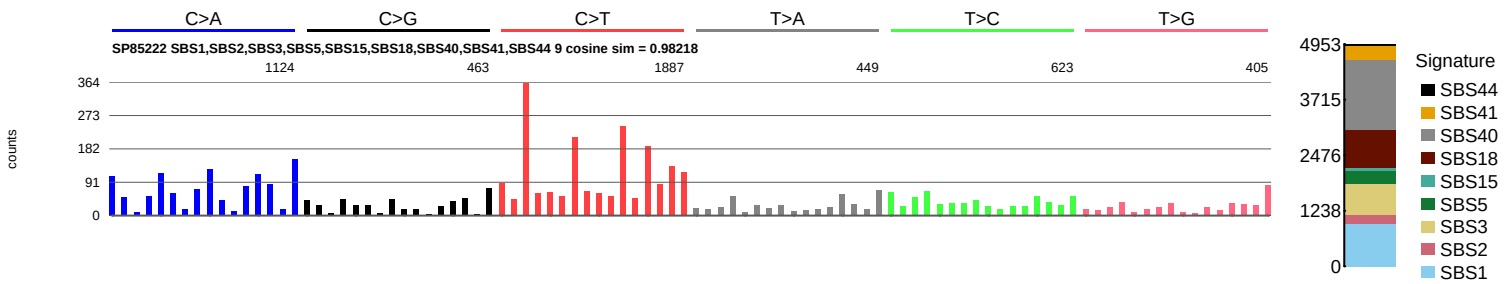
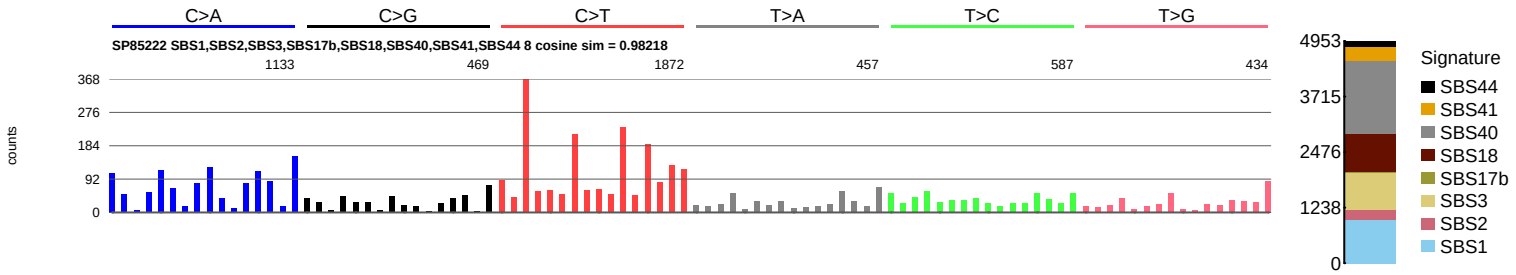
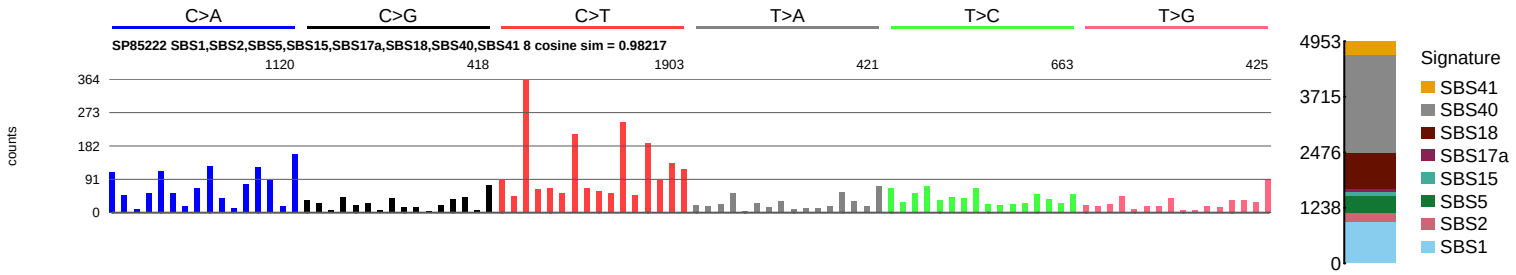
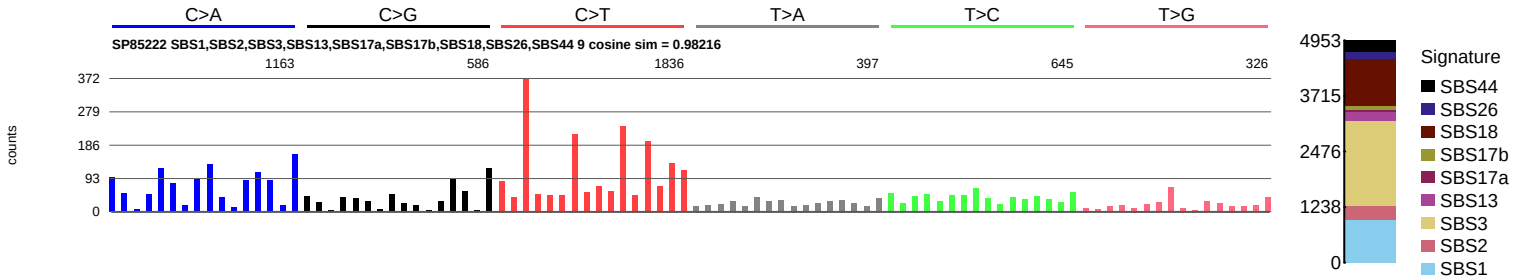
SBS3
SBS2
SBS1

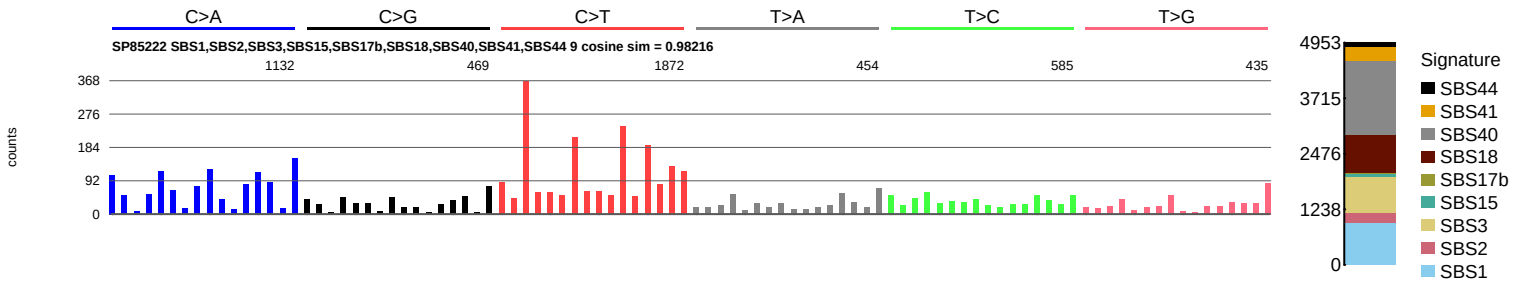

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98219
1133
466
1869
453
603
433
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1

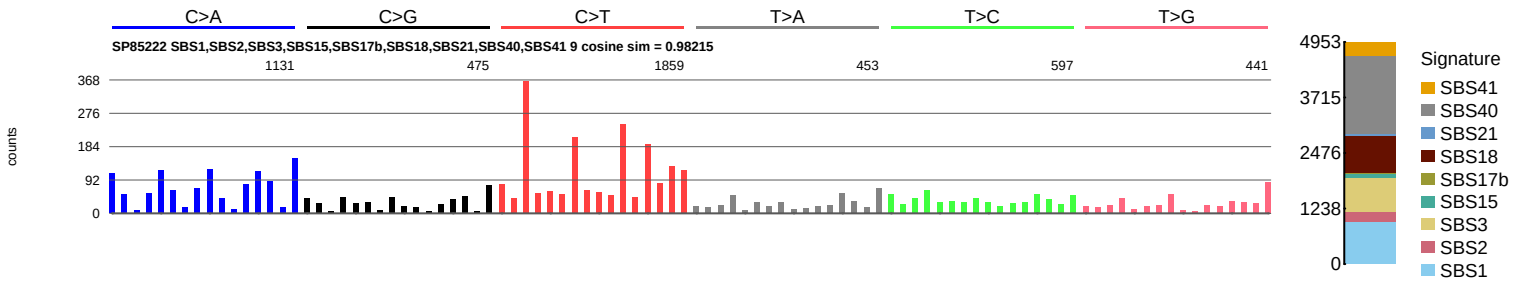

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98218
1124
463
1887
449
623
405
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS15
SBS5
SBS3
SBS2
SBS1

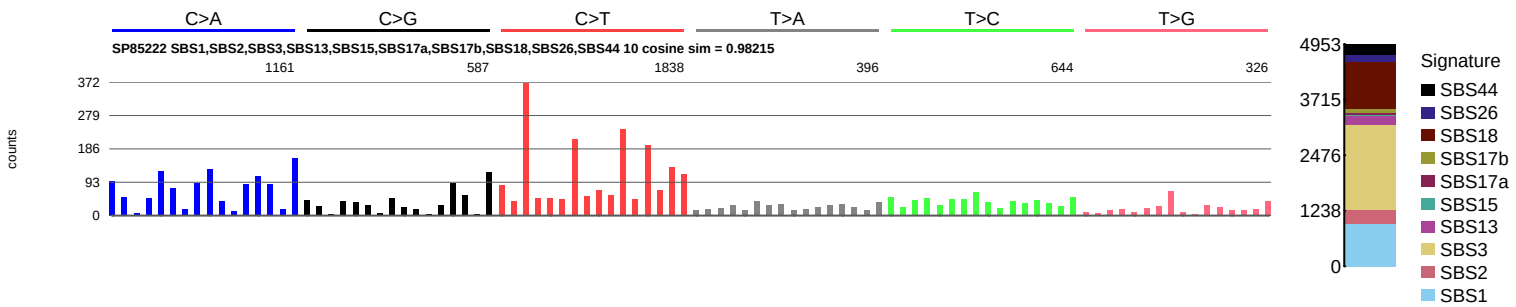

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98218
1133
469
1872
457
587
434
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS3
SBS2
SBS1

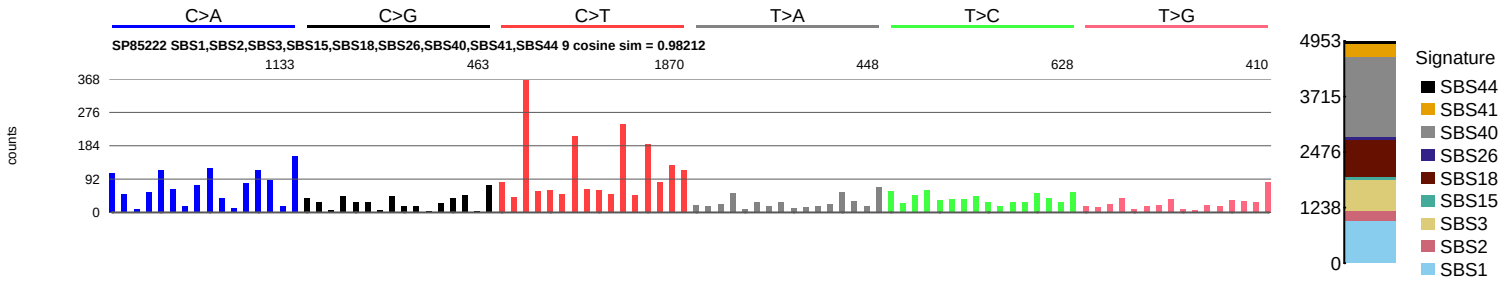

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98217
1120
418
1903
421
663
425
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1

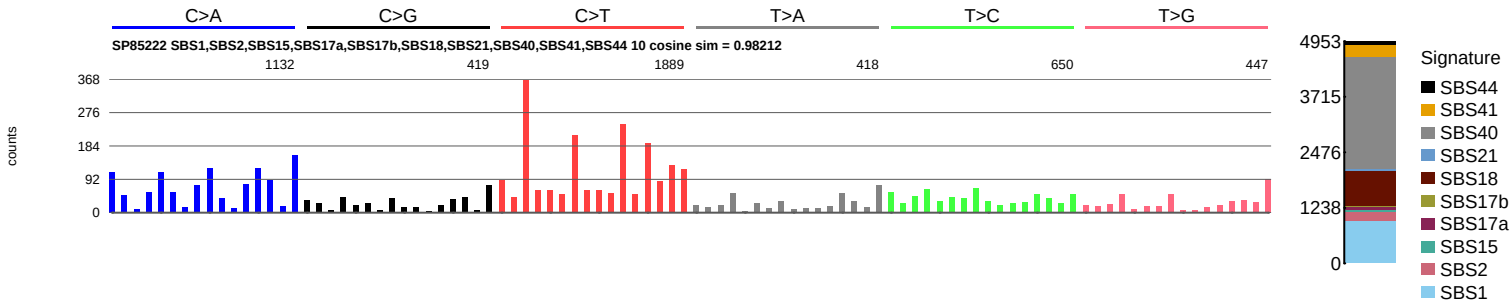

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98216
1163
586
1836
397
645
326
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

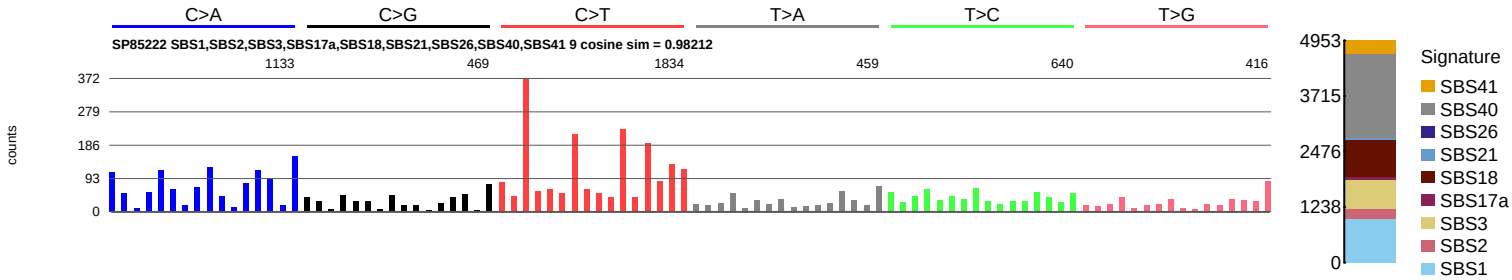

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98216
1132
469
1872
454
585
435
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

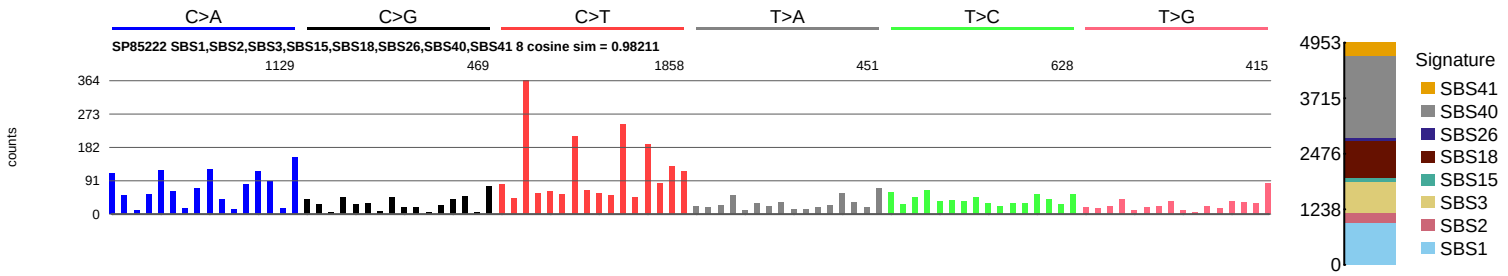

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98215
1131
475
1859
453
597
441
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

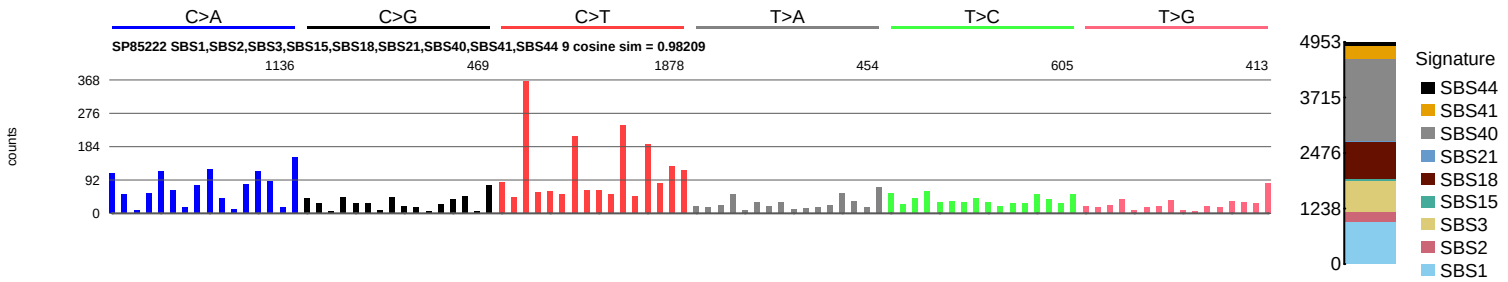

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98215
1161
587
1838
396
644
326
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

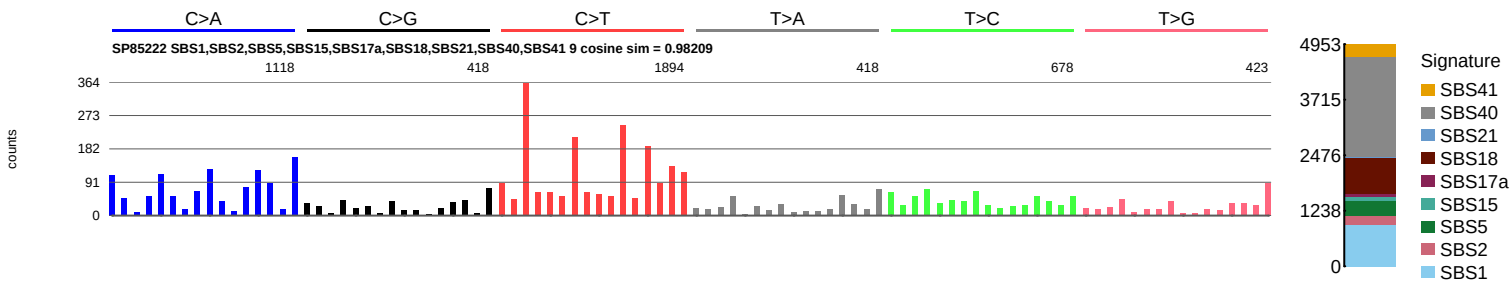

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98212
1133
463
1870
448
628
410
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS15
SBS3
SBS2
SBS1

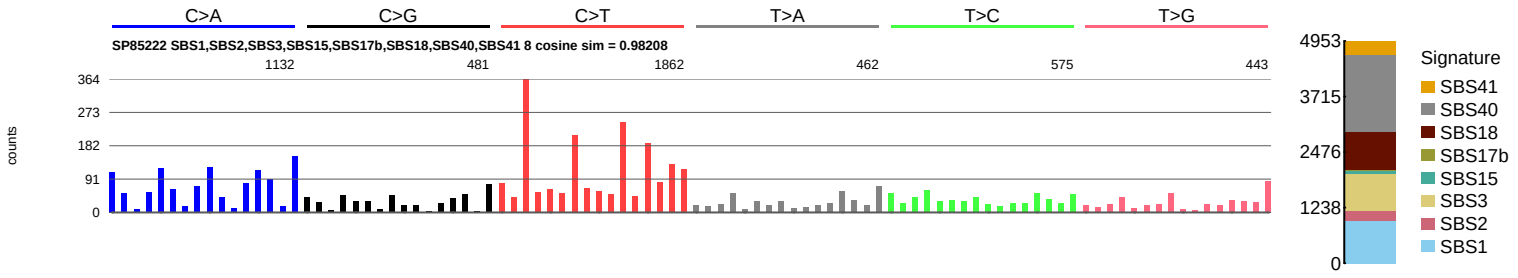

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98212
1132
419
1889
418
650
447
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

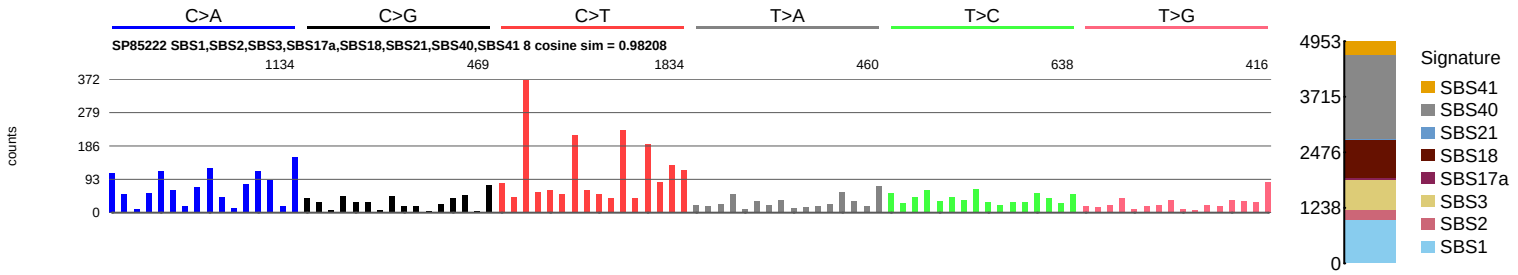

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98212
1133
469
1834
459
640
416
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS3
SBS2
SBS1

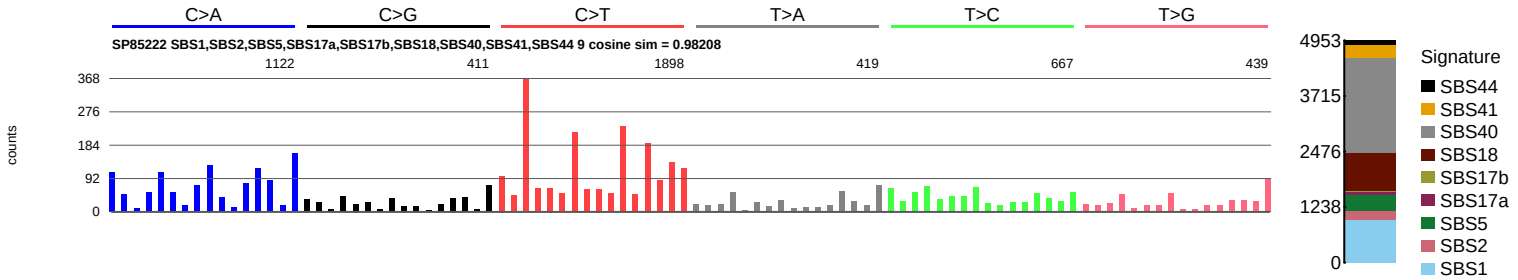

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98211
1129
469
1858
451
628
415
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS3
SBS2
SBS1

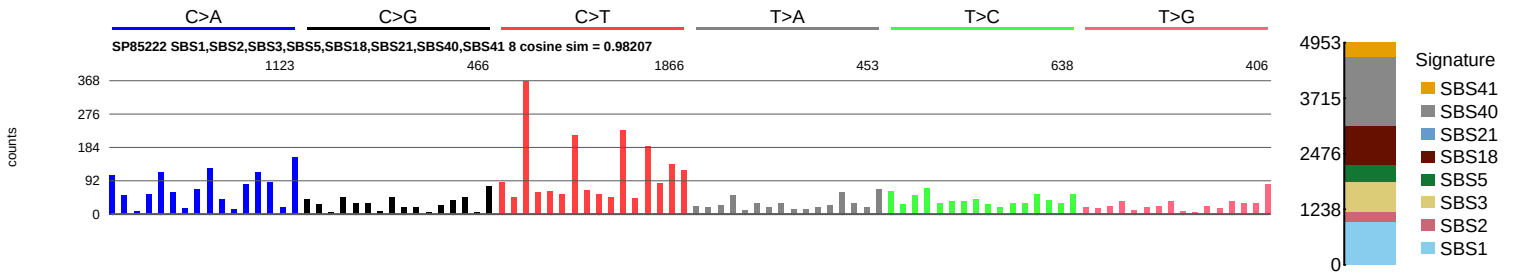

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98209
1136
469
1878
454
605
413
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98209
1118
418
1894
418
678
423
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98208
1132
481
1862
462
575
443
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98208
1134
469
1834
460
638
416
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98208
1122
411
1898
419
667
439
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98207
1123
466
1866
453
638
406
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
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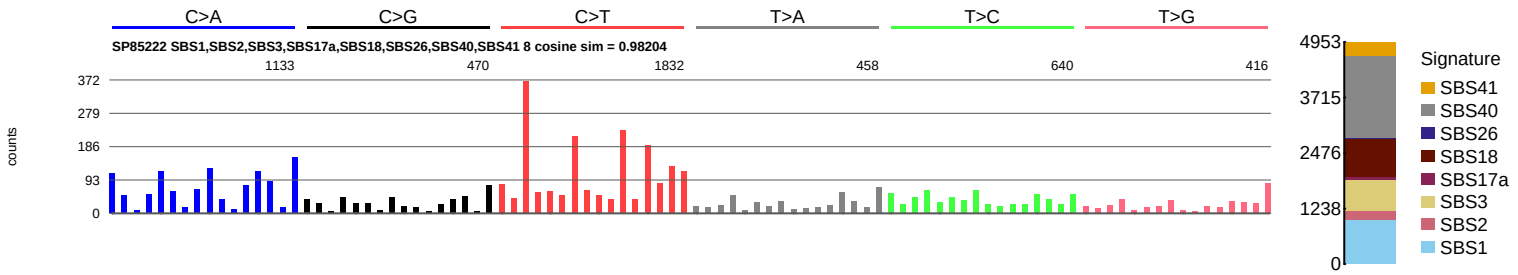
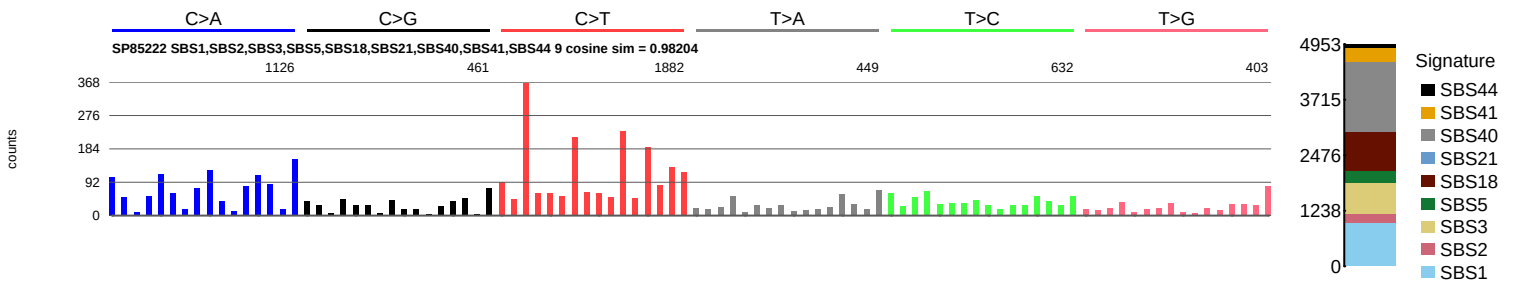
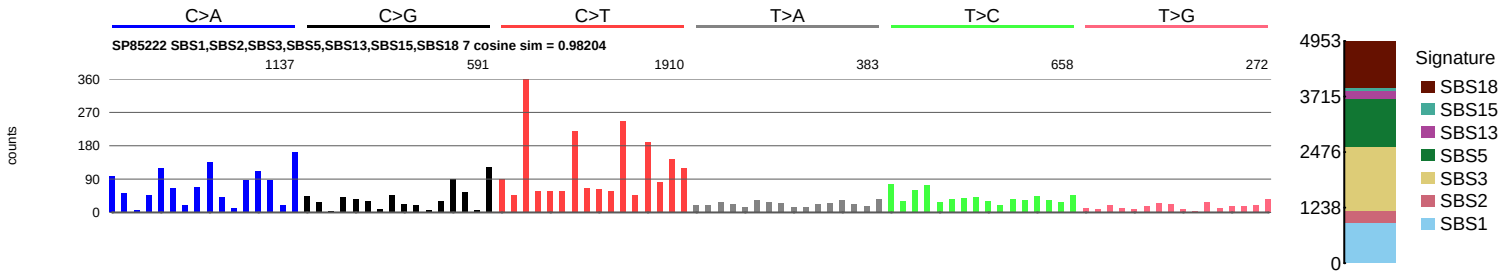
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98204
1133
470
1832
458
640
416
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98204
1126
461
1882
449
632
403
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS5
SBS3
SBS2
SBS1


C>A
C>G
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T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18 7 cosine sim = 0.98204
1137
591
1910
383
658
272
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS18
SBS15
SBS13
SBS5
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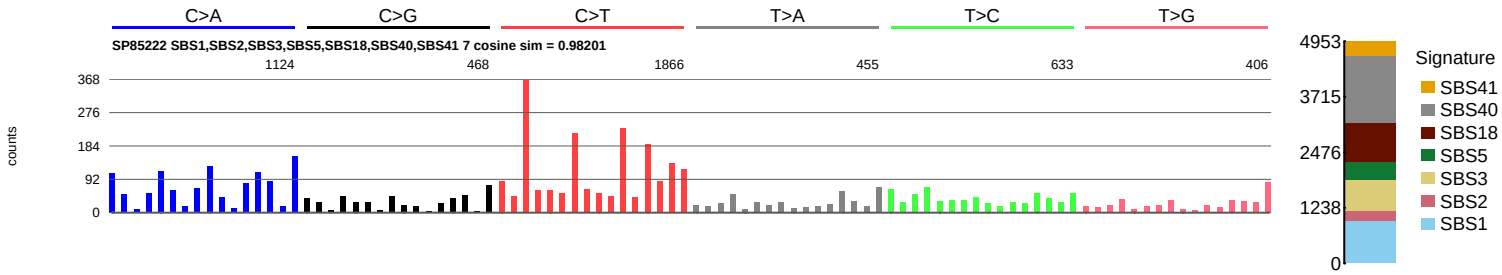
SBS2
SBS1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C>G
C>T
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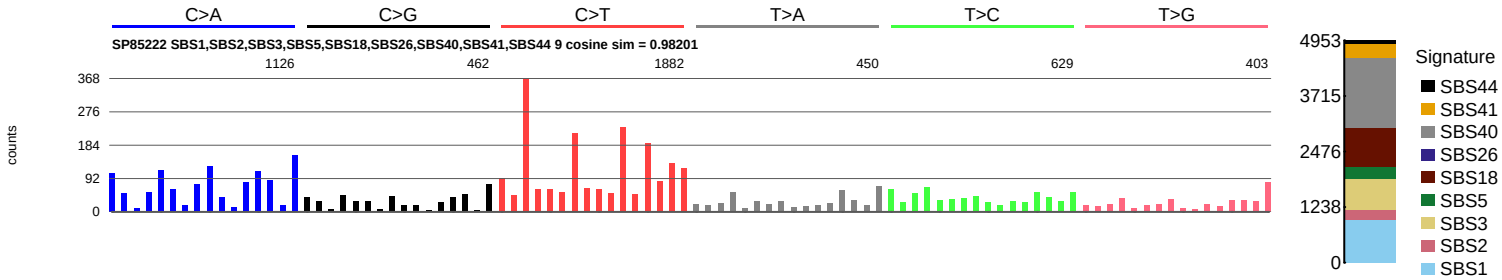
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS18,SBS40,SBS41 7 cosine sim = 0.98201
1124
468
1866
455
633
406
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS5
SBS3
SBS2
SBS1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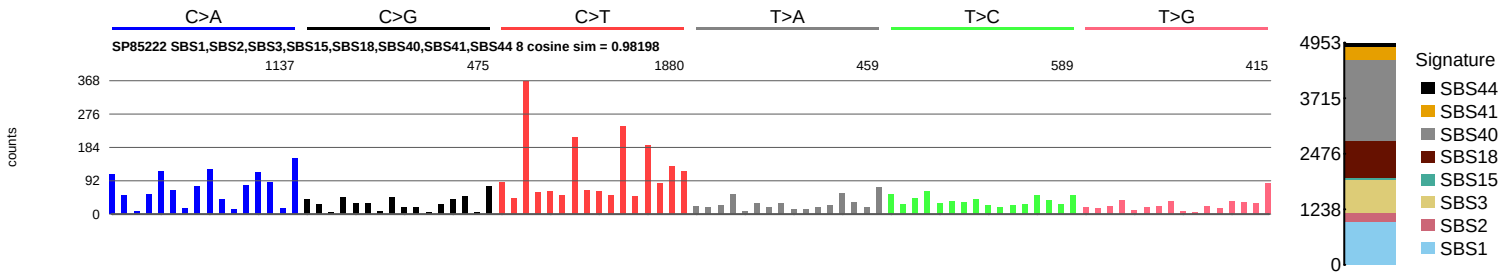
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98201
1126
462
1882
450
629
403
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS5
SBS3
SBS2
SBS1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C>G
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T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98198
1137
475
1880
459
589
415
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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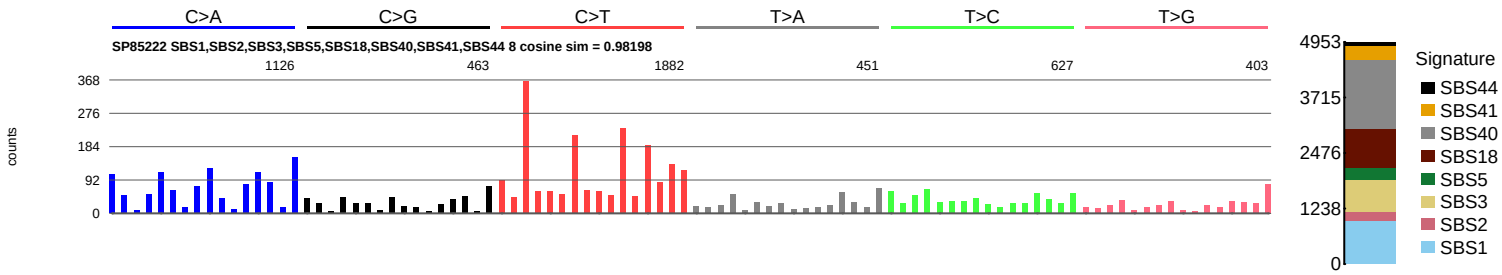
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98198
1126
463
1882
451
627
403
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
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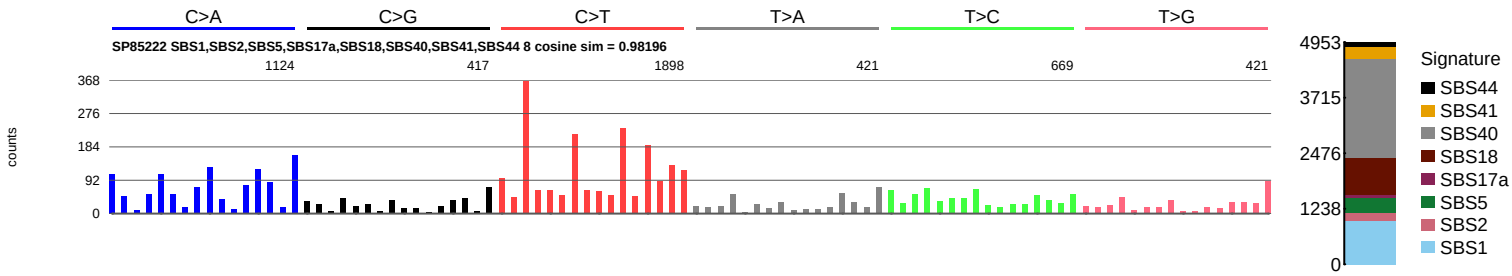
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98196
1124
417
1898
421
669
421
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
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SBS17a
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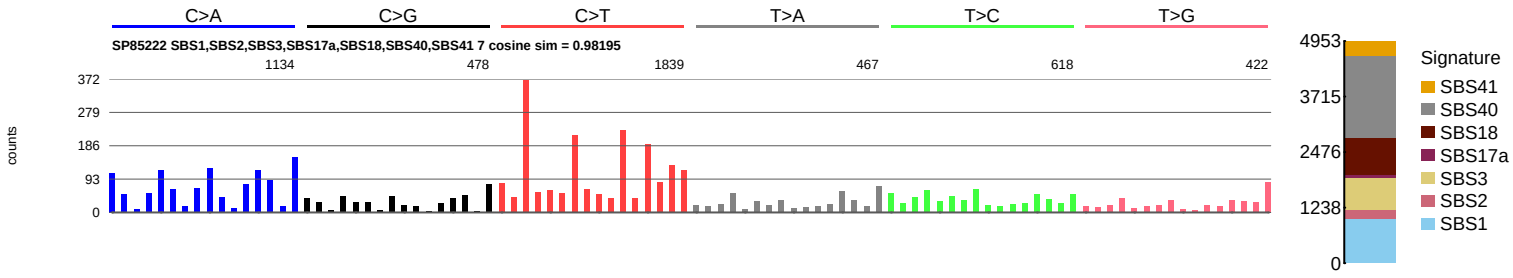
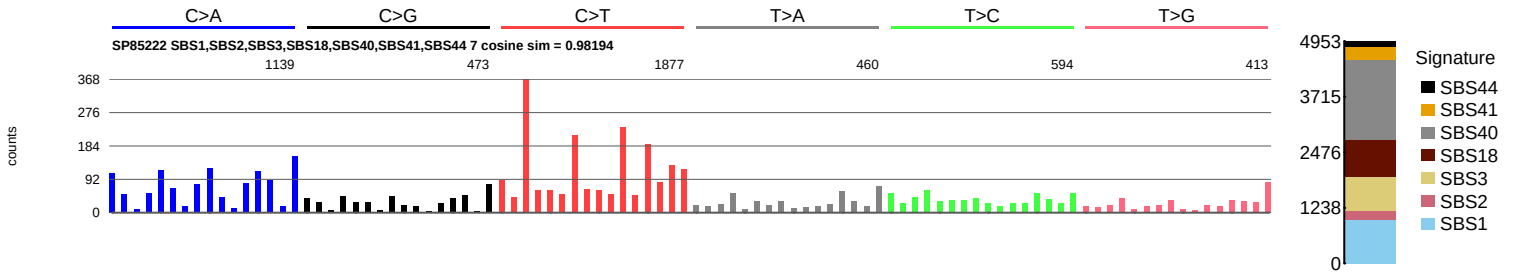
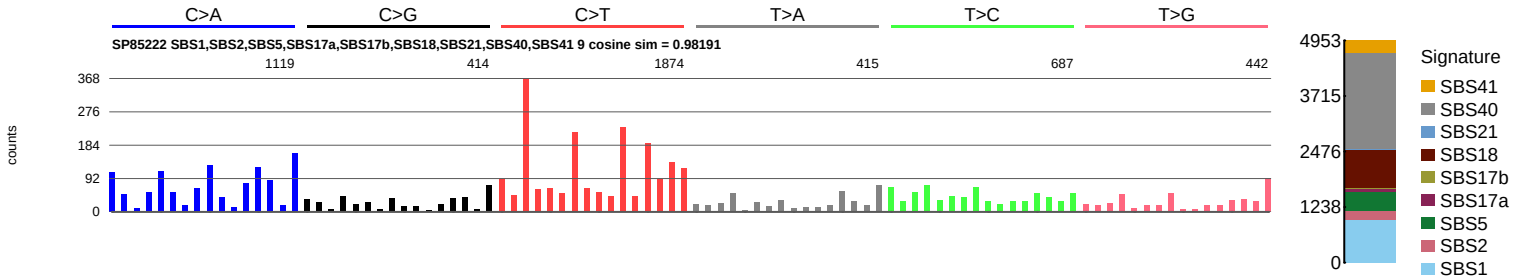
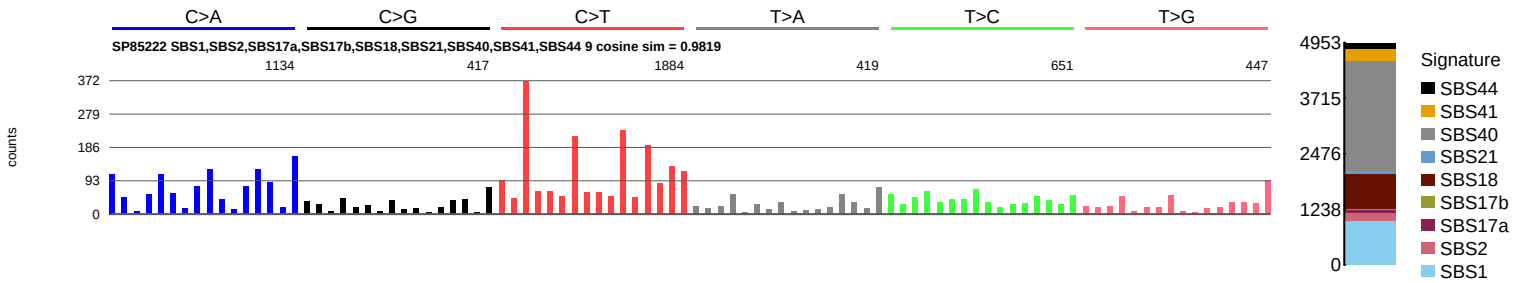
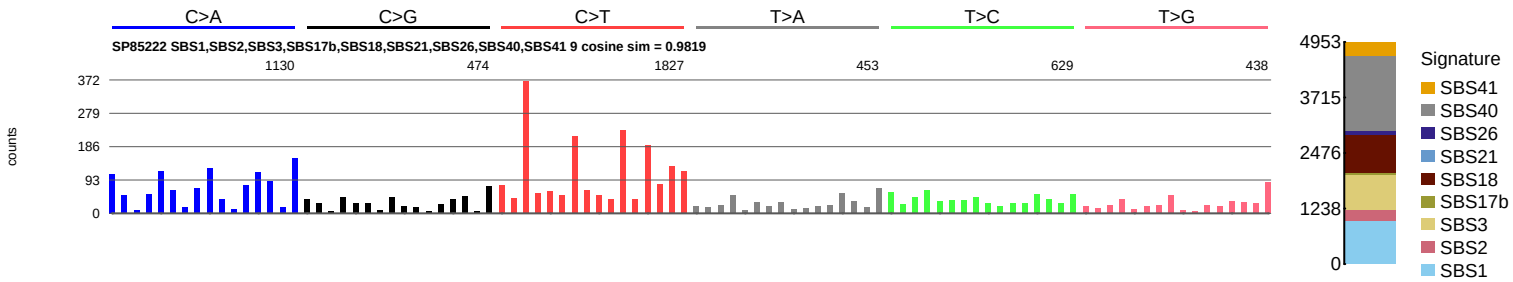
SBS2
SBS1

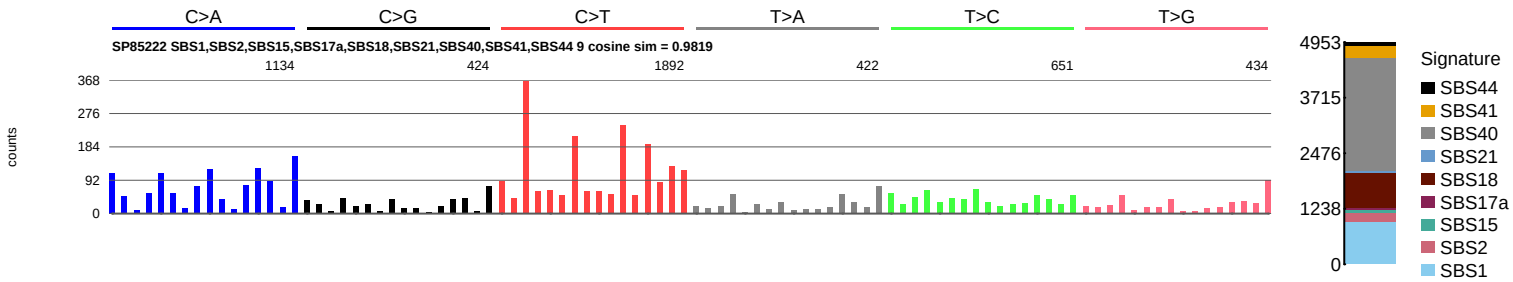

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS40,SBS41 7 cosine sim = 0.98195
1134
478
1839
467
618
422
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98194
1139
473
1877
460
594
413
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98191
1119
414
1874
415
687
442
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9819
1134
417
1884
419
651
447
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.9819
1130
474
1827
453
629
438
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9819
1134
424
1892
422
651
434
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
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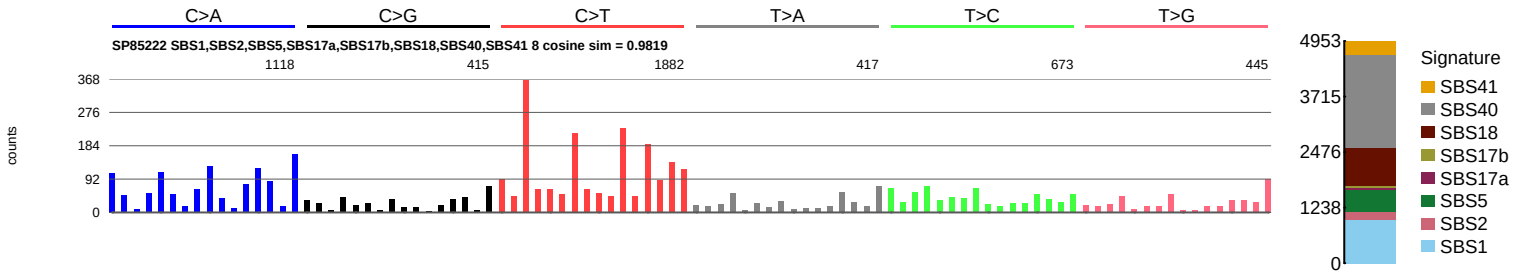
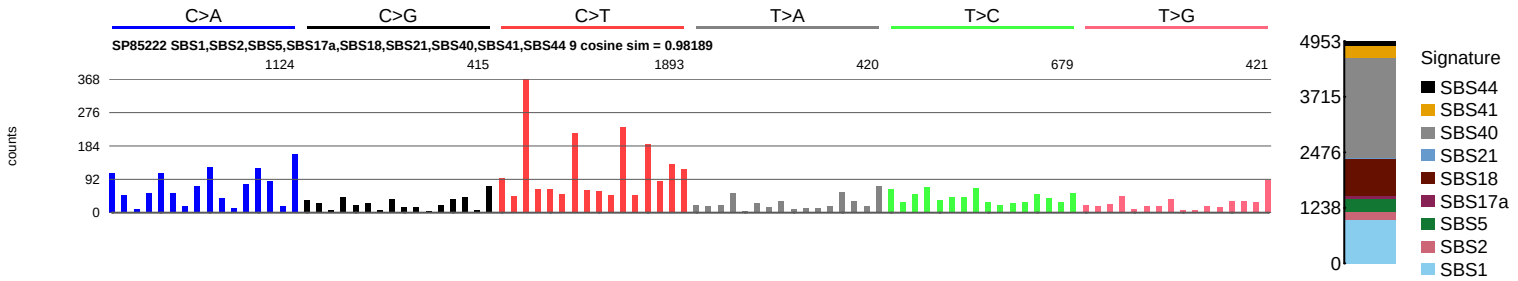
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.9819
1118
415
1882
417
673
445
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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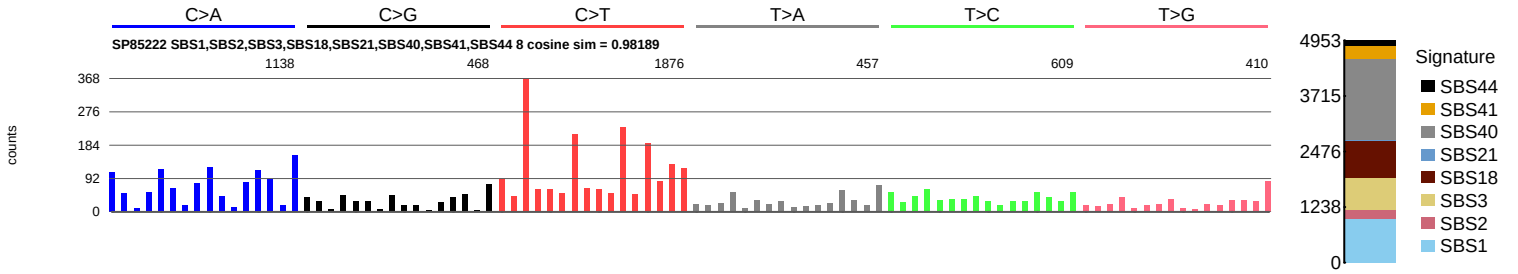
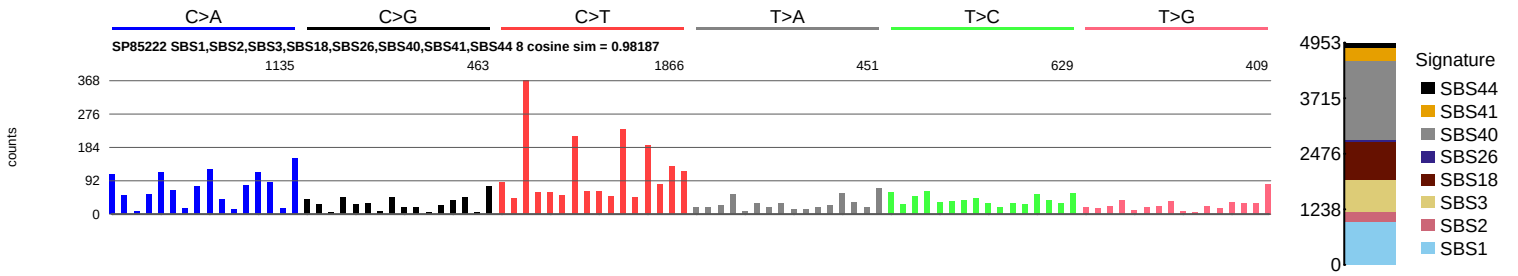
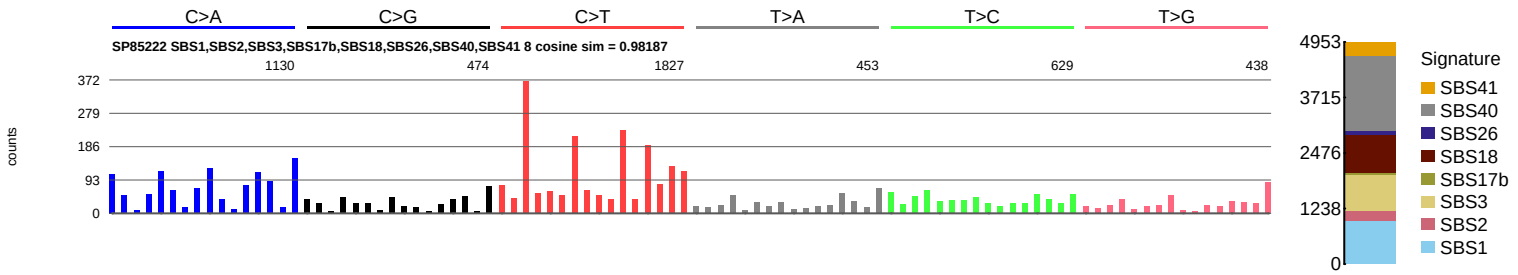
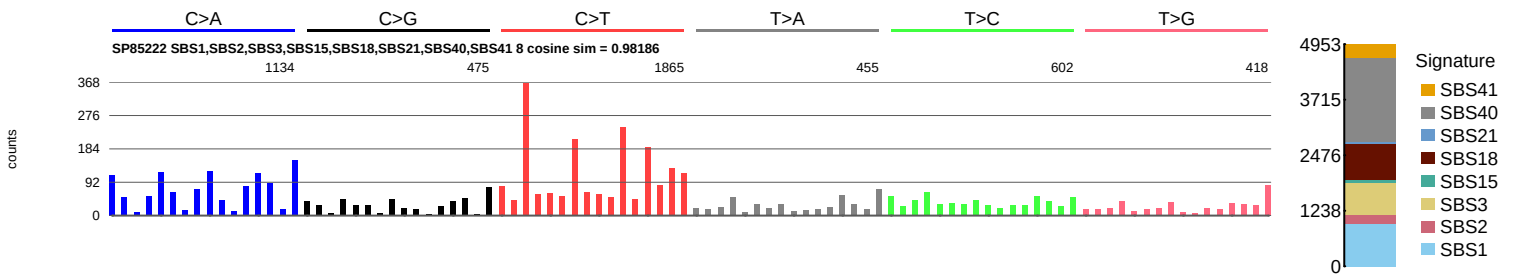
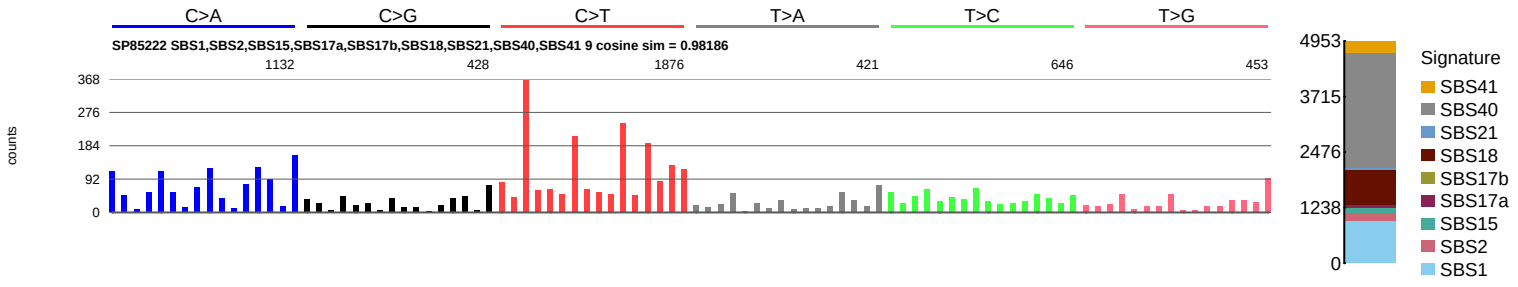
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98189
1124
415
1893
420
679
421
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS2
SBS1

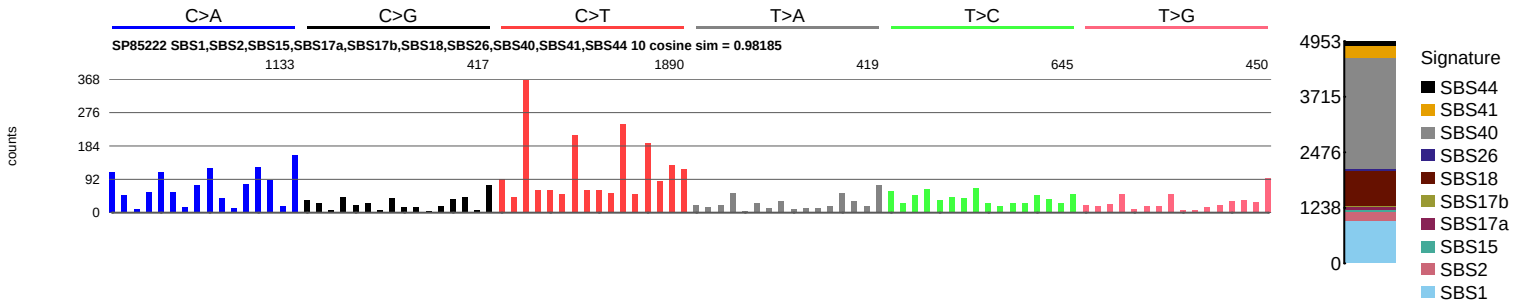

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98189
1138
468
1876
457
609
410
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS3
SBS2
SBS1

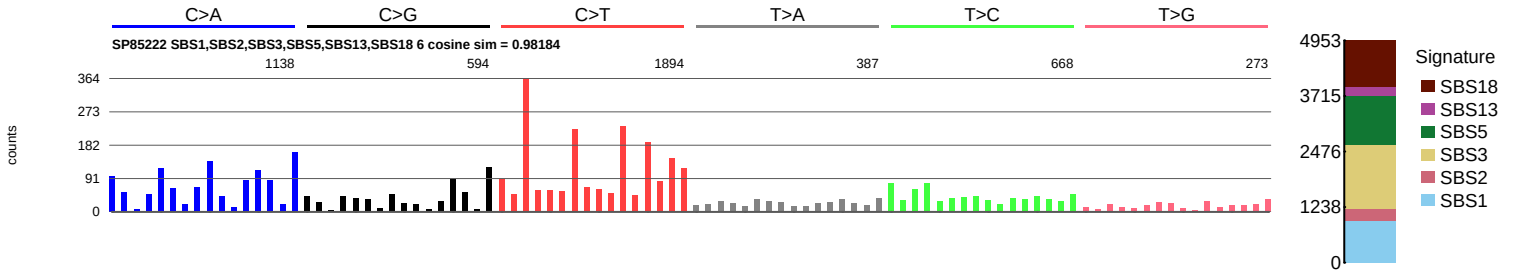

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98187
1135
463
1866
451
629
409
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS3
SBS2
SBS1

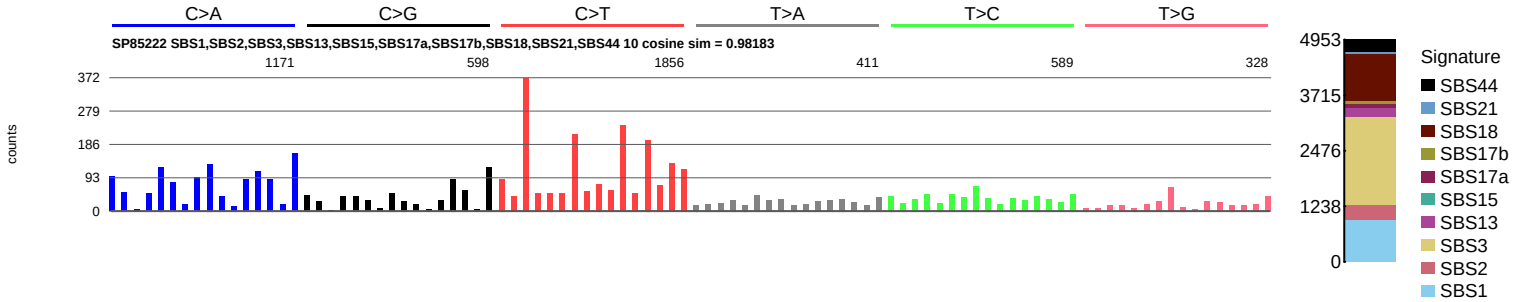

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98187
1130
474
1827
453
629
438
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1

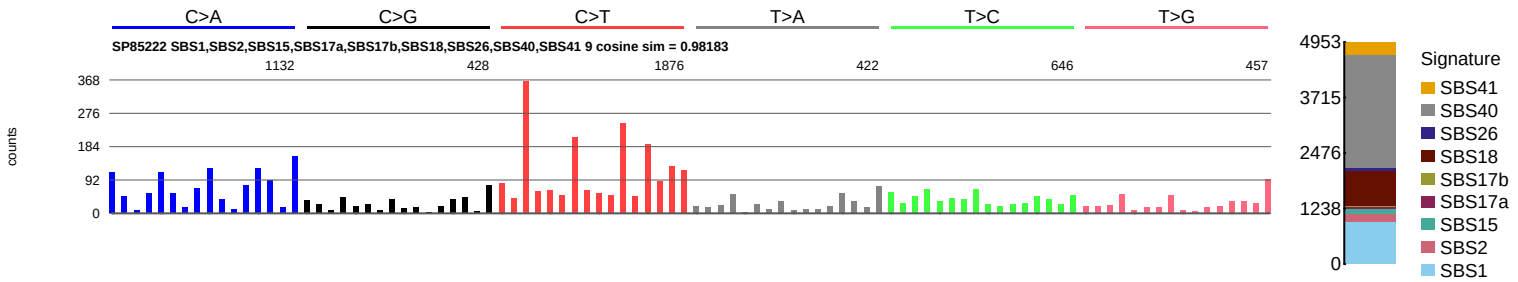

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98186
1134
475
1865
455
602
418
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98186
1132
428
1876
421
646
453
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98185
1133
417
1890
419
645
450
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18 6 cosine sim = 0.98184
1138
594
1894
387
668
273
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98183
1171
598
1856
411
589
328
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98183
1132
428
1876
422
646
457
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
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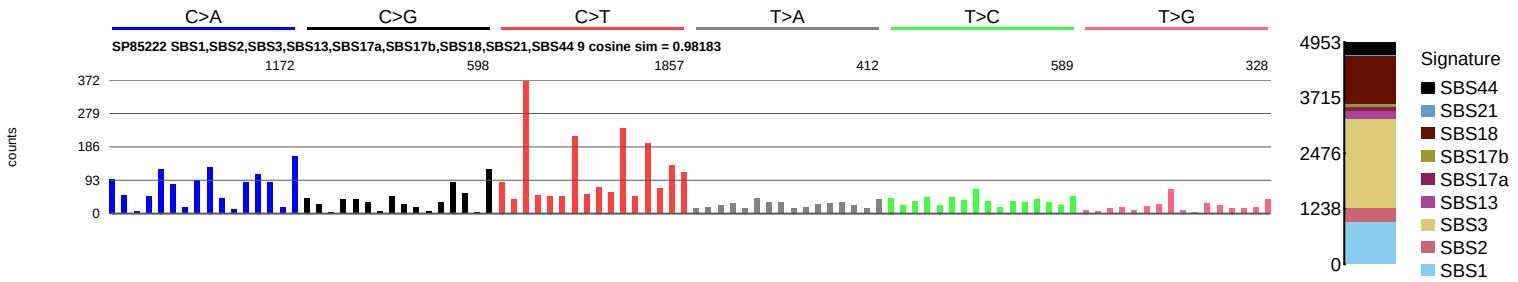
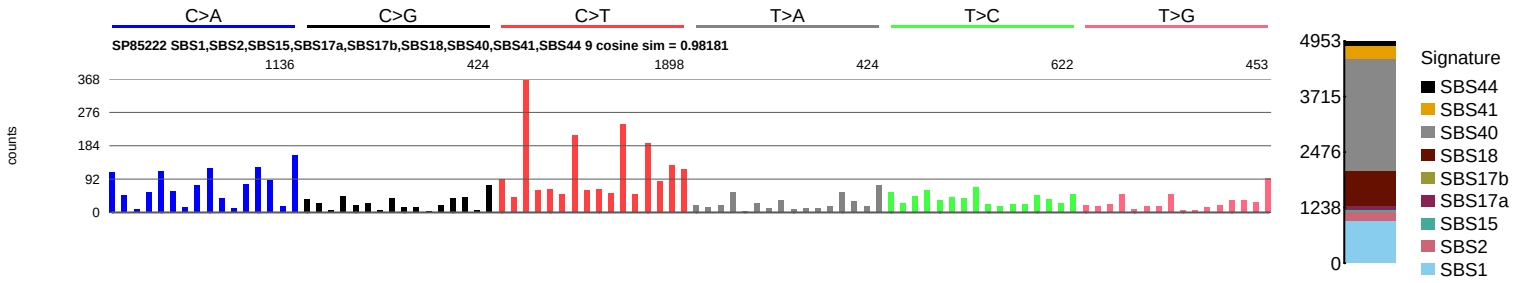
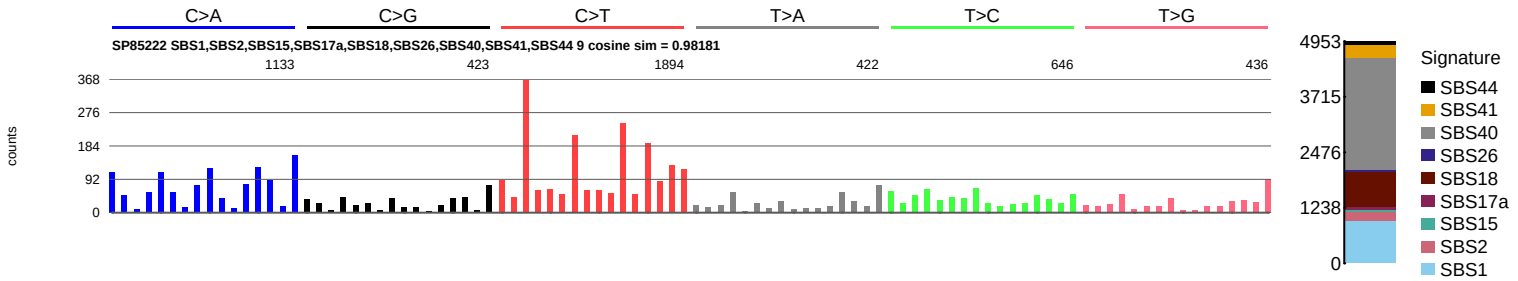
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98183
1172
598
1857
412
589
328
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98181
1136
424
1898
424
622
453
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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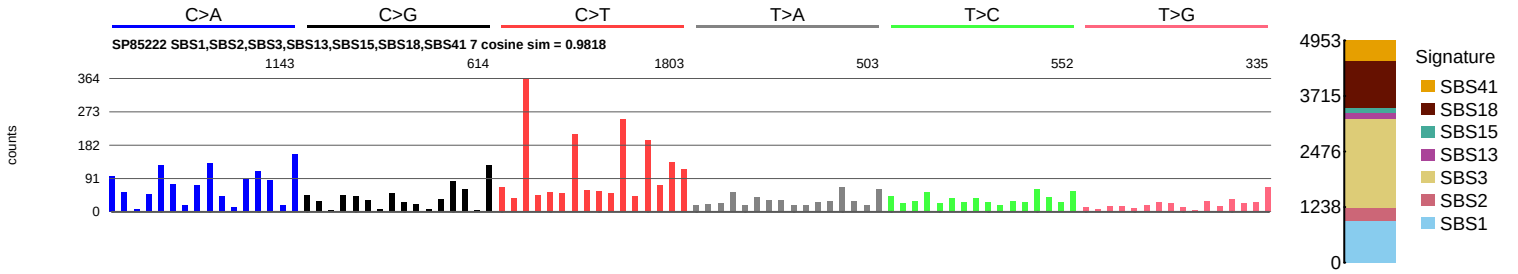
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98181
1133
423
1894
422
646
436
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS2
SBS1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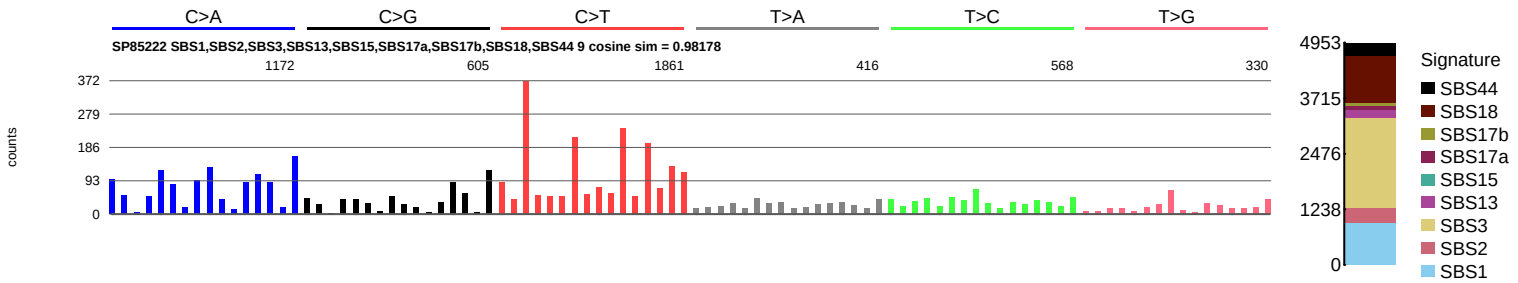
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS41 7 cosine sim = 0.9818
1143
614
1803
503
552
335
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1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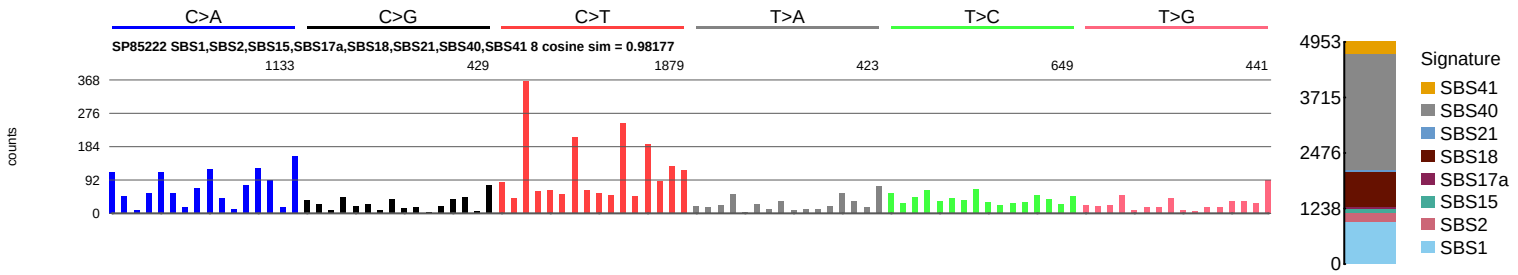
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98178
1172
605
1861
416
568
330
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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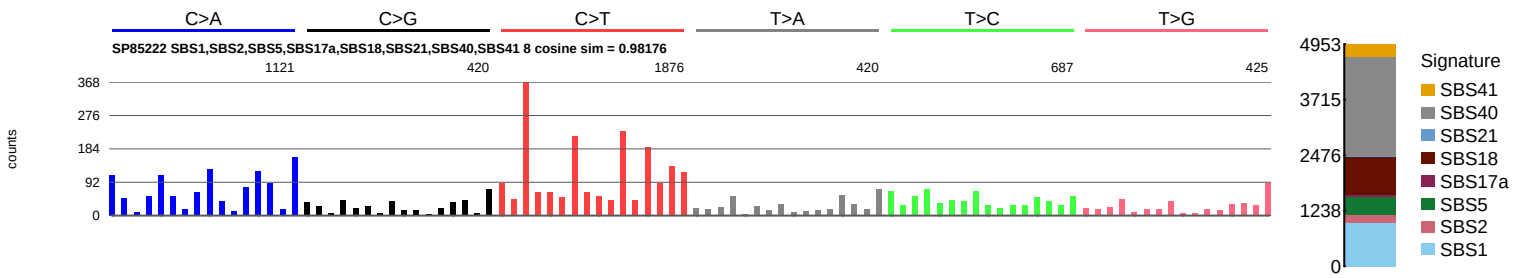
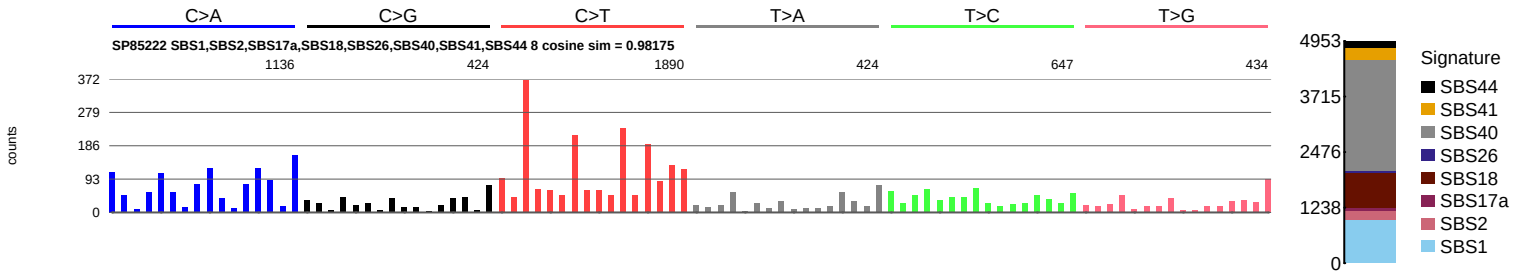
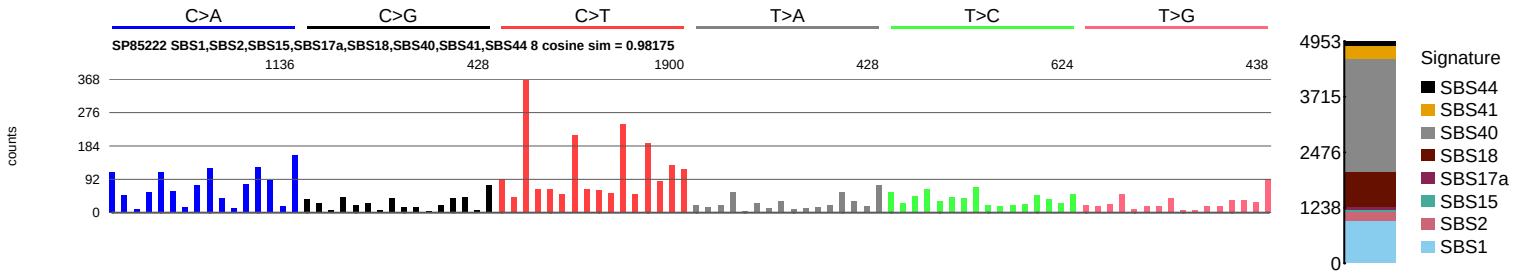
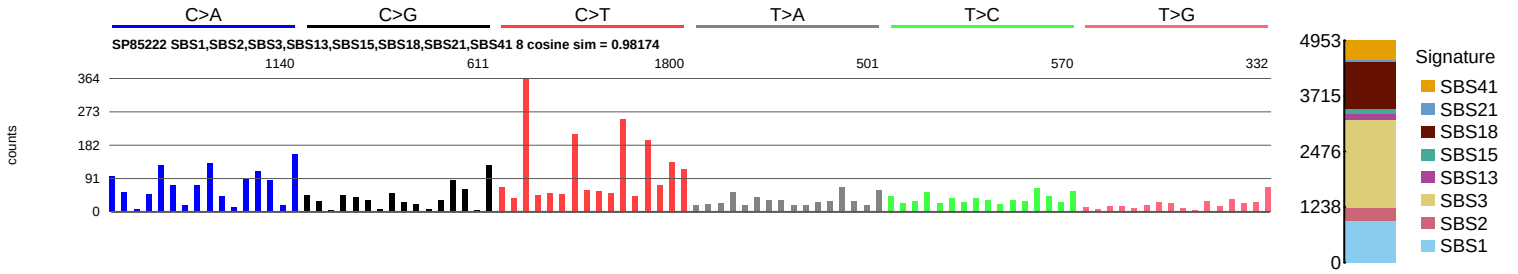
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98177
1133
429
1879
423
649
441
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98176
1121
420
1876
420
687
425
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98175
1136
424
1890
424
647
434
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98175
1136
428
1900
428
624
438
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS41 8 cosine sim = 0.98174
1140
611
1800
501
570
332
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS21
SBS18
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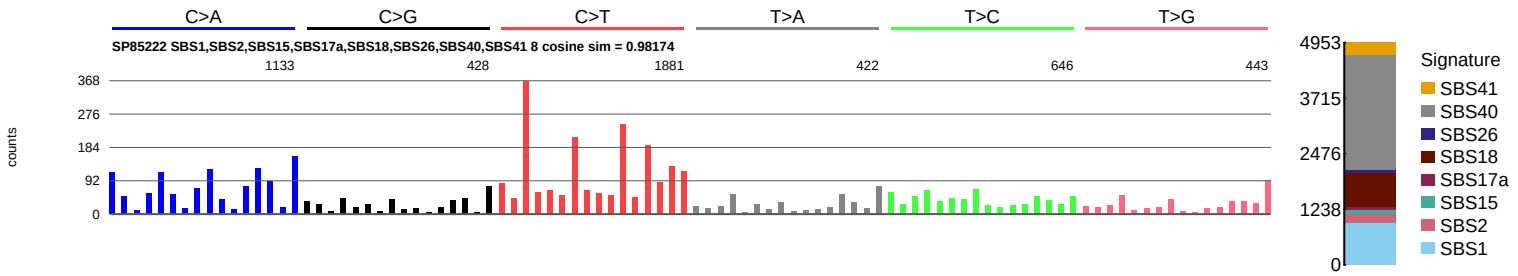
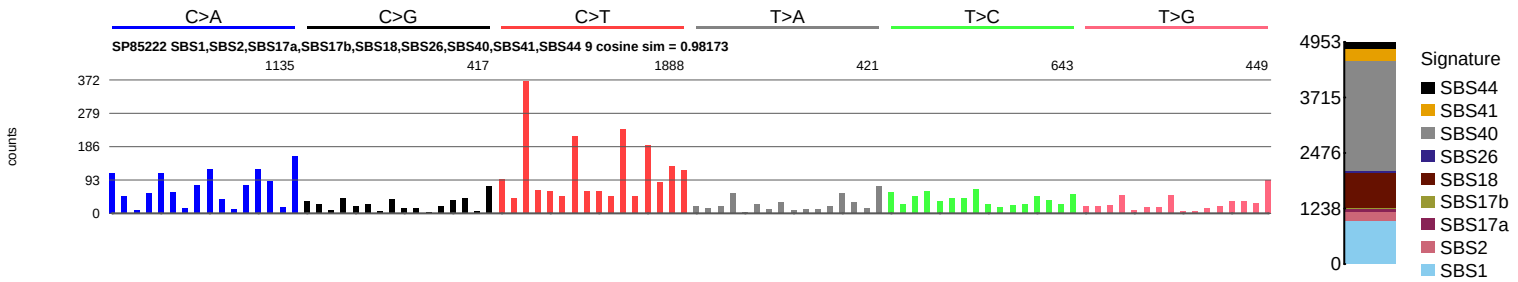
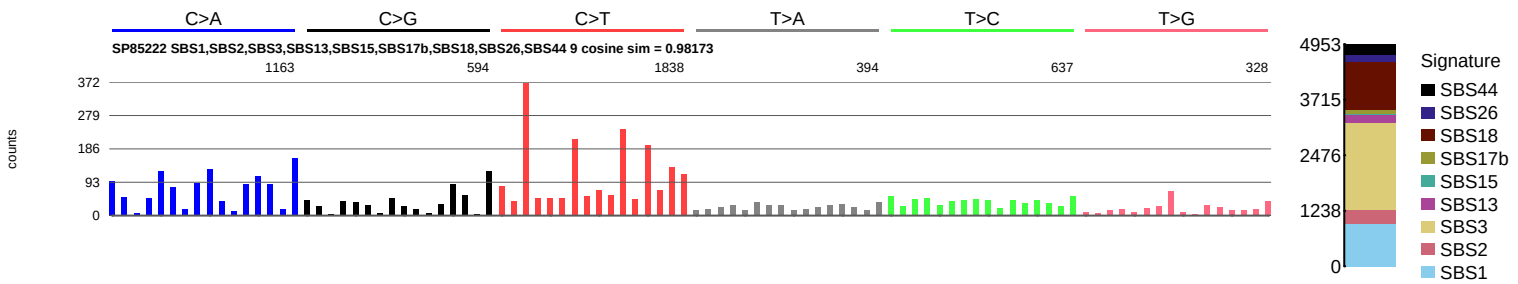
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98174
1133
428
1881
422
646
443
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98173
1135
417
1888
421
643
449
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98173
1163
594
1838
394
637
328
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS26
SBS18
SBS17b
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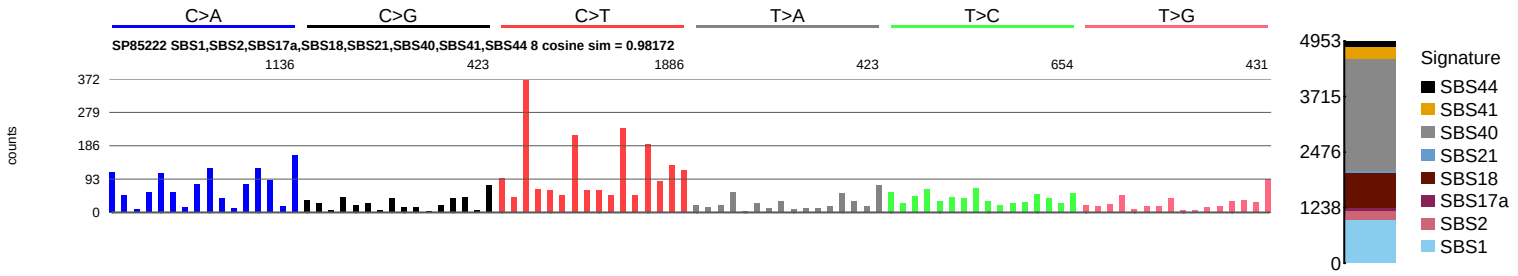
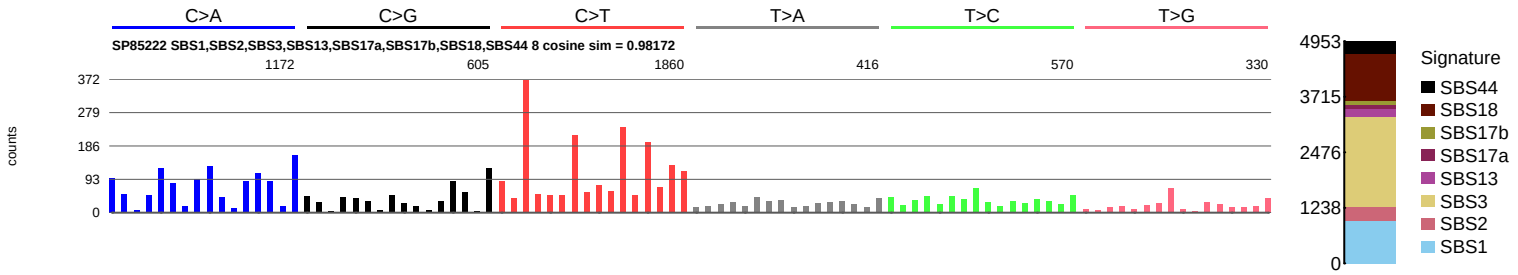
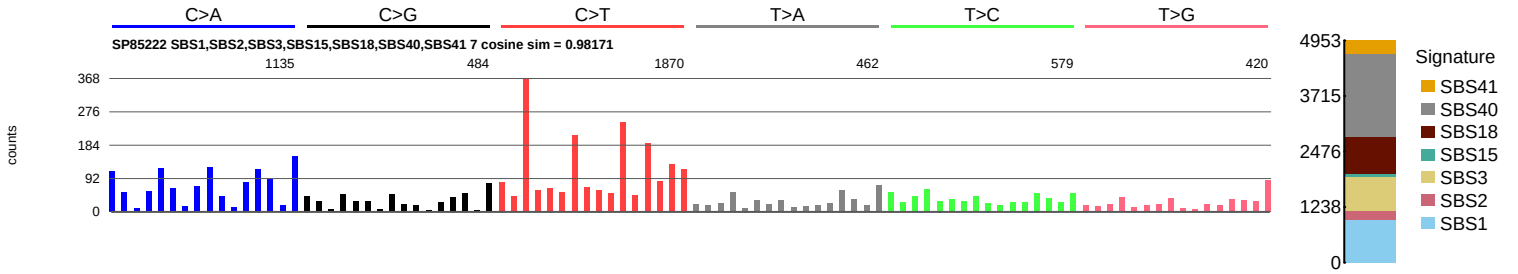
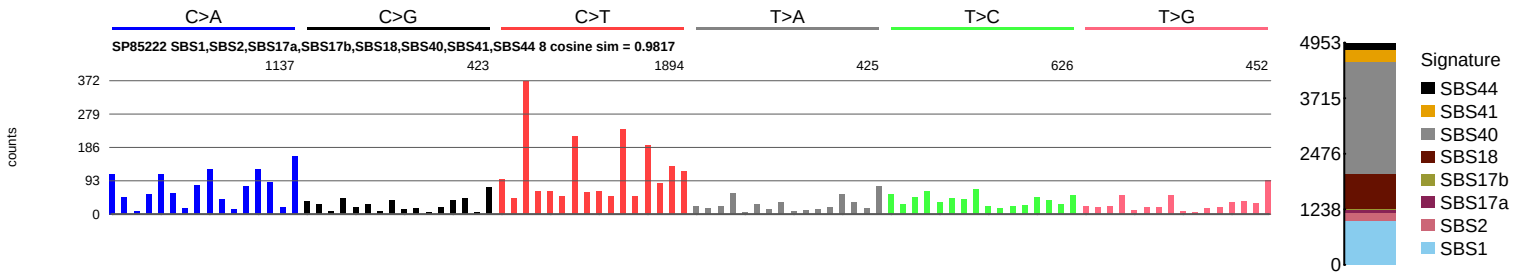
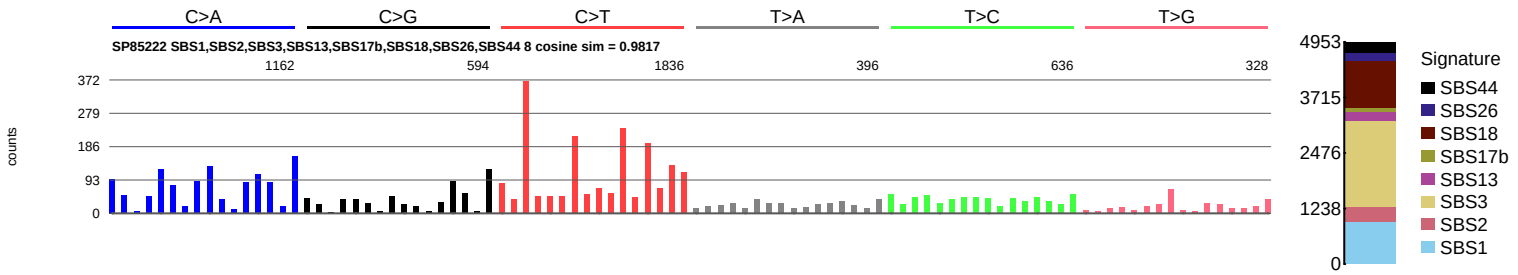
SBS3
SBS2
SBS1

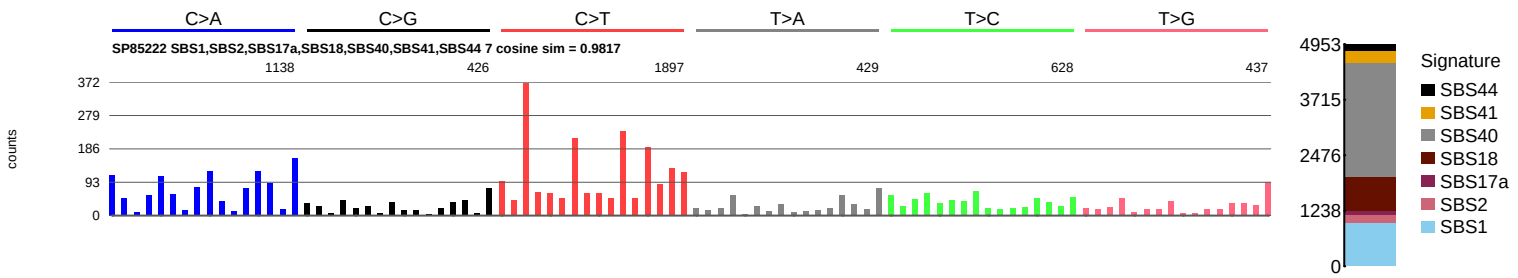

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98172
1136
423
1886
423
654
431
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS2
SBS1

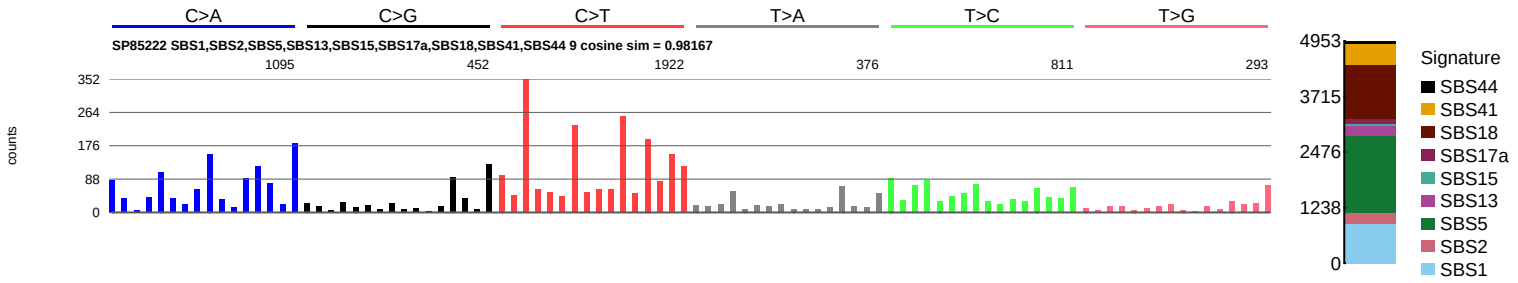

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98172
1172
605
1860
416
570
330
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

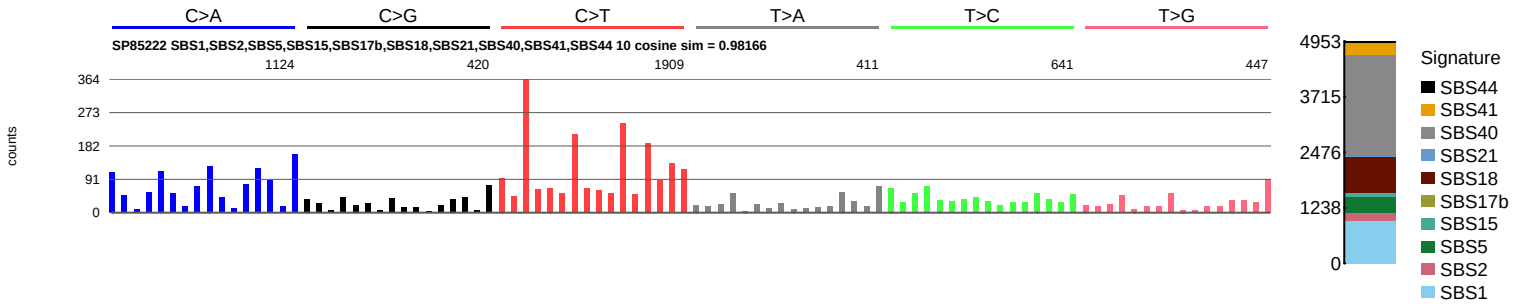

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS18,SBS40,SBS41 7 cosine sim = 0.98171
1135
484
1870
462
579
420
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.9817
1137
423
1894
425
626
452
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.9817
1162
594
1836
396
636
328
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17a,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.9817
1138
426
1897
429
628
437
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41,SBS44 9 cosine sim = 0.98167
1095
452
1922
376
811
293
counts
352
264
176
88
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98166
1124
420
1909
411
641
447
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
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SBS18
SBS17b
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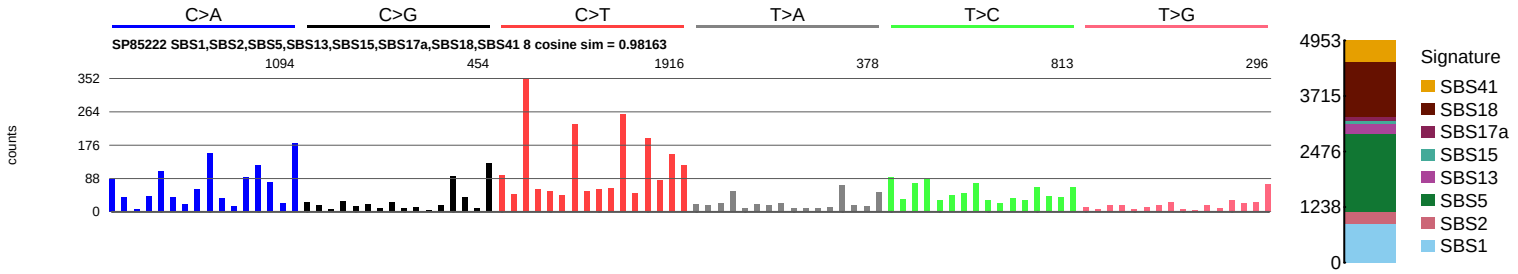
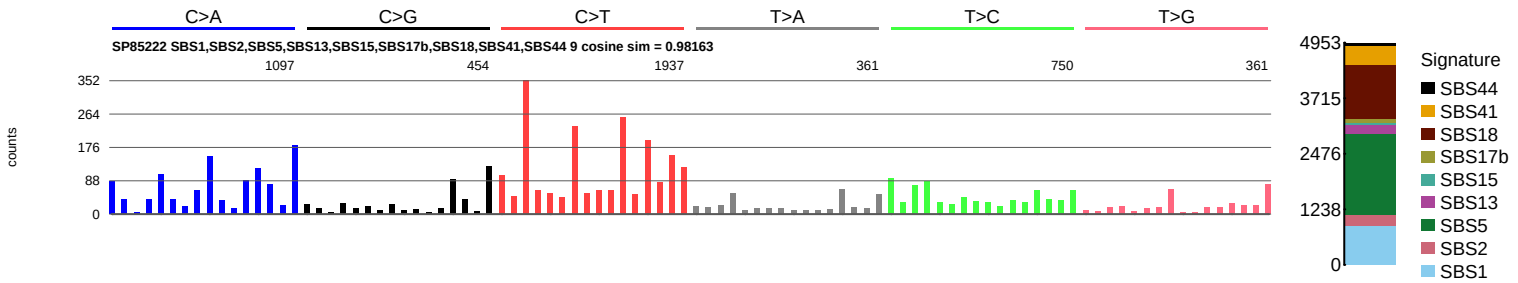
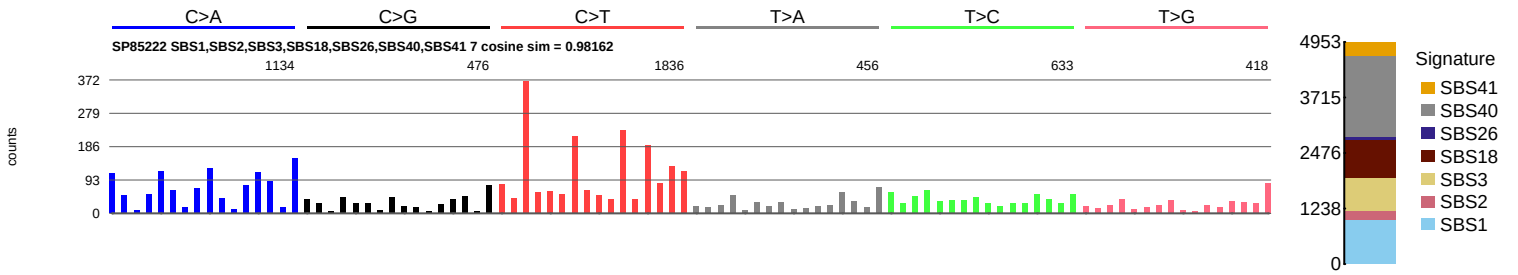
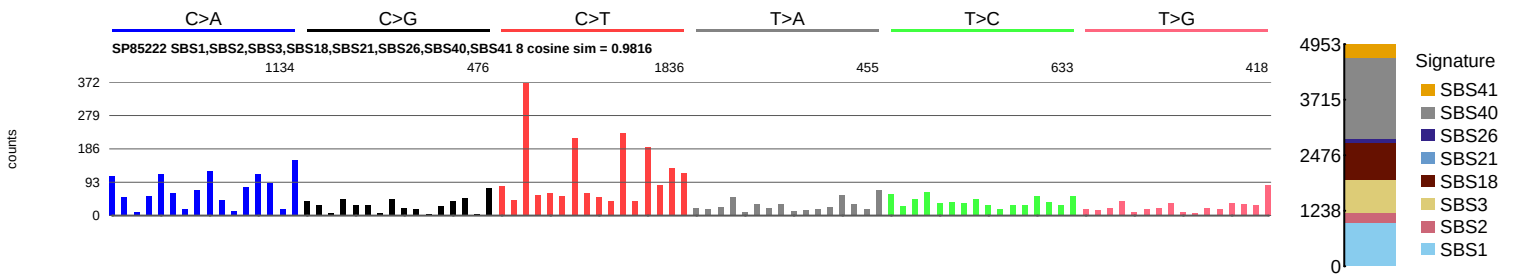
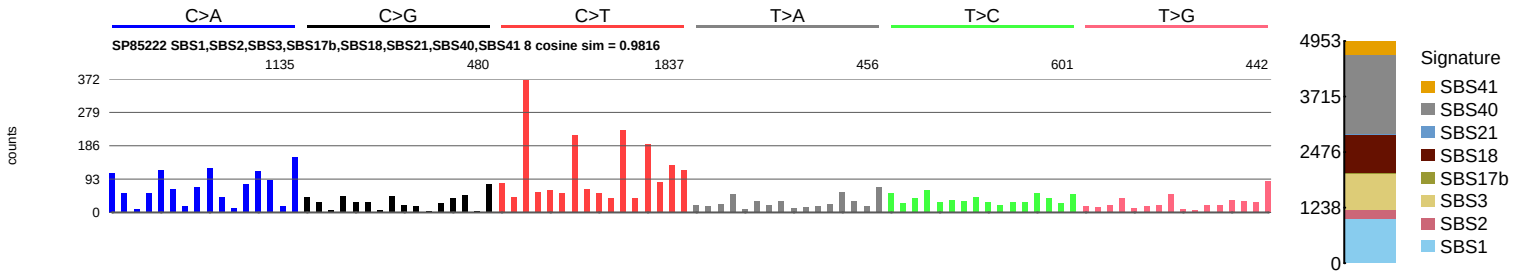
SBS5
SBS2
SBS1

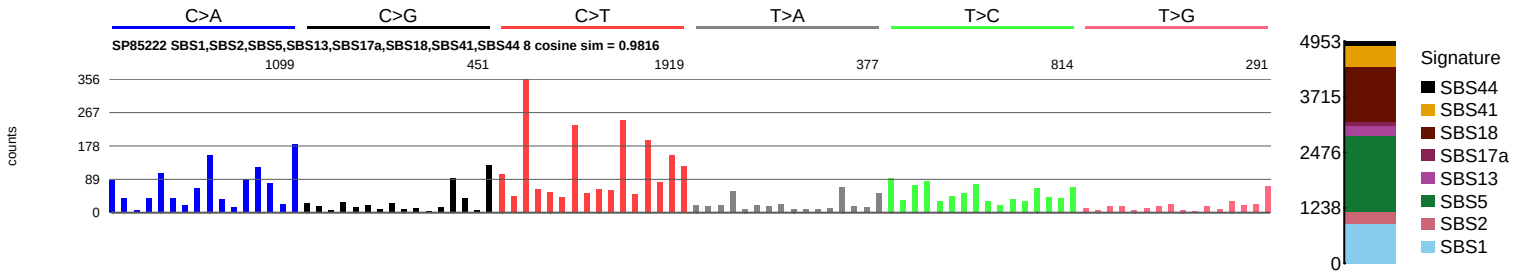

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41 8 cosine sim = 0.98163
1094
454
1916
378
813
296
counts
352
264
176
88
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

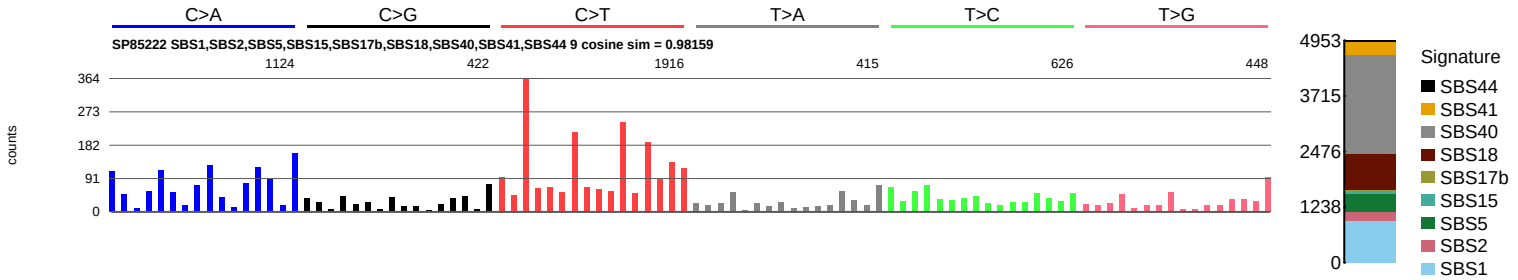

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98163
1097
454
1937
361
750
361
counts
352
264
176
88
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98162
1134
476
1836
456
633
418
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.9816
1134
476
1836
455
633
418
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.9816
1135
480
1837
456
601
442
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS41,SBS44 8 cosine sim = 0.9816
1099
451
1919
377
814
291
counts
356
267
178
89
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1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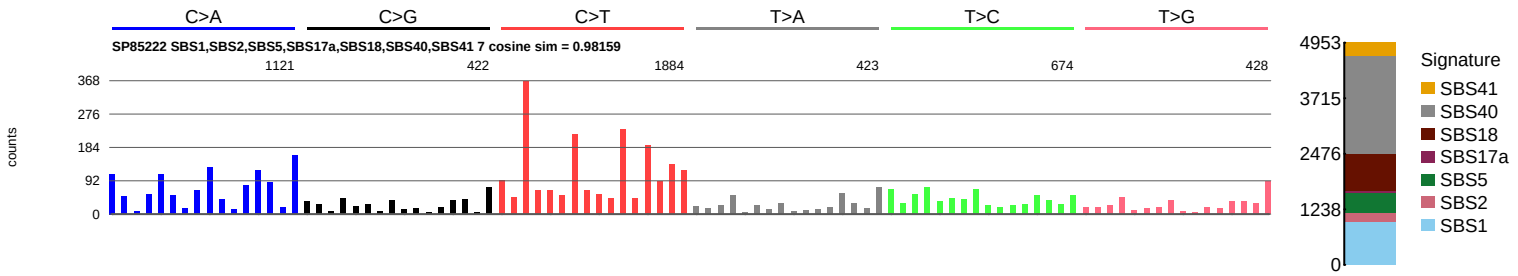
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98159
1124
422
1916
415
626
448
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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C>G
C>T
T>A
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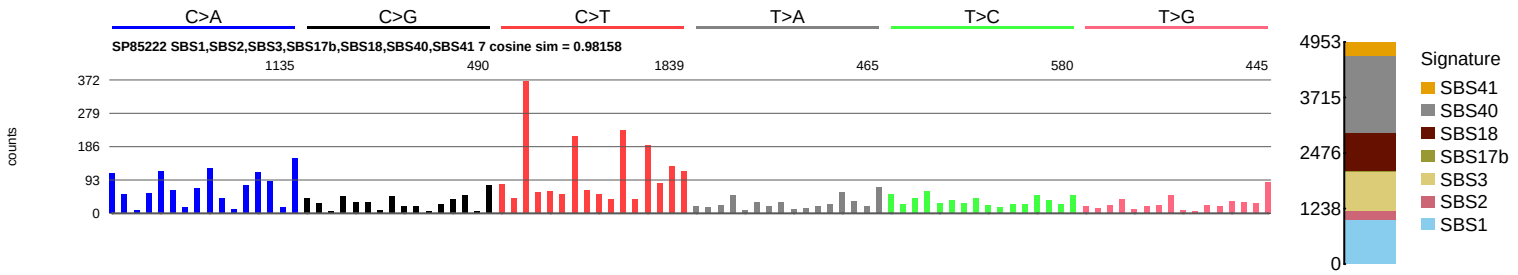
SP85222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS40,SBS41 7 cosine sim = 0.98159
1121
422
1884
423
674
428
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS5
SBS2
SBS1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C>G
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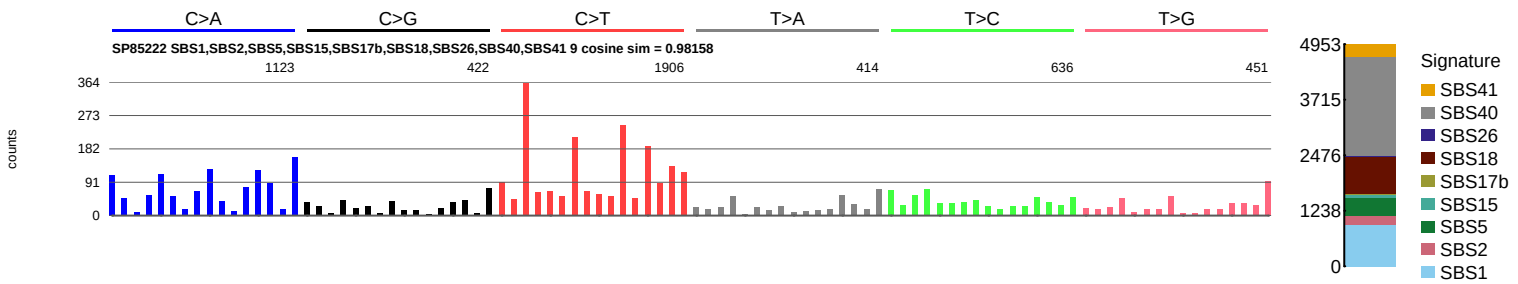
SP85222 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98158
1135
490
1839
465
580
445
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS3
SBS2
SBS1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T>A
T>C
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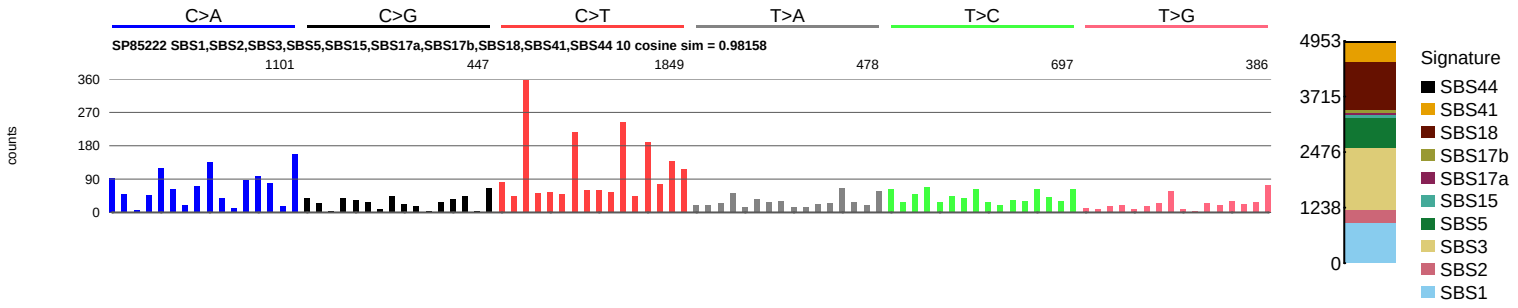
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98158
1123
422
1906
414
636
451
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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C>G
C>T
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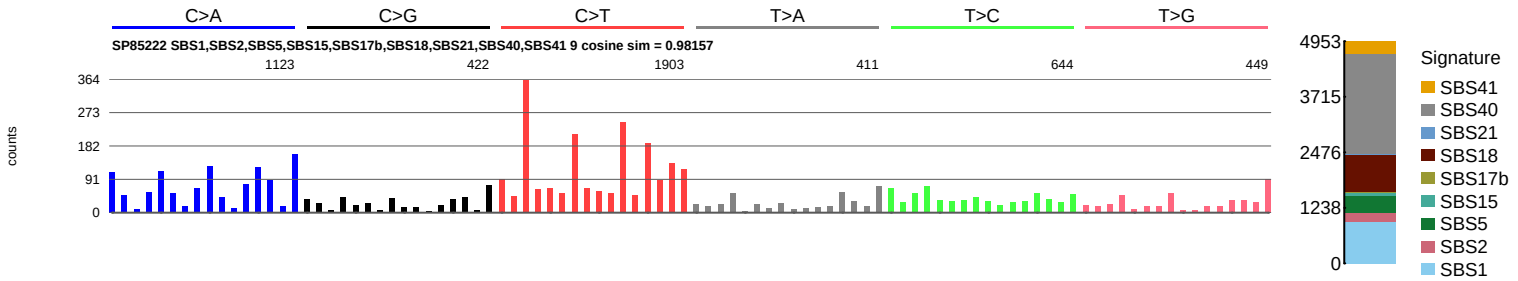
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98158
1101
447
1849
478
697
386
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98157
1123
422
1903
411
644
449
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
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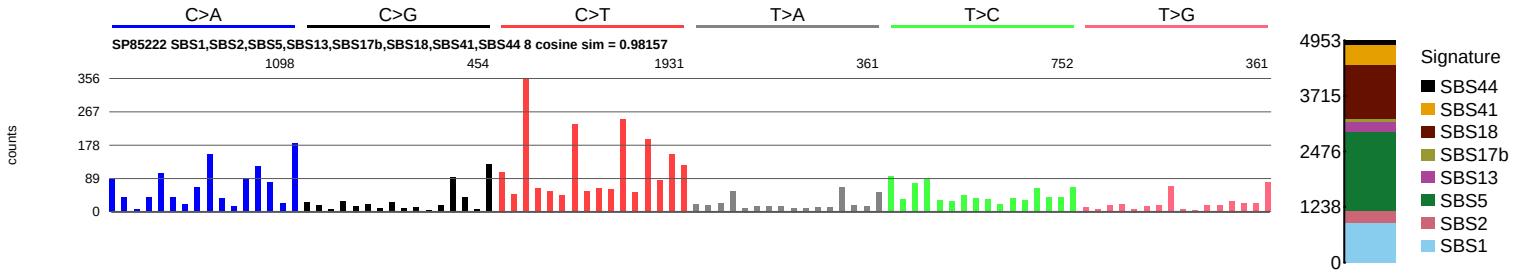
SBS2
SBS1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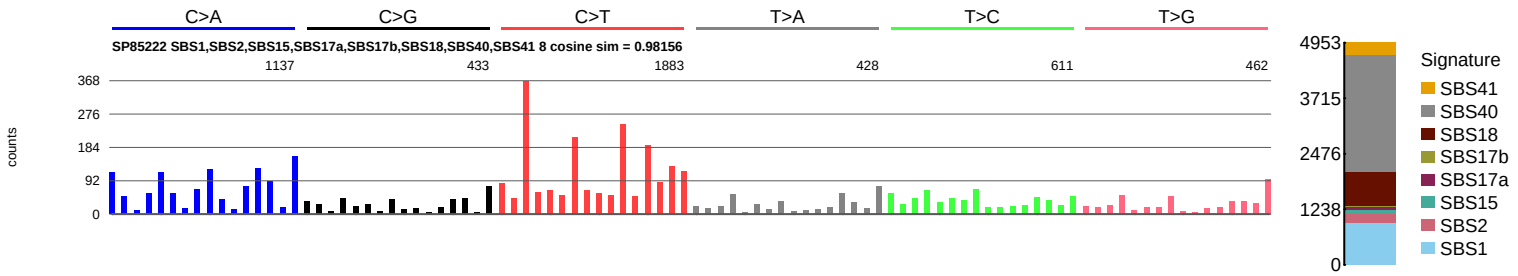
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98157
1098
454
1931
361
752
361
counts
356
267
178
89
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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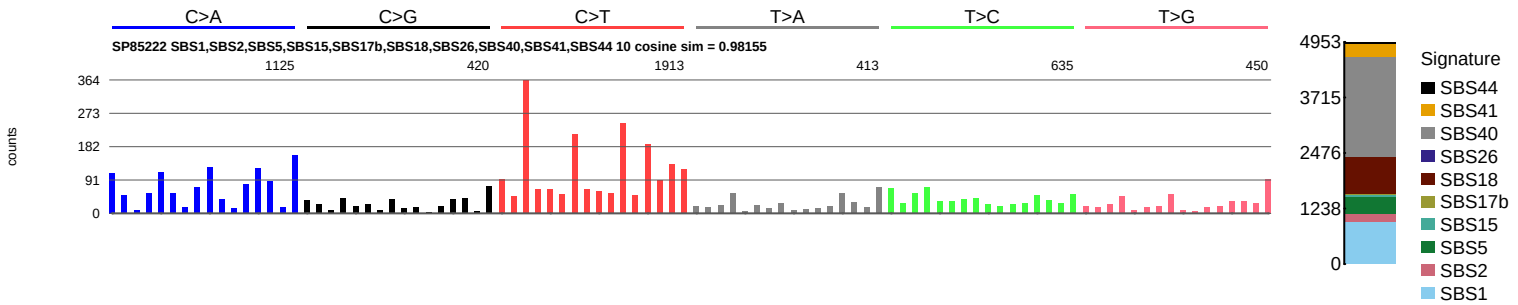
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98156
1137
433
1883
428
611
462
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98155
1125
420
1913
413
635
450
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
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SBS17b
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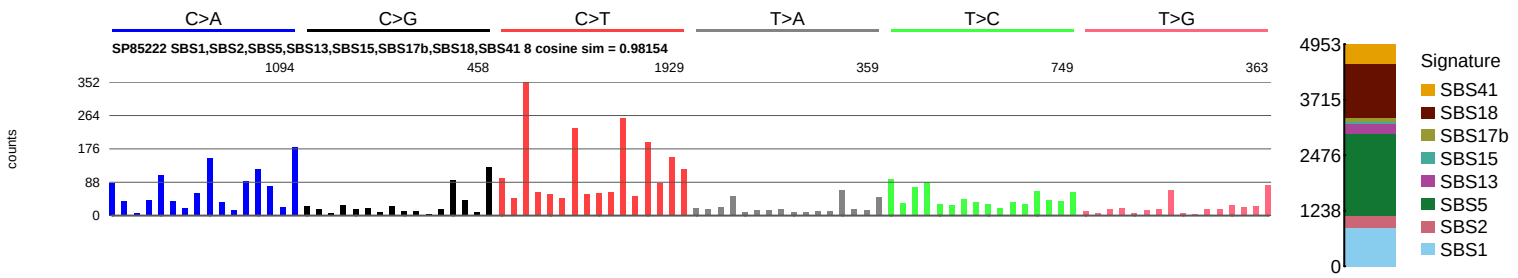
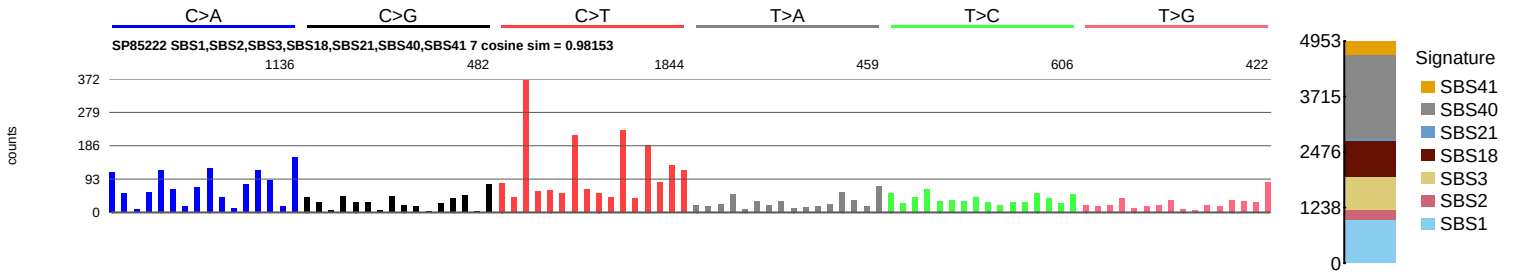
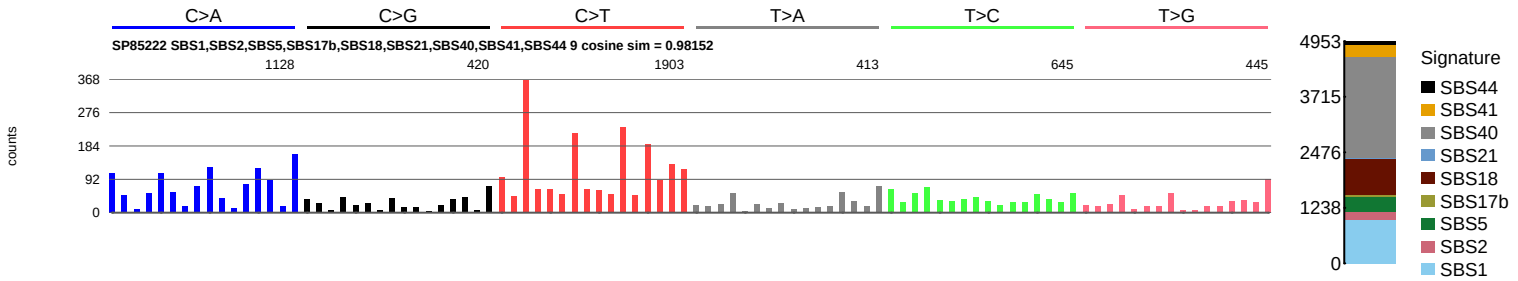
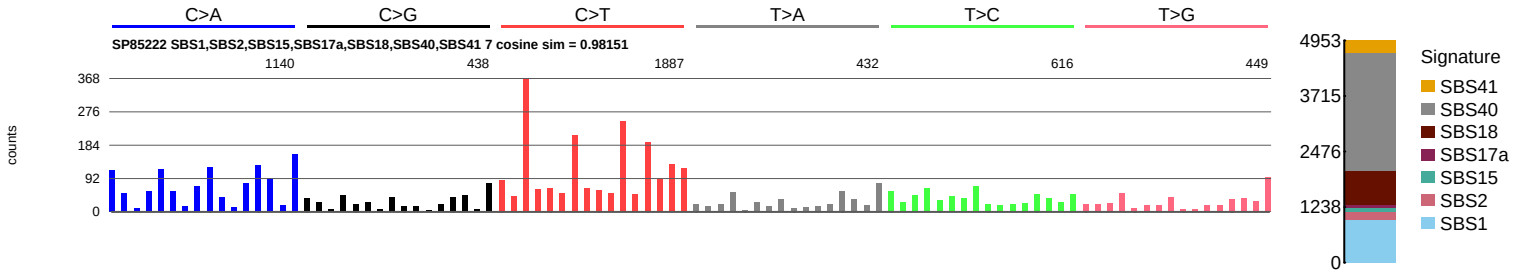
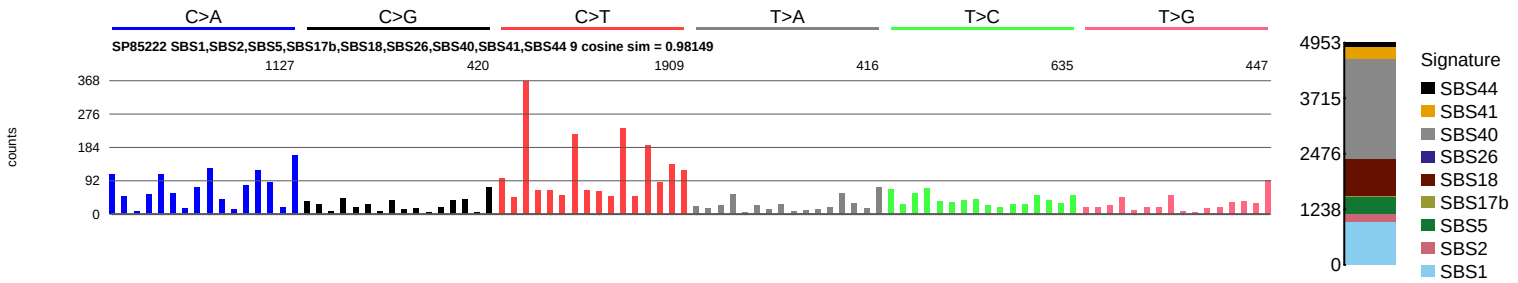
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.98154
1094
458
1929
359
749
363
counts
352
264
176
88
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98153
1136
482
1844
459
606
422
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98152
1128
420
1903
413
645
445
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS40,SBS41 7 cosine sim = 0.98151
1140
438
1887
432
616
449
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS2
SBS1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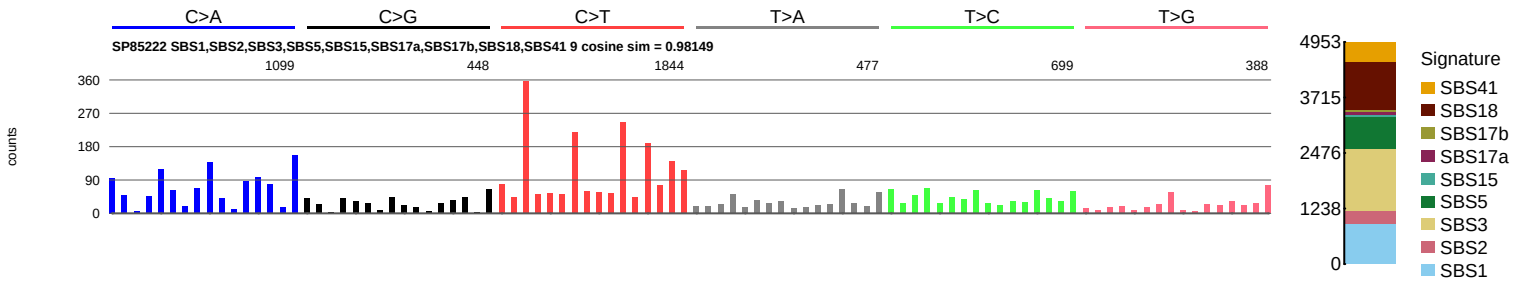
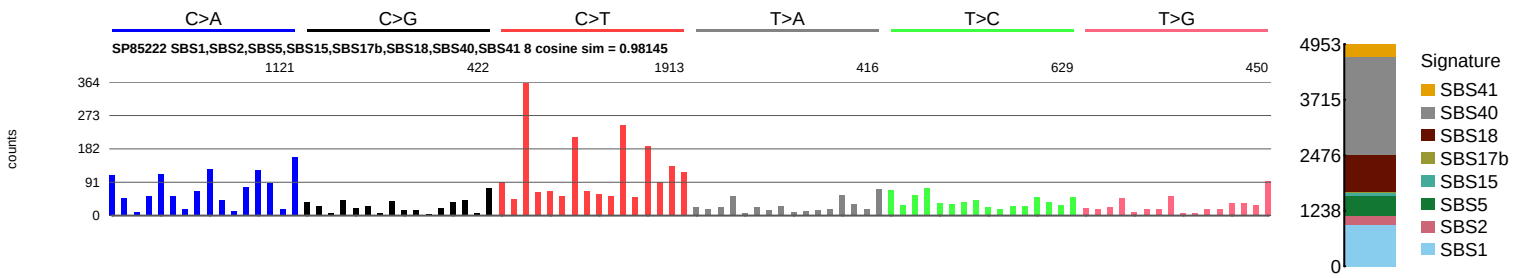
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98149
1127
420
1909
416
635
447
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98149
1099
448
1844
477
699
388
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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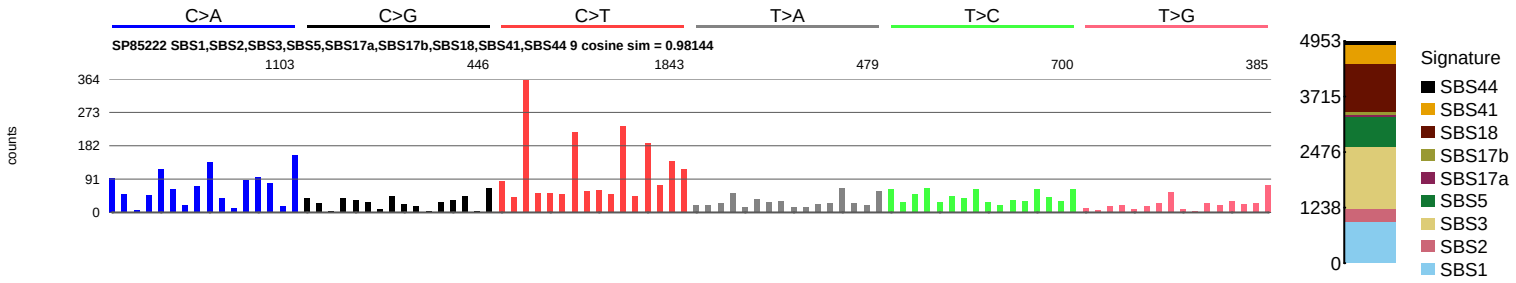
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98145
1121
422
1913
416
629
450
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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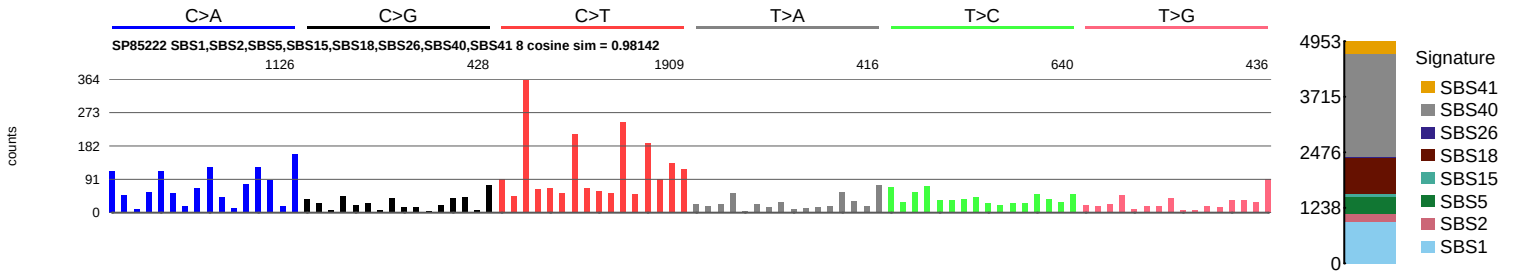
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98144
1103
446
1843
479
700
385
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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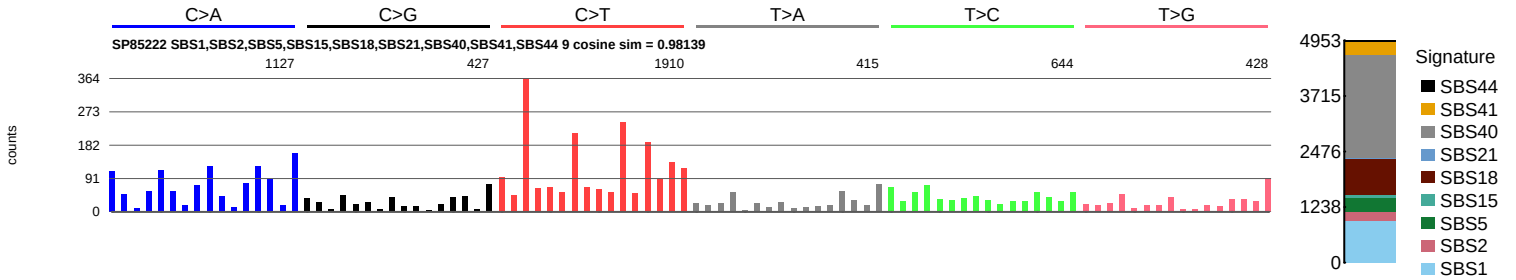
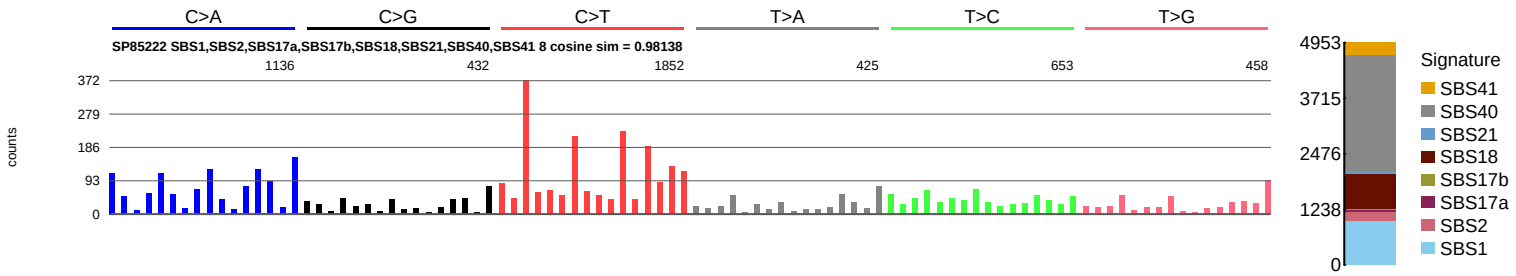
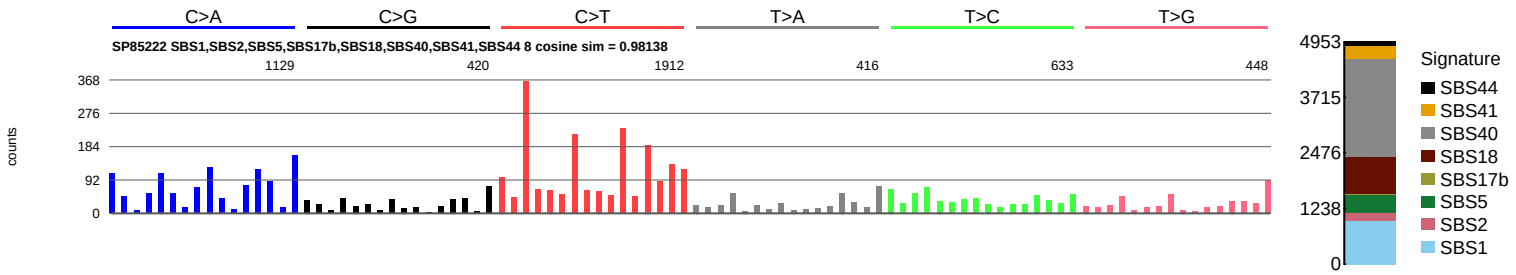
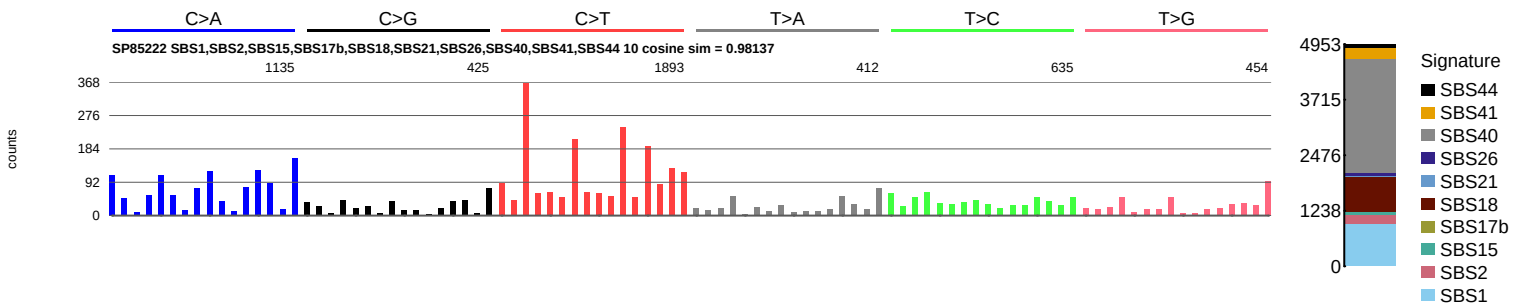
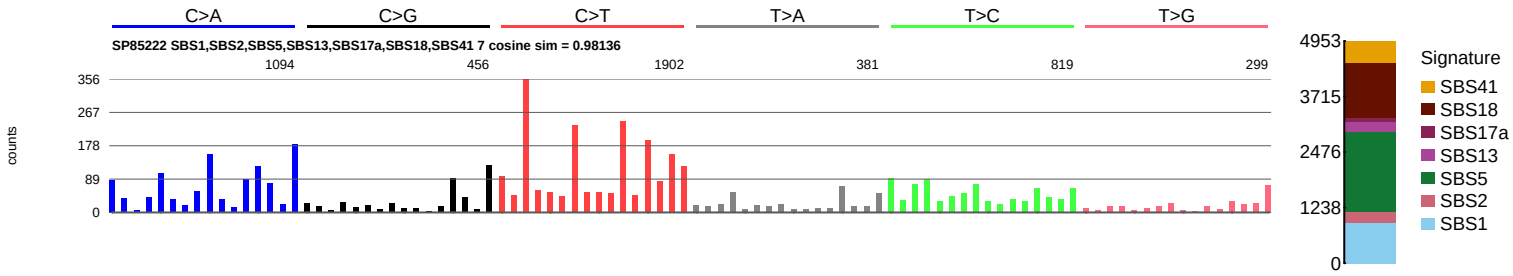
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98142
1126
428
1909
416
640
436
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS5
SBS2
SBS1

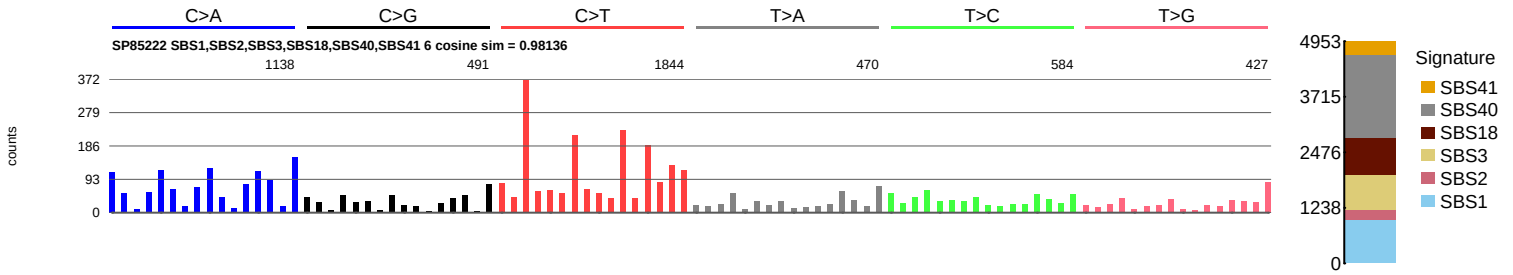

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98139
1127
427
1910
415
644
428
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98138
1136
432
1852
425
653
458
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98138
1129
420
1912
416
633
448
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98137
1135
425
1893
412
635
454
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS41 7 cosine sim = 0.98136
1094
456
1902
381
819
299
counts
356
267
178
89
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS18,SBS40,SBS41 6 cosine sim = 0.98136
1138
491
1844
470
584
427
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
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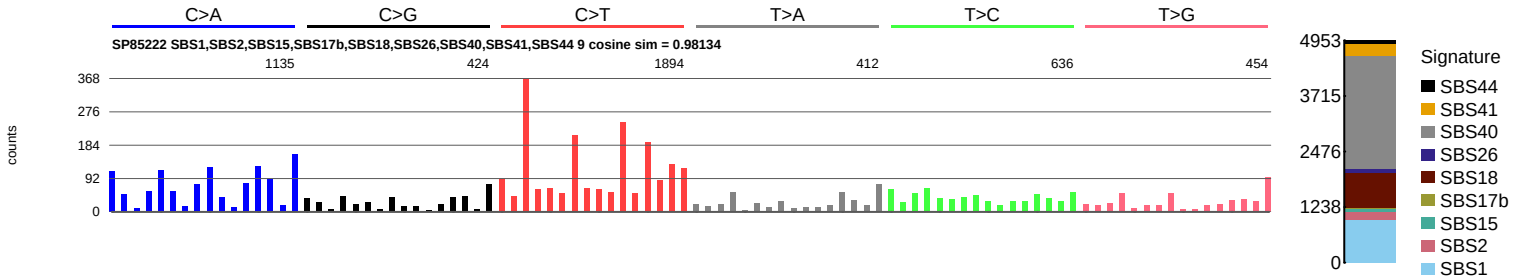
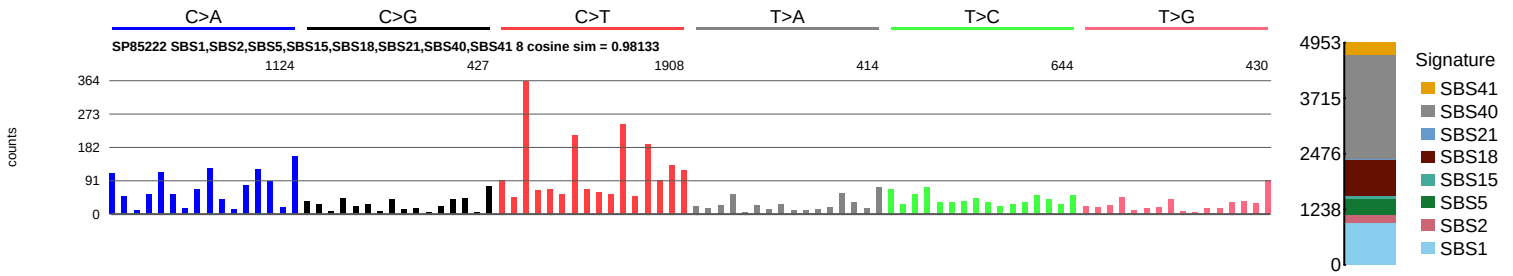
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98134
1135
424
1894
412
636
454
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1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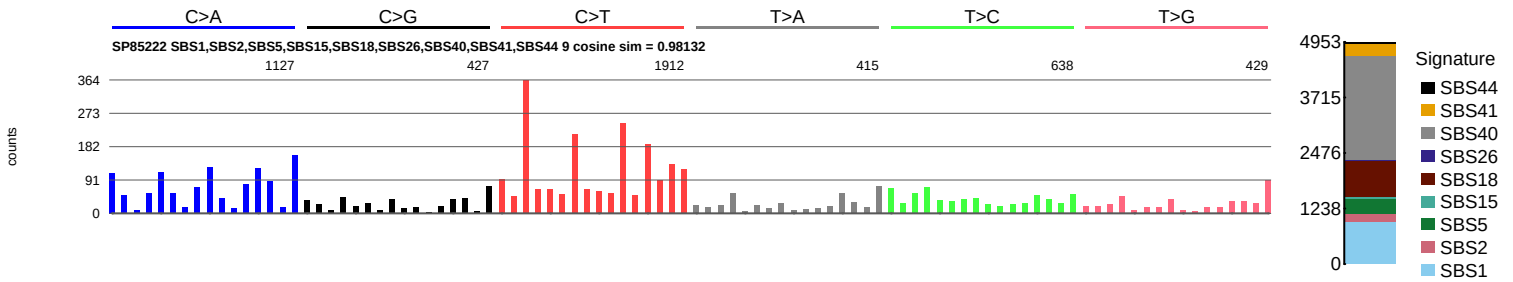
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98133
1124
427
1908
414
644
430
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS5
SBS2
SBS1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C>G
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T>A
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T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98132
1127
427
1912
415
638
429
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
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SBS40
SBS26
SBS18
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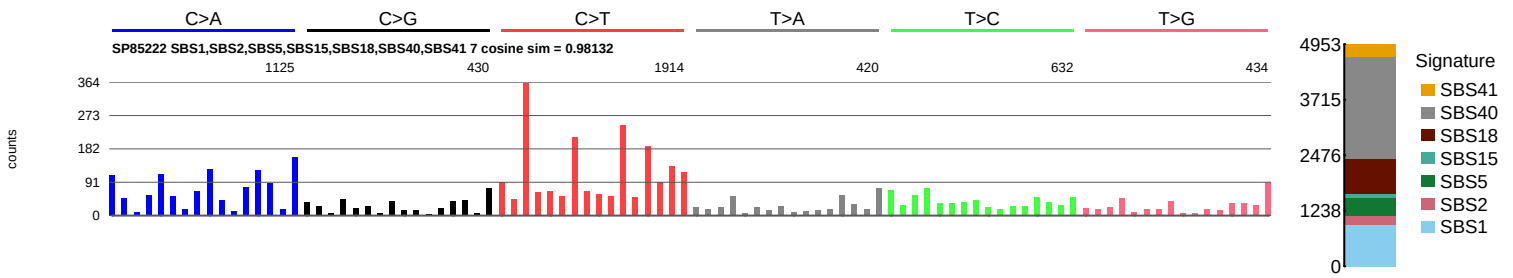
SBS5
SBS2
SBS1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C>G
C>T
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T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS40,SBS41 7 cosine sim = 0.98132
1125
430
1914
420
632
434
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
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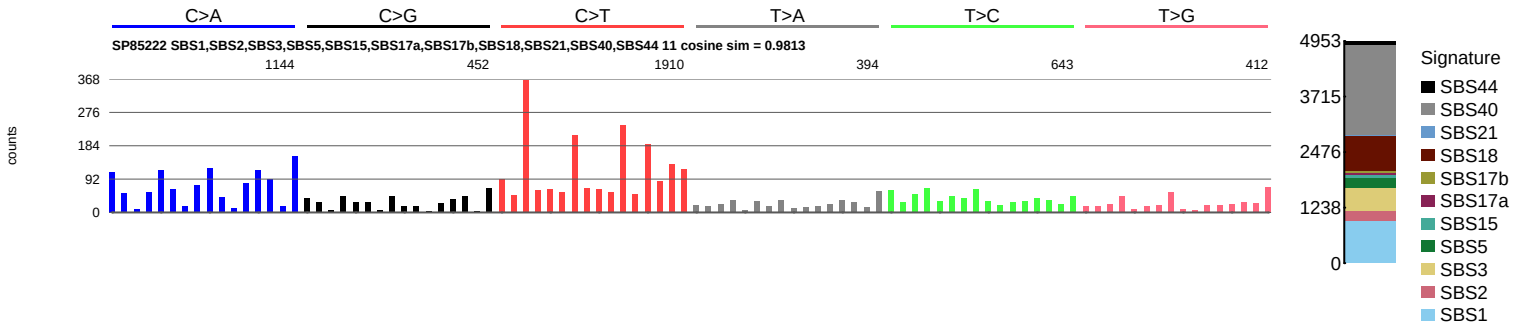
SBS2
SBS1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C>G
C>T
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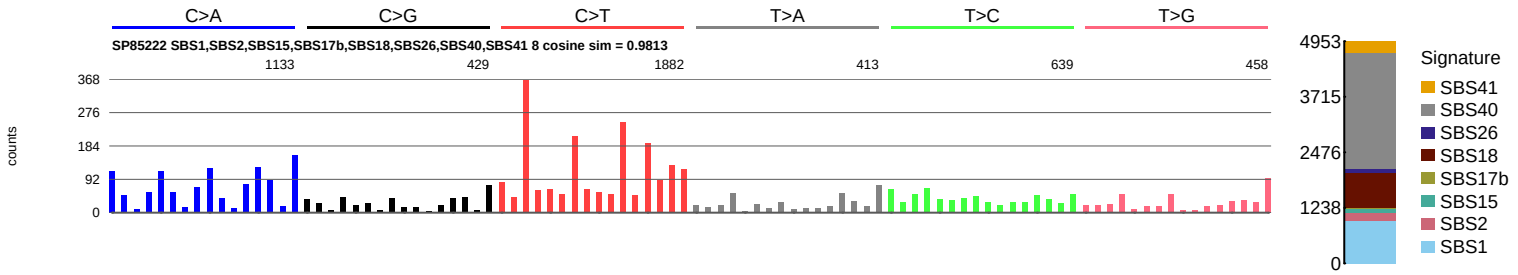
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.9813
1144
452
1910
394
643
412
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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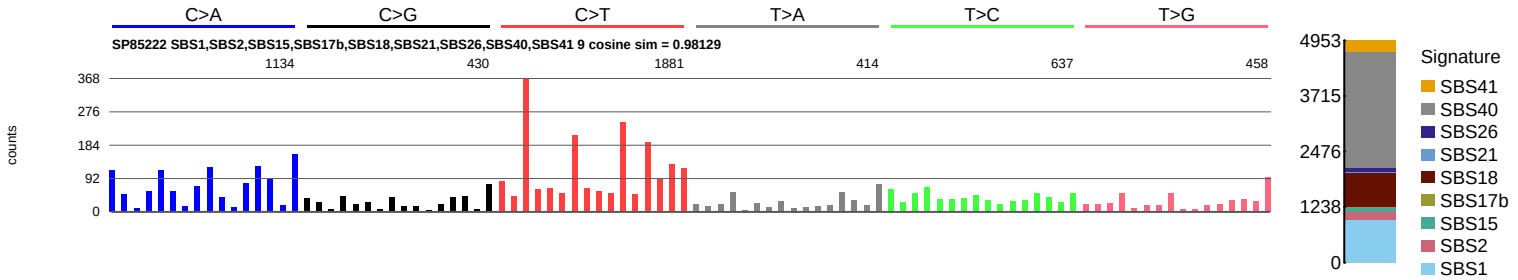
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.9813
1133
429
1882
413
639
458
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1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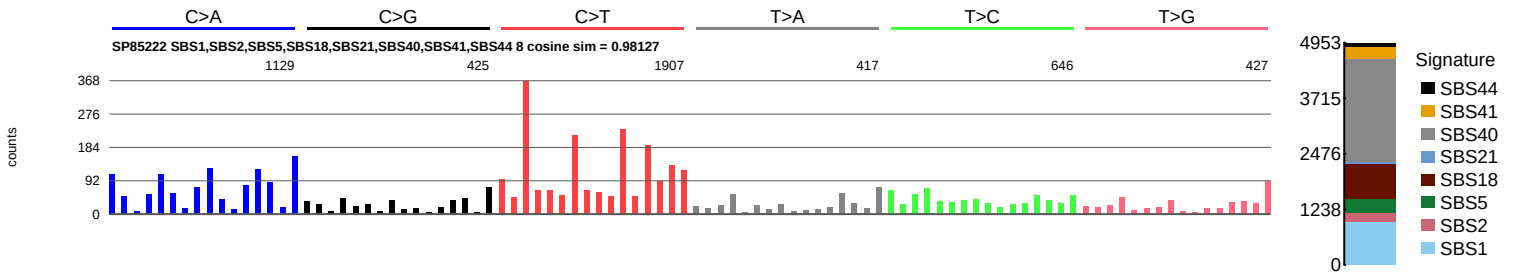
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98129
1134
430
1881
414
637
458
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1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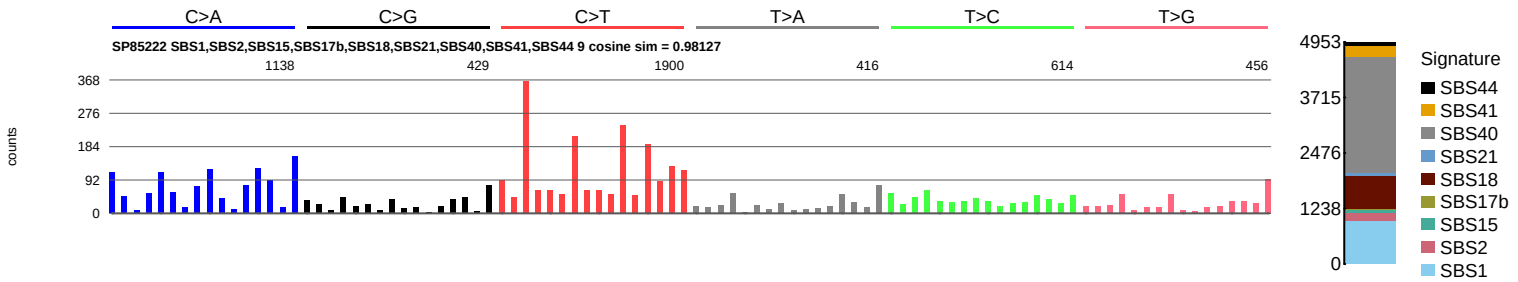
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98127
1129
425
1907
417
646
427
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS5
SBS2
SBS1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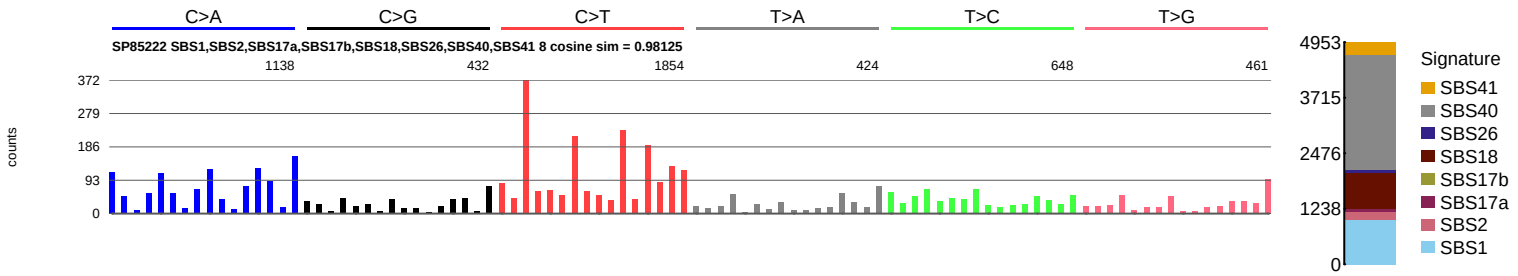
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98127
1138
429
1900
416
614
456
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1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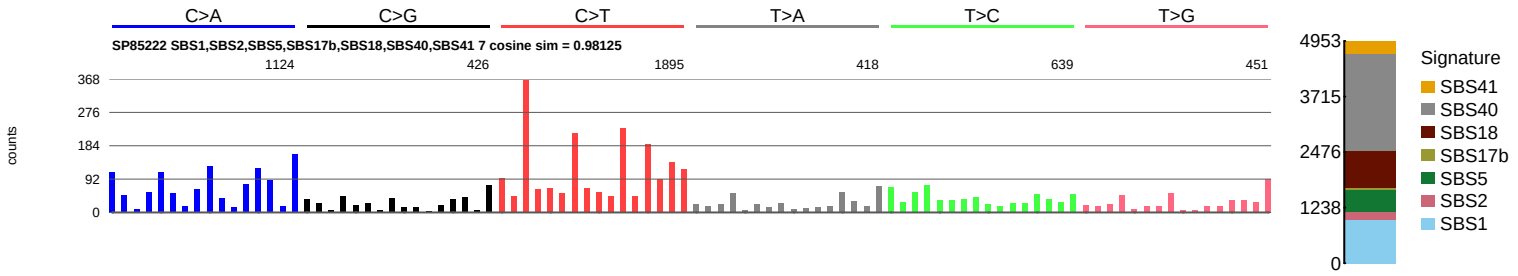
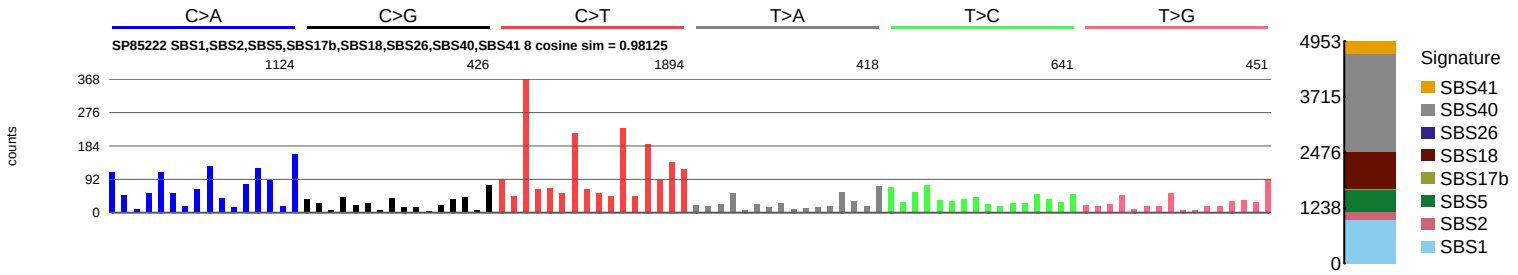
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98125
1138
432
1854
424
648
461
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98125
1124
426
1895
418
639
451
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1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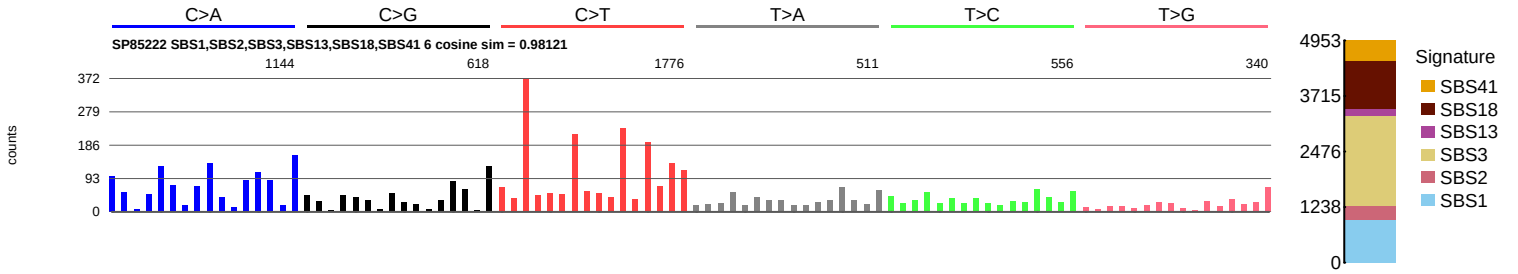
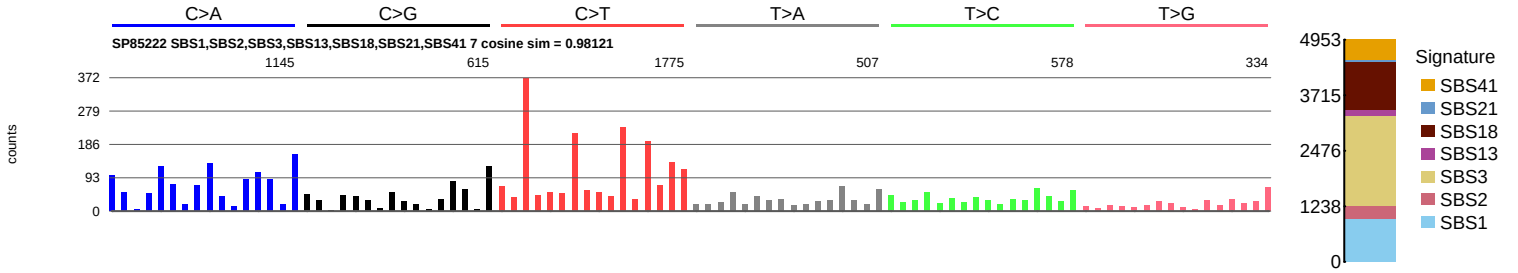
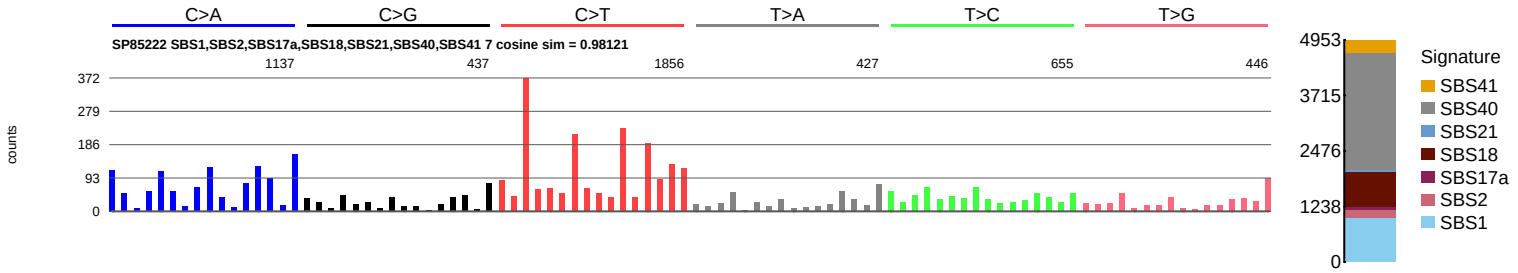
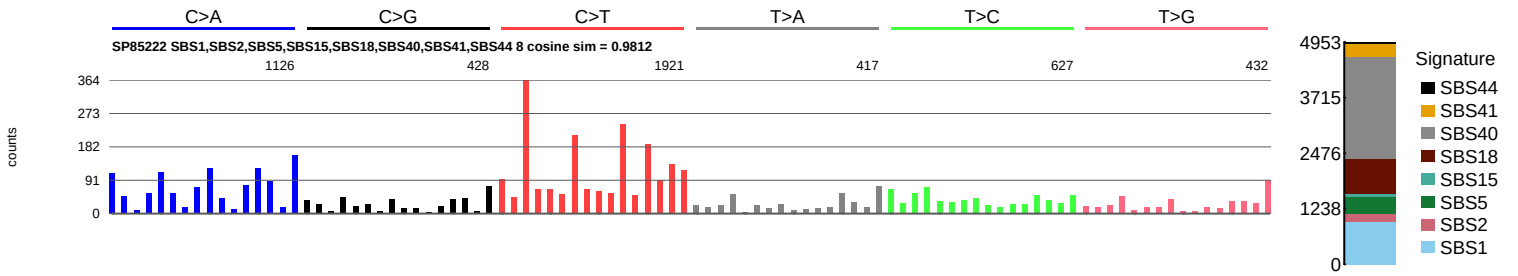
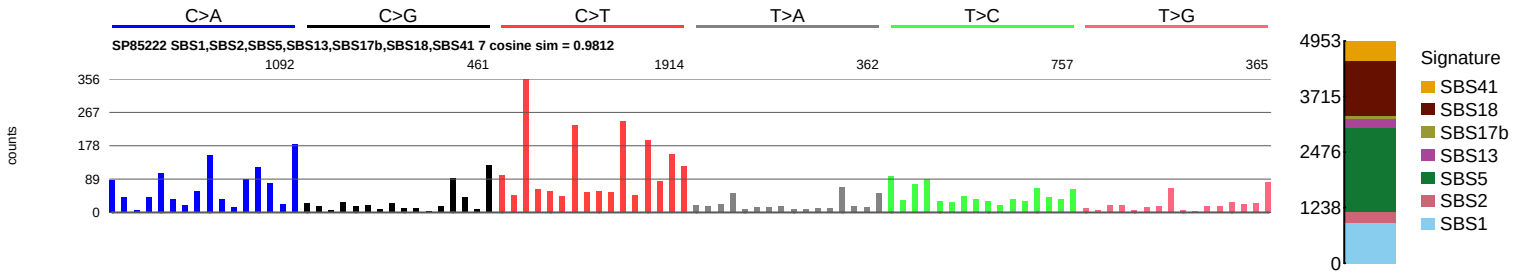
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98125
1124
426
1894
418
641
451
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1

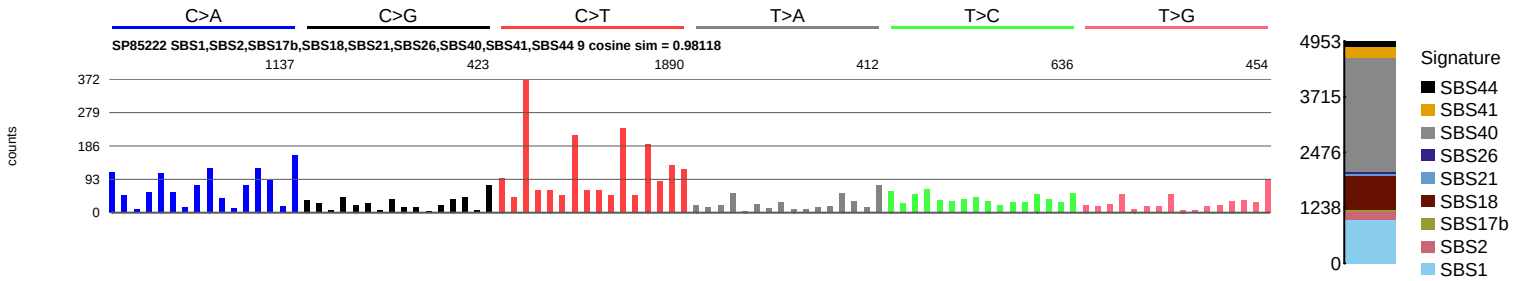

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS41 6 cosine sim = 0.98121
1144
618
1776
511
556
340
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS41 7 cosine sim = 0.98121
1145
615
1775
507
578
334
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98121
1137
437
1856
427
655
446
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.9812
1126
428
1921
417
627
432
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS41 7 cosine sim = 0.9812
1092
461
1914
362
757
365
counts
356
267
178
89
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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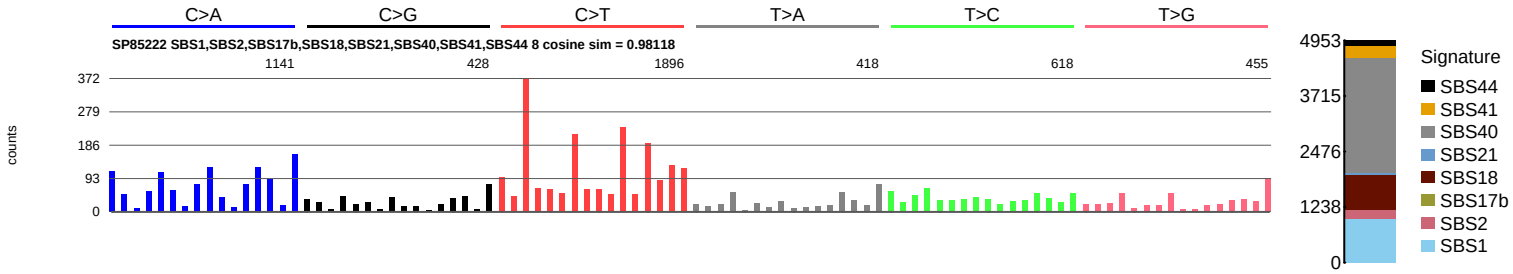
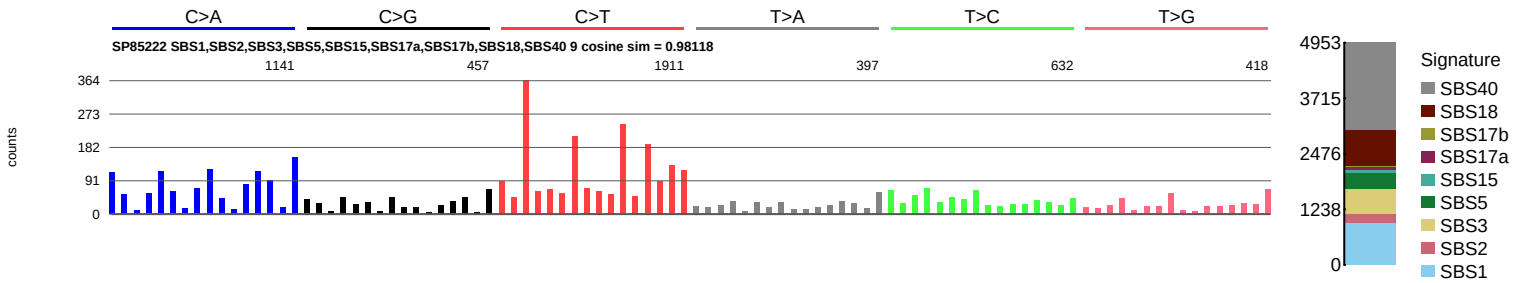
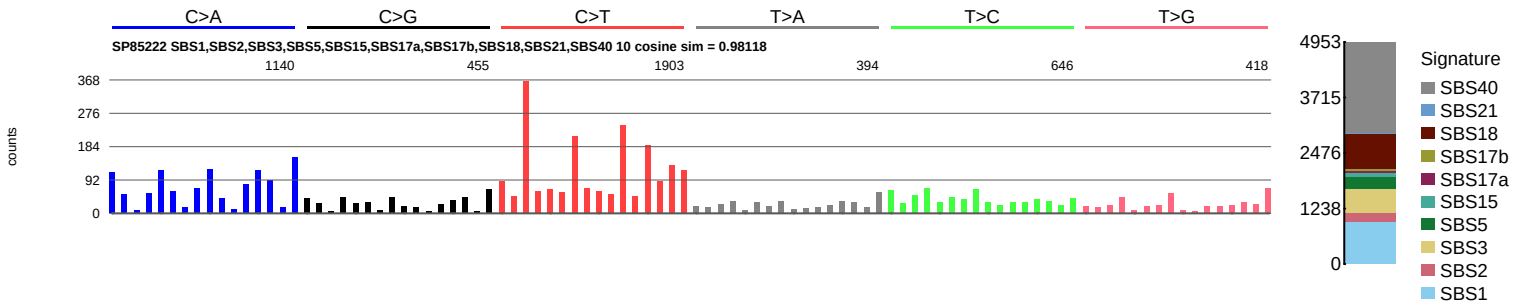
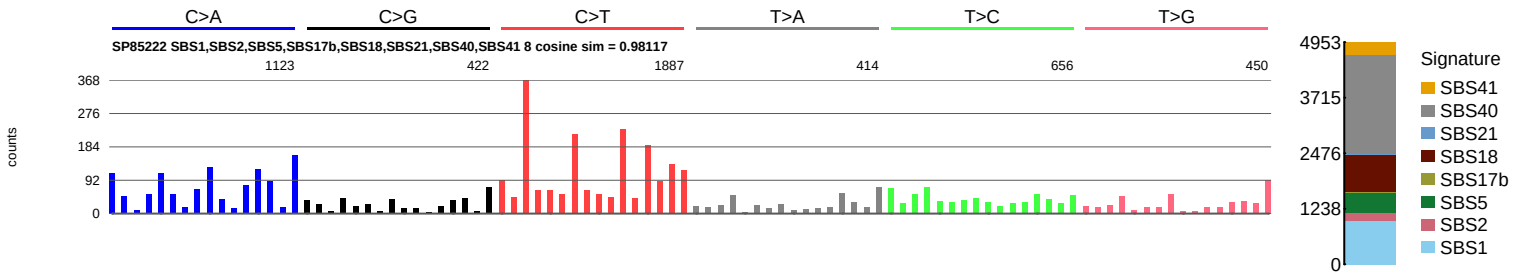
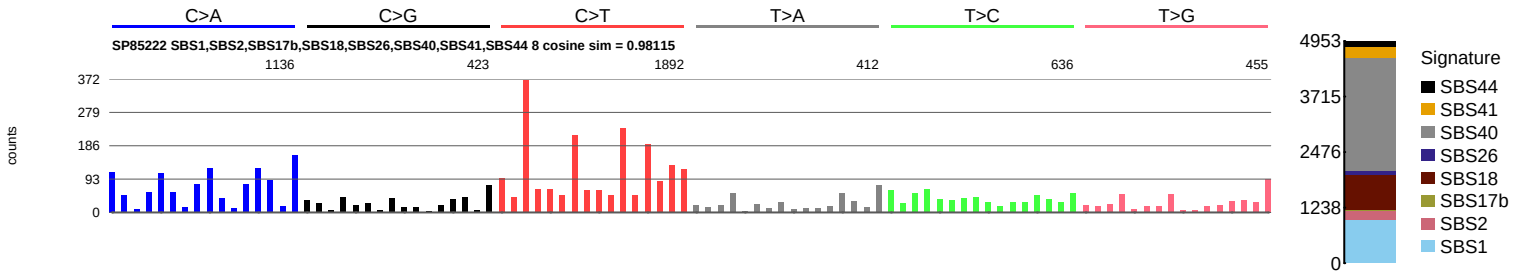
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98118
1137
423
1890
412
636
454
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS2
SBS1

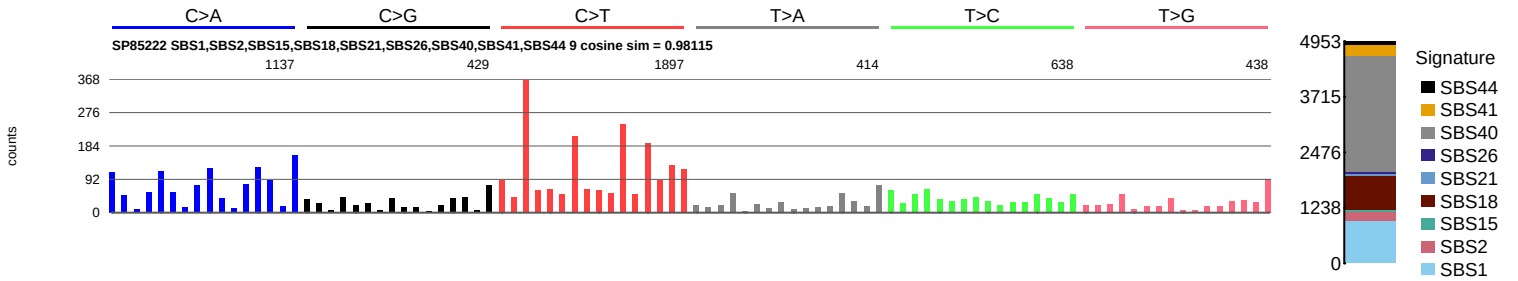

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98118
1141
428
1896
418
618
455
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1

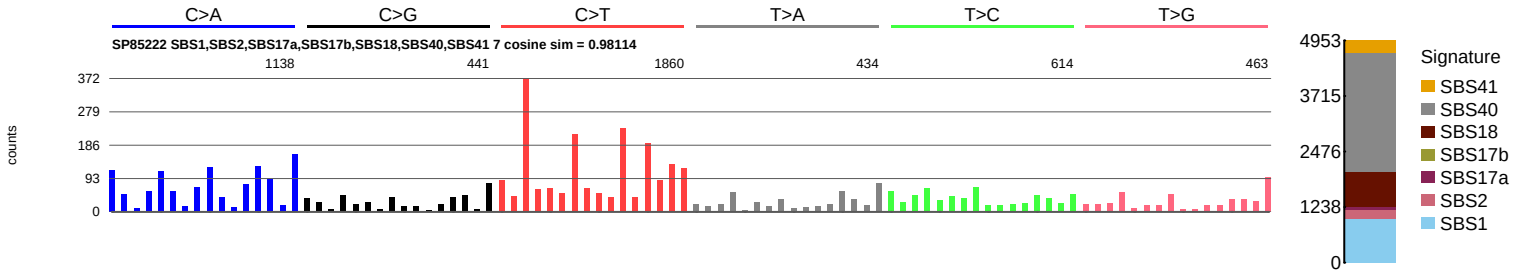

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98118
1141
457
1911
397
632
418
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

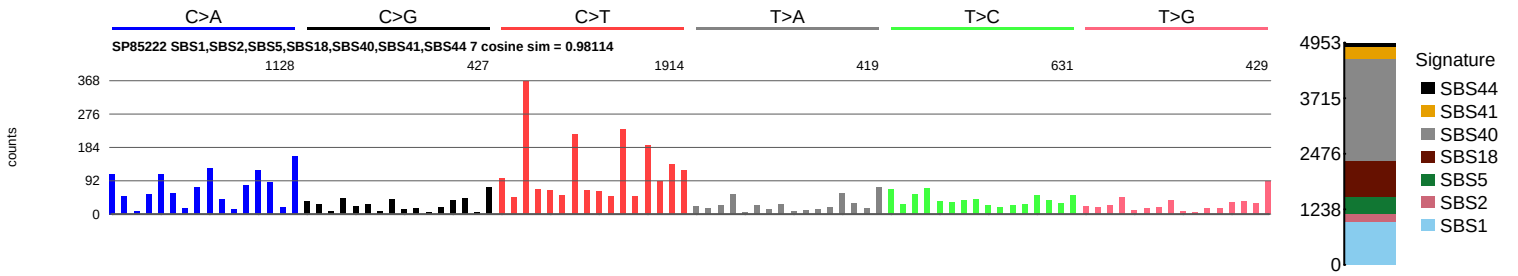

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98118
1140
455
1903
394
646
418
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

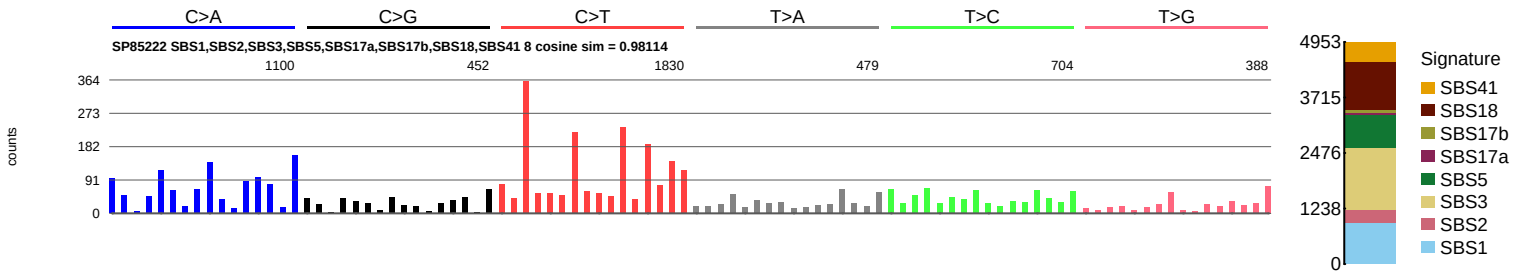

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98117
1123
422
1887
414
656
450
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98115
1136
423
1892
412
636
455
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98115
1137
429
1897
414
638
438
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98114
1138
441
1860
434
614
463
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98114
1128
427
1914
419
631
429
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS5
SBS2
SBS1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C>G
C>T
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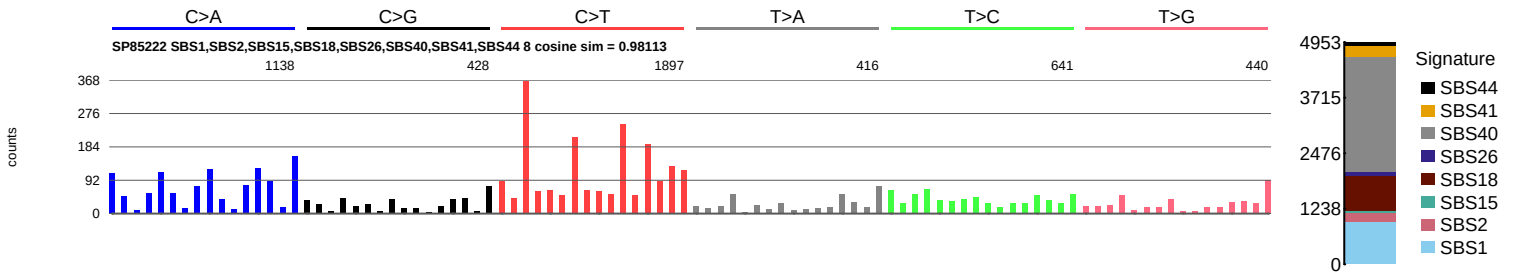
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98114
1100
452
1830
479
704
388
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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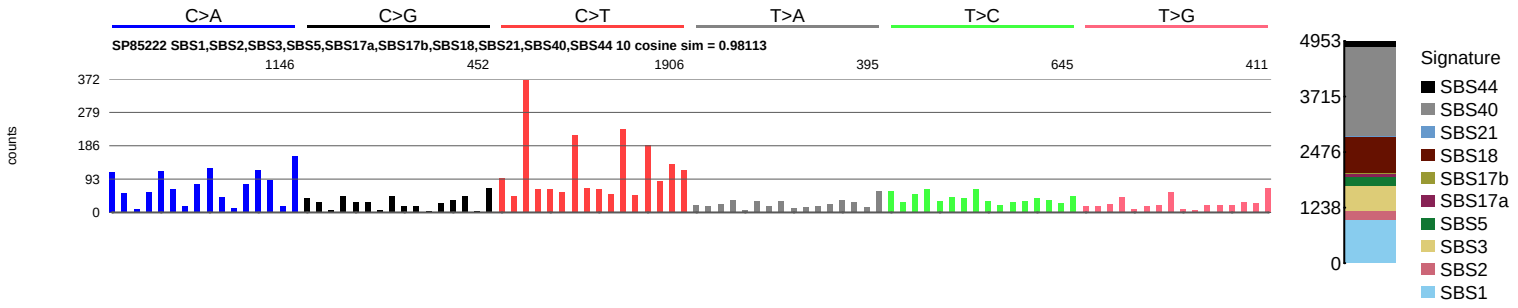
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98113
1138
428
1897
416
641
440
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS15
SBS2
SBS1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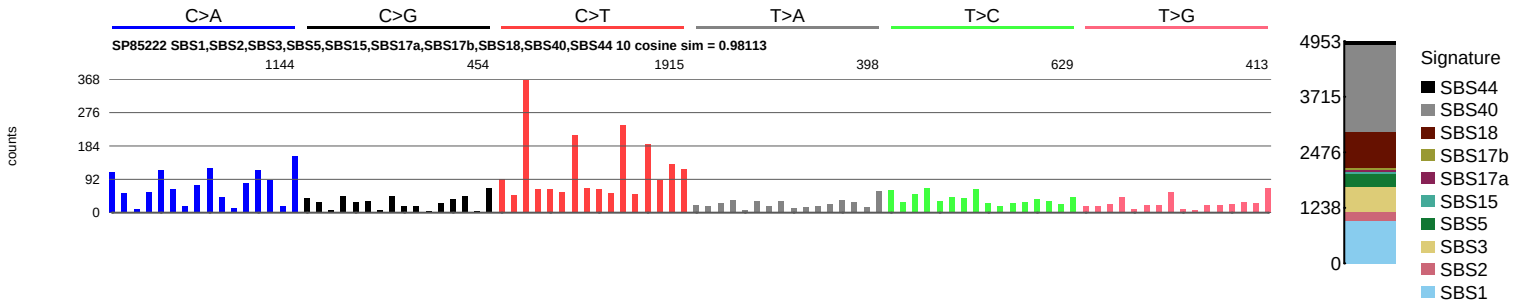
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98113
1146
452
1906
395
645
411
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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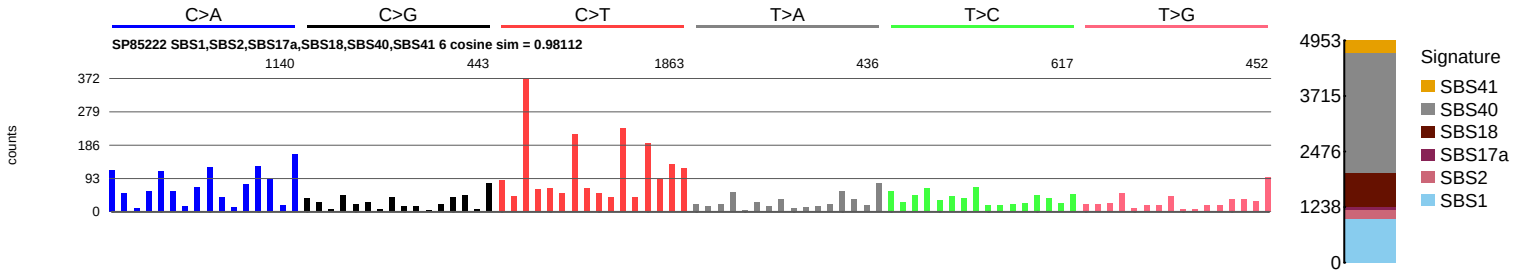
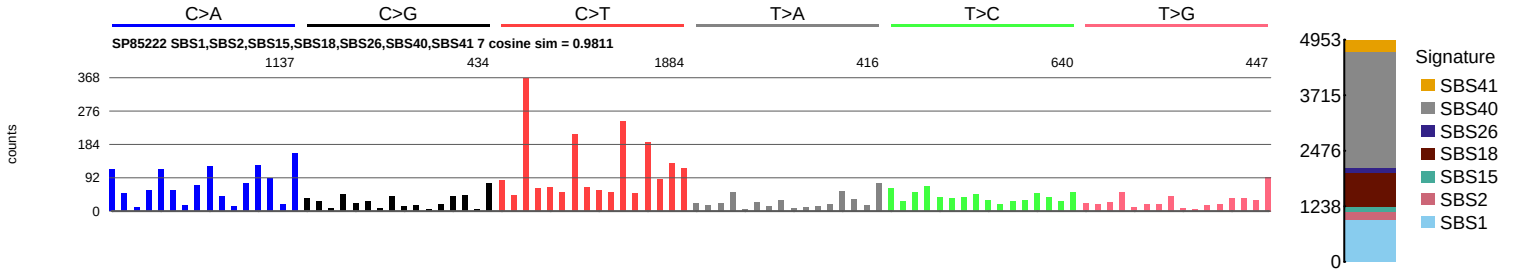
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98113
1144
454
1915
398
629
413
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17a,SBS18,SBS40,SBS41 6 cosine sim = 0.98112
1140
443
1863
436
617
452
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS2
SBS1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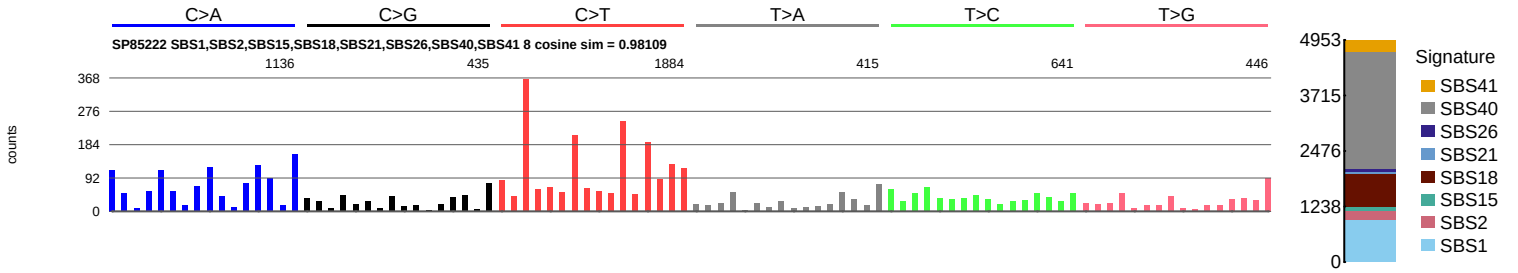
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.9811
1137
434
1884
416
640
447
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS2
SBS1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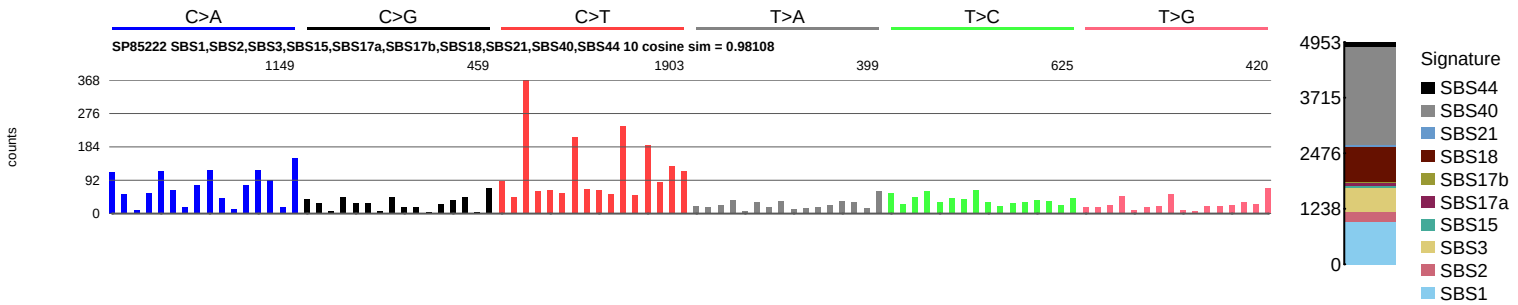
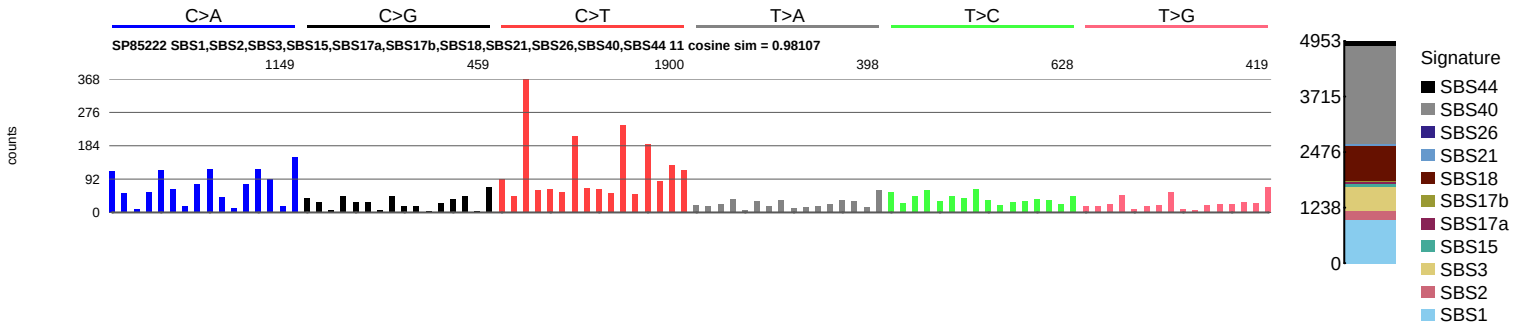
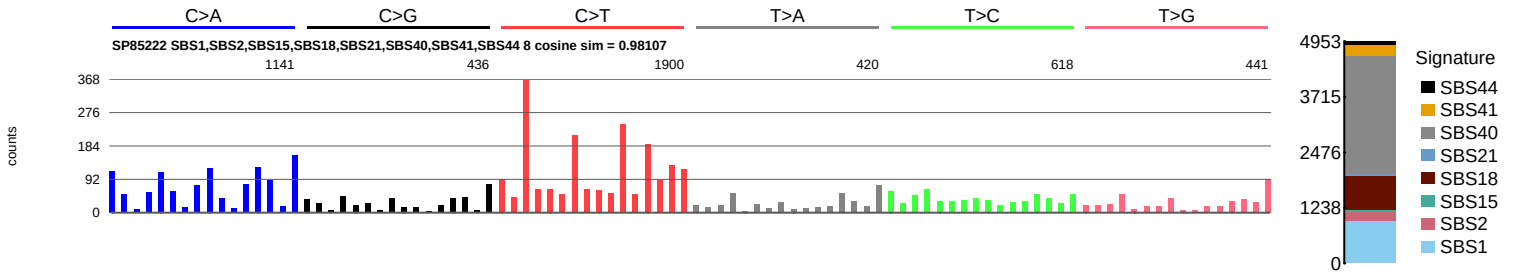
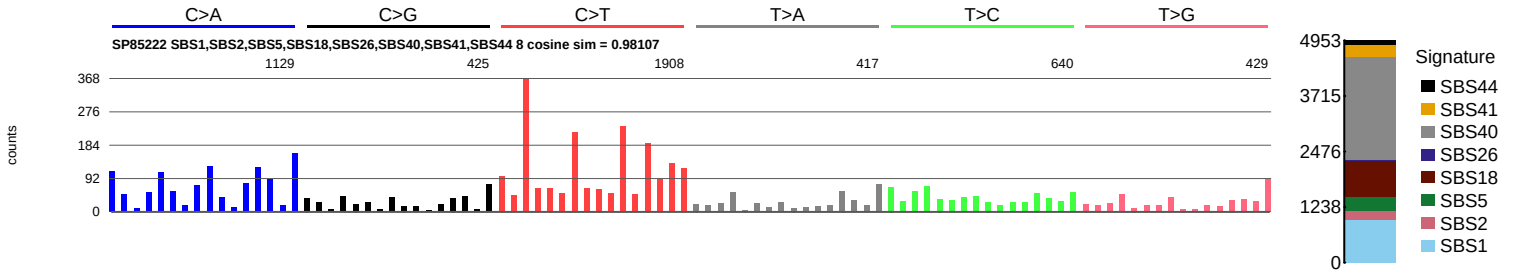
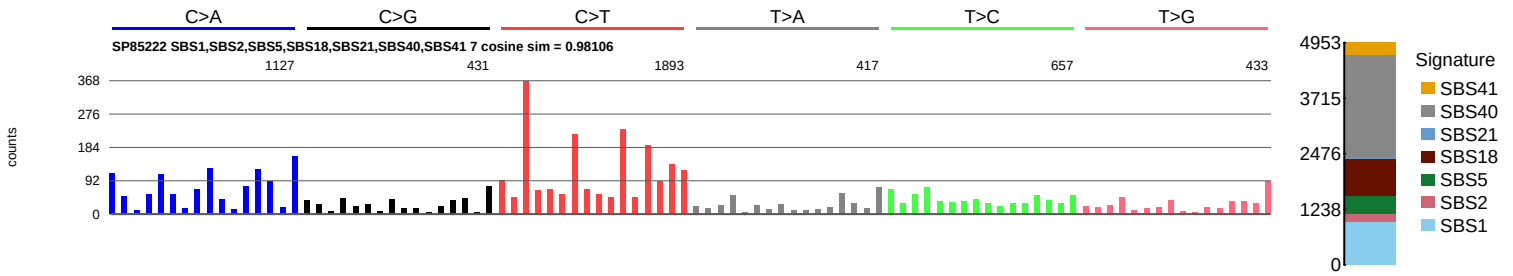
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98109
1136
435
1884
415
641
446
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS2
SBS1

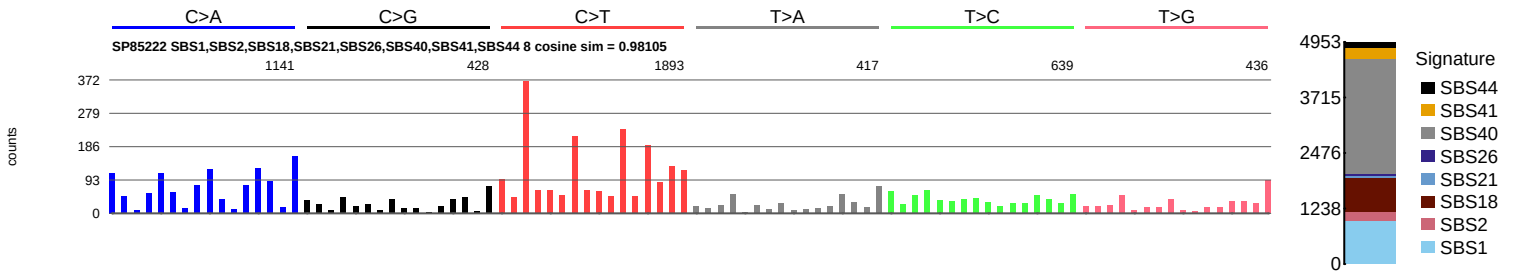

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98108
1149
459
1903
399
625
420
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

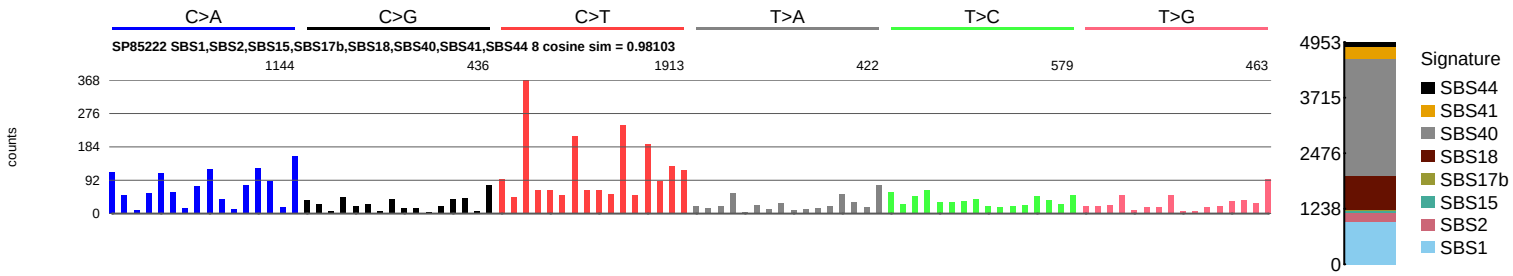

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98107
1149
459
1900
398
628
419
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

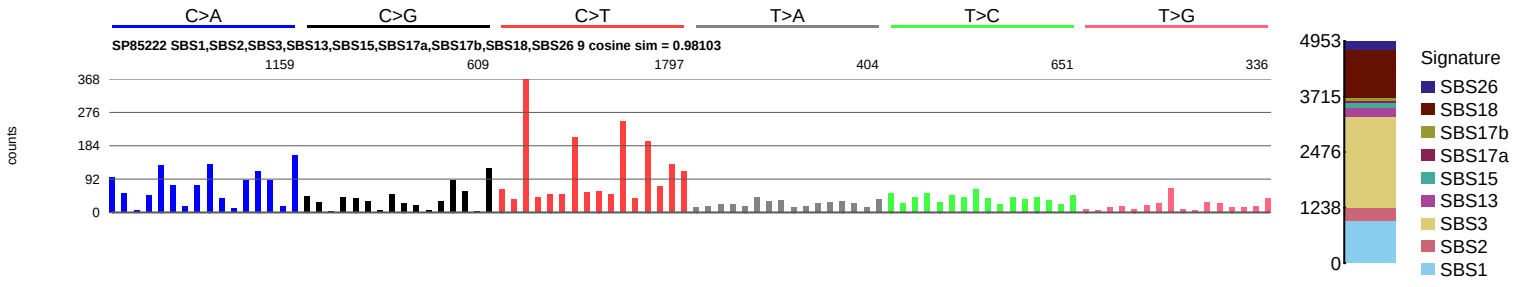

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98107
1141
436
1900
420
618
441
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS2
SBS1

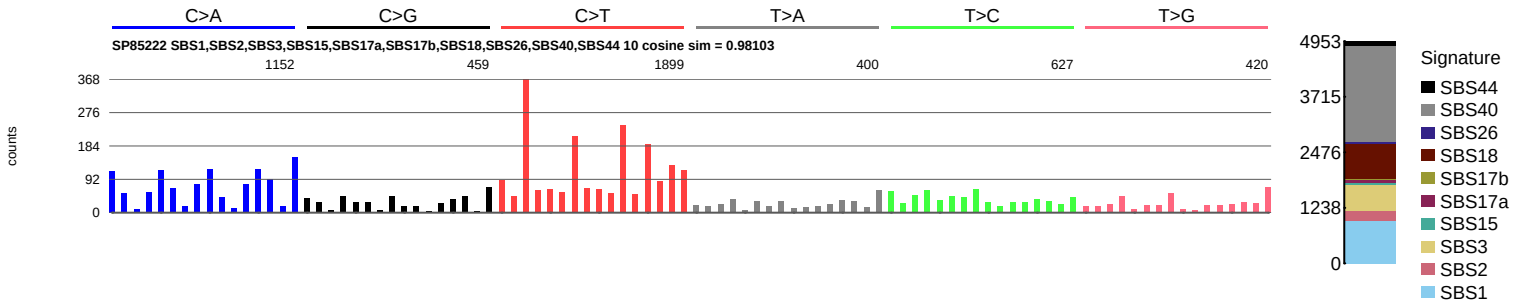

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98107
1129
425
1908
417
640
429
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS5
SBS2
SBS1

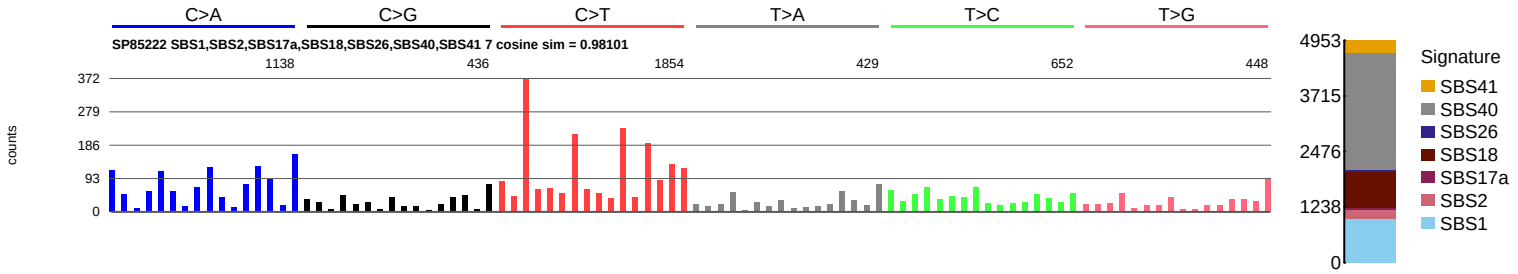

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98106
1127
431
1893
417
657
433
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS5
SBS2
SBS1

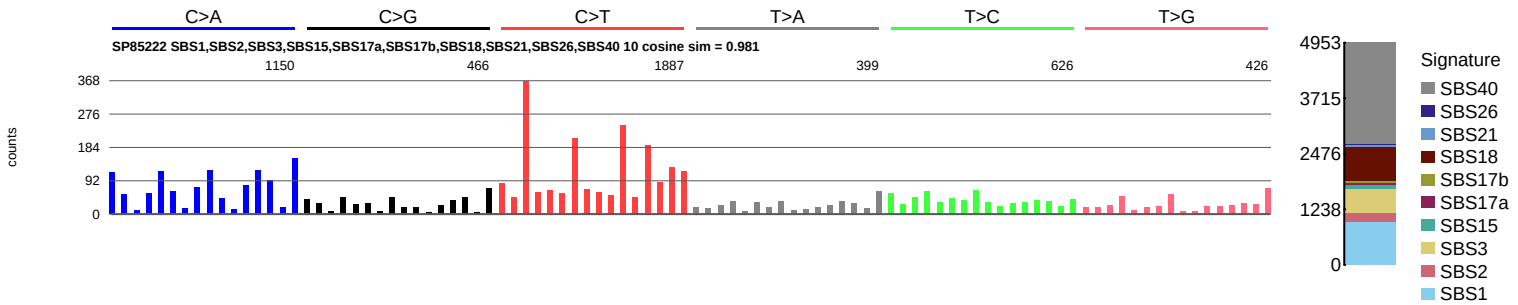

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98105
1141
428
1893
417
639
436
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98103
1144
436
1913
422
579
463
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98103
1159
609
1797
404
651
336
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98103
1152
459
1899
400
627
420
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98101
1138
436
1854
429
652
448
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.981
1150
466
1887
399
626
426
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
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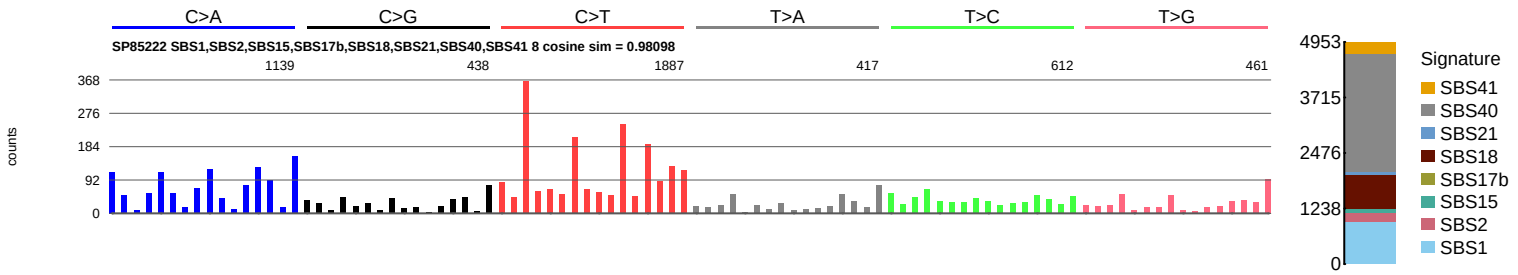
SBS2
SBS1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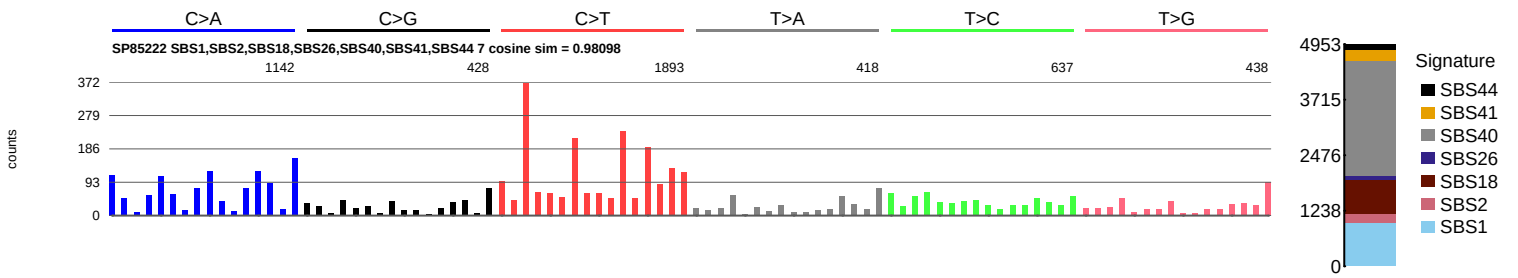
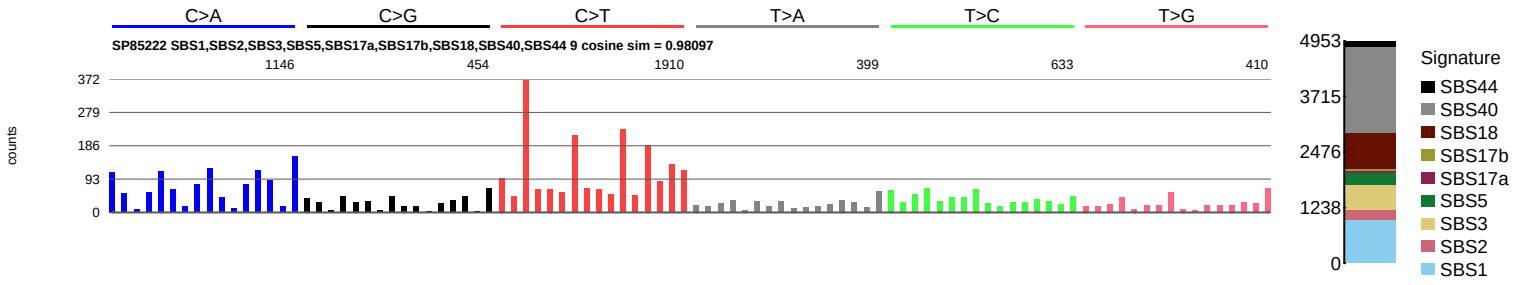
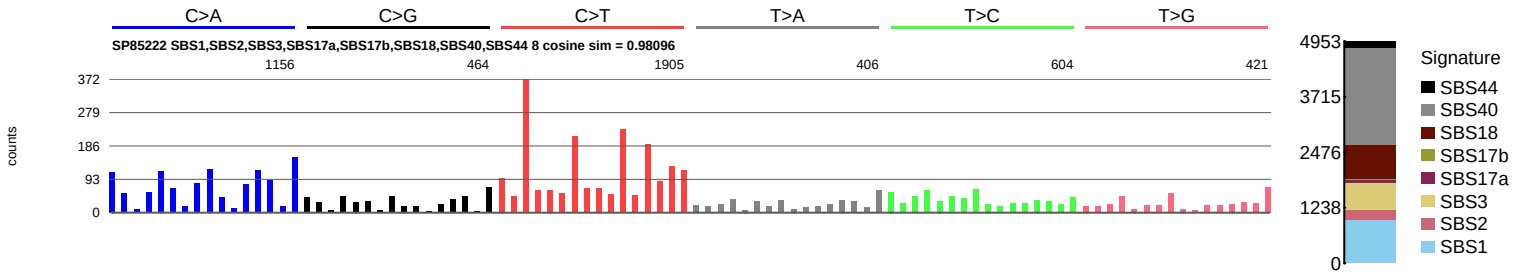
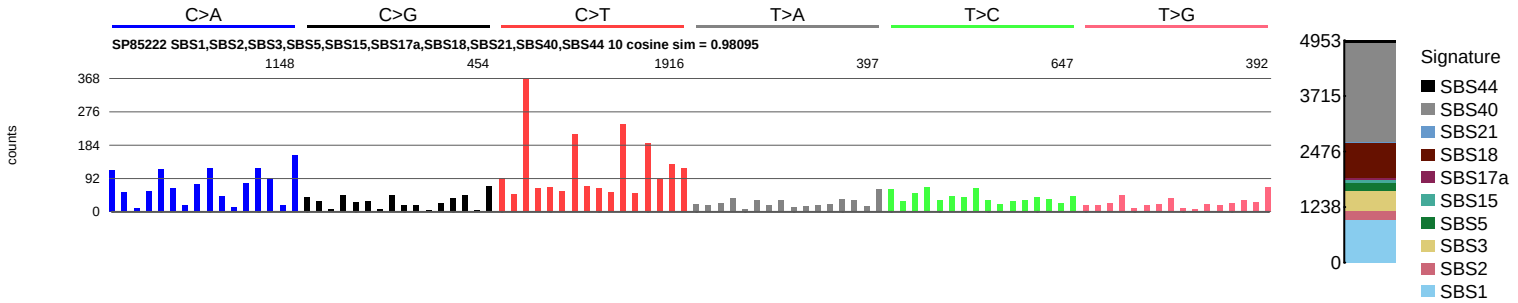
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98098
1139
438
1887
417
612
461
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS18,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98098
1142
428
1893
418
637
438
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98097
1146
454
1910
399
633
410
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98096
1156
464
1905
406
604
421
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98095
1148
454
1916
397
647
392
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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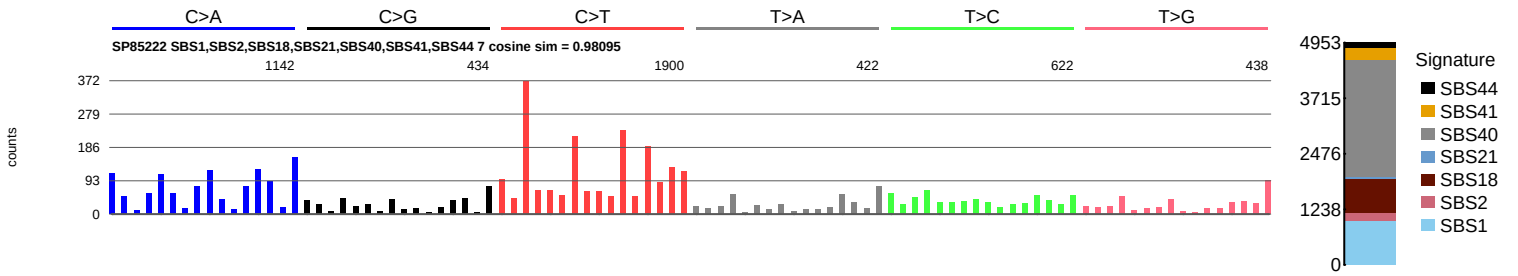
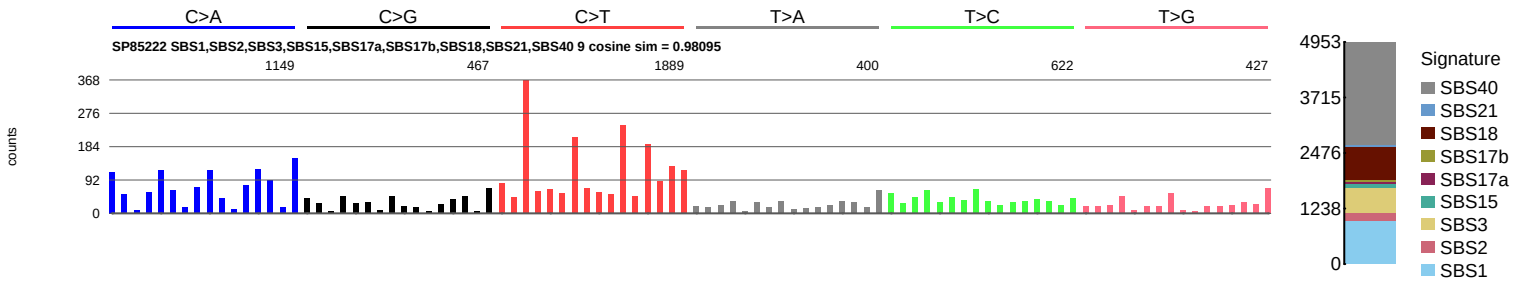
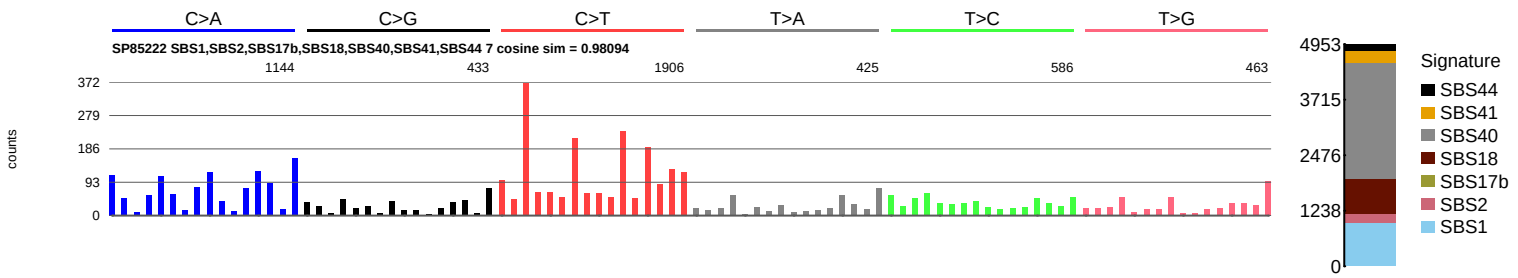
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS18,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98095
1142
434
1900
422
622
438
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98095
1149
467
1889
400
622
427
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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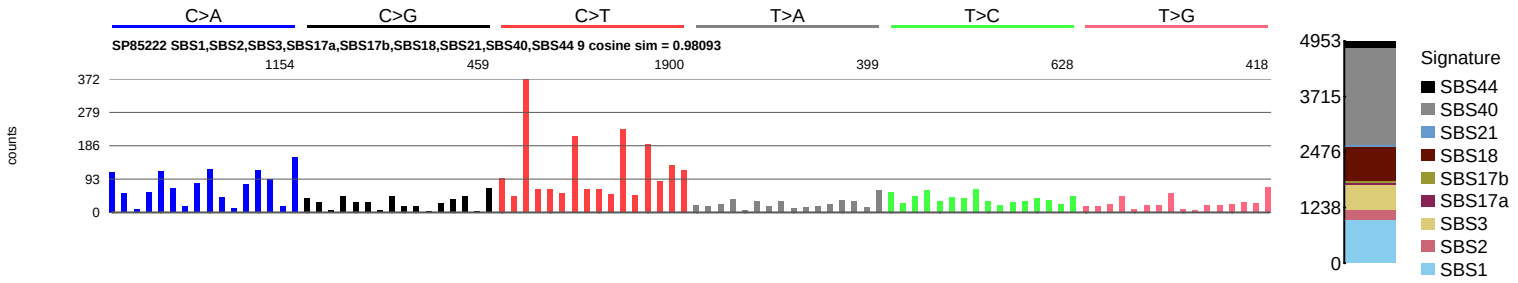
T>C
T>G
SP85222 SBS1,SBS2,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98094
1144
433
1906
425
586
463
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS2
SBS1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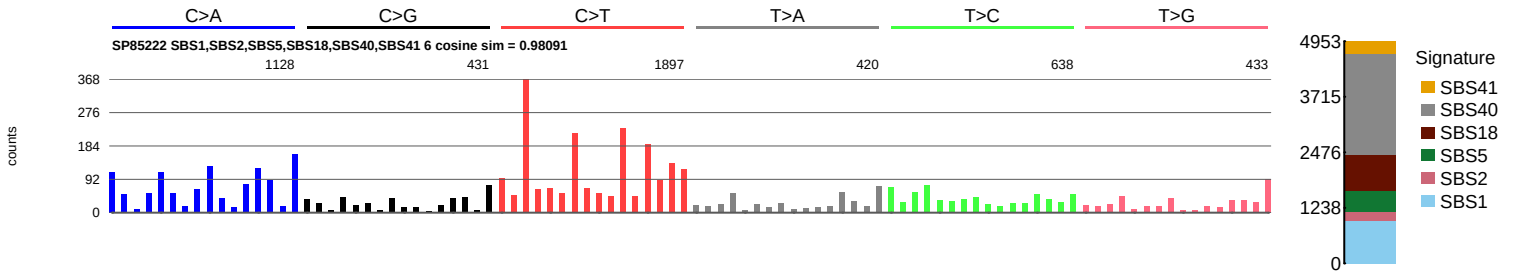
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98093
1154
459
1900
399
628
418
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
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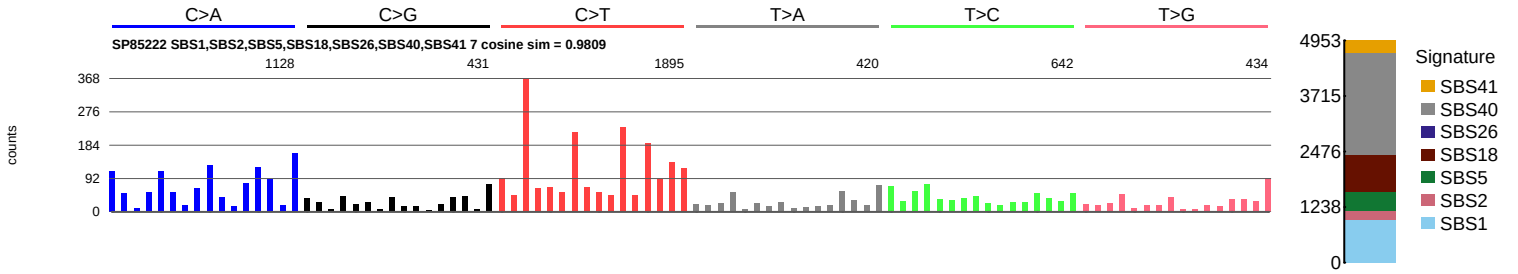
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS18,SBS40,SBS41 6 cosine sim = 0.98091
1128
431
1897
420
638
433
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS5
SBS2
SBS1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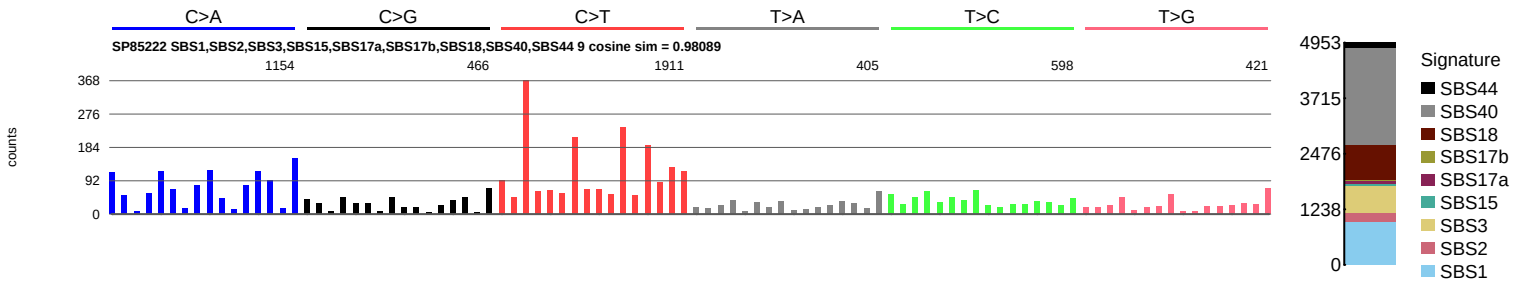
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.9809
1128
431
1895
420
642
434
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS5
SBS2
SBS1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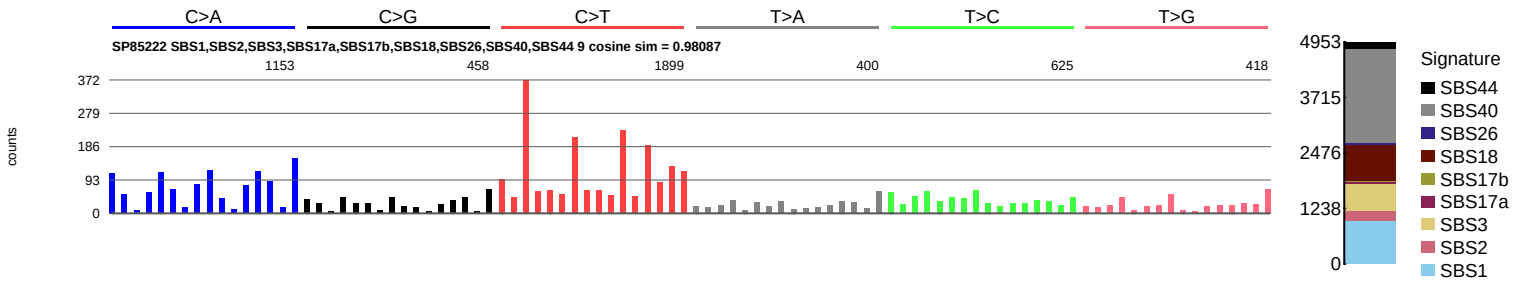
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98089
1154
466
1911
405
598
421
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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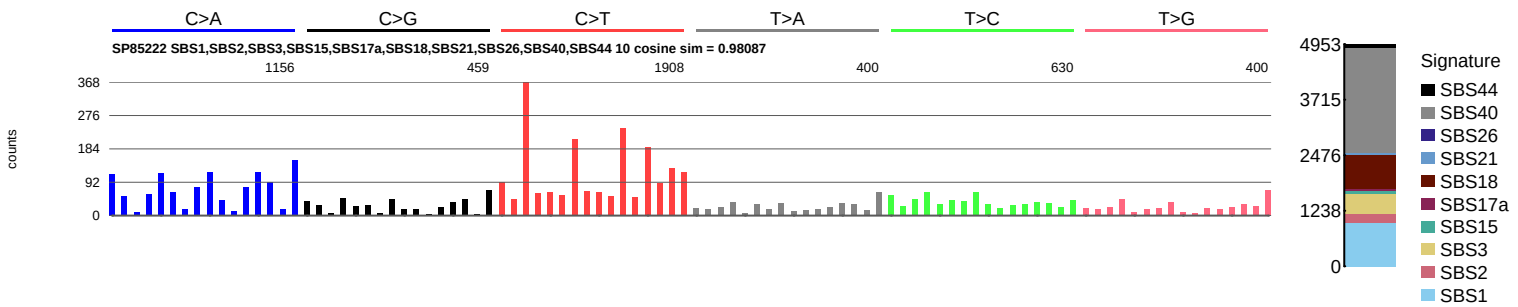
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98087
1153
458
1899
400
625
418
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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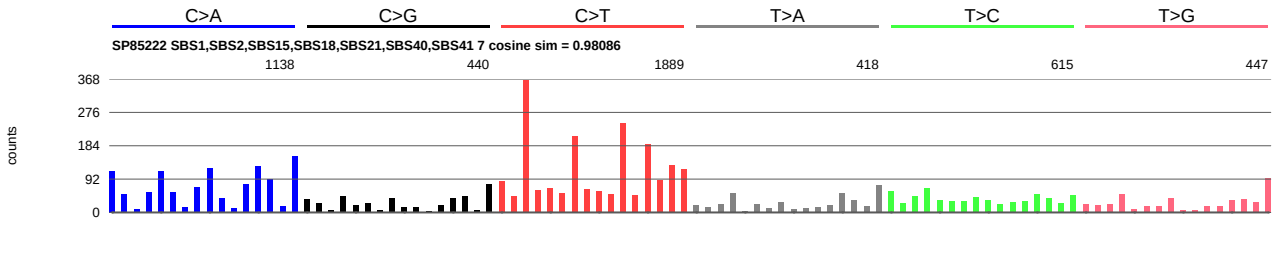
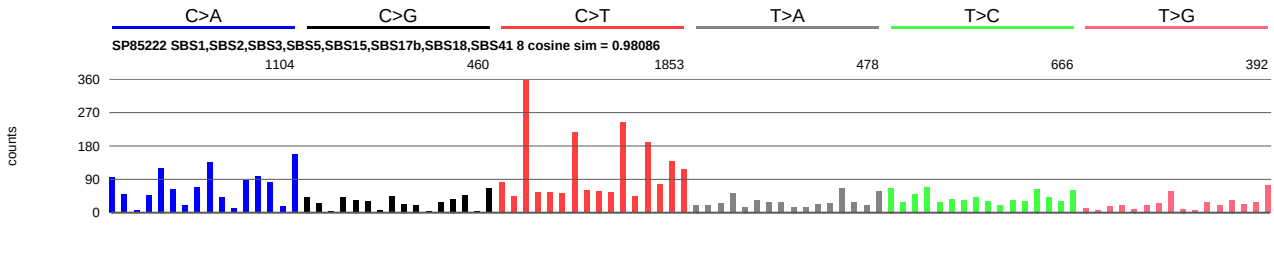
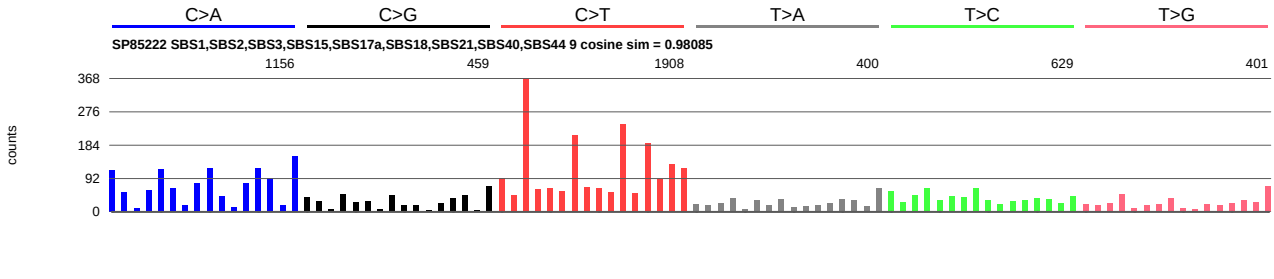
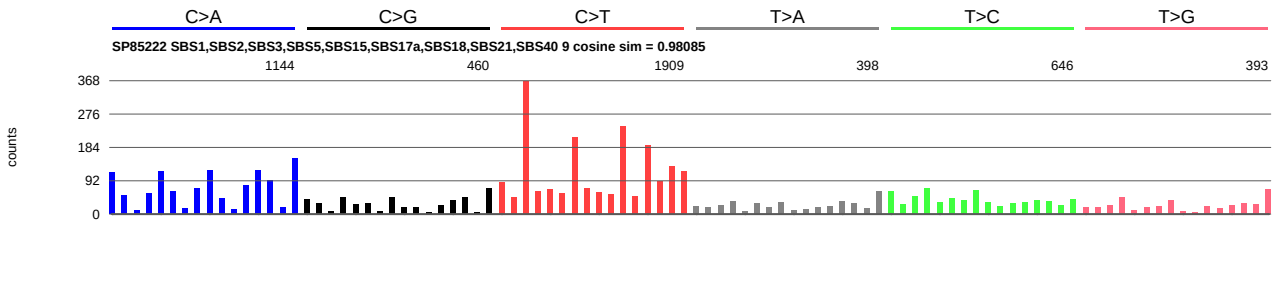
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98087
1156
459
1908
400
630
400
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98086
1138
440
1889
418
615
447
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.98086
1104
460
1853
478
666
392
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98085
1156
459
1908
400
629
401
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40 9 cosine sim = 0.98085
1144
460
1909
398
646
393
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17a
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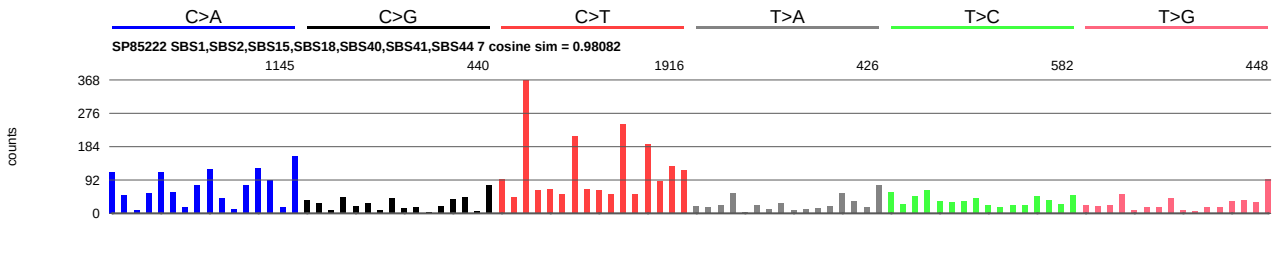
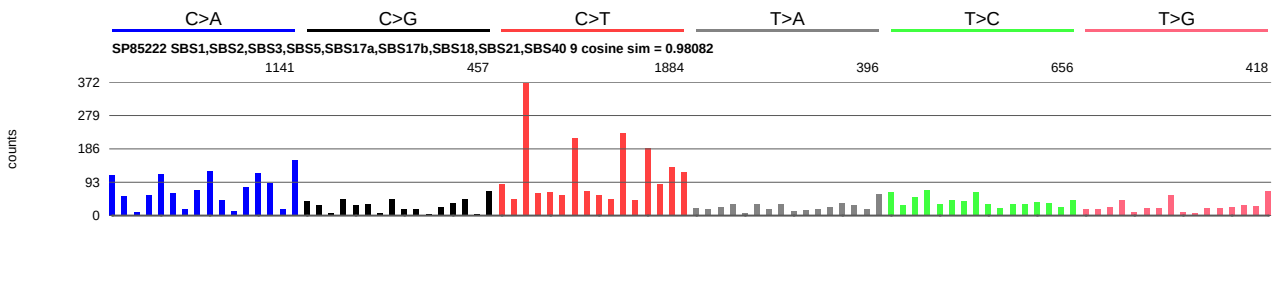
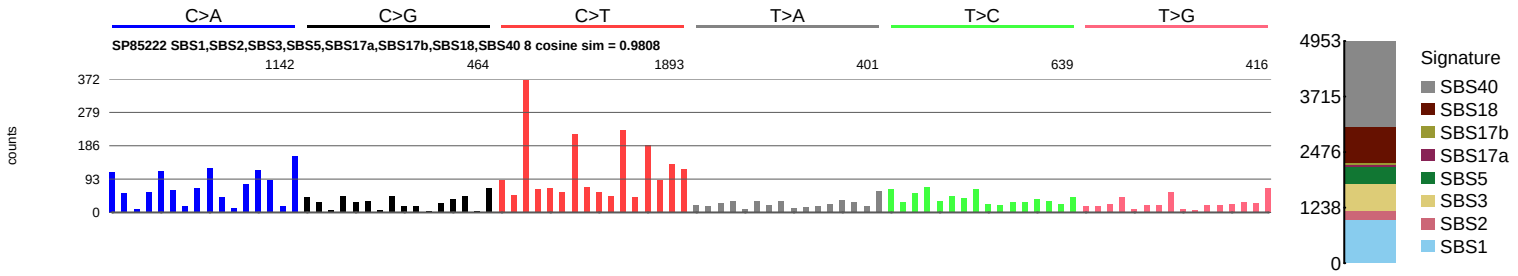
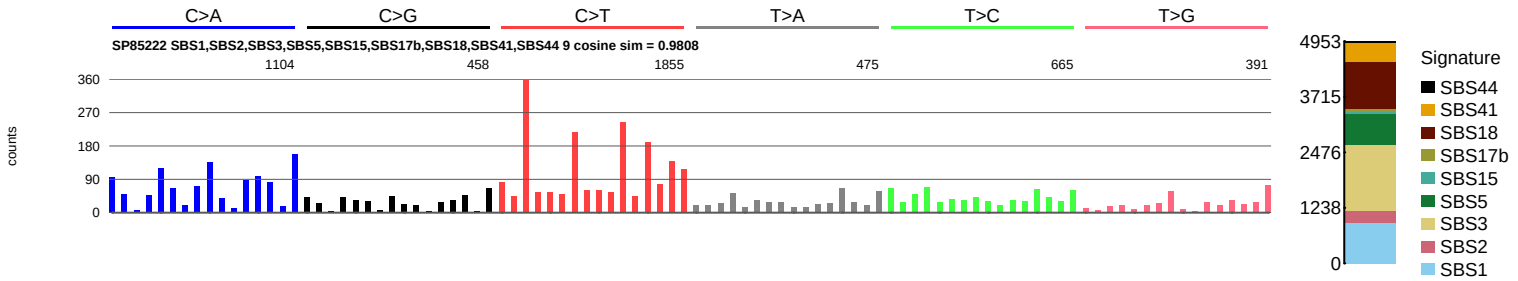
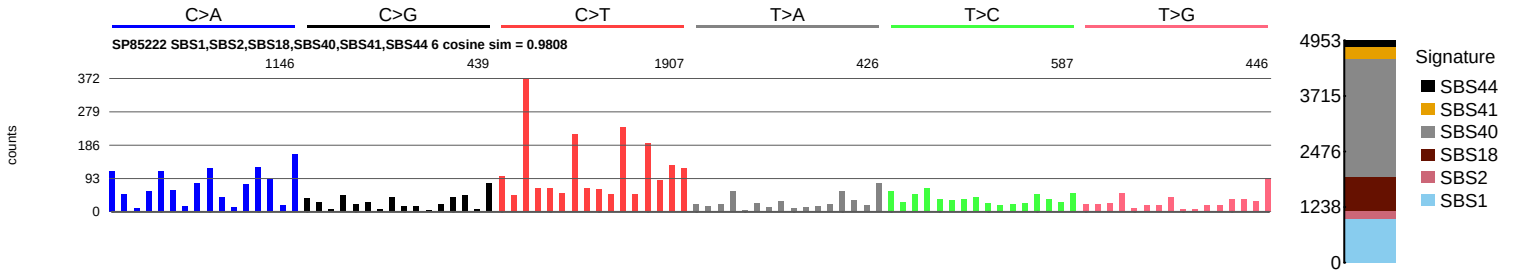
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98082
1145
440
1916
426
582
448
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98082
1141
457
1884
396
656
418
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.9808
1142
464
1893
401
639
416
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.9808
1104
458
1855
475
665
391
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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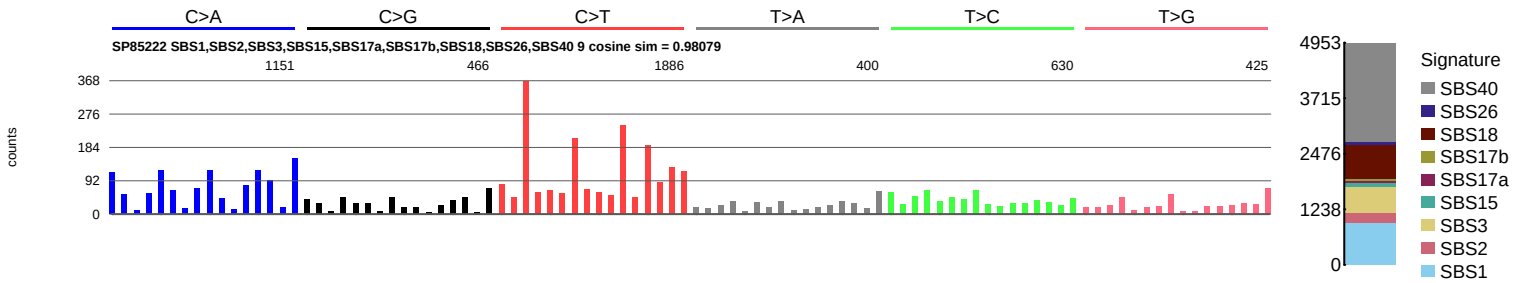
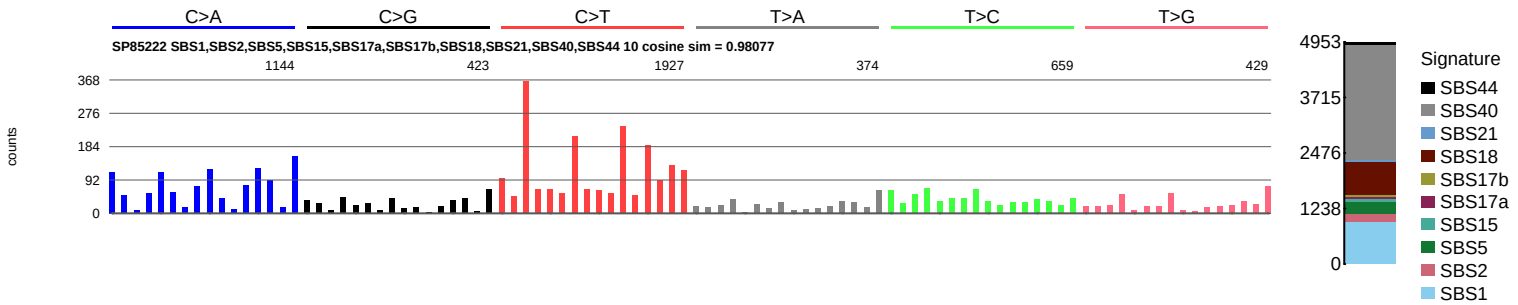
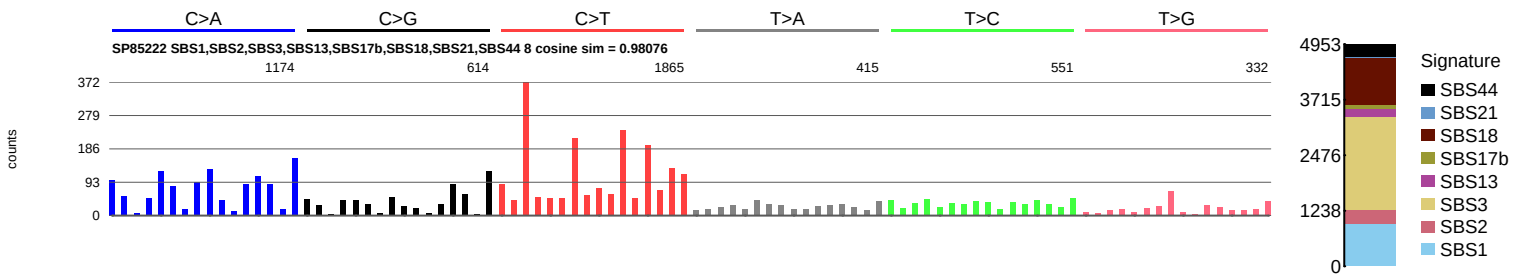
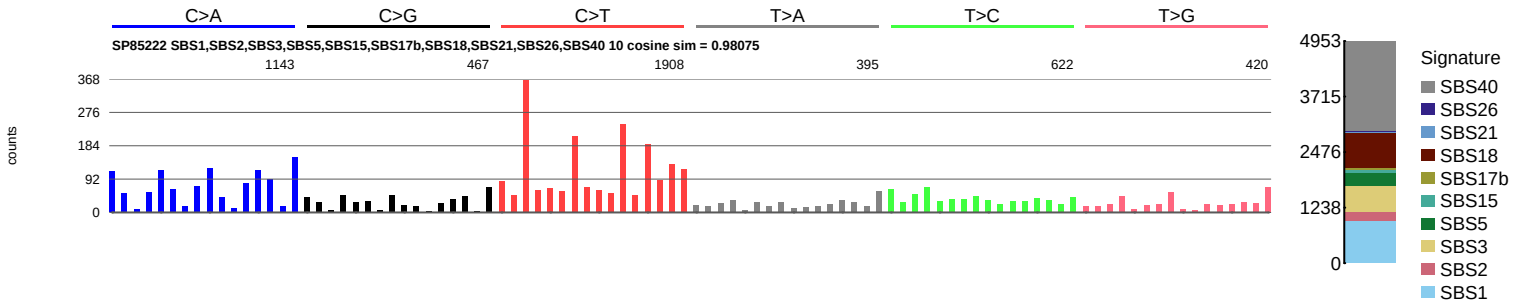
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS18,SBS40,SBS41,SBS44 6 cosine sim = 0.9808
1146
439
1907
426
587
446
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98079
1151
466
1886
400
630
425
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98077
1144
423
1927
374
659
429
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98076
1174
614
1865
415
551
332
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98075
1143
467
1908
395
622
420
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
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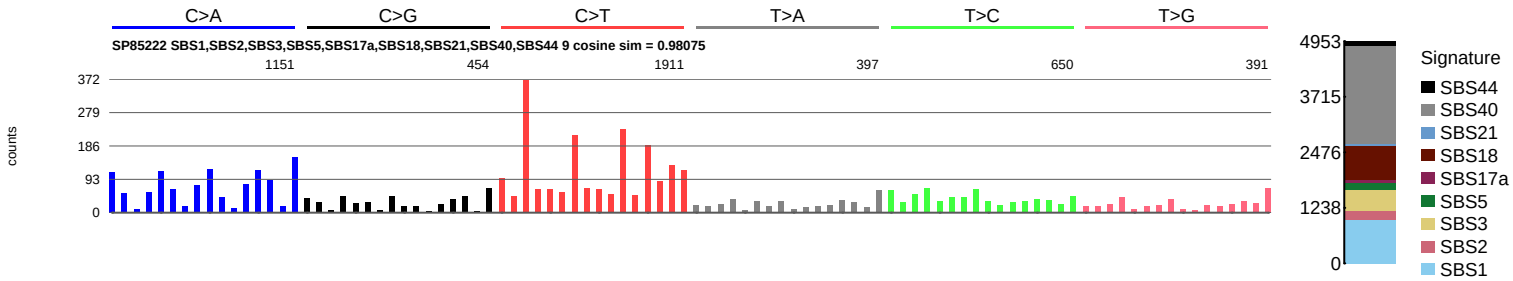
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98075
1151
454
1911
397
650
391
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
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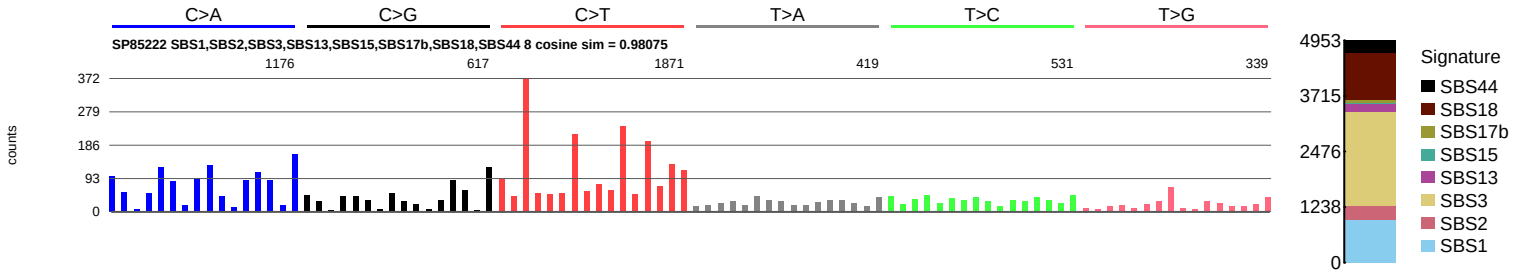
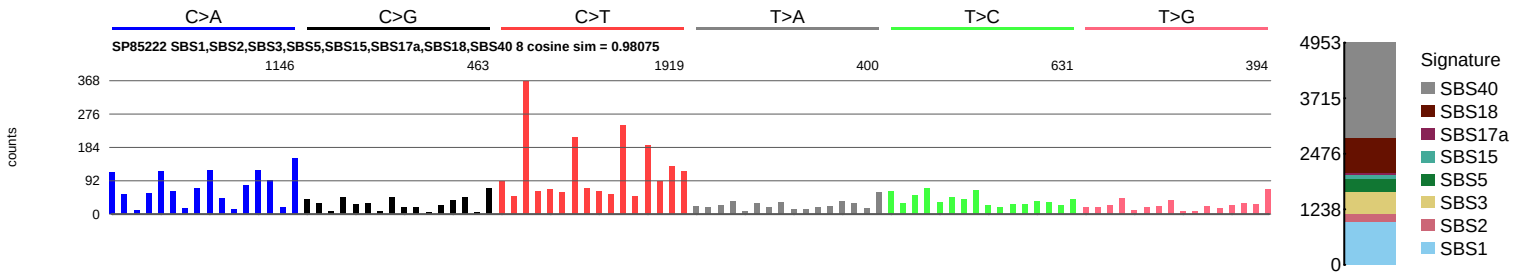
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS44 8 cosine sim = 0.98075
1176
617
1871
419
531
339
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS40 8 cosine sim = 0.98075
1146
463
1919
400
631
394
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17a
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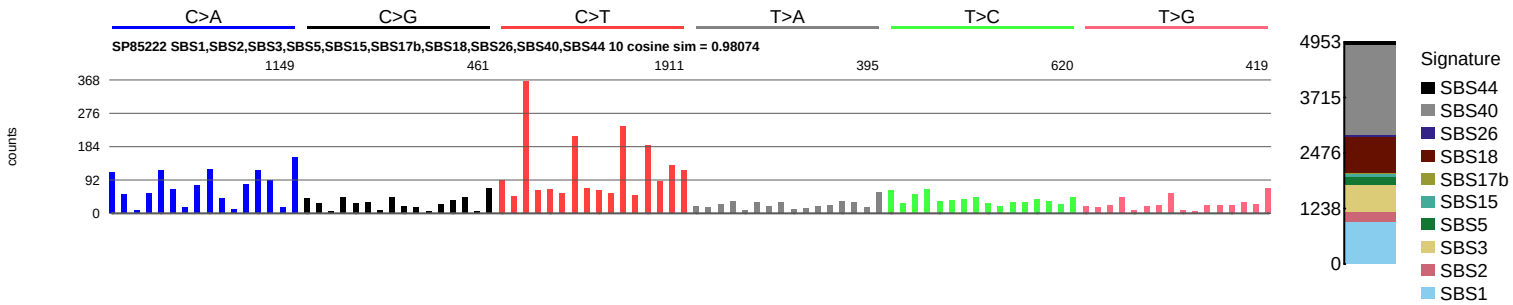
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98074
1149
461
1911
395
620
419
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
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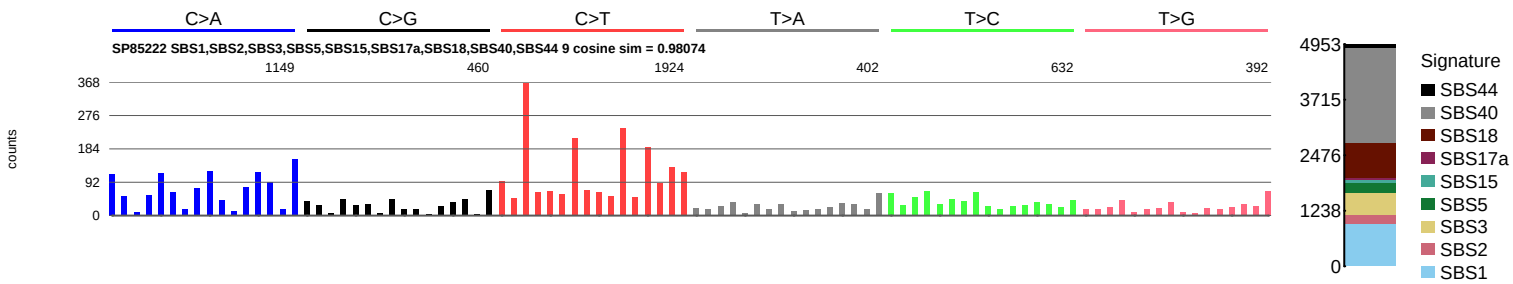
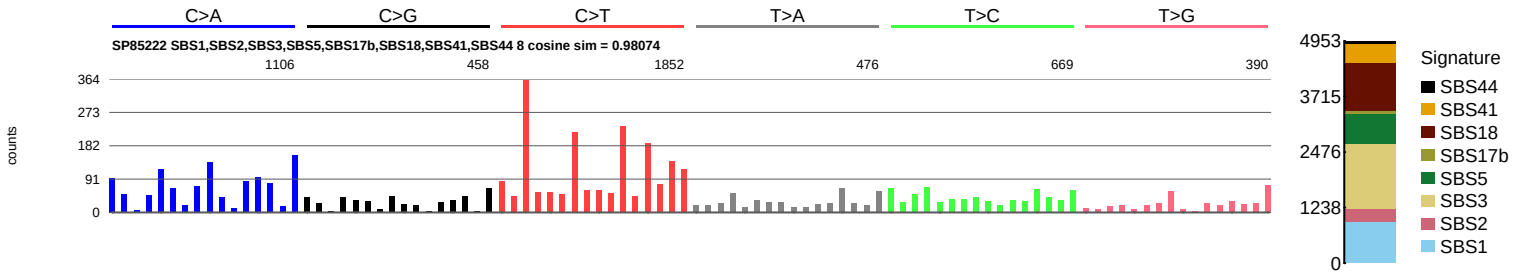
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS40,SBS44 9 cosine sim = 0.98074
1149
460
1924
402
632
392
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
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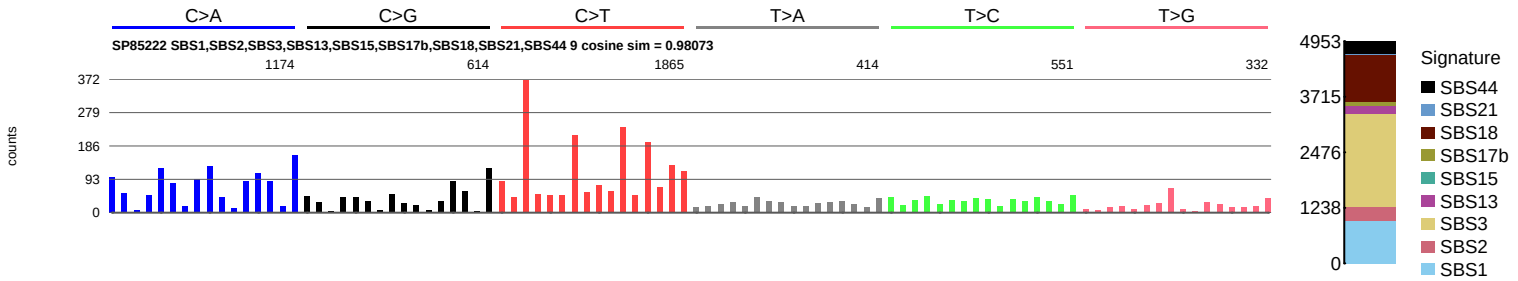
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98074
1106
458
1852
476
669
390
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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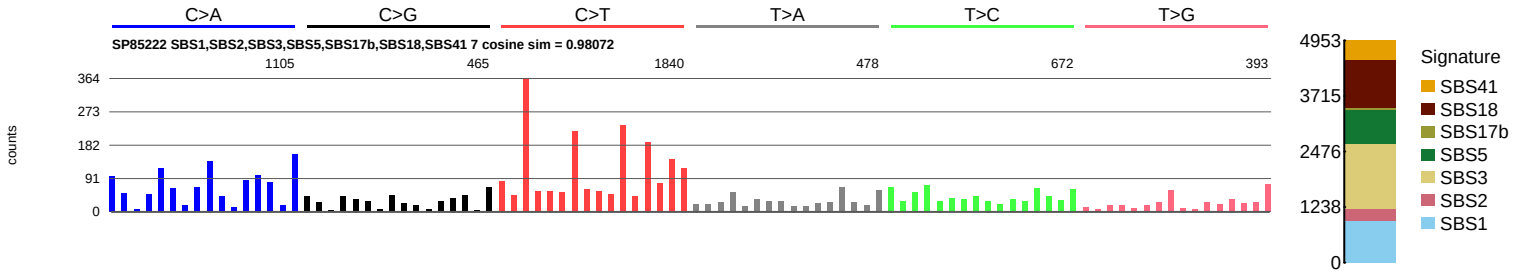
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98073
1174
614
1865
414
551
332
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1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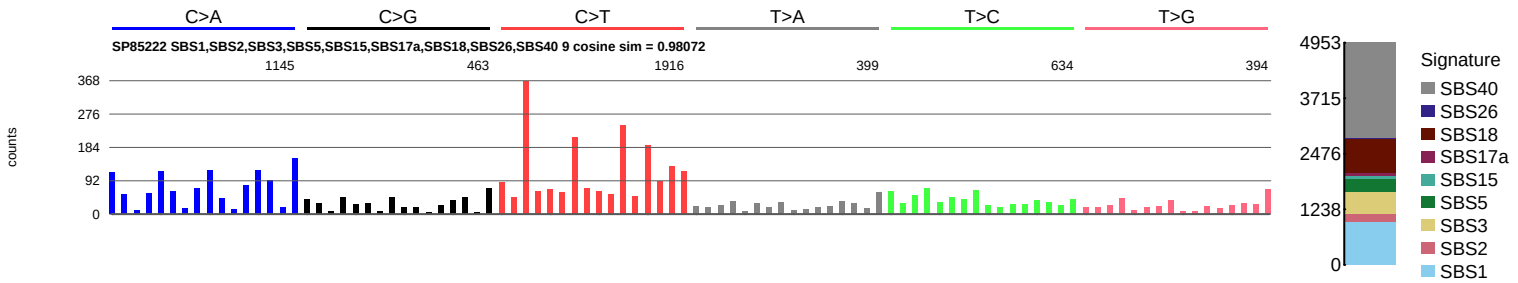
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS41 7 cosine sim = 0.98072
1105
465
1840
478
672
393
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
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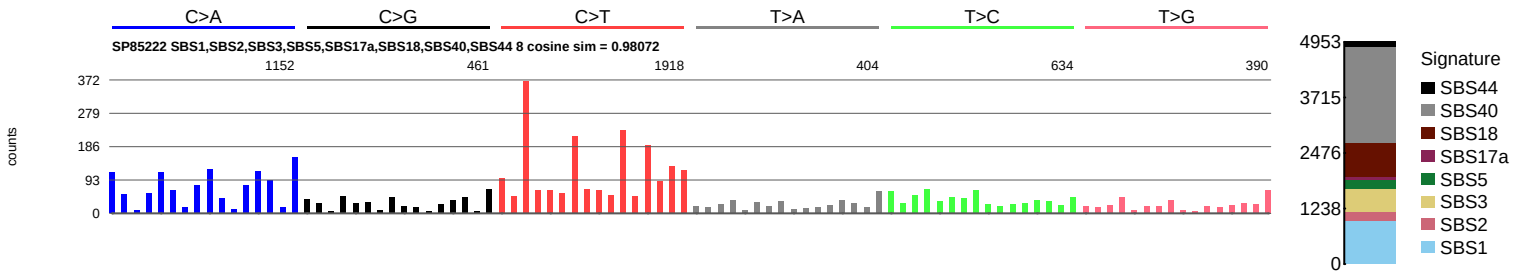
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS40 9 cosine sim = 0.98072
1145
463
1916
399
634
394
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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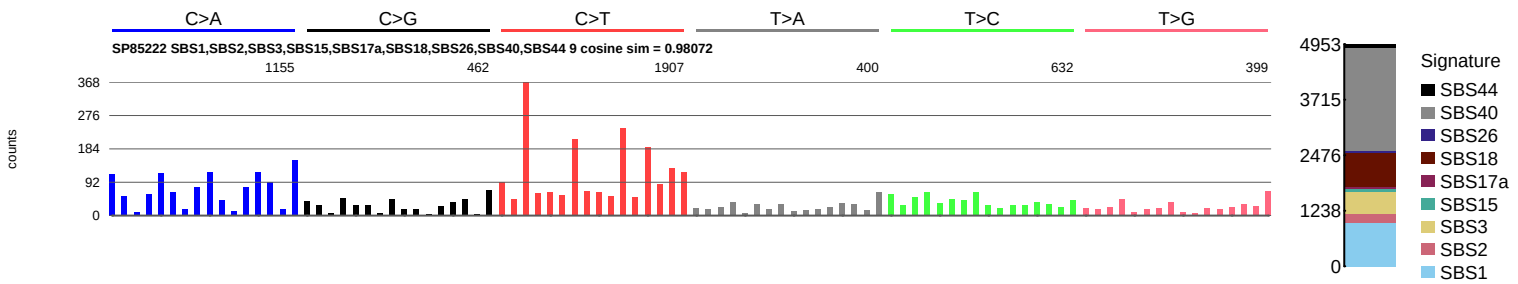
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.98072
1152
461
1918
404
634
390
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1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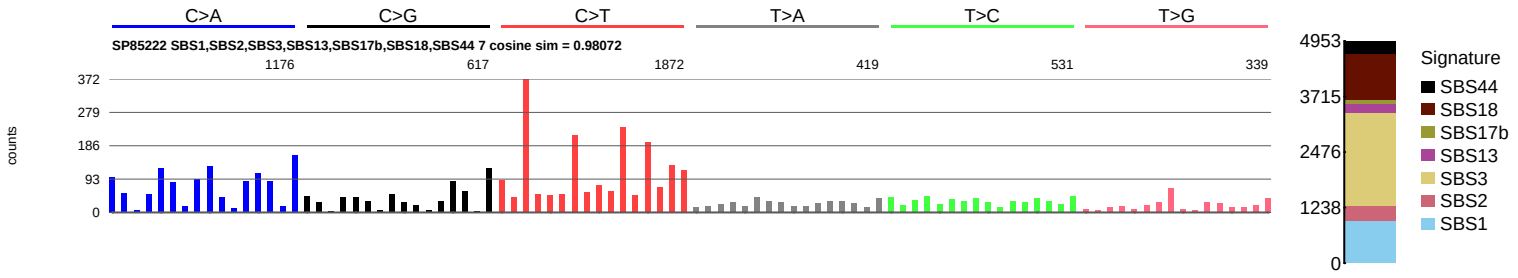
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98072
1155
462
1907
400
632
399
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS44 7 cosine sim = 0.98072
1176
617
1872
419
531
339
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS18
SBS17b
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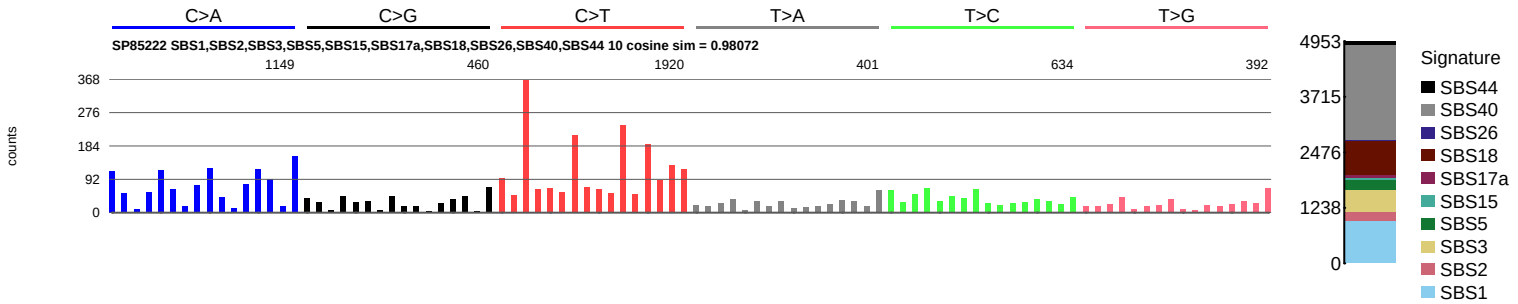
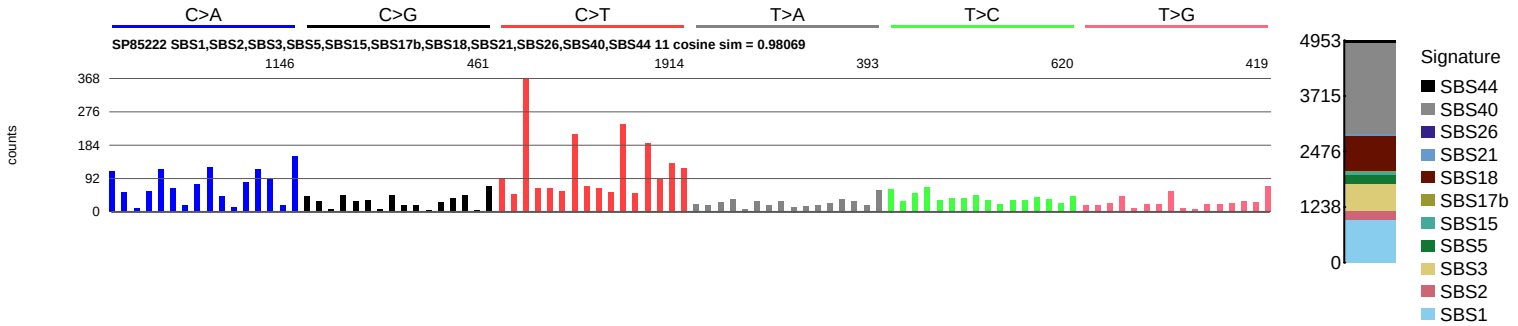
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98072
1149
460
1920
401
634
392
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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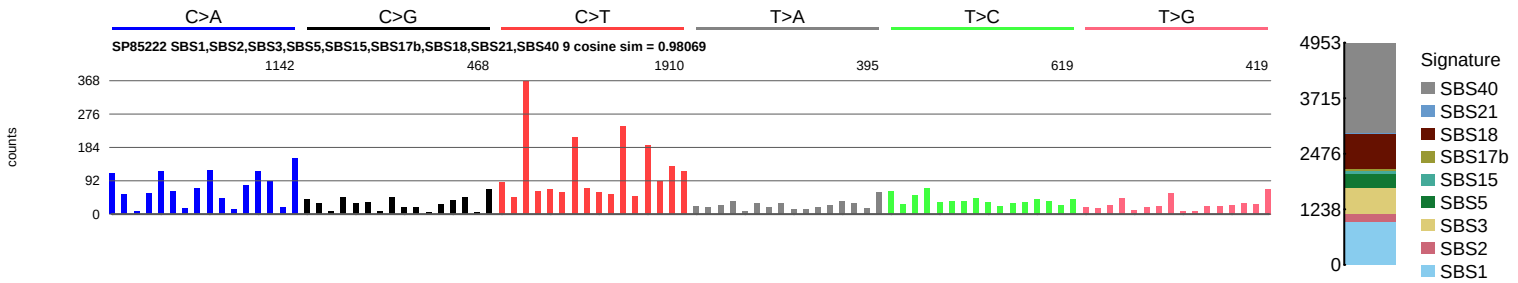
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98069
1146
461
1914
393
620
419
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98069
1142
468
1910
395
619
419
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
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SBS17b
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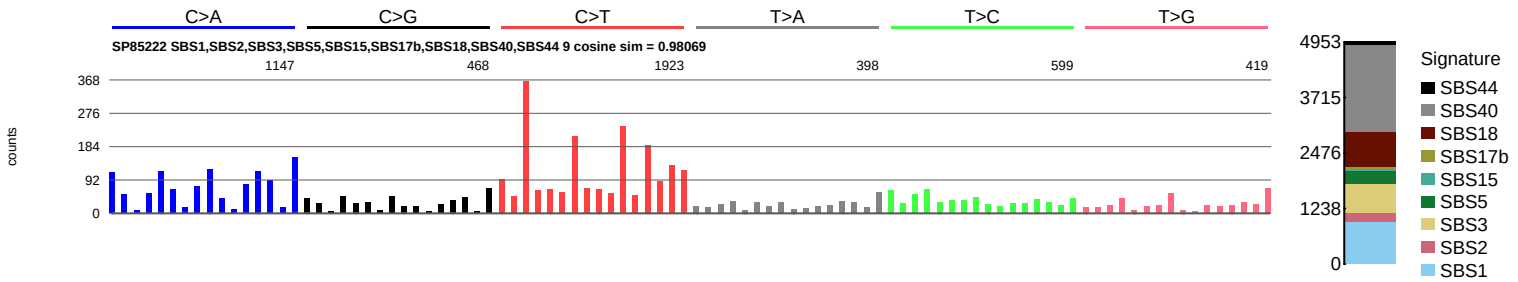
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98069
1147
468
1923
398
599
419
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
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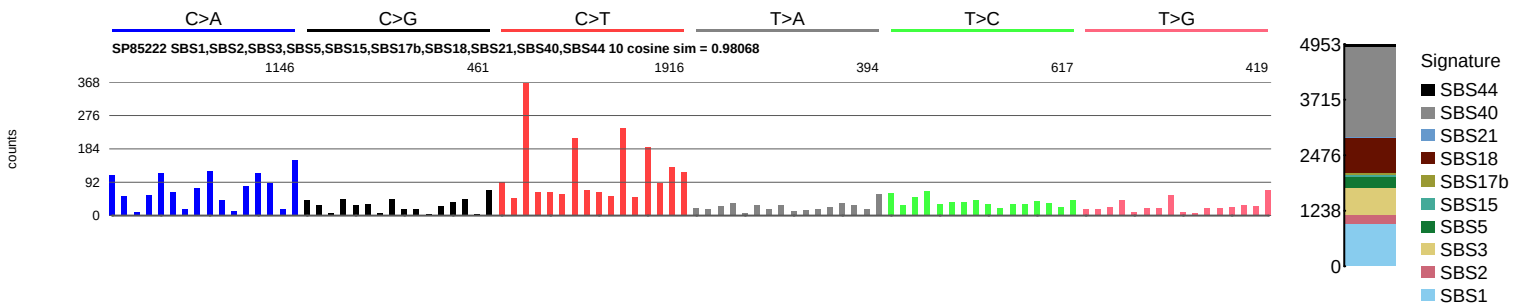
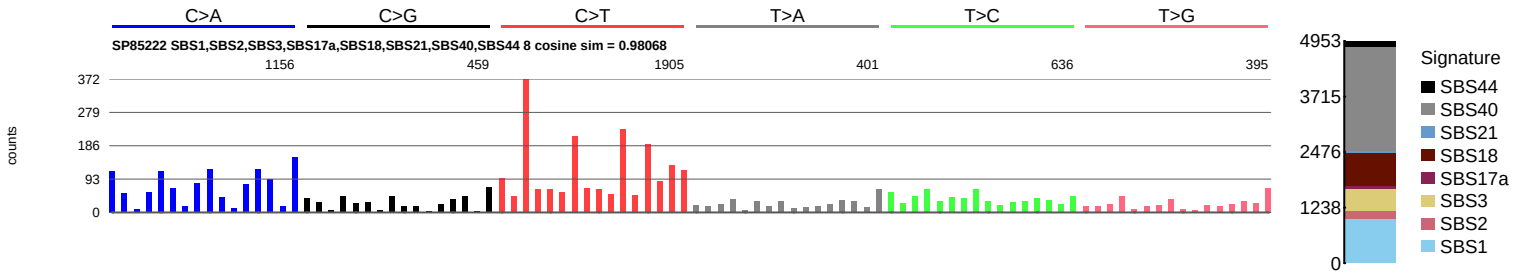
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98068
1146
461
1916
394
617
419
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98068
1156
459
1905
401
636
395
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
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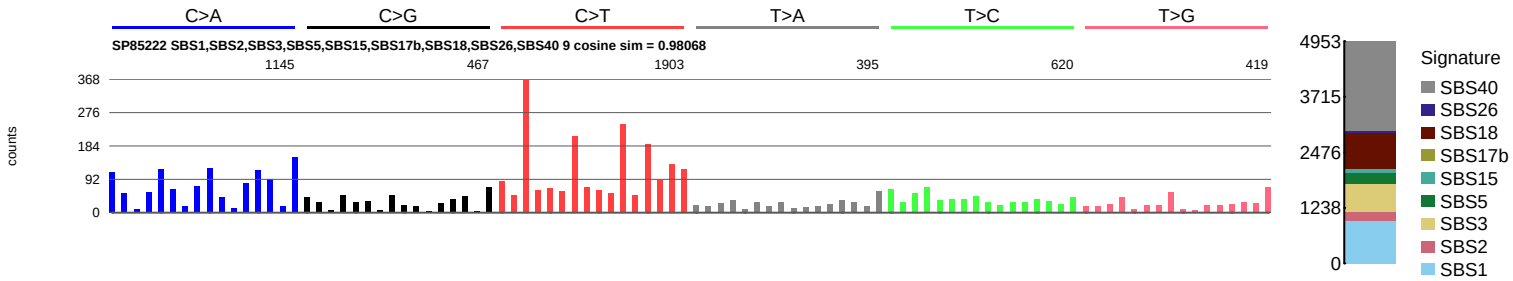
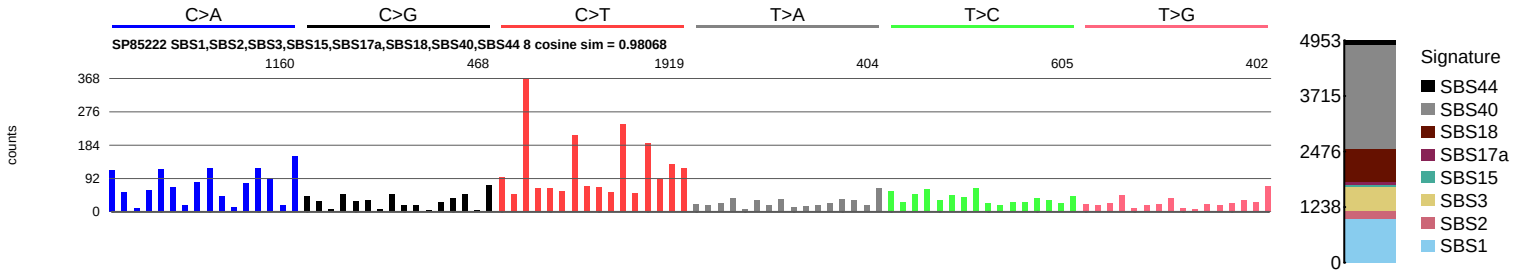
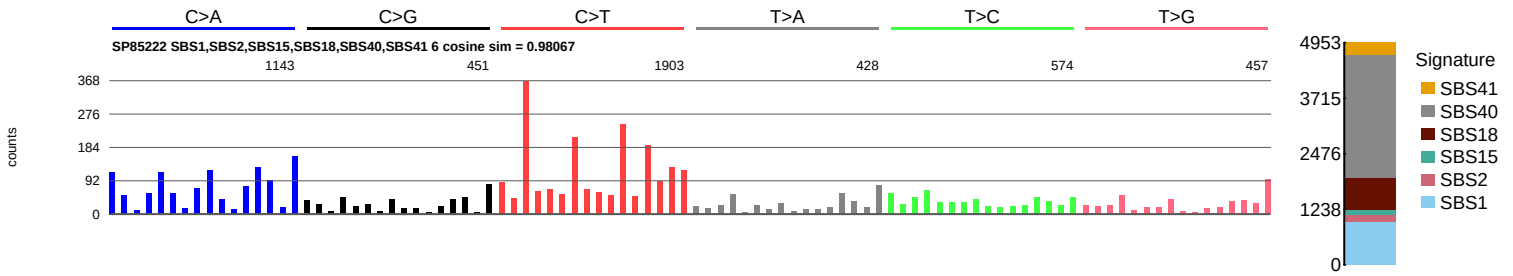
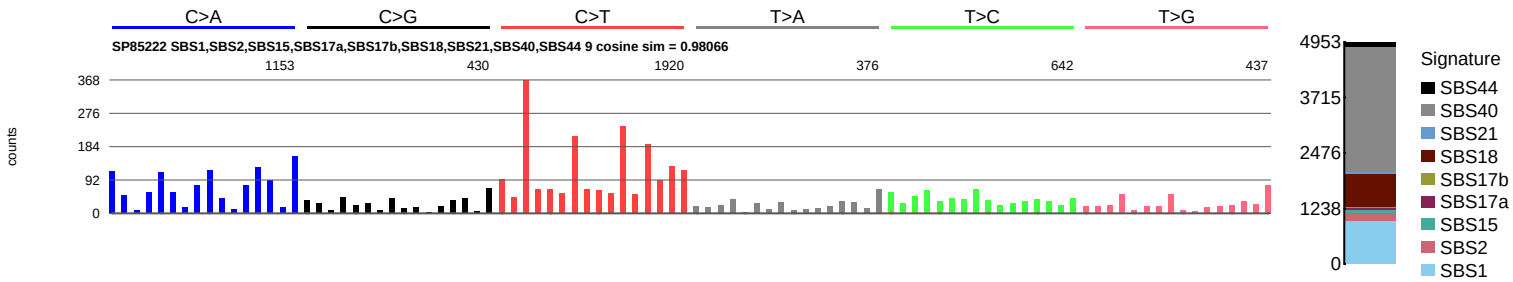
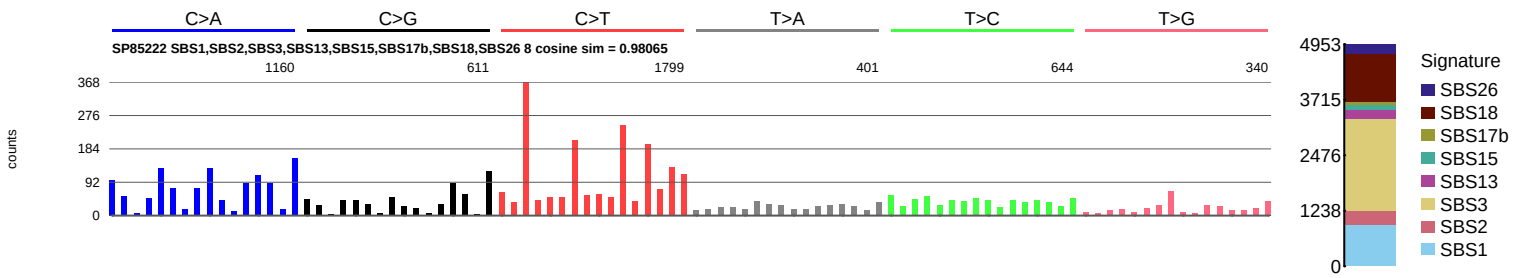
SBS2
SBS1

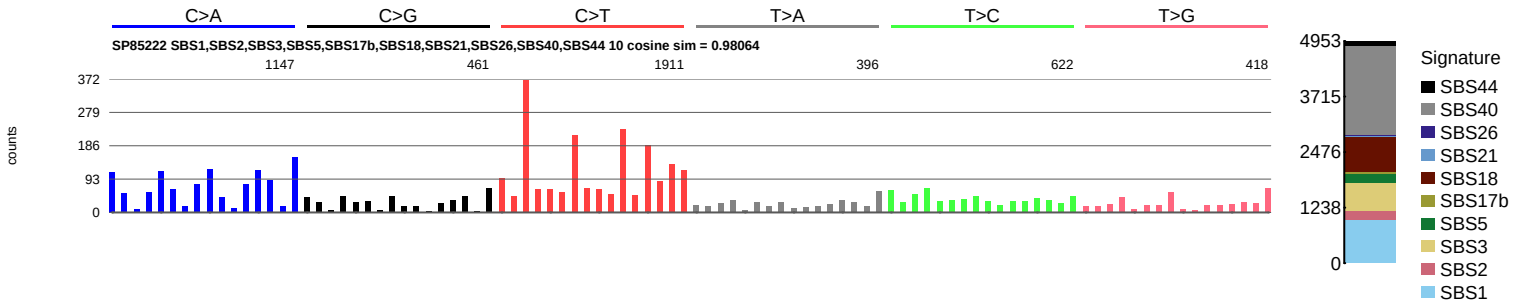

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98068
1145
467
1903
395
620
419
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

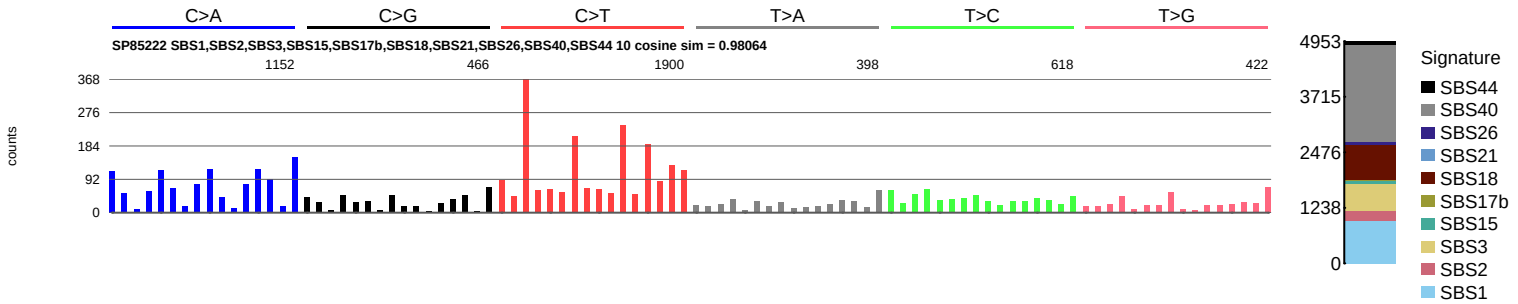

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.98068
1160
468
1919
404
605
402
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS18,SBS40,SBS41 6 cosine sim = 0.98067
1143
451
1903
428
574
457
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98066
1153
430
1920
376
642
437
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26 8 cosine sim = 0.98065
1160
611
1799
401
644
340
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98064
1147
461
1911
396
622
418
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98064
1152
466
1900
398
618
422
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
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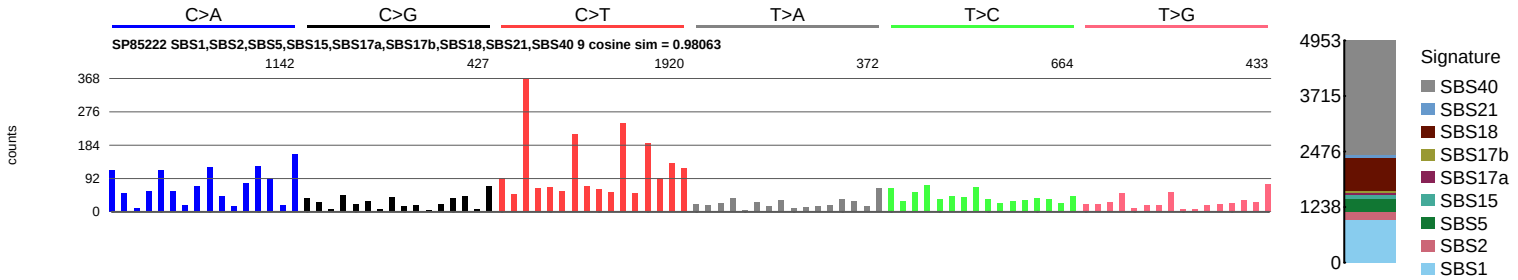
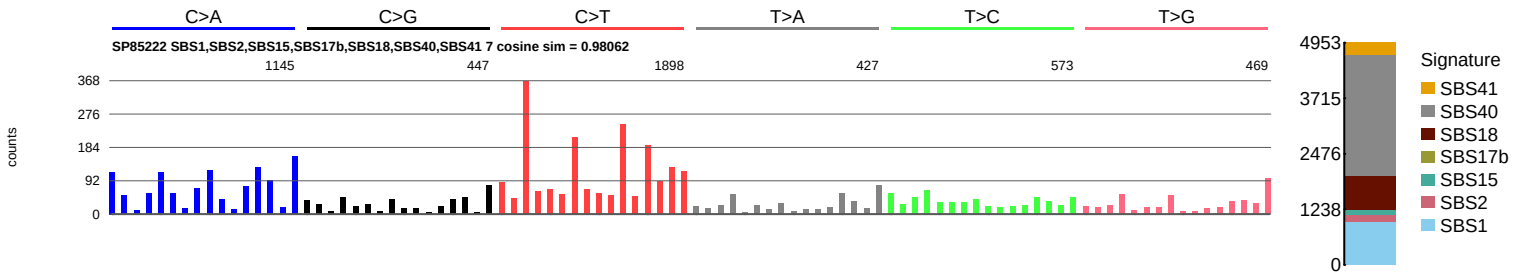
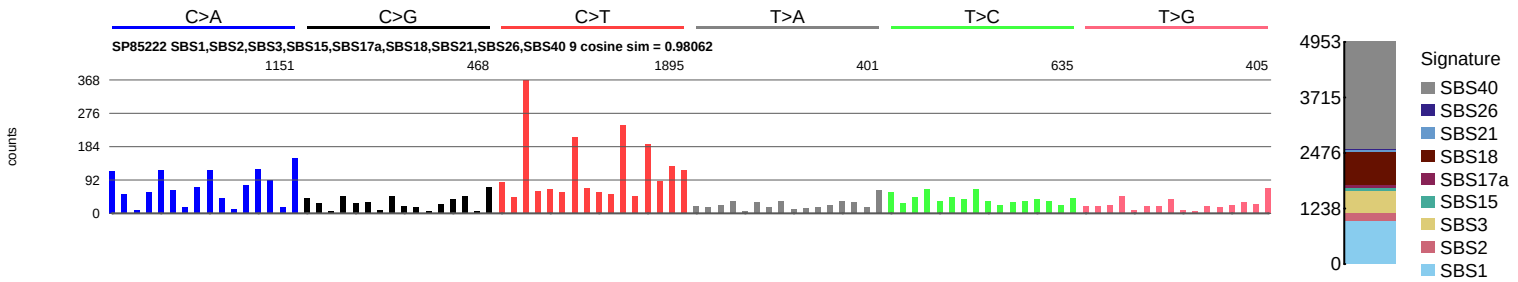
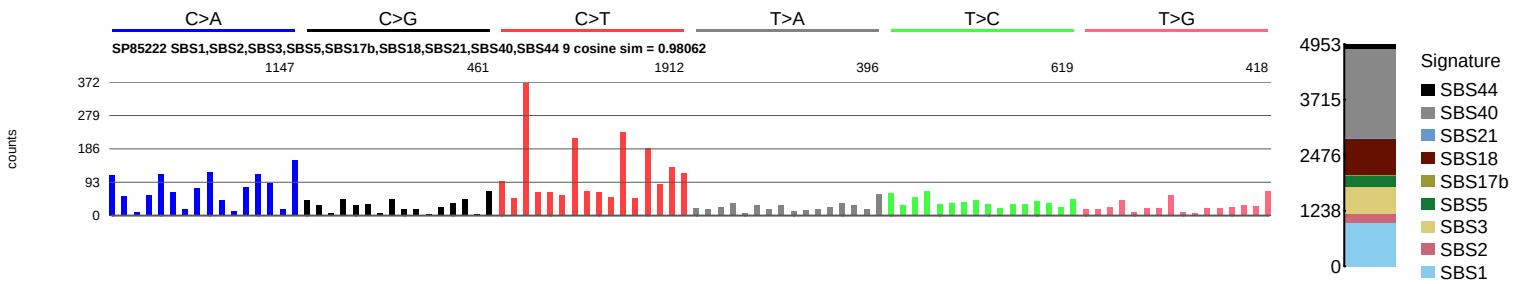
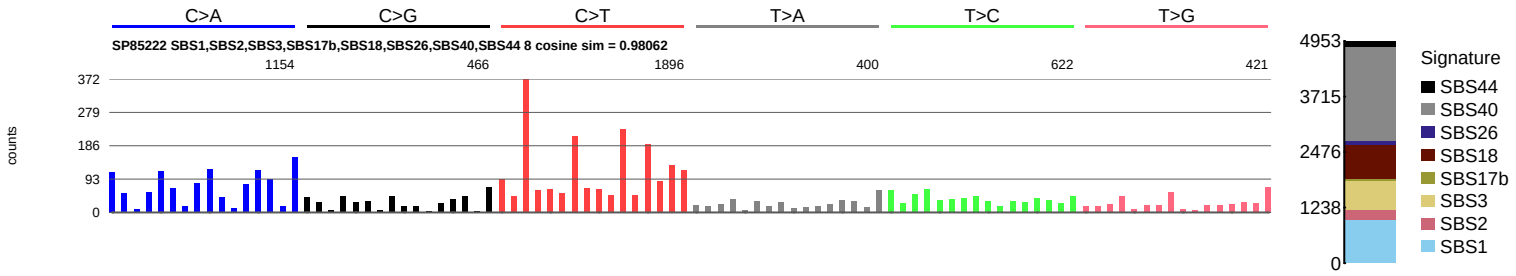
SBS3
SBS2
SBS1

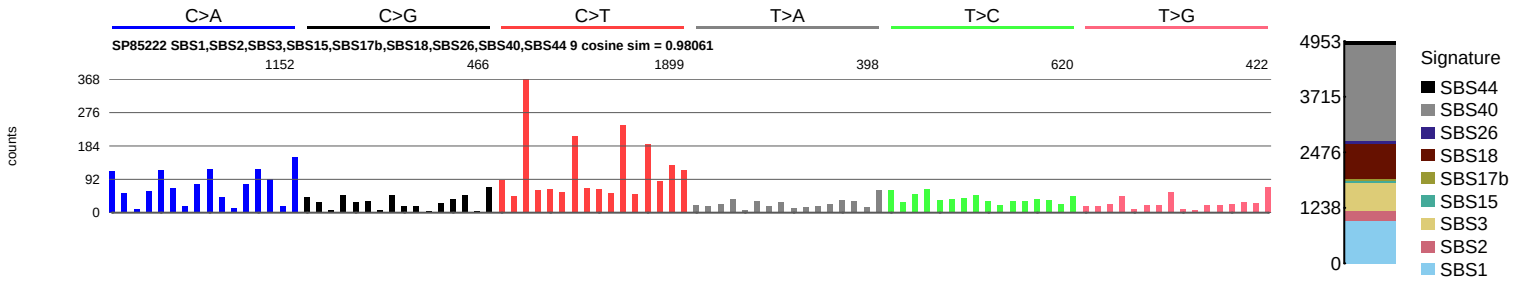

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98063
1142
427
1920
372
664
433
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98062
1145
447
1898
427
573
469
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98062
1151
468
1895
401
635
405
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98062
1147
461
1912
396
619
418
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98062
1154
466
1896
400
622
421
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1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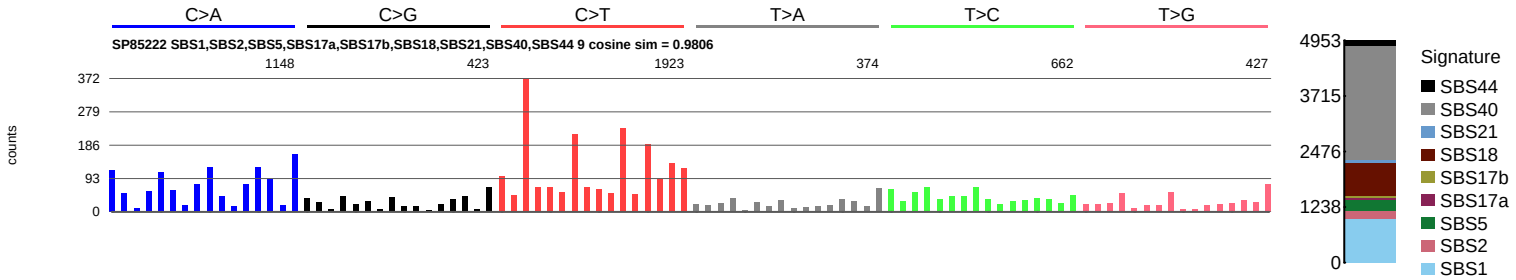
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98061
1152
466
1899
398
620
422
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1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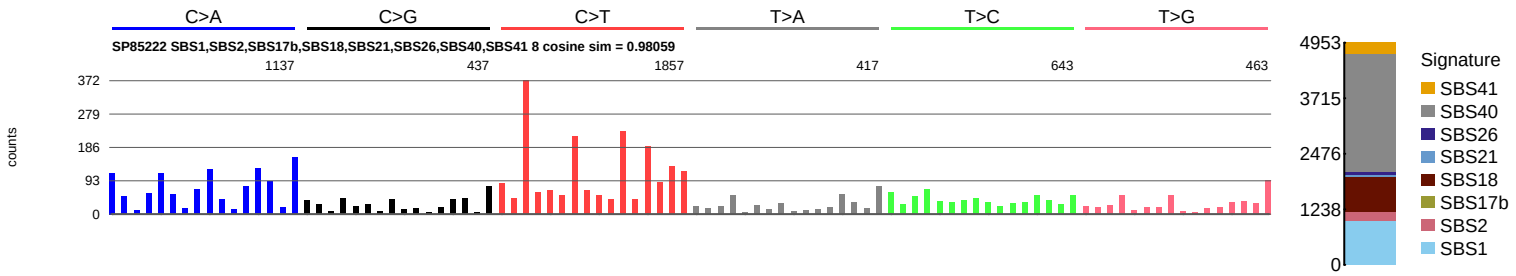
SP85222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.9806
1148
423
1923
374
662
427
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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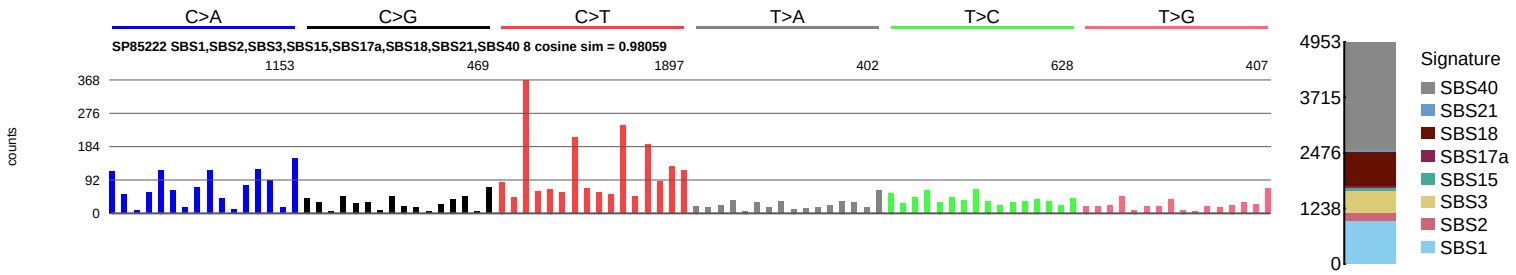
SP85222 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98059
1137
437
1857
417
643
463
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS2
SBS1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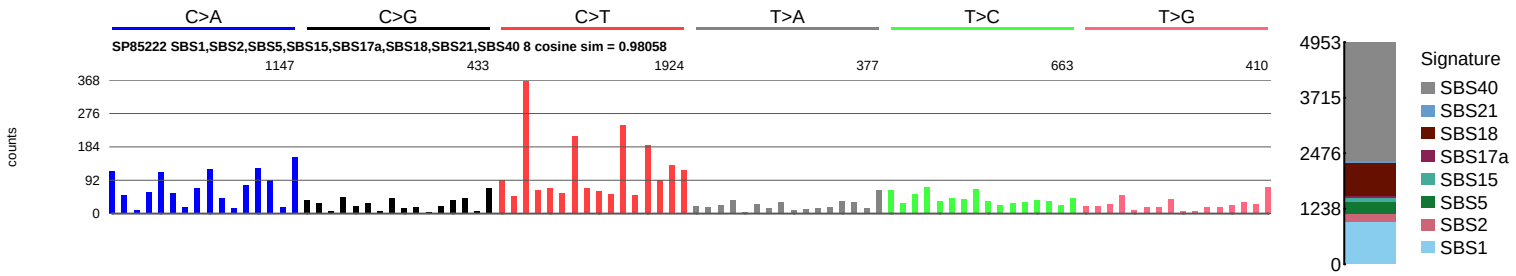
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS21,SBS40 8 cosine sim = 0.98059
1153
469
1897
402
628
407
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1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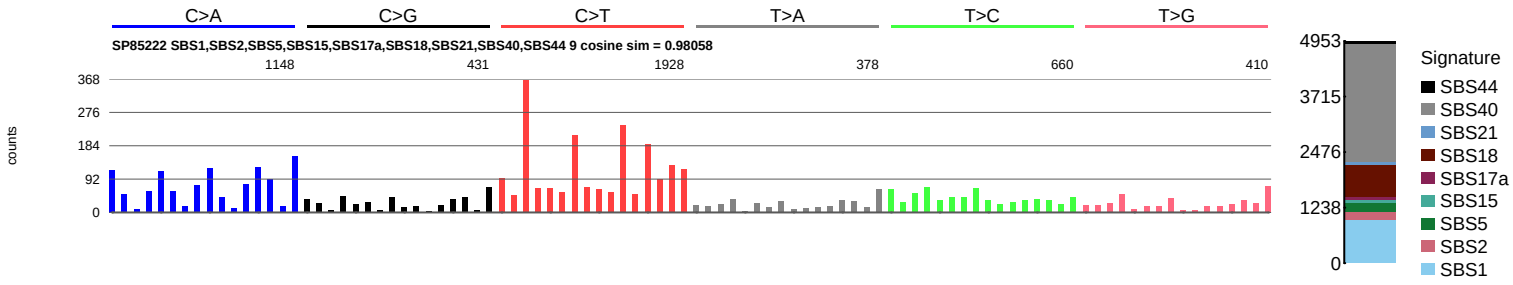
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40 8 cosine sim = 0.98058
1147
433
1924
377
663
410
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98058
1148
431
1928
378
660
410
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
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SBS18
SBS17a
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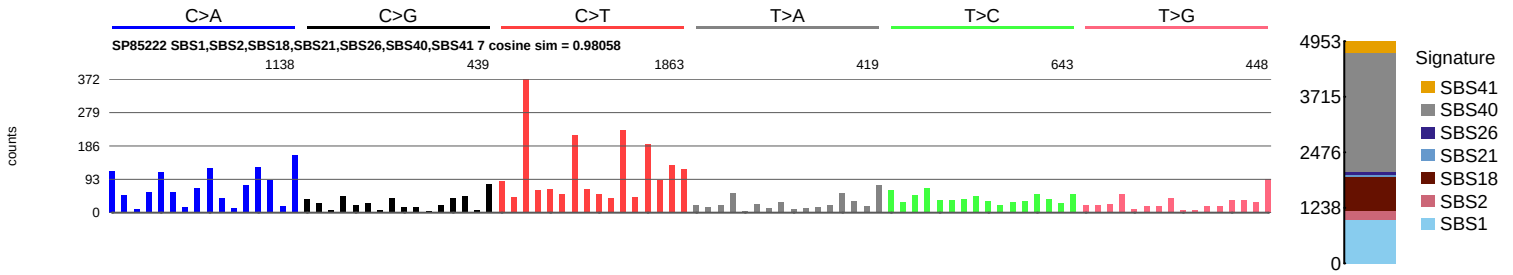
SBS2
SBS1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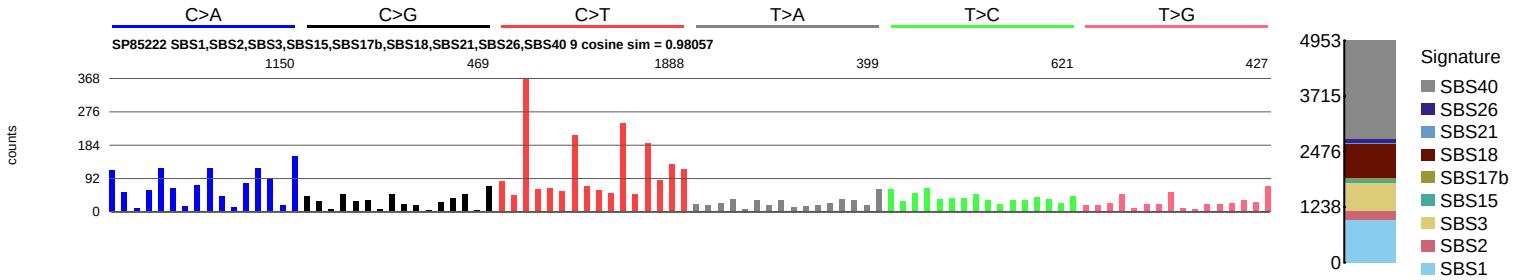
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS18,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.98058
1138
439
1863
419
643
448
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98057
1150
469
1888
399
621
427
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
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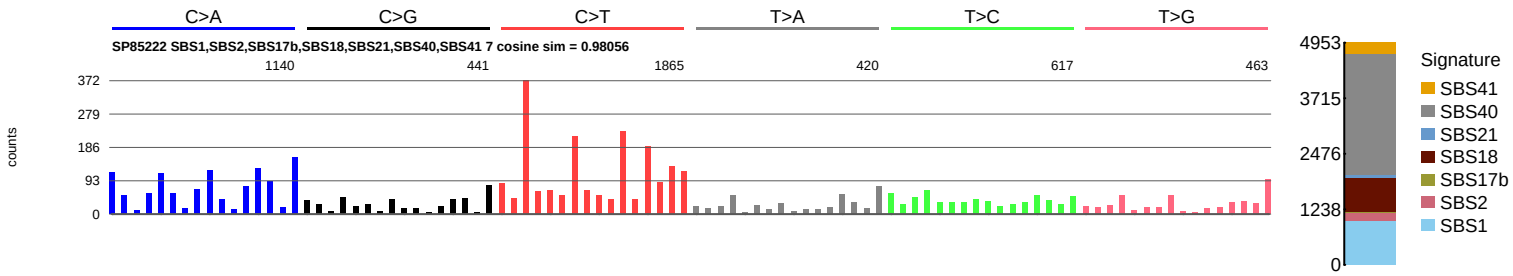
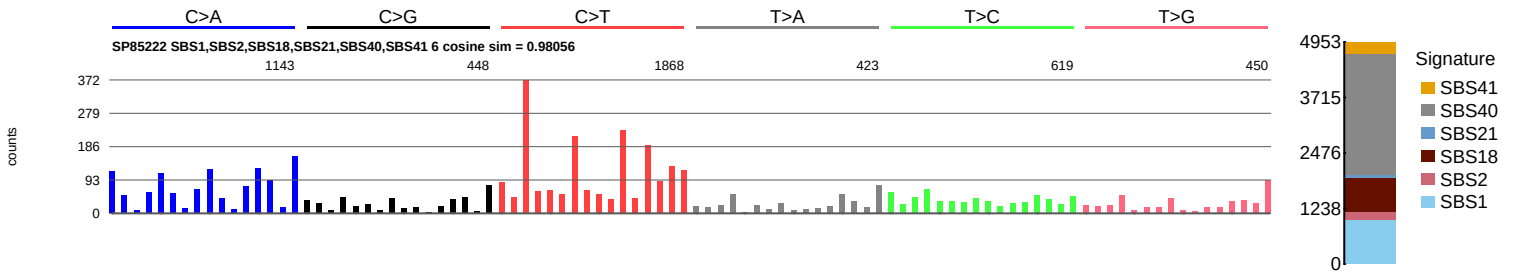
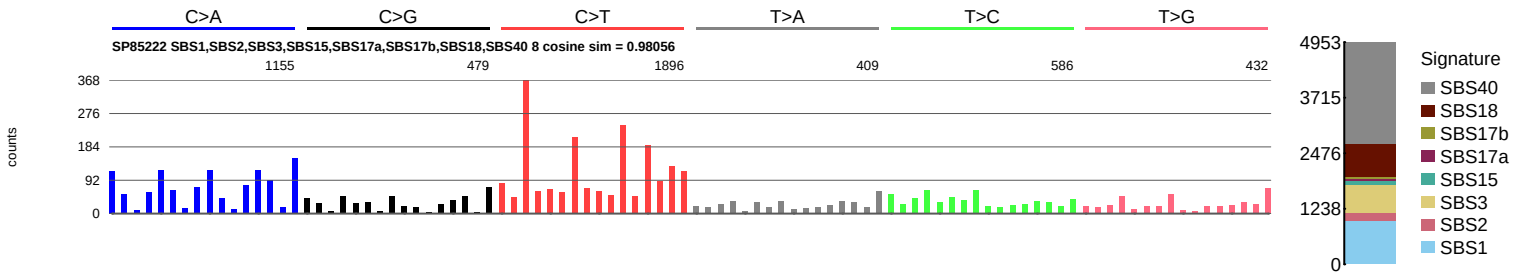
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98056
1140
441
1865
420
617
463
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS18,SBS21,SBS40,SBS41 6 cosine sim = 0.98056
1143
448
1868
423
619
450
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS2
SBS1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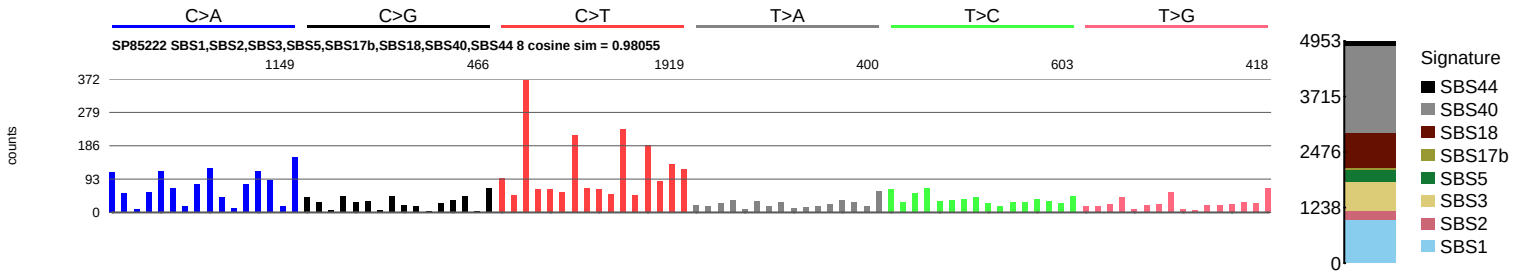
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98056
1155
479
1896
409
586
432
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98055
1149
466
1919
400
603
418
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
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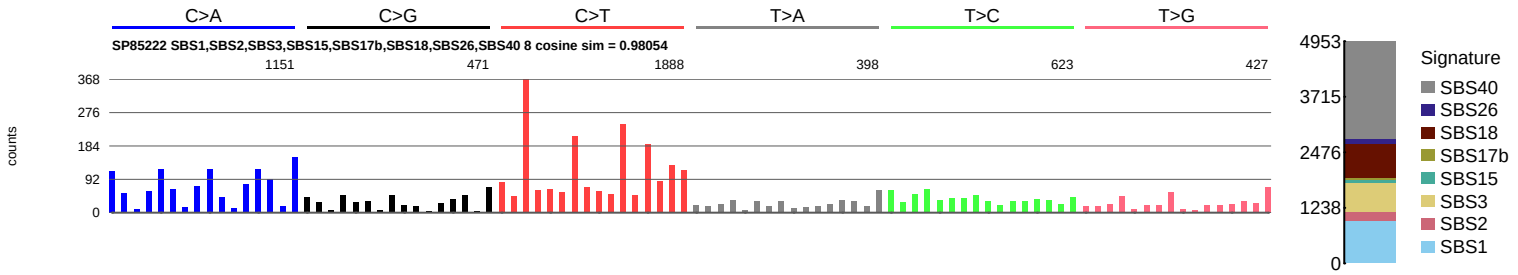
SBS2
SBS1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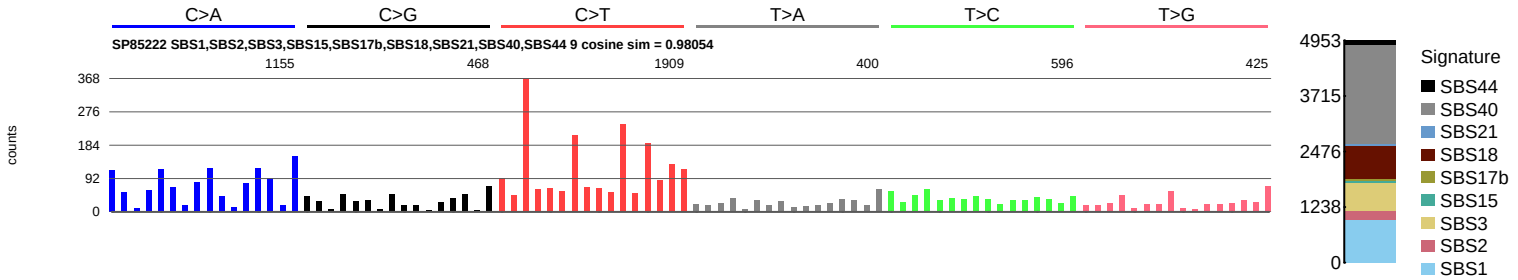
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98054
1151
471
1888
398
623
427
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1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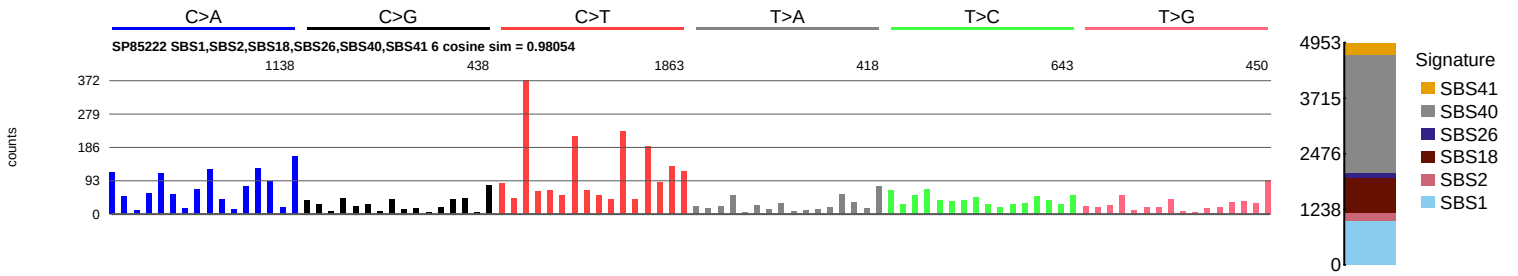
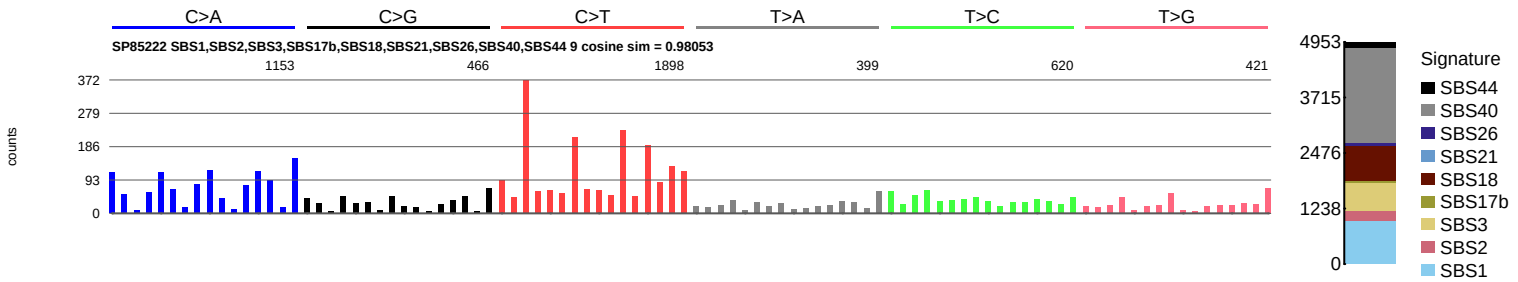
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98054
1155
468
1909
400
596
425
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS18,SBS26,SBS40,SBS41 6 cosine sim = 0.98054
1138
438
1863
418
643
450
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS2
SBS1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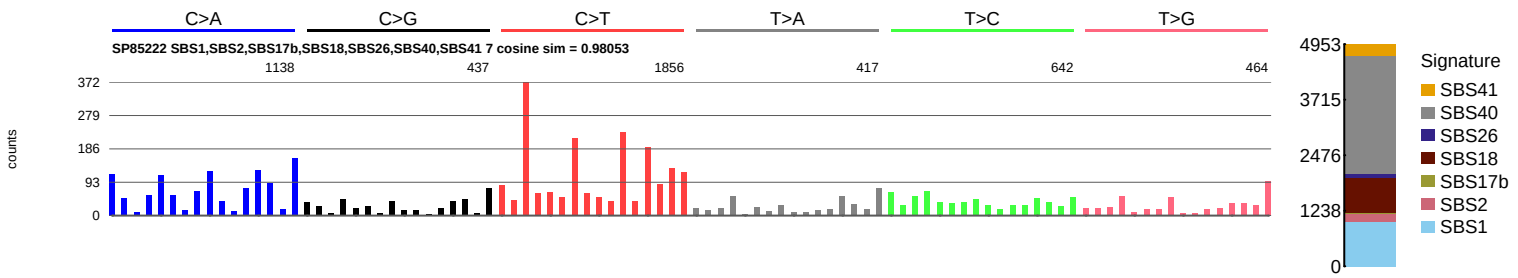
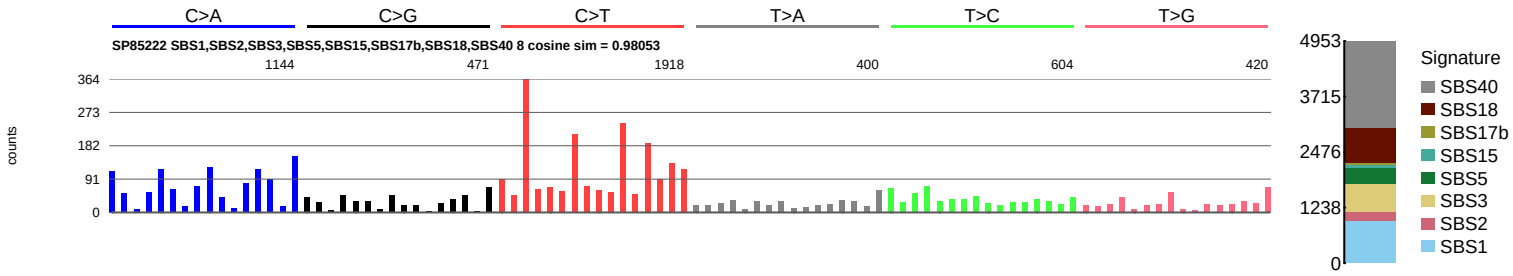
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98053
1153
466
1898
399
620
421
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98053
1138
437
1856
417
642
464
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1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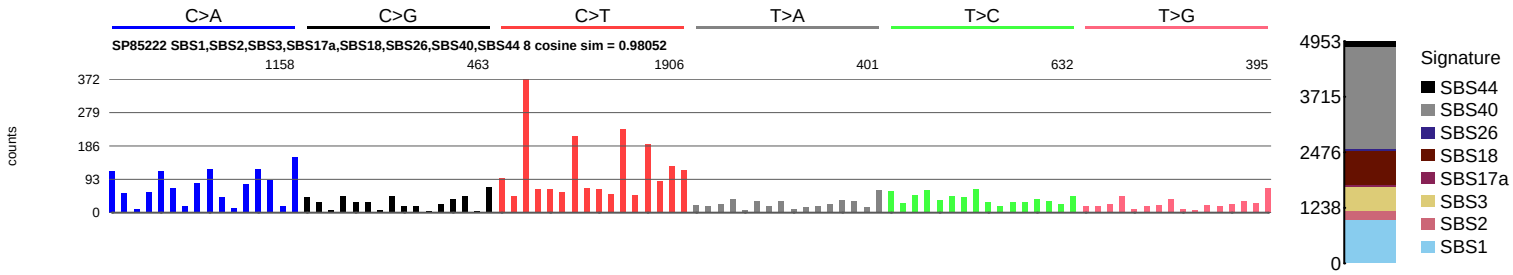
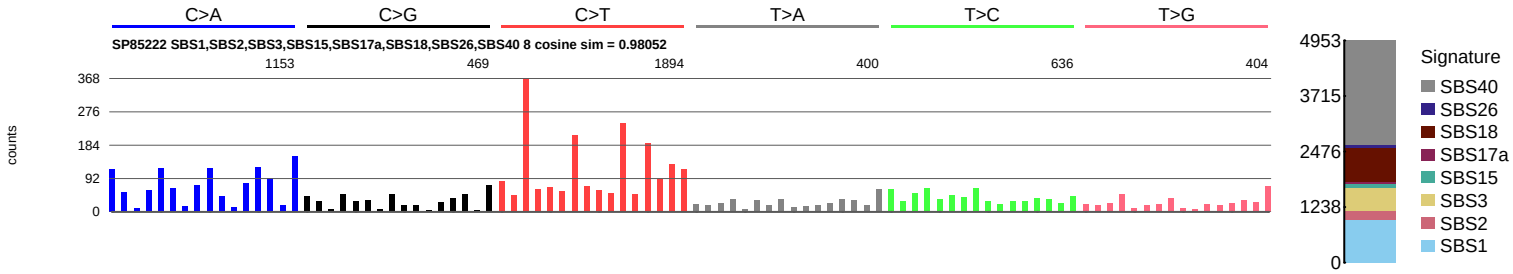
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.98053
1144
471
1918
400
604
420
364
273
182
91
0
counts
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98052
1158
463
1906
401
632
395
372
279
186
93
0
counts
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS3
SBS2
SBS1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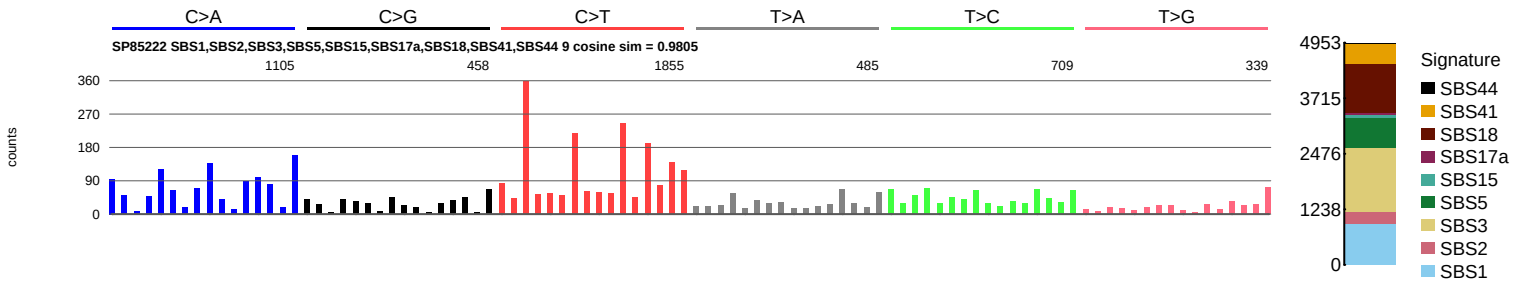
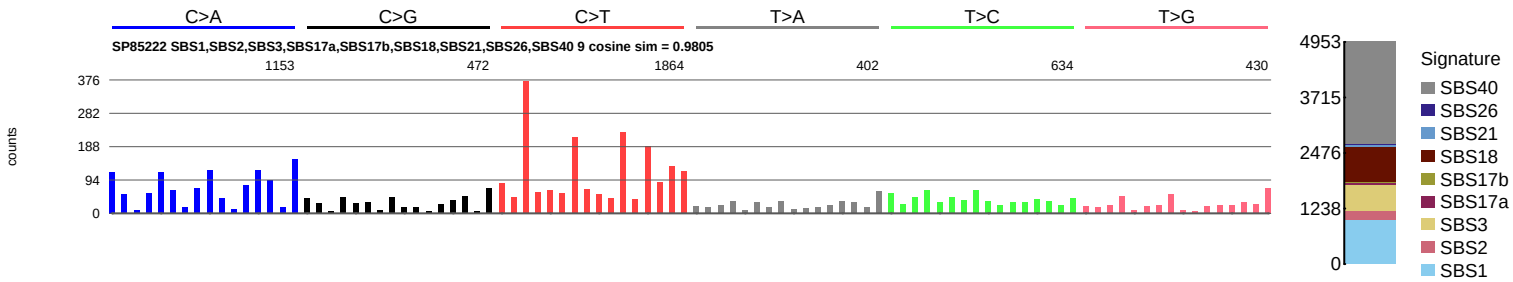
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS26,SBS40 8 cosine sim = 0.98052
1153
469
1894
400
636
404
368
276
184
92
0
counts
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS41,SBS44 9 cosine sim = 0.9805
1105
458
1855
485
709
339
360
270
180
90
0
counts
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.9805
1153
472
1864
402
634
430
376
282
188
94
0
counts
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
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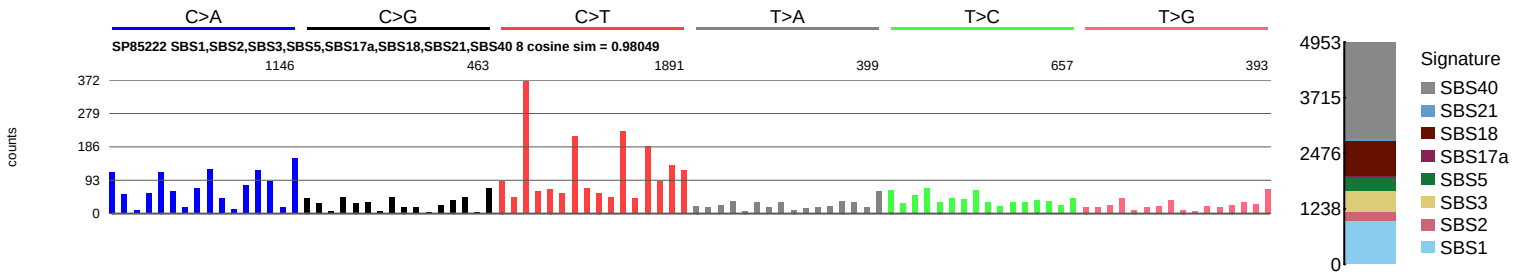
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS21,SBS40 8 cosine sim = 0.98049
1146
463
1891
399
657
393
372
279
186
93
0
counts
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17a
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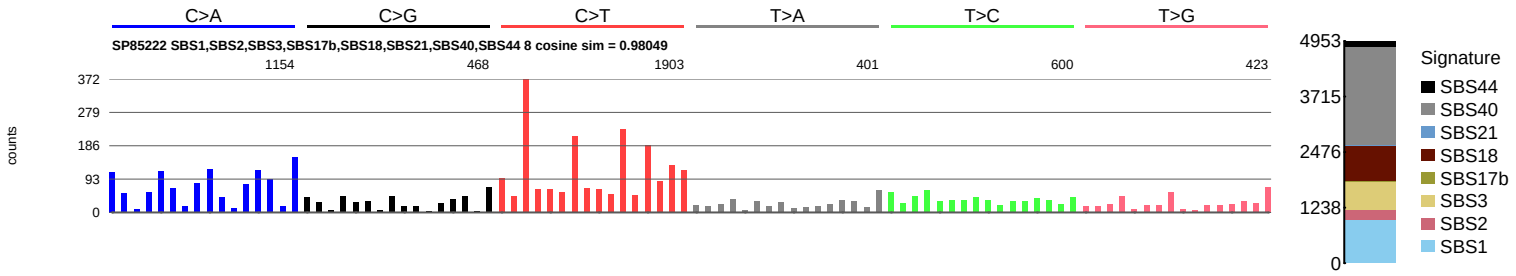
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98049
1154
468
1903
401
600
423
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
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SBS21
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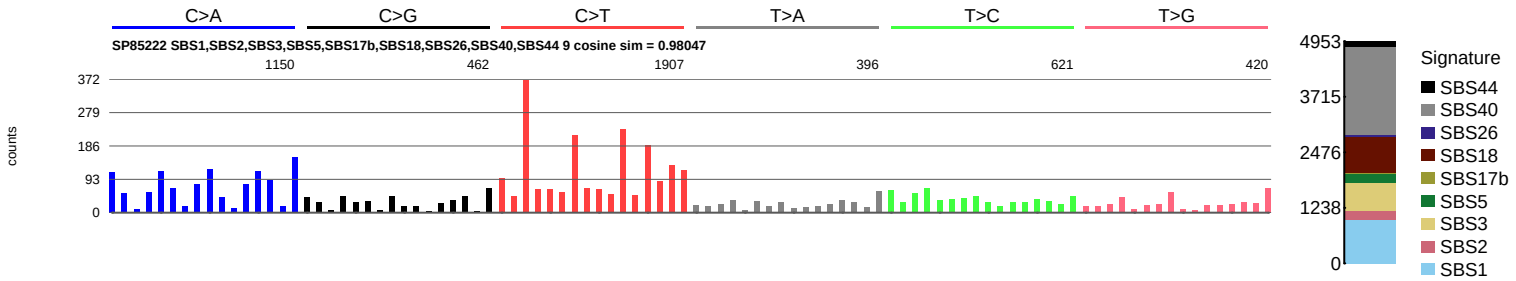
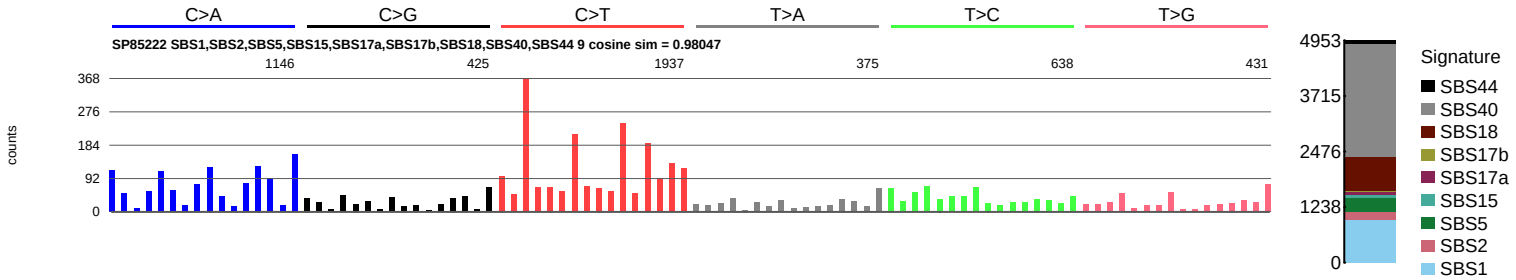
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98047
1150
462
1907
396
621
420
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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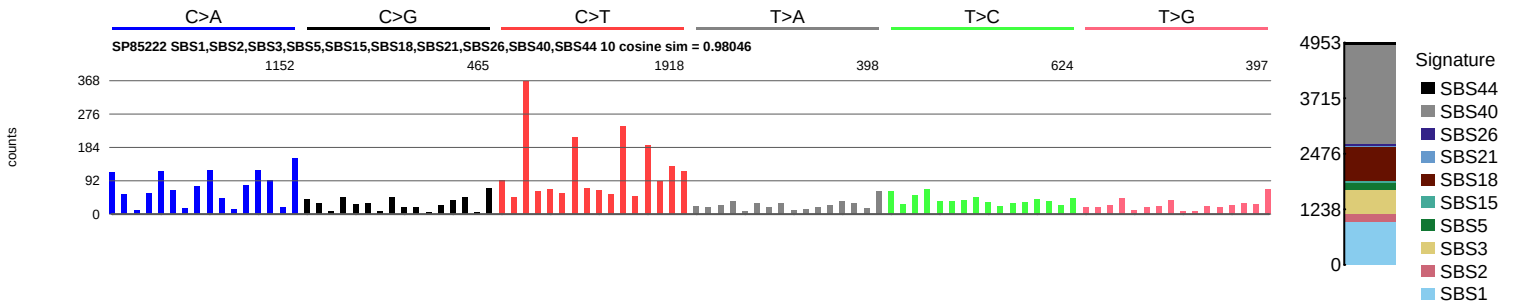
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98047
1146
425
1937
375
638
431
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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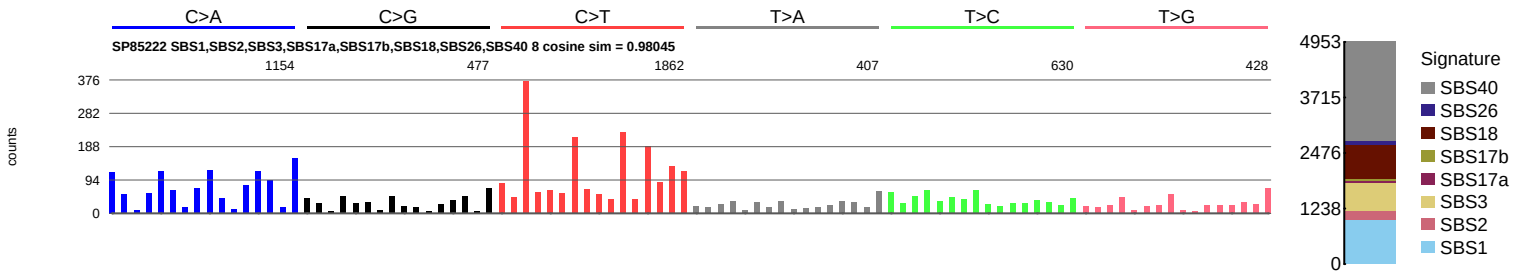
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98046
1152
465
1918
398
624
397
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS15
SBS5
SBS3
SBS2
SBS1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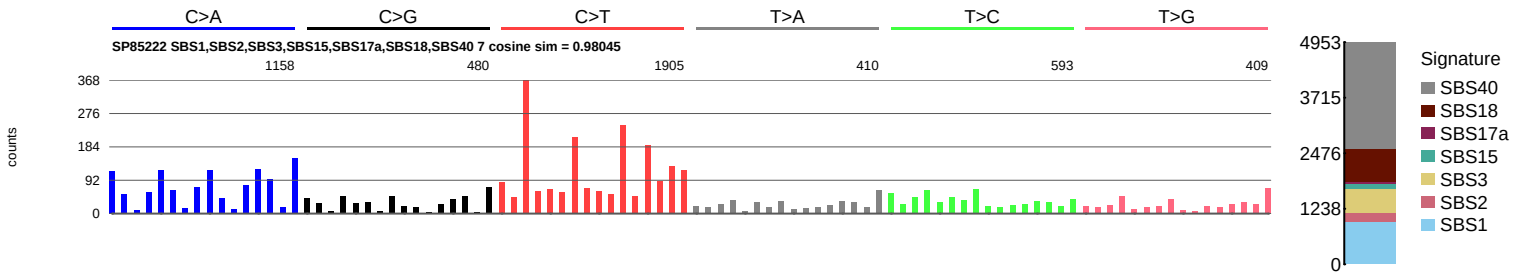
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98045
1154
477
1862
407
630
428
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
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T>C
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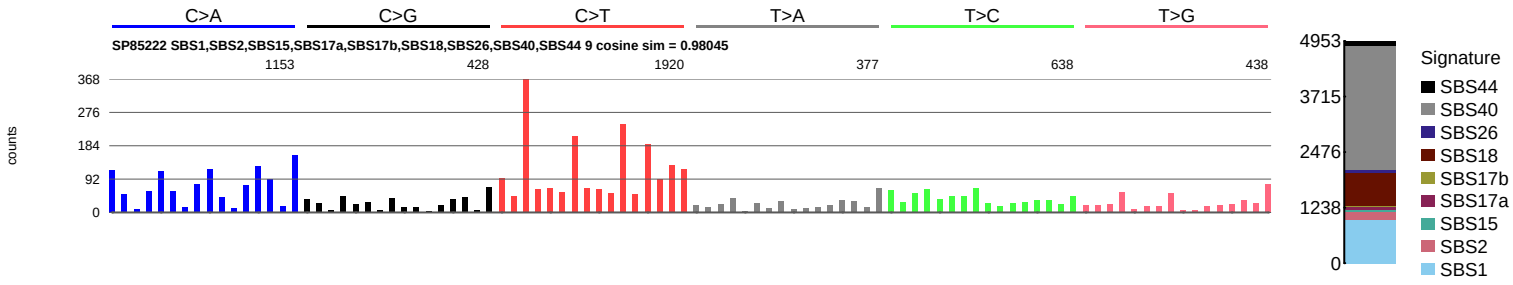
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS40 7 cosine sim = 0.98045
1158
480
1905
410
593
409
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1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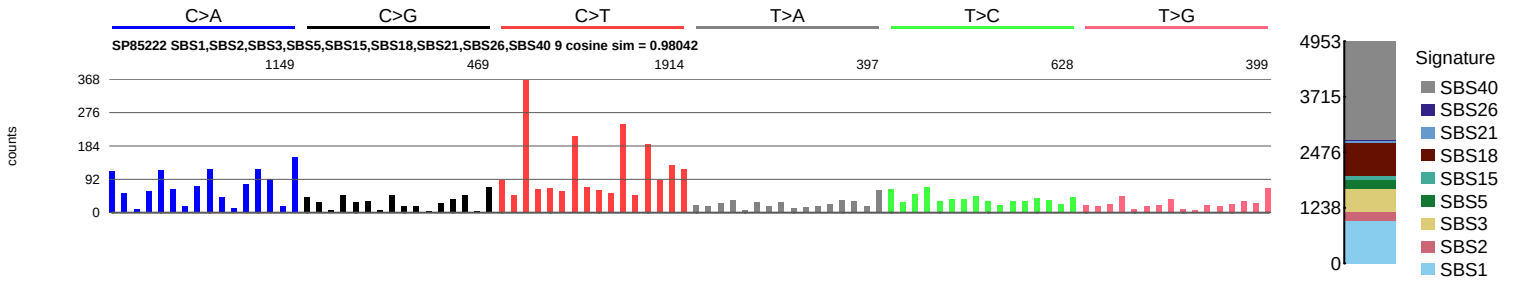
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98045
1153
428
1920
377
638
438
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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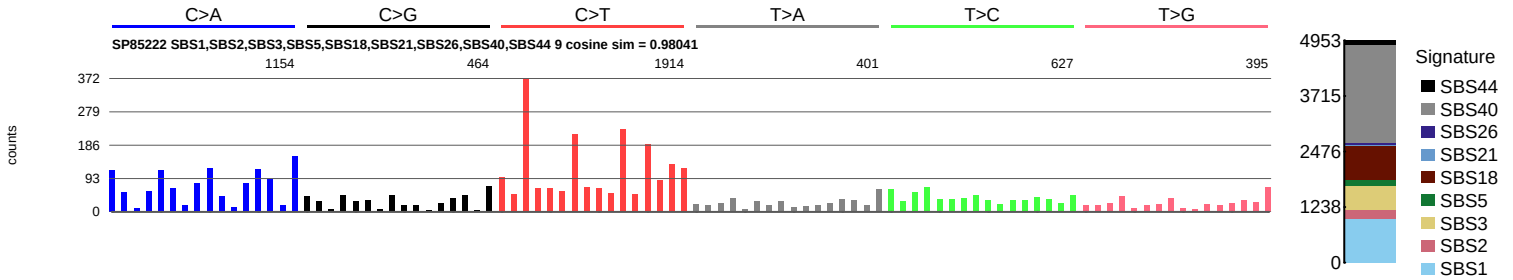
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98042
1149
469
1914
397
628
399
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS15
SBS5
SBS3
SBS2
SBS1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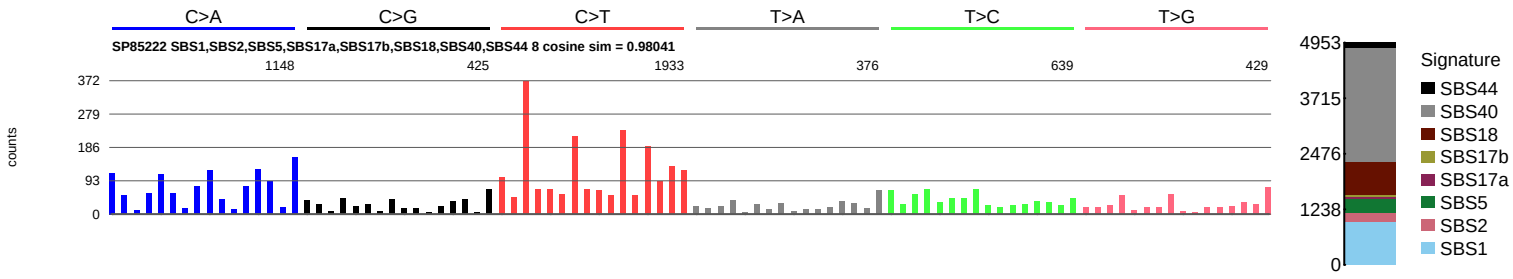
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98041
1154
464
1914
401
627
395
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS5
SBS3
SBS2
SBS1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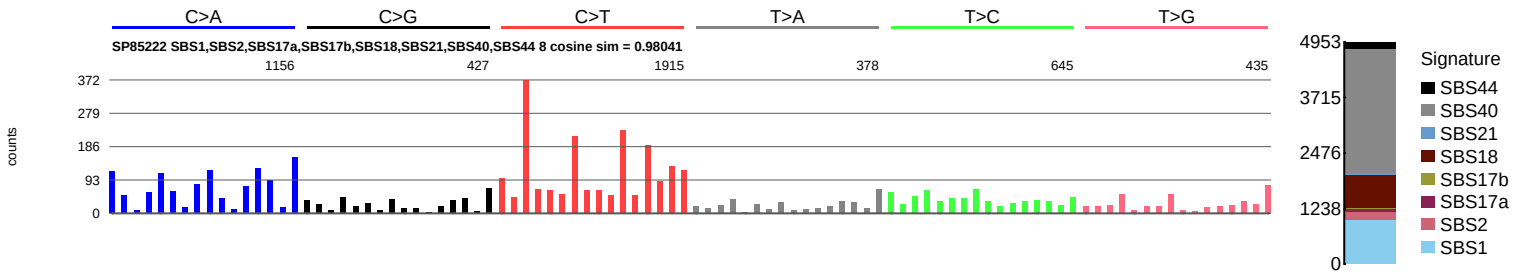
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98041
1148
425
1933
376
639
429
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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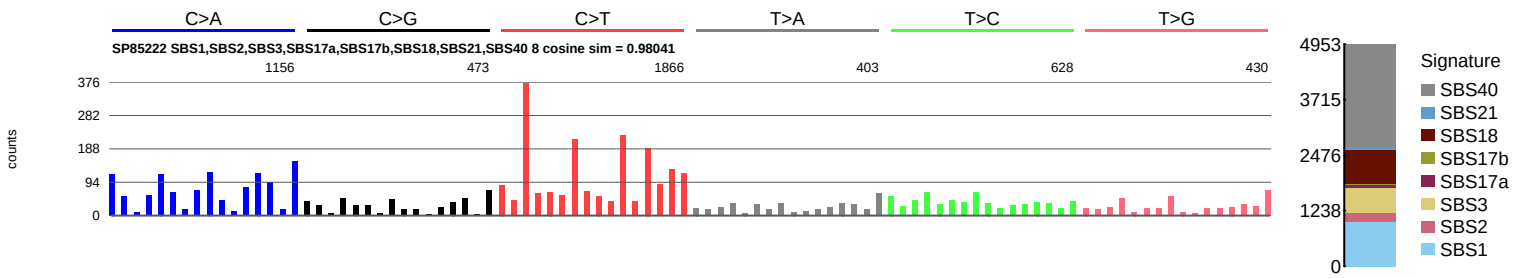
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98041
1156
427
1915
378
645
435
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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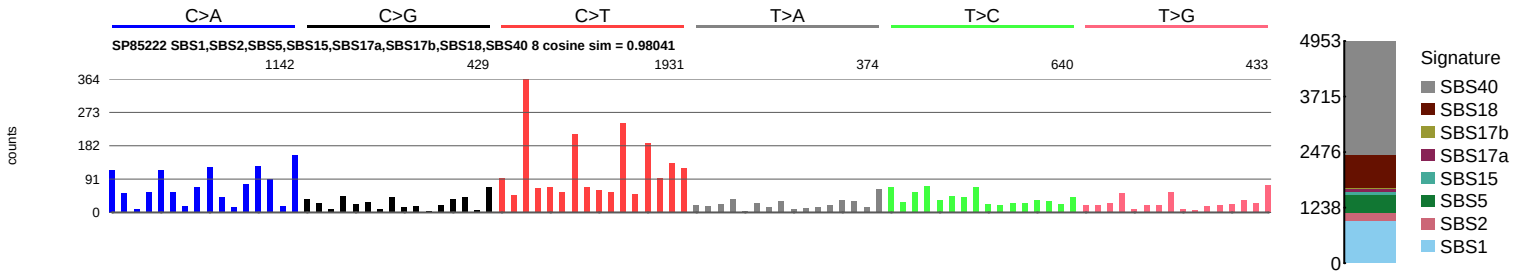
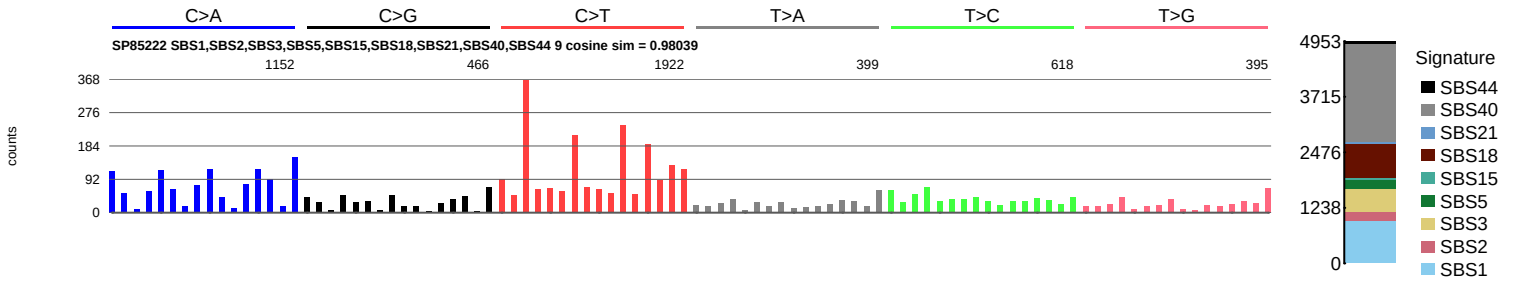
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98041
1156
473
1866
403
628
430
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98041
1142
429
1931
374
640
433
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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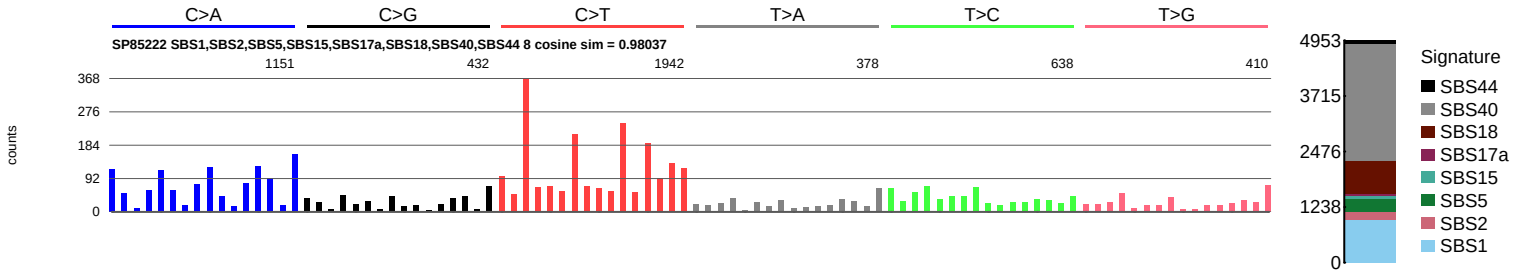
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98039
1152
466
1922
399
618
395
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS15
SBS5
SBS3
SBS2
SBS1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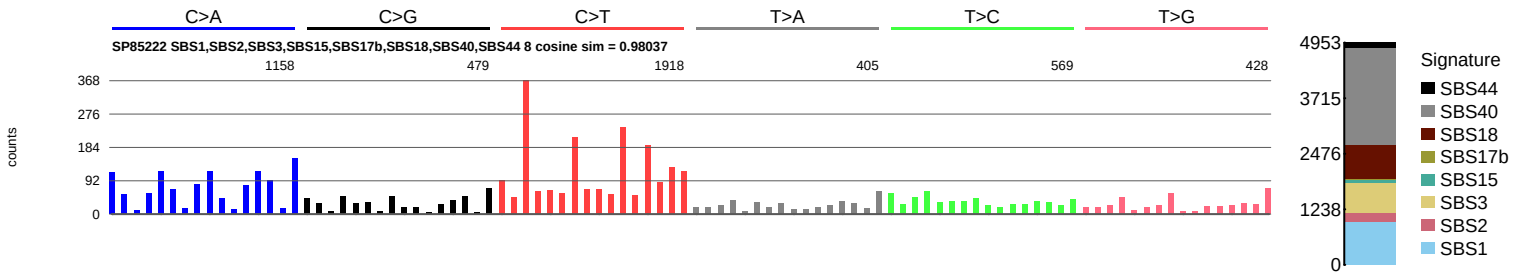
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.98037
1151
432
1942
378
638
410
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98037
1158
479
1918
405
569
428
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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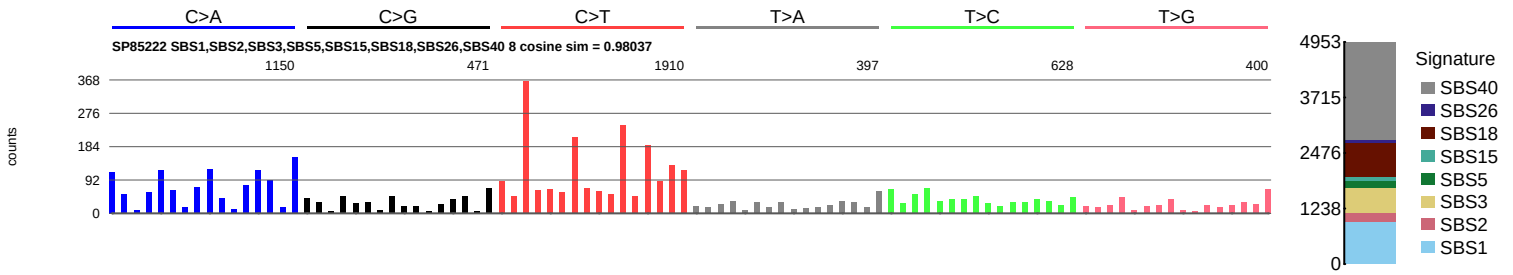
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS26,SBS40 8 cosine sim = 0.98037
1150
471
1910
397
628
400
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
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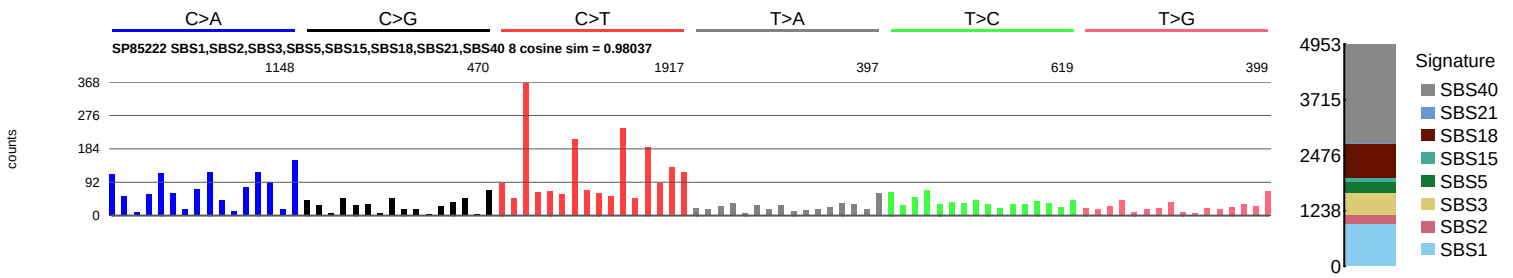
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS21,SBS40 8 cosine sim = 0.98037
1148
470
1917
397
619
399
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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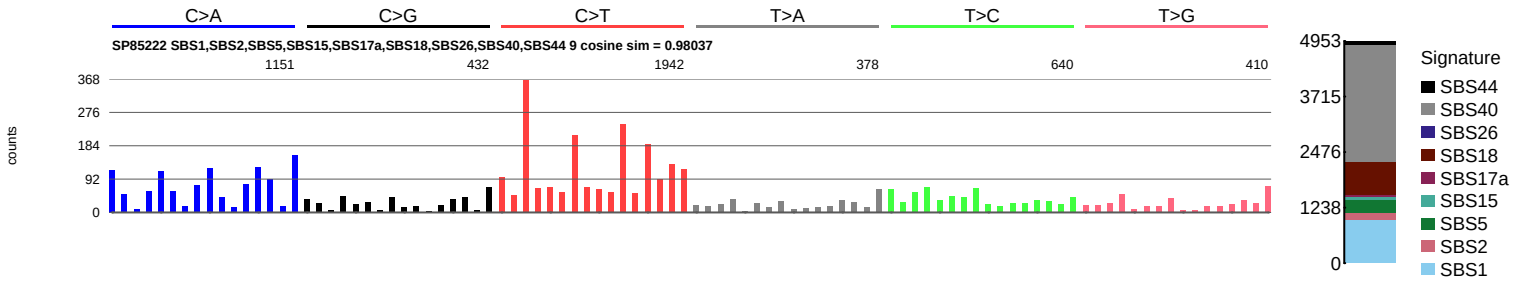
SBS2
SBS1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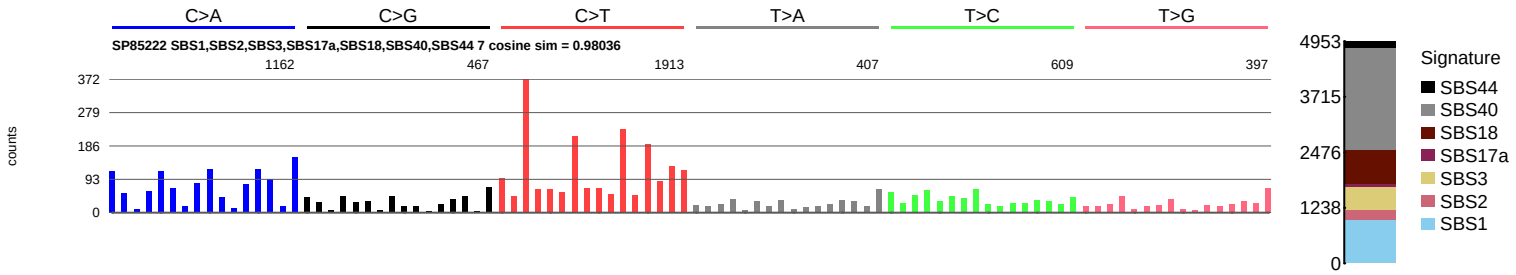
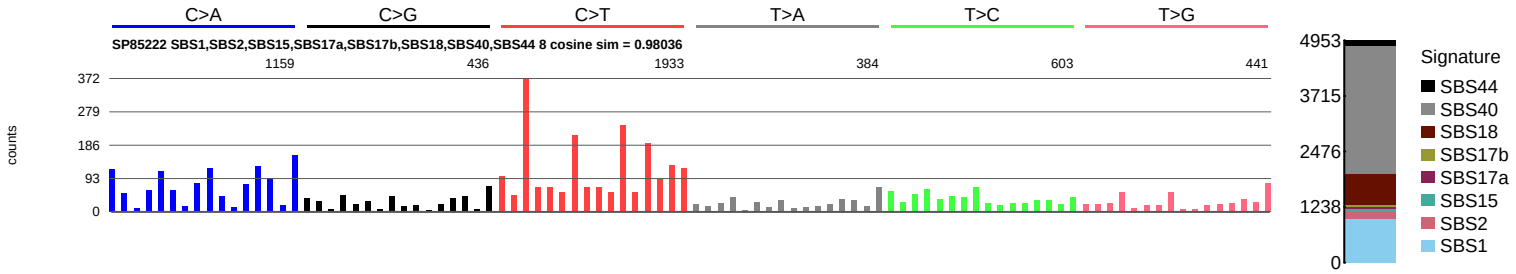
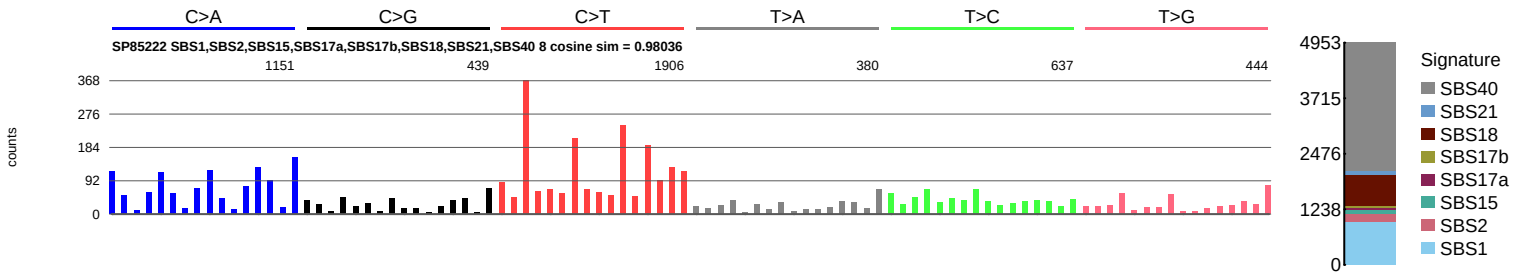
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98037
1151
432
1942
378
640
410
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS40,SBS44 7 cosine sim = 0.98036
1162
467
1913
407
609
397
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98036
1159
436
1933
384
603
441
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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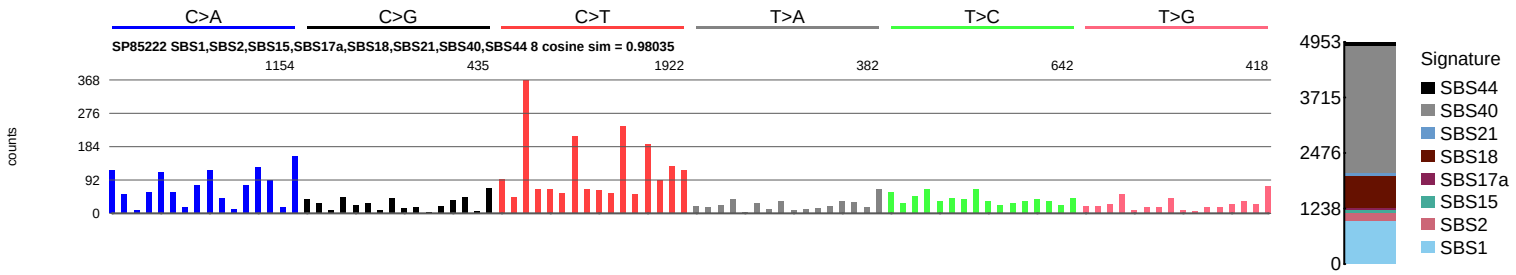
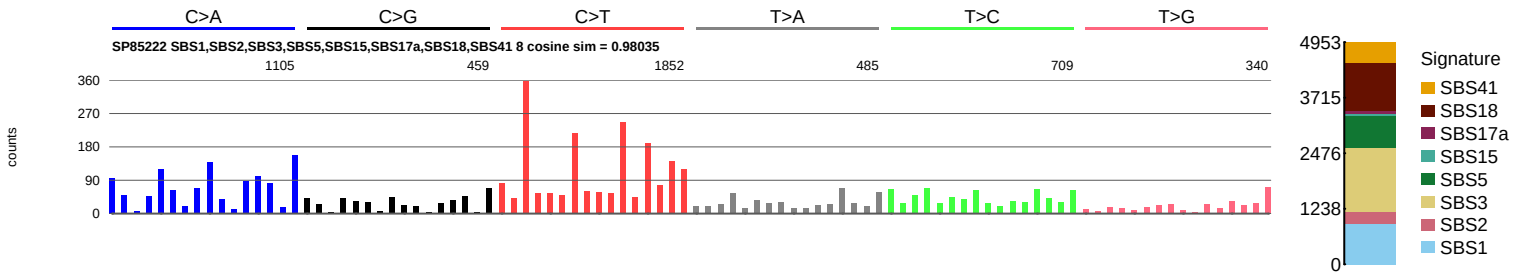
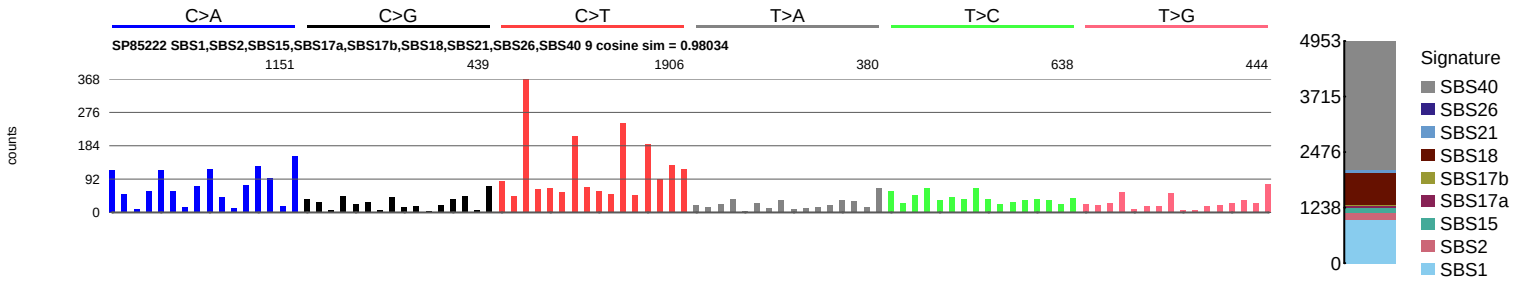
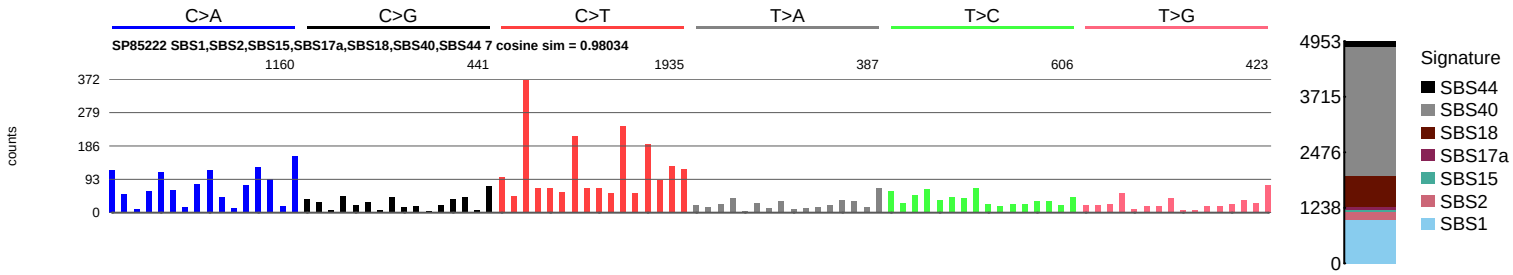
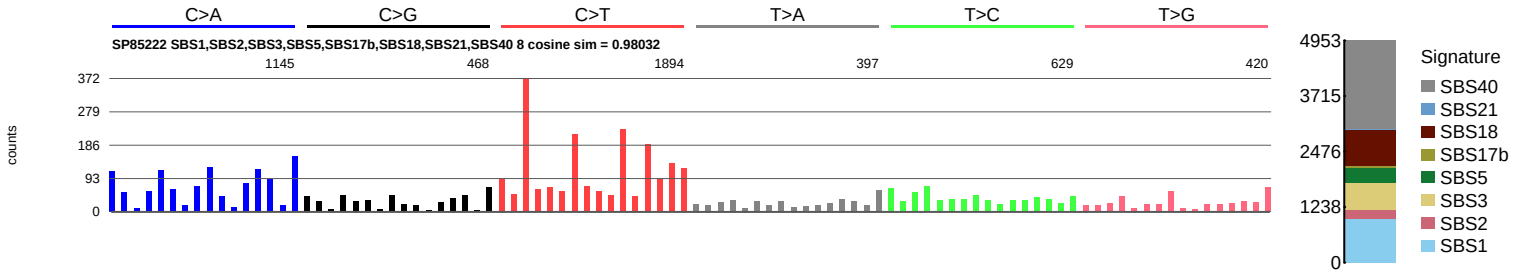
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98036
1151
439
1906
380
637
444
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

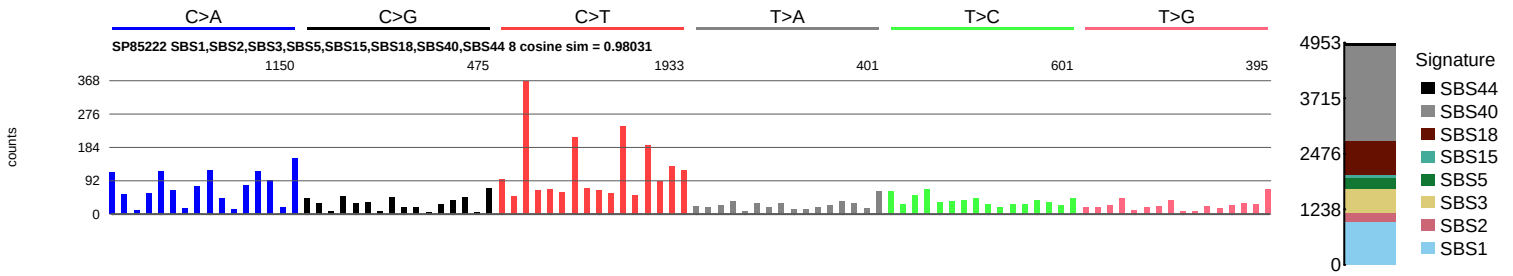

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98035
1154
435
1922
382
642
418
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS41 8 cosine sim = 0.98035
1105
459
1852
485
709
340
counts
360
270
180
90
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98034
1151
439
1906
380
638
444
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS40,SBS44 7 cosine sim = 0.98034
1160
441
1935
387
606
423
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98032
1145
468
1894
397
629
420
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS40,SBS44 8 cosine sim = 0.98031
1150
475
1933
401
601
395
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS5
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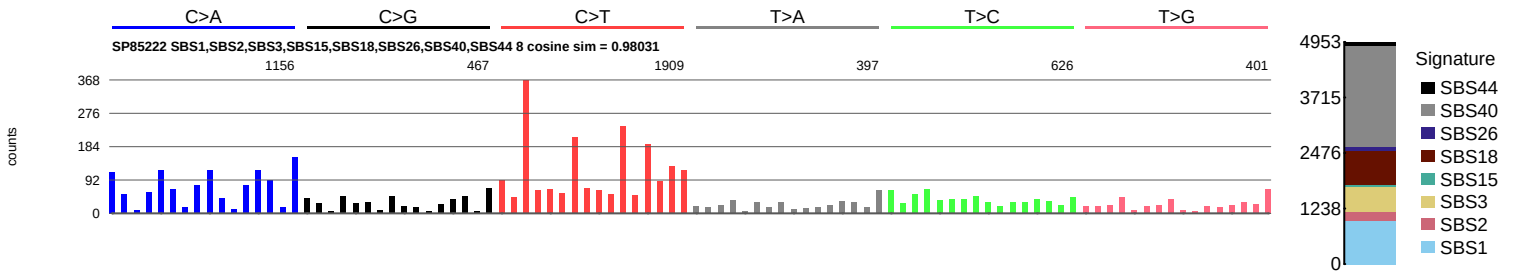
SBS2
SBS1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SP85222 SBS1,SBS2,SBS3,SBS15,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98031
1156
467
1909
397
626
401
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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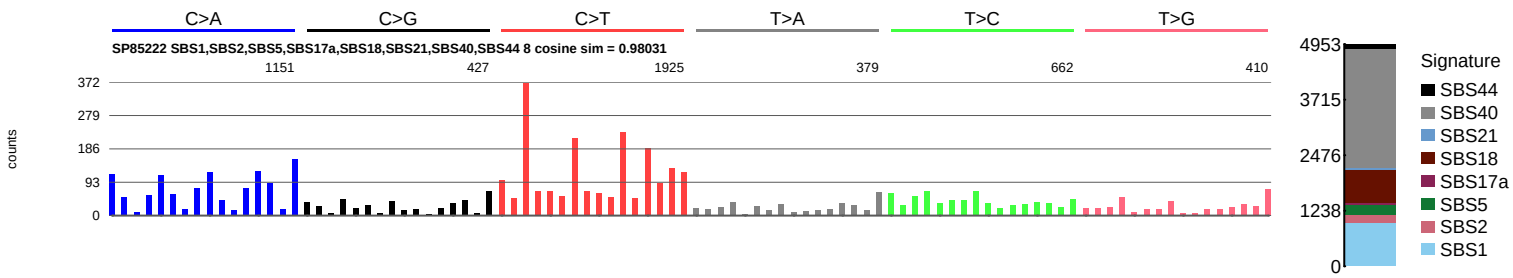
SBS2
SBS1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SP85222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98031
1151
427
1925
379
662
410
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
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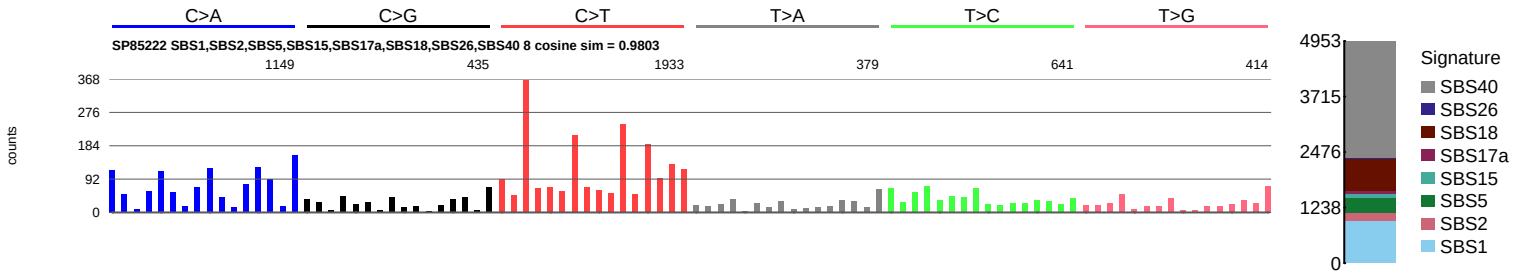
SBS21
SBS18
SBS17a
SBS5
SBS2
SBS1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1149
435
1933
379
641
414
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
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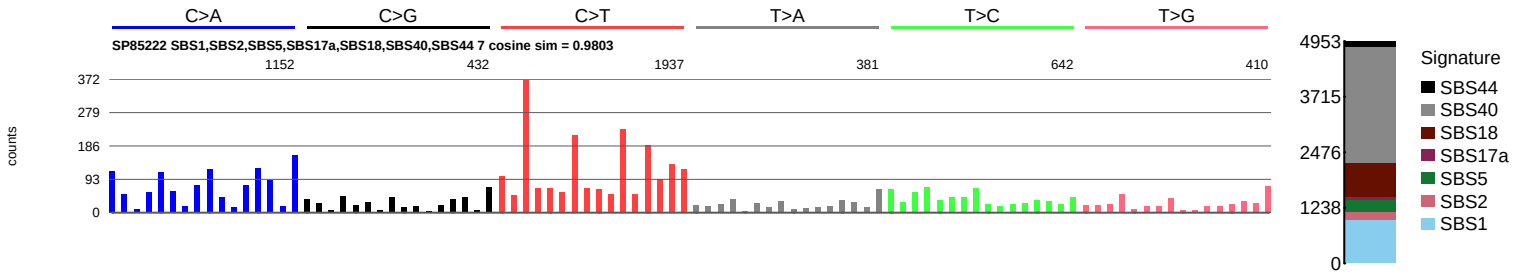
SBS2
SBS1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C>G
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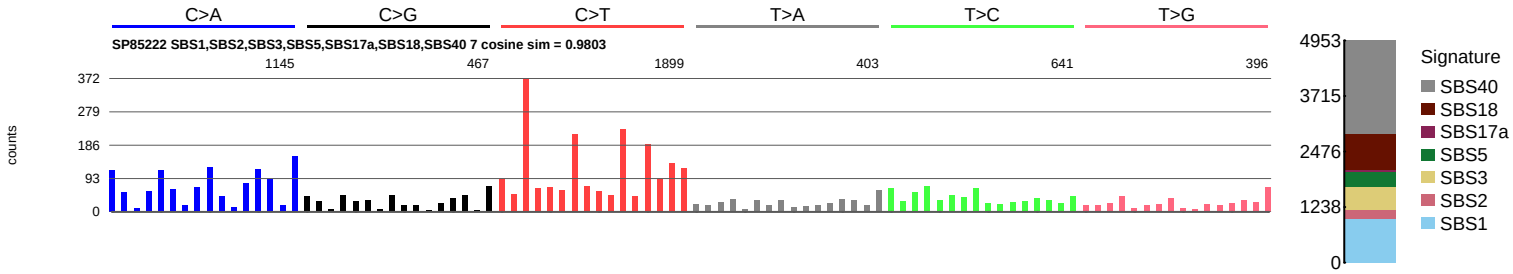
SP85222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS40,SBS44 7 cosine sim = 0.9803
1152
432
1937
381
642
410
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS5
SBS2
SBS1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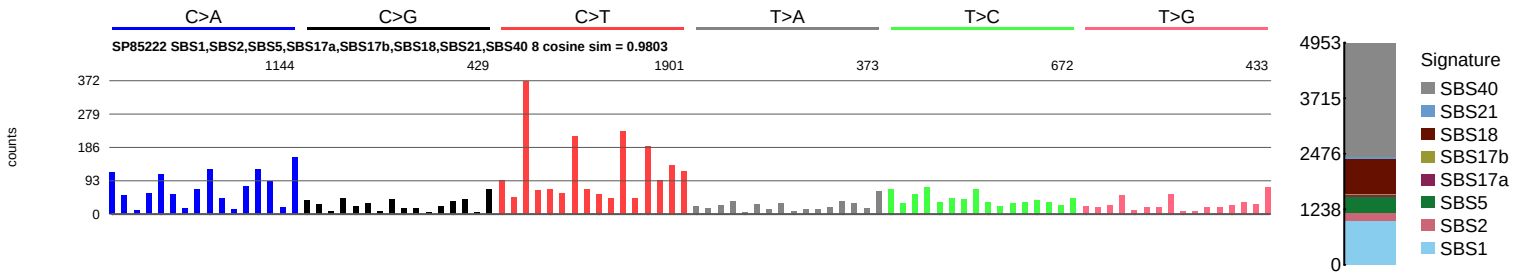
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS40 7 cosine sim = 0.9803
1145
467
1899
403
641
396
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
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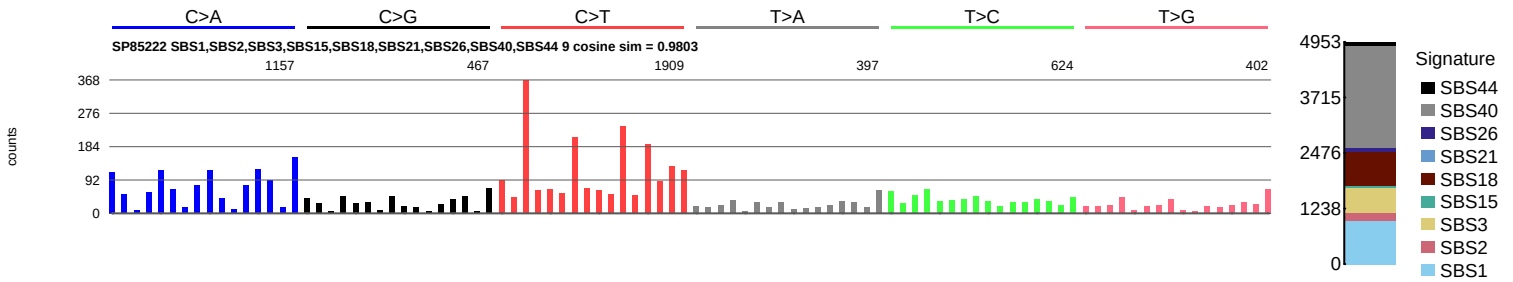
T>G
SP85222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.9803
1144
429
1901
373
672
433
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
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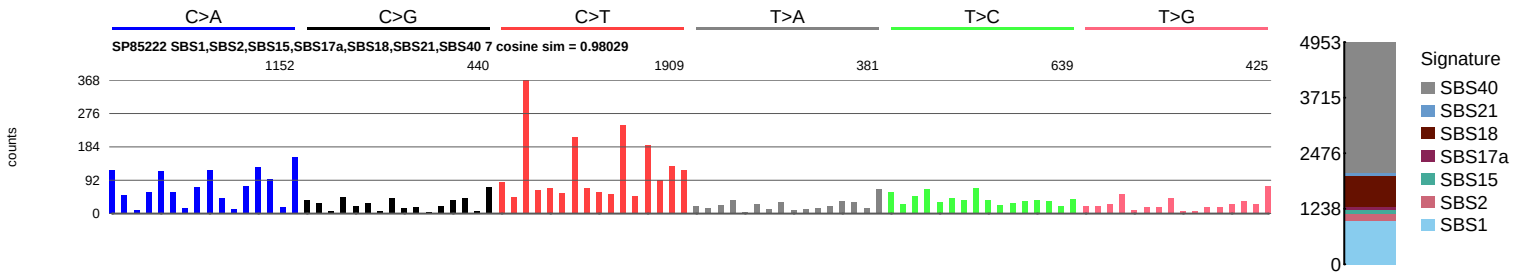
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.9803
1157
467
1909
397
624
402
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS15
SBS3
SBS2
SBS1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C>G
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T>A
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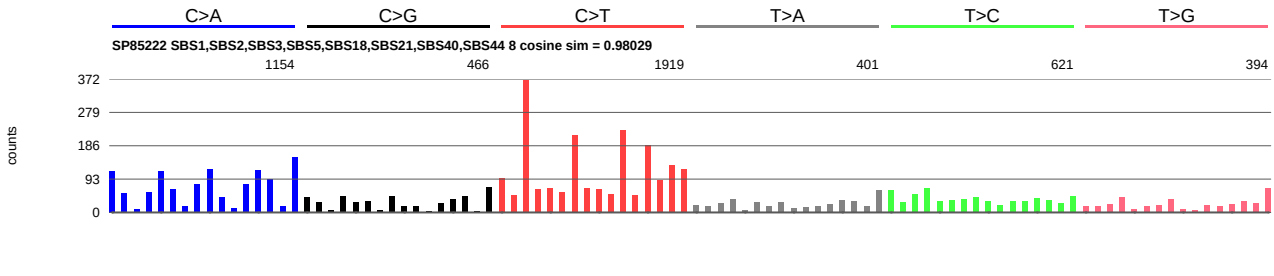
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS40 7 cosine sim = 0.98029
1152
440
1909
381
639
425
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98029
1154
466
1919
401
621
394
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
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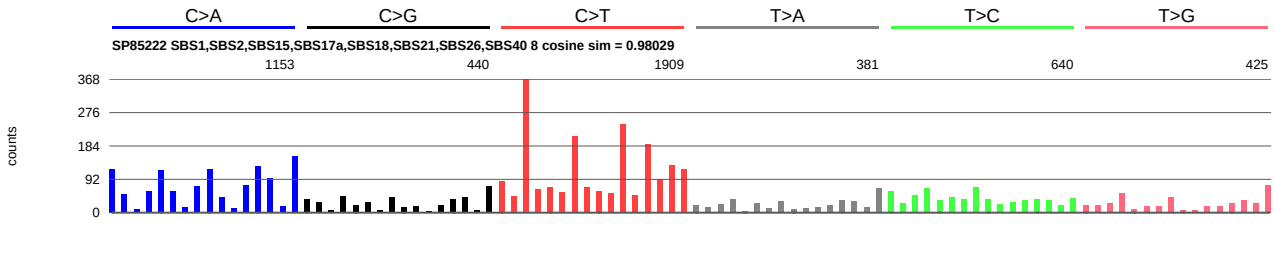
SBS2
SBS1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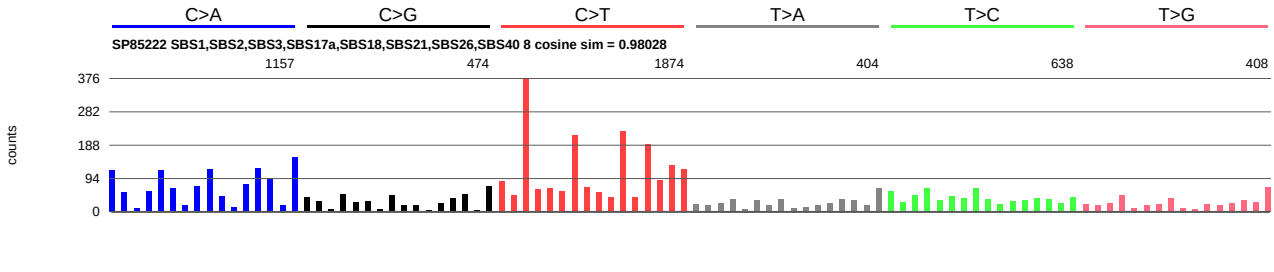
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98029
1153
440
1909
381
640
425
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98028
1157
474
1874
404
638
408
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
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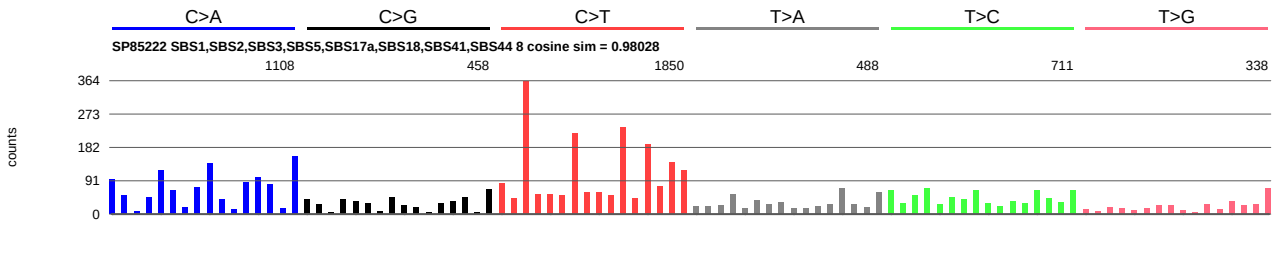
SBS2
SBS1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C>G
C>T
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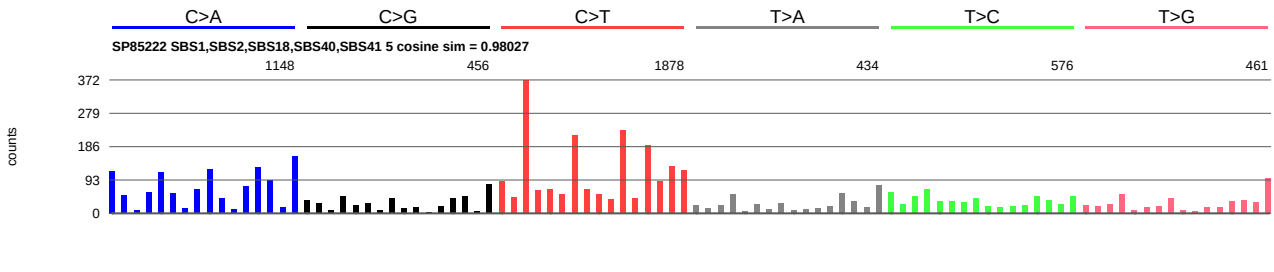
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS41,SBS44 8 cosine sim = 0.98028
1108
458
1850
488
711
338
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
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T>G
SP85222 SBS1,SBS2,SBS18,SBS40,SBS41 5 cosine sim = 0.98027
1148
456
1878
434
576
461
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
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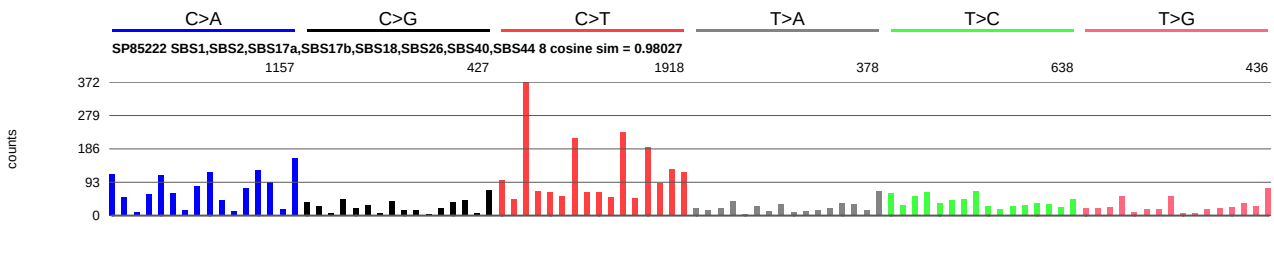
SBS18
SBS2
SBS1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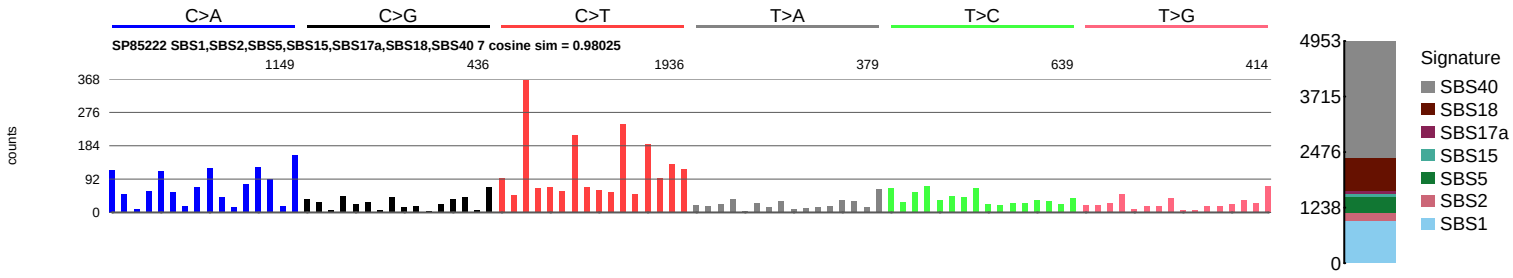
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98027
1157
427
1918
378
638
436
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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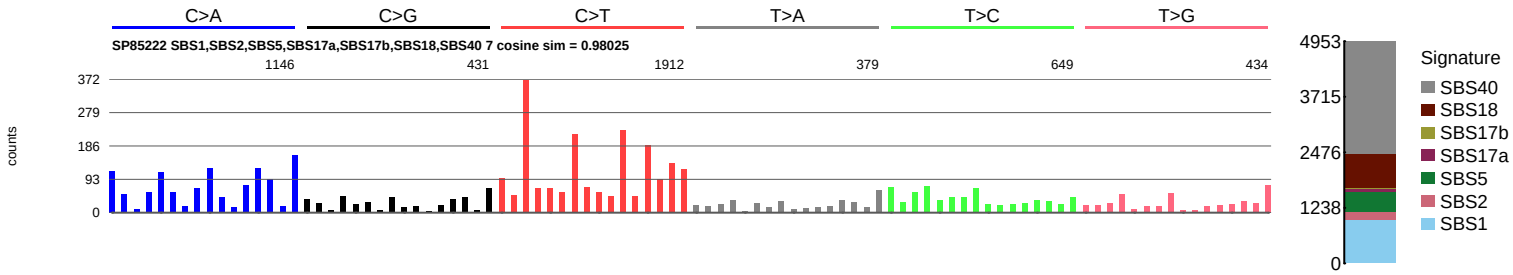
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS40 7 cosine sim = 0.98025
1149
436
1936
379
639
414
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
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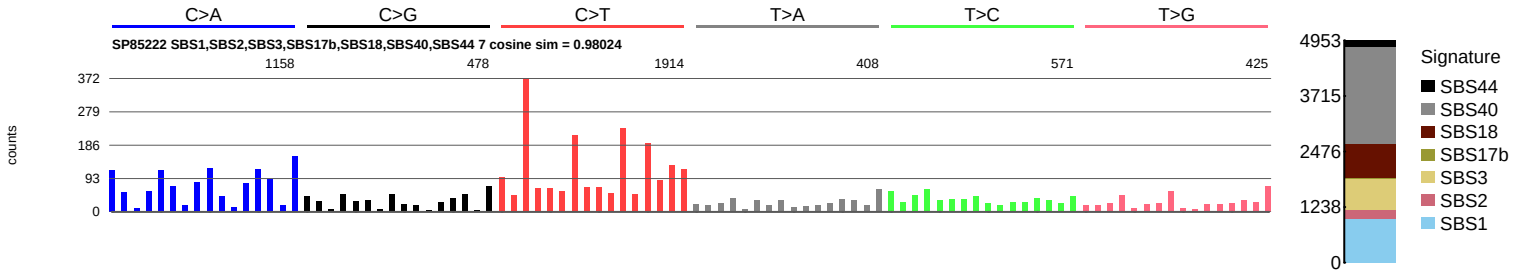
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98025
1146
431
1912
379
649
434
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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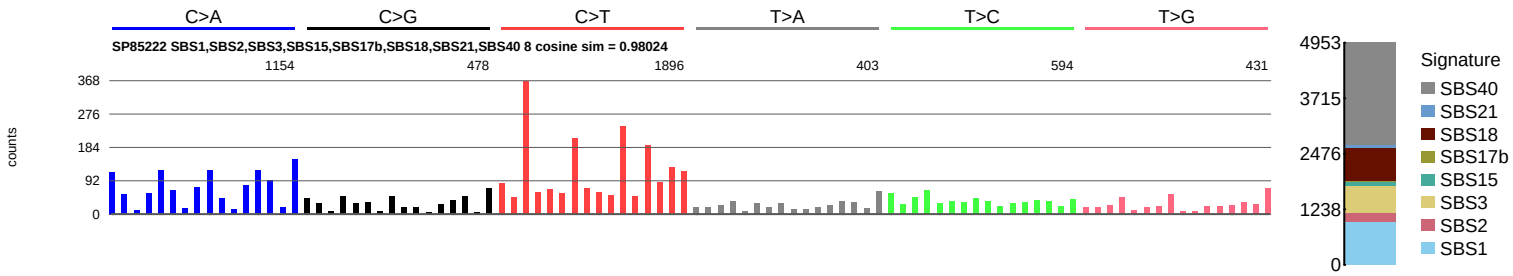
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98024
1158
478
1914
408
571
425
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98024
1154
478
1896
403
594
431
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
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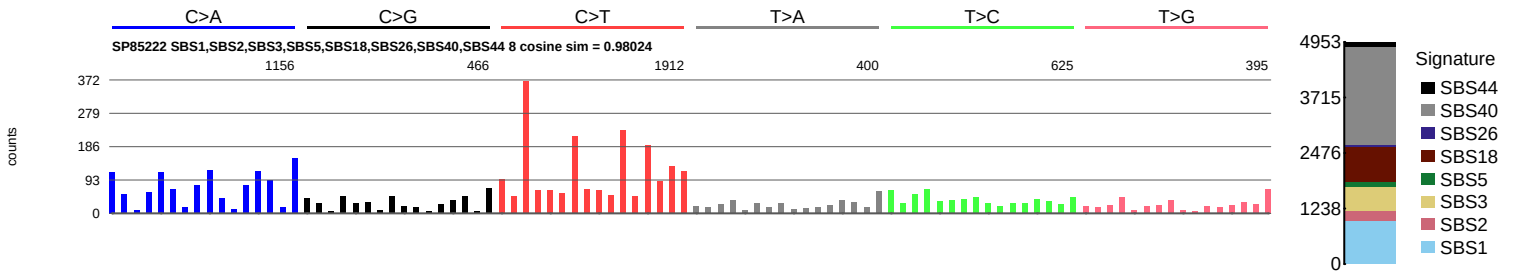
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98024
1156
466
1912
400
625
395
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
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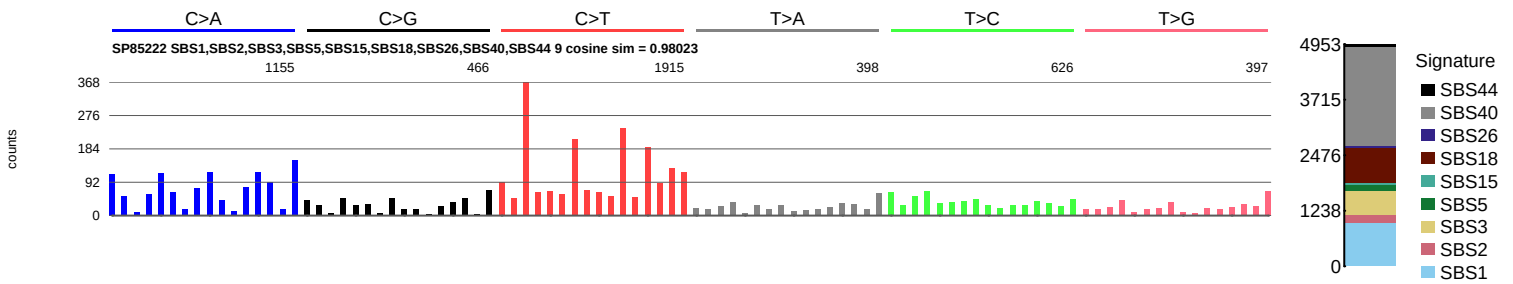
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98023
1155
466
1915
398
626
397
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS15
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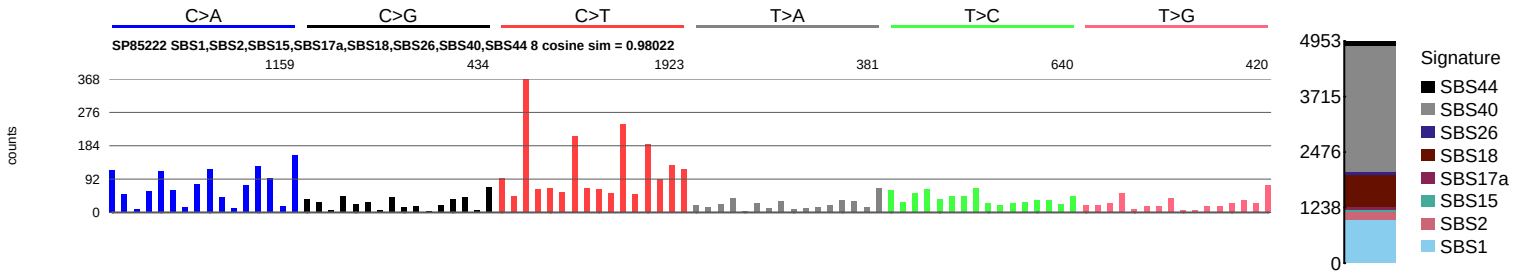
SBS3
SBS2
SBS1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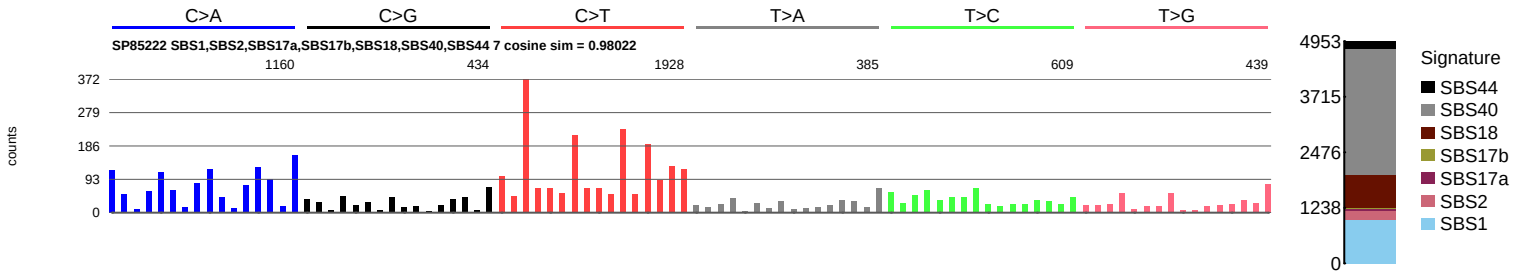
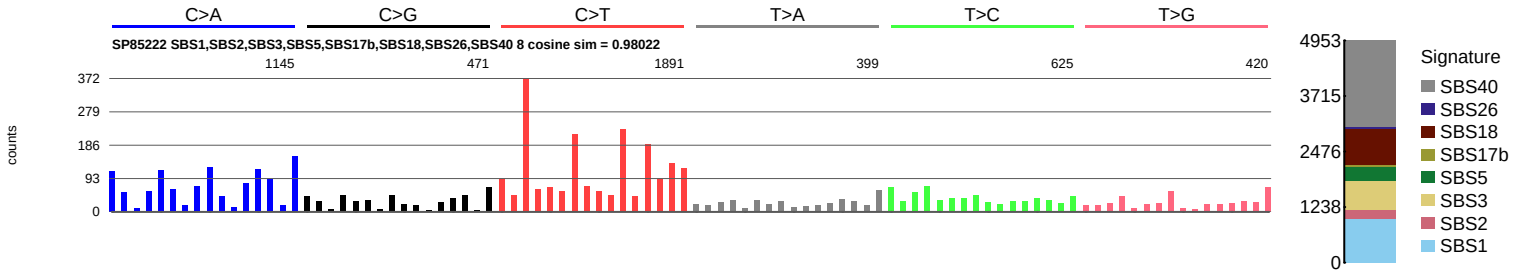
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98022
1159
434
1923
381
640
420
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98022
1160
434
1928
385
609
439
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98022
1145
471
1891
399
625
420
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
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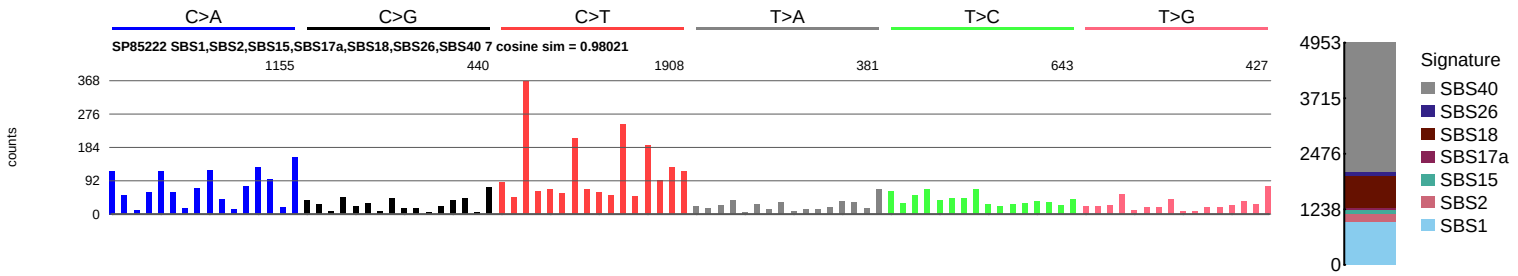
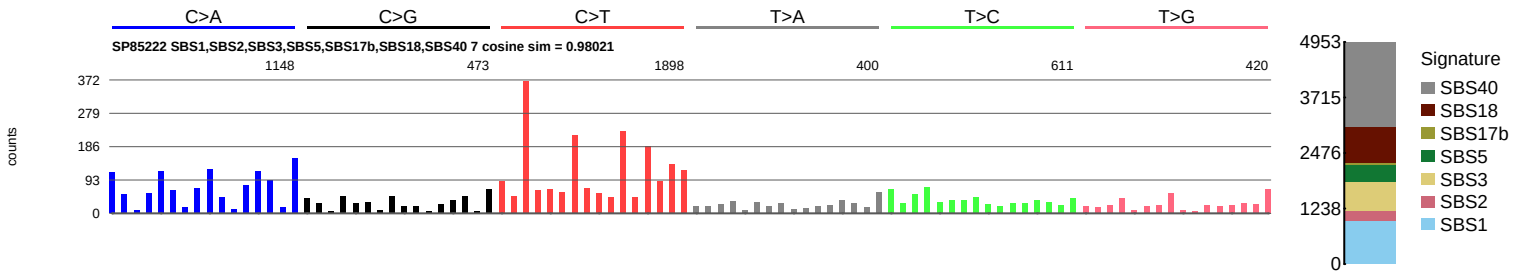
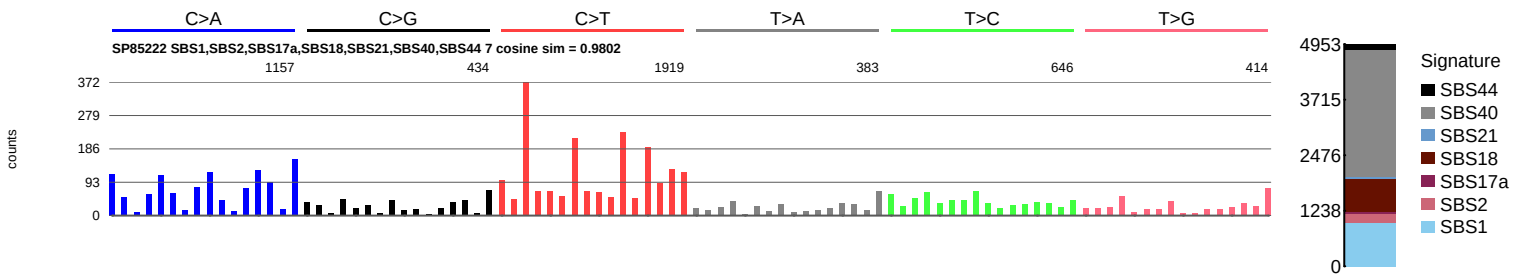
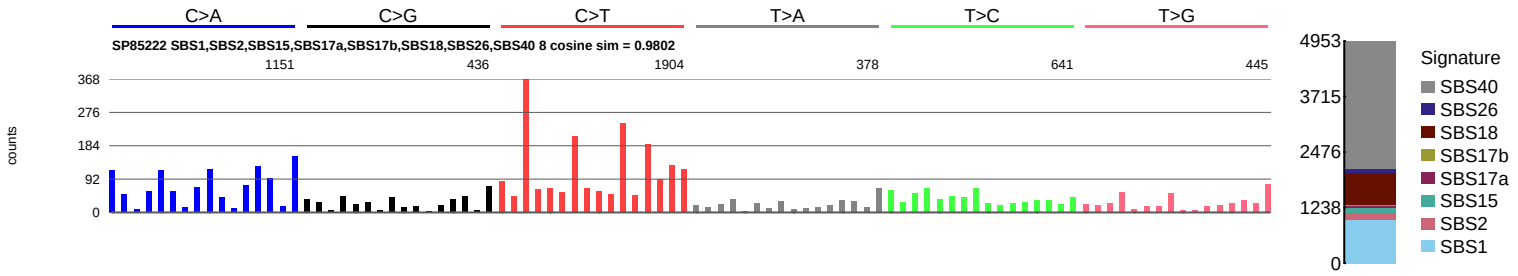
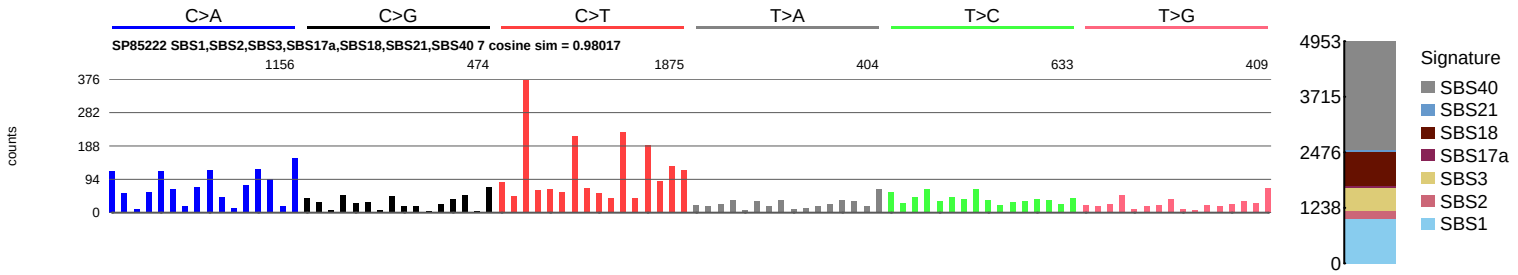
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS40 7 cosine sim = 0.98021
1155
440
1908
381
643
427
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40 7 cosine sim = 0.98021
1148
473
1898
400
611
420
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.9802
1157
434
1919
383
646
414
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.9802
1151
436
1904
378
641
445
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS21,SBS40 7 cosine sim = 0.98017
1156
474
1875
404
633
409
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
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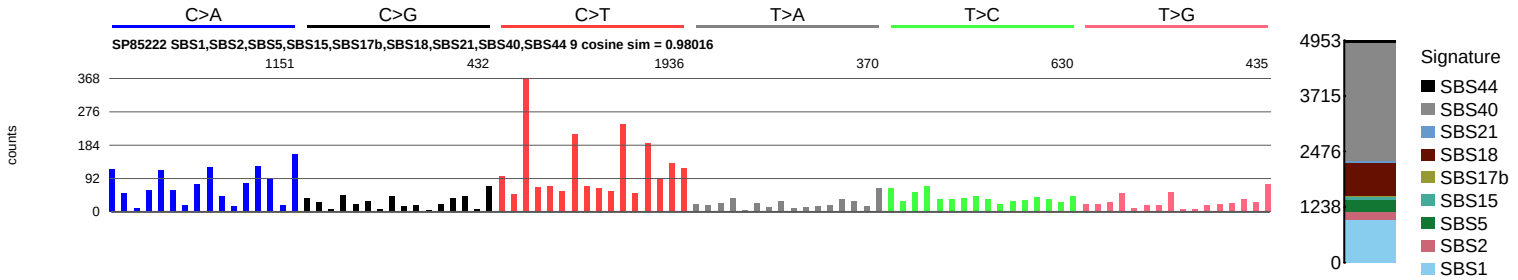
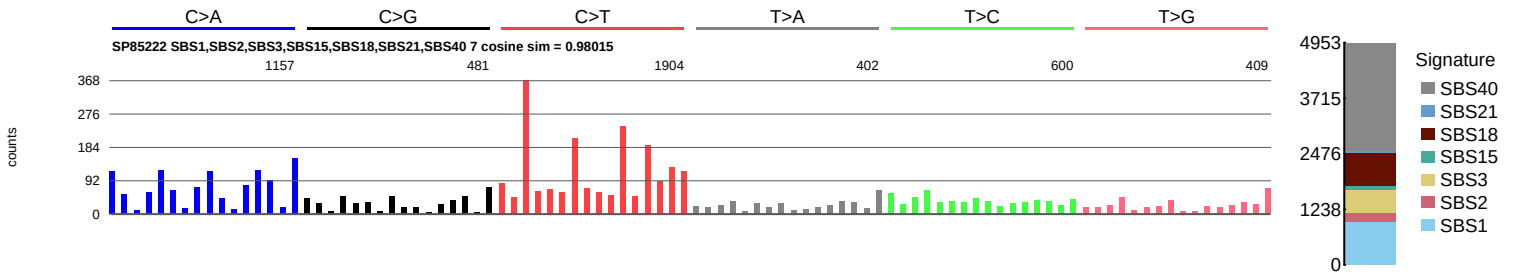
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98016
1151
432
1936
370
630
435
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS18,SBS21,SBS40 7 cosine sim = 0.98015
1157
481
1904
402
600
409
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
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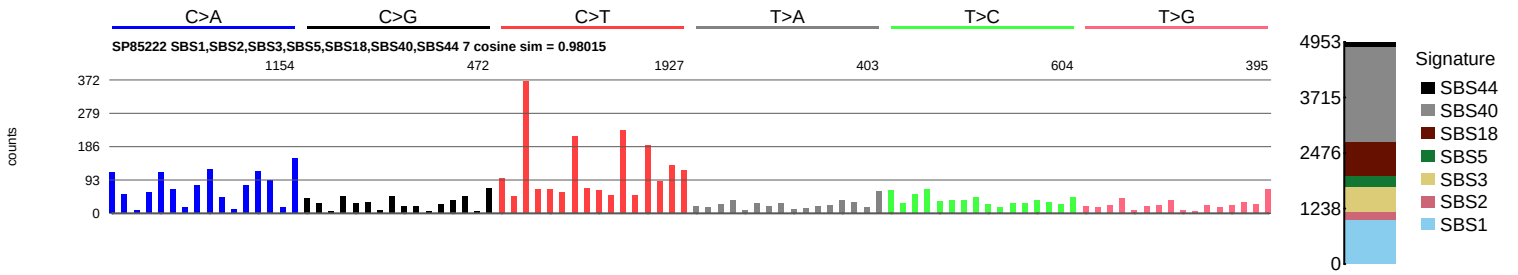
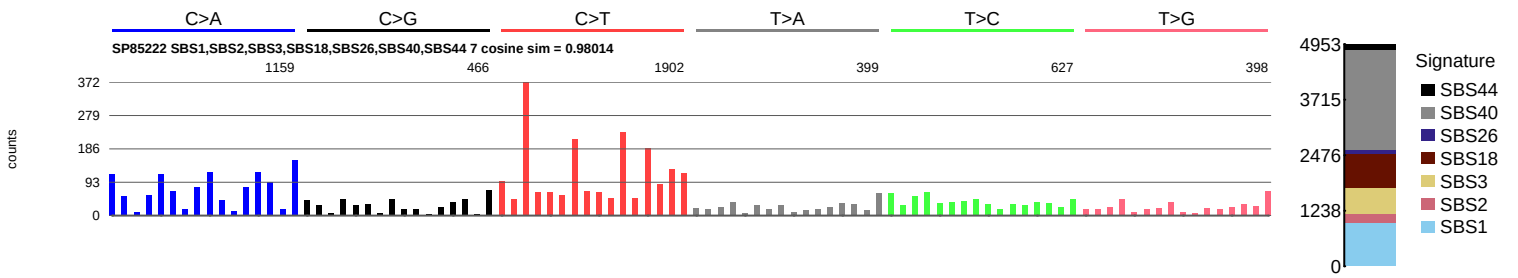
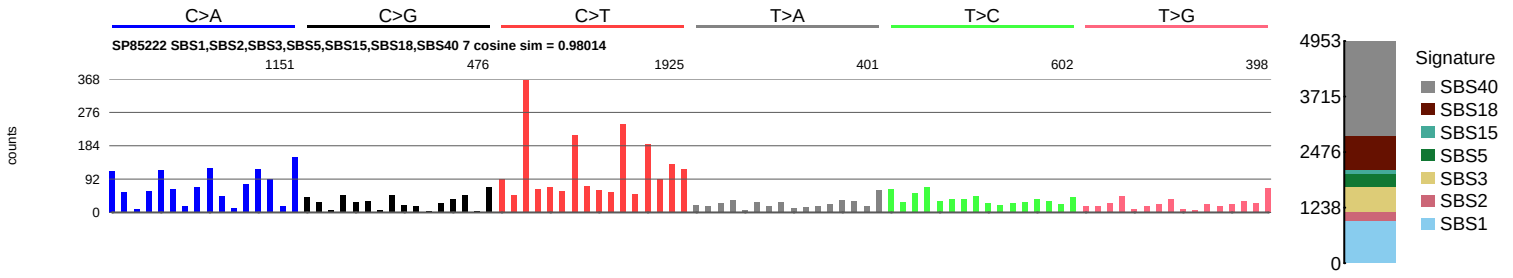
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS18,SBS40,SBS44 7 cosine sim = 0.98015
1154
472
1927
403
604
395
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98014
1159
466
1902
399
627
398
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS40 7 cosine sim = 0.98014
1151
476
1925
401
602
398
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
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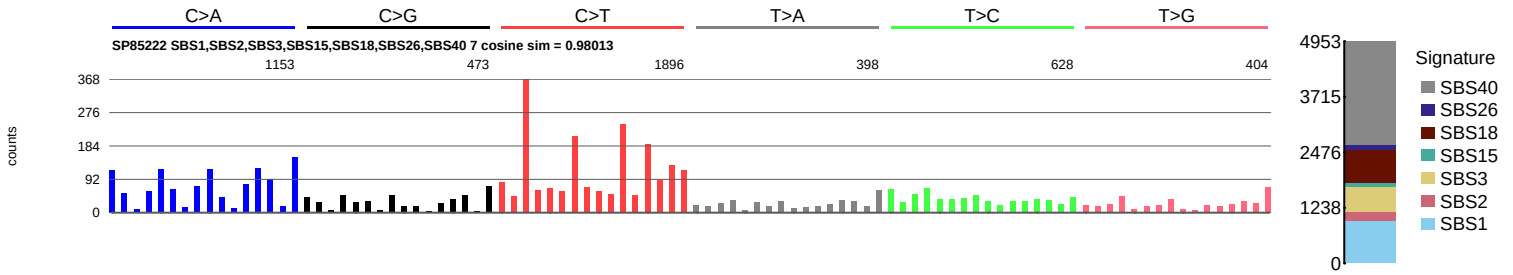
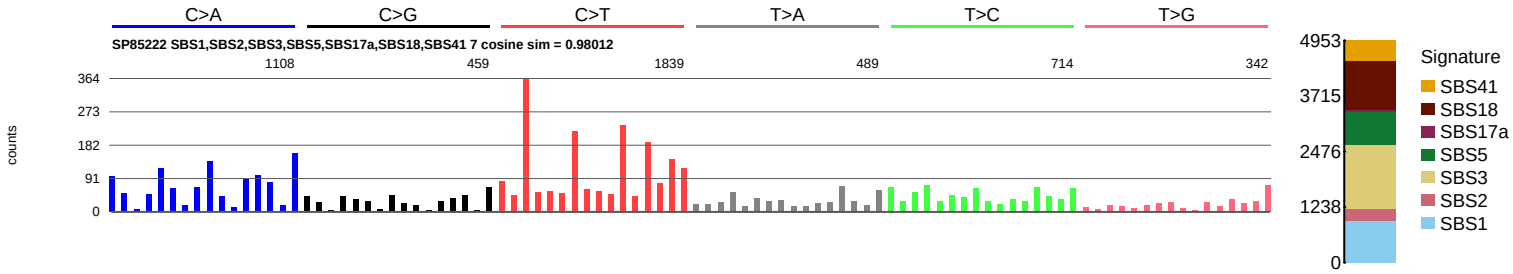
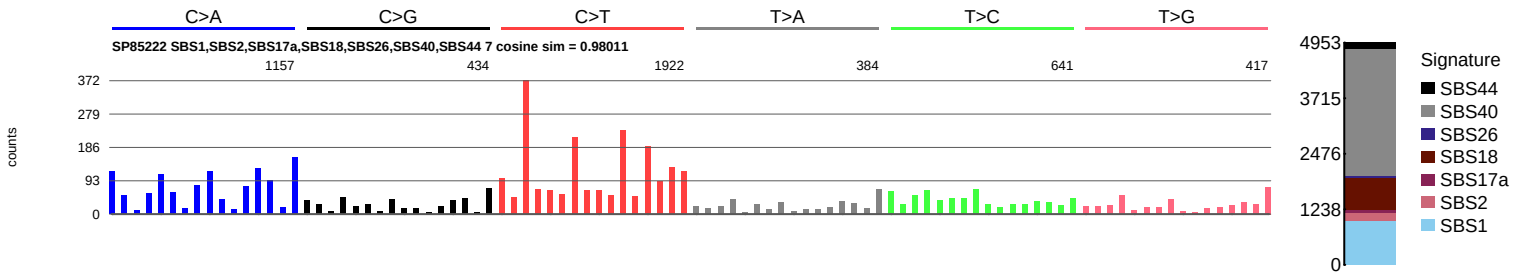
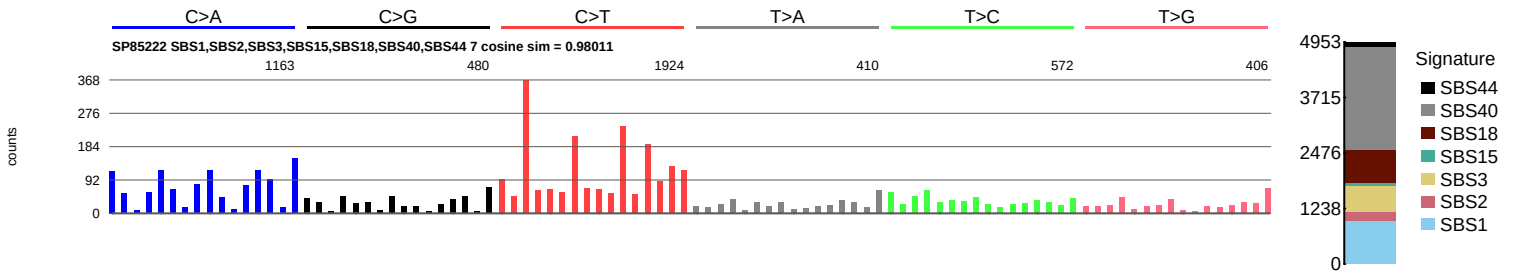
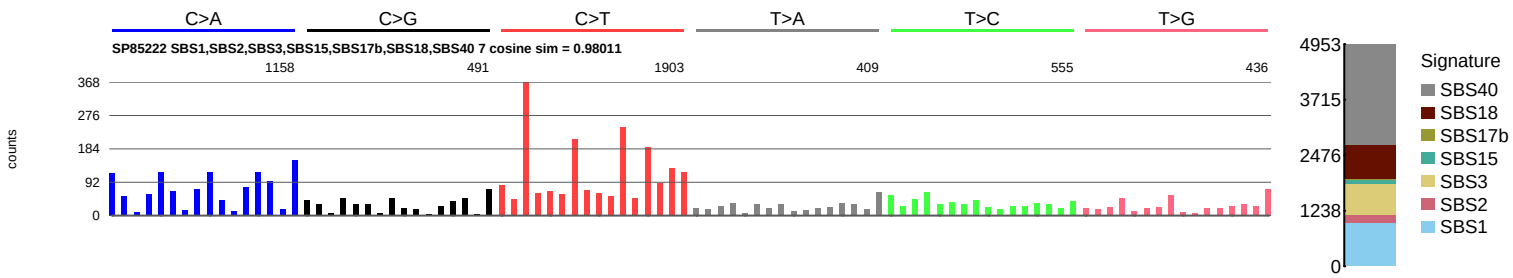
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS18,SBS26,SBS40 7 cosine sim = 0.98013
1153
473
1896
398
628
404
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS41 7 cosine sim = 0.98012
1108
459
1839
489
714
342
counts
364
273
182
91
0
4953
3715
2476
1238
0
Signature
SBS41
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98011
1157
434
1922
384
641
417
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS18,SBS40,SBS44 7 cosine sim = 0.98011
1163
480
1924
410
572
406
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS40 7 cosine sim = 0.98011
1158
491
1903
409
555
436
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
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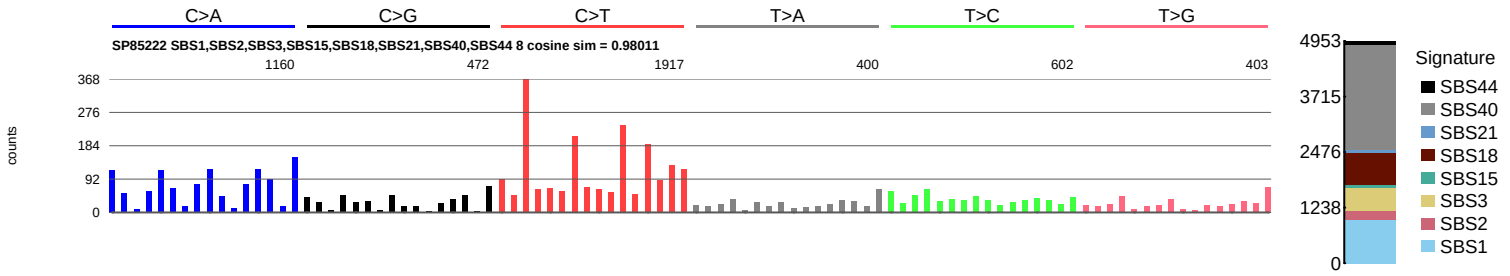
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98011
1160
472
1917
400
602
403
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS44
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SBS21
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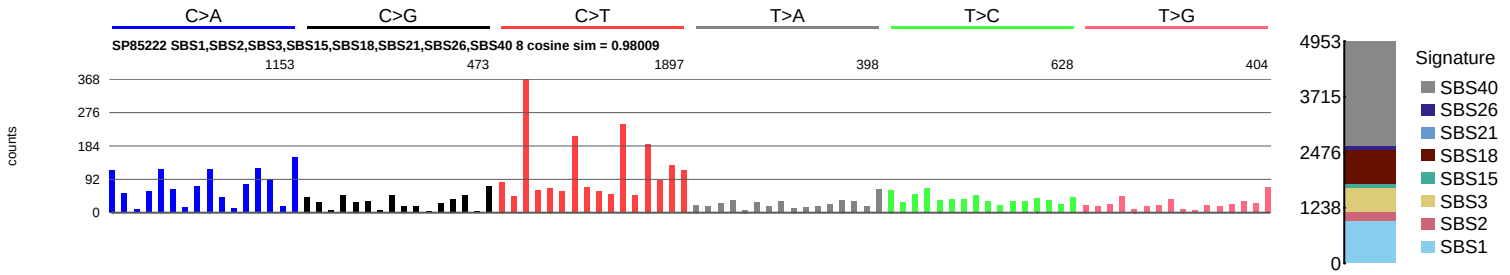
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS15,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98009
1153
473
1897
398
628
404
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
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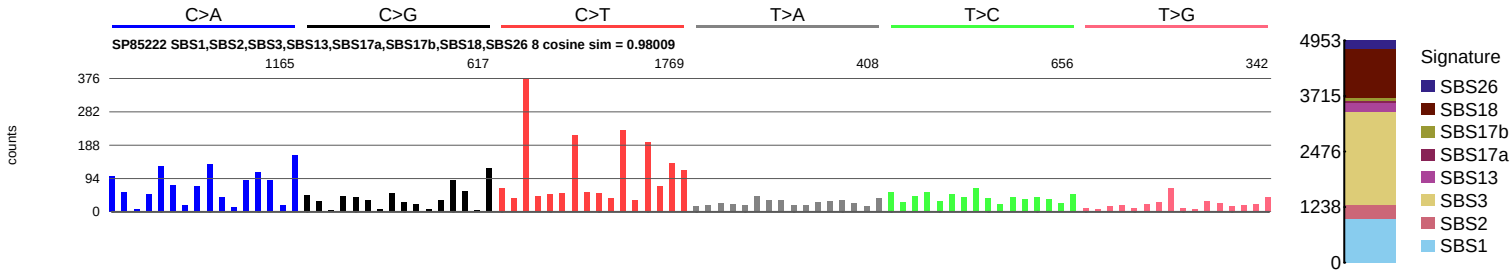
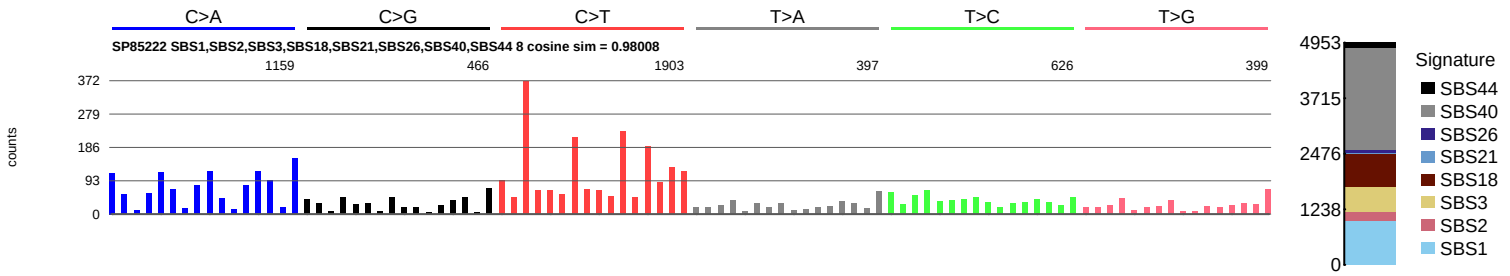
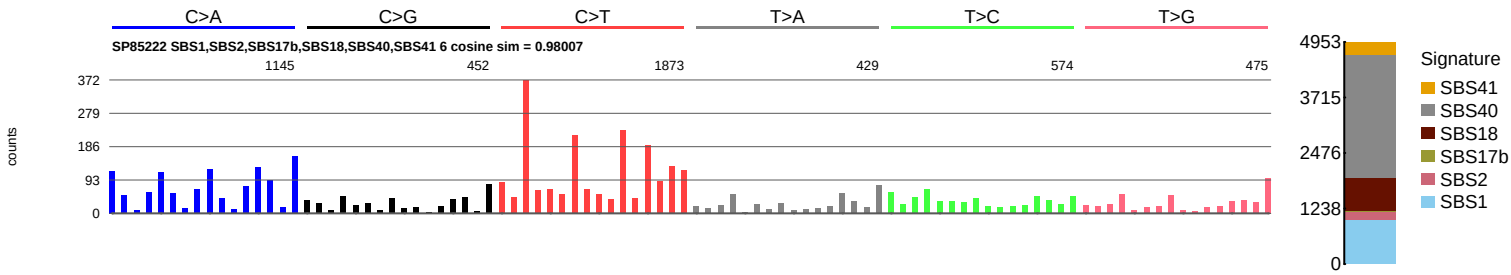
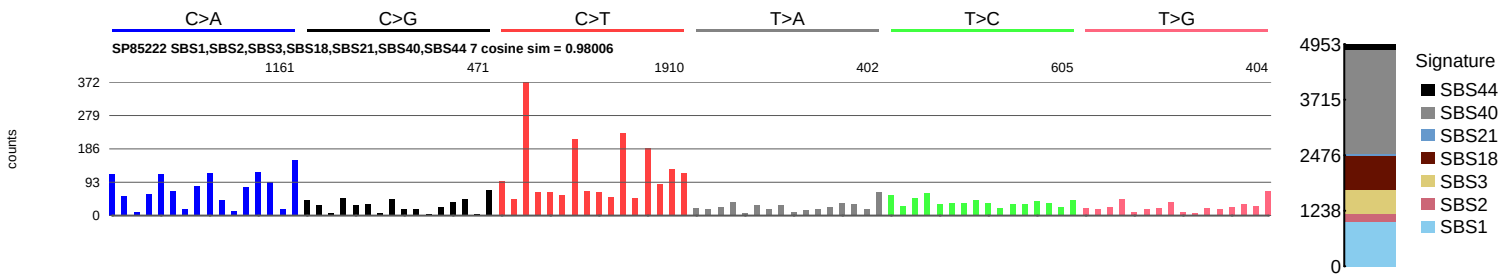
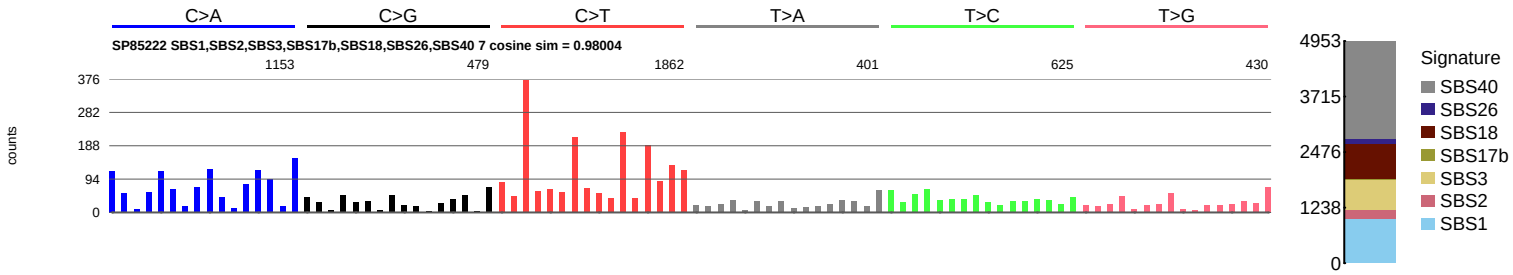
SBS3
SBS2
SBS1

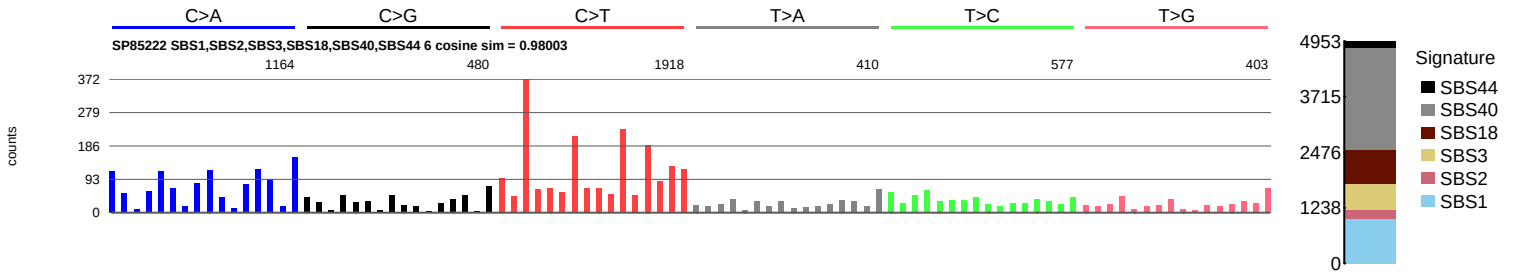

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98009
1165
617
1769
408
656
342
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

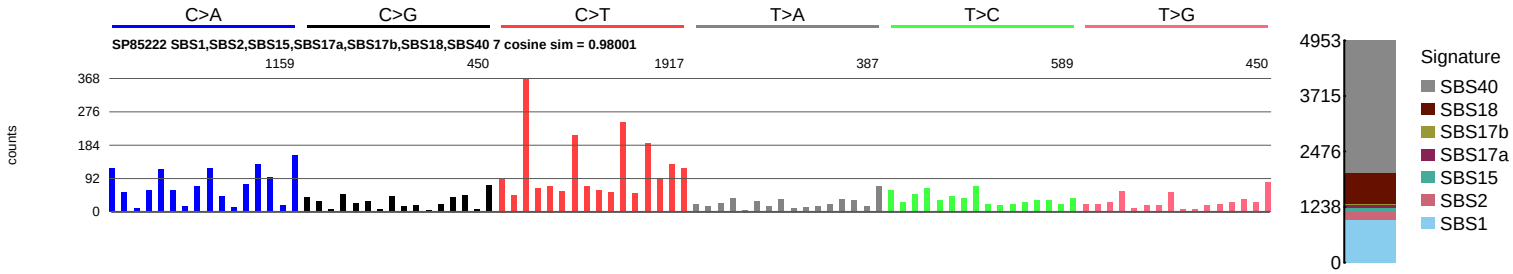

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98008
1159
466
1903
397
626
399
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS3
SBS2
SBS1

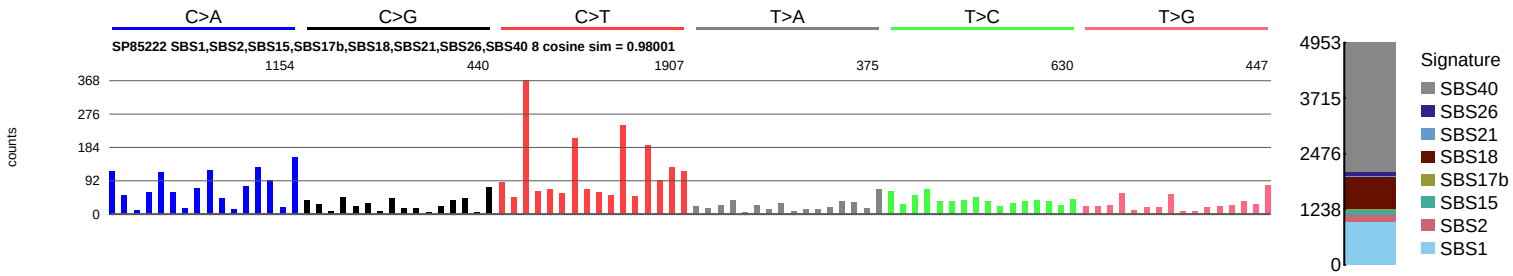

C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98007
1145
452
1873
429
574
475
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98006
1161
471
1910
402
605
404
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98004
1153
479
1862
401
625
430
counts
376
282
188
94
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS3,SBS18,SBS40,SBS44 6 cosine sim = 0.98003
1164
480
1918
410
577
403
counts
372
279
186
93
0
4953
3715
2476
1238
0
Signature
SBS44
SBS40
SBS18
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98001
1159
450
1917
387
589
450
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98001
1154
440
1907
375
630
447
counts
368
276
184
92
0
4953
3715
2476
1238
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1